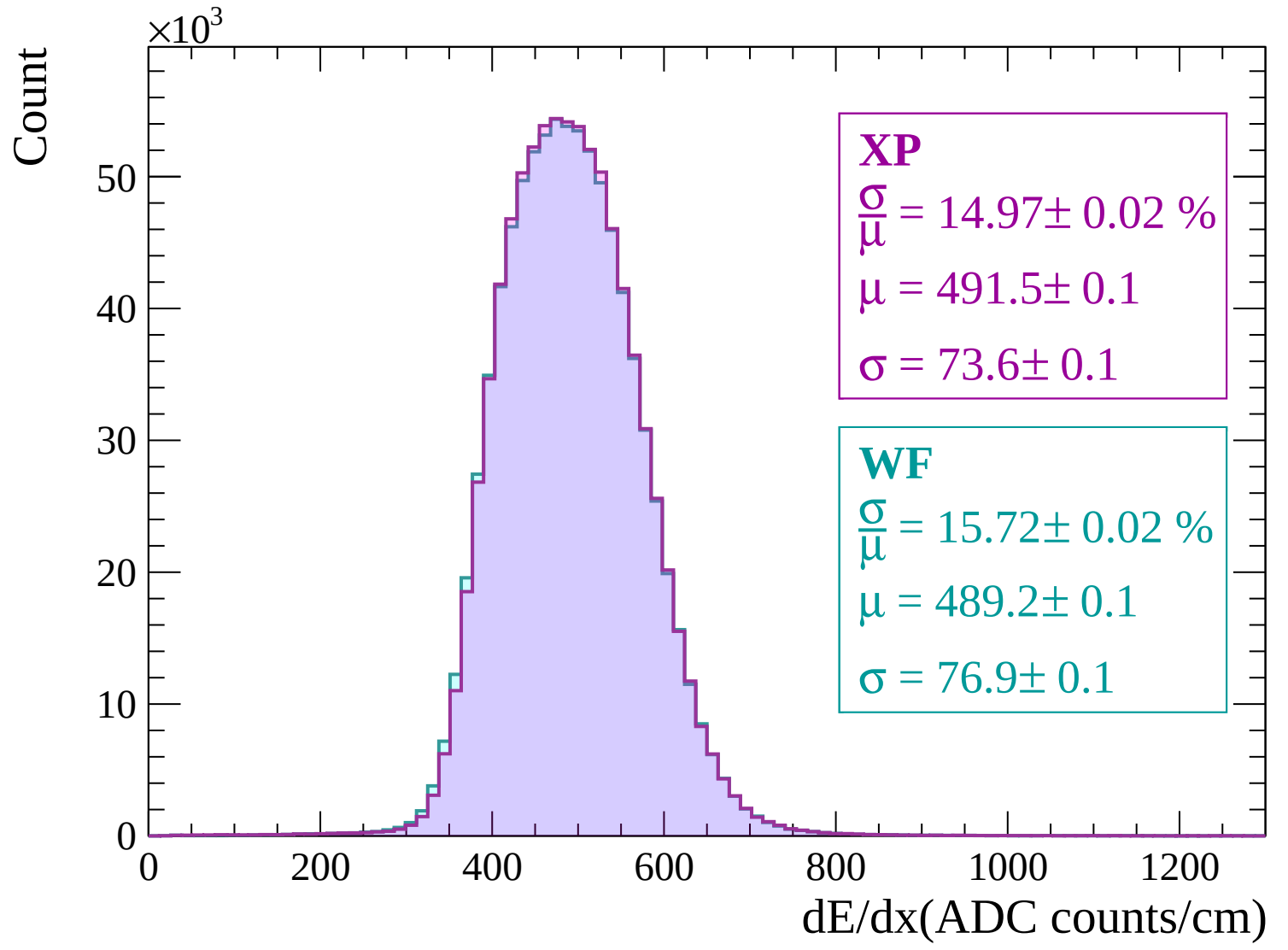
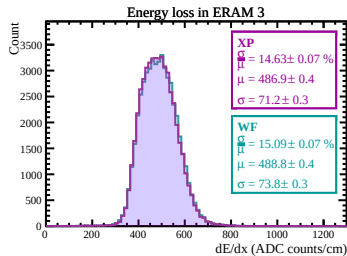
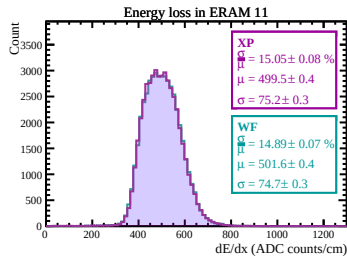
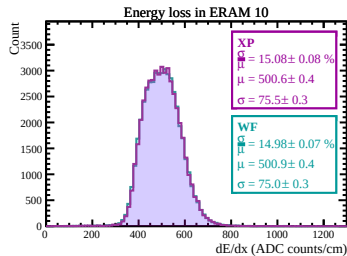
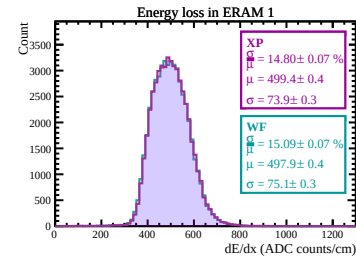
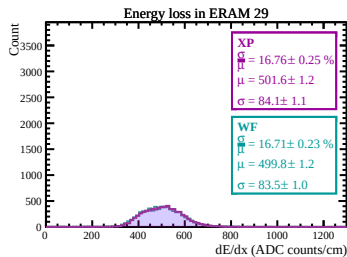
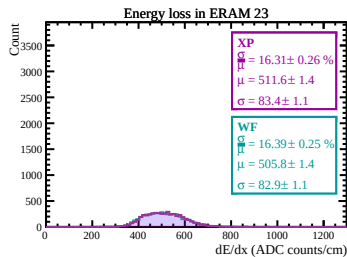
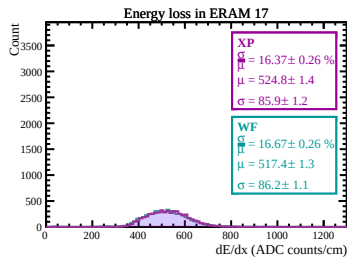
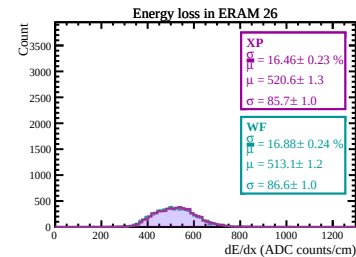
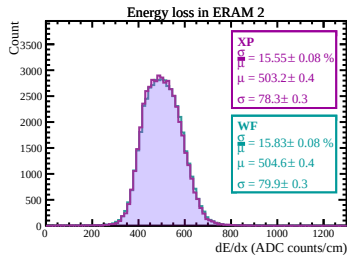
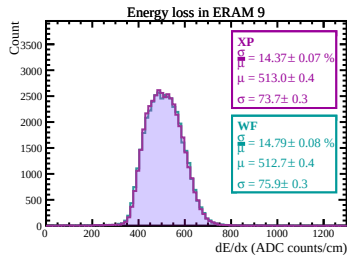
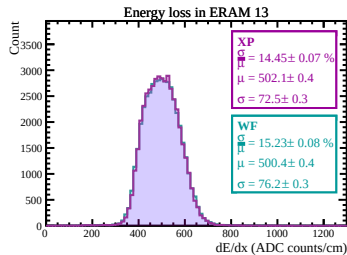
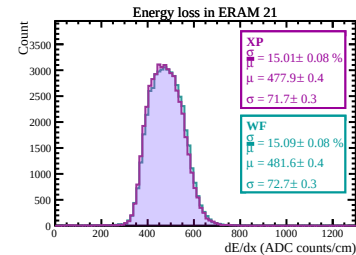
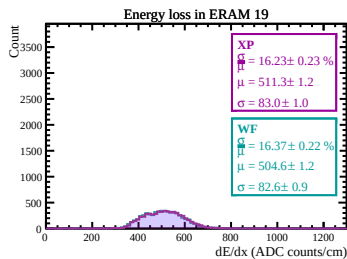
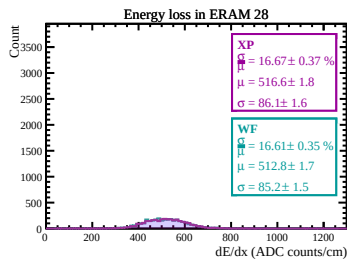
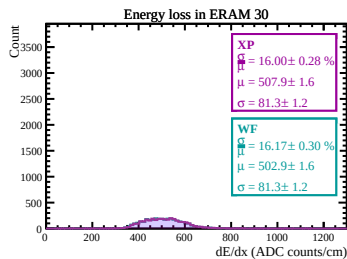
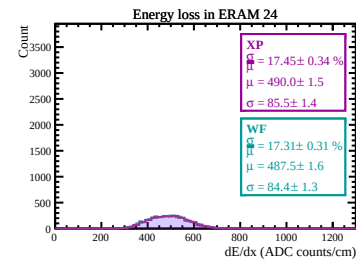
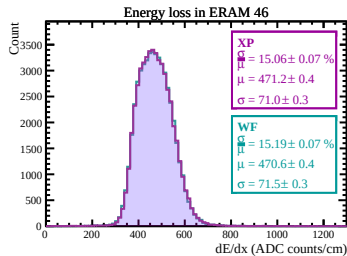
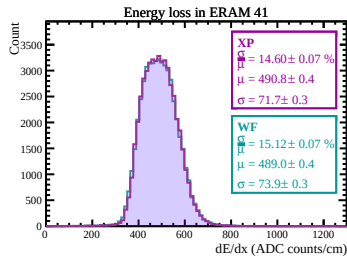
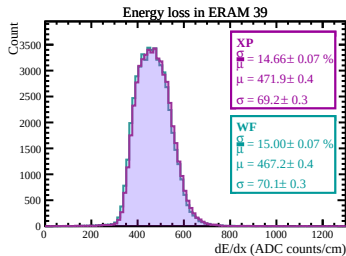
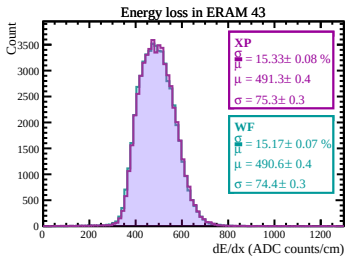
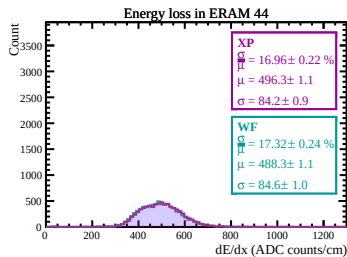
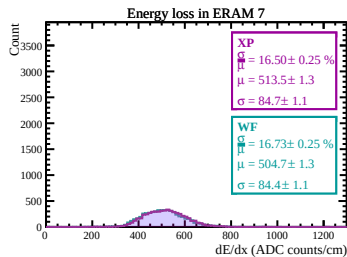
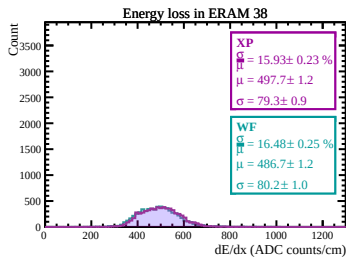
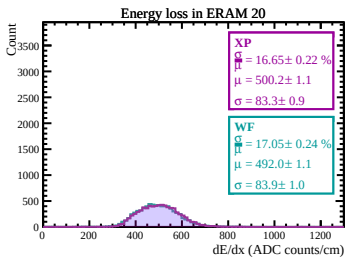
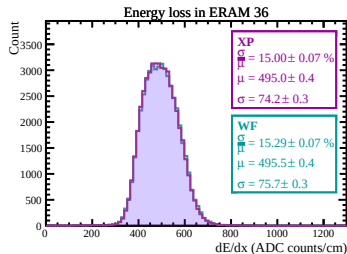
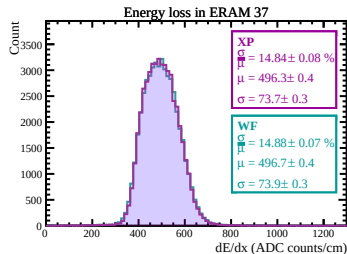
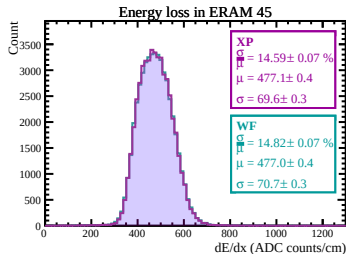
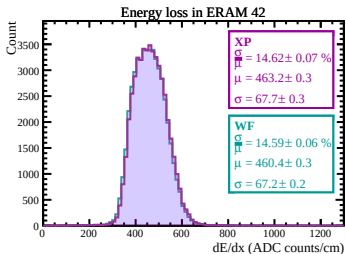
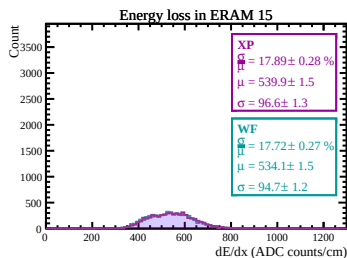
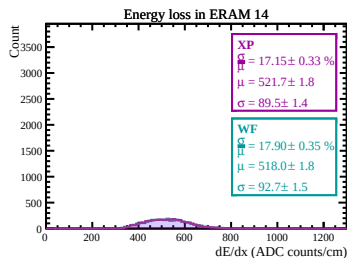
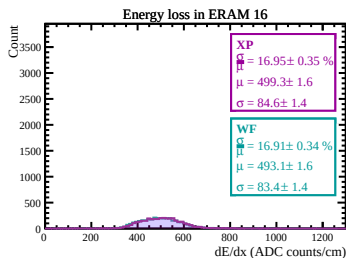
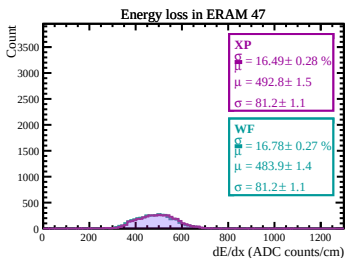
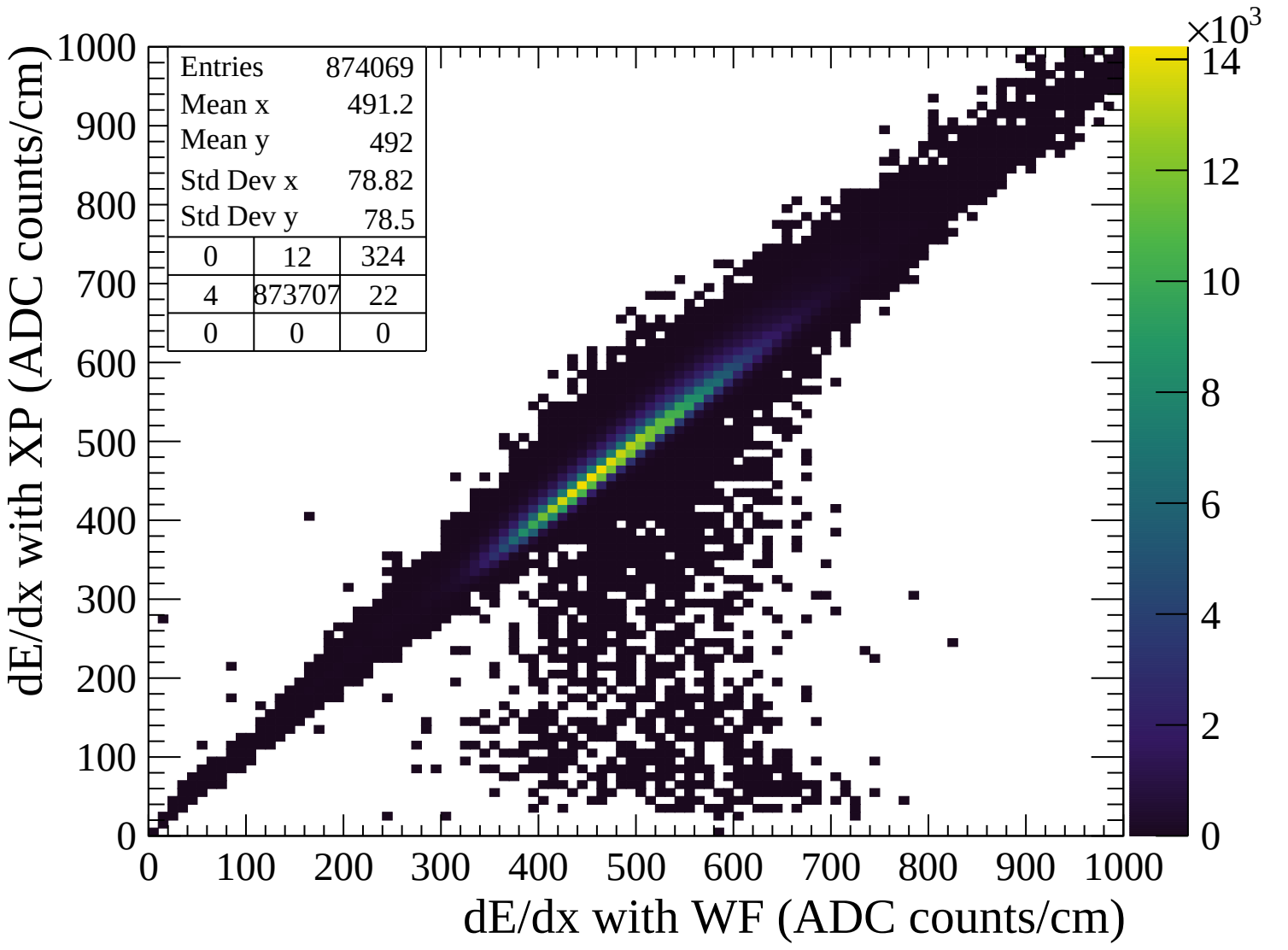
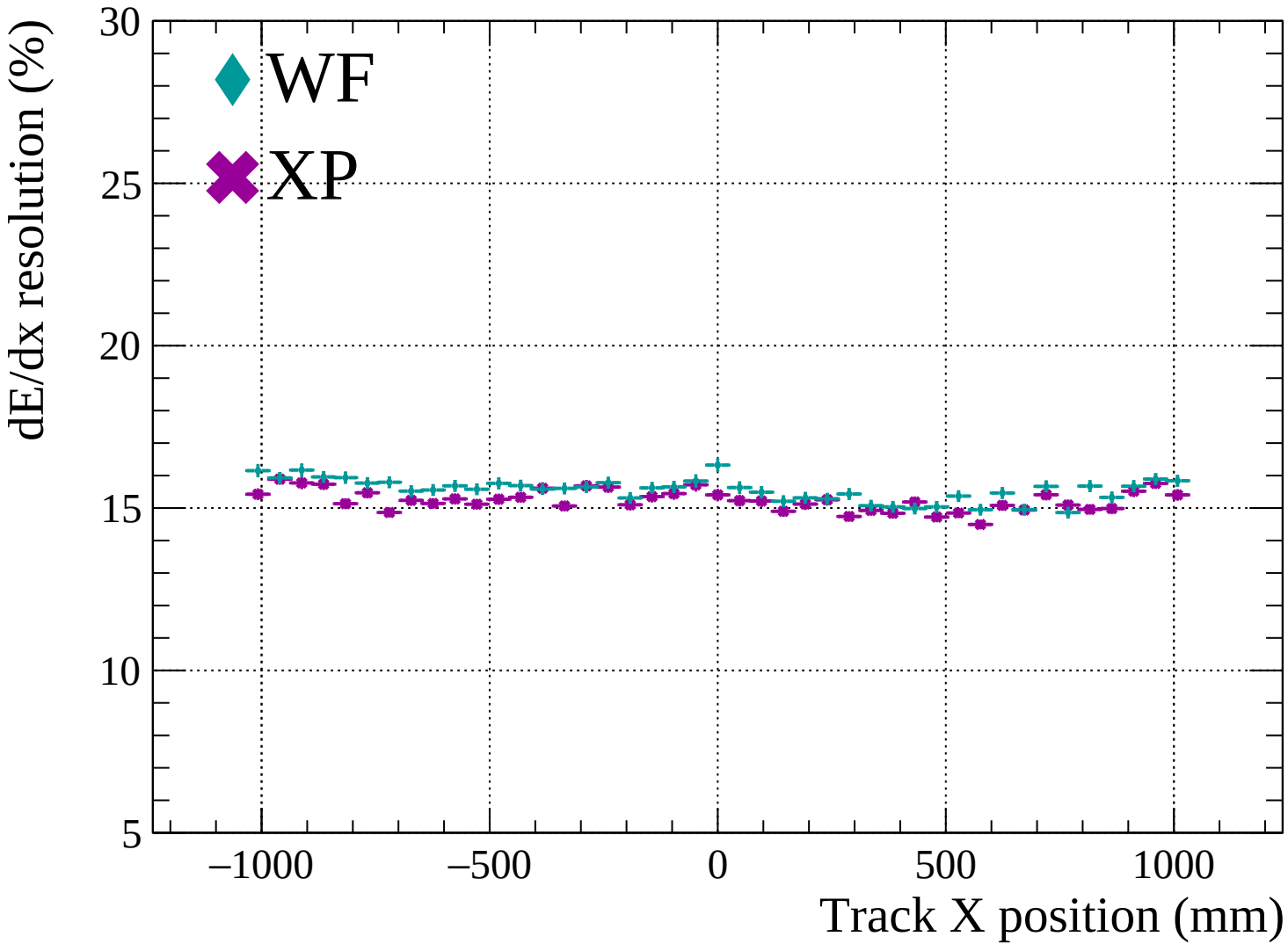


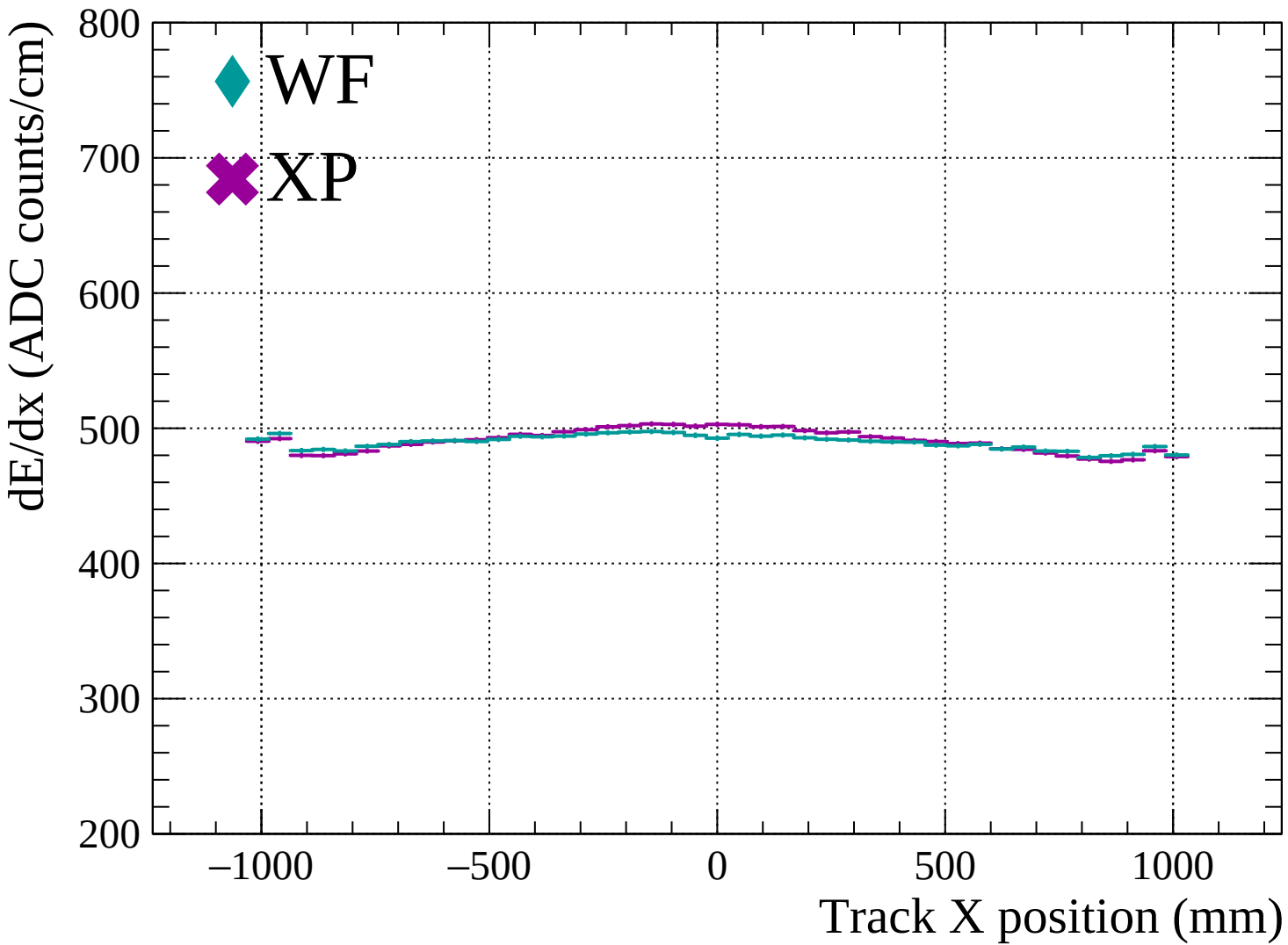


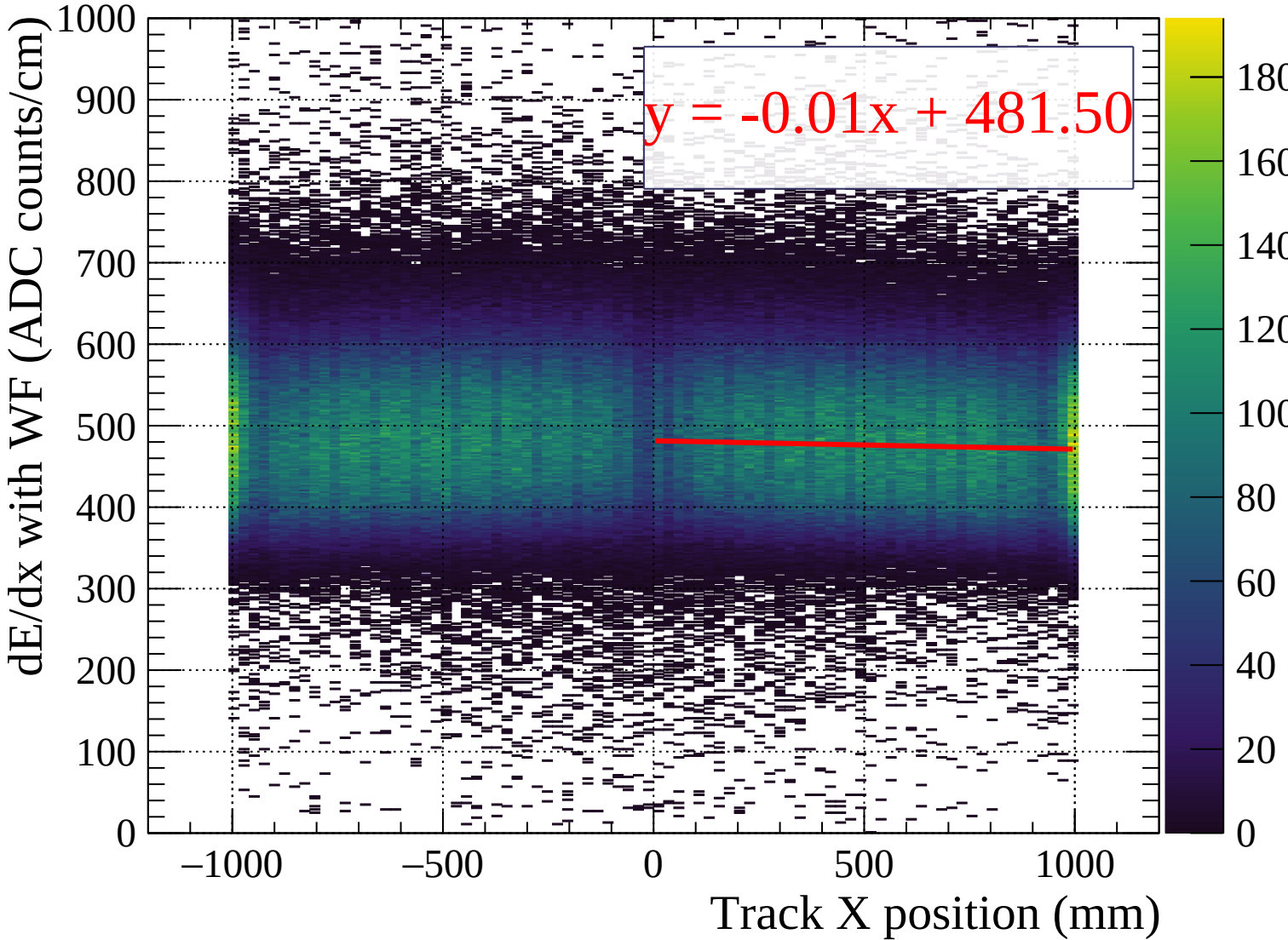


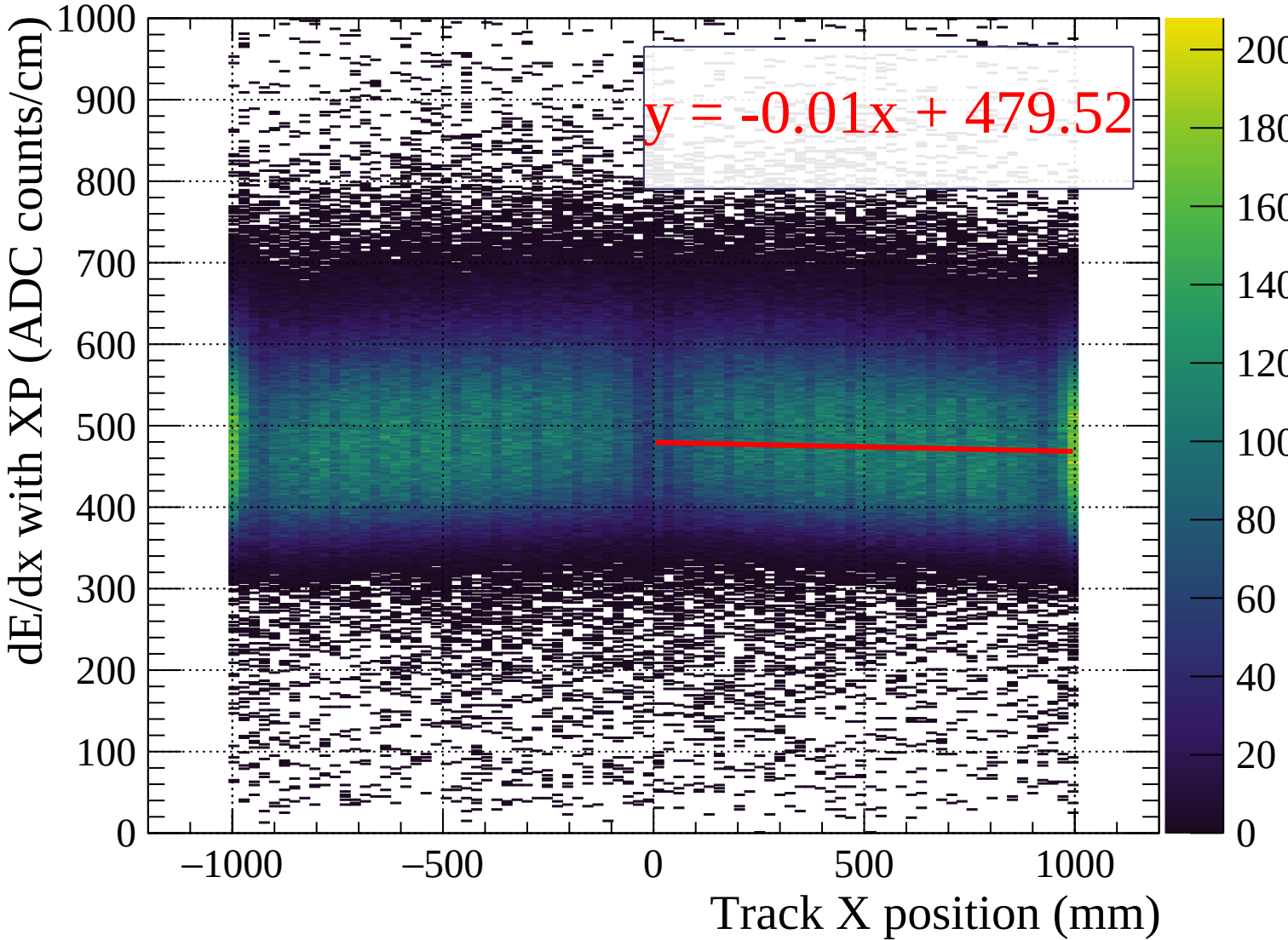


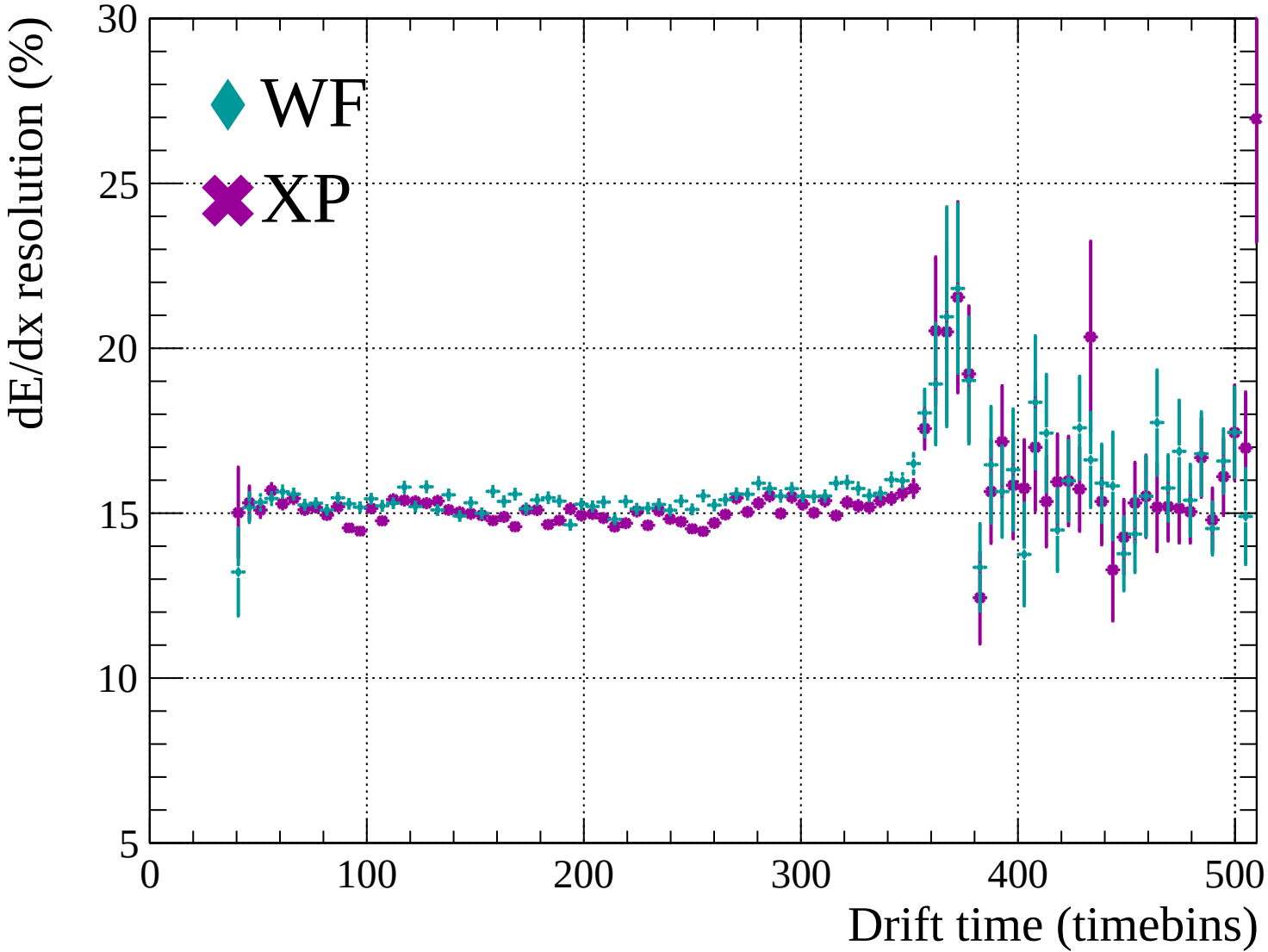


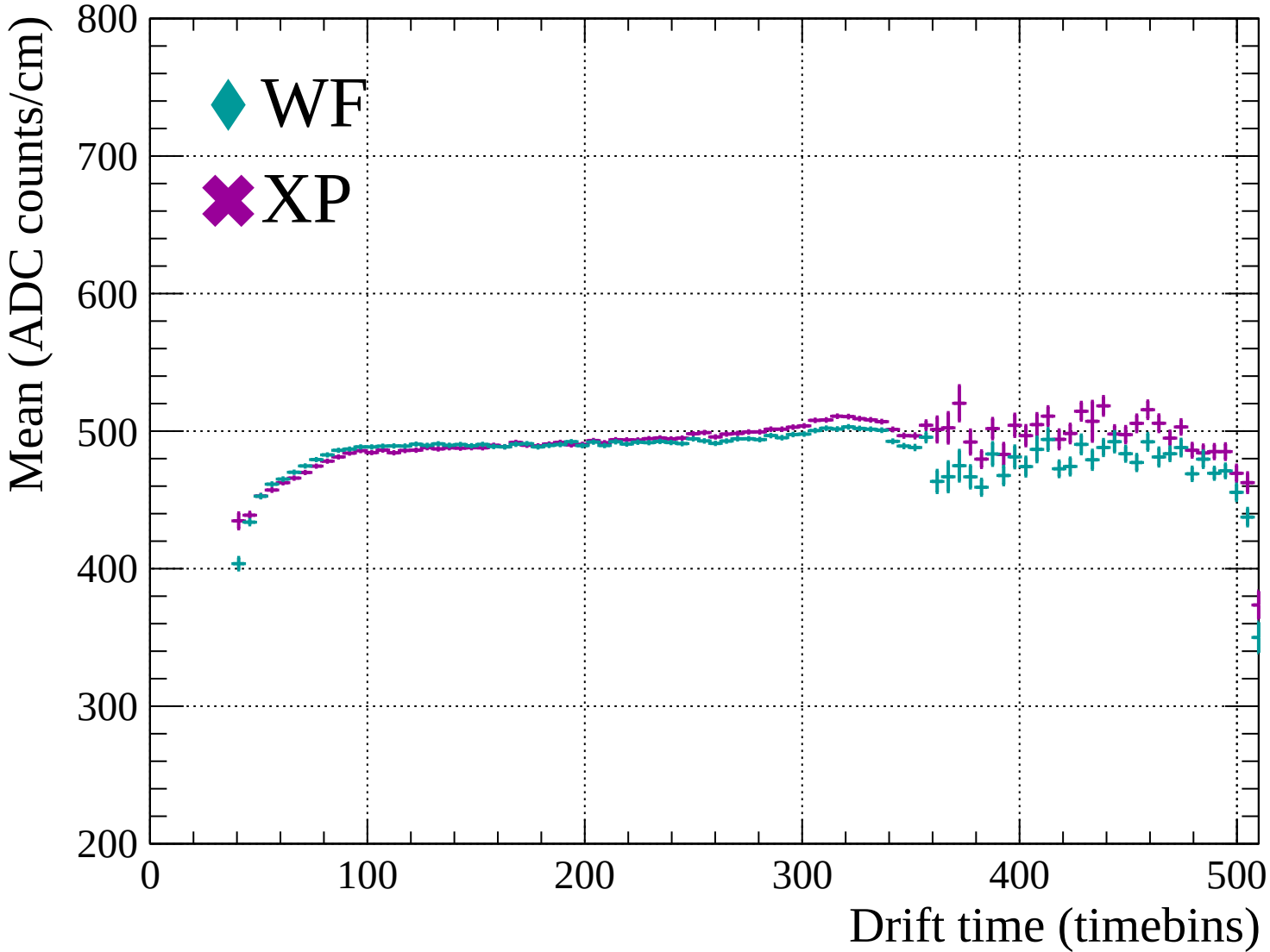


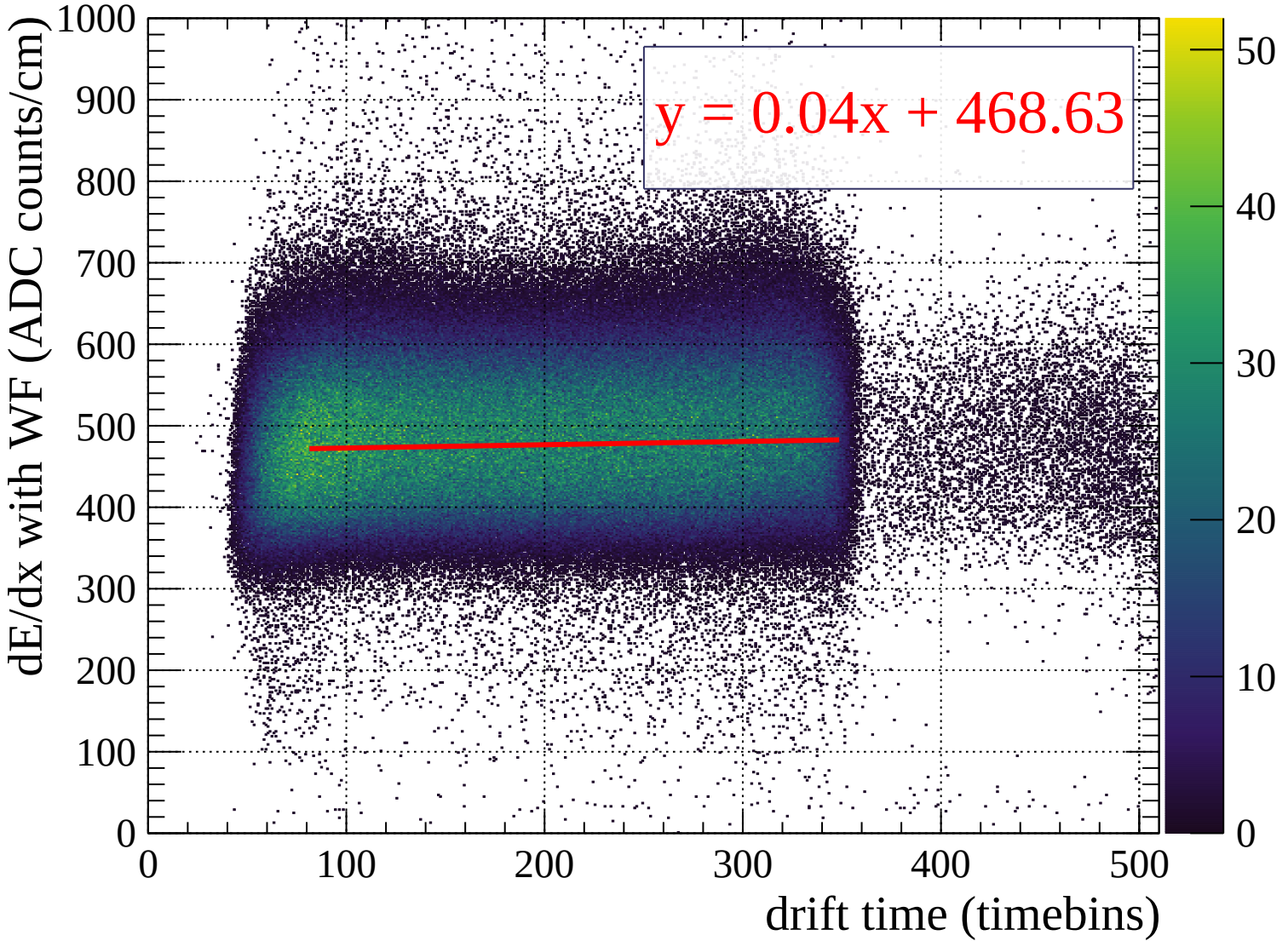


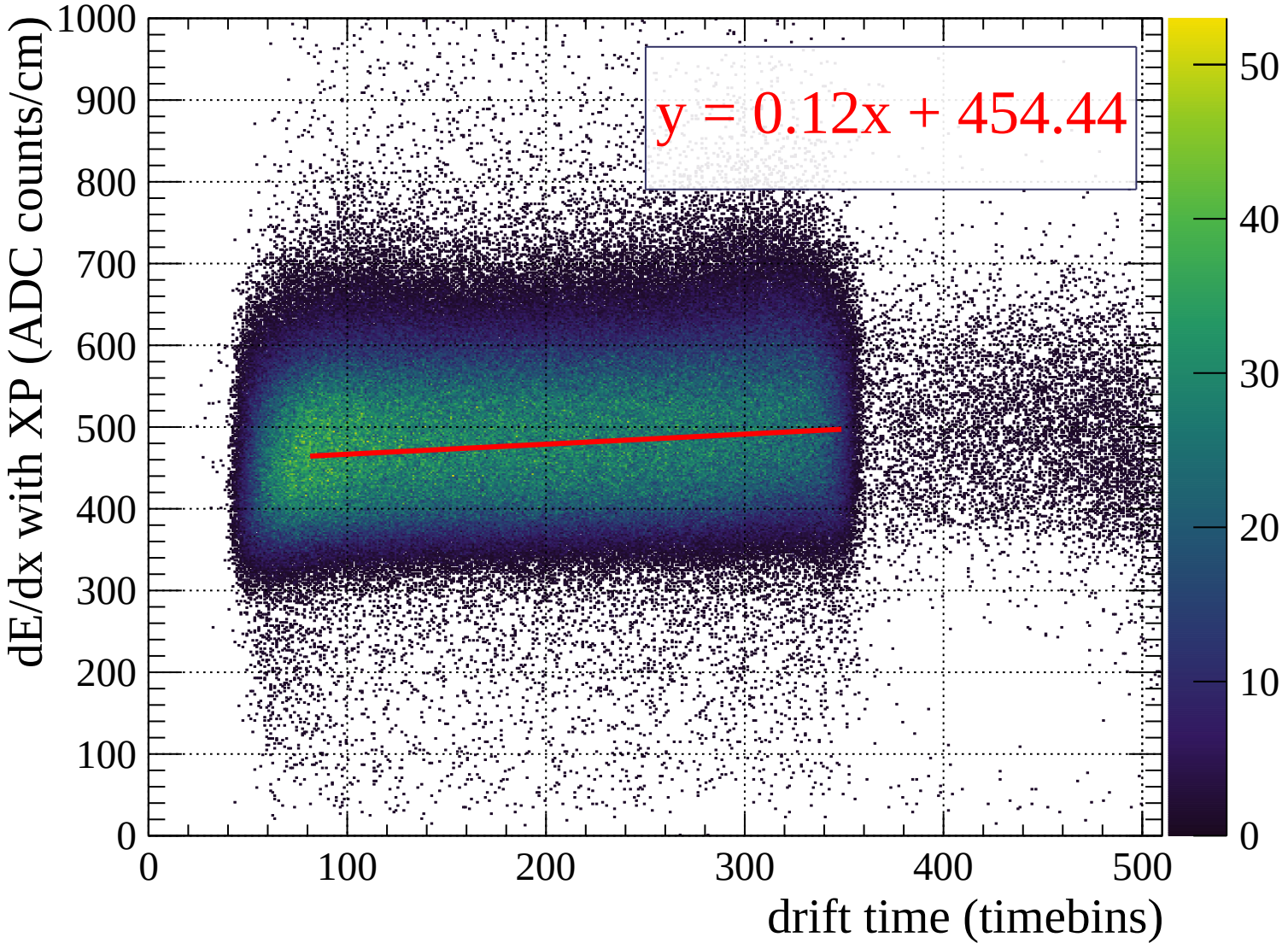


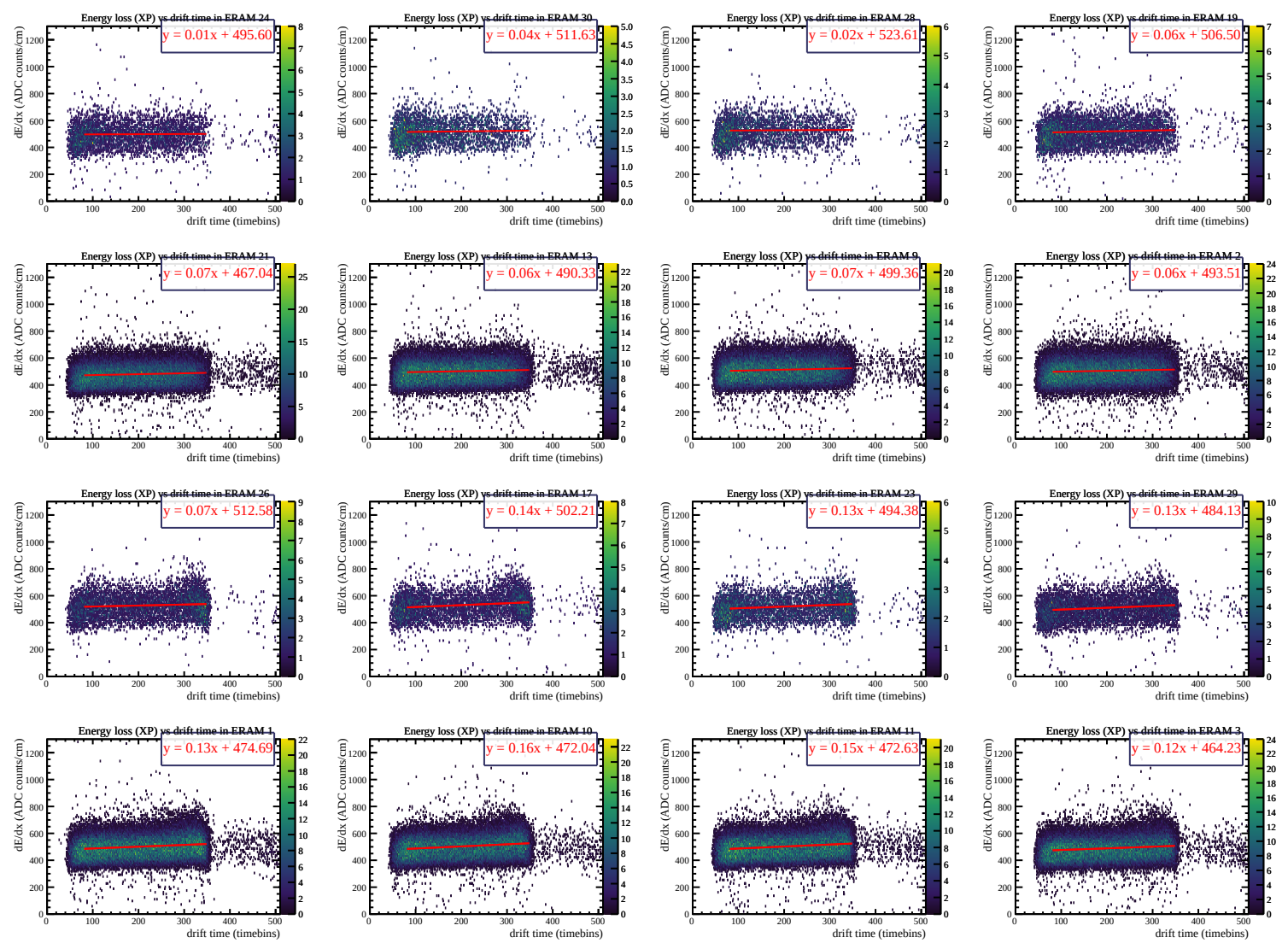


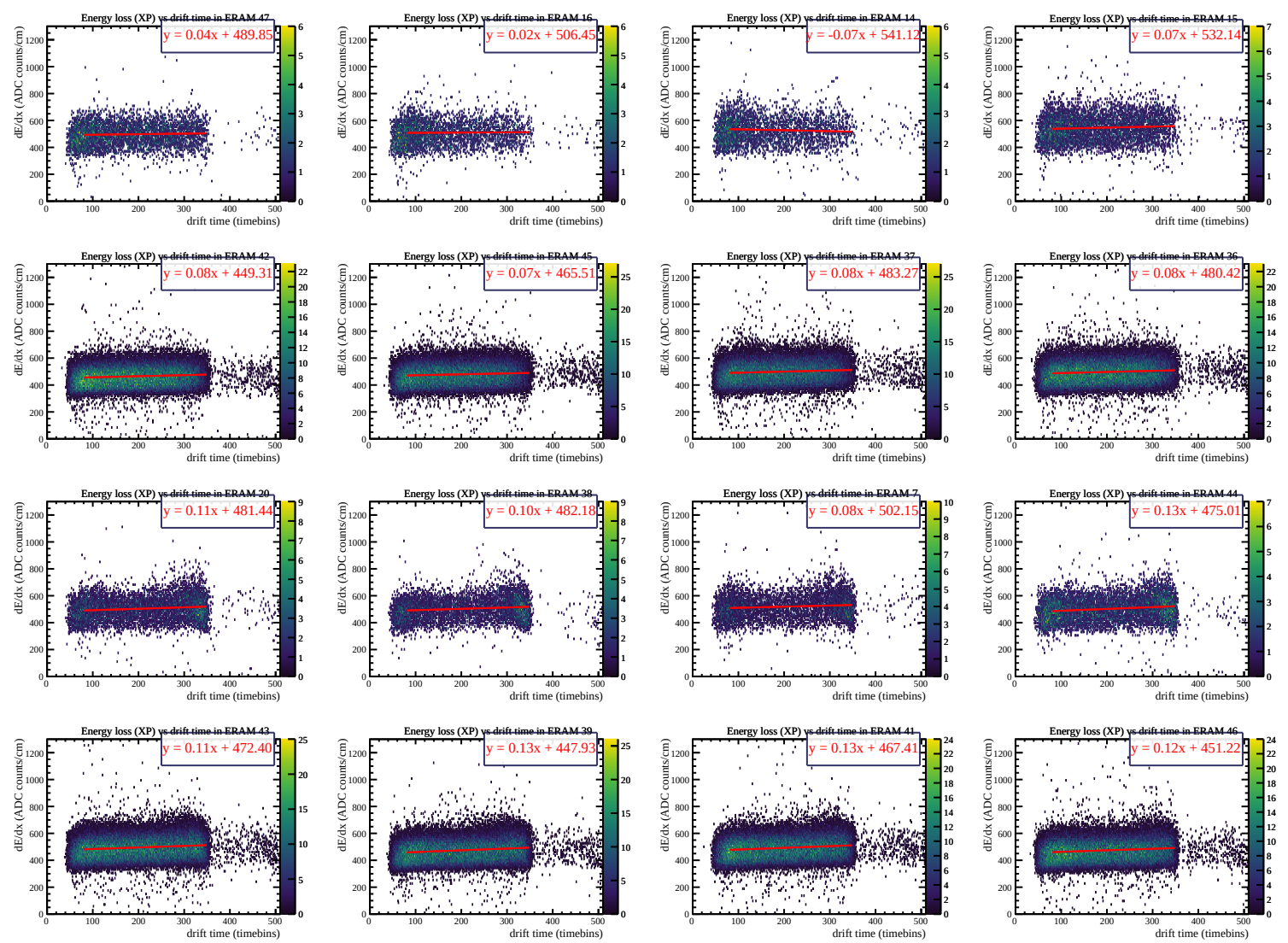


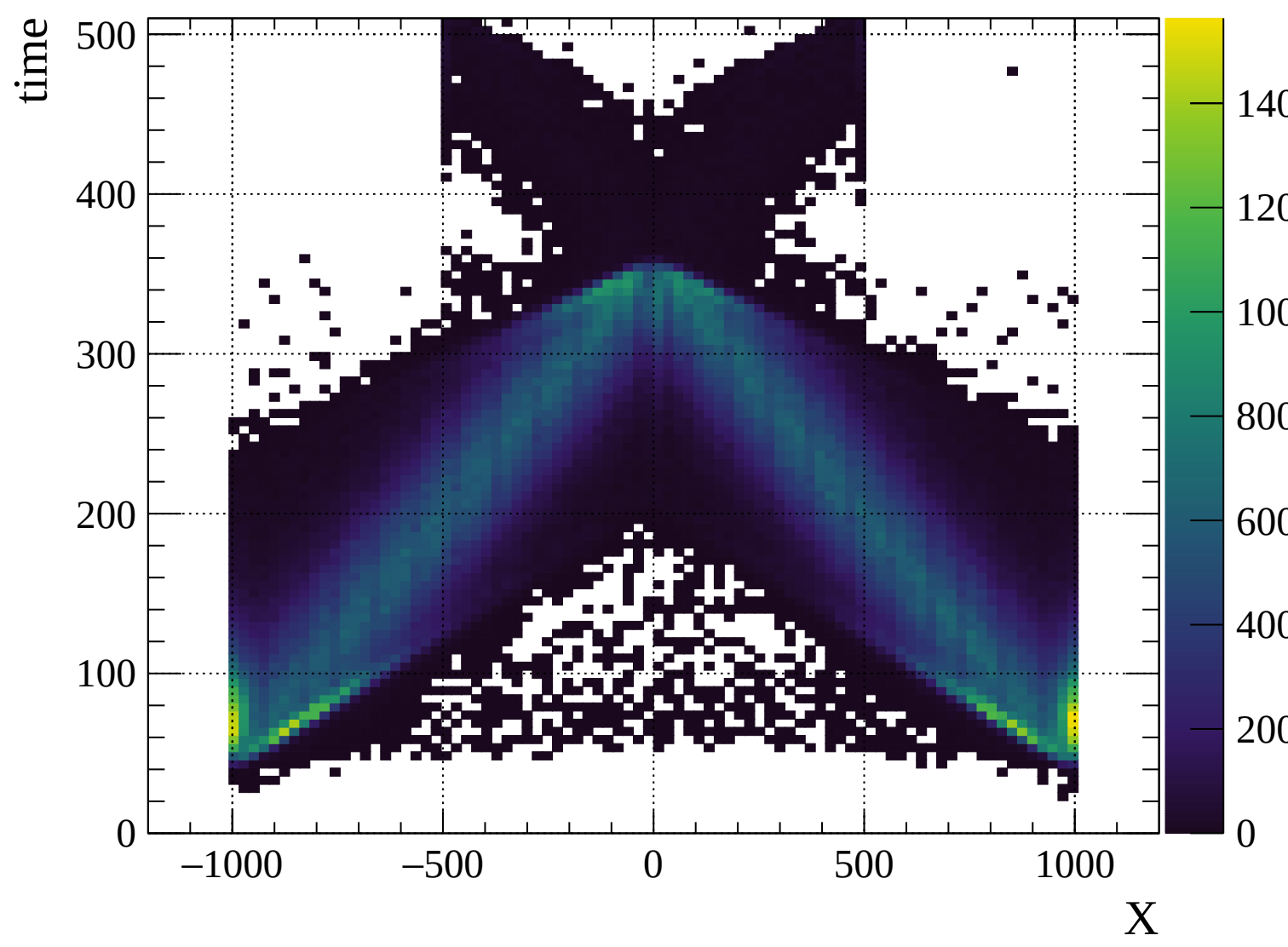


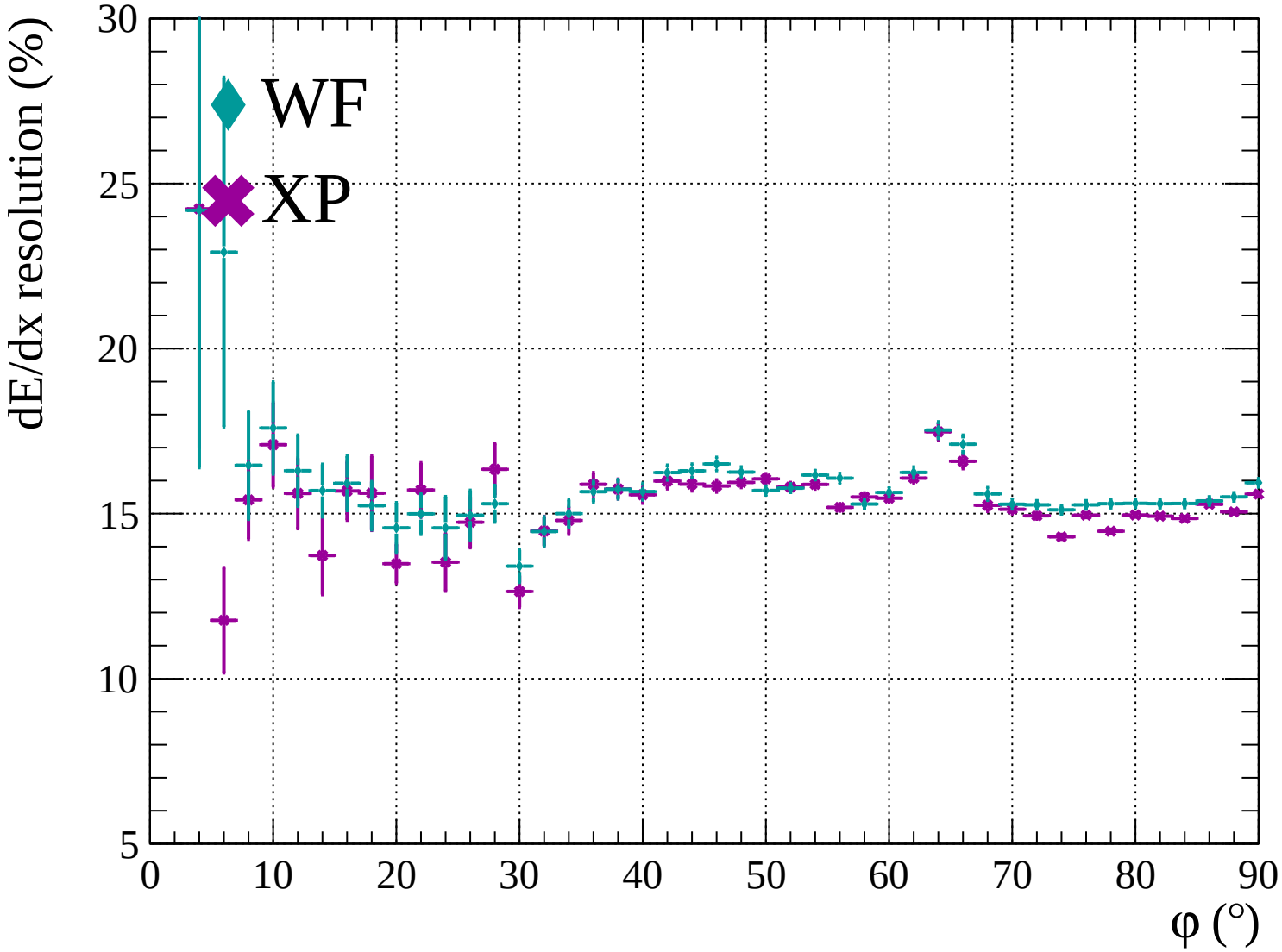


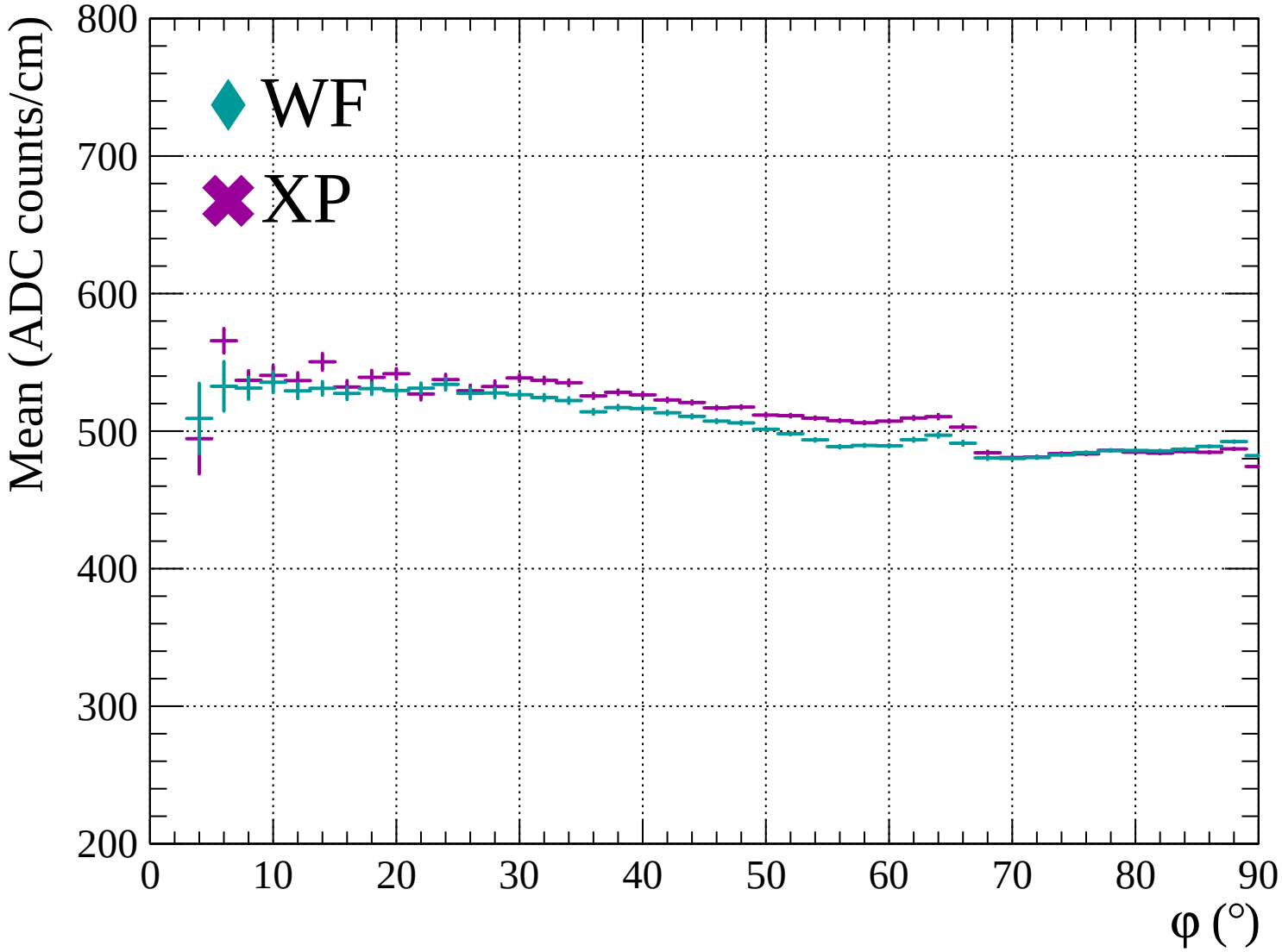


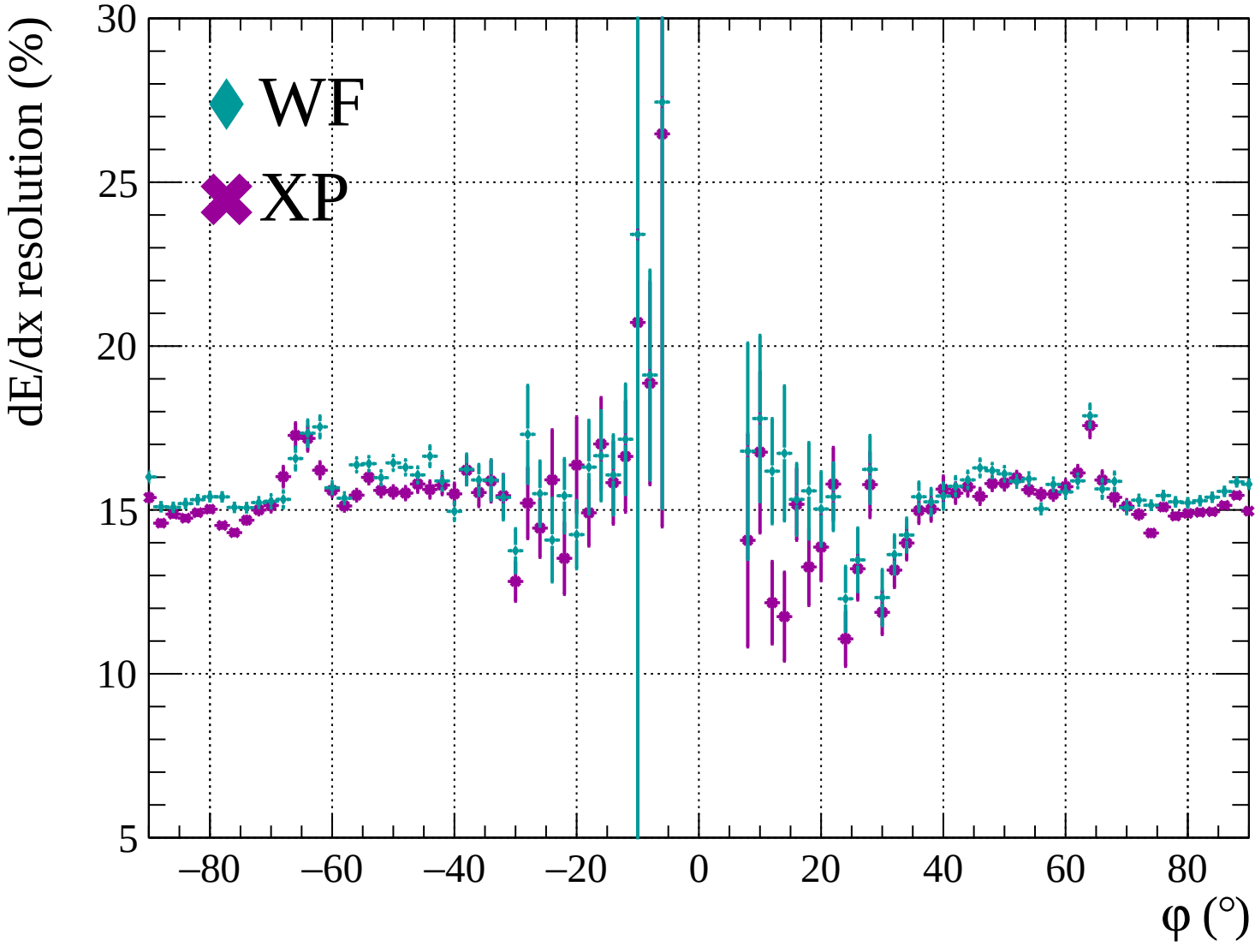


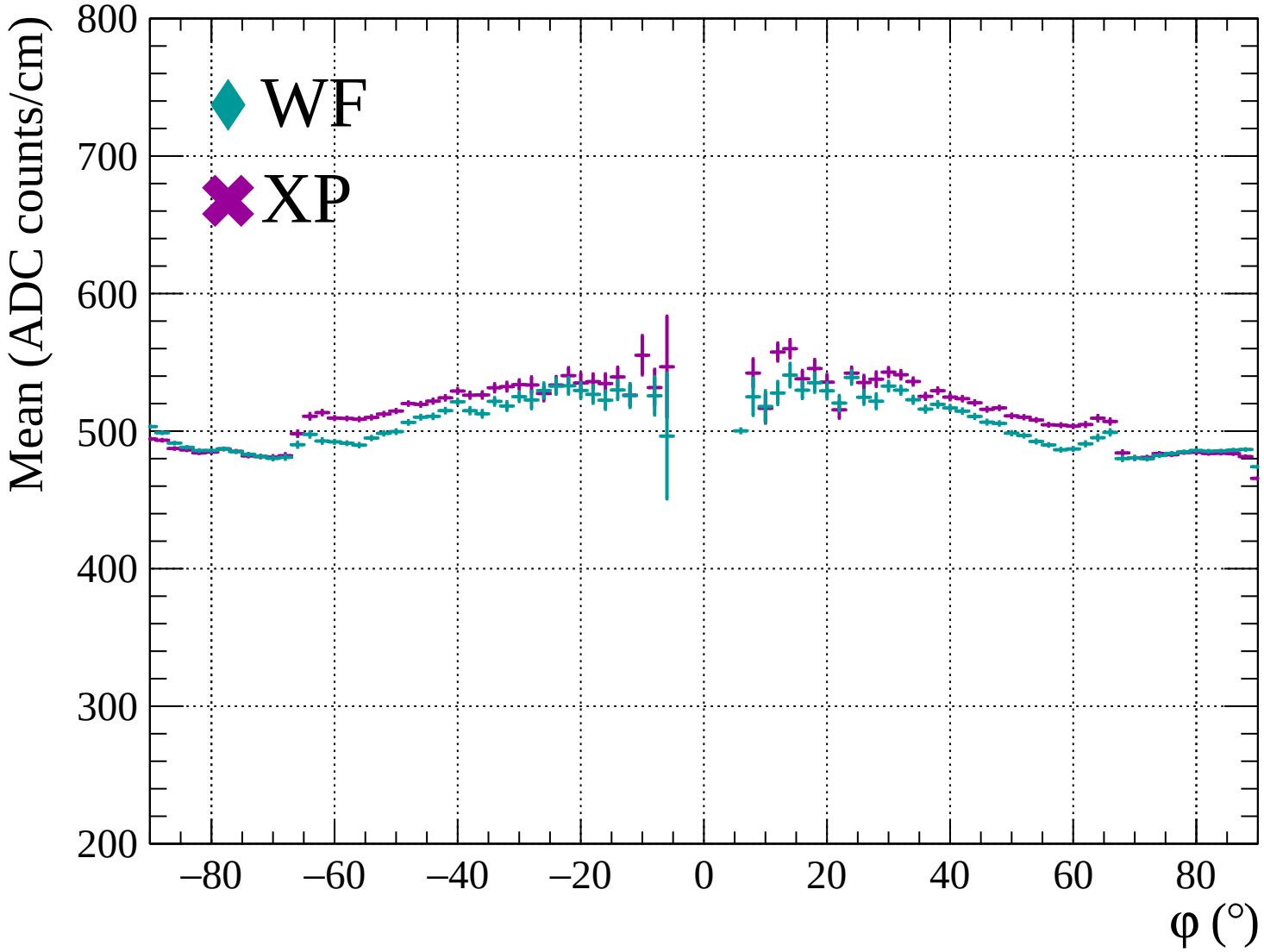


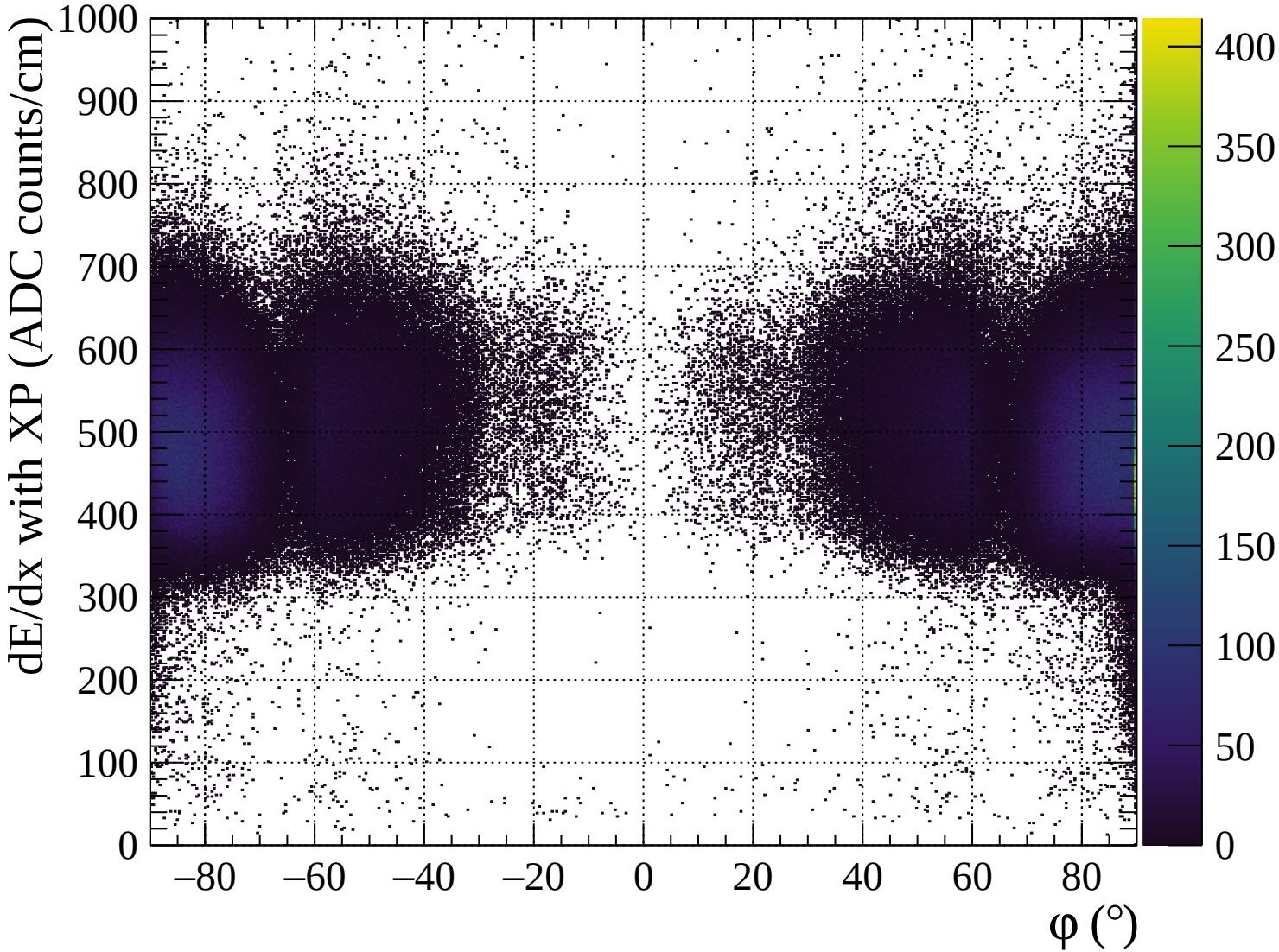


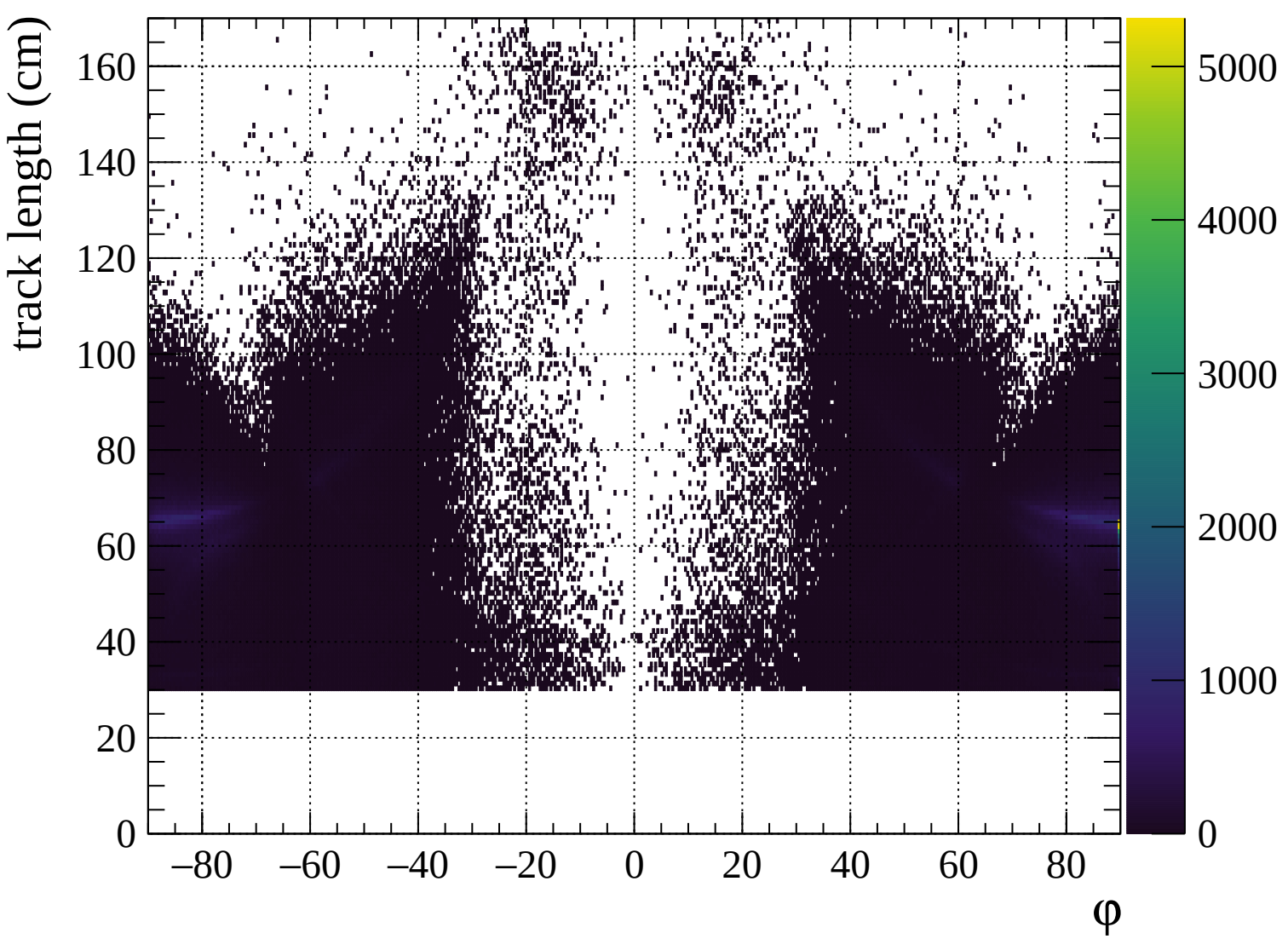


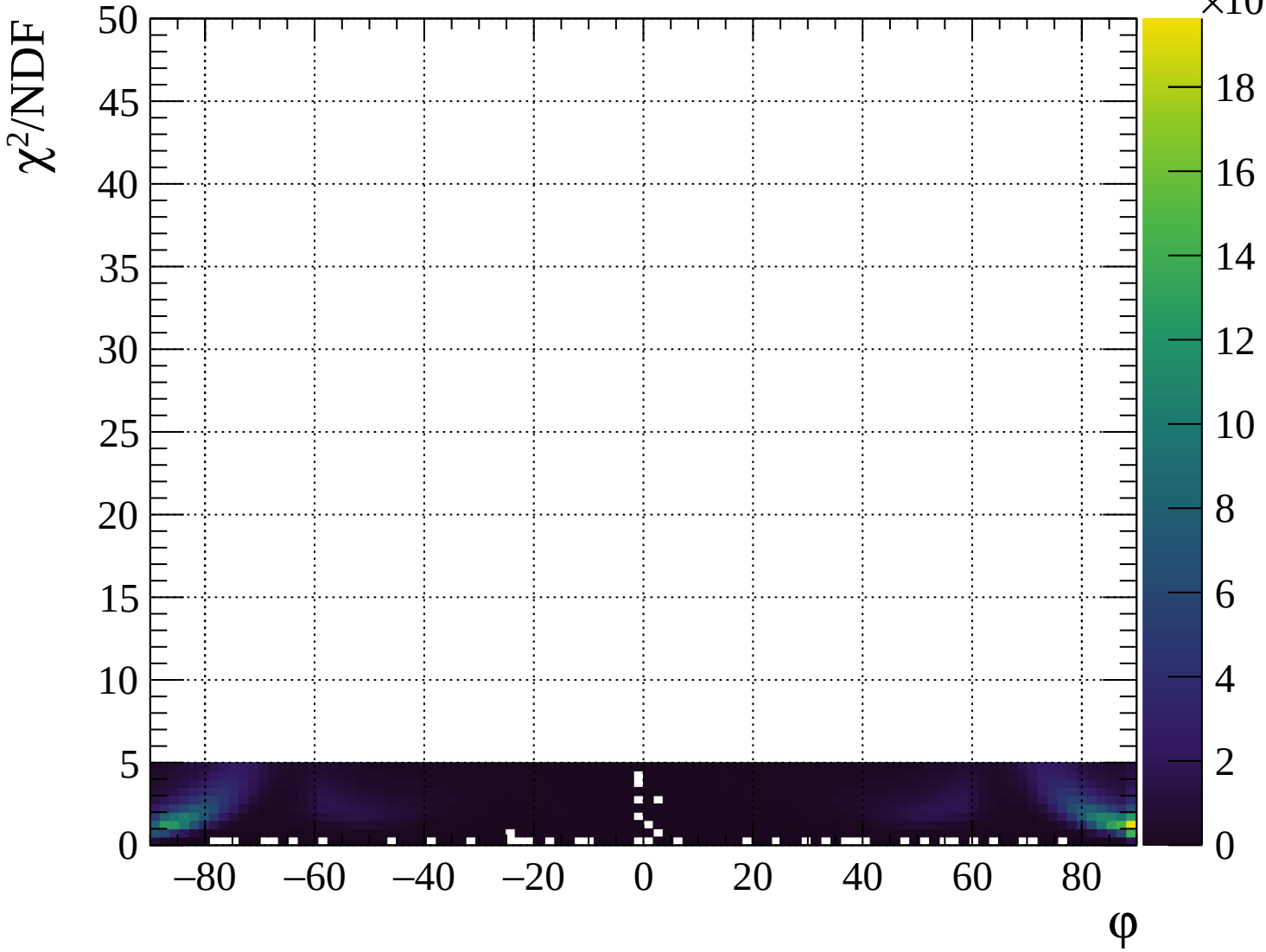


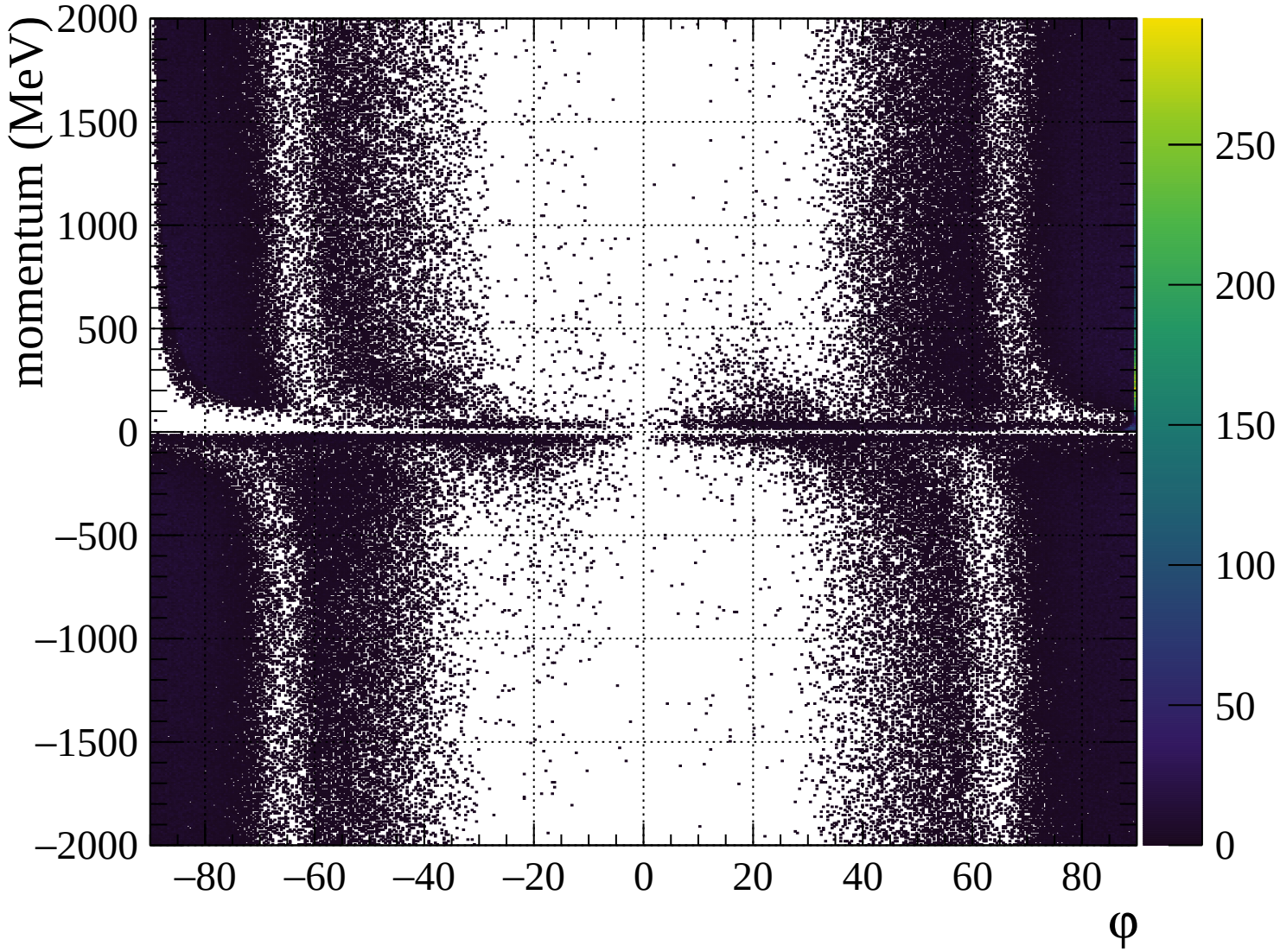


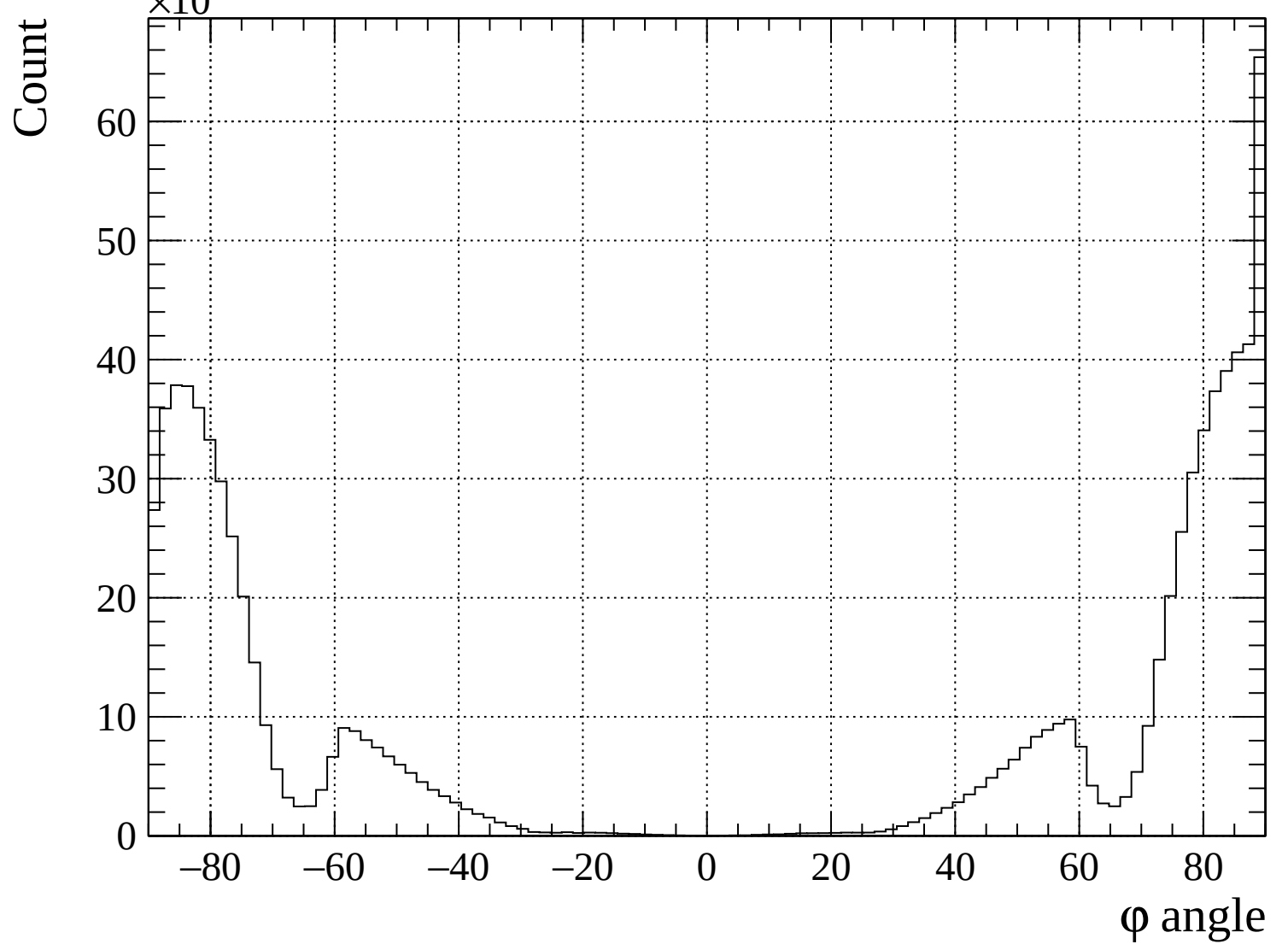


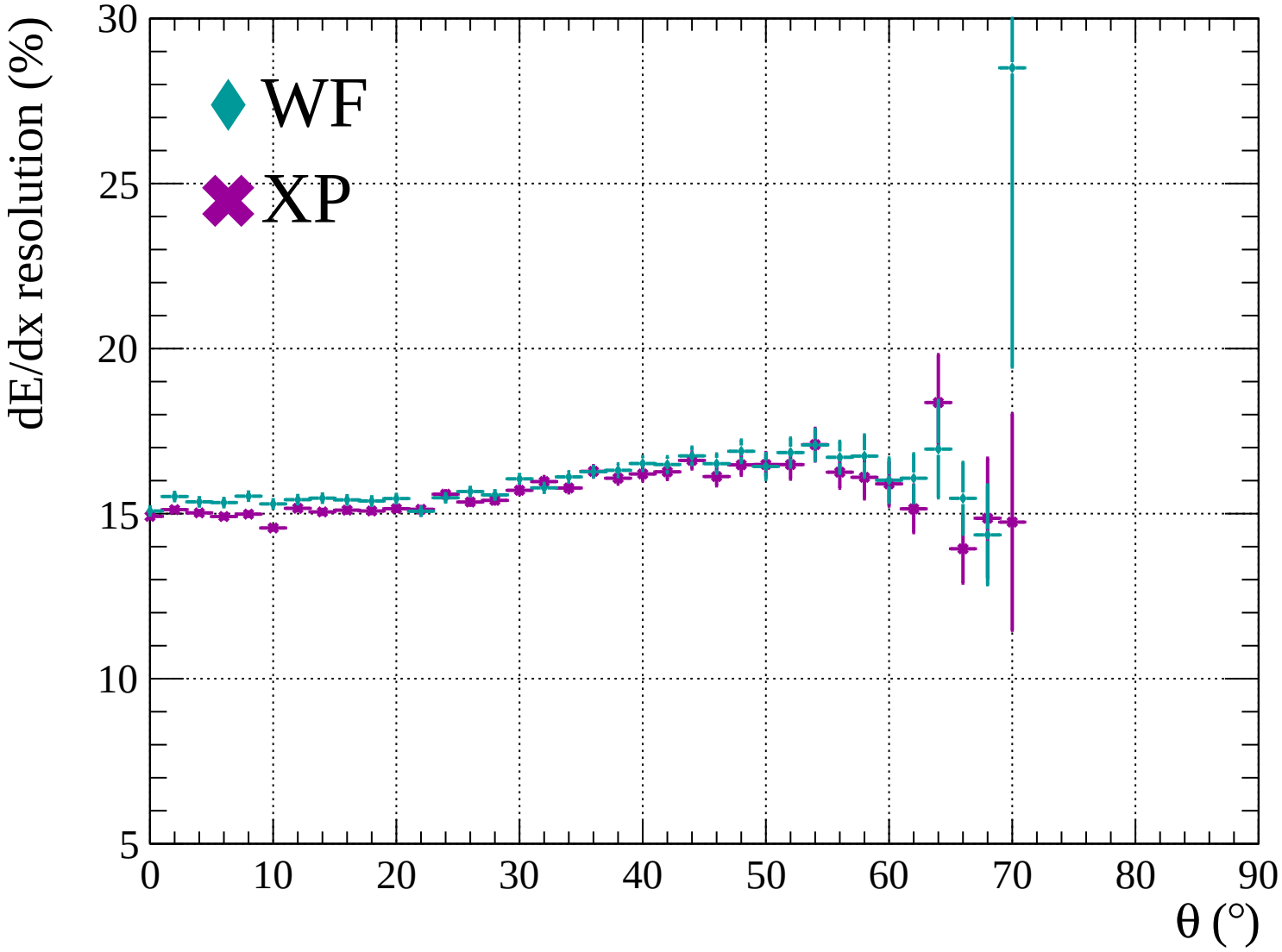


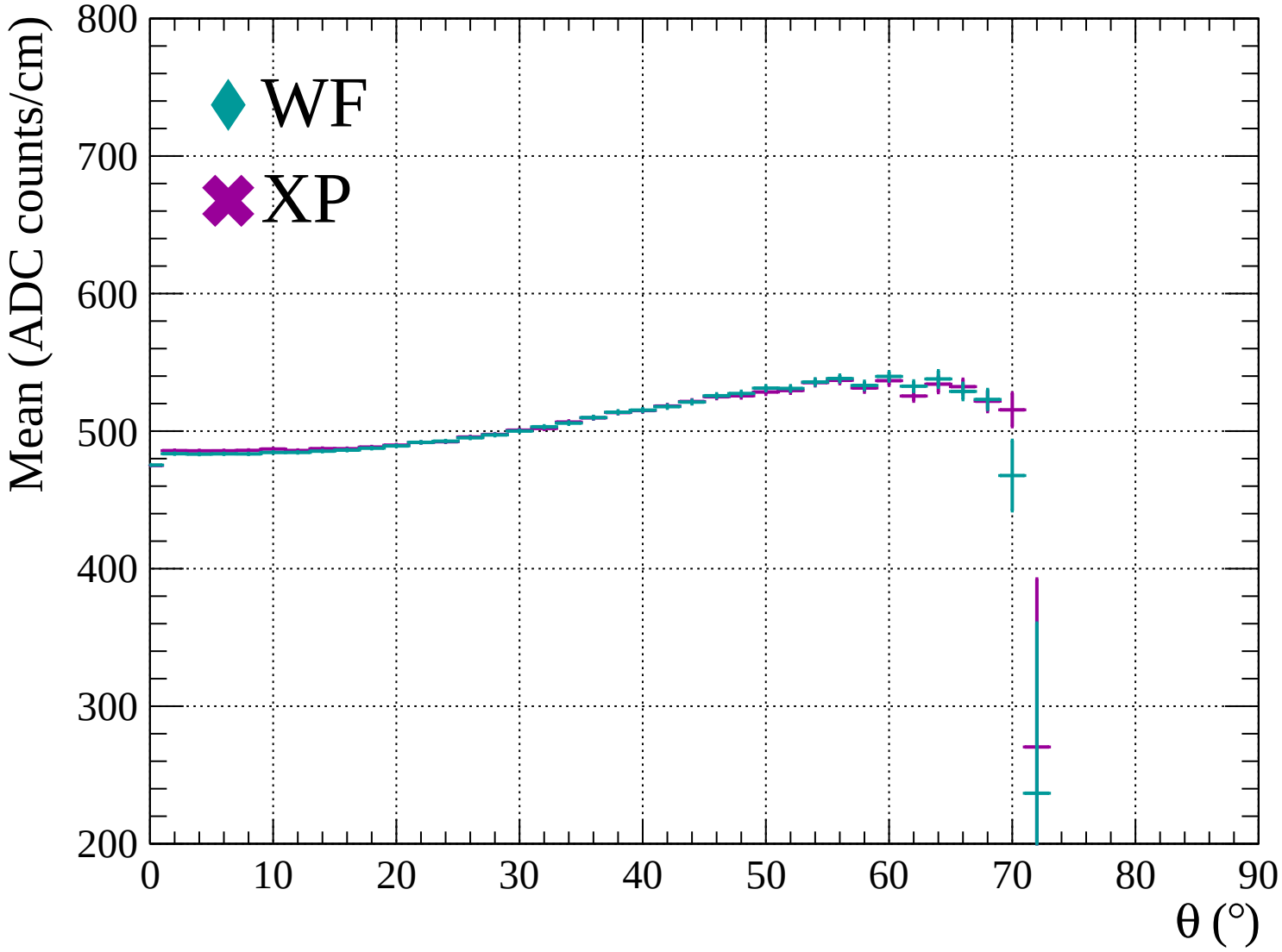


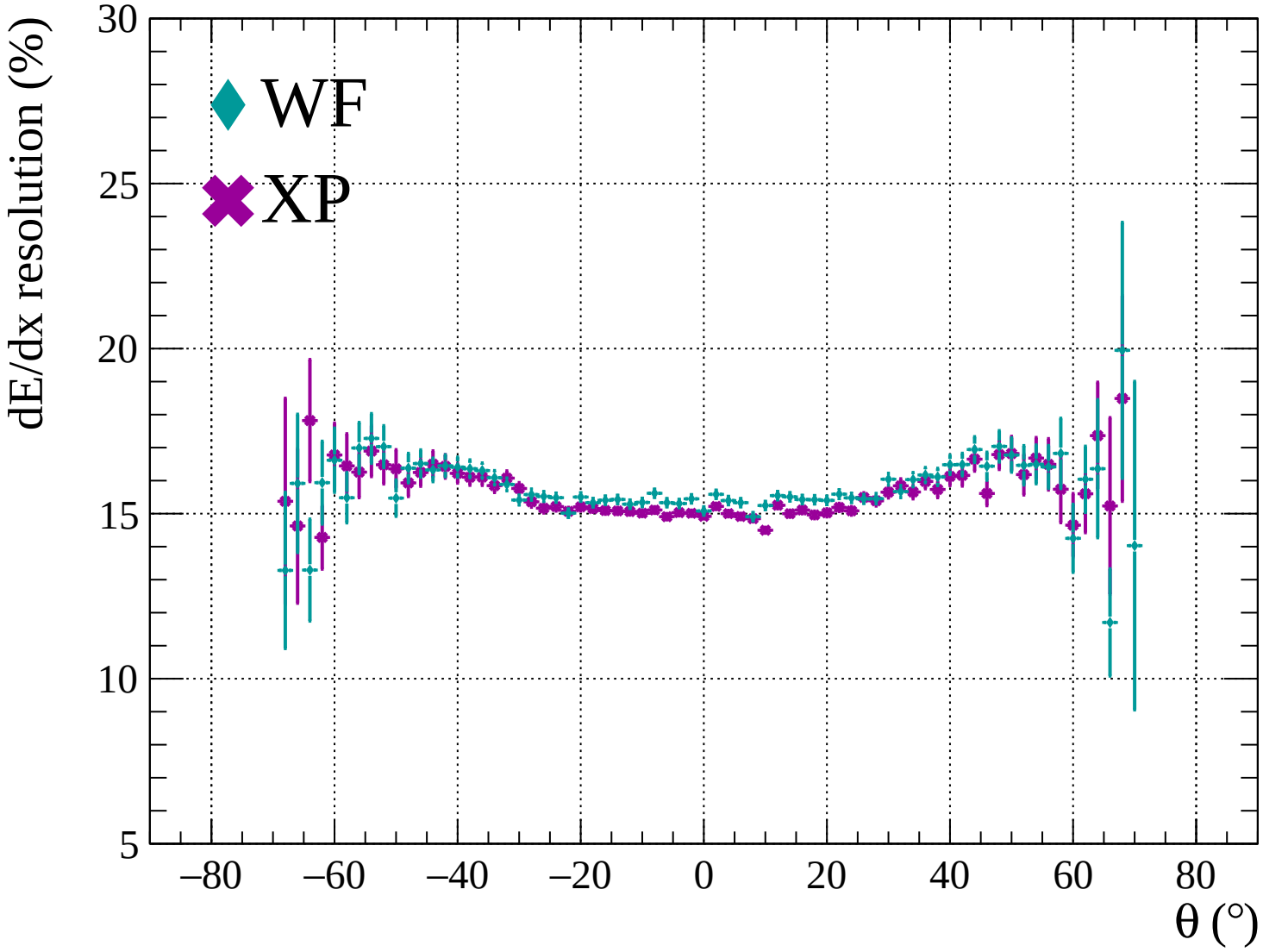


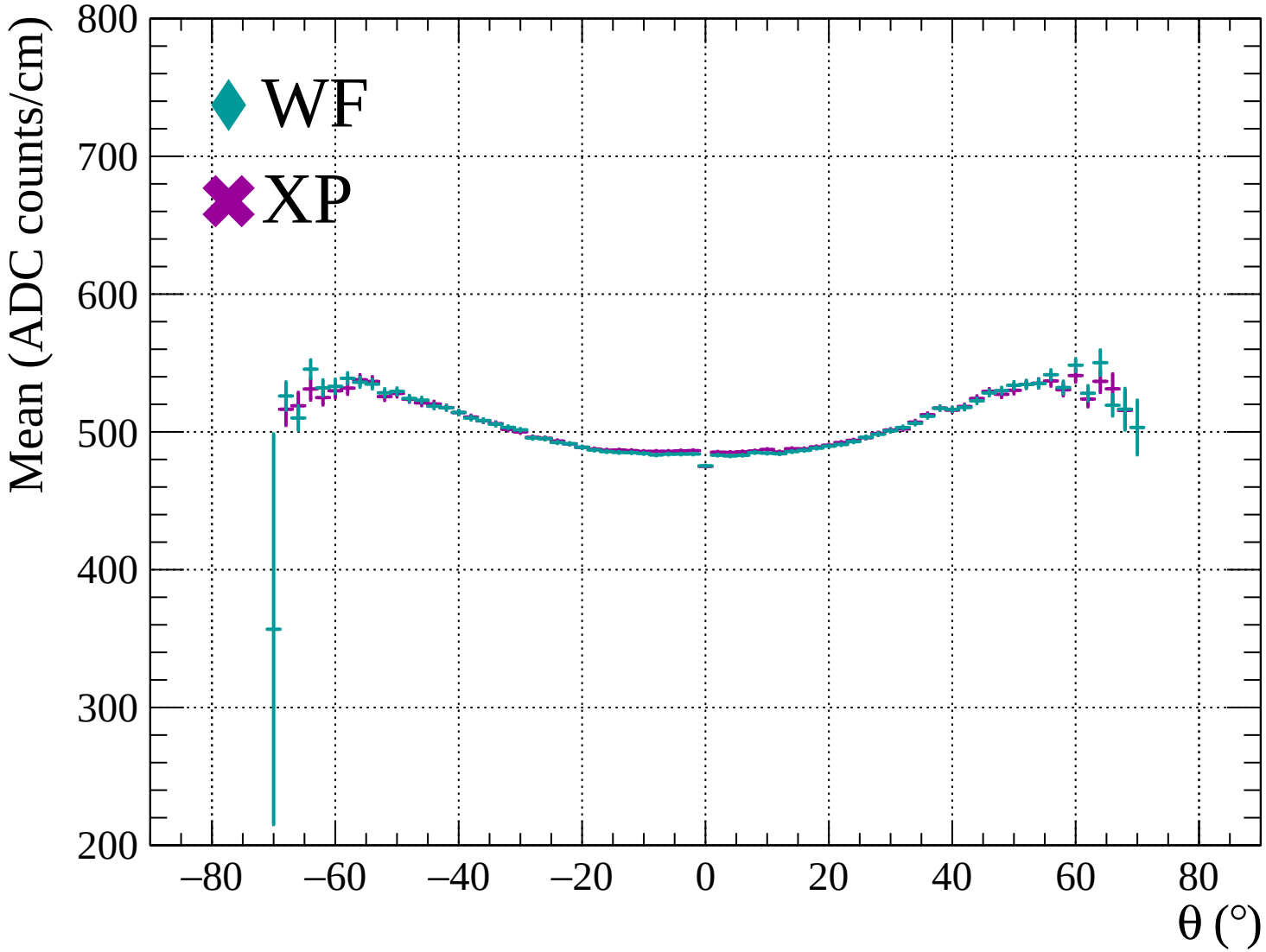


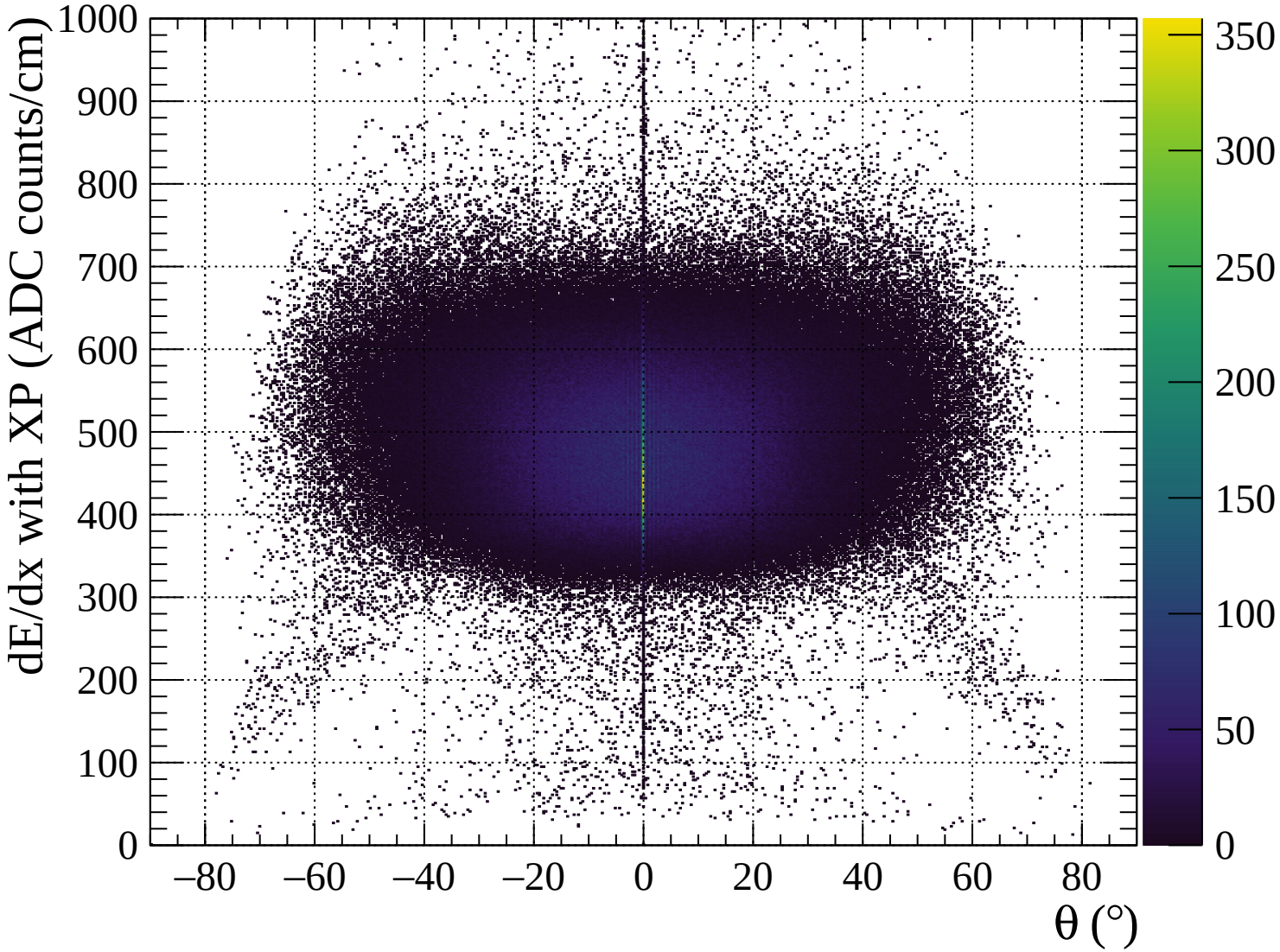


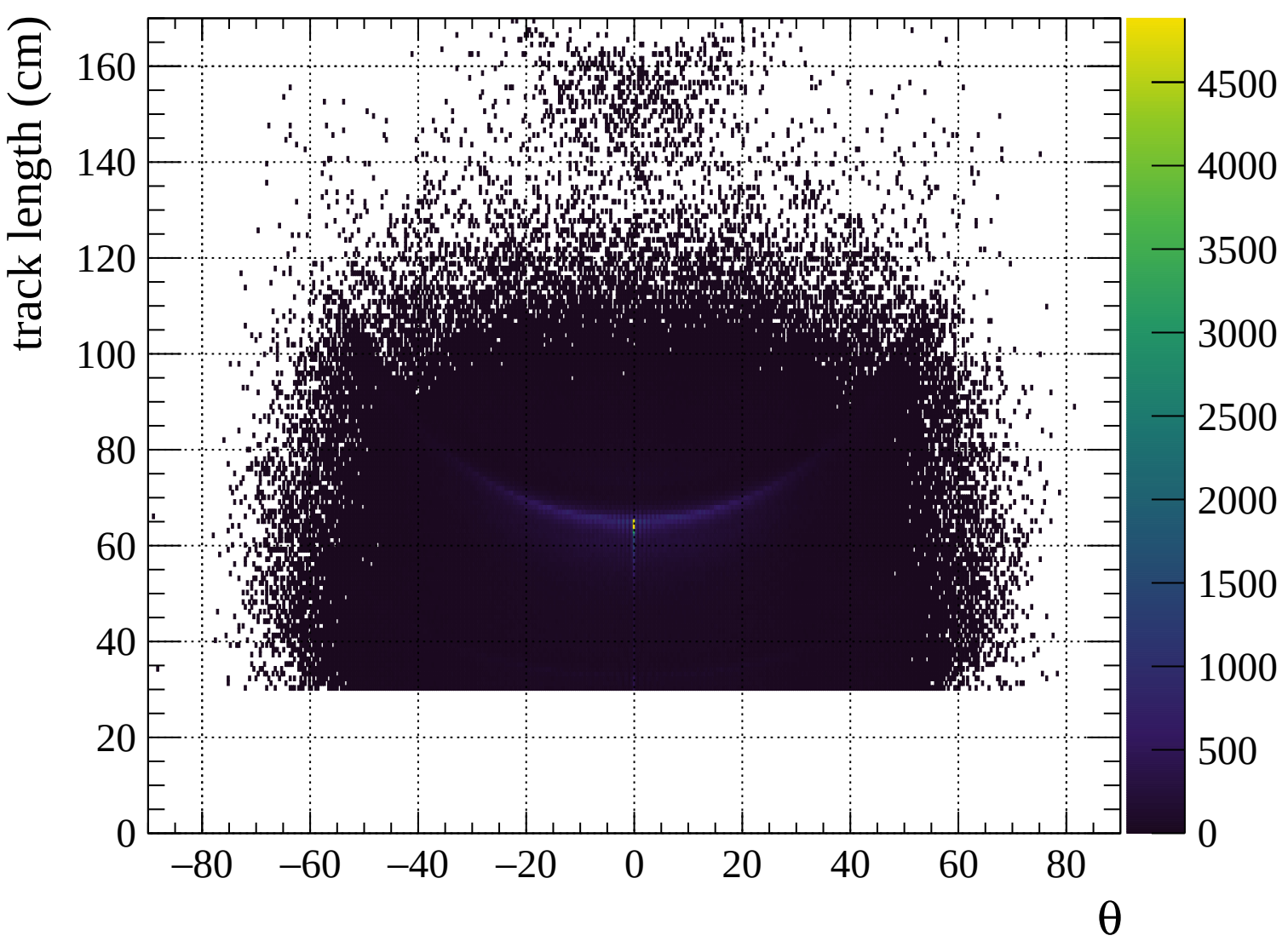


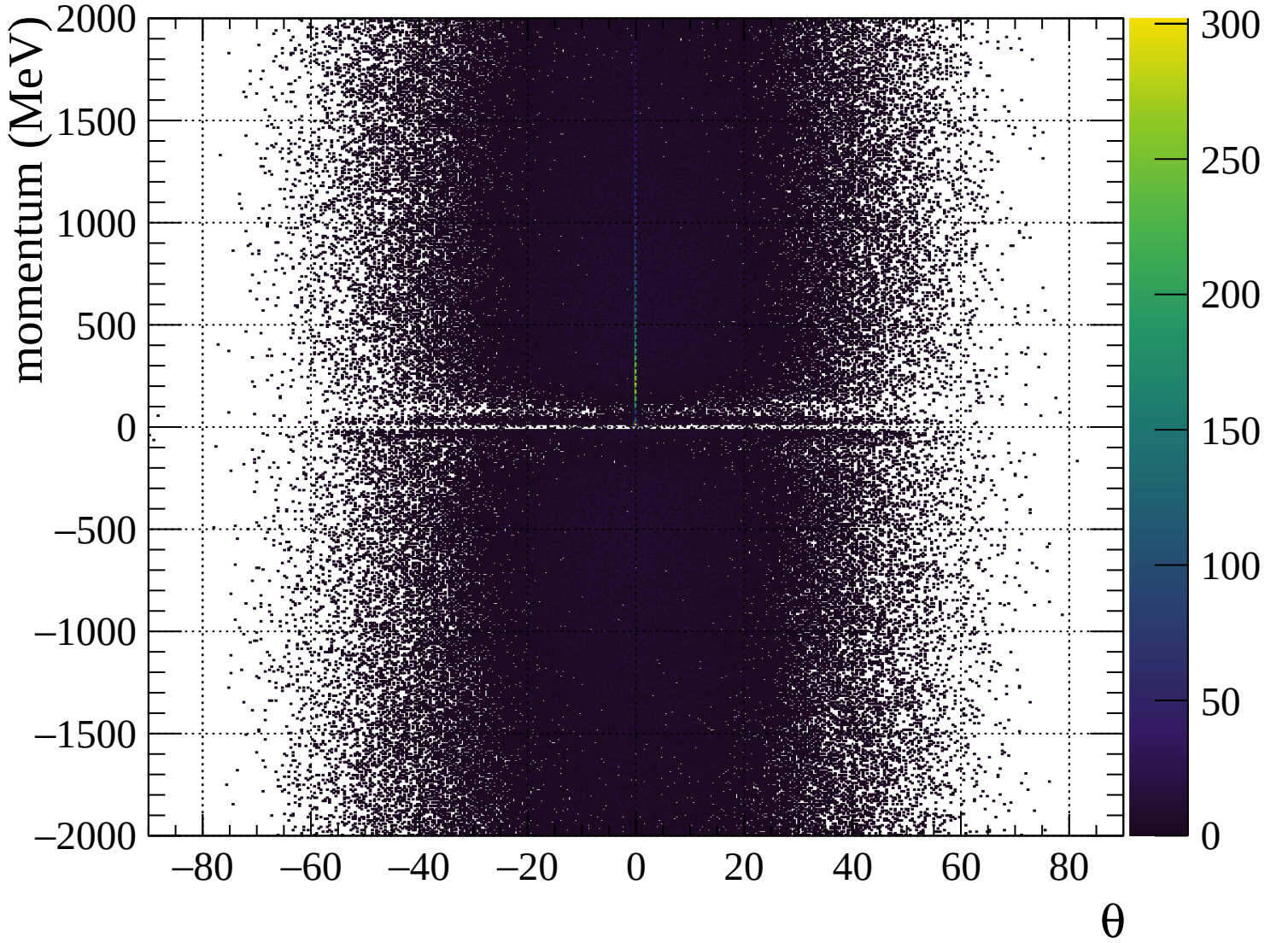


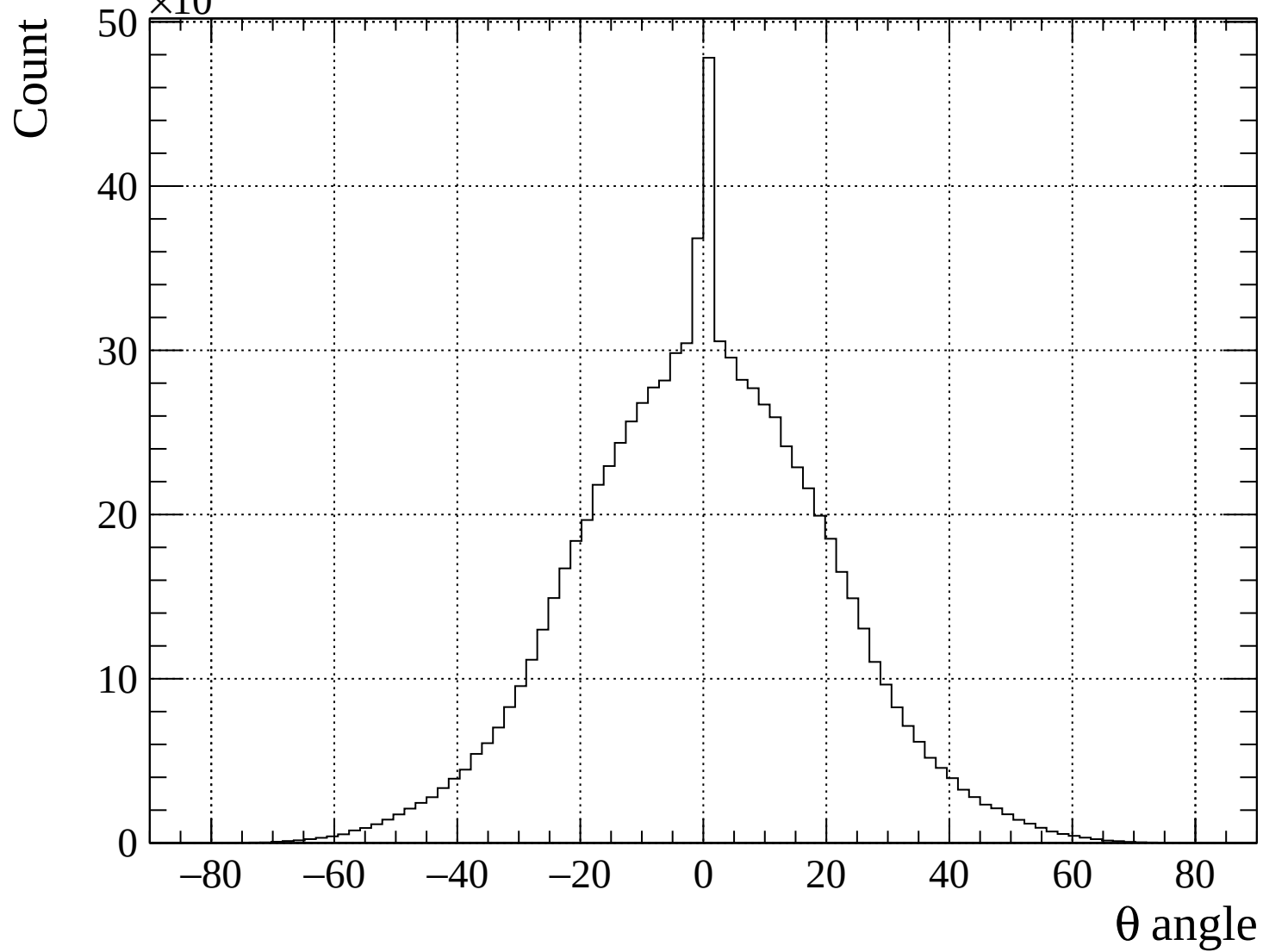


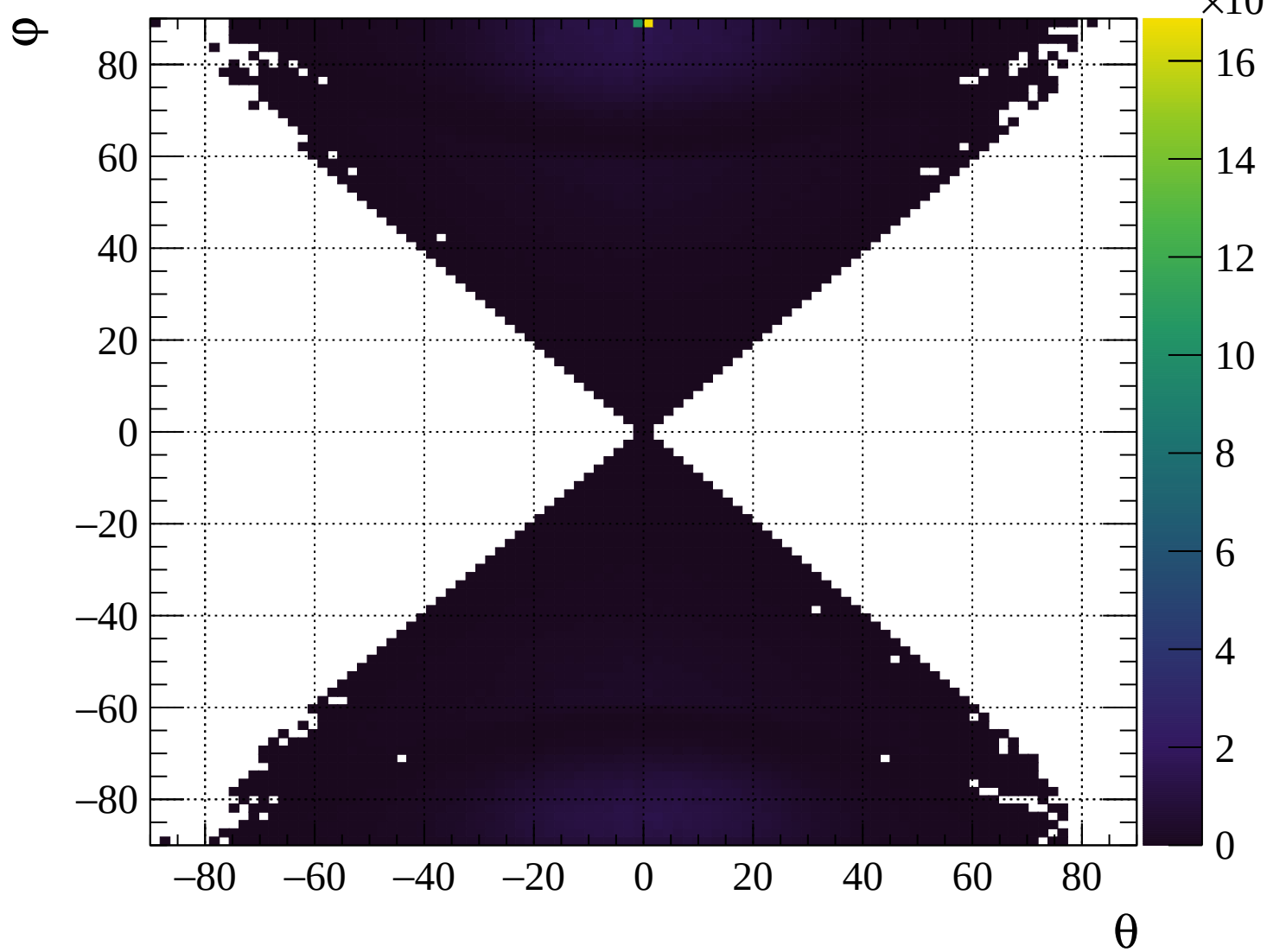


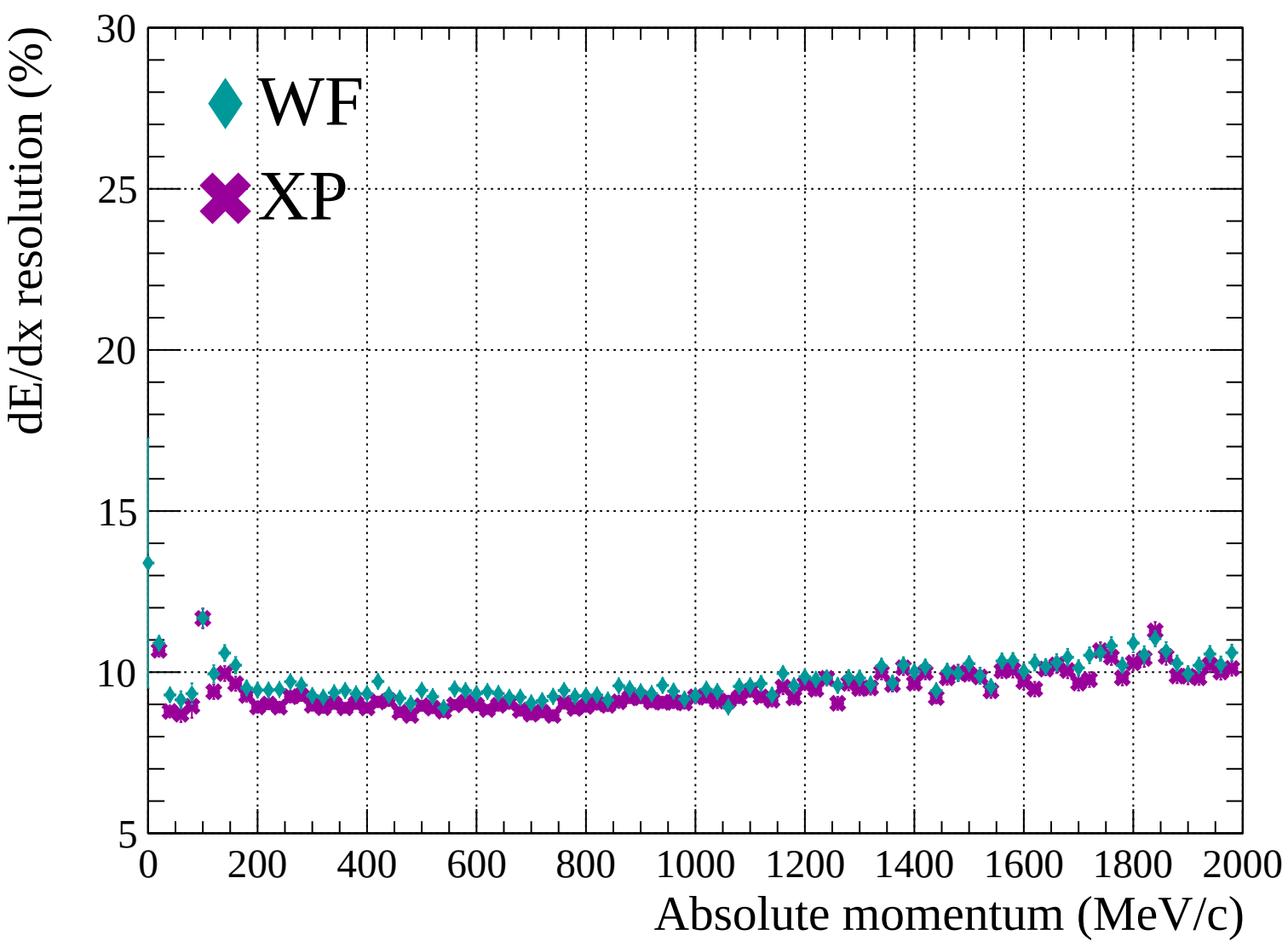


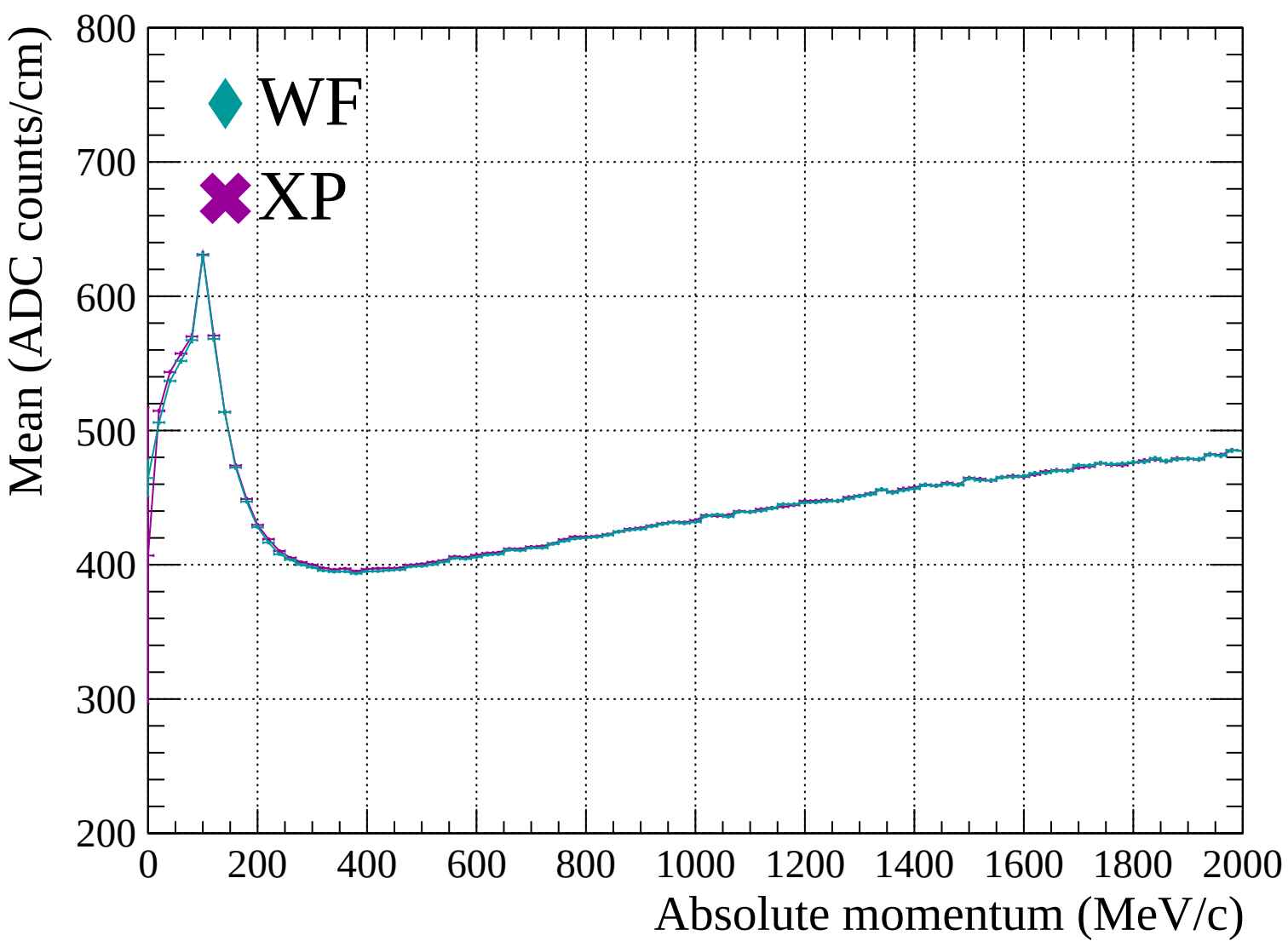


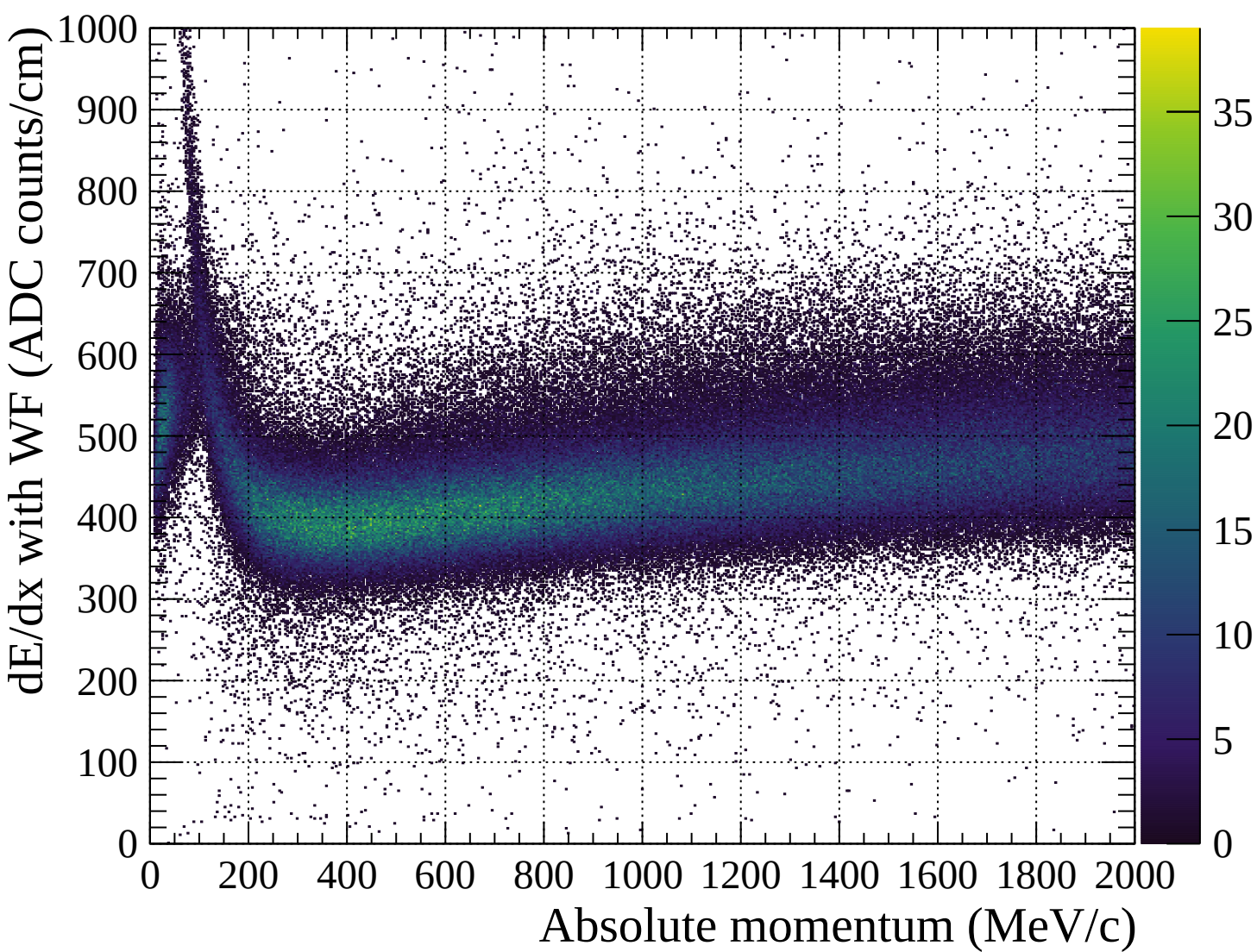


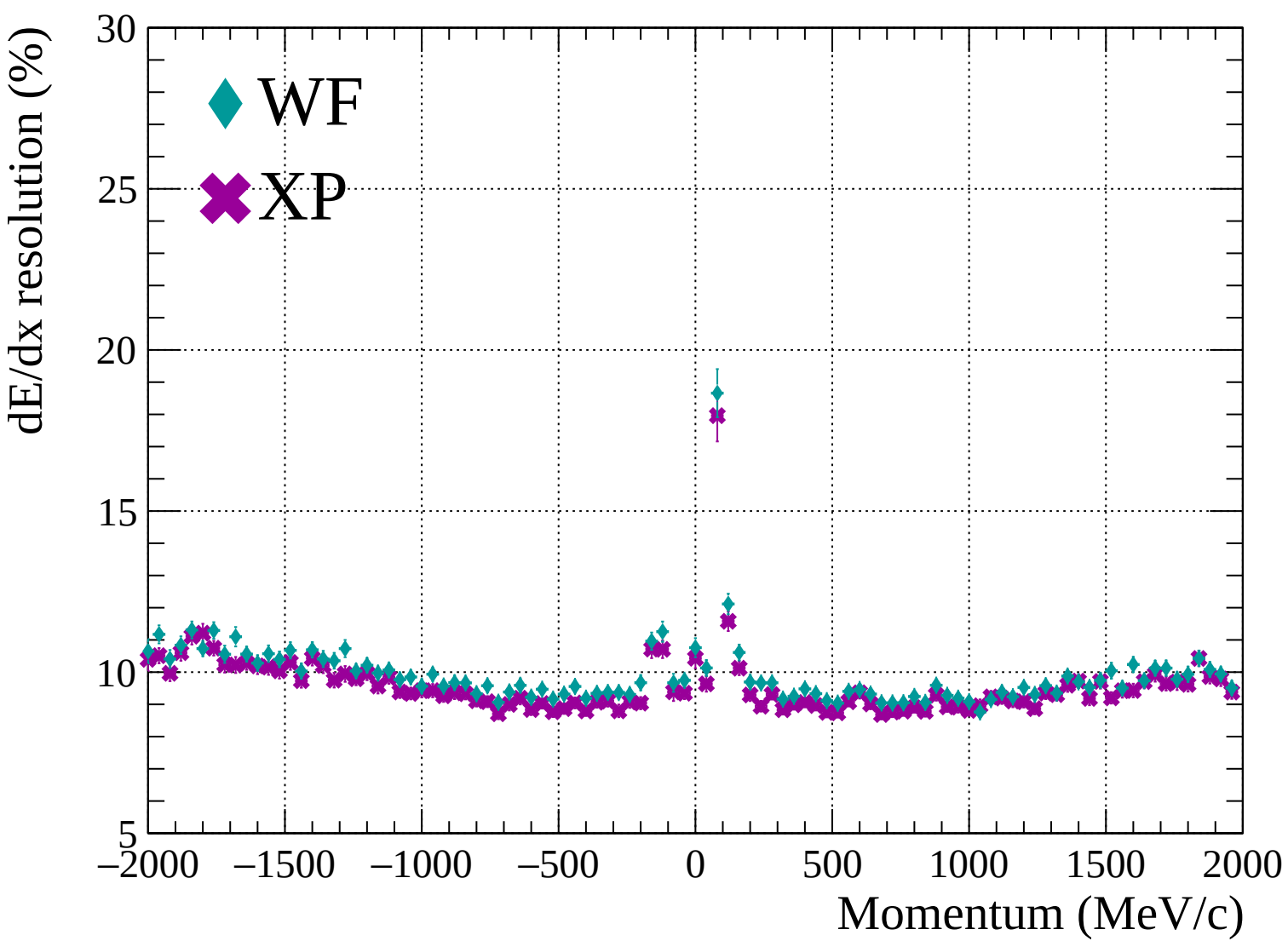


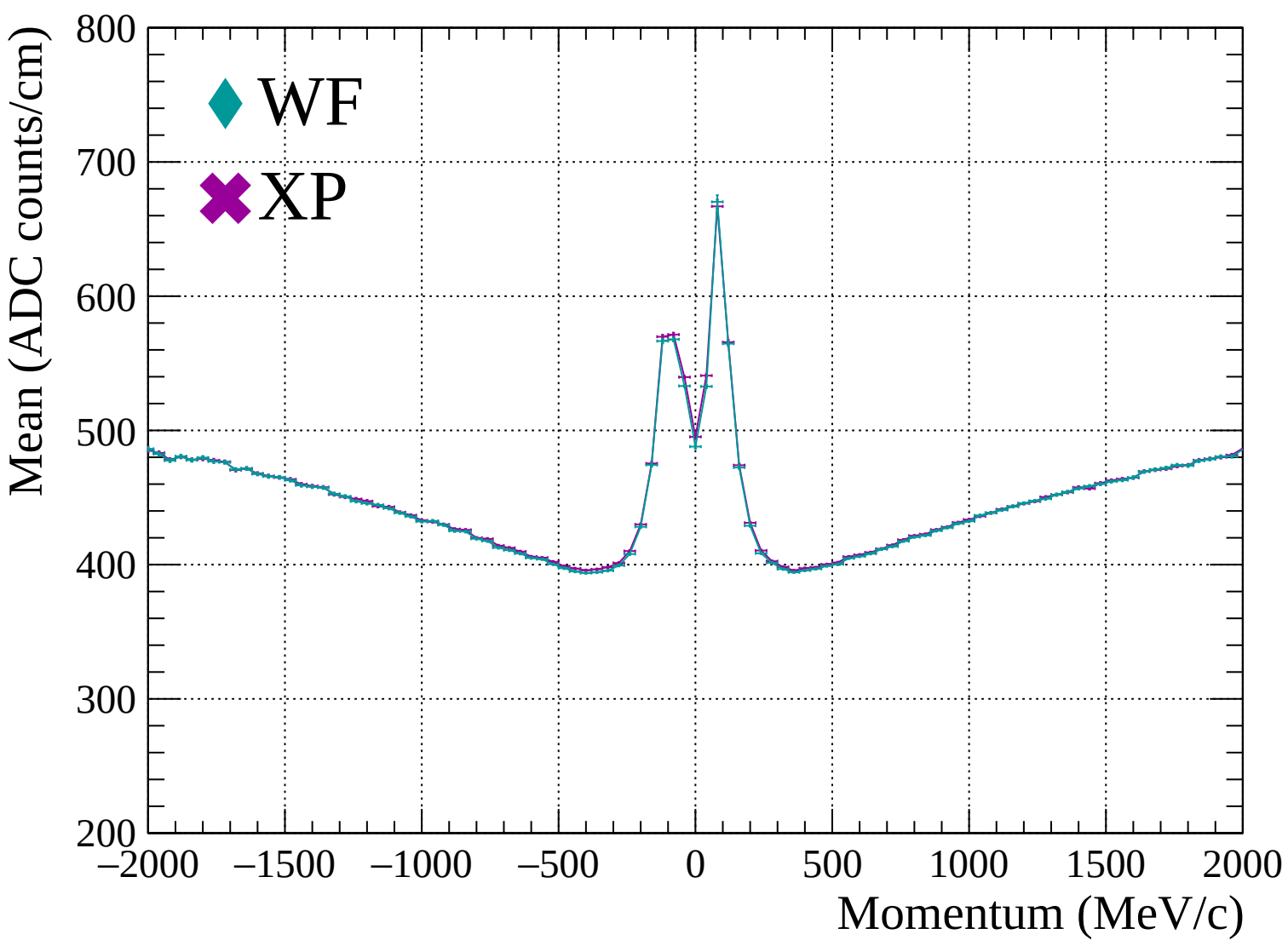


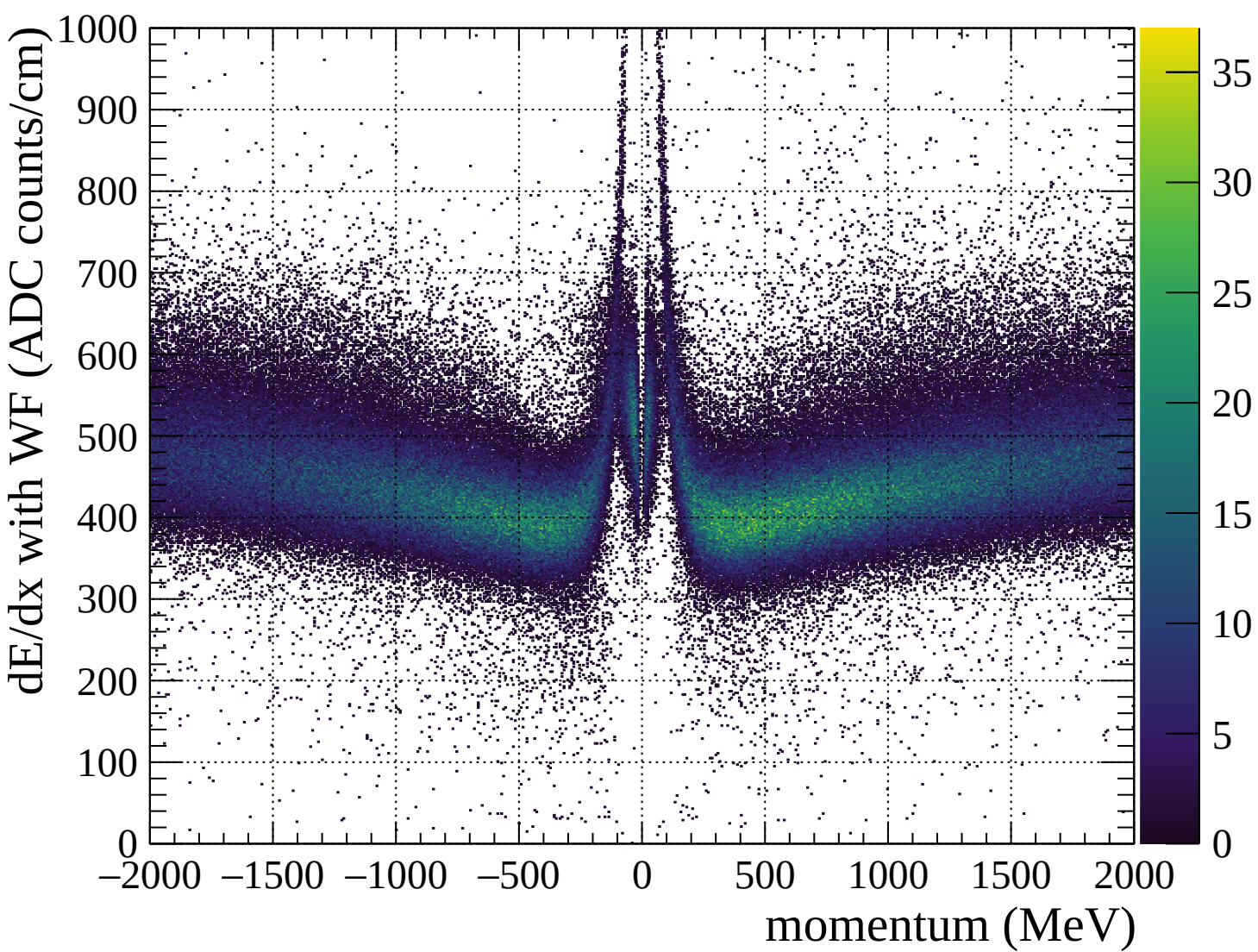


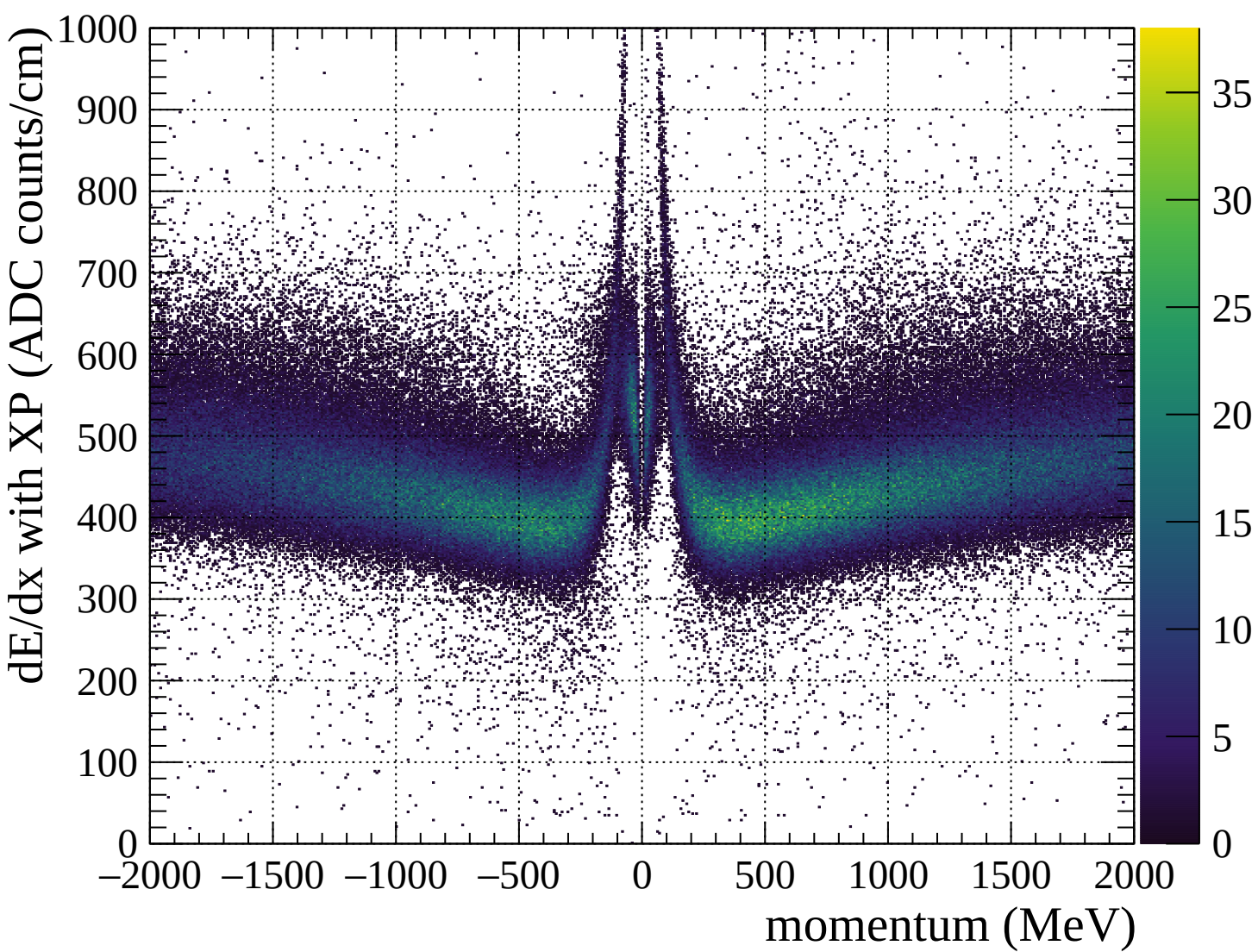


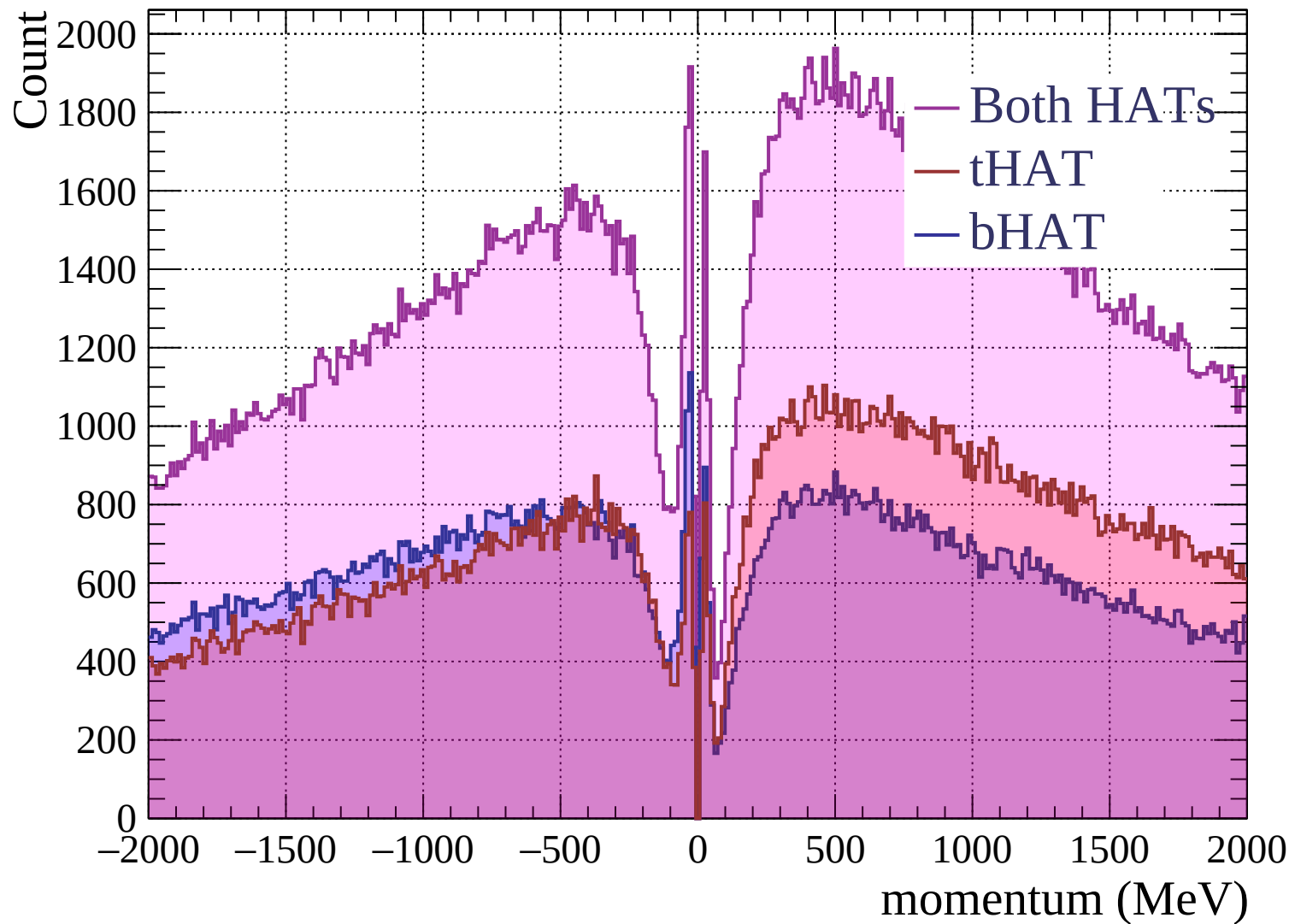


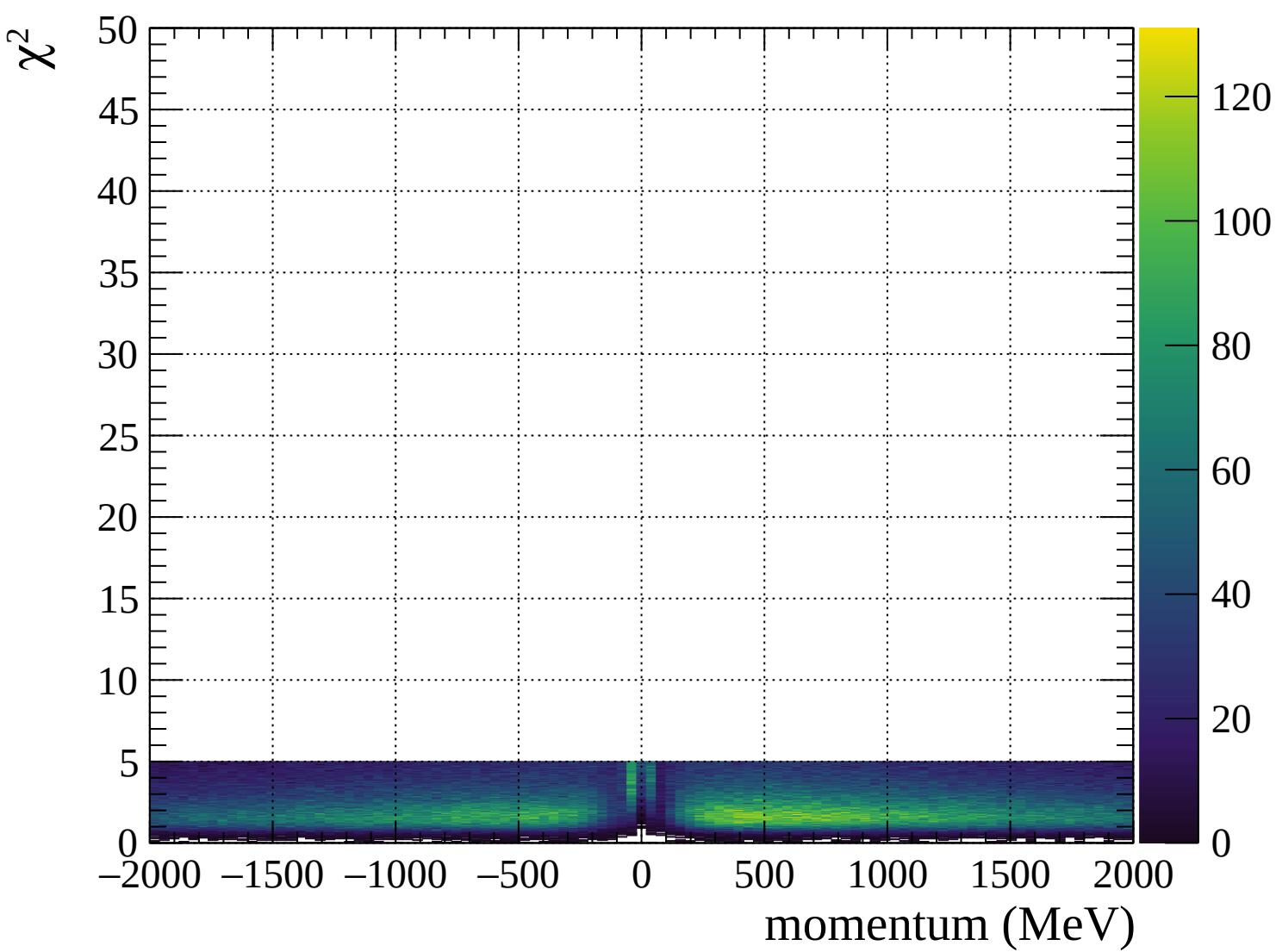




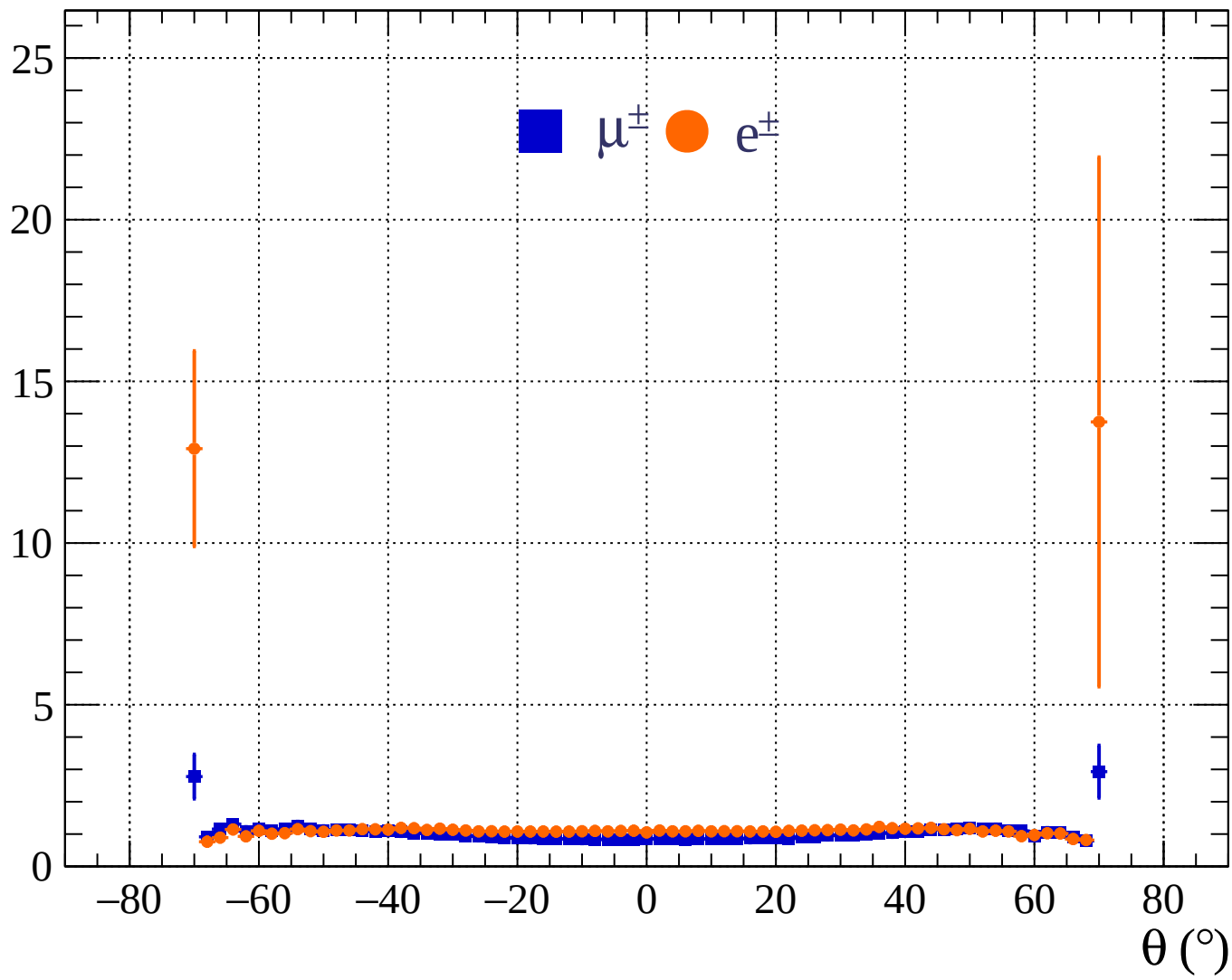


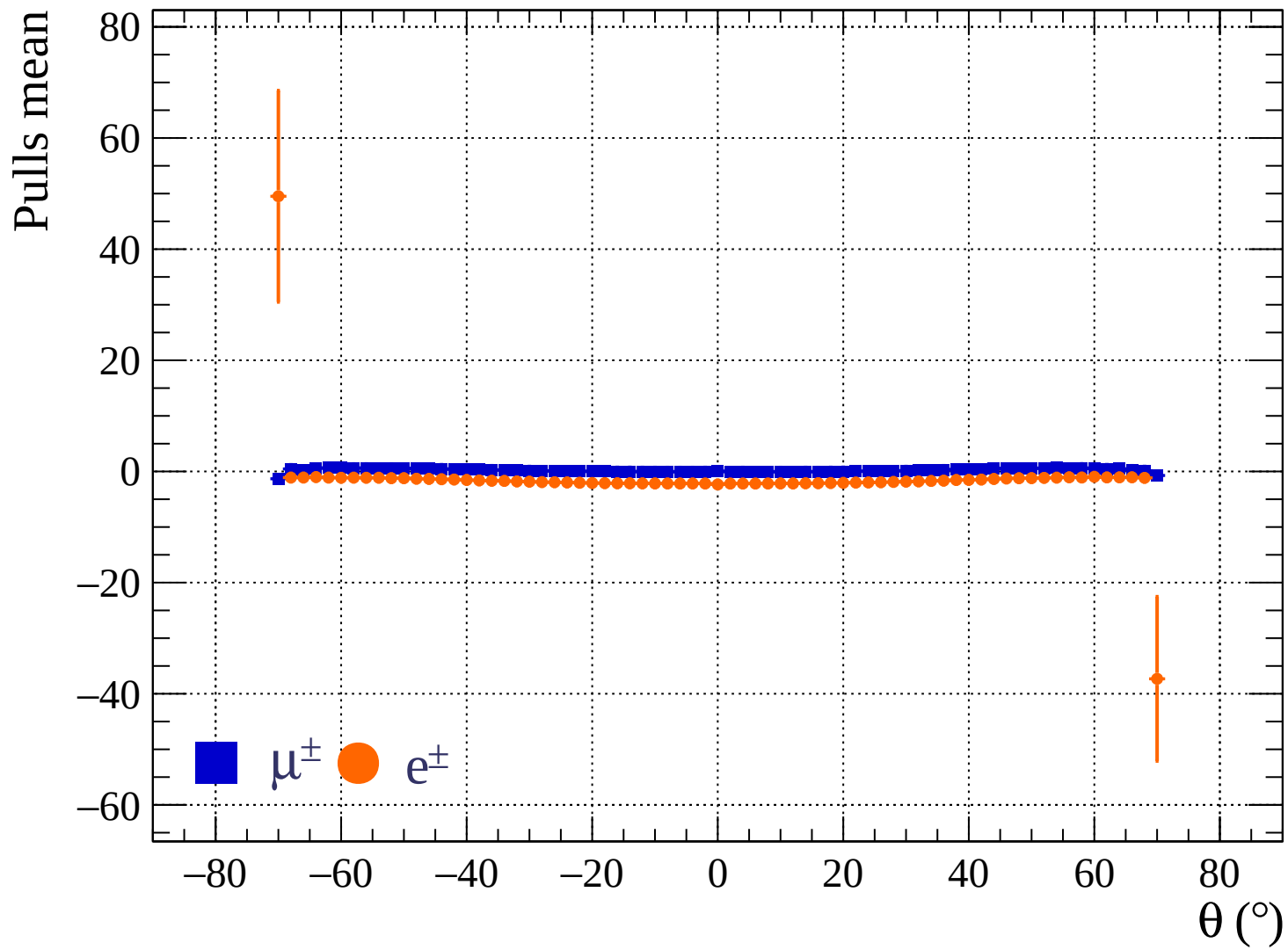






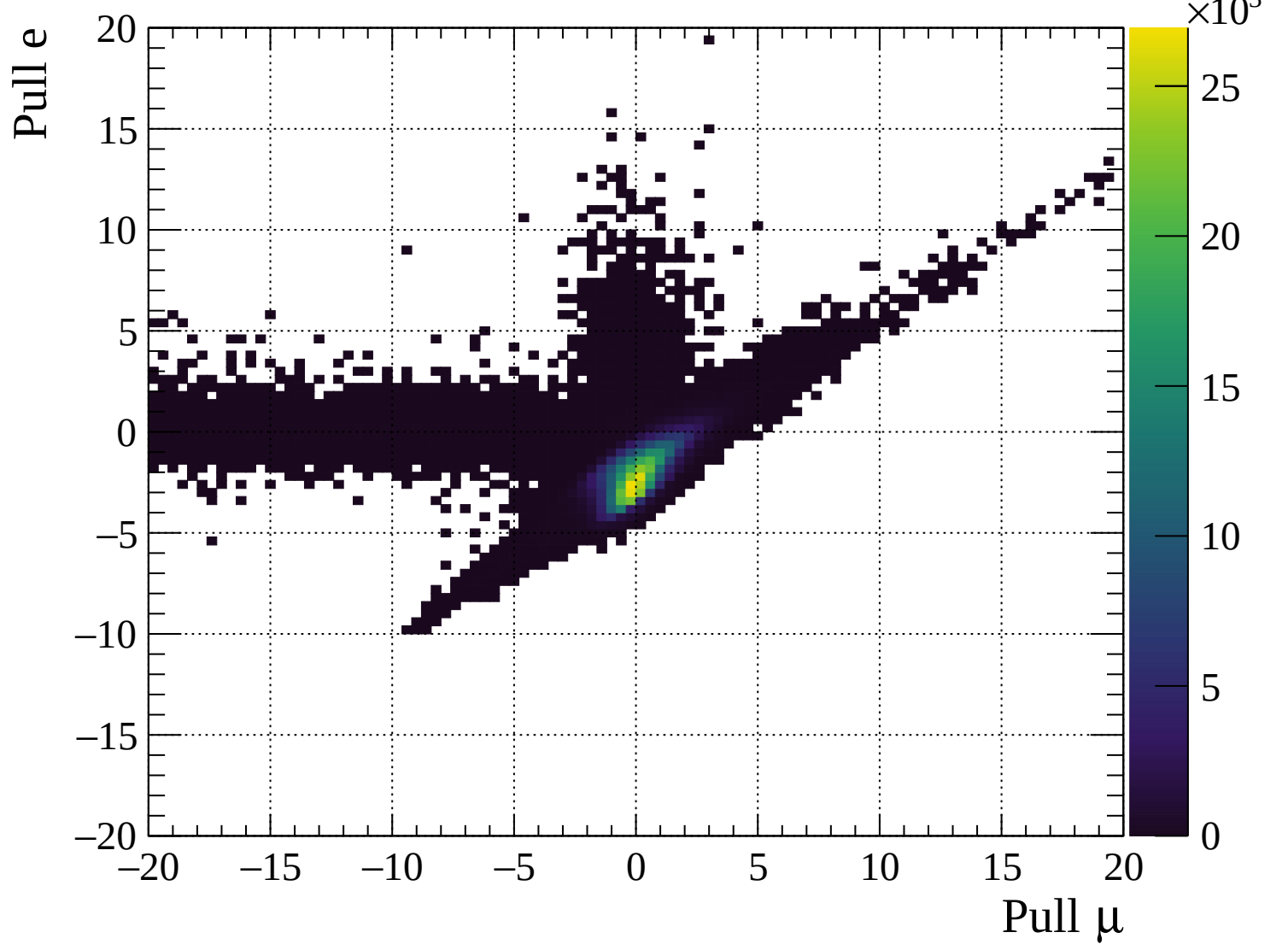
Pulls standard deviation

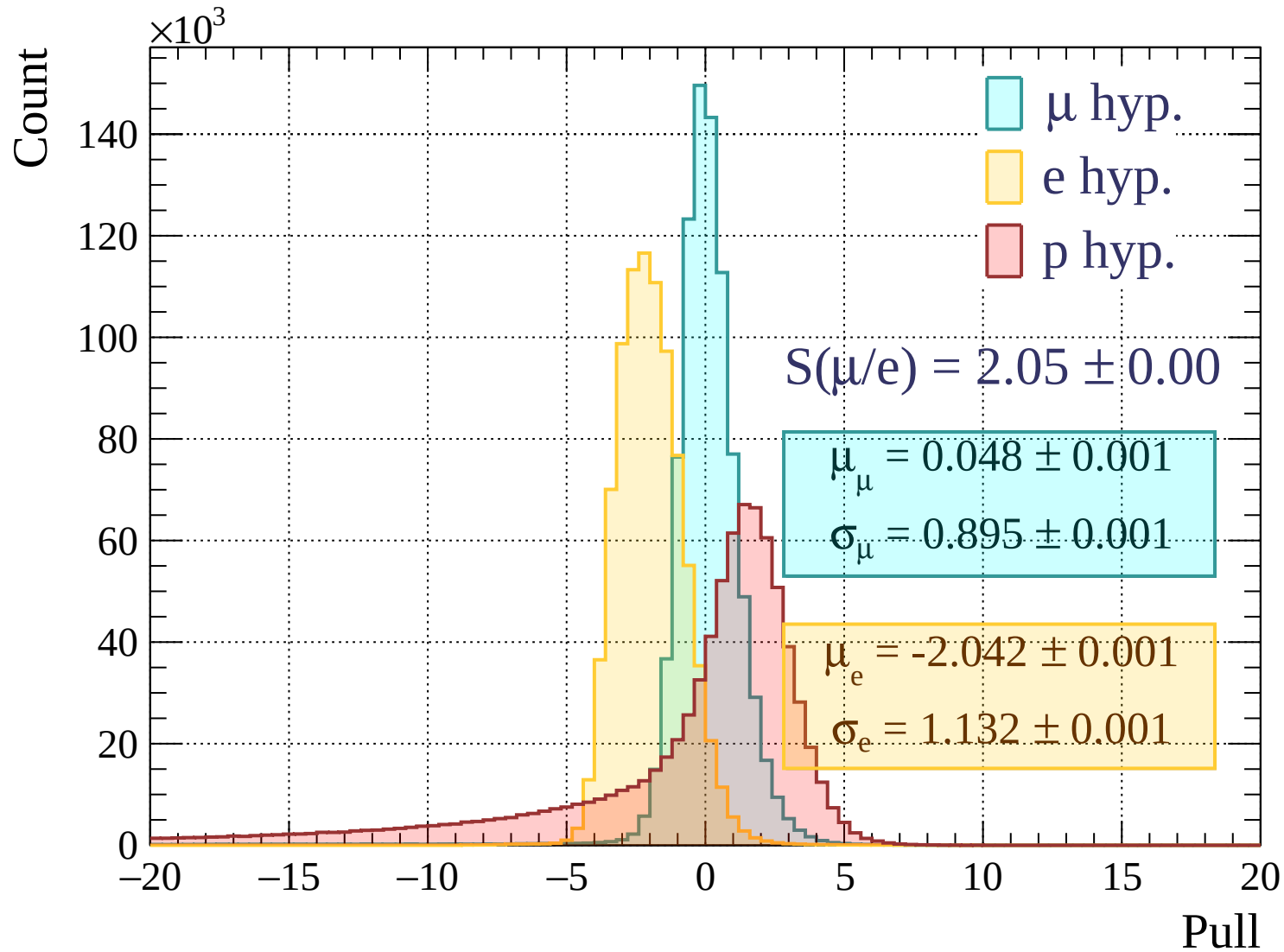


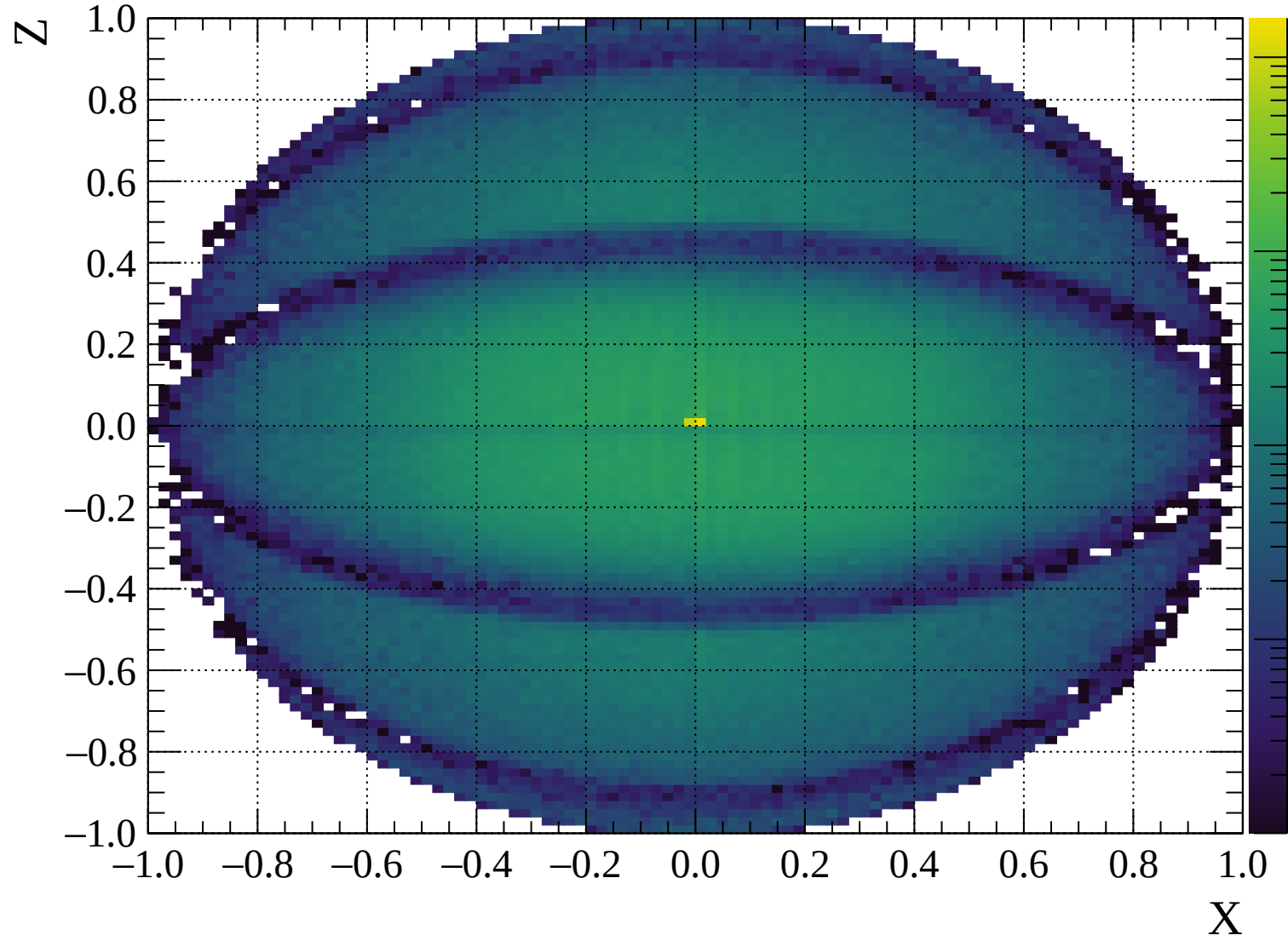


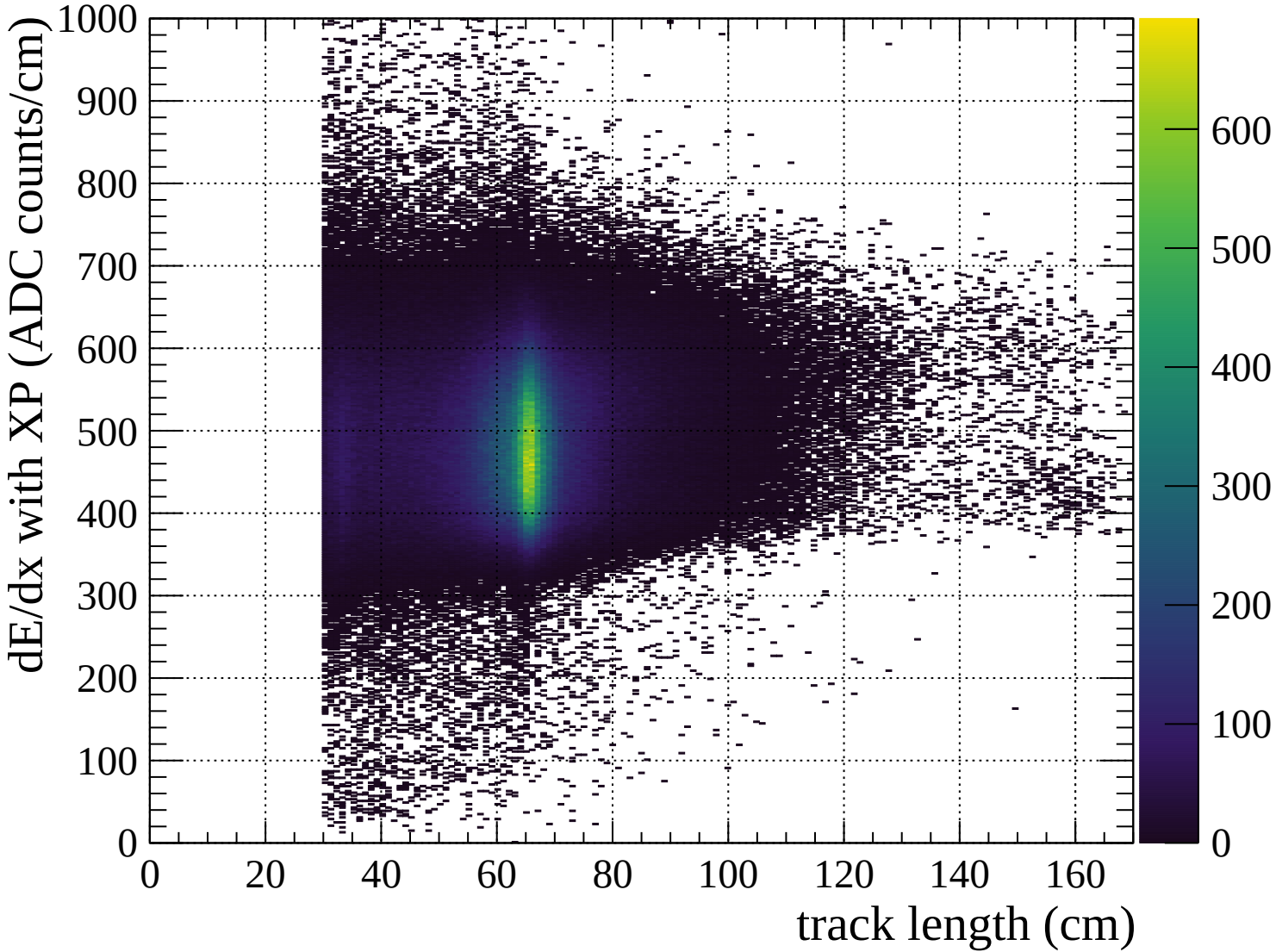
■ $\mu^\pm$ ● $e^\pm$

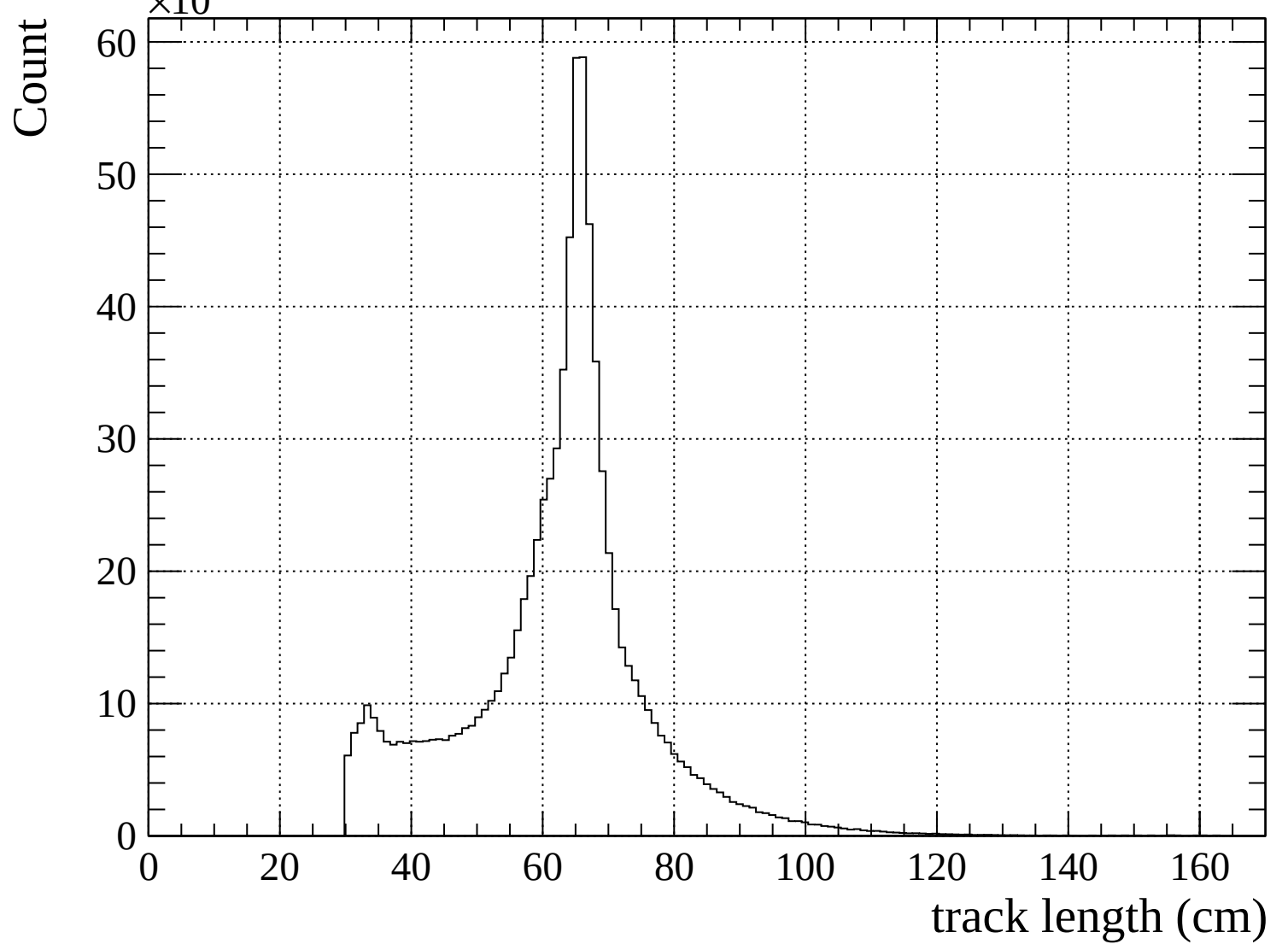
■ $\mu^\pm$ ● $e^\pm$

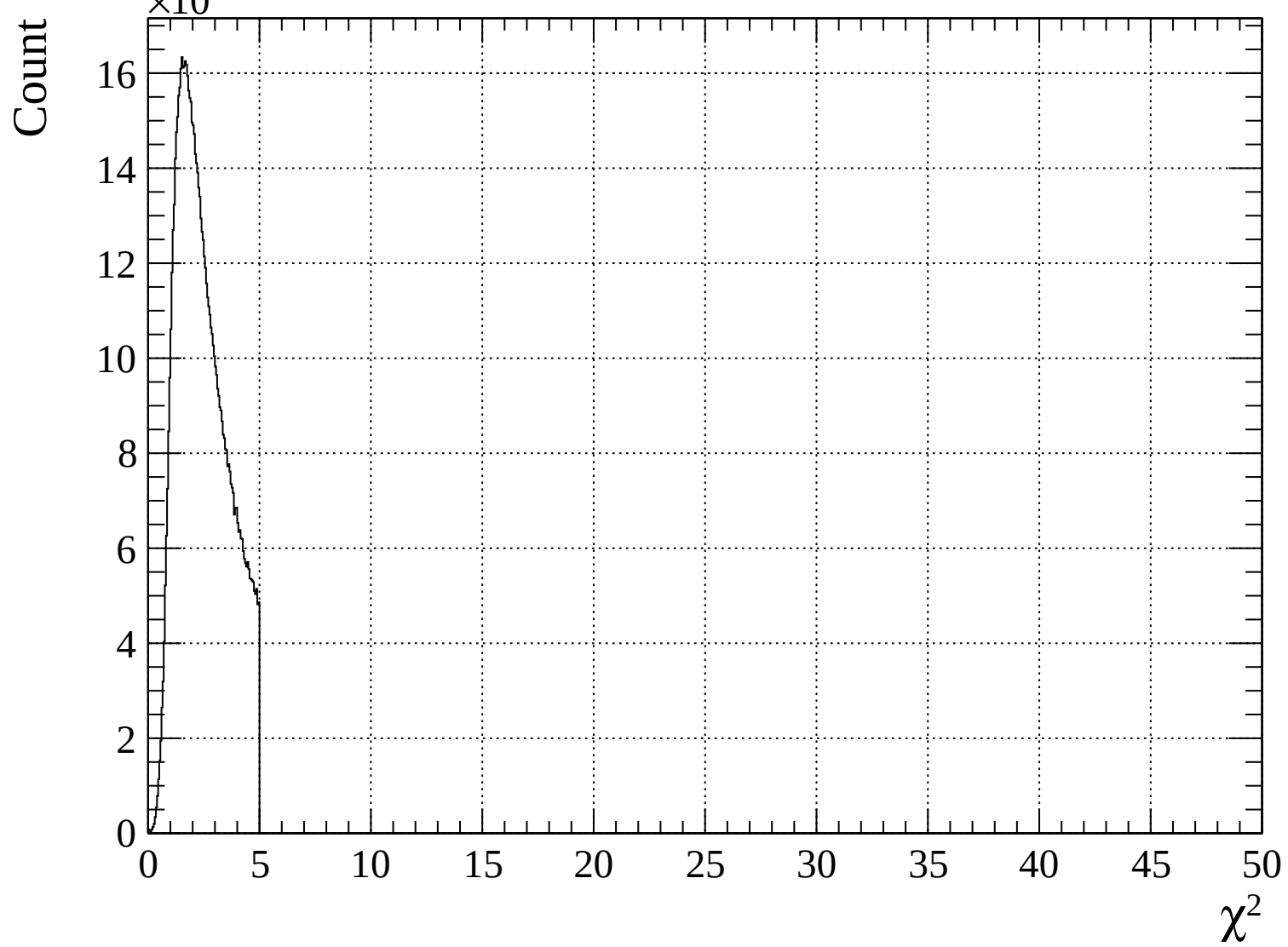


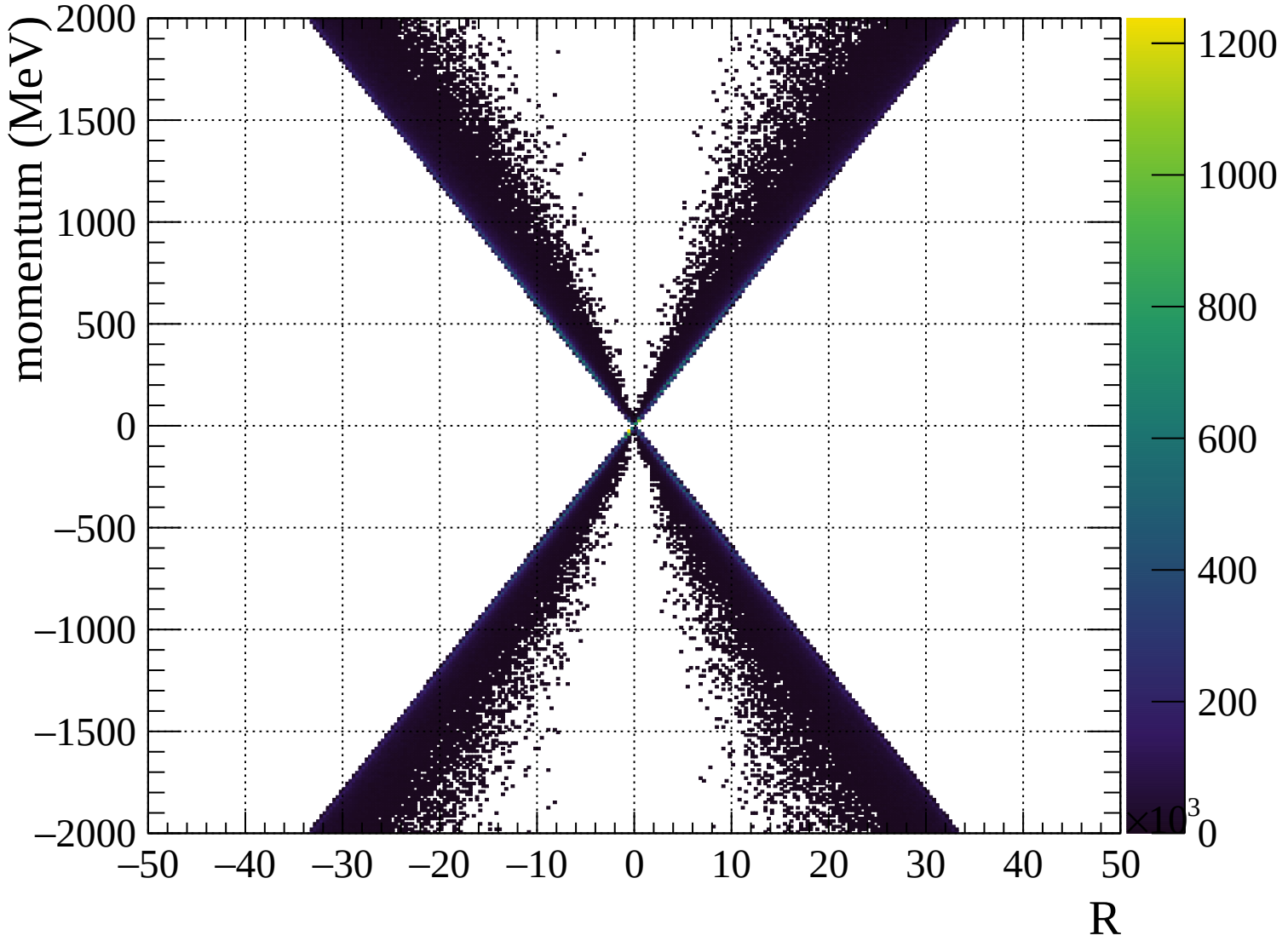


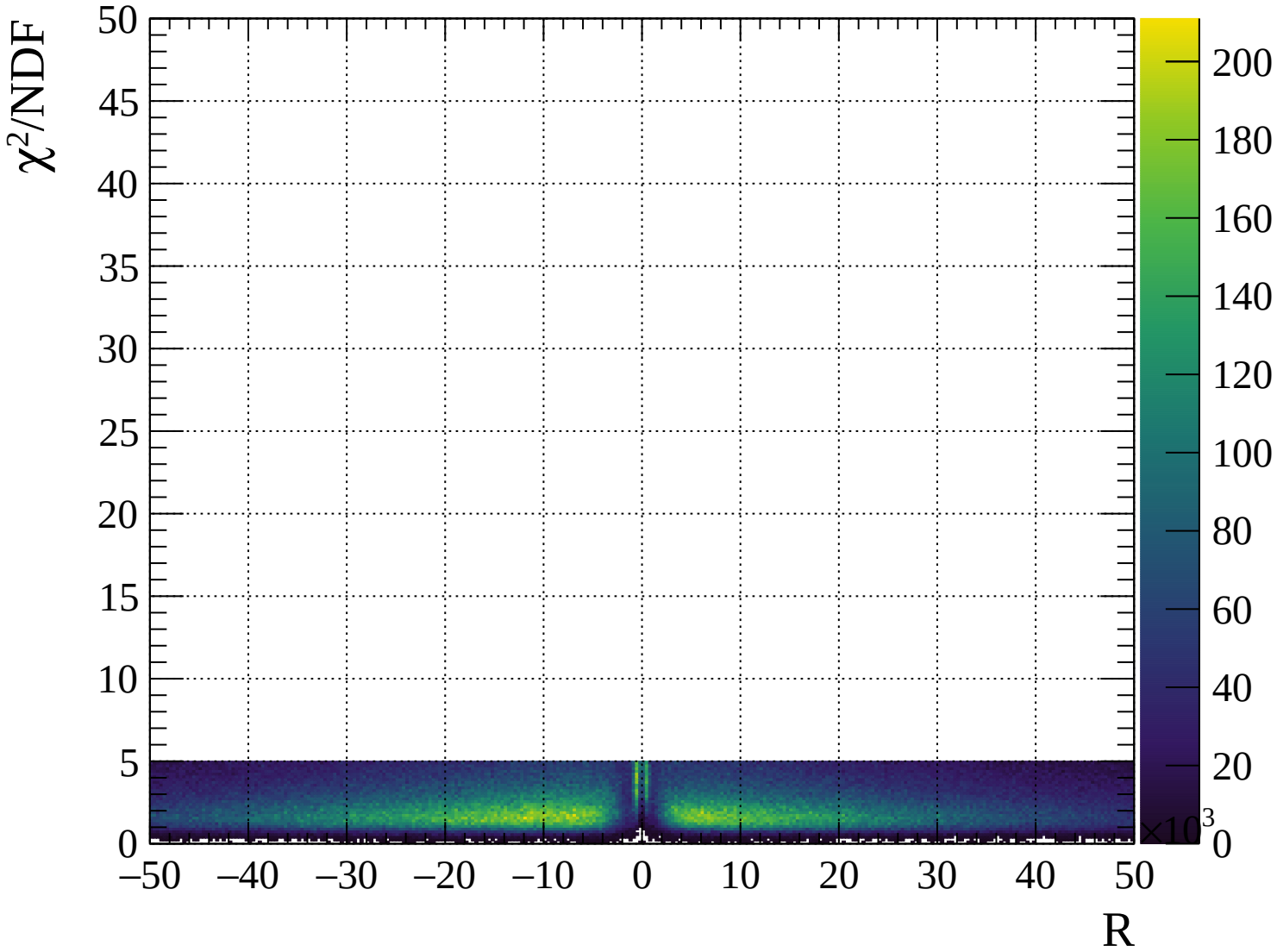






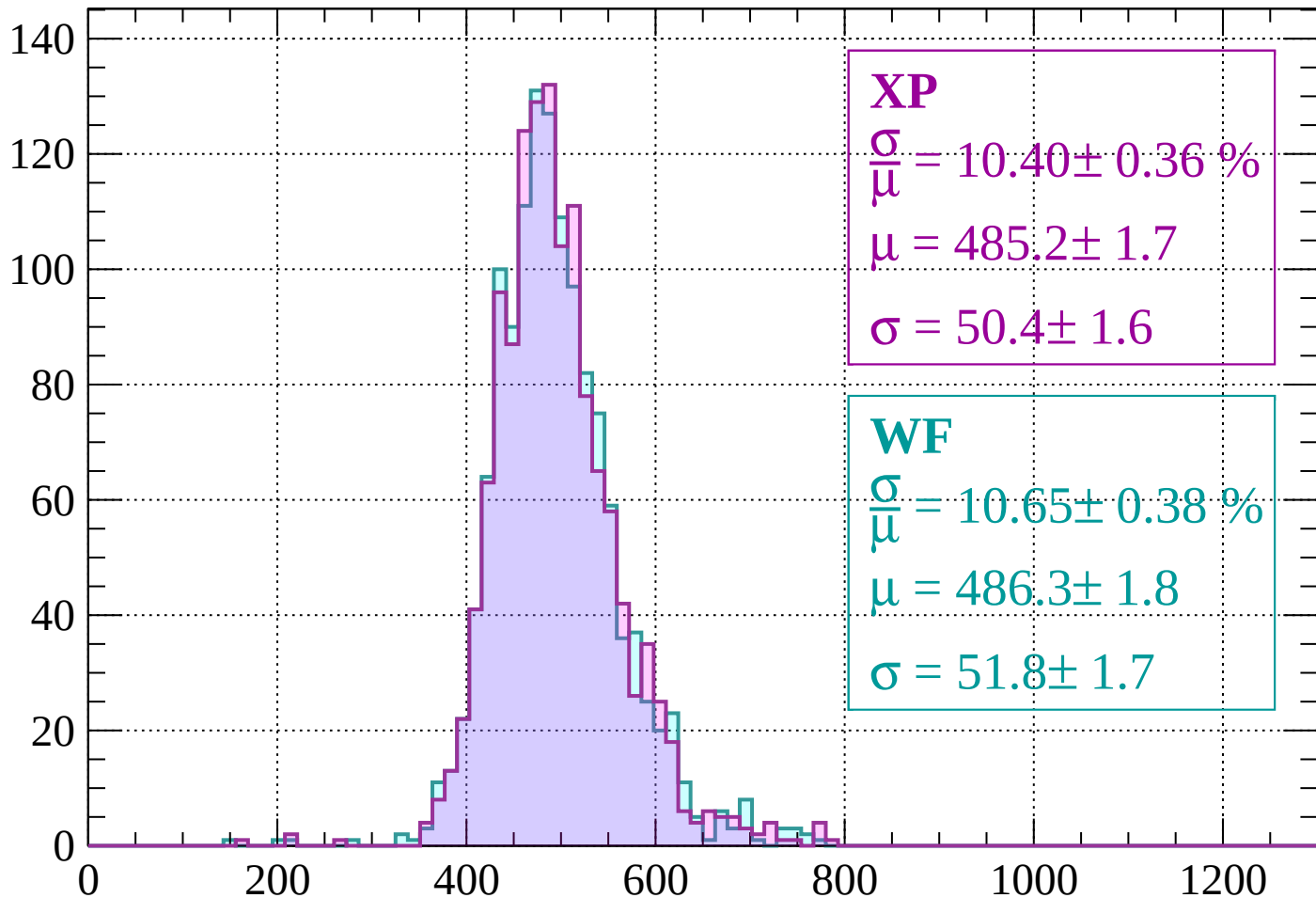






Energy loss | -2000 < p < -1960

Count



XP

$$\frac{\sigma}{\mu} = 10.40 \pm 0.36 \%$$

$$\mu = 485.2 \pm 1.7$$

$$\sigma = 50.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 10.65 \pm 0.38 \%$$

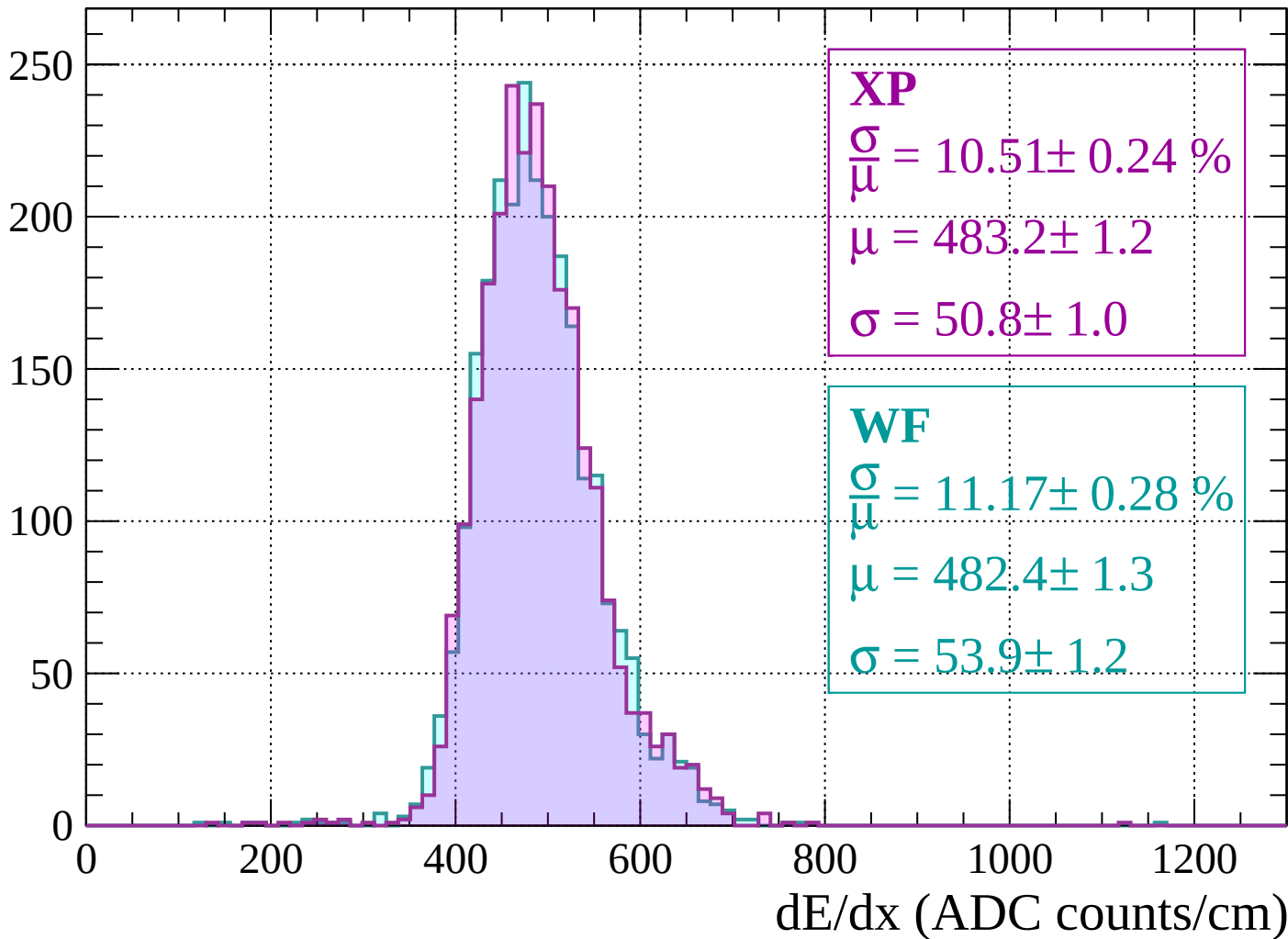
$$\mu = 486.3 \pm 1.8$$

$$\sigma = 51.8 \pm 1.7$$

dE/dx (ADC counts/cm)

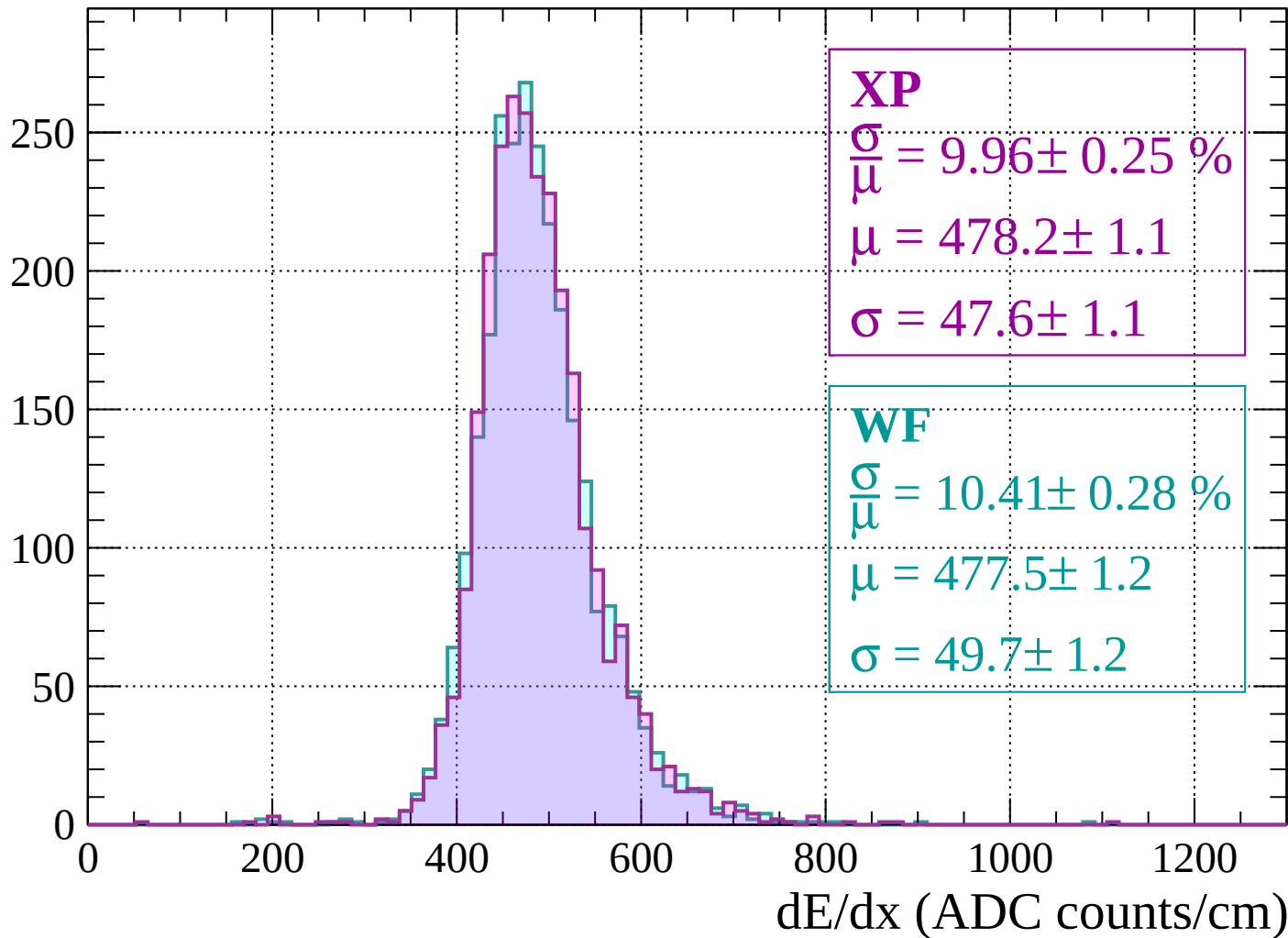
Energy loss | -1960 < p < -1920

Count



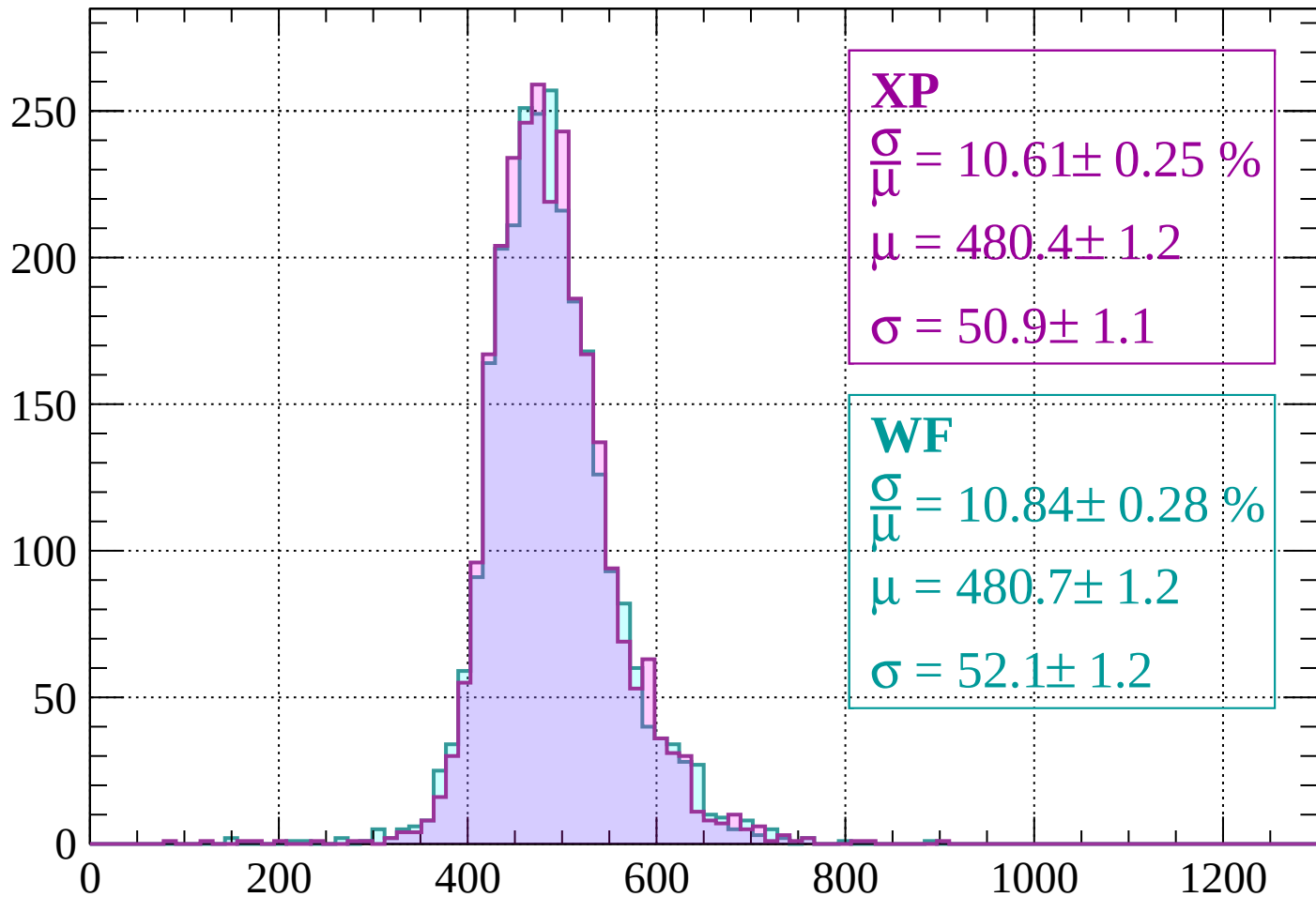
Energy loss | $-1920 < p < -1880$

Count



Energy loss | -1880 < p < -1840

Count



XP

$$\frac{\sigma}{\mu} = 10.61 \pm 0.25 \%$$

$$\mu = 480.4 \pm 1.2$$

$$\sigma = 50.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.84 \pm 0.28 \%$$

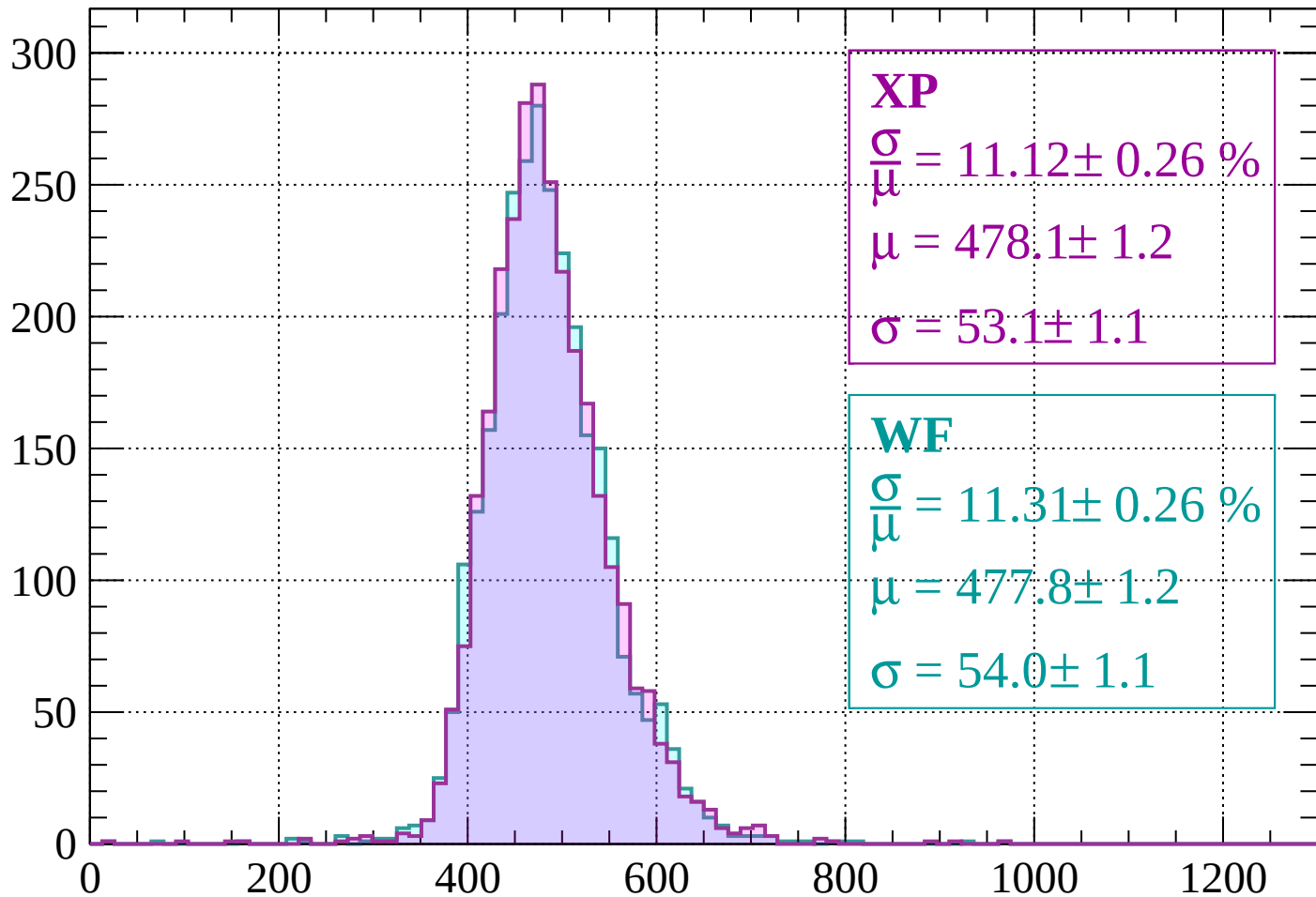
$$\mu = 480.7 \pm 1.2$$

$$\sigma = 52.1 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | -1840 < p < -1800

Count



XP

$$\frac{\sigma}{\mu} = 11.12 \pm 0.26 \%$$

$$\mu = 478.1 \pm 1.2$$

$$\sigma = 53.1 \pm 1.1$$

WF

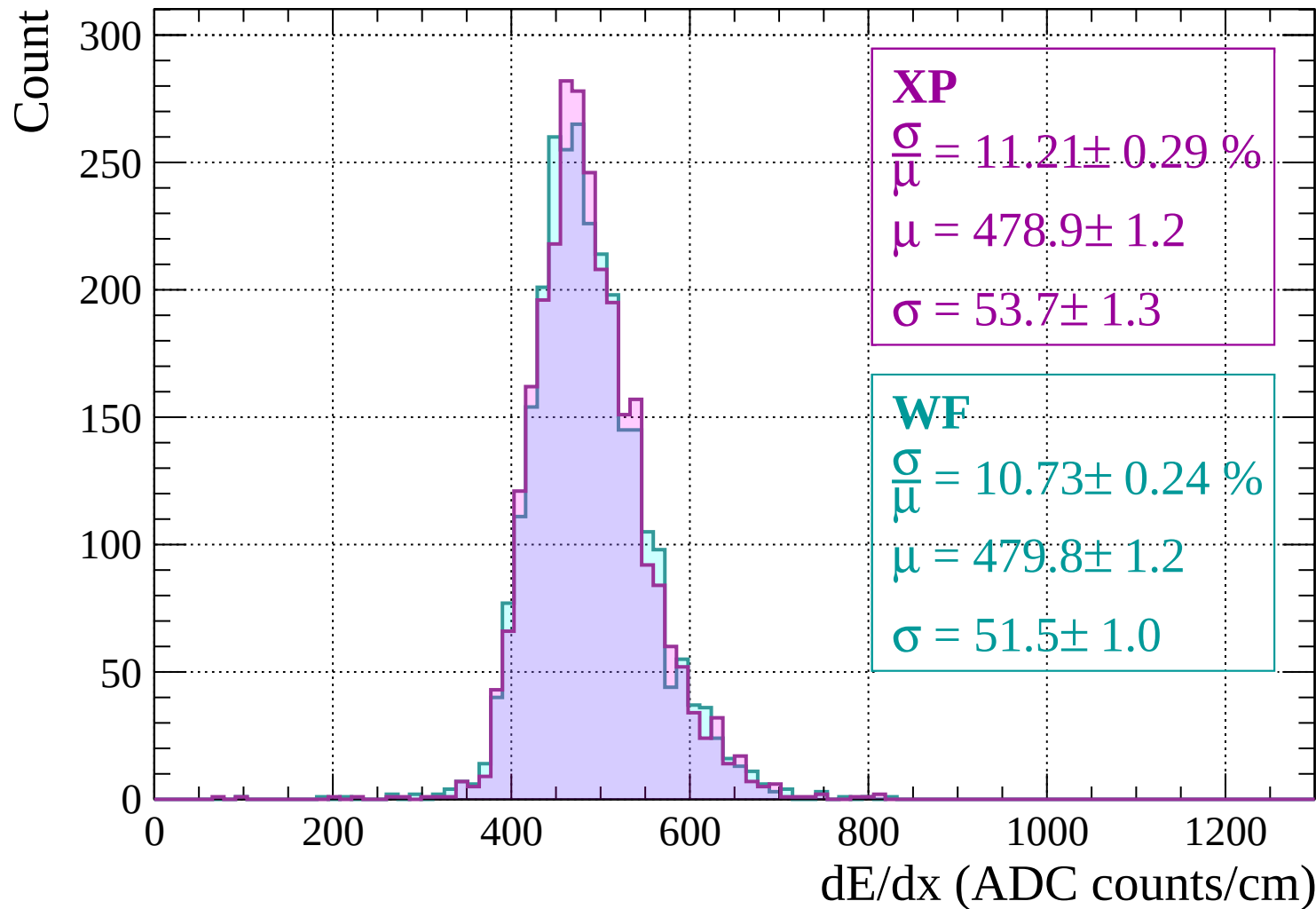
$$\frac{\sigma}{\mu} = 11.31 \pm 0.26 \%$$

$$\mu = 477.8 \pm 1.2$$

$$\sigma = 54.0 \pm 1.1$$

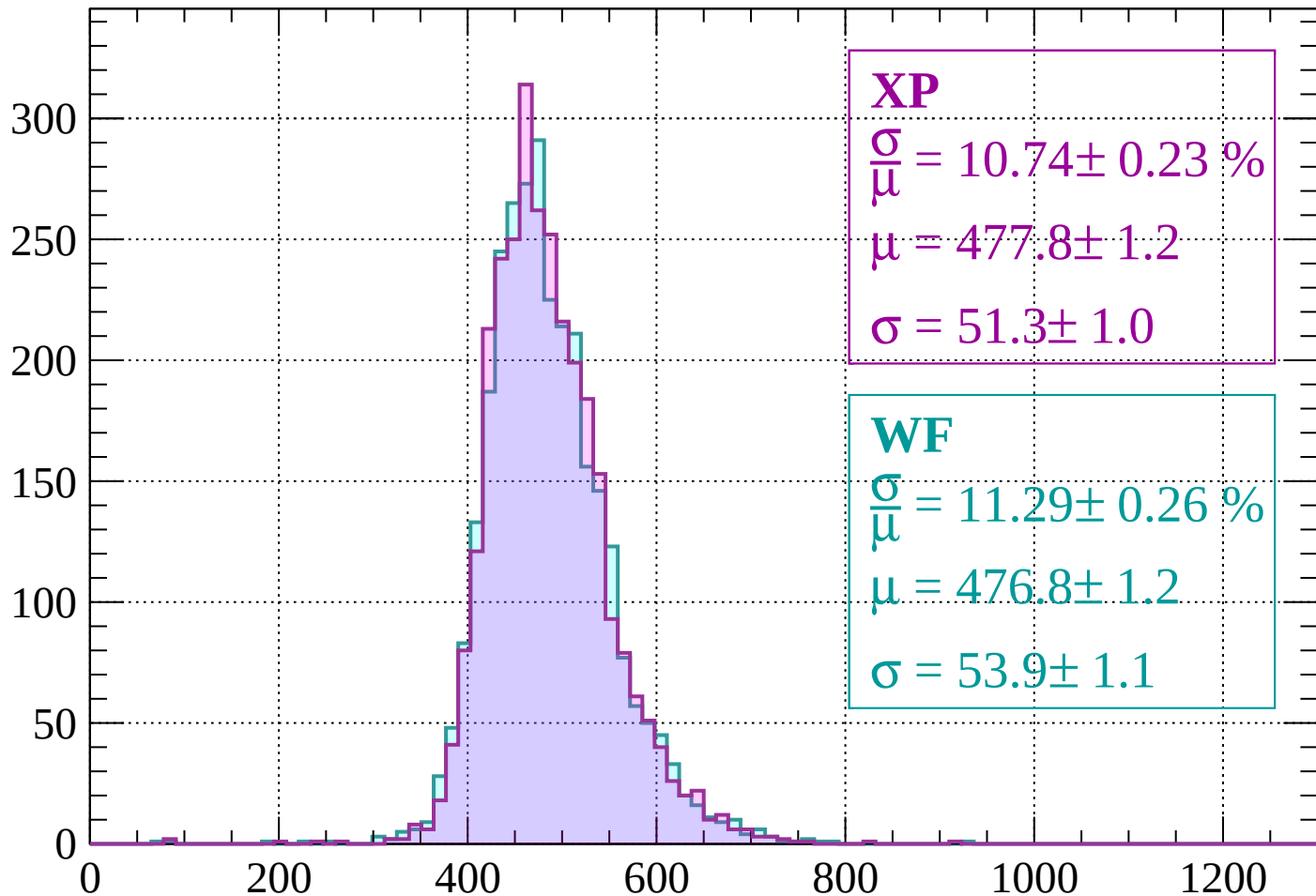
dE/dx (ADC counts/cm)

Energy loss | -1800 < p < -1760



Energy loss | -1760 < p < -1720

Count



XP

$$\frac{\sigma}{\mu} = 10.74 \pm 0.23 \%$$

$$\mu = 477.8 \pm 1.2$$

$$\sigma = 51.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.29 \pm 0.26 \%$$

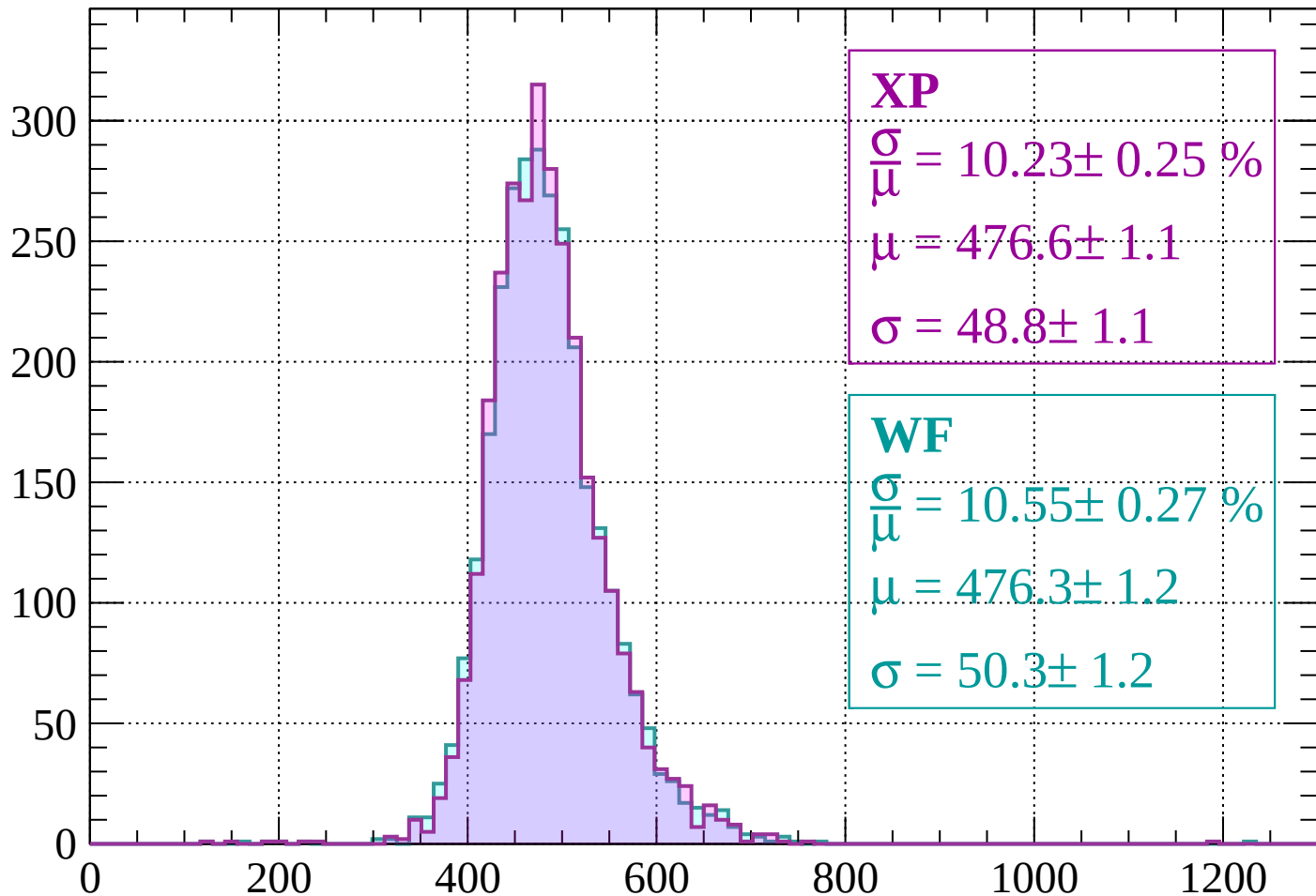
$$\mu = 476.8 \pm 1.2$$

$$\sigma = 53.9 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 10.23 \pm 0.25 \%$$

$$\mu = 476.6 \pm 1.1$$

$$\sigma = 48.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.55 \pm 0.27 \%$$

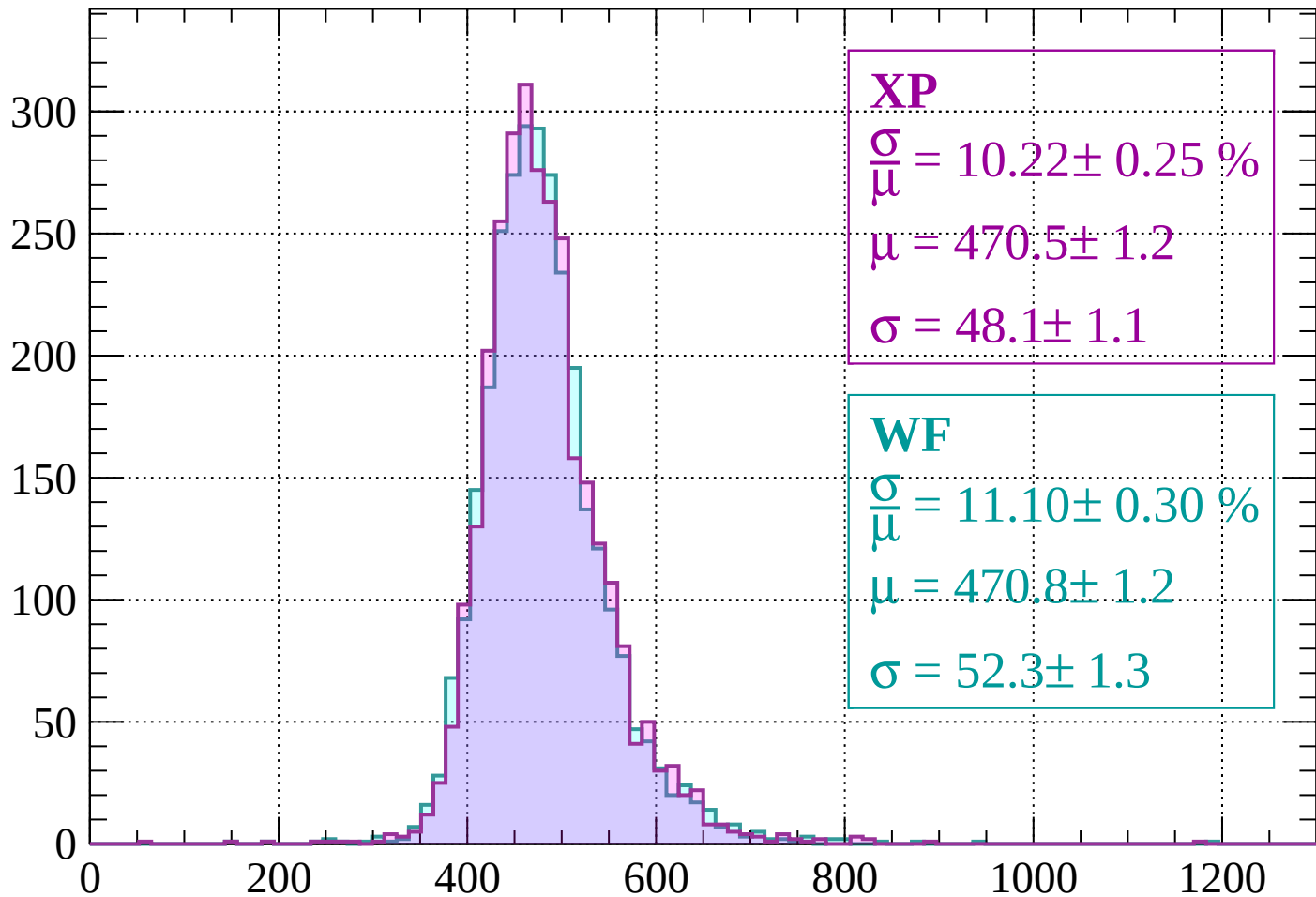
$$\mu = 476.3 \pm 1.2$$

$$\sigma = 50.3 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1640

Count



XP

$$\frac{\sigma}{\mu} = 10.22 \pm 0.25 \%$$

$$\mu = 470.5 \pm 1.2$$

$$\sigma = 48.1 \pm 1.1$$

WF

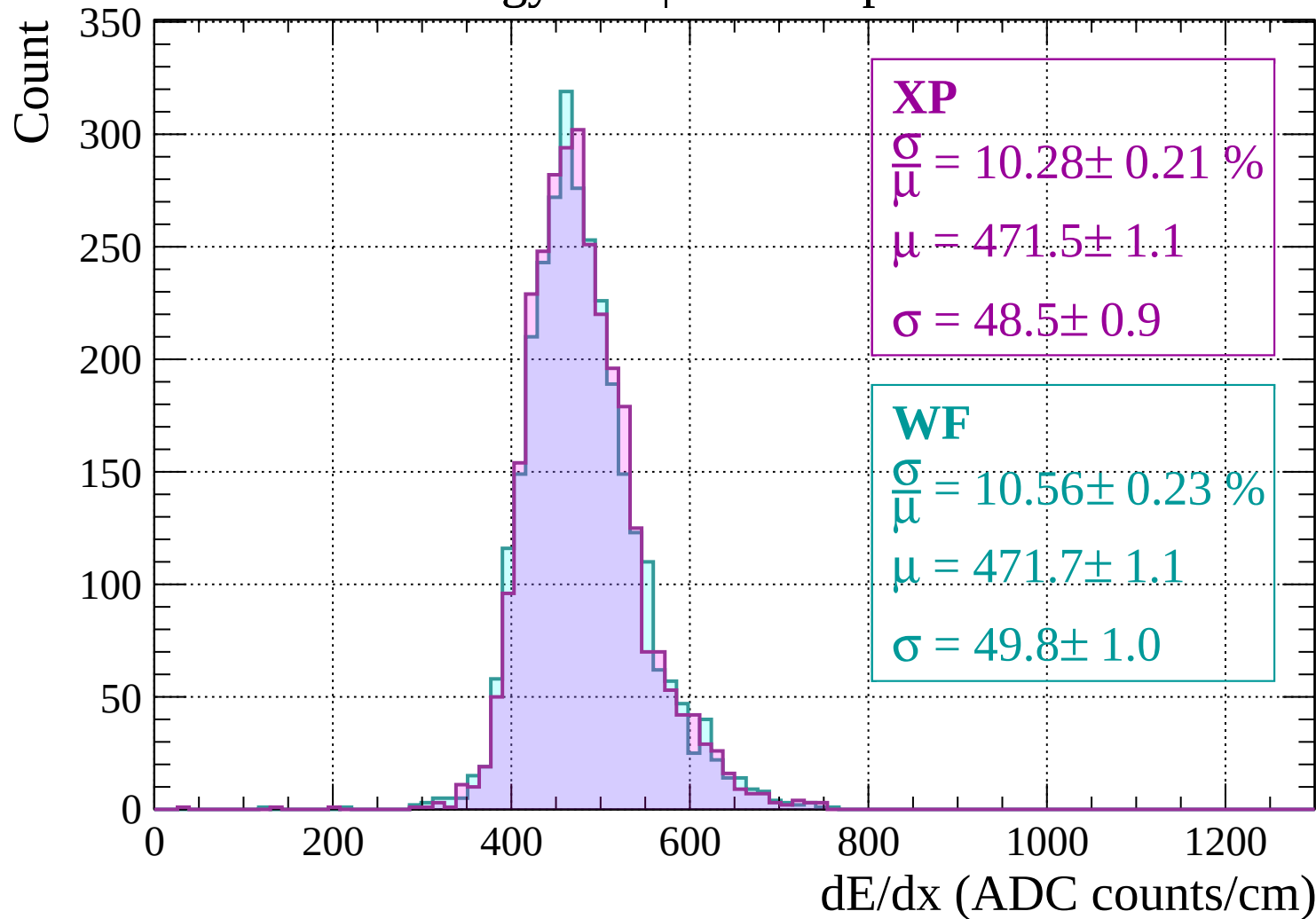
$$\frac{\sigma}{\mu} = 11.10 \pm 0.30 \%$$

$$\mu = 470.8 \pm 1.2$$

$$\sigma = 52.3 \pm 1.3$$

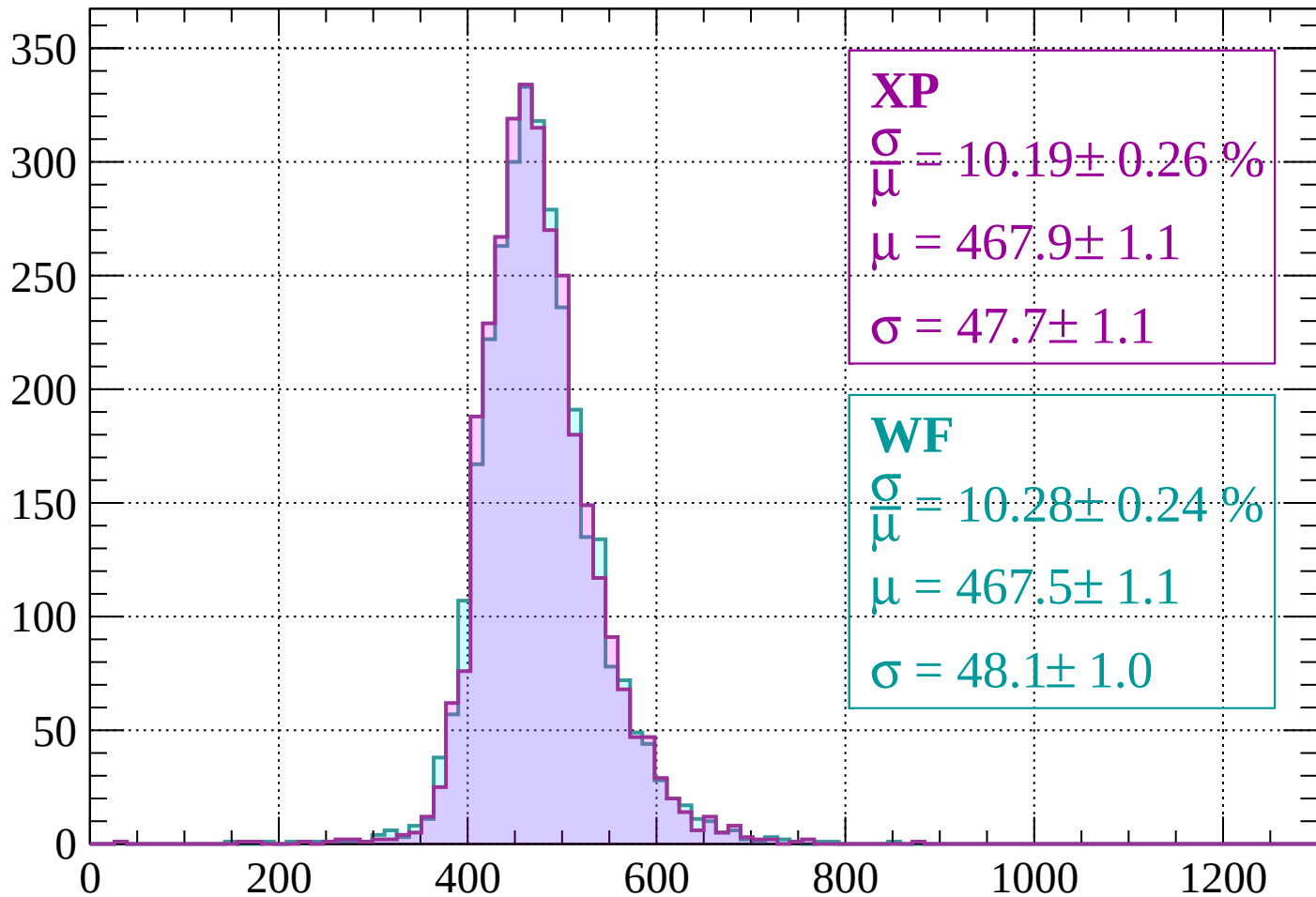
dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$



Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 10.19 \pm 0.26 \%$$

$$\mu = 467.9 \pm 1.1$$

$$\sigma = 47.7 \pm 1.1$$

WF

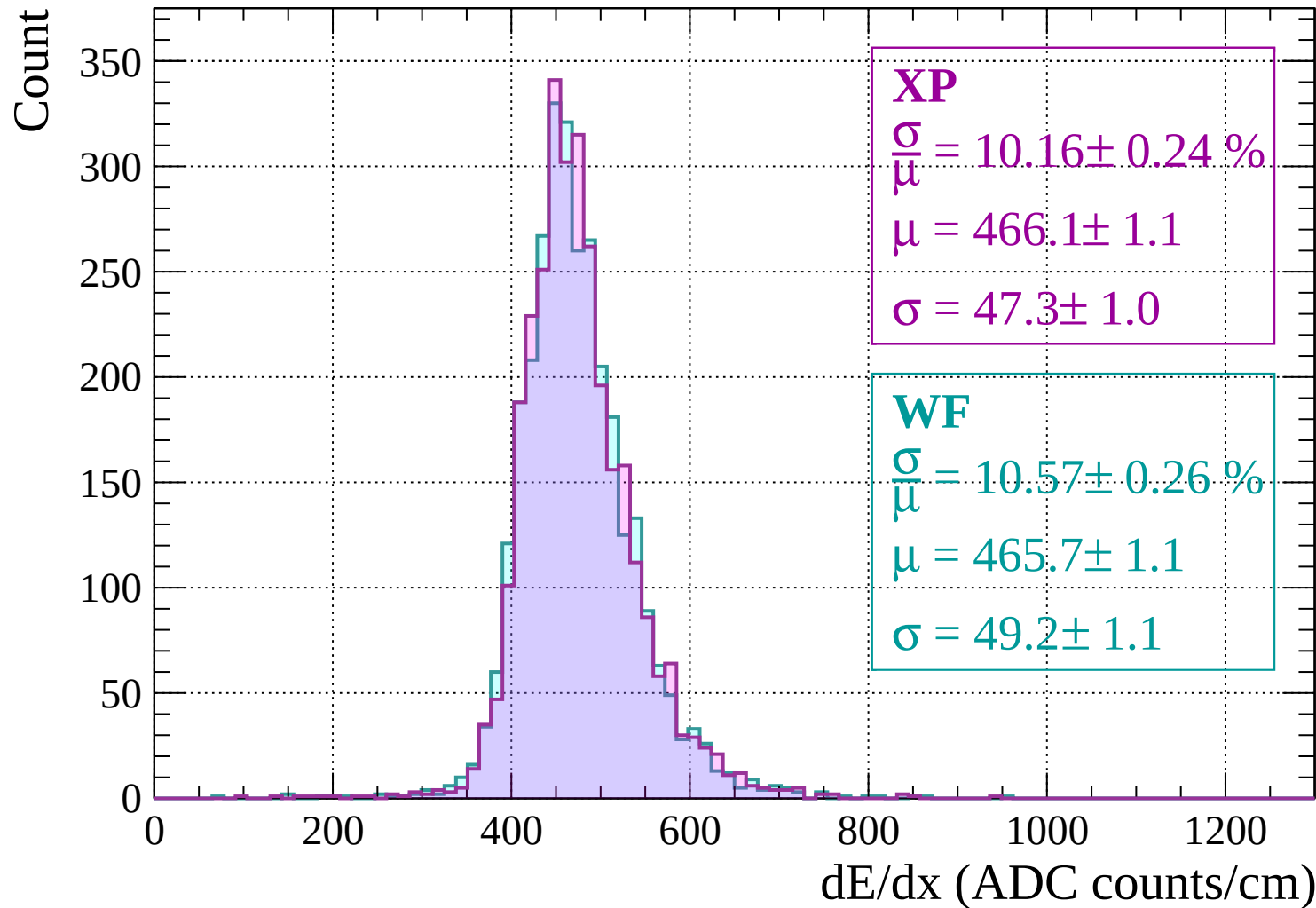
$$\frac{\sigma}{\mu} = 10.28 \pm 0.24 \%$$

$$\mu = 467.5 \pm 1.1$$

$$\sigma = 48.1 \pm 1.0$$

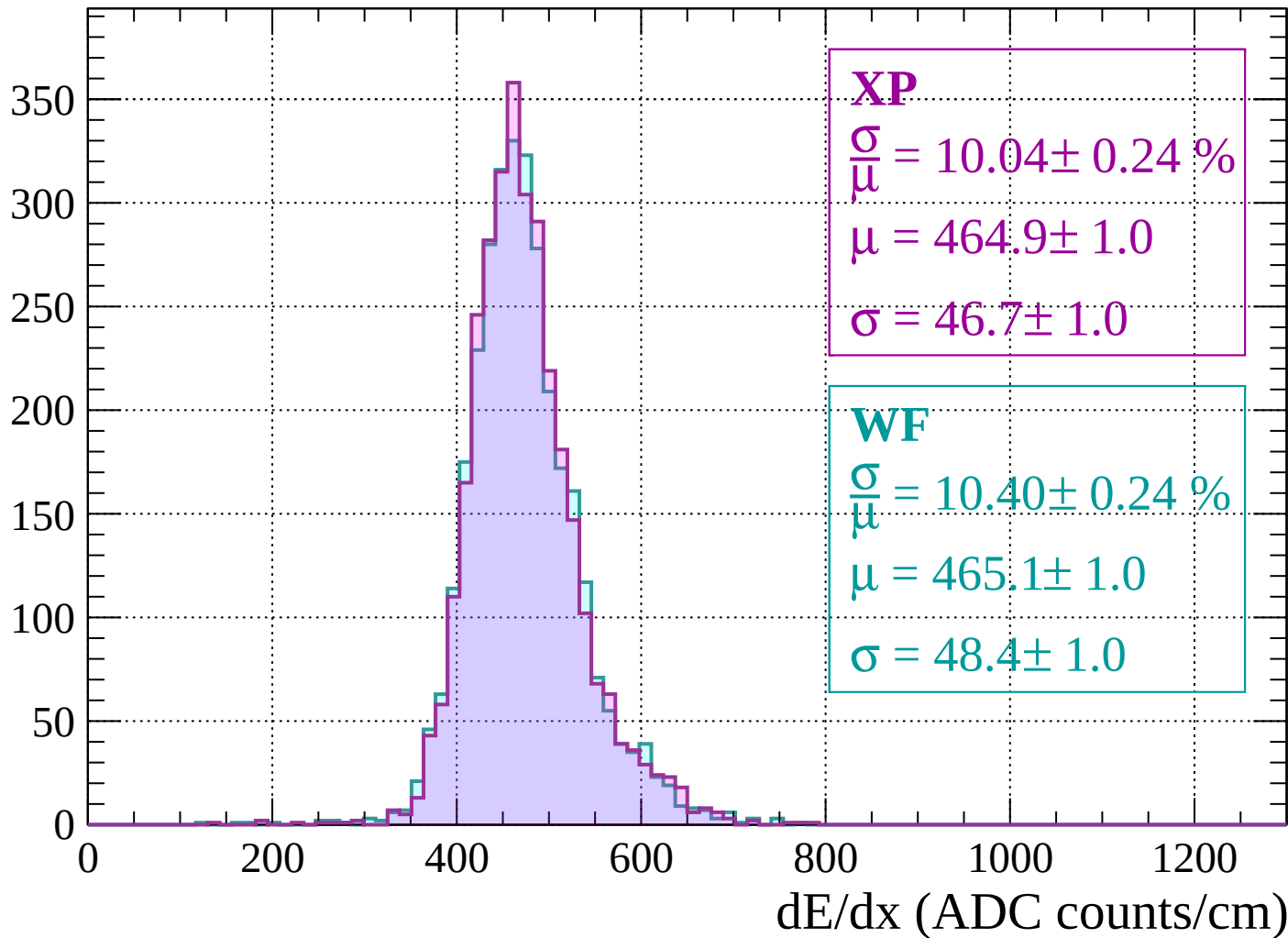
dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1520$



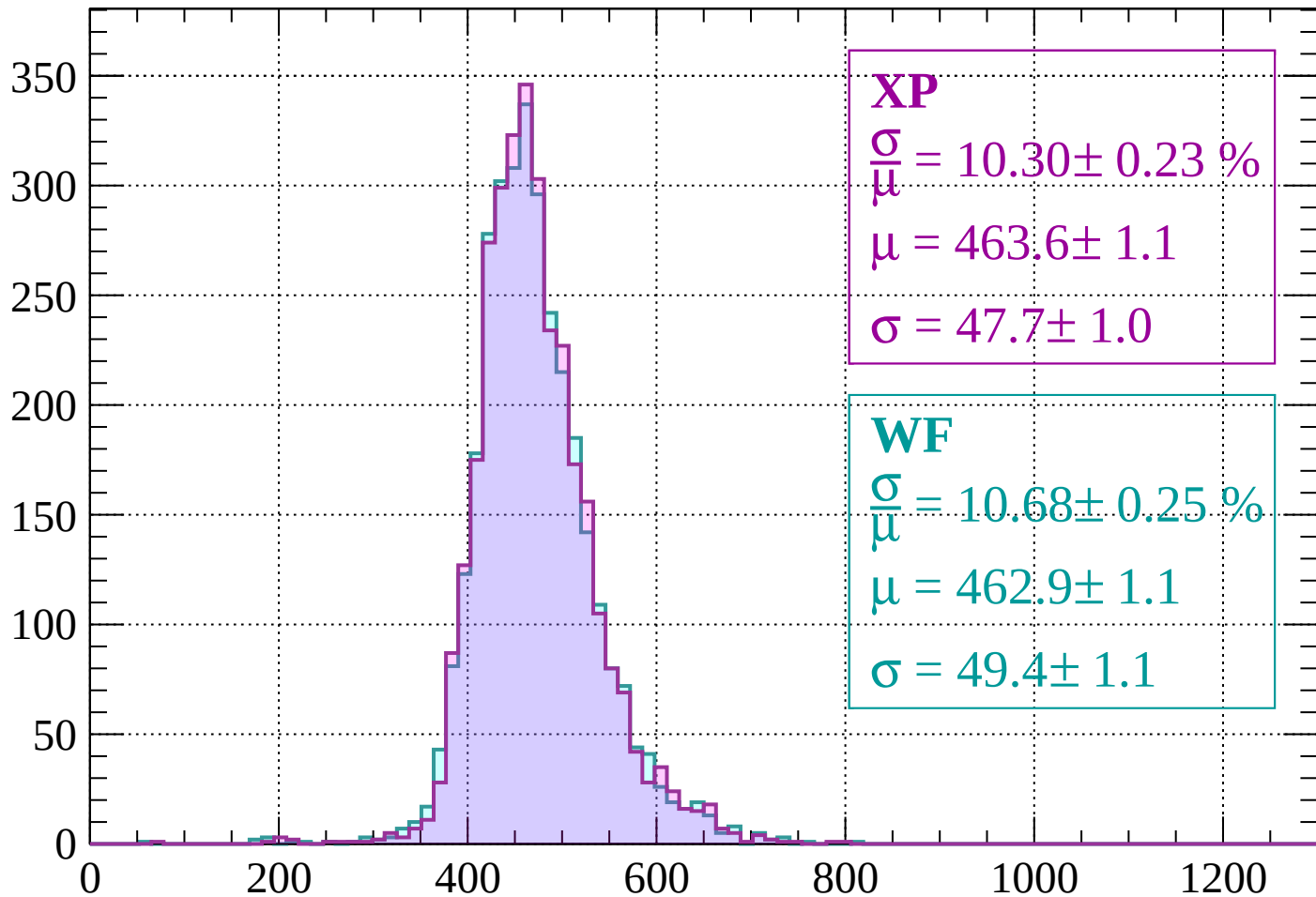
Energy loss | -1520 < p < -1480

Count



Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 10.30 \pm 0.23 \%$$

$$\mu = 463.6 \pm 1.1$$

$$\sigma = 47.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.68 \pm 0.25 \%$$

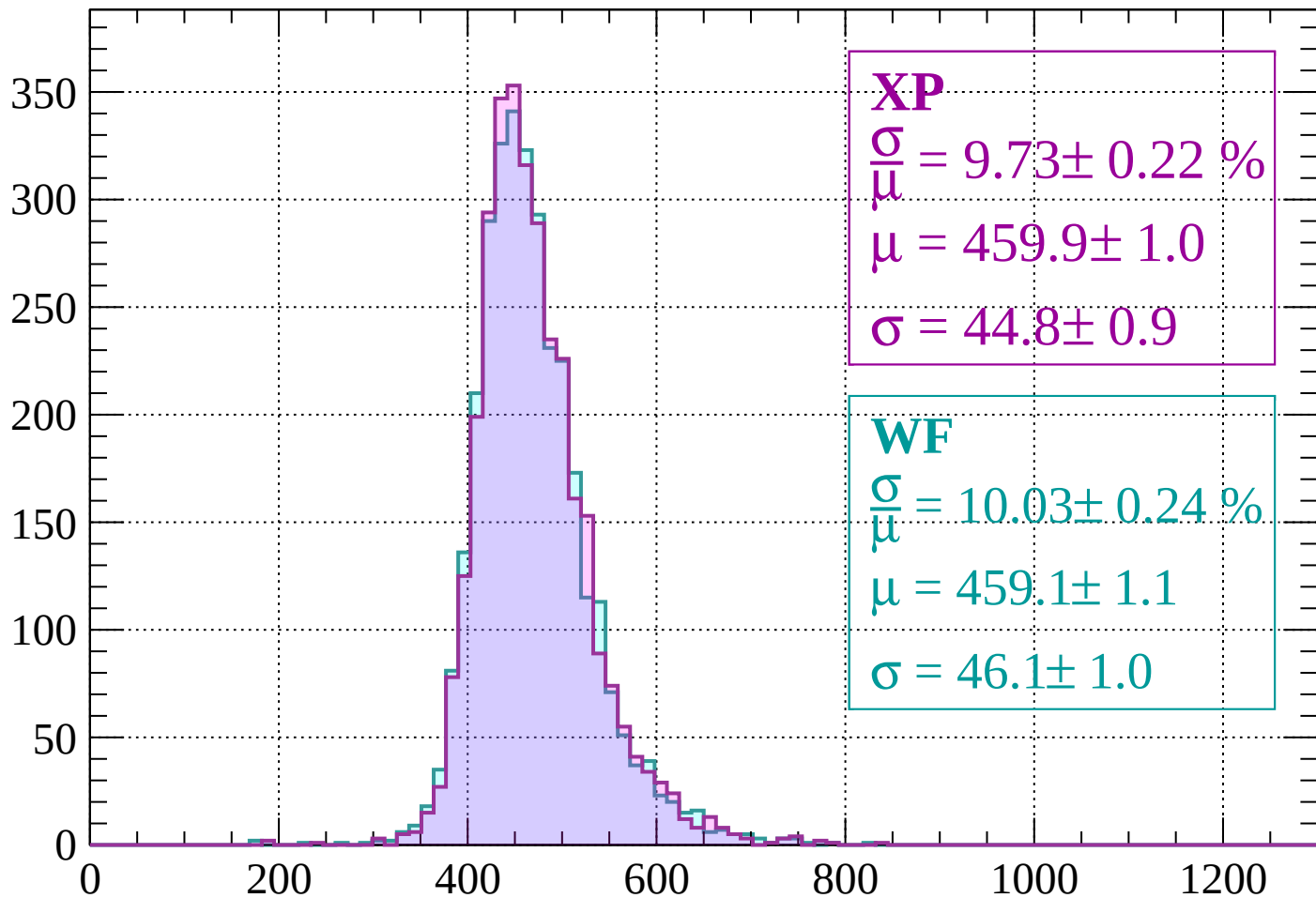
$$\mu = 462.9 \pm 1.1$$

$$\sigma = 49.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1440 < p < -1400

Count



XP

$$\frac{\sigma}{\mu} = 9.73 \pm 0.22 \%$$

$$\mu = 459.9 \pm 1.0$$

$$\sigma = 44.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.03 \pm 0.24 \%$$

$$\mu = 459.1 \pm 1.1$$

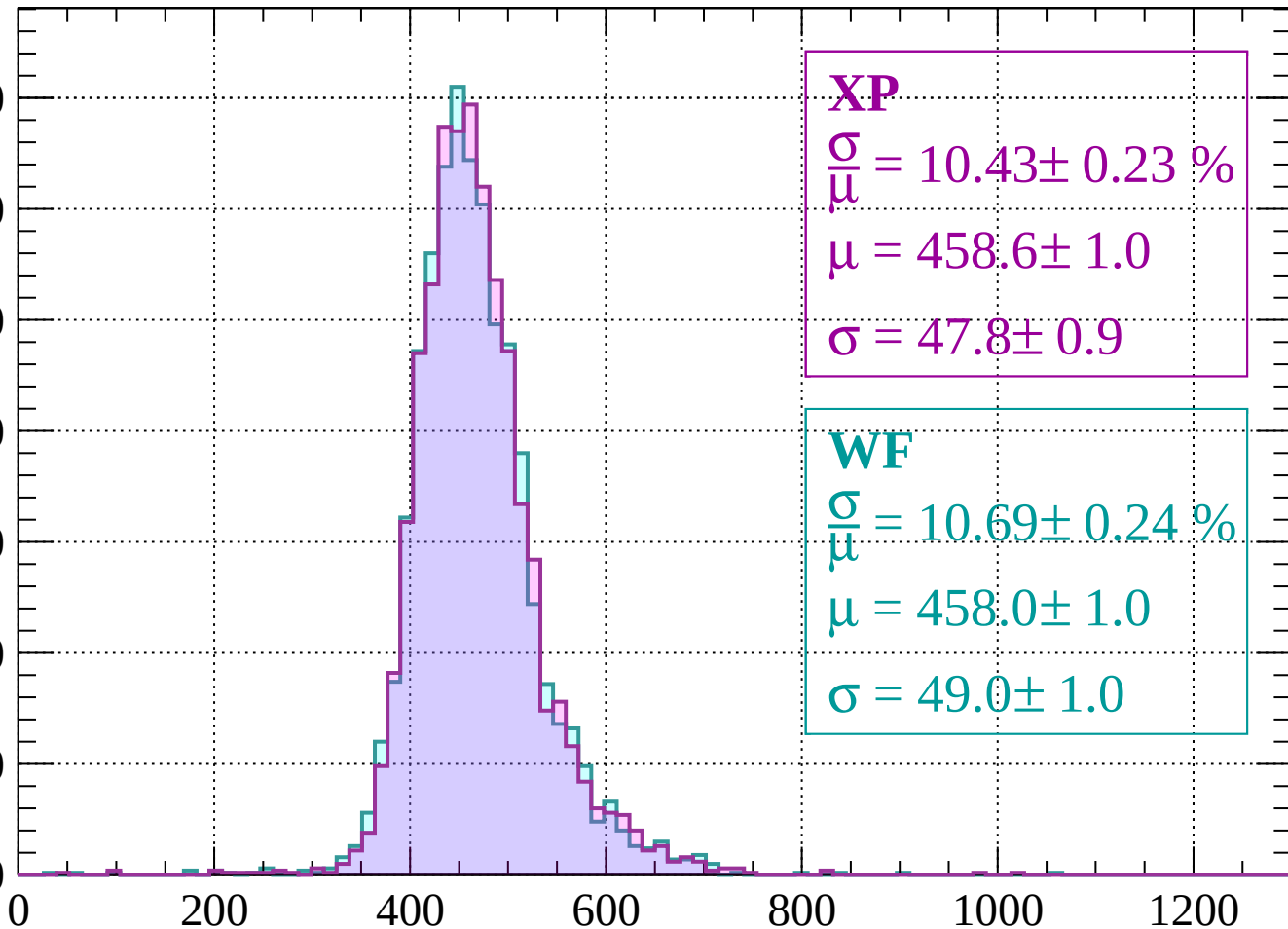
$$\sigma = 46.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.43 \pm 0.23 \%$$

$$\mu = 458.6 \pm 1.0$$

$$\sigma = 47.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.69 \pm 0.24 \%$$

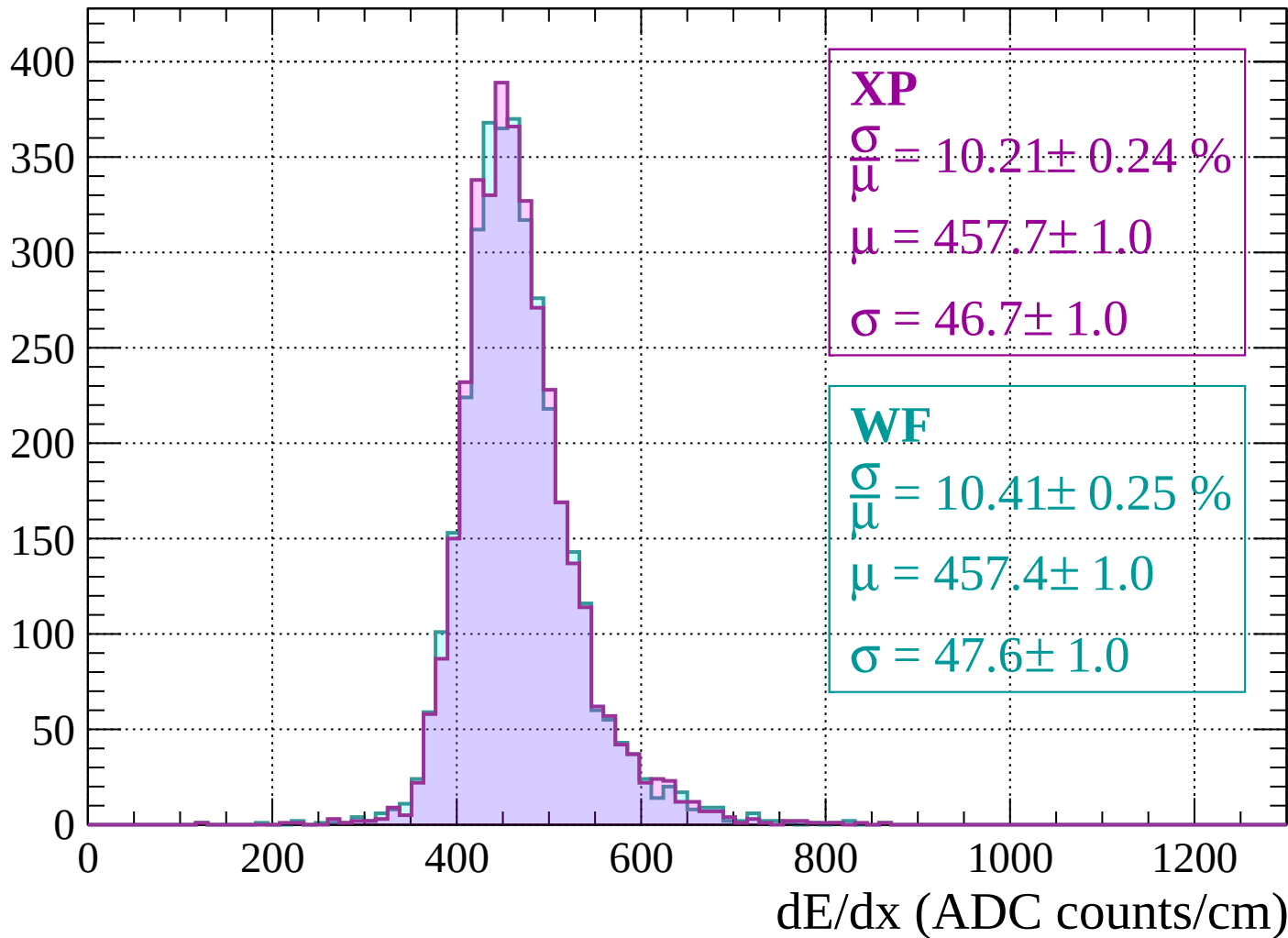
$$\mu = 458.0 \pm 1.0$$

$$\sigma = 49.0 \pm 1.0$$

dE/dx (ADC counts/cm)

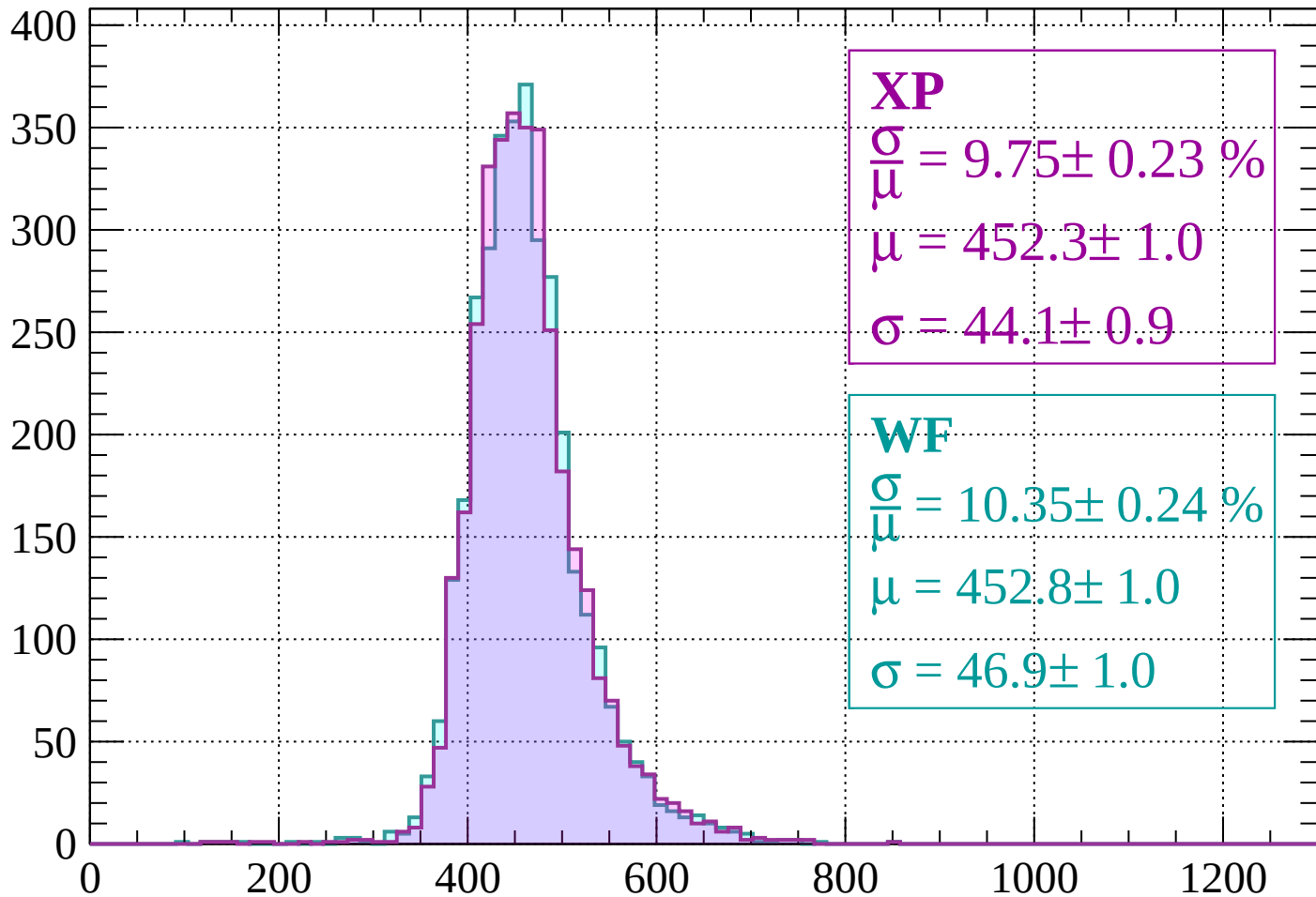
Energy loss | $-1360 < p < -1320$

Count



Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 9.75 \pm 0.23 \%$$

$$\mu = 452.3 \pm 1.0$$

$$\sigma = 44.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.35 \pm 0.24 \%$$

$$\mu = 452.8 \pm 1.0$$

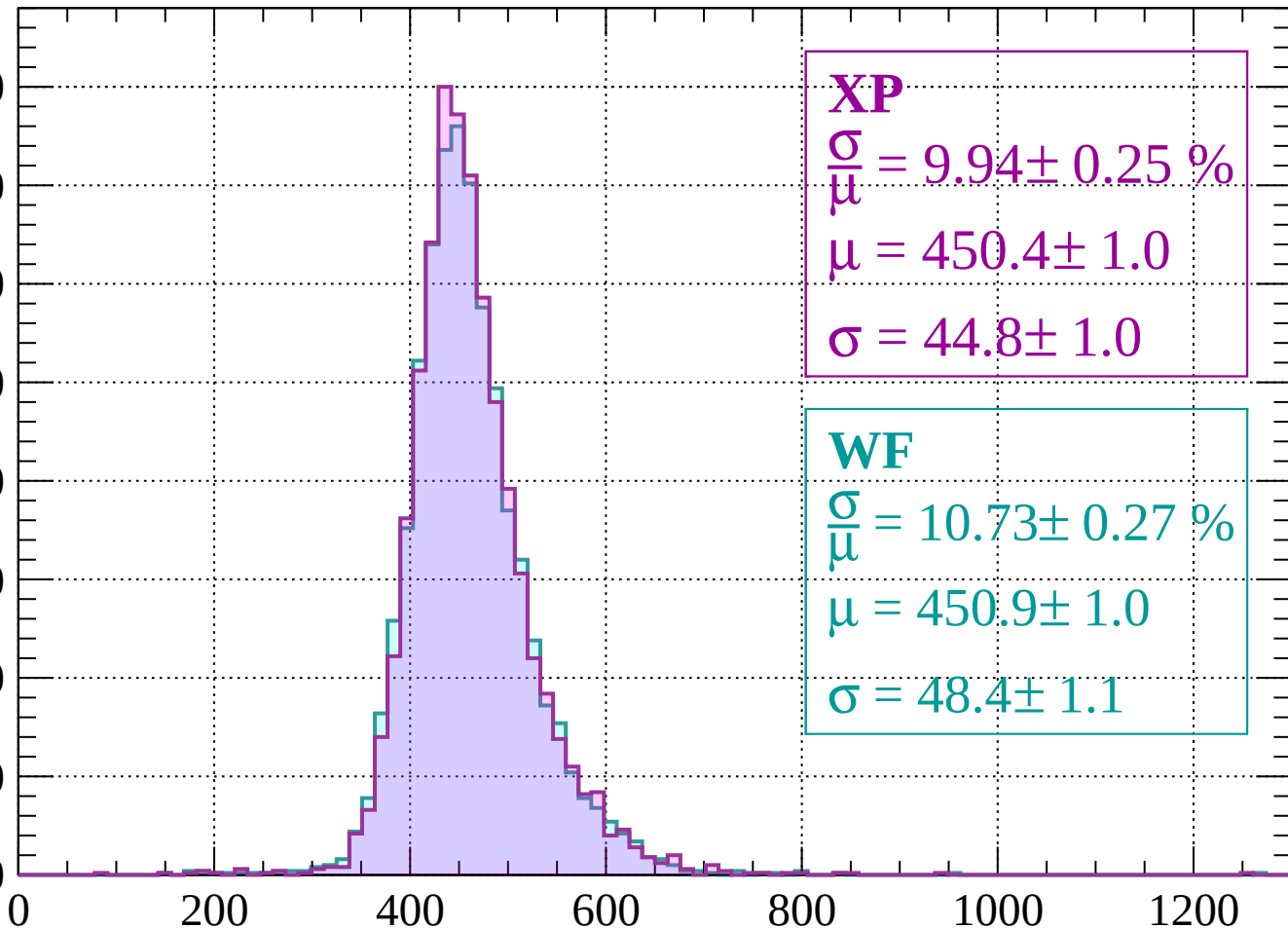
$$\sigma = 46.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count

400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.94 \pm 0.25 \%$$

$$\mu = 450.4 \pm 1.0$$

$$\sigma = 44.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.73 \pm 0.27 \%$$

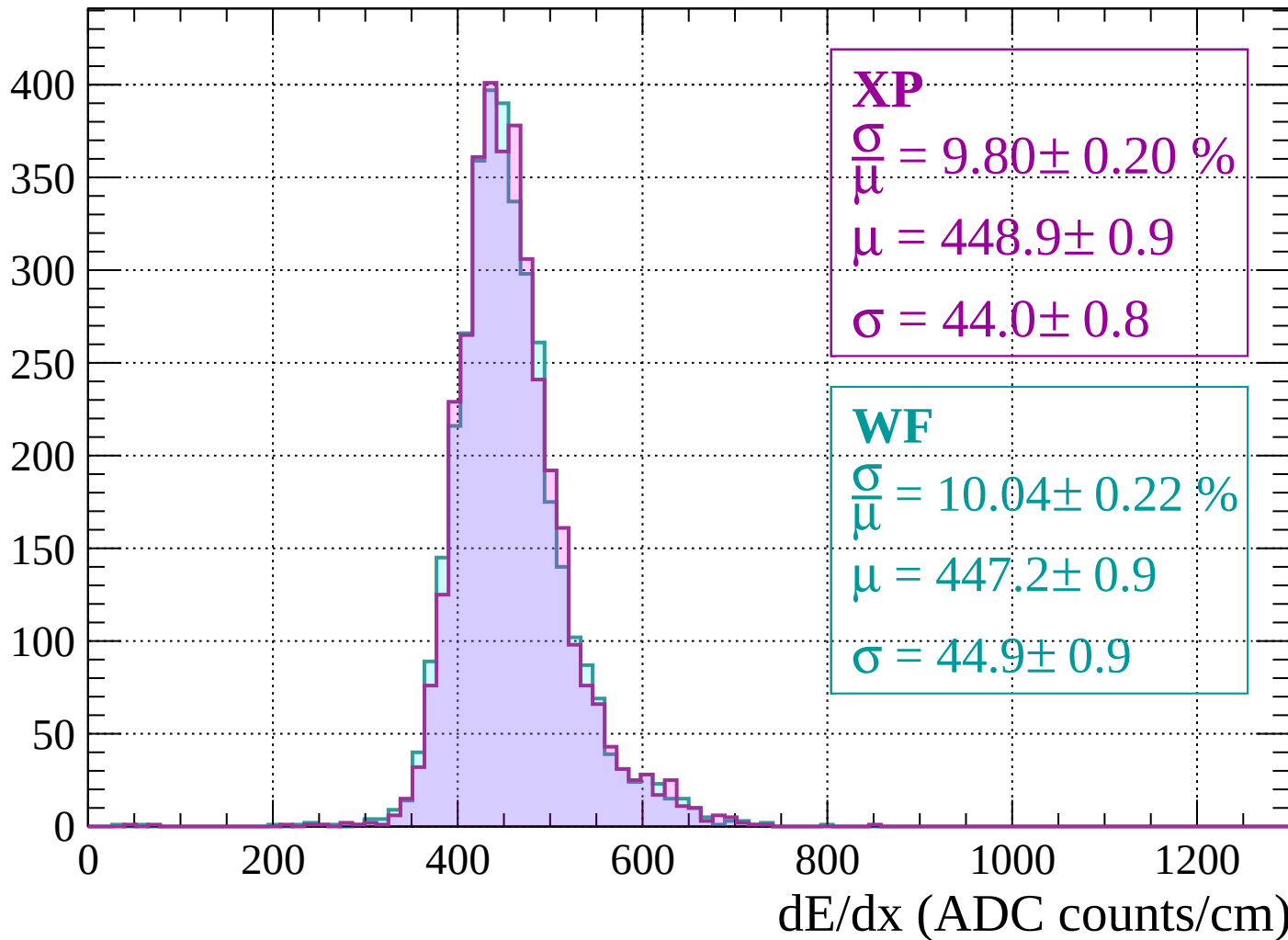
$$\mu = 450.9 \pm 1.0$$

$$\sigma = 48.4 \pm 1.1$$

dE/dx (ADC counts/cm)

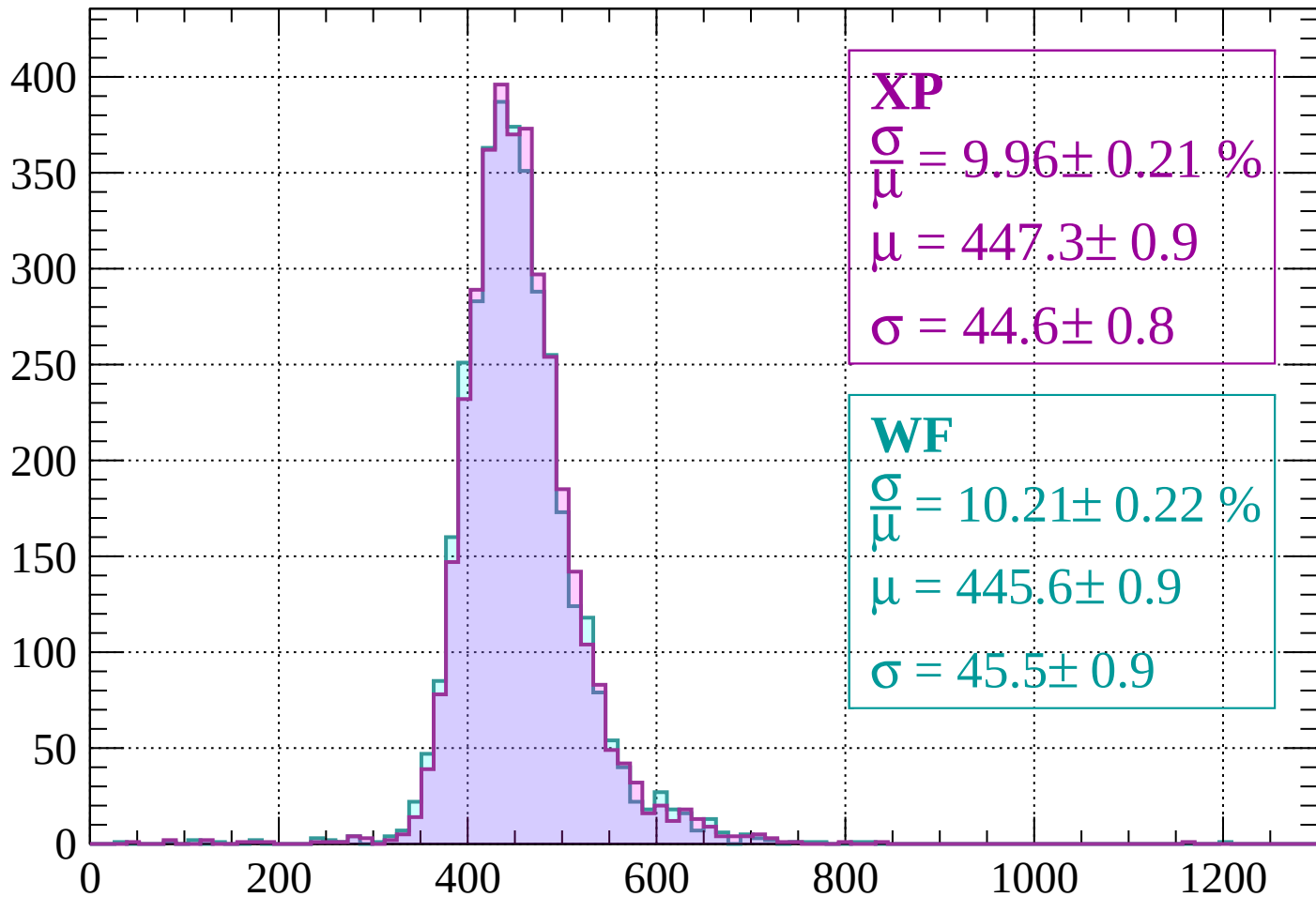
Energy loss | $-1240 < p < -1200$

Count



Energy loss | $-1200 < p < -1160$

Count



XP

$$\frac{\sigma}{\mu} = 9.96 \pm 0.21 \%$$

$$\mu = 447.3 \pm 0.9$$

$$\sigma = 44.6 \pm 0.8$$

WF

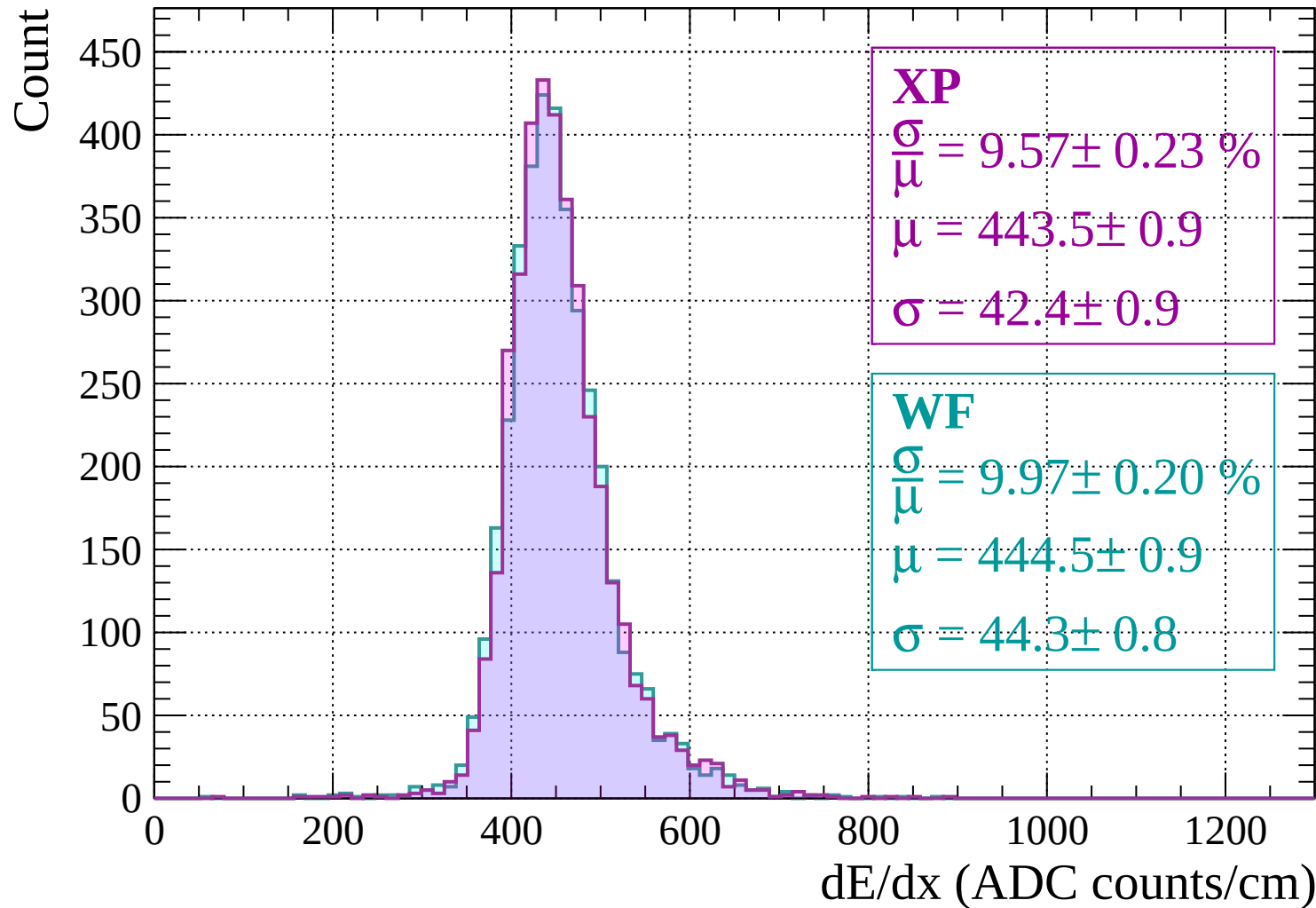
$$\frac{\sigma}{\mu} = 10.21 \pm 0.22 \%$$

$$\mu = 445.6 \pm 0.9$$

$$\sigma = 45.5 \pm 0.9$$

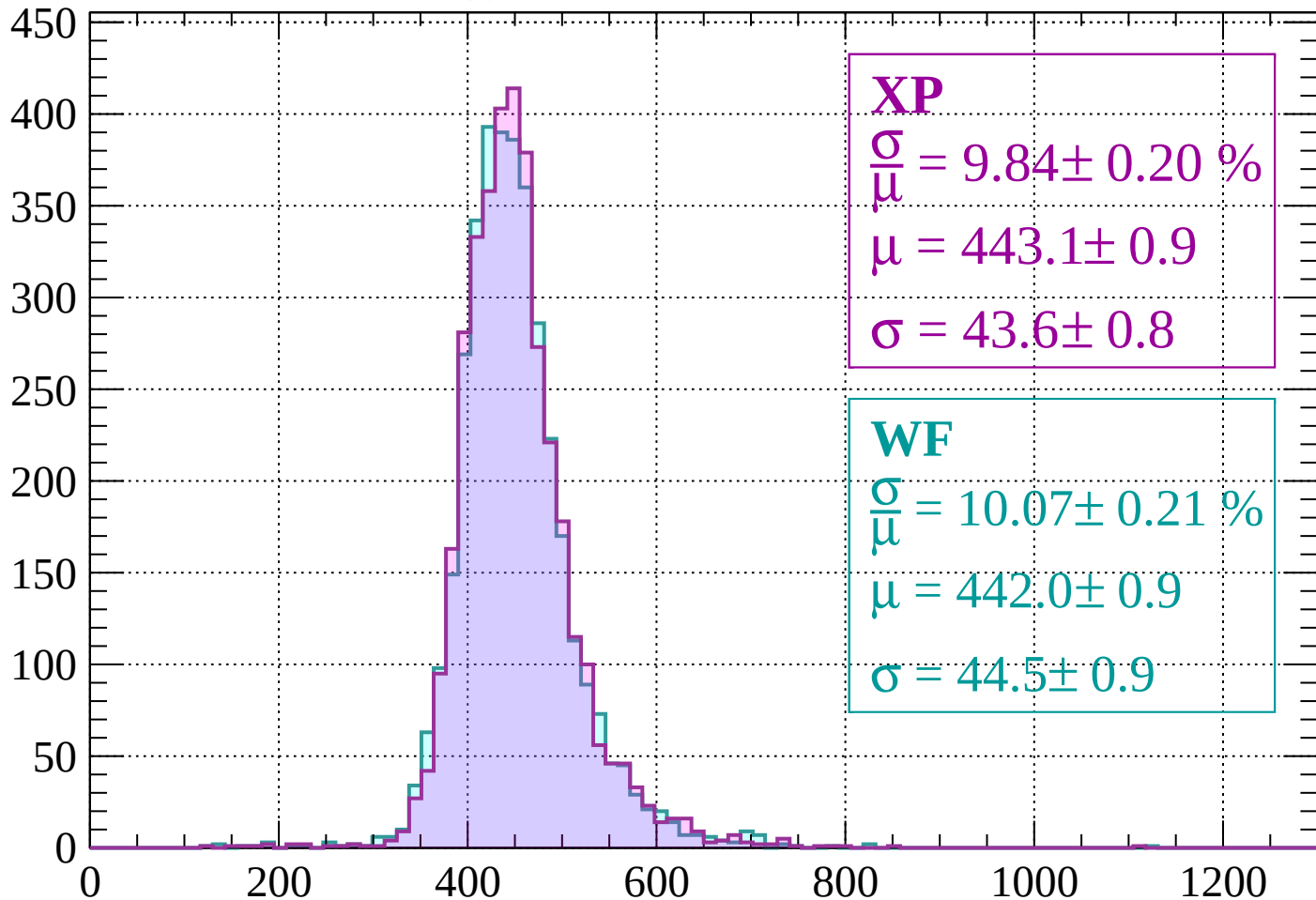
dE/dx (ADC counts/cm)

Energy loss | -1160 < p < -1120



Energy loss | -1120 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 9.84 \pm 0.20 \%$$

$$\mu = 443.1 \pm 0.9$$

$$\sigma = 43.6 \pm 0.8$$

WF

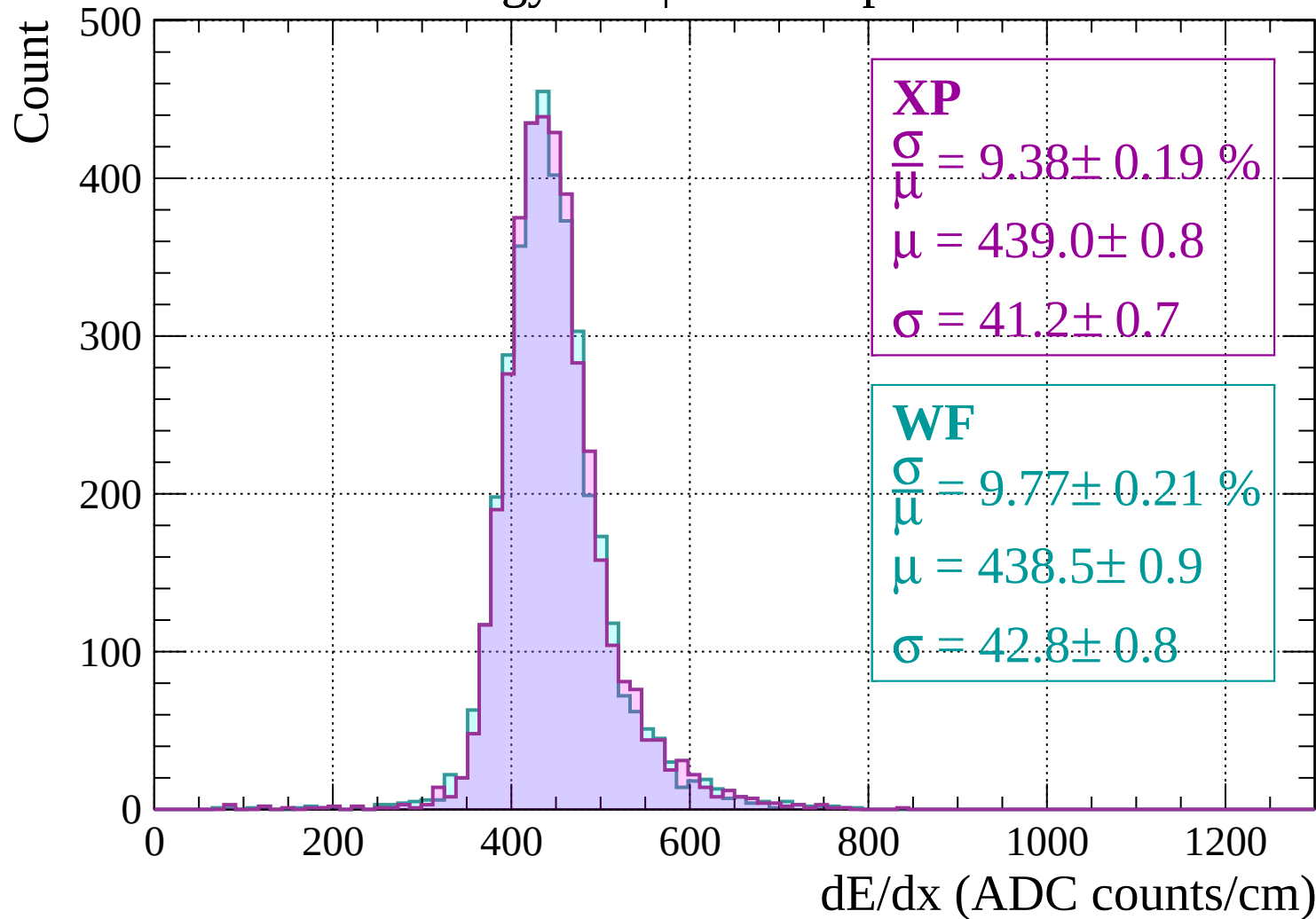
$$\frac{\sigma}{\mu} = 10.07 \pm 0.21 \%$$

$$\mu = 442.0 \pm 0.9$$

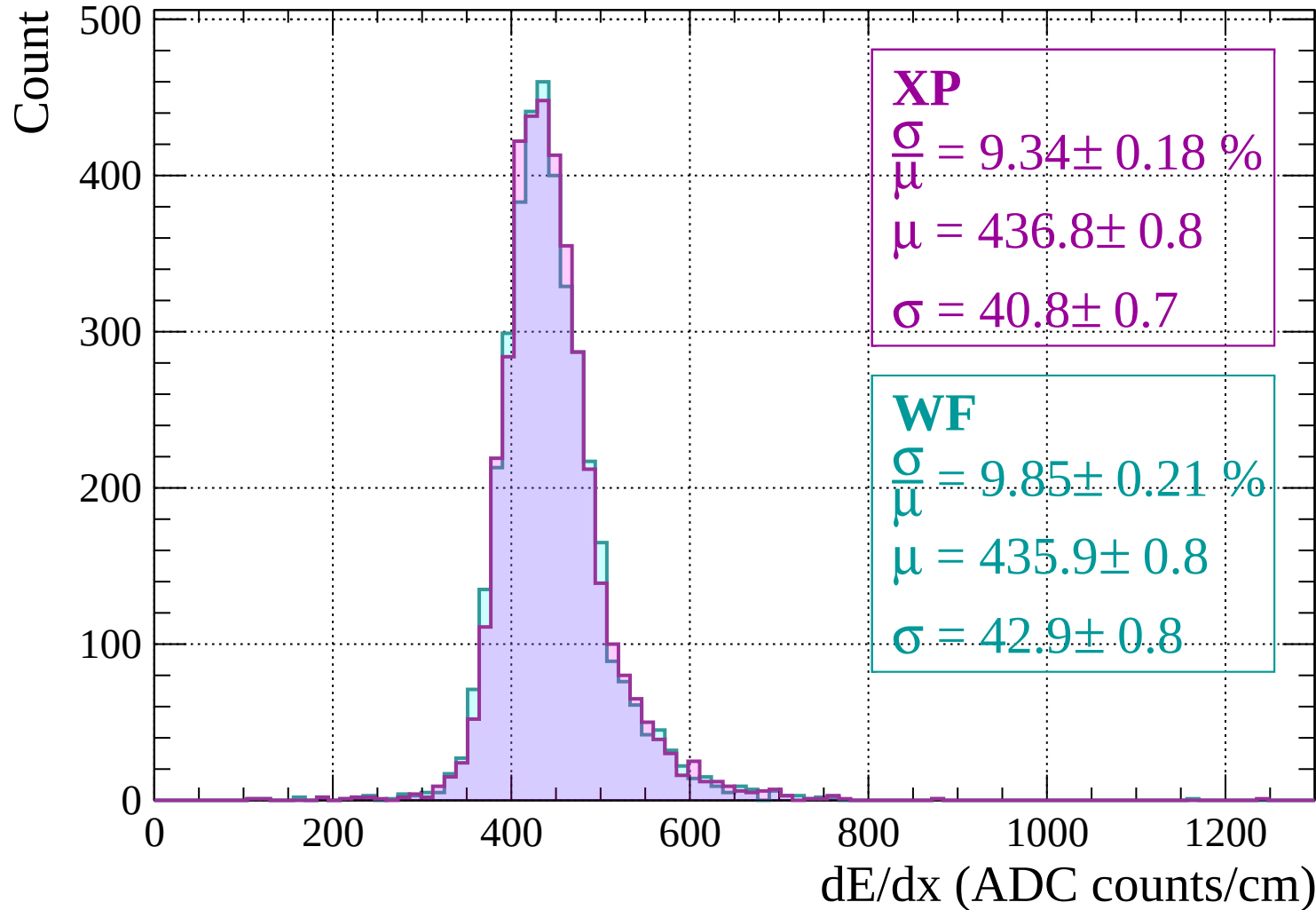
$$\sigma = 44.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

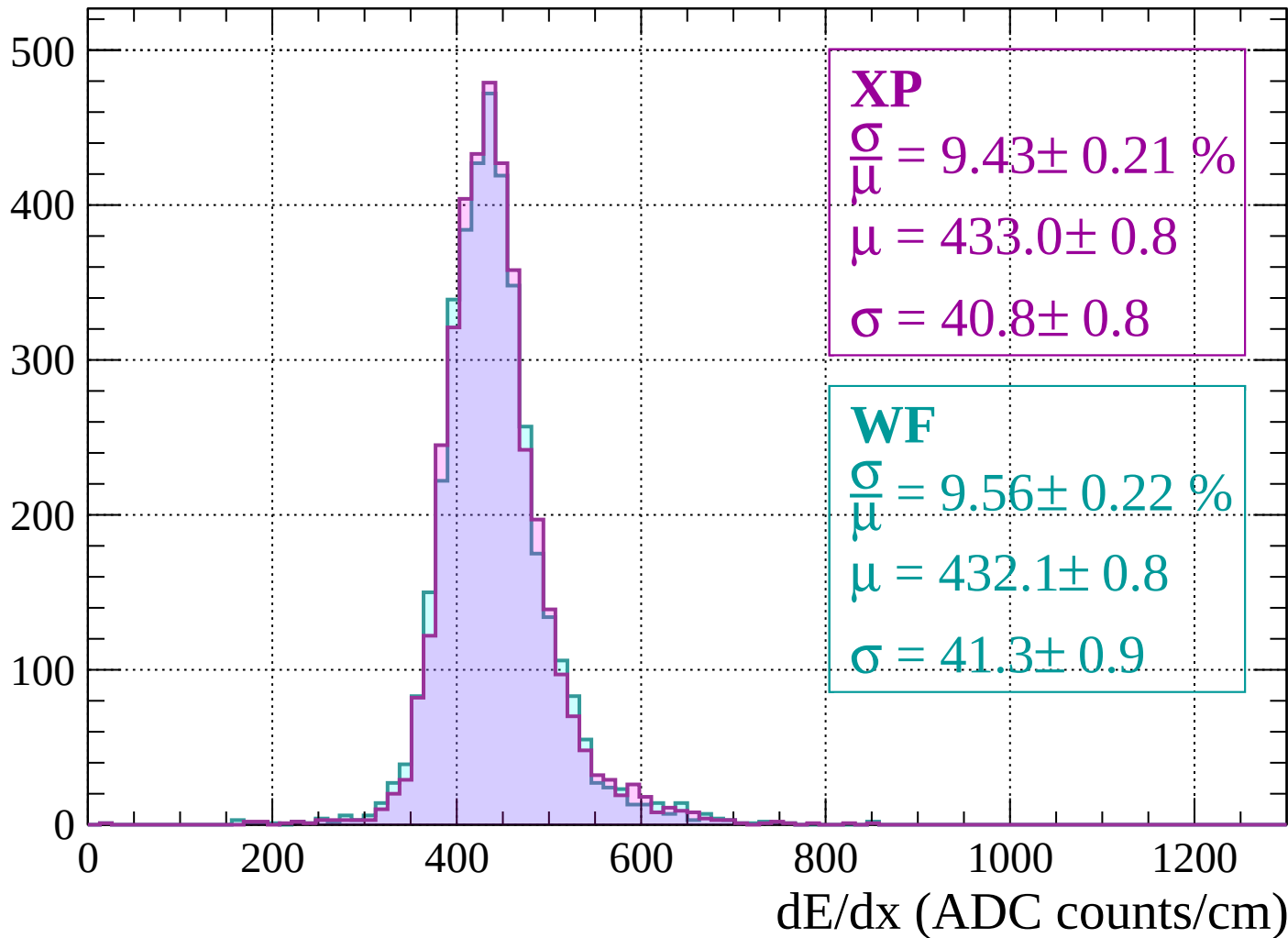


Energy loss | $-1040 < p < -1000$



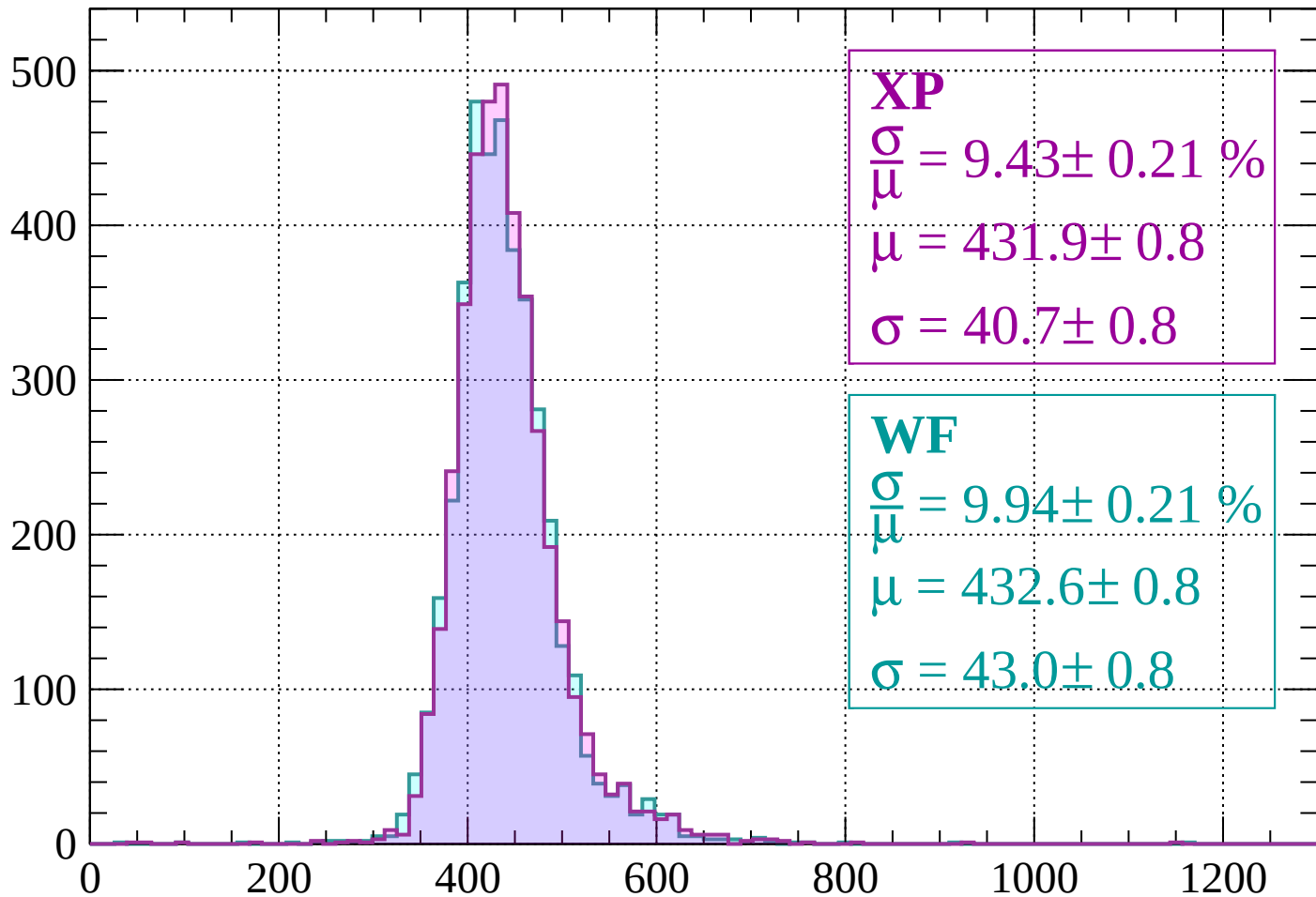
Energy loss | $-1000 < p < -960$

Count



Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 9.43 \pm 0.21 \%$$

$$\mu = 431.9 \pm 0.8$$

$$\sigma = 40.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.94 \pm 0.21 \%$$

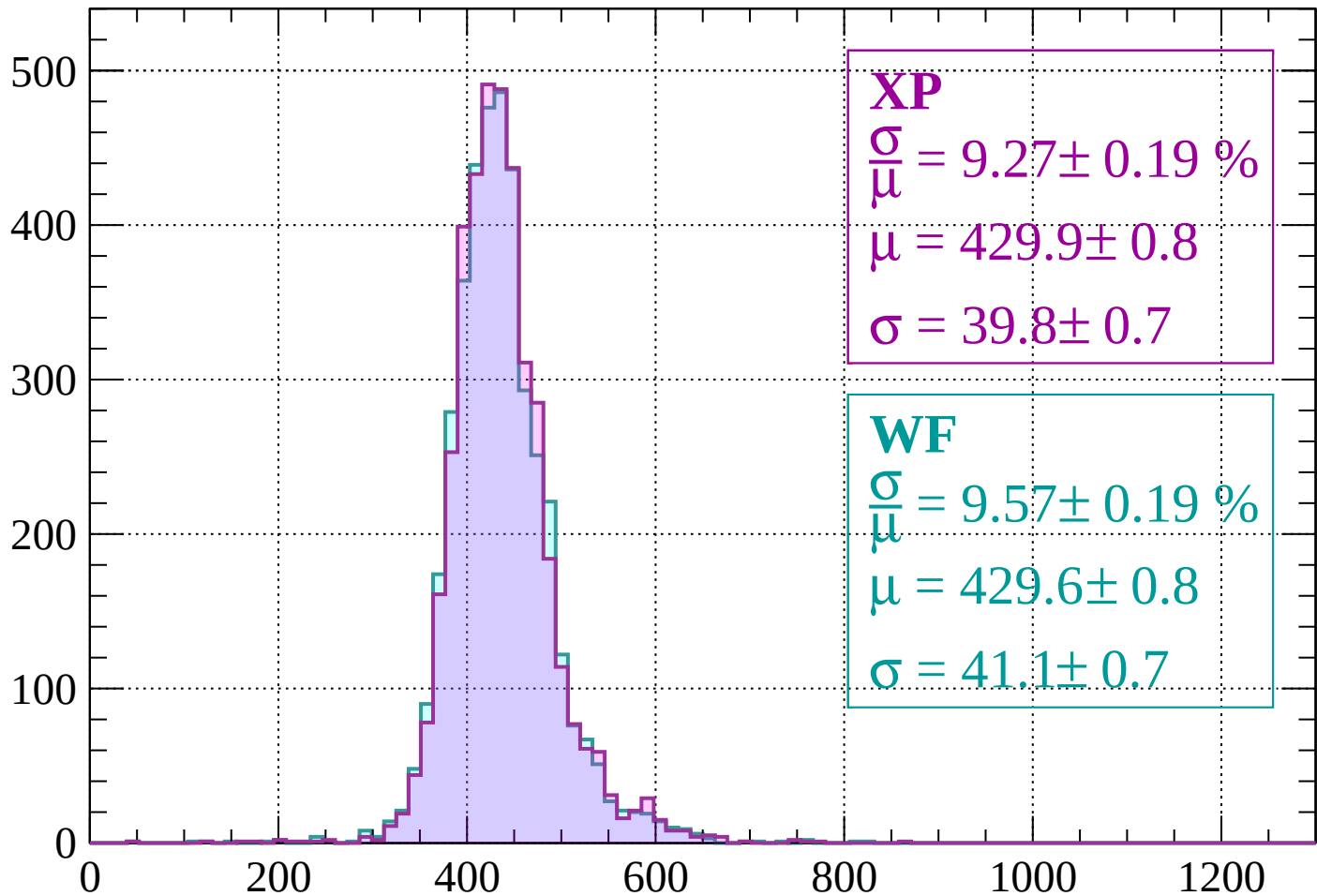
$$\mu = 432.6 \pm 0.8$$

$$\sigma = 43.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 9.27 \pm 0.19 \%$$

$$\mu = 429.9 \pm 0.8$$

$$\sigma = 39.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.57 \pm 0.19 \%$$

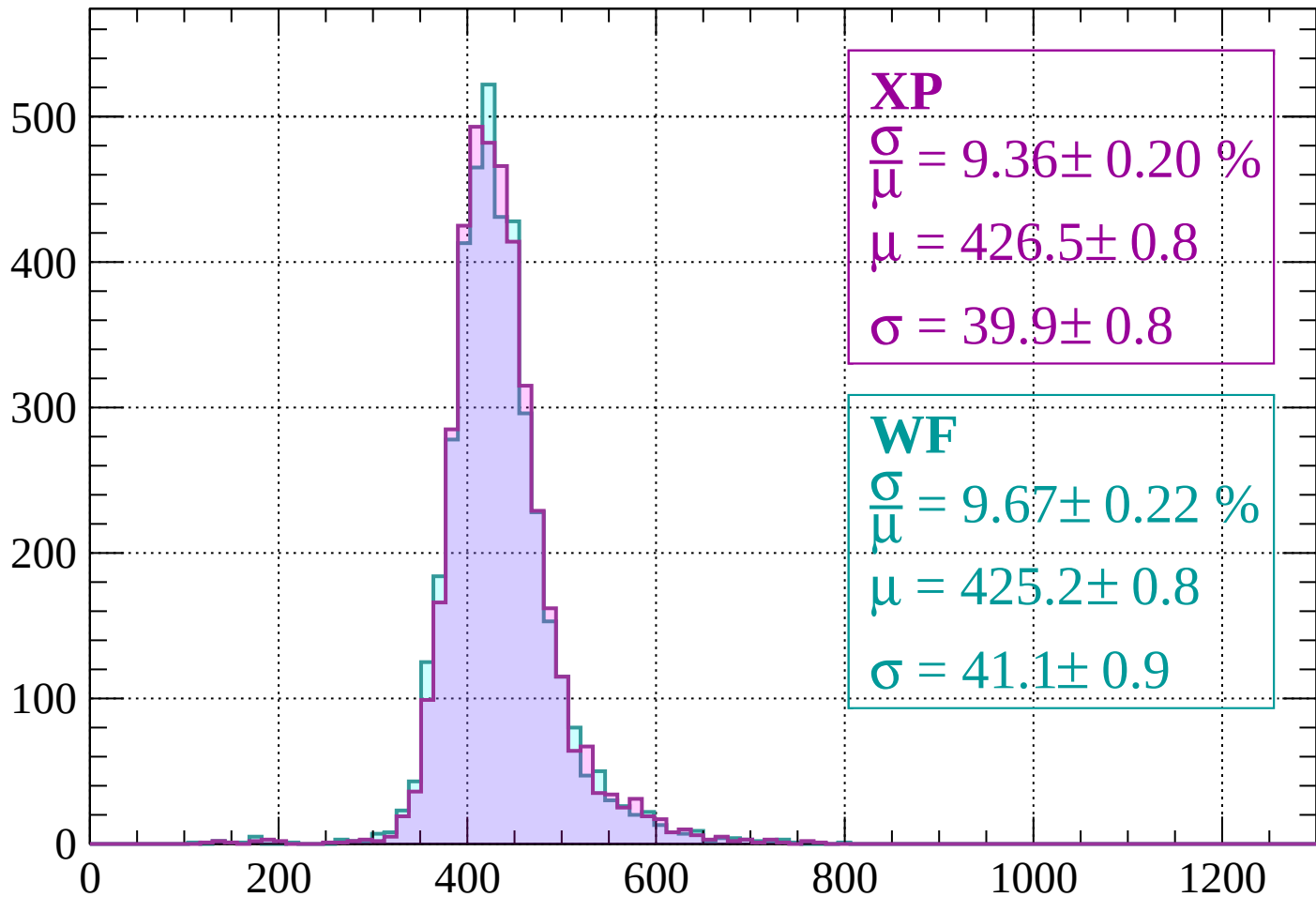
$$\mu = 429.6 \pm 0.8$$

$$\sigma = 41.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.36 \pm 0.20 \%$$

$$\mu = 426.5 \pm 0.8$$

$$\sigma = 39.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.22 \%$$

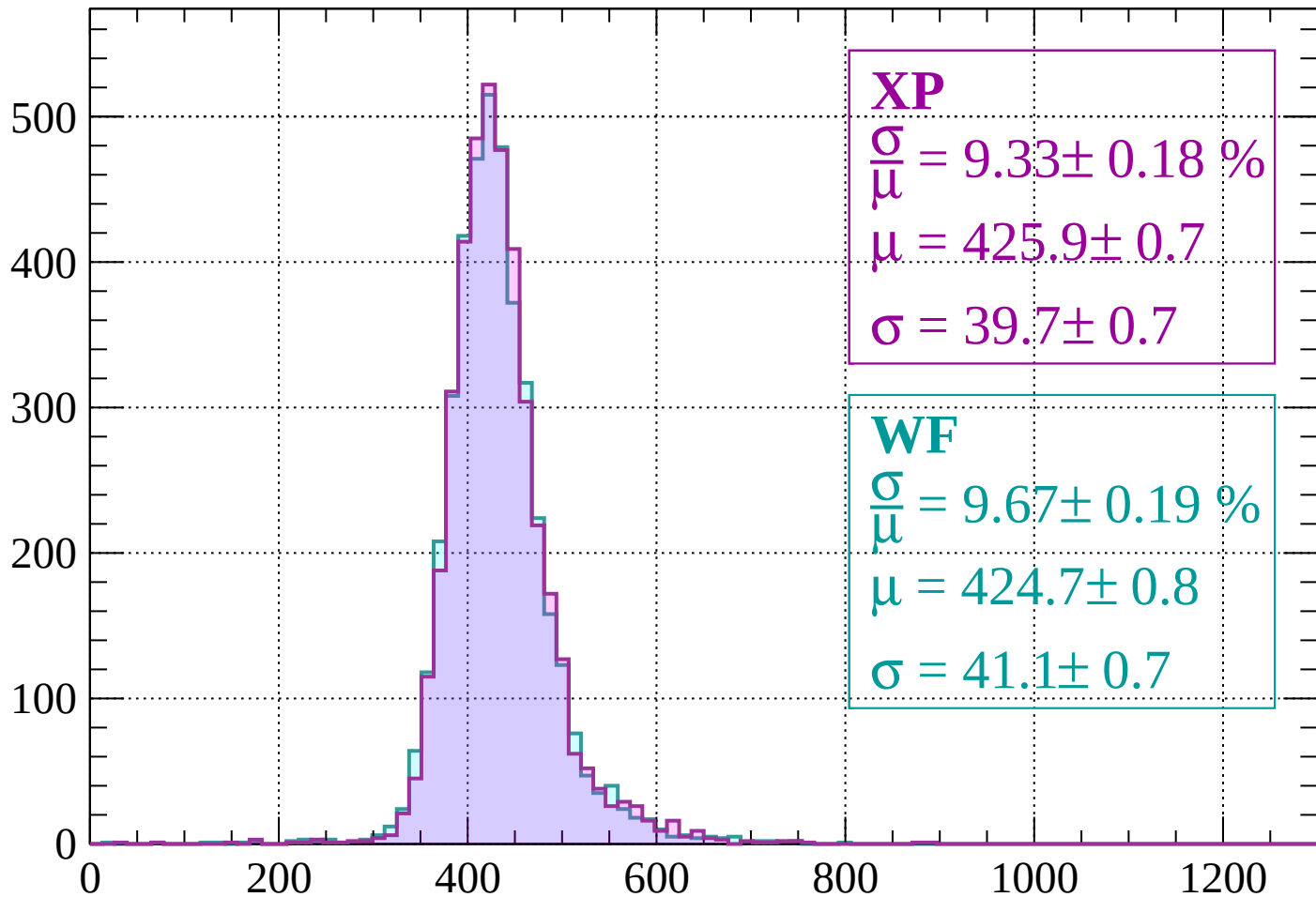
$$\mu = 425.2 \pm 0.8$$

$$\sigma = 41.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 9.33 \pm 0.18 \%$$

$$\mu = 425.9 \pm 0.7$$

$$\sigma = 39.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.19 \%$$

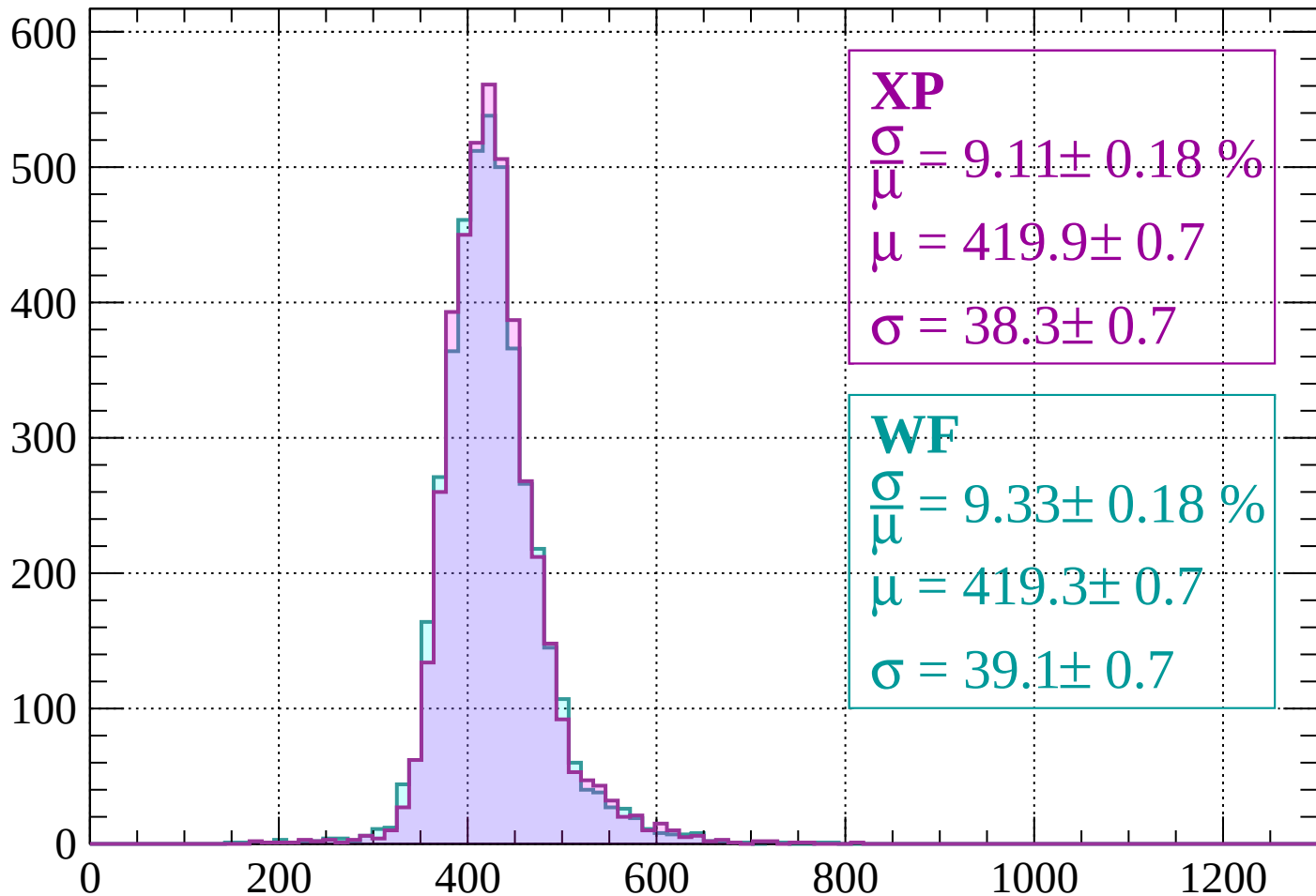
$$\mu = 424.7 \pm 0.8$$

$$\sigma = 41.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.18 \%$$

$$\mu = 419.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.18 \%$$

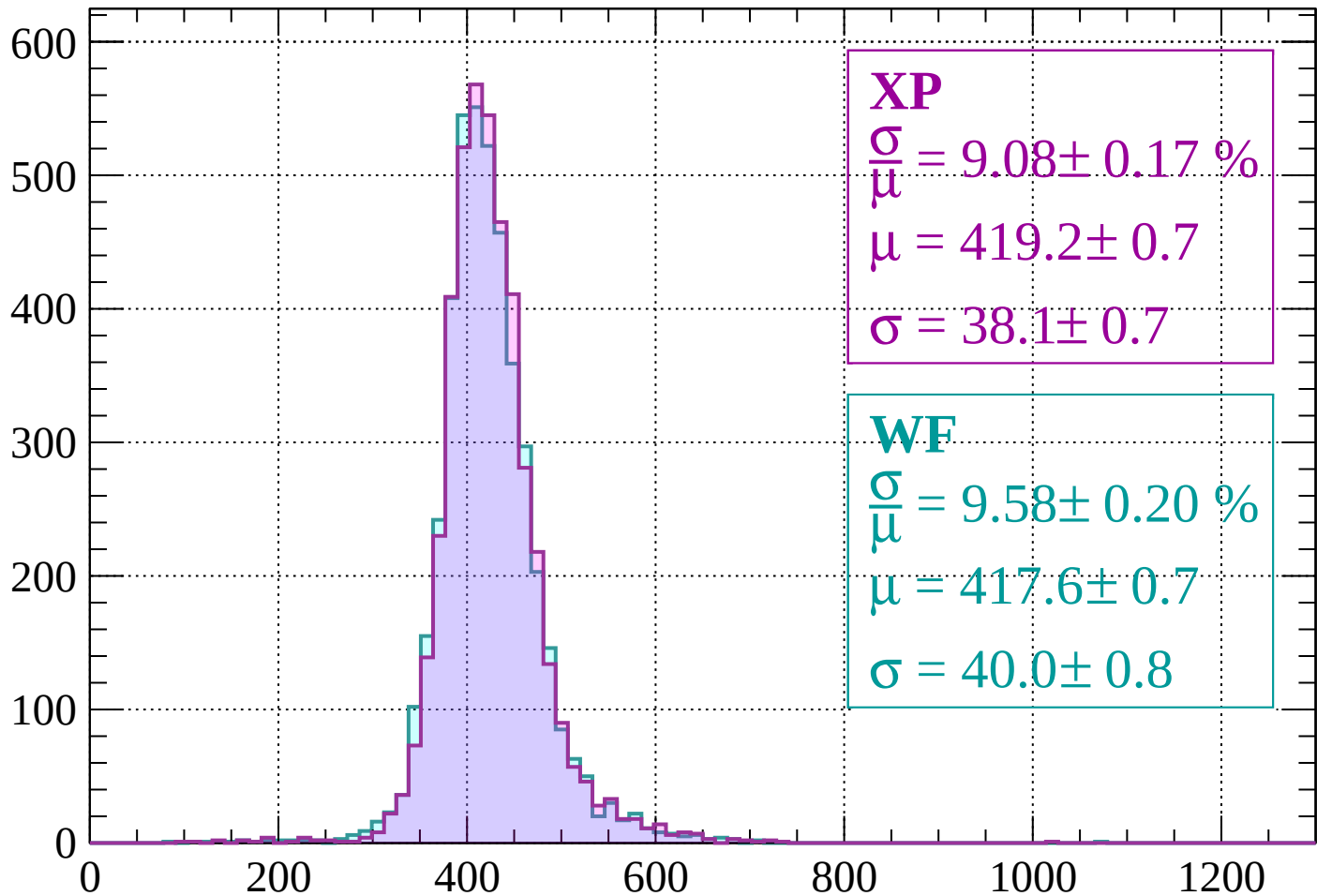
$$\mu = 419.3 \pm 0.7$$

$$\sigma = 39.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.08 \pm 0.17 \%$$

$$\mu = 419.2 \pm 0.7$$

$$\sigma = 38.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.58 \pm 0.20 \%$$

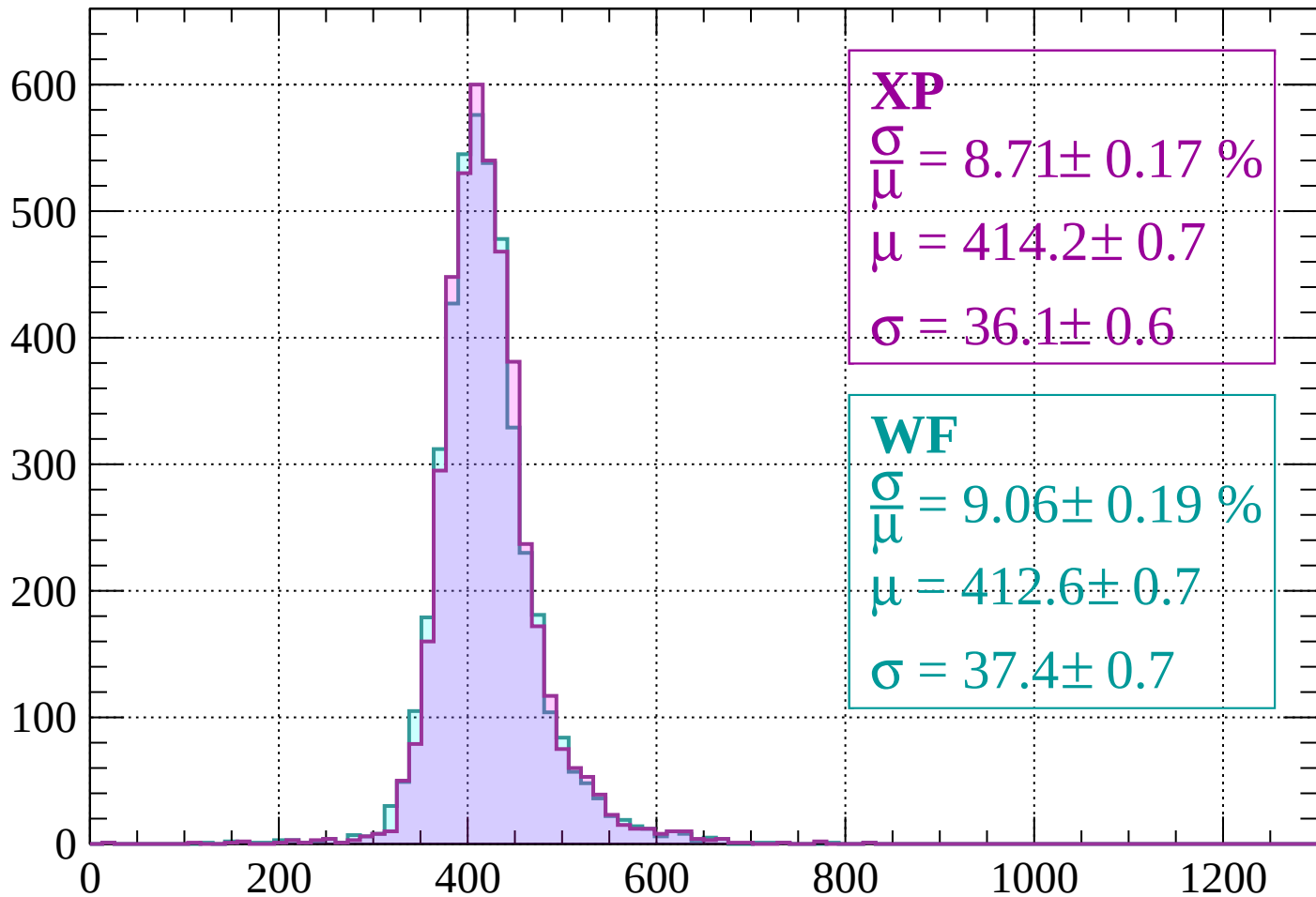
$$\mu = 417.6 \pm 0.7$$

$$\sigma = 40.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.71 \pm 0.17 \%$$

$$\mu = 414.2 \pm 0.7$$

$$\sigma = 36.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.19 \%$$

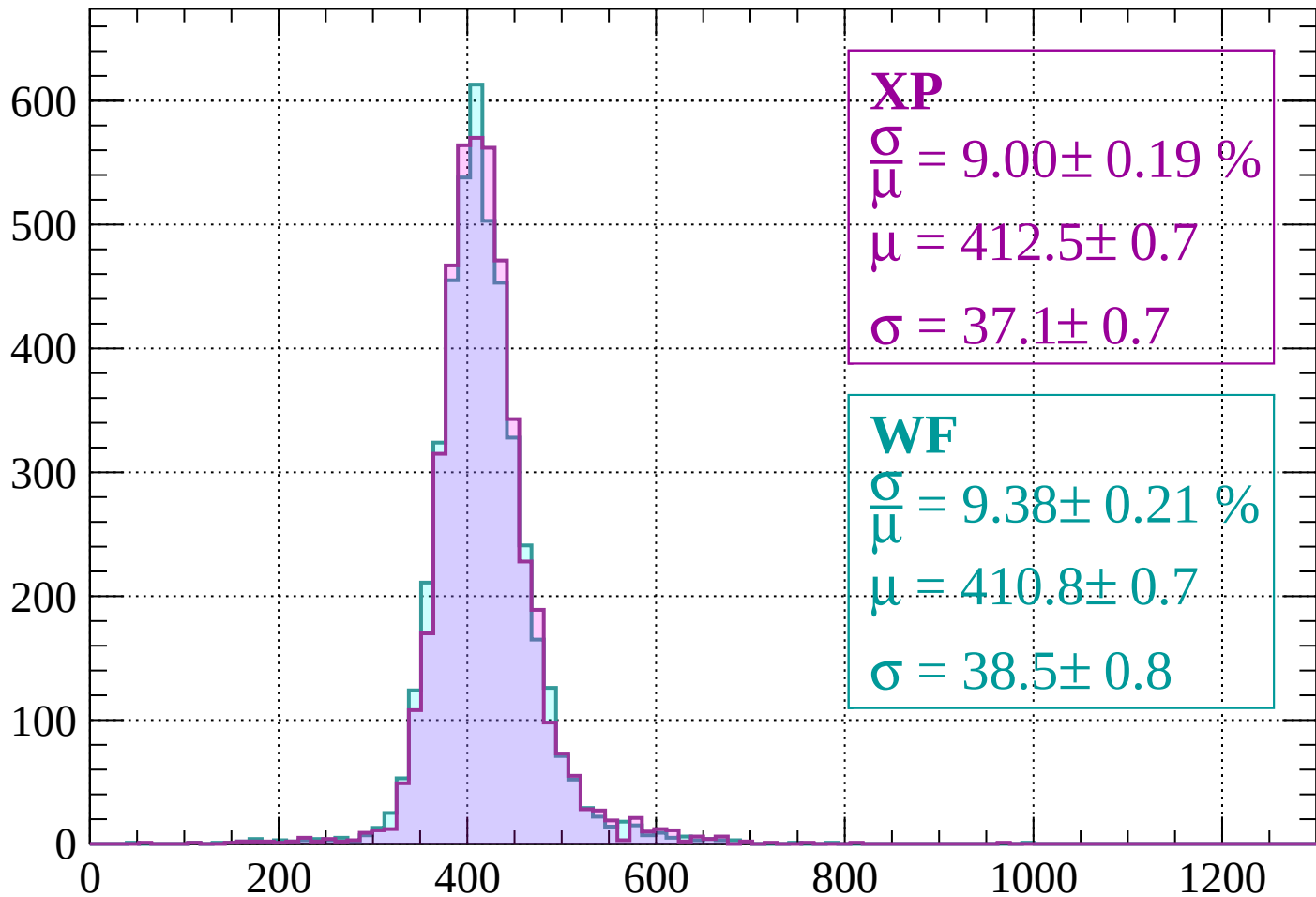
$$\mu = 412.6 \pm 0.7$$

$$\sigma = 37.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.00 \pm 0.19 \%$$

$$\mu = 412.5 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.38 \pm 0.21 \%$$

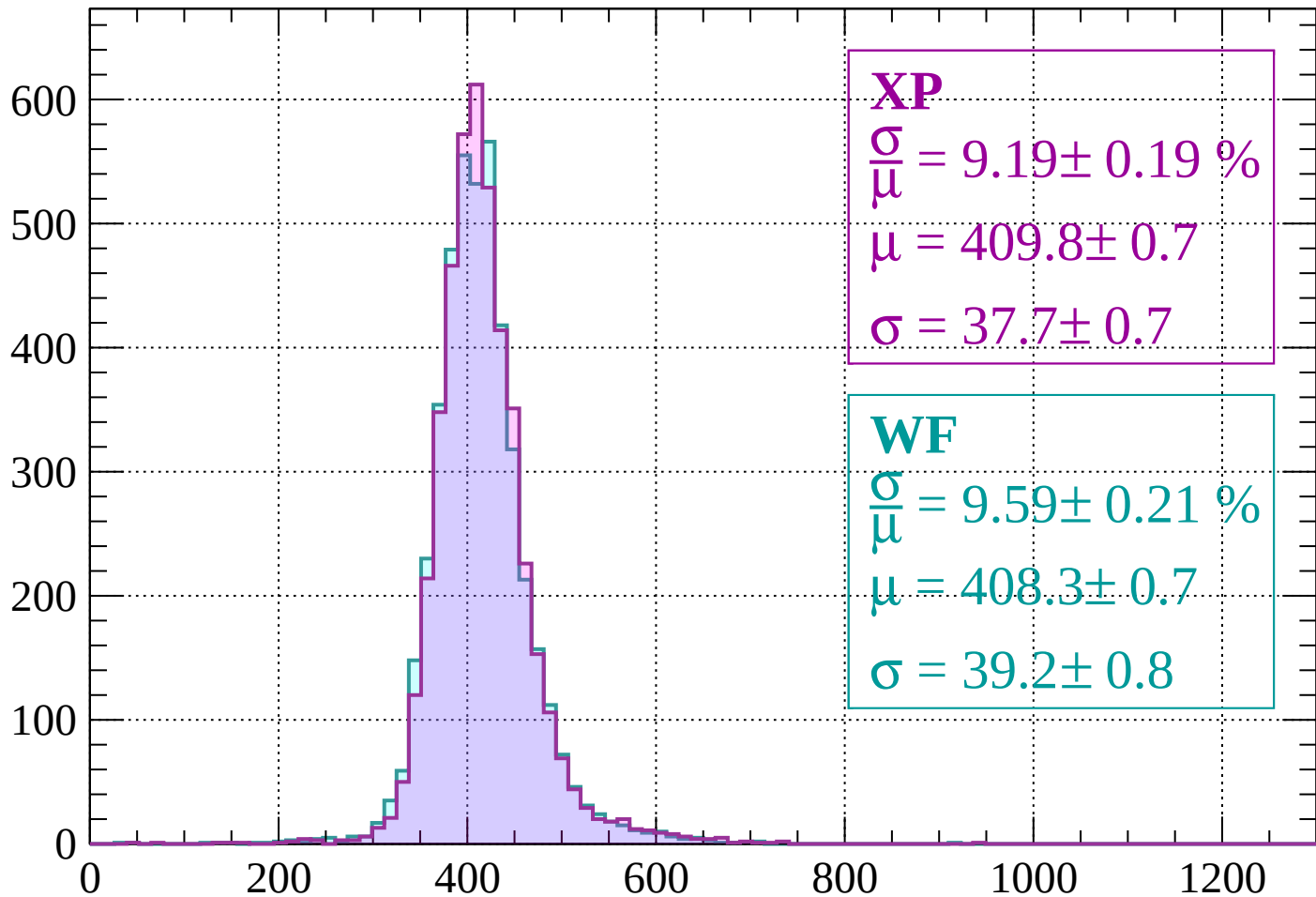
$$\mu = 410.8 \pm 0.7$$

$$\sigma = 38.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.19 \pm 0.19 \%$$

$$\mu = 409.8 \pm 0.7$$

$$\sigma = 37.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.21 \%$$

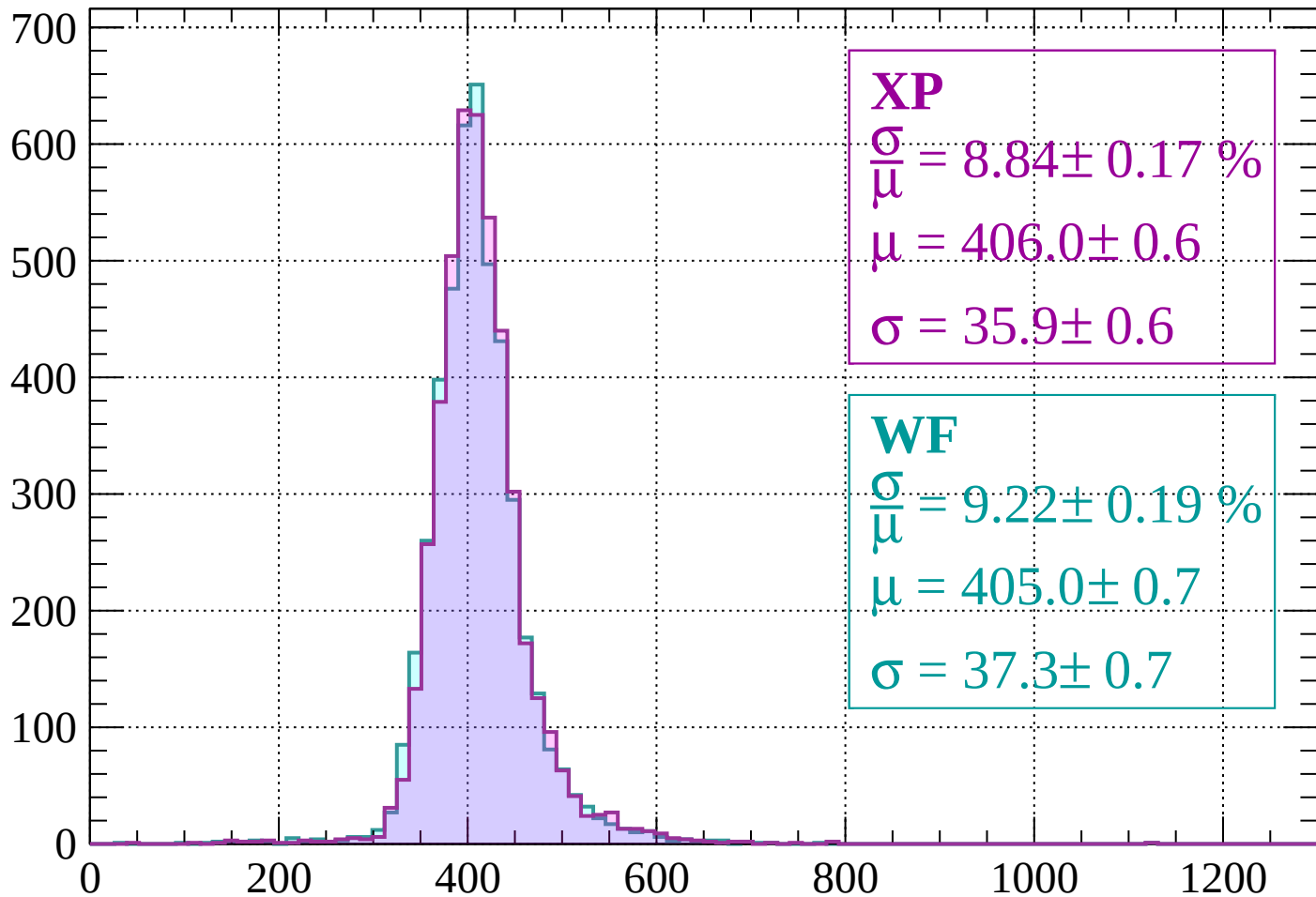
$$\mu = 408.3 \pm 0.7$$

$$\sigma = 39.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.17 \%$$

$$\mu = 406.0 \pm 0.6$$

$$\sigma = 35.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.19 \%$$

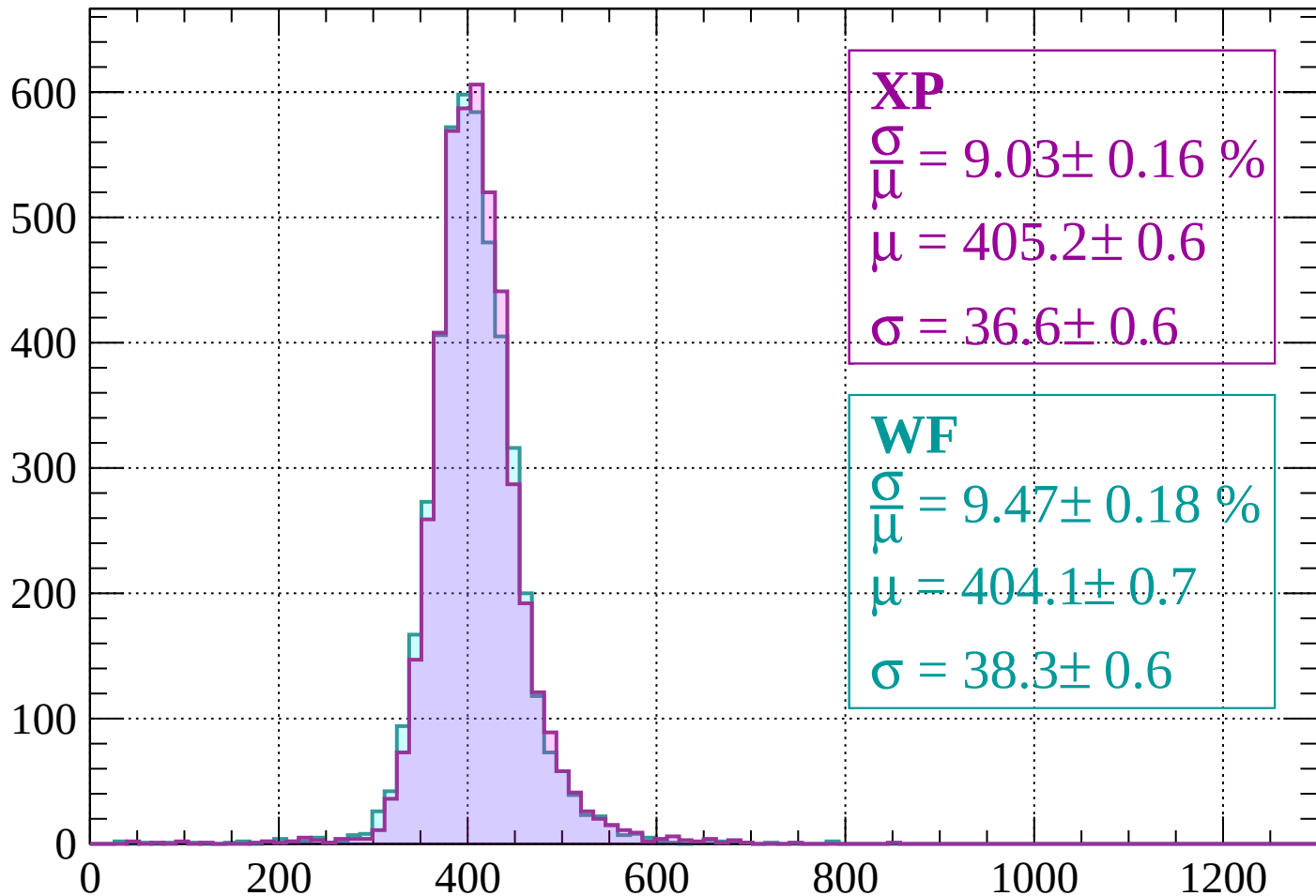
$$\mu = 405.0 \pm 0.7$$

$$\sigma = 37.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.16 \%$$

$$\mu = 405.2 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.47 \pm 0.18 \%$$

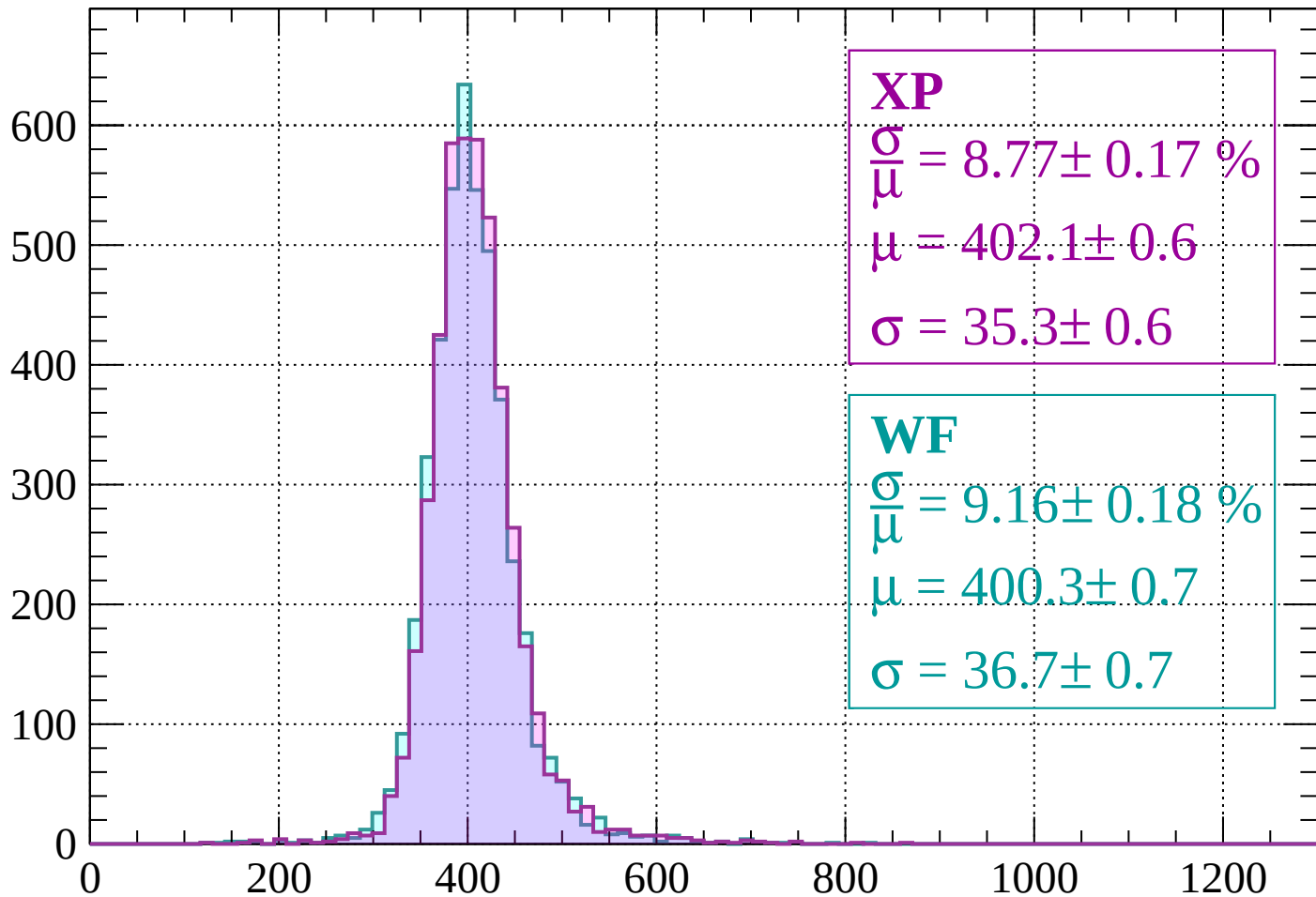
$$\mu = 404.1 \pm 0.7$$

$$\sigma = 38.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.77 \pm 0.17 \%$$

$$\mu = 402.1 \pm 0.6$$

$$\sigma = 35.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.16 \pm 0.18 \%$$

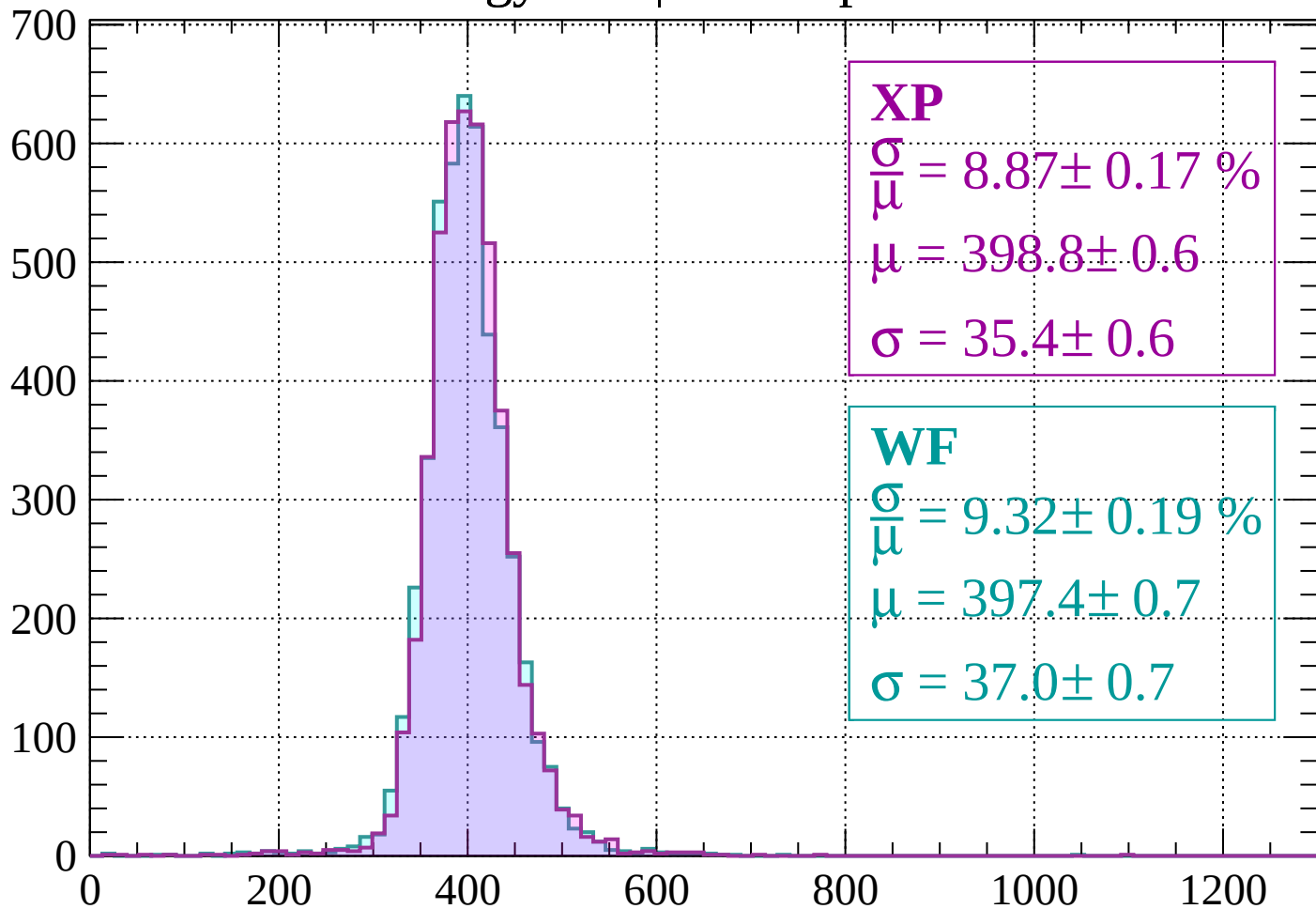
$$\mu = 400.3 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 8.87 \pm 0.17 \%$$

$$\mu = 398.8 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.19 \%$$

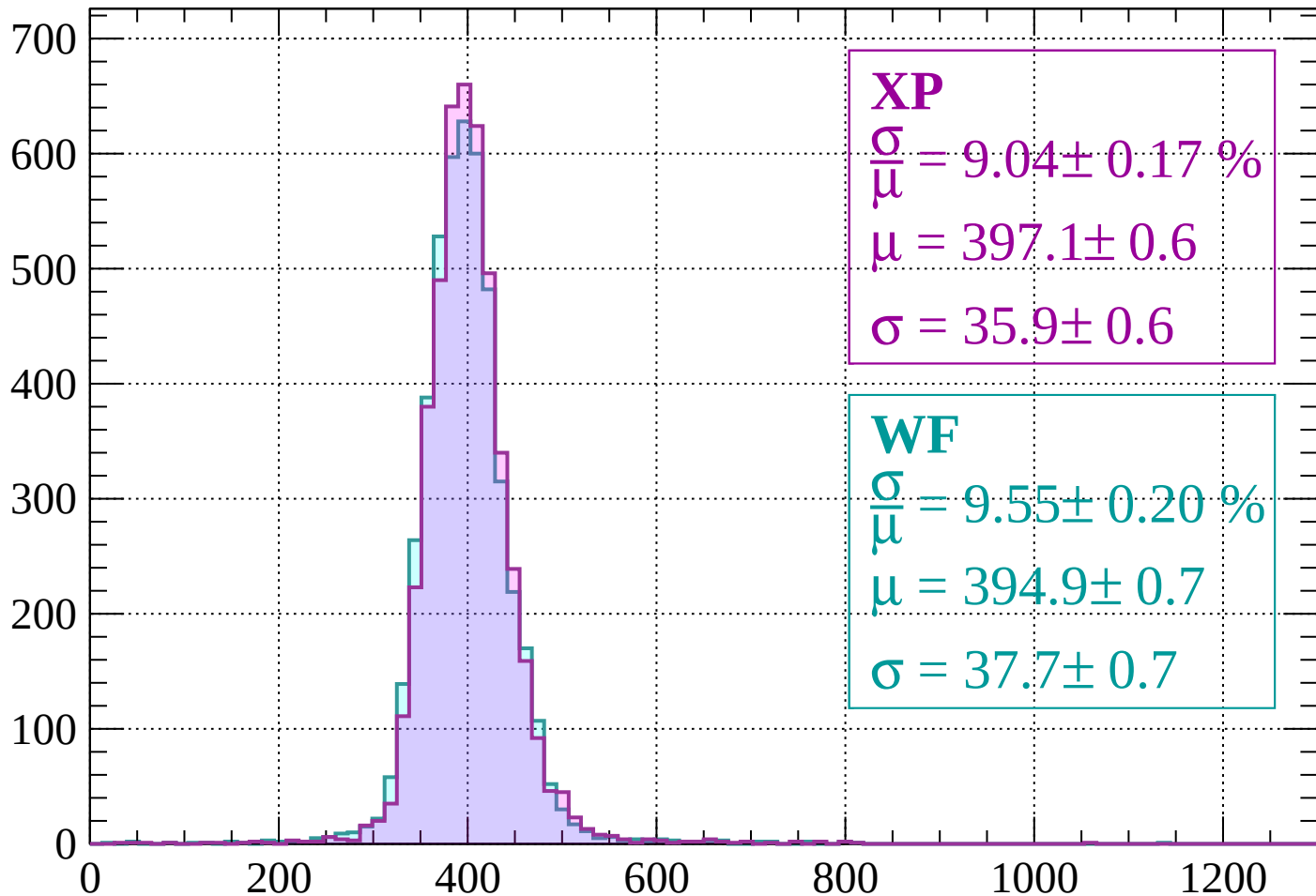
$$\mu = 397.4 \pm 0.7$$

$$\sigma = 37.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.04 \pm 0.17 \%$$

$$\mu = 397.1 \pm 0.6$$

$$\sigma = 35.9 \pm 0.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.55 \pm 0.20 \%$$

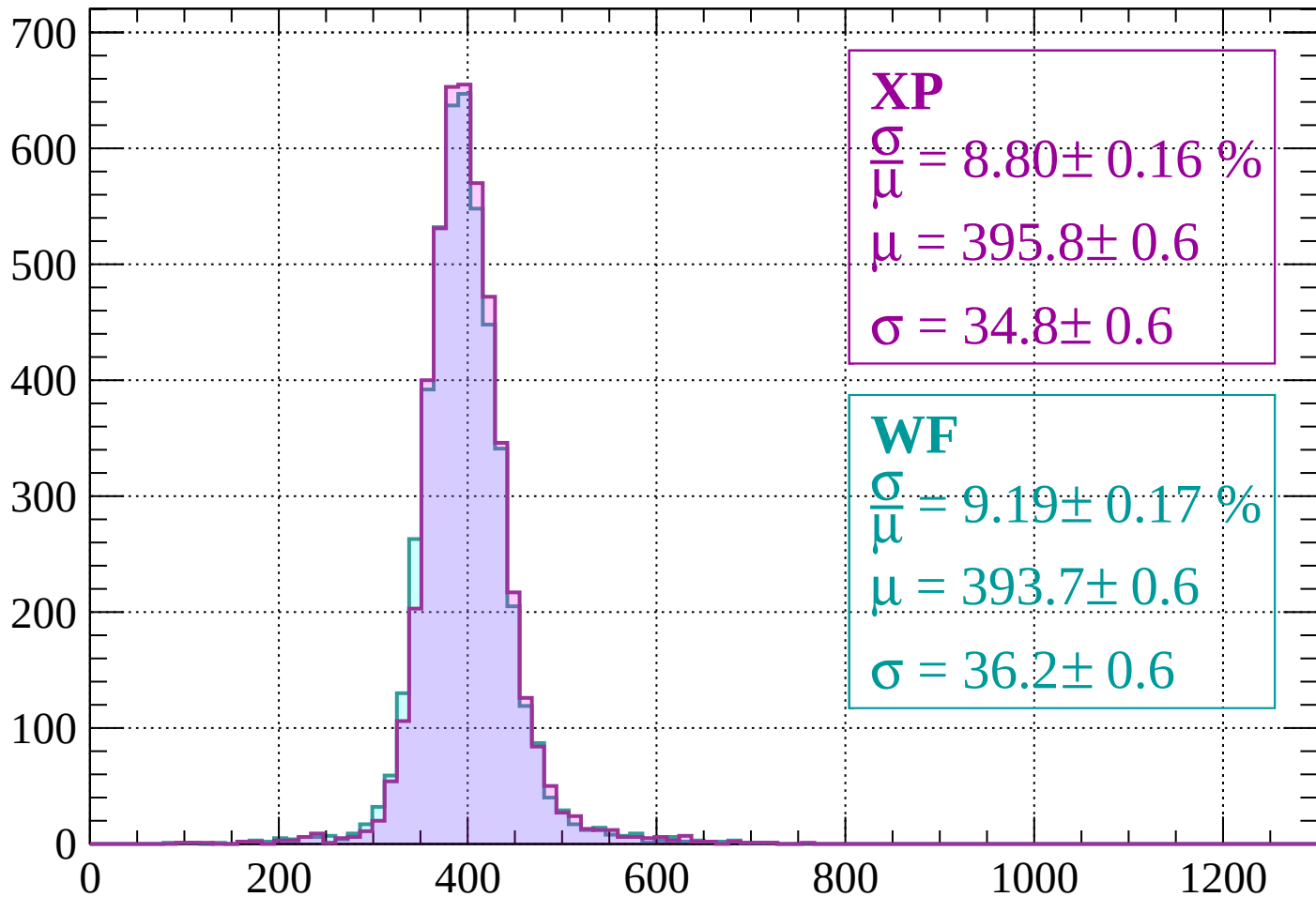
$$\mu = 394.9 \pm 0.7$$

$$\sigma = 37.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.80 \pm 0.16 \%$$

$$\mu = 395.8 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.17 \%$$

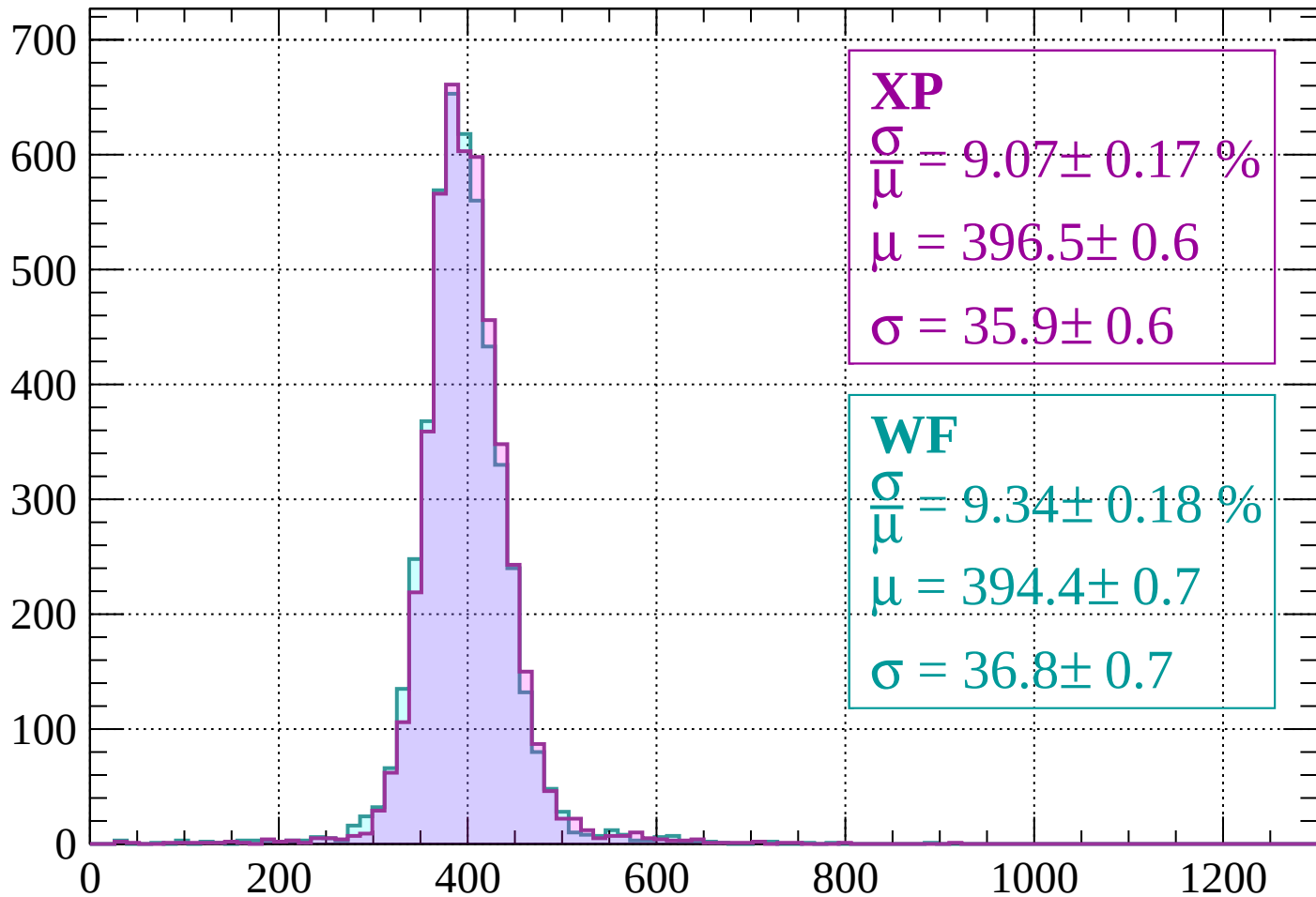
$$\mu = 393.7 \pm 0.6$$

$$\sigma = 36.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.07 \pm 0.17 \%$$

$$\mu = 396.5 \pm 0.6$$

$$\sigma = 35.9 \pm 0.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.34 \pm 0.18 \%$$

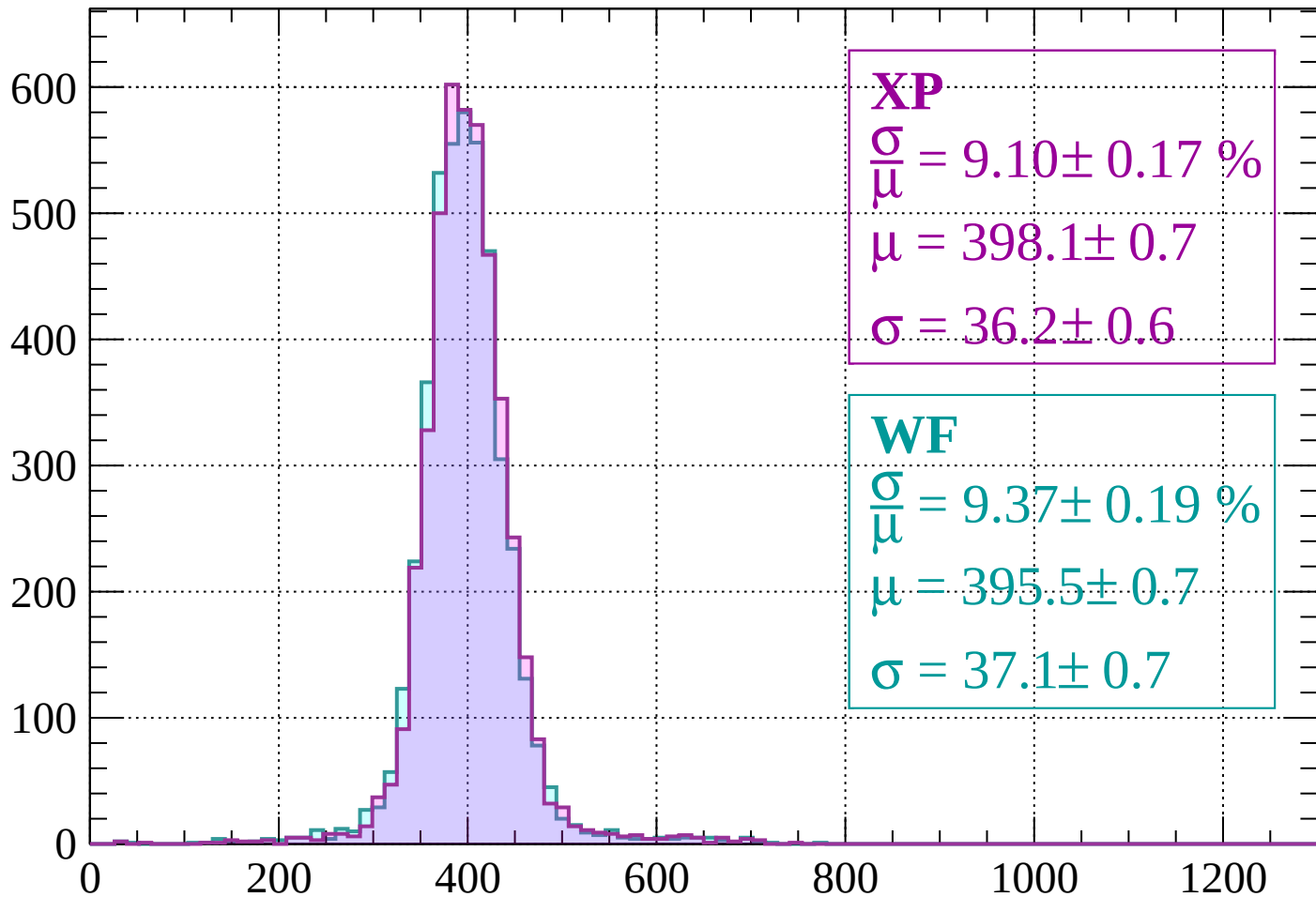
$$\mu = 394.4 \pm 0.7$$

$$\sigma = 36.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.17 \%$$

$$\mu = 398.1 \pm 0.7$$

$$\sigma = 36.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.19 \%$$

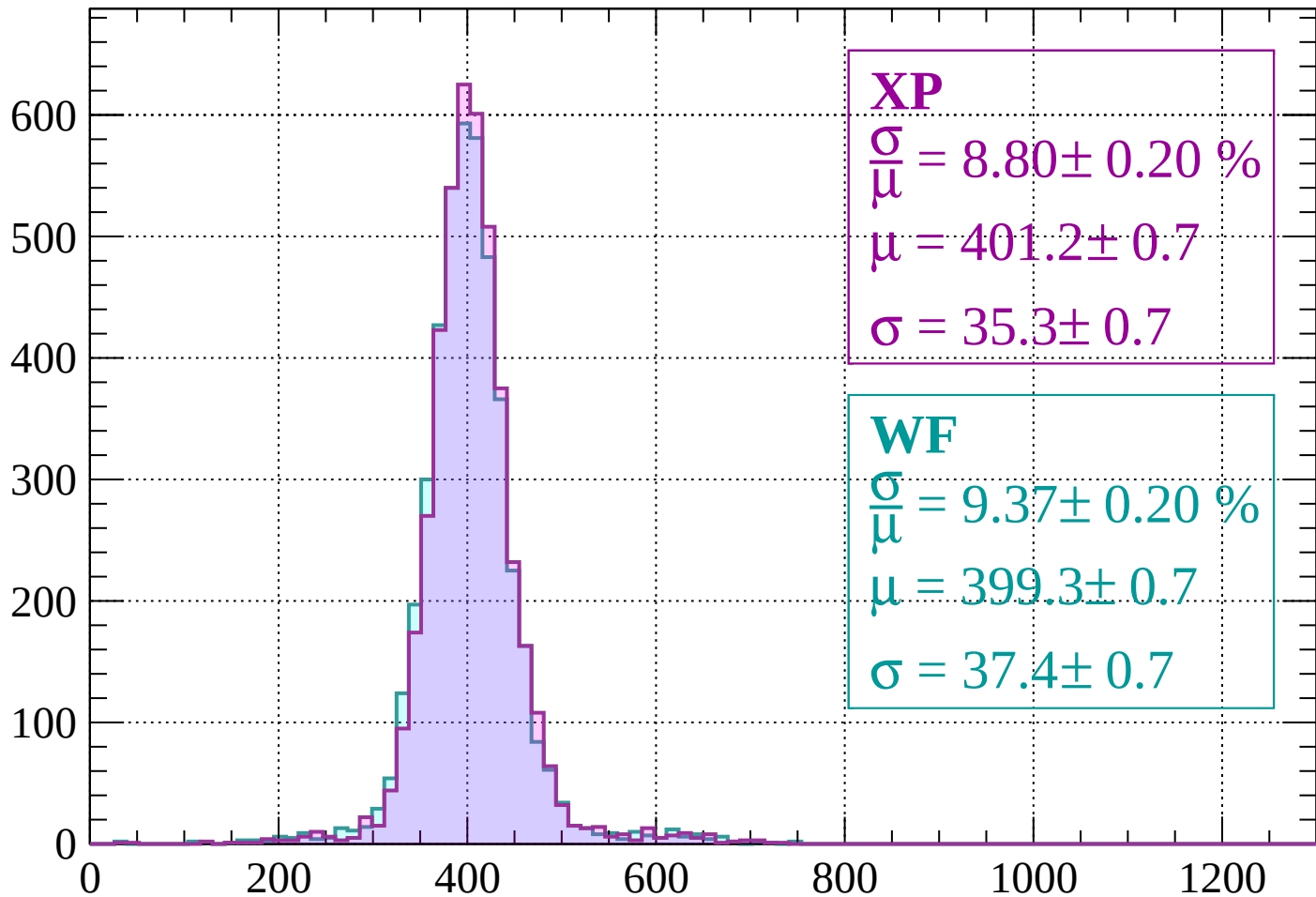
$$\mu = 395.5 \pm 0.7$$

$$\sigma = 37.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.80 \pm 0.20 \%$$

$$\mu = 401.2 \pm 0.7$$

$$\sigma = 35.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.20 \%$$

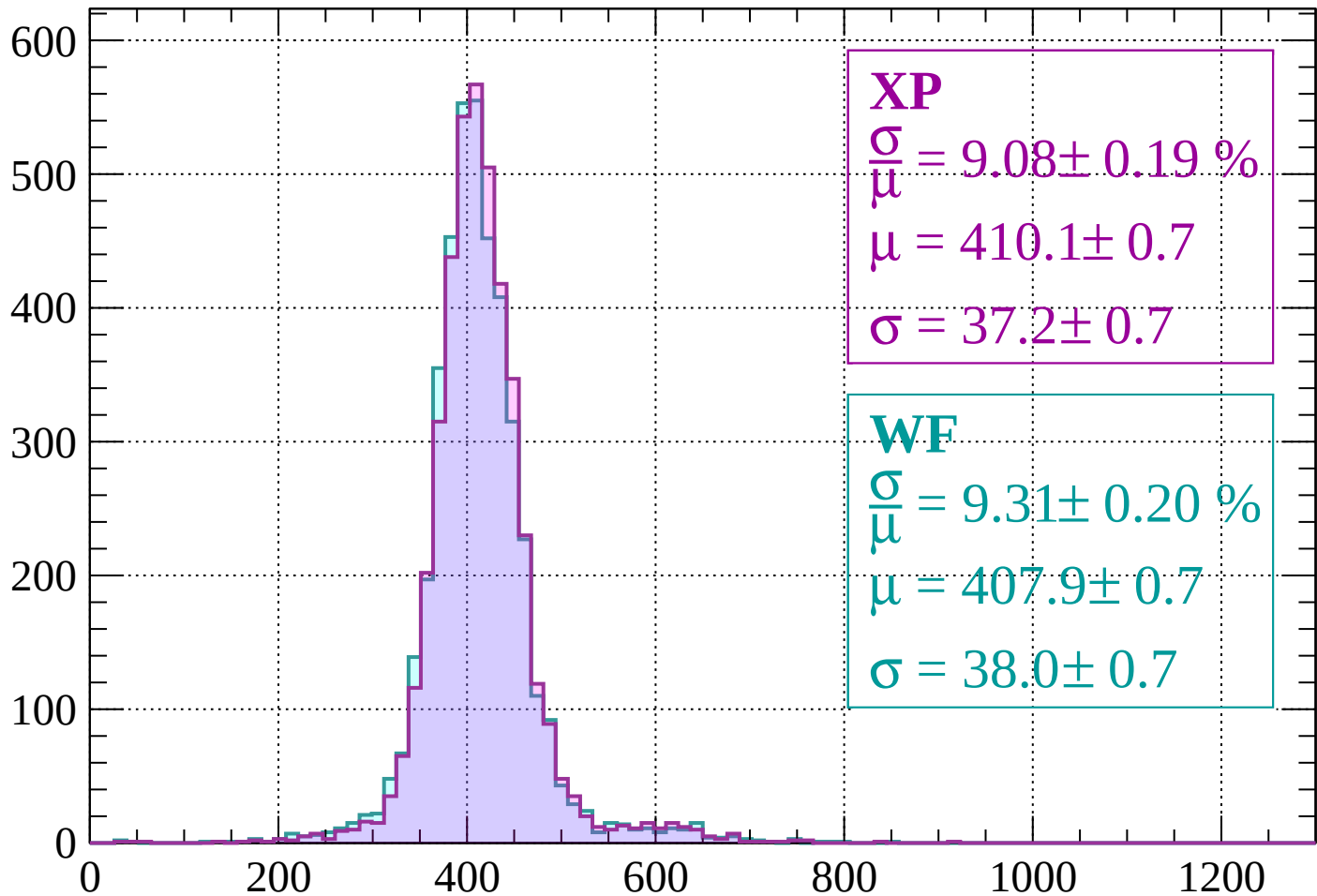
$$\mu = 399.3 \pm 0.7$$

$$\sigma = 37.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 9.08 \pm 0.19 \%$$

$$\mu = 410.1 \pm 0.7$$

$$\sigma = 37.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.31 \pm 0.20 \%$$

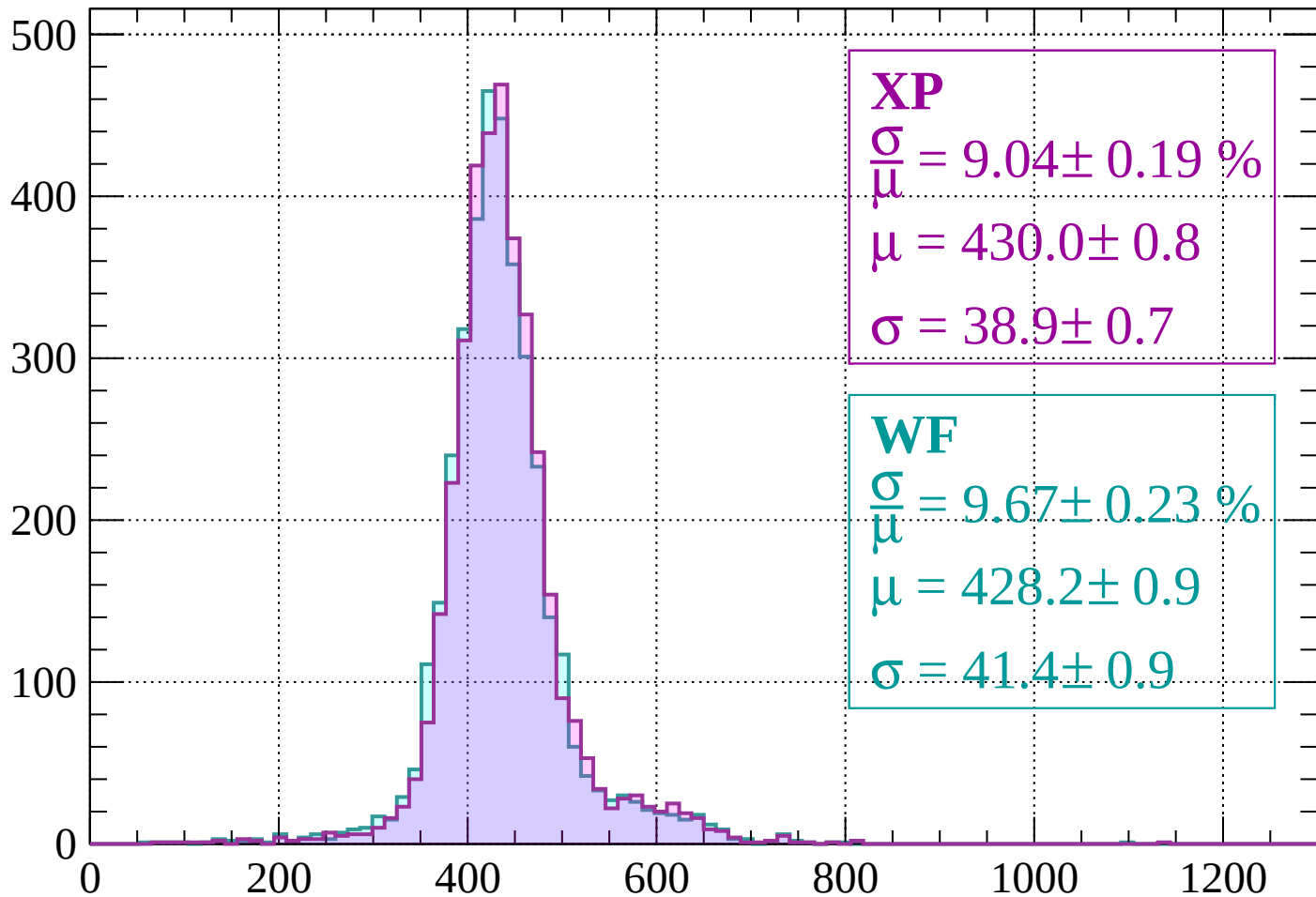
$$\mu = 407.9 \pm 0.7$$

$$\sigma = 38.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.19 \%$$

$$\mu = 430.0 \pm 0.8$$

$$\sigma = 38.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.23 \%$$

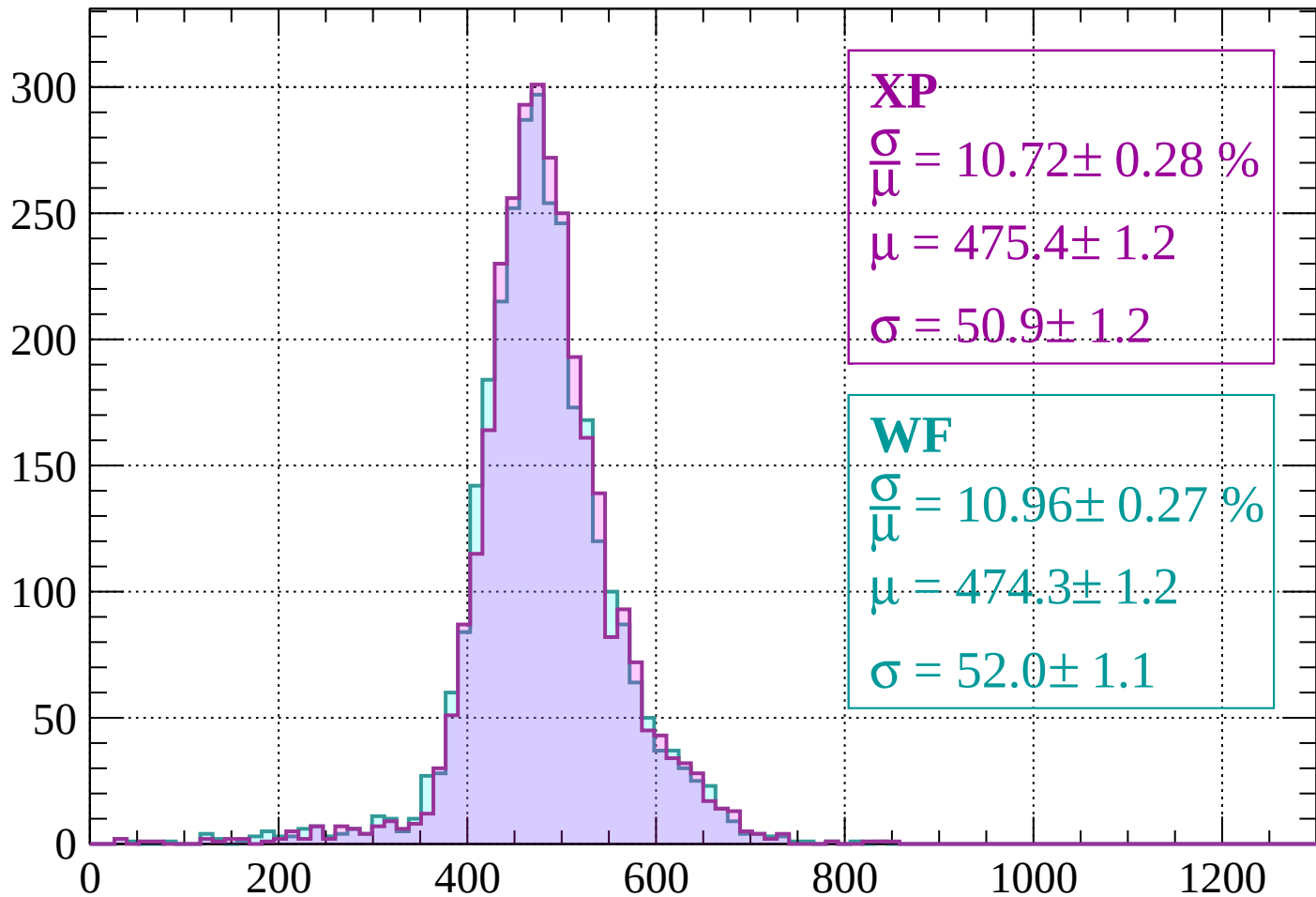
$$\mu = 428.2 \pm 0.9$$

$$\sigma = 41.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 10.72 \pm 0.28 \%$$

$$\mu = 475.4 \pm 1.2$$

$$\sigma = 50.9 \pm 1.2$$

WF

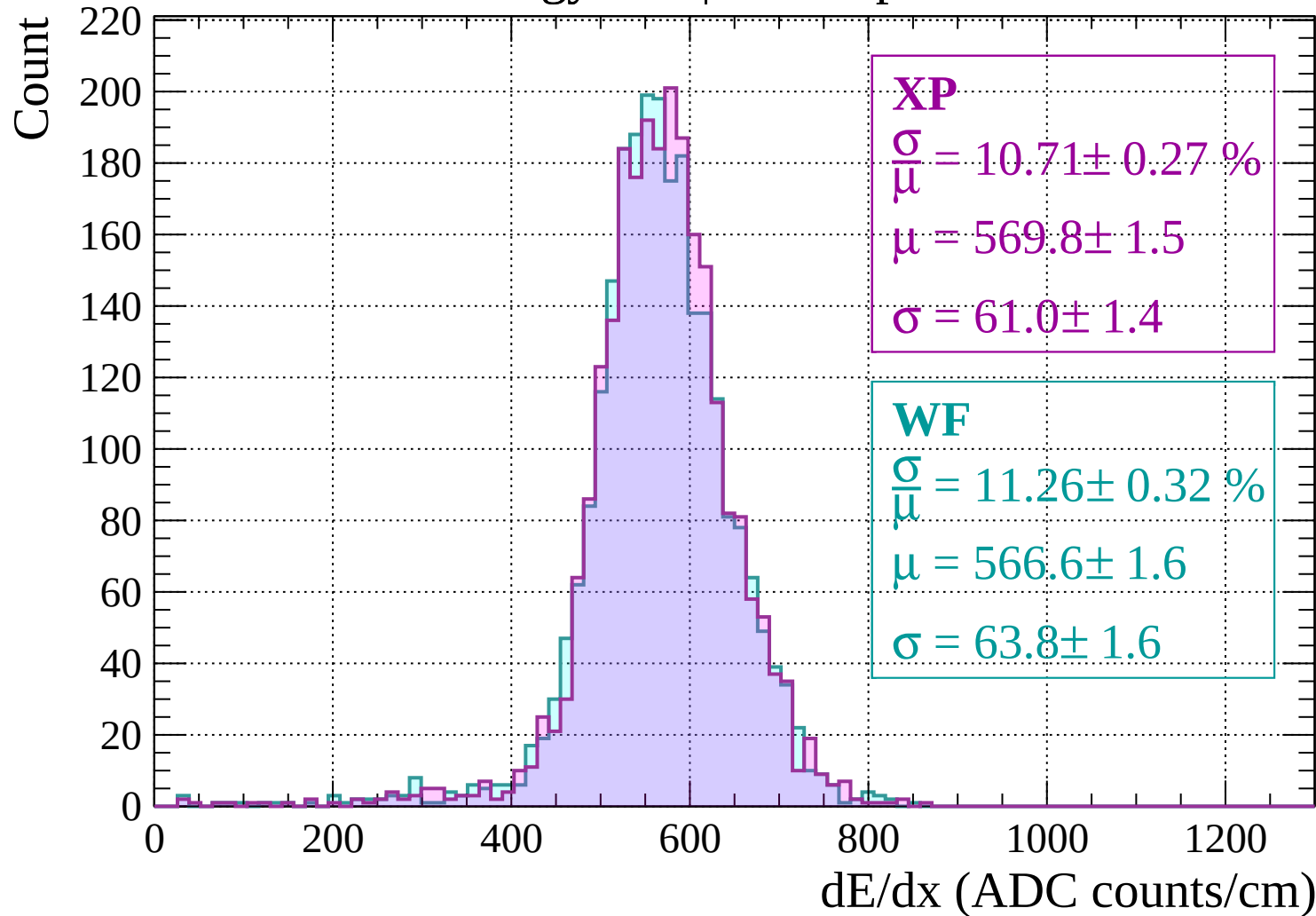
$$\frac{\sigma}{\mu} = 10.96 \pm 0.27 \%$$

$$\mu = 474.3 \pm 1.2$$

$$\sigma = 52.0 \pm 1.1$$

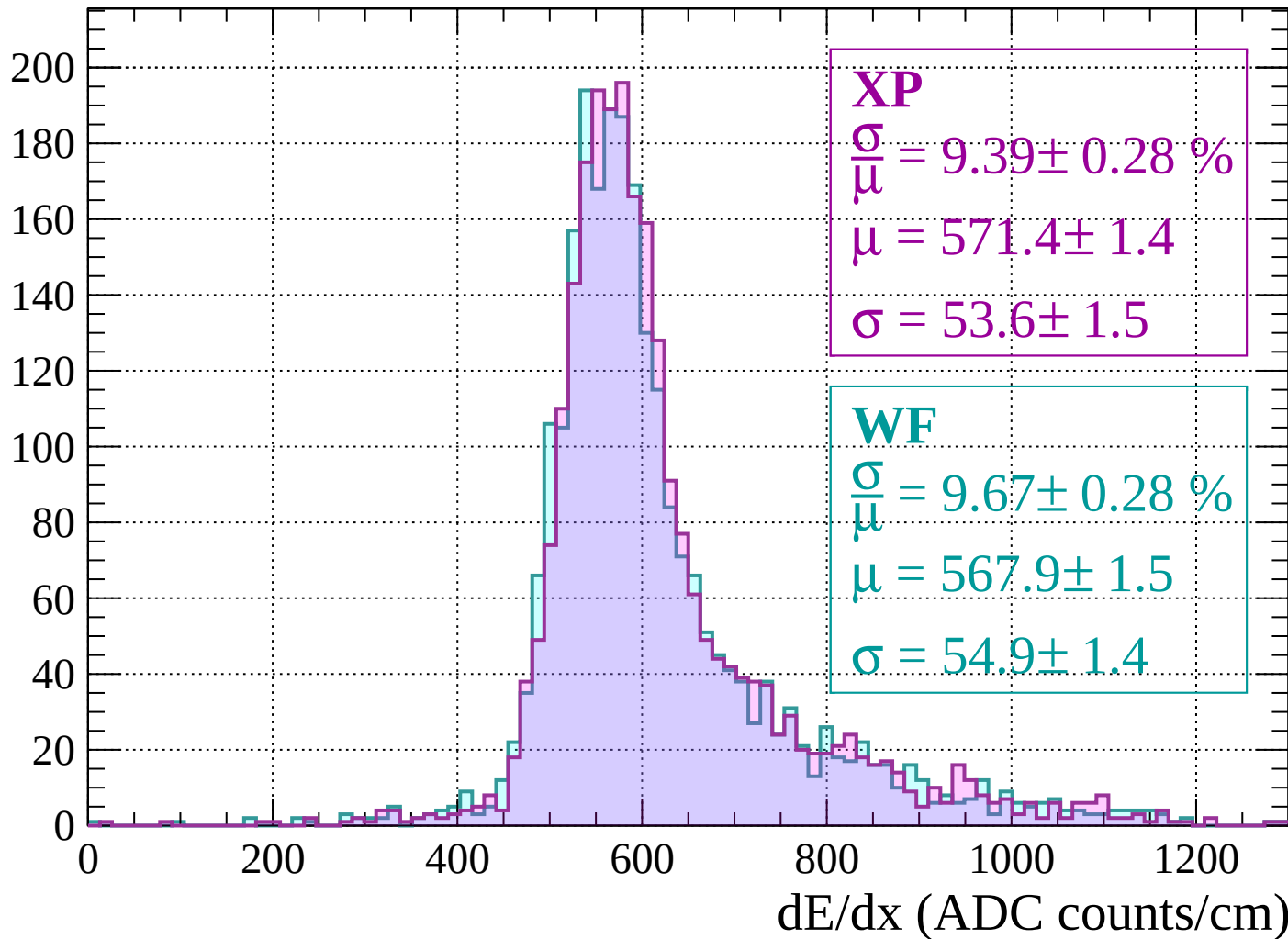
dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$



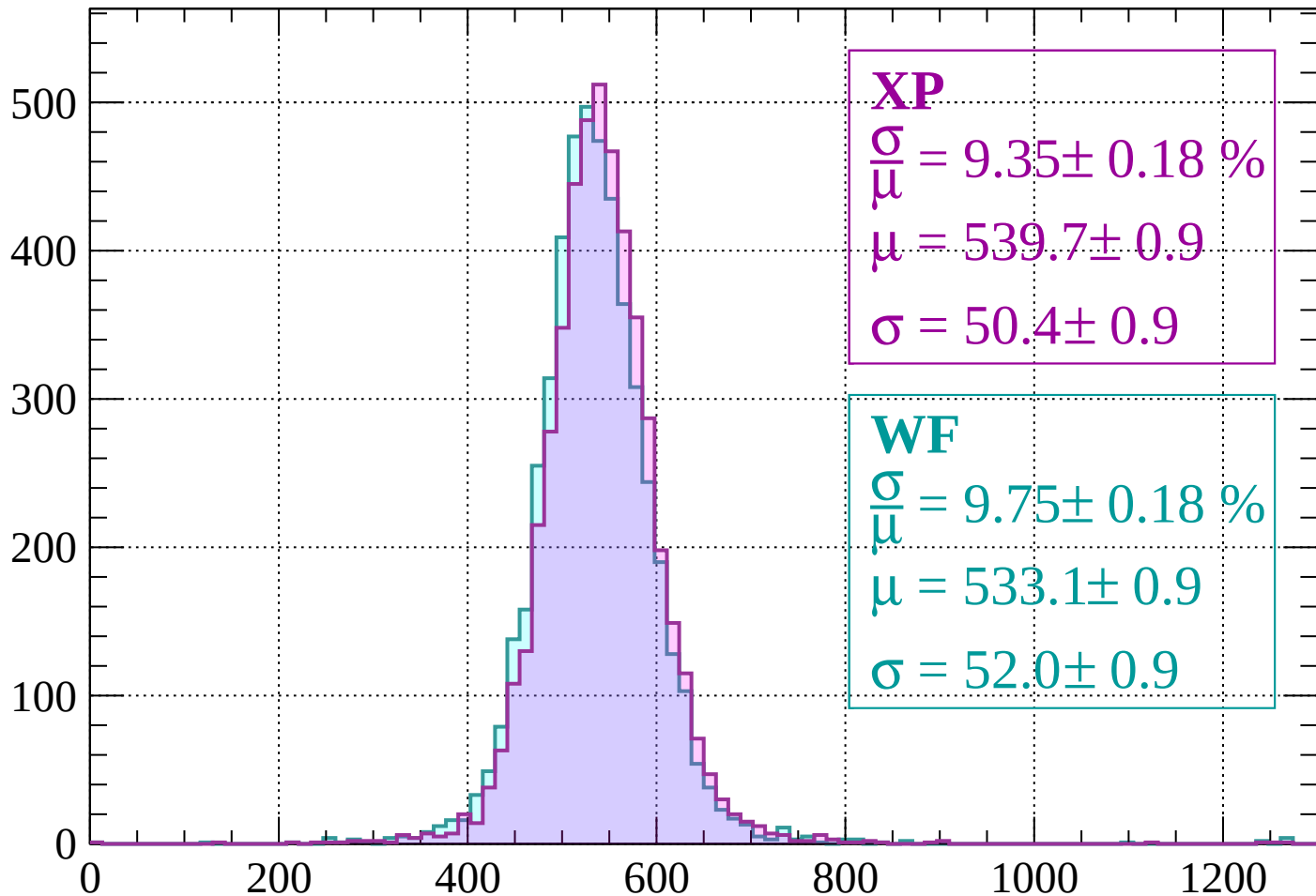
Energy loss | $-80 < p < -40$

Count



Energy loss | $-40 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 9.35 \pm 0.18 \%$$

$$\mu = 539.7 \pm 0.9$$

$$\sigma = 50.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.18 \%$$

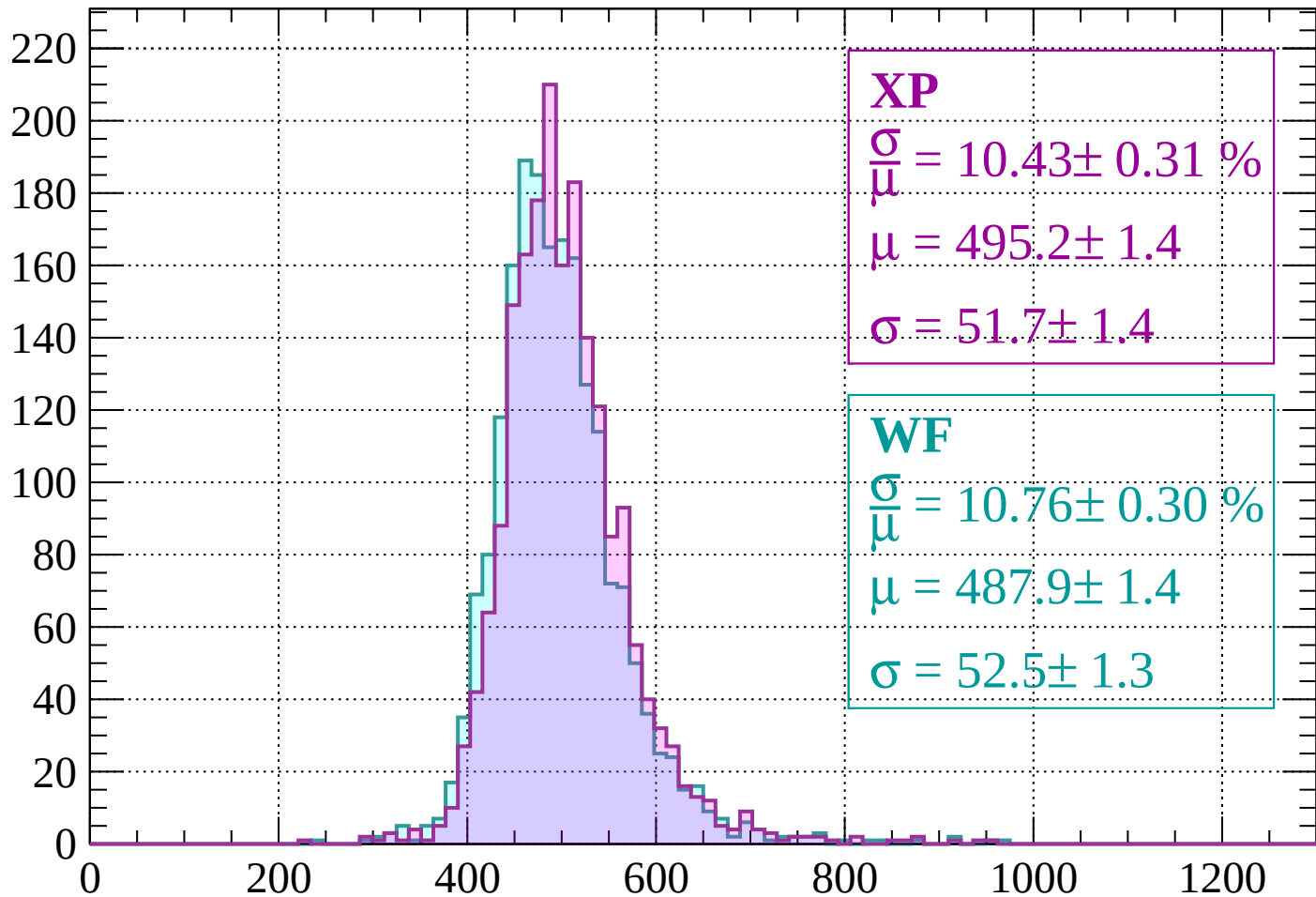
$$\mu = 533.1 \pm 0.9$$

$$\sigma = 52.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $0 < p < 40$

Count



XP

$$\frac{\sigma}{\mu} = 10.43 \pm 0.31 \%$$

$$\mu = 495.2 \pm 1.4$$

$$\sigma = 51.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.76 \pm 0.30 \%$$

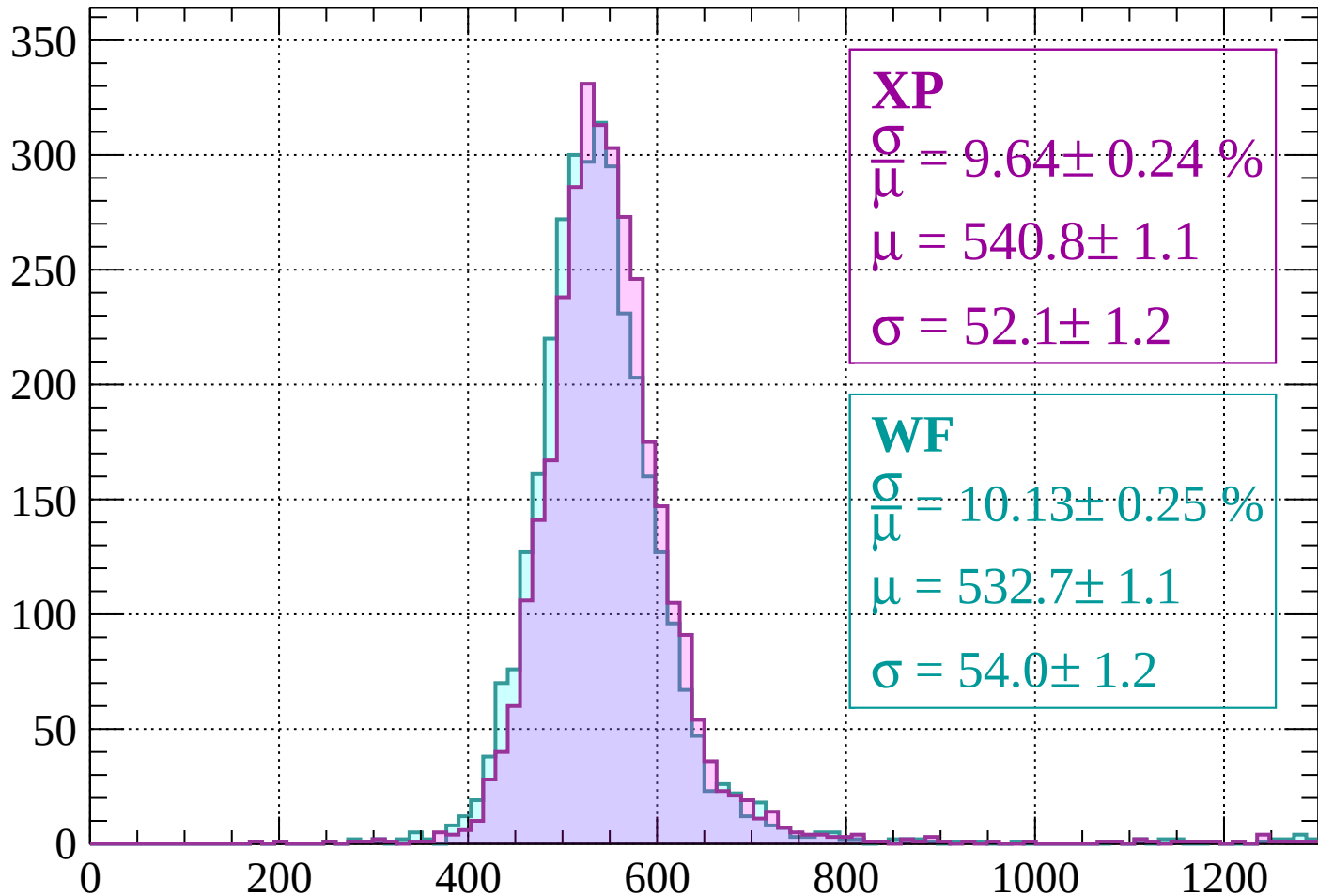
$$\mu = 487.9 \pm 1.4$$

$$\sigma = 52.5 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 9.64 \pm 0.24 \%$$

$$\mu = 540.8 \pm 1.1$$

$$\sigma = 52.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.25 \%$$

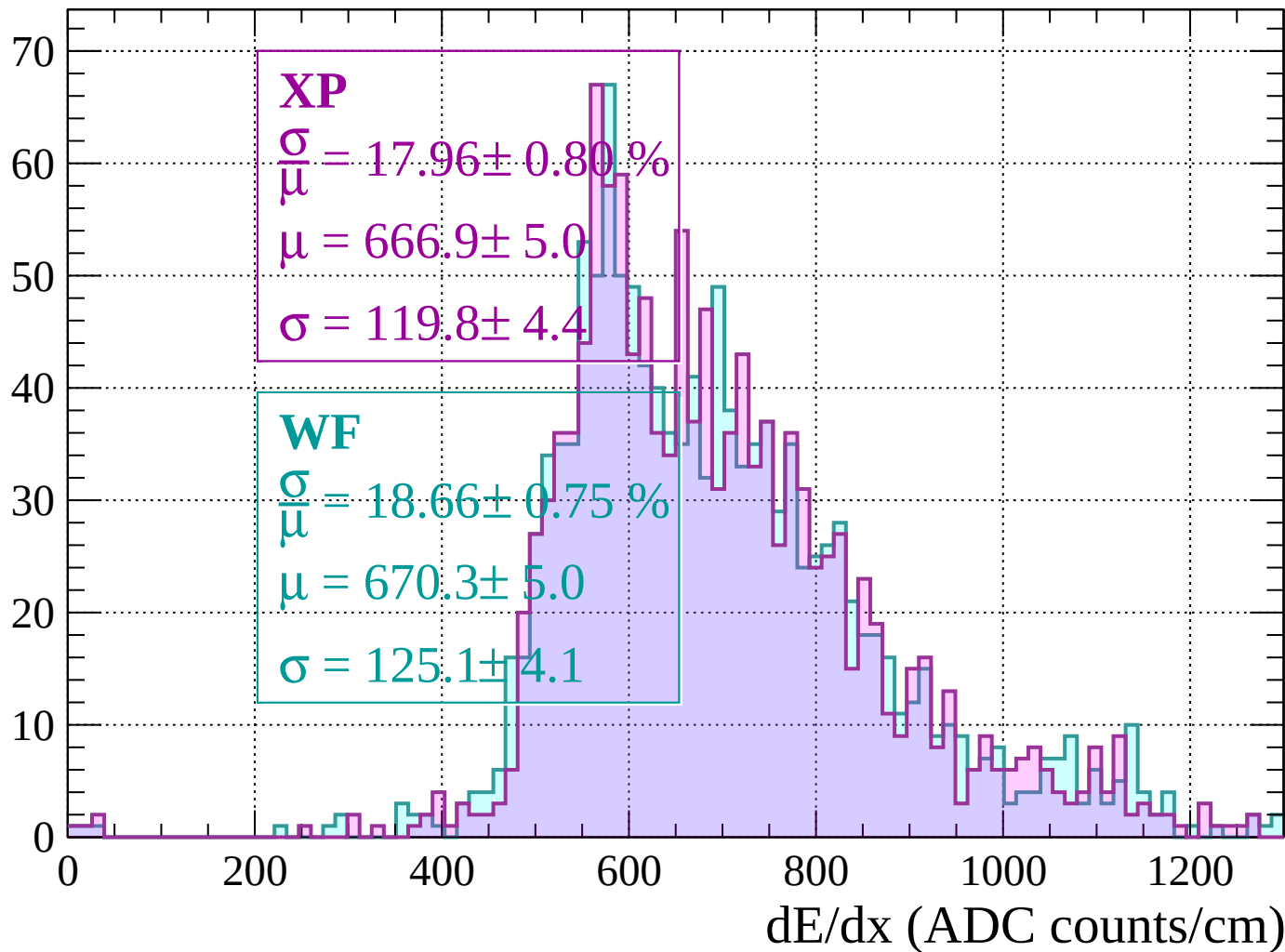
$$\mu = 532.7 \pm 1.1$$

$$\sigma = 54.0 \pm 1.2$$

dE/dx (ADC counts/cm)

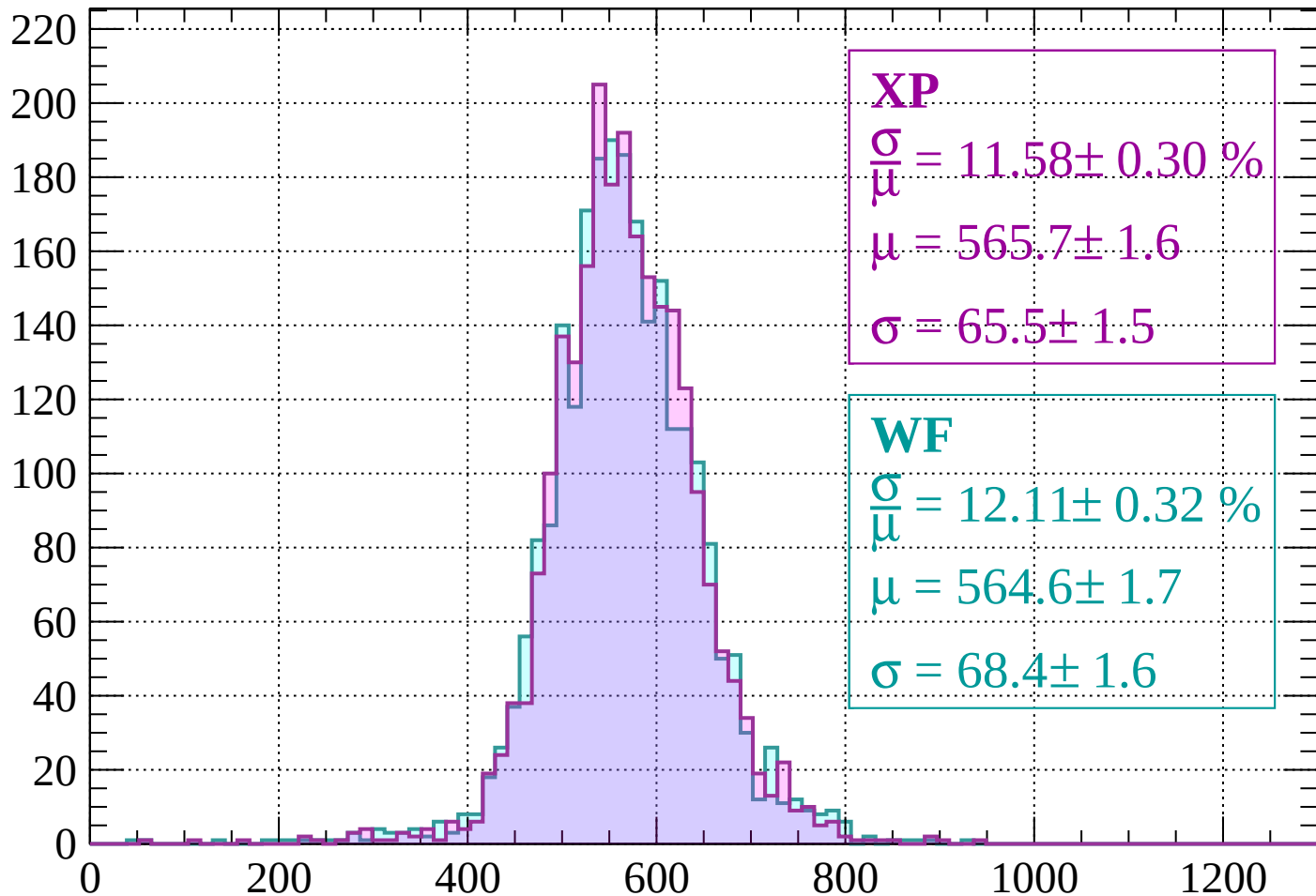
Energy loss | $80 < p < 120$

Count



Energy loss | $120 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 11.58 \pm 0.30 \%$$

$$\mu = 565.7 \pm 1.6$$

$$\sigma = 65.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 12.11 \pm 0.32 \%$$

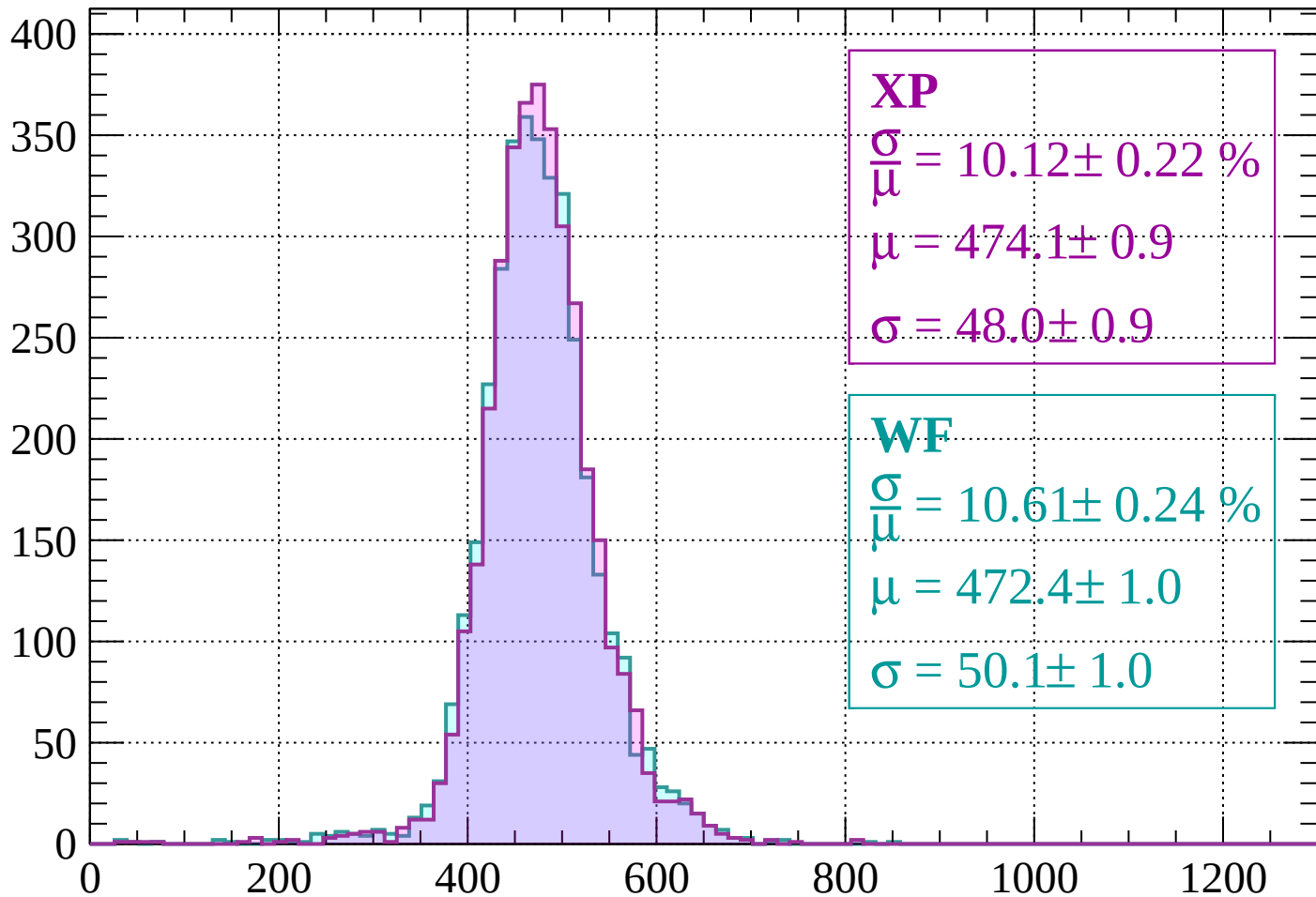
$$\mu = 564.6 \pm 1.7$$

$$\sigma = 68.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $160 < p < 200$

Count



XP

$$\frac{\sigma}{\mu} = 10.12 \pm 0.22 \%$$

$$\mu = 474.1 \pm 0.9$$

$$\sigma = 48.0 \pm 0.9$$

WF

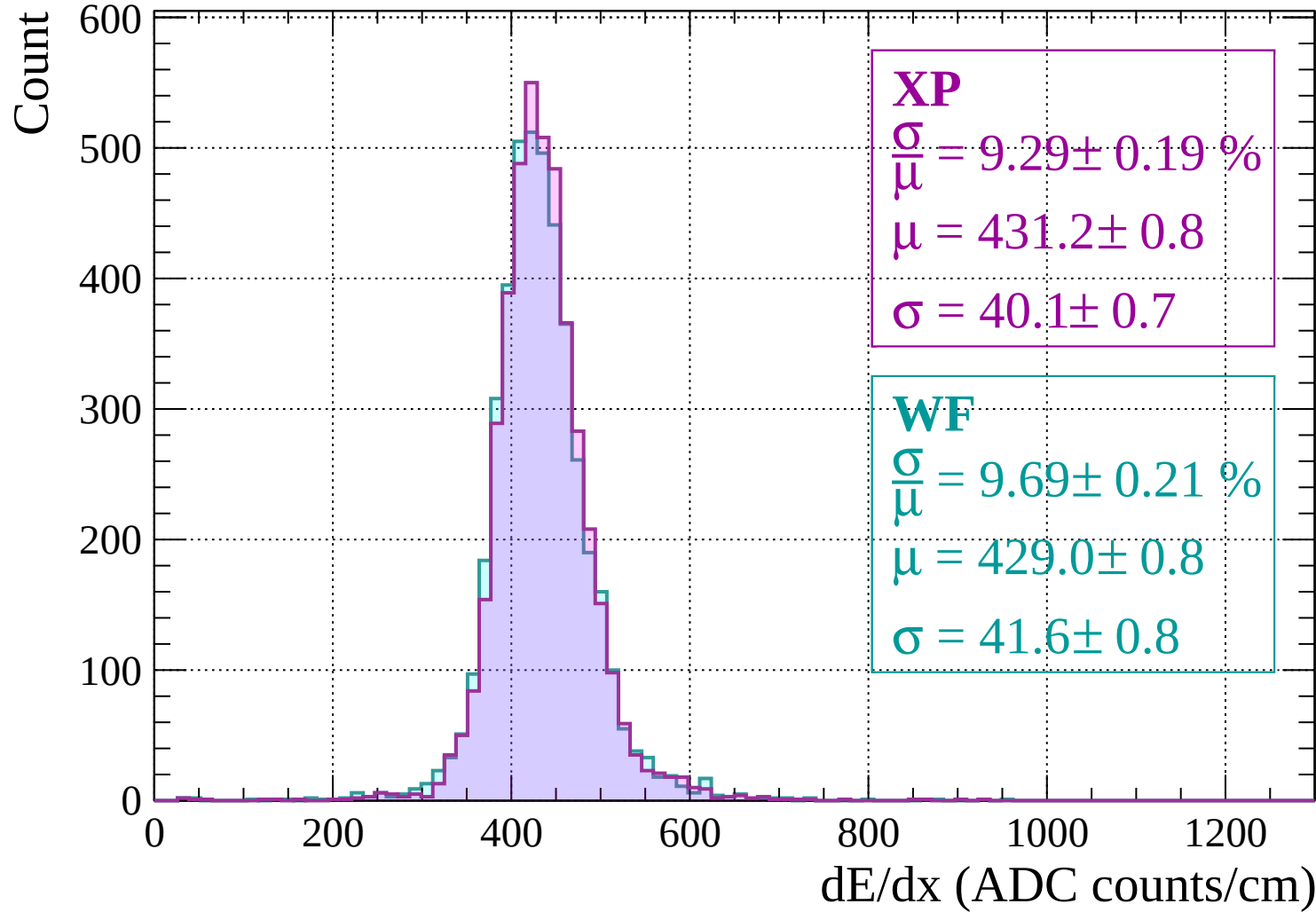
$$\frac{\sigma}{\mu} = 10.61 \pm 0.24 \%$$

$$\mu = 472.4 \pm 1.0$$

$$\sigma = 50.1 \pm 1.0$$

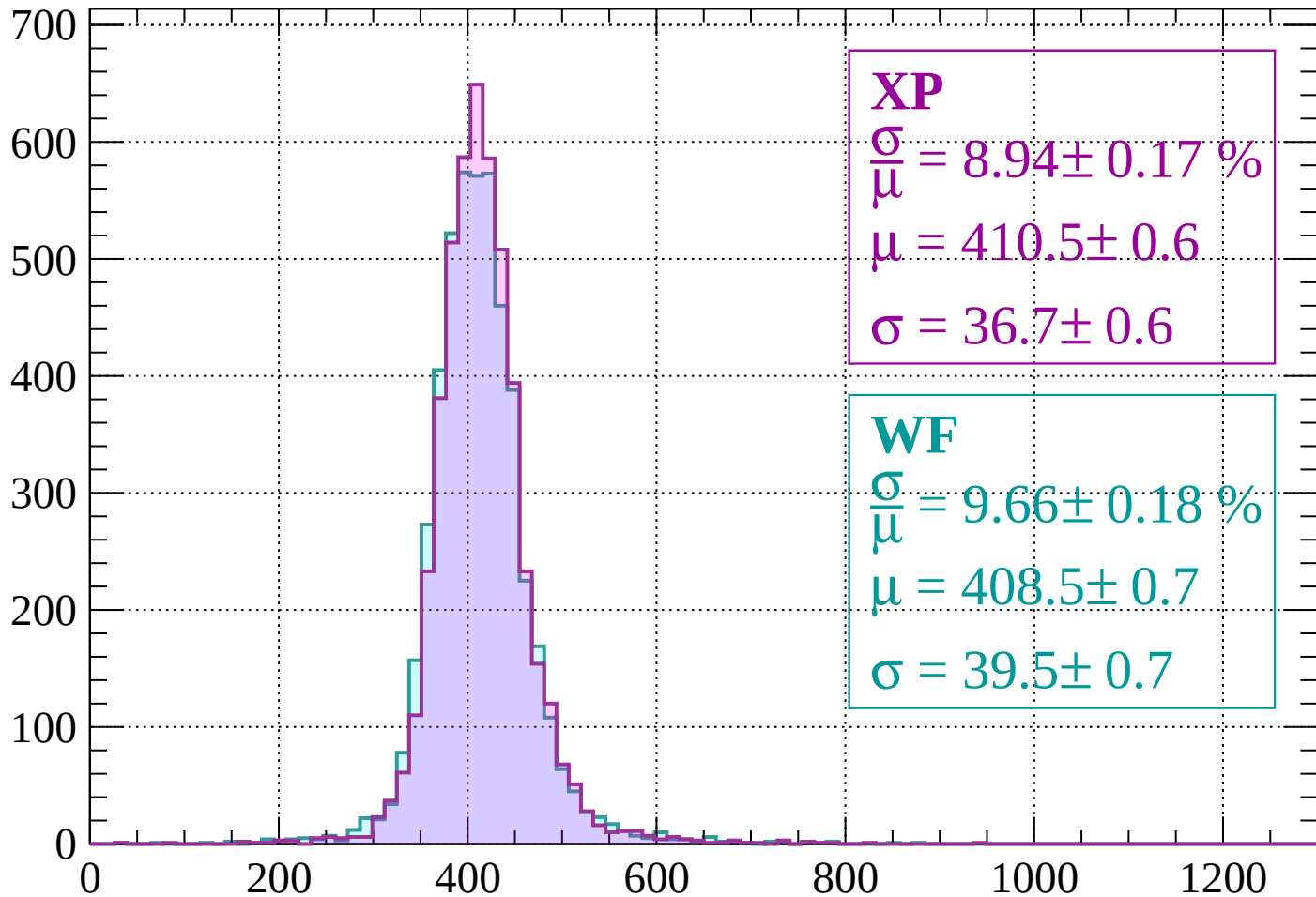
dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$



Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 8.94 \pm 0.17 \%$$

$$\mu = 410.5 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.18 \%$$

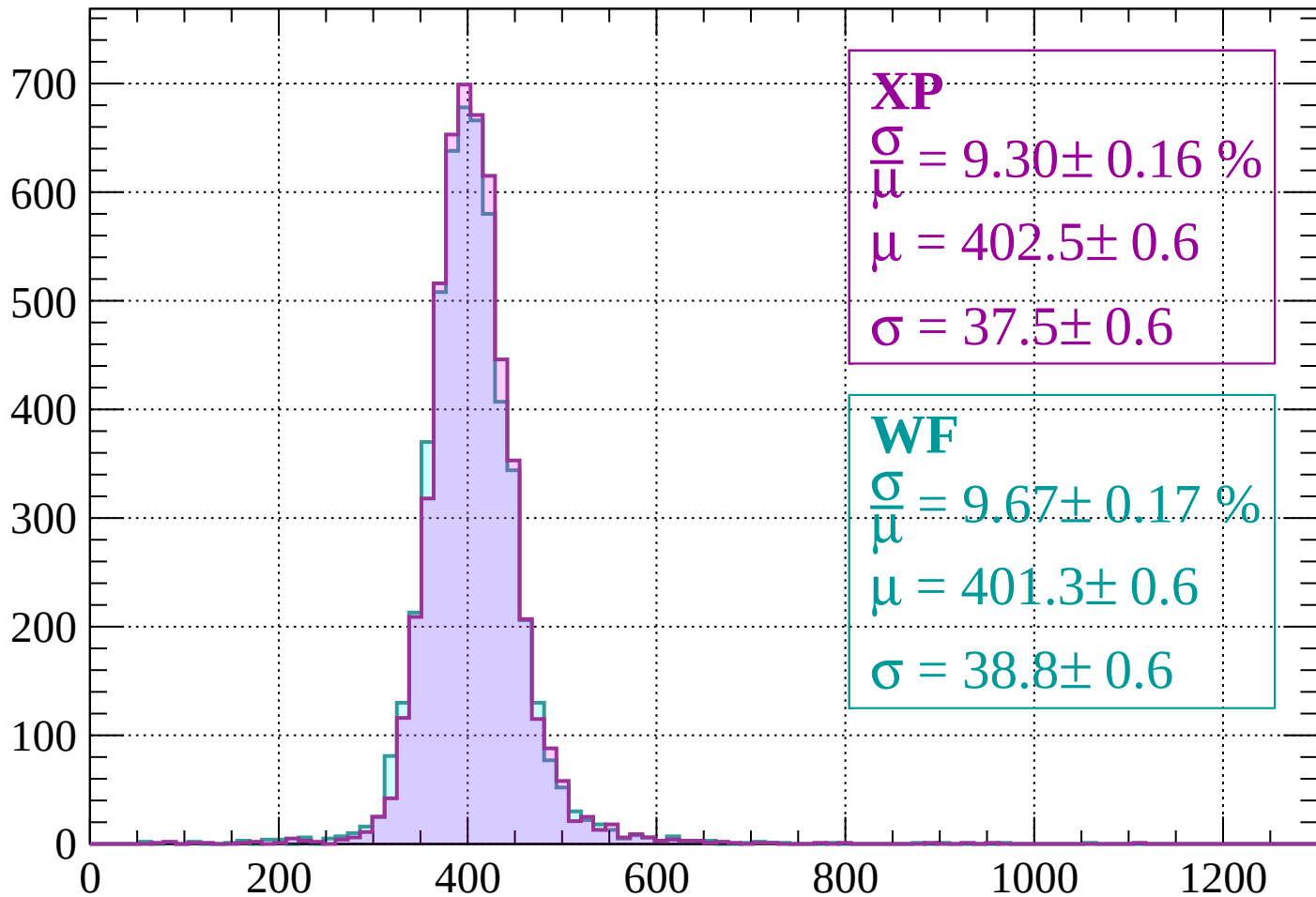
$$\mu = 408.5 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.16 \%$$

$$\mu = 402.5 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.17 \%$$

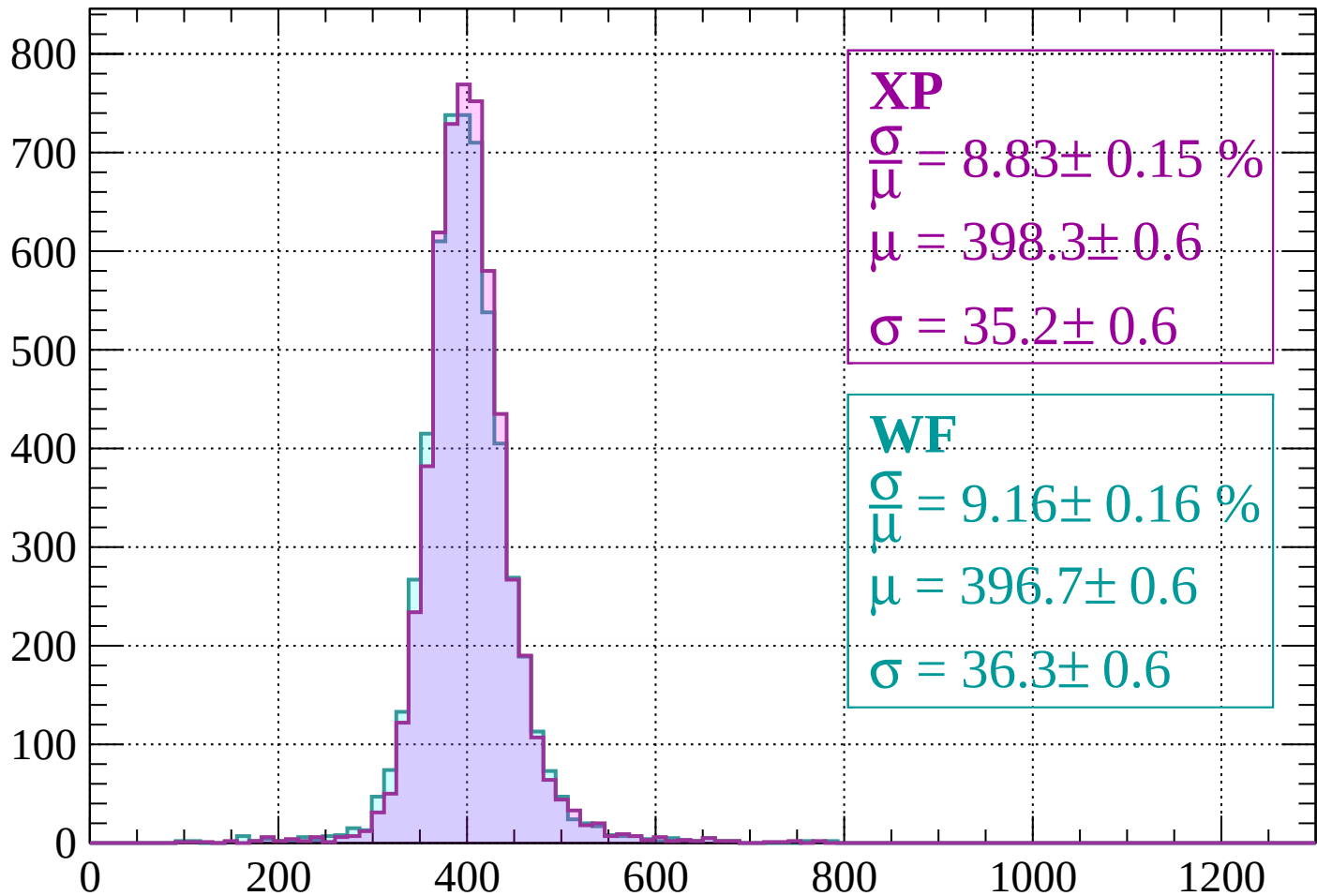
$$\mu = 401.3 \pm 0.6$$

$$\sigma = 38.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.15 \%$$

$$\mu = 398.3 \pm 0.6$$

$$\sigma = 35.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.16 \pm 0.16 \%$$

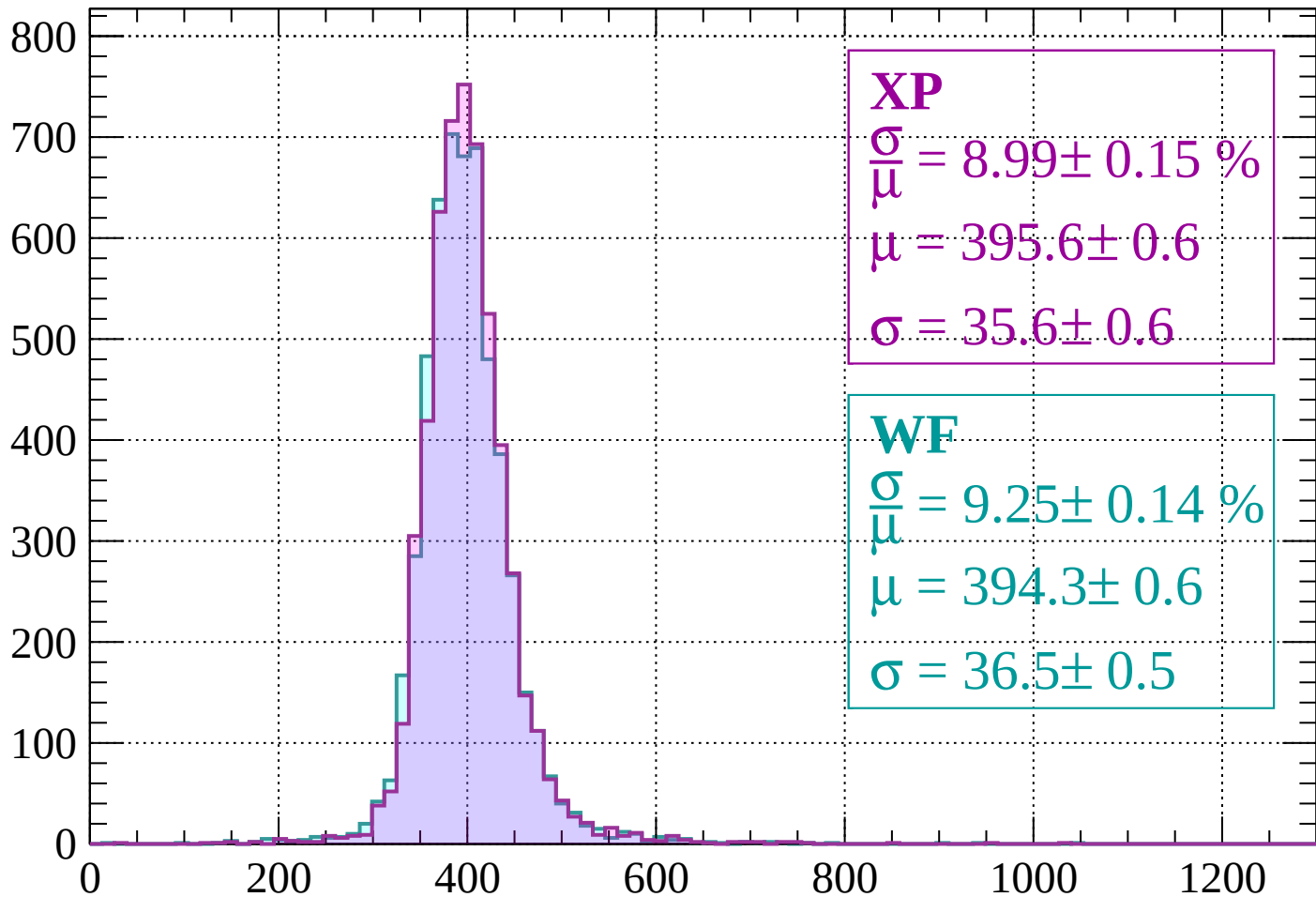
$$\mu = 396.7 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.15 \%$$

$$\mu = 395.6 \pm 0.6$$

$$\sigma = 35.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.25 \pm 0.14 \%$$

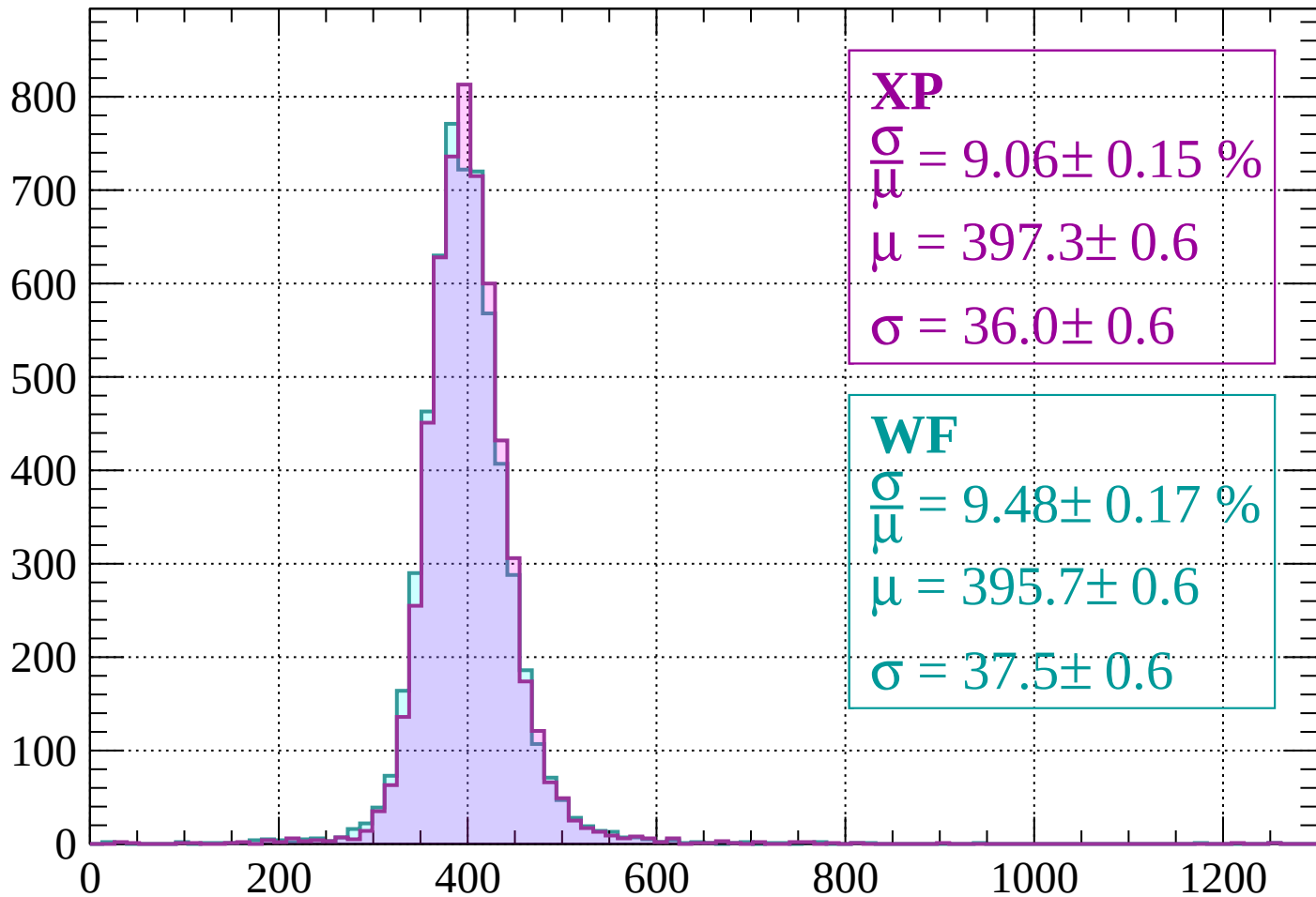
$$\mu = 394.3 \pm 0.6$$

$$\sigma = 36.5 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.15 \%$$

$$\mu = 397.3 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.17 \%$$

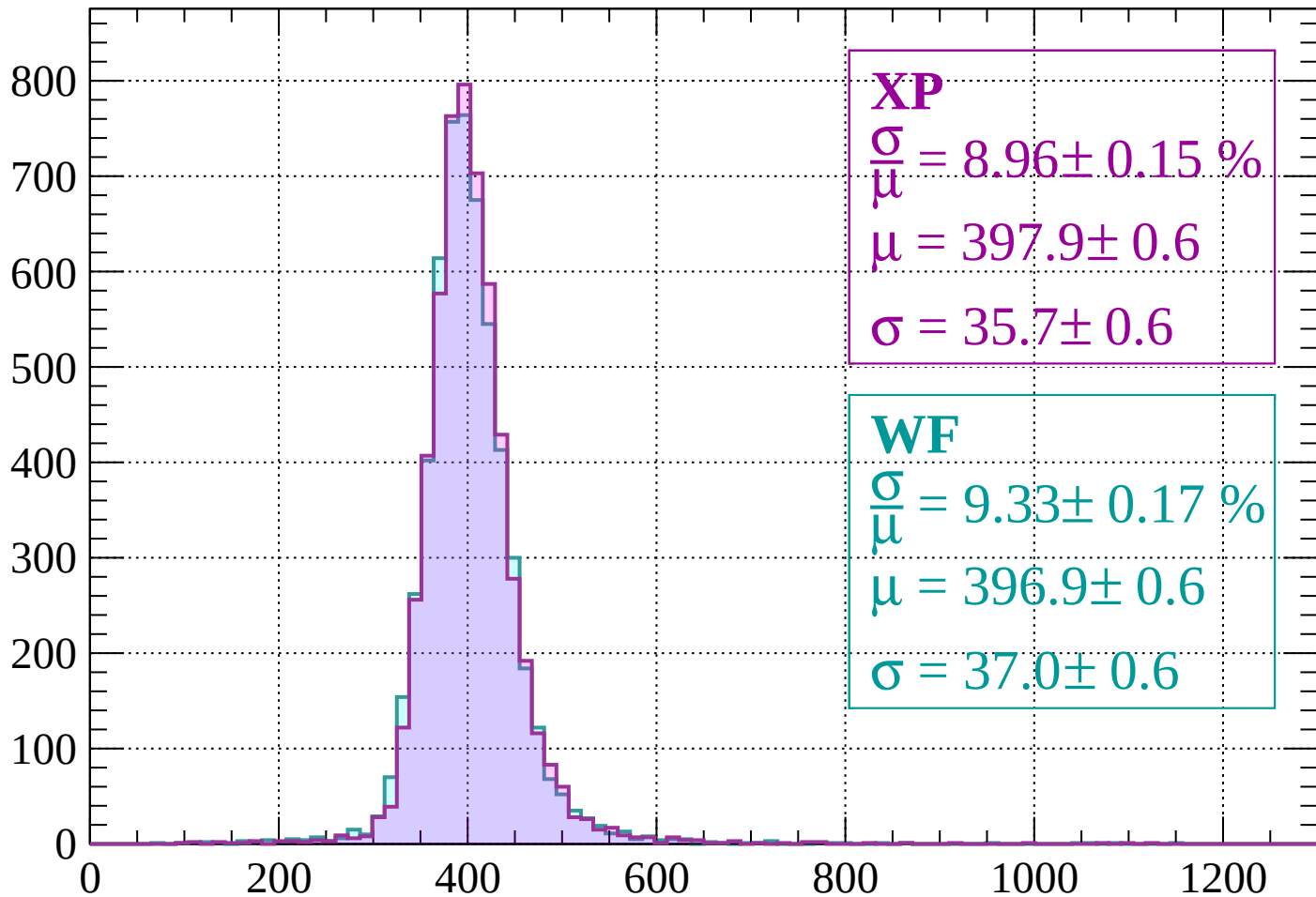
$$\mu = 395.7 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.96 \pm 0.15 \%$$

$$\mu = 397.9 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

WF

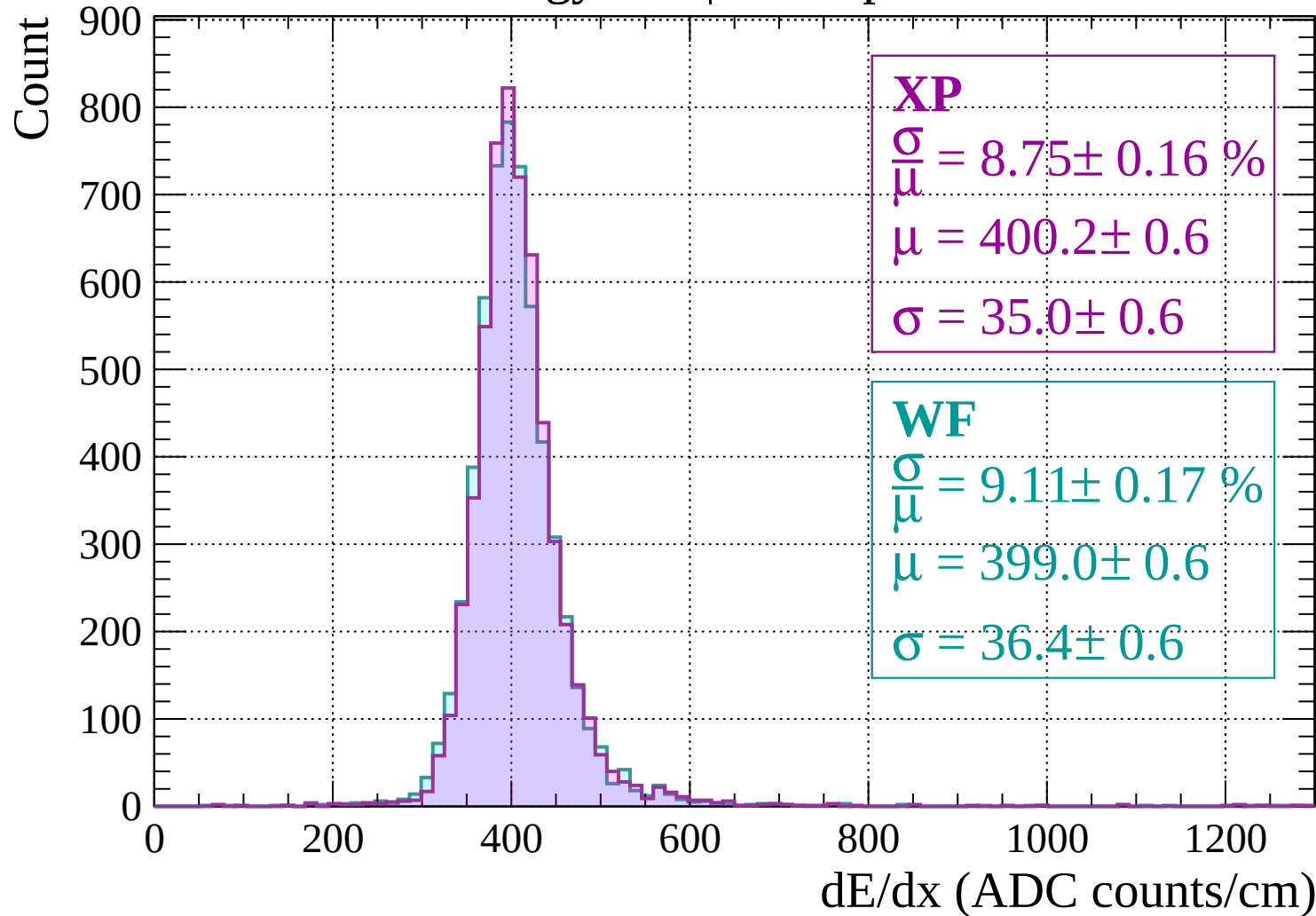
$$\frac{\sigma}{\mu} = 9.33 \pm 0.17 \%$$

$$\mu = 396.9 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

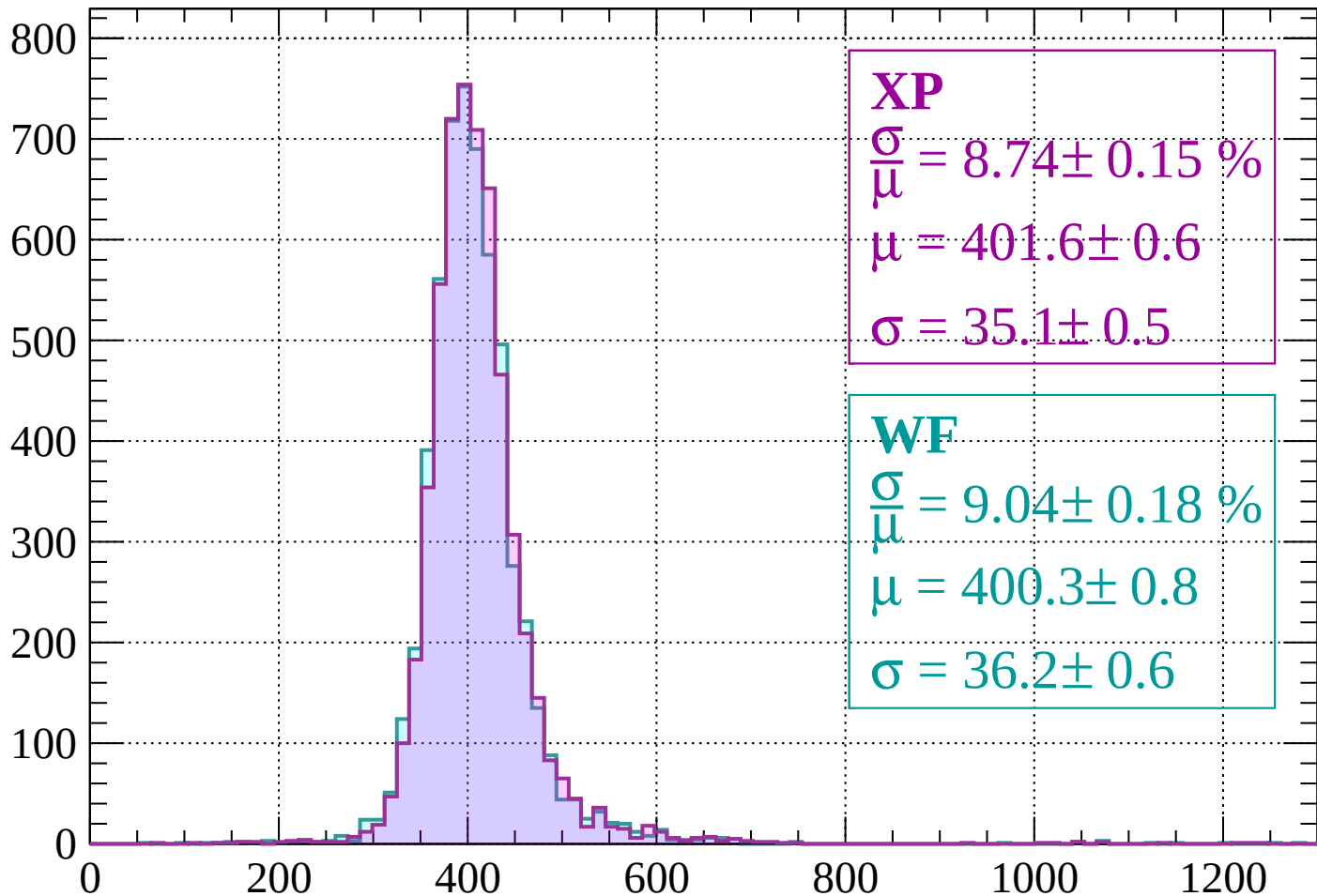
dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$



Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.15 \%$$

$$\mu = 401.6 \pm 0.6$$

$$\sigma = 35.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.04 \pm 0.18 \%$$

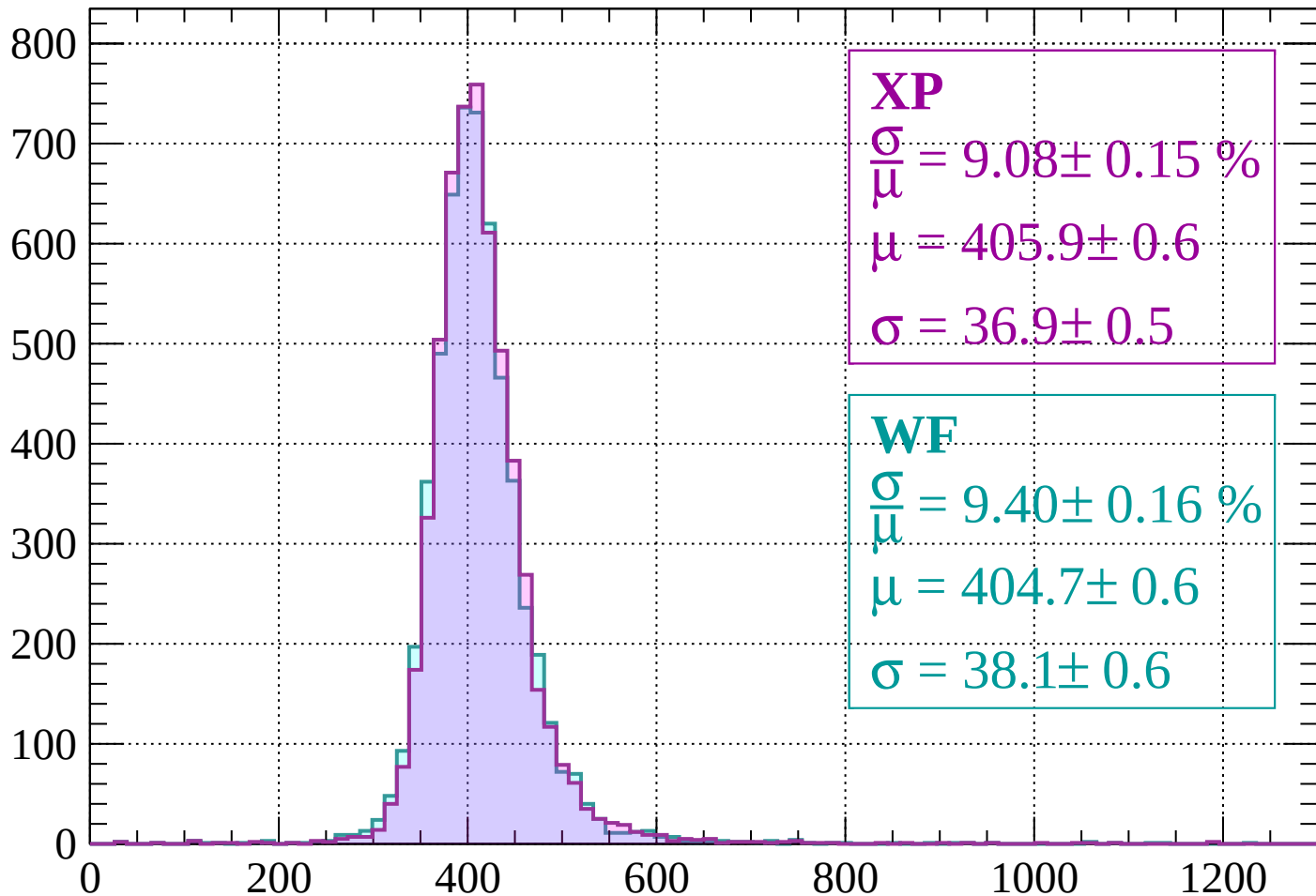
$$\mu = 400.3 \pm 0.8$$

$$\sigma = 36.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 9.08 \pm 0.15 \%$$

$$\mu = 405.9 \pm 0.6$$

$$\sigma = 36.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.16 \%$$

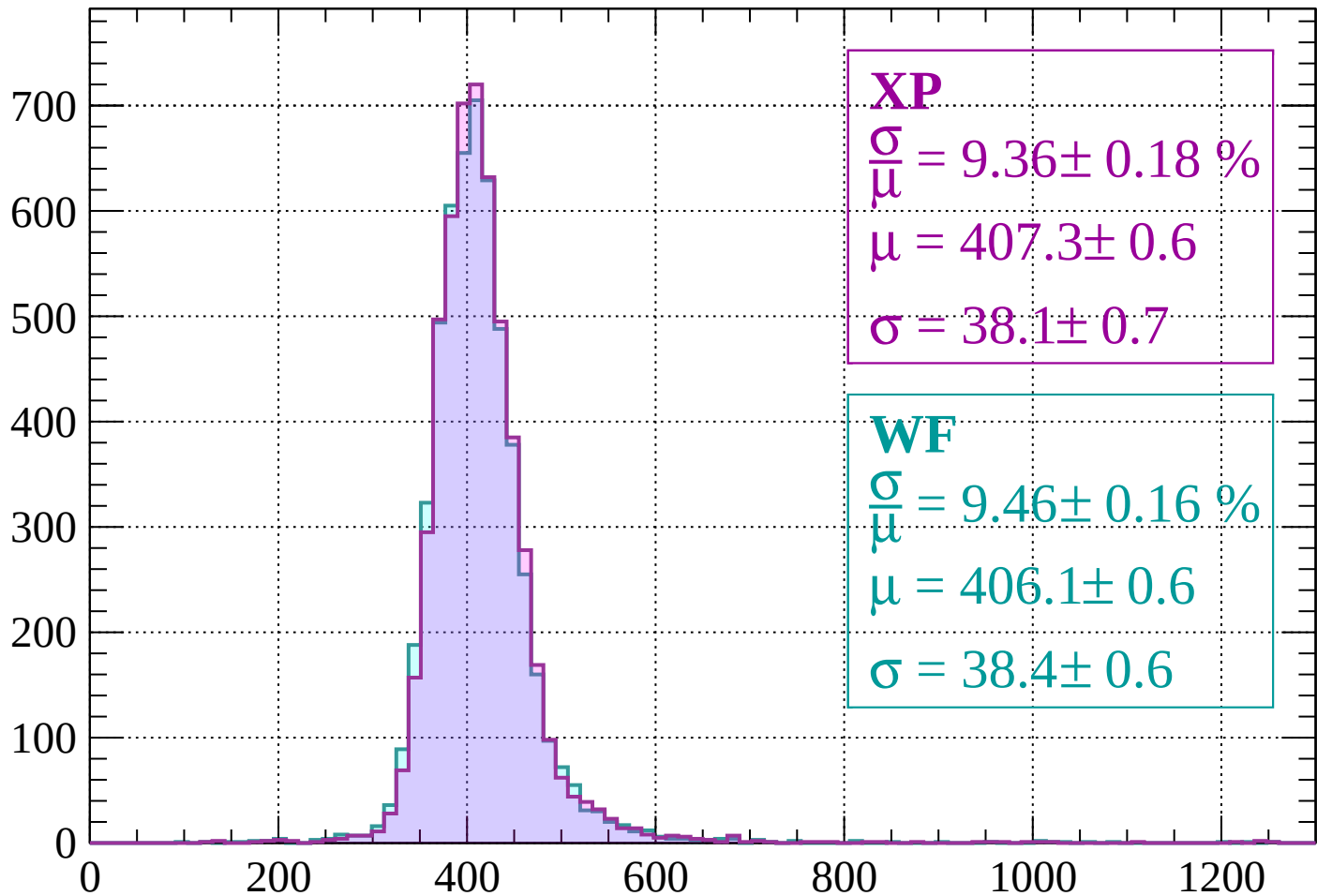
$$\mu = 404.7 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 9.36 \pm 0.18 \%$$

$$\mu = 407.3 \pm 0.6$$

$$\sigma = 38.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.16 \%$$

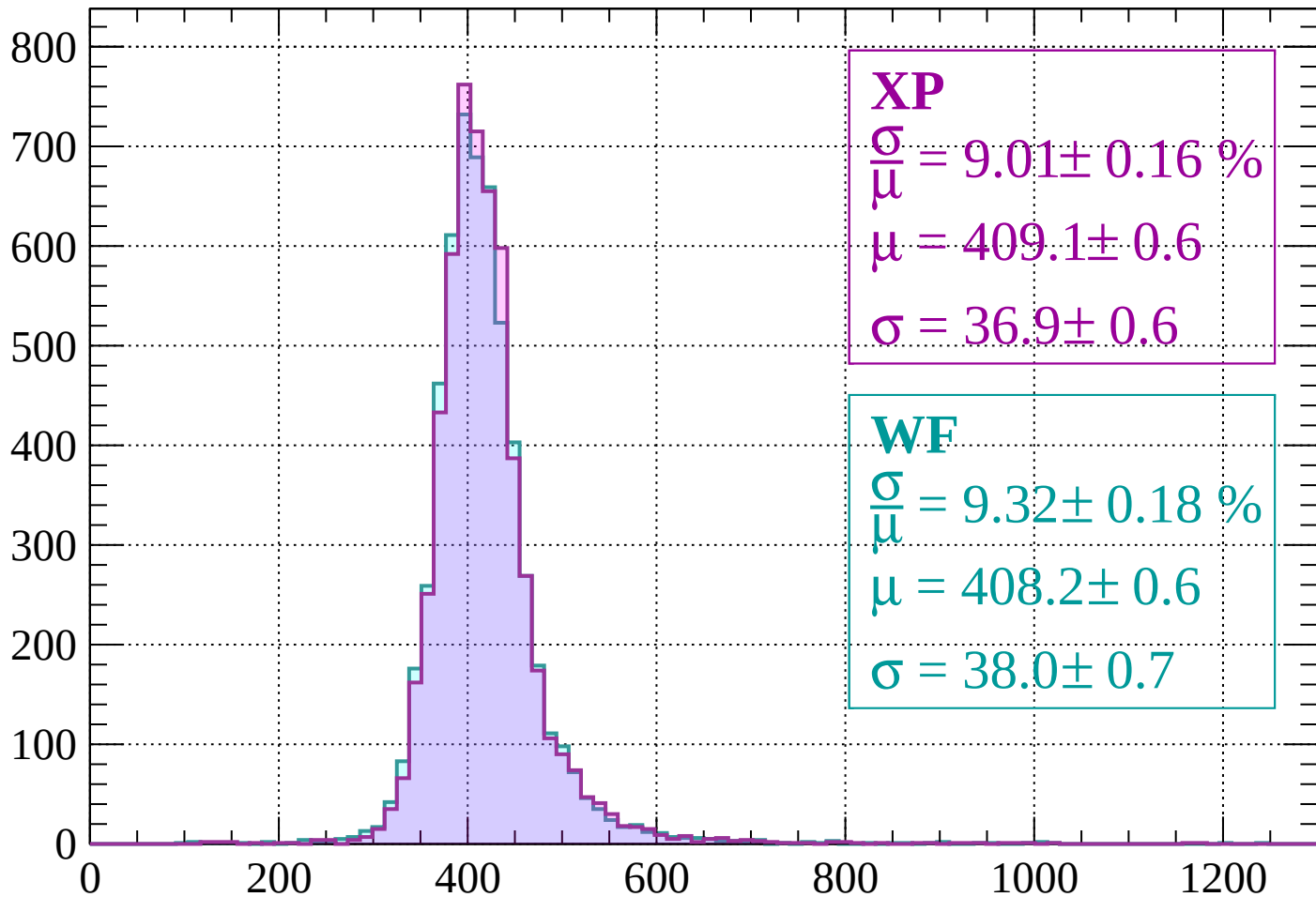
$$\mu = 406.1 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.16 \%$$

$$\mu = 409.1 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.18 \%$$

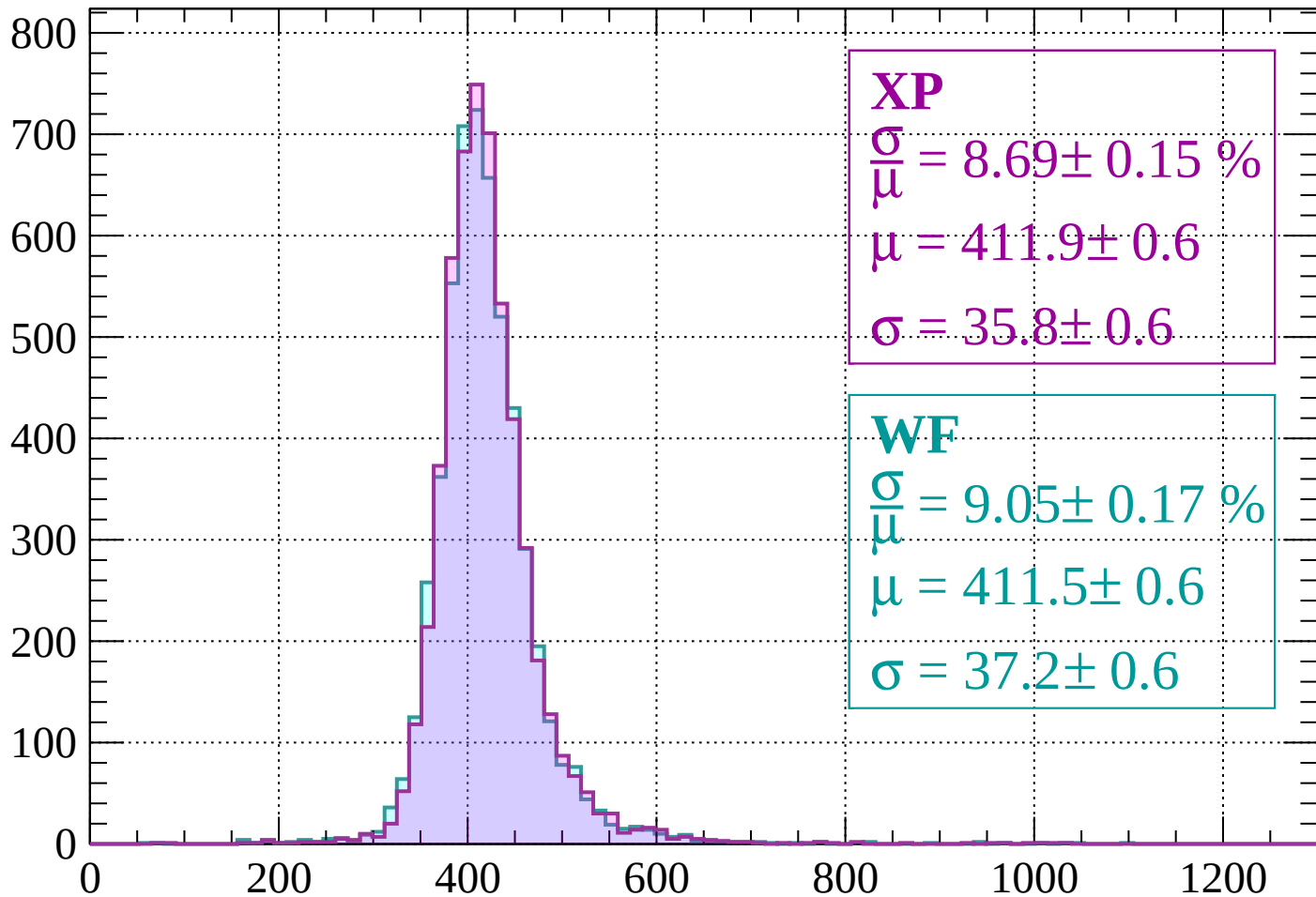
$$\mu = 408.2 \pm 0.6$$

$$\sigma = 38.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.69 \pm 0.15 \%$$

$$\mu = 411.9 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.17 \%$$

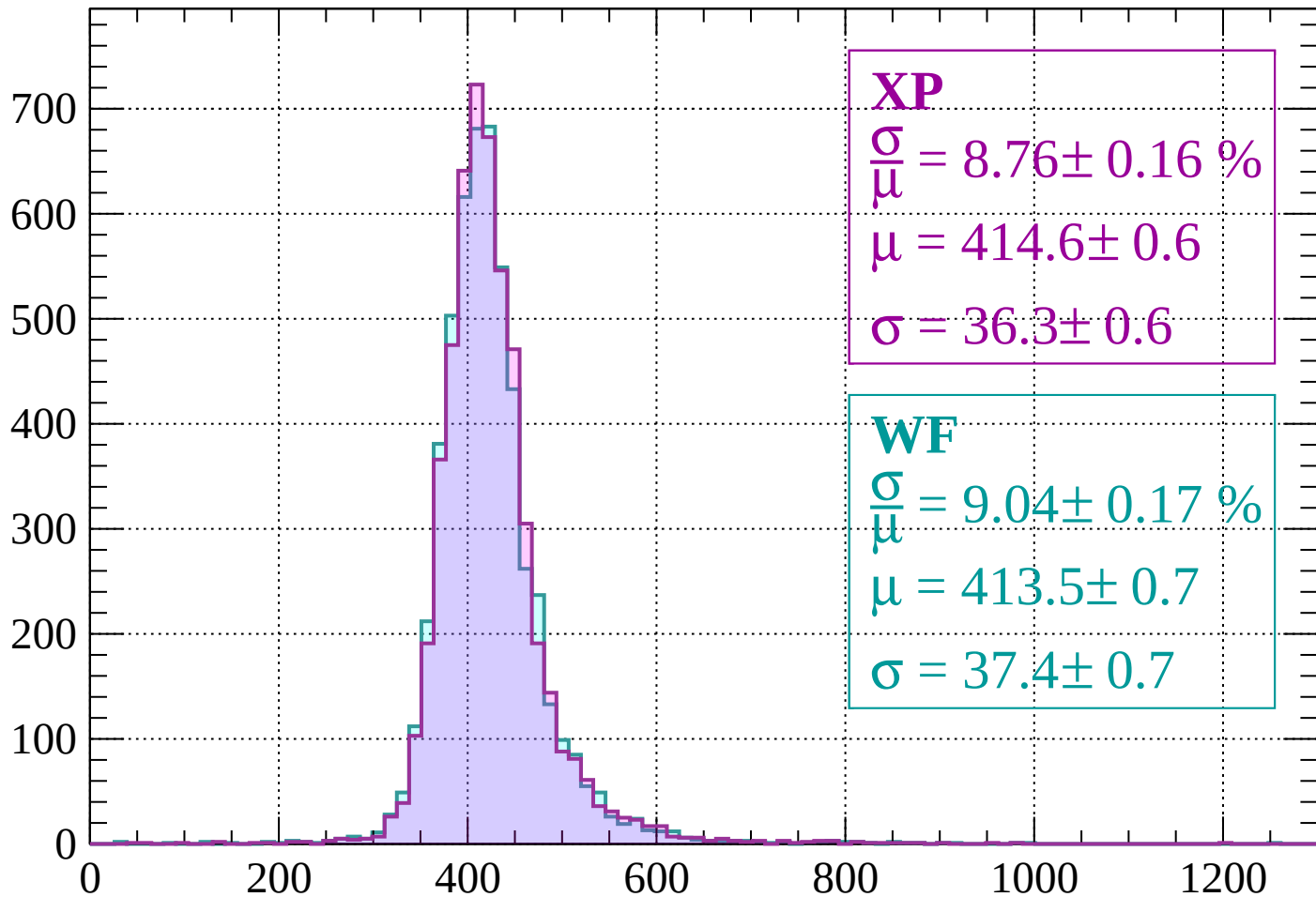
$$\mu = 411.5 \pm 0.6$$

$$\sigma = 37.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.16 \%$$

$$\mu = 414.6 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.04 \pm 0.17 \%$$

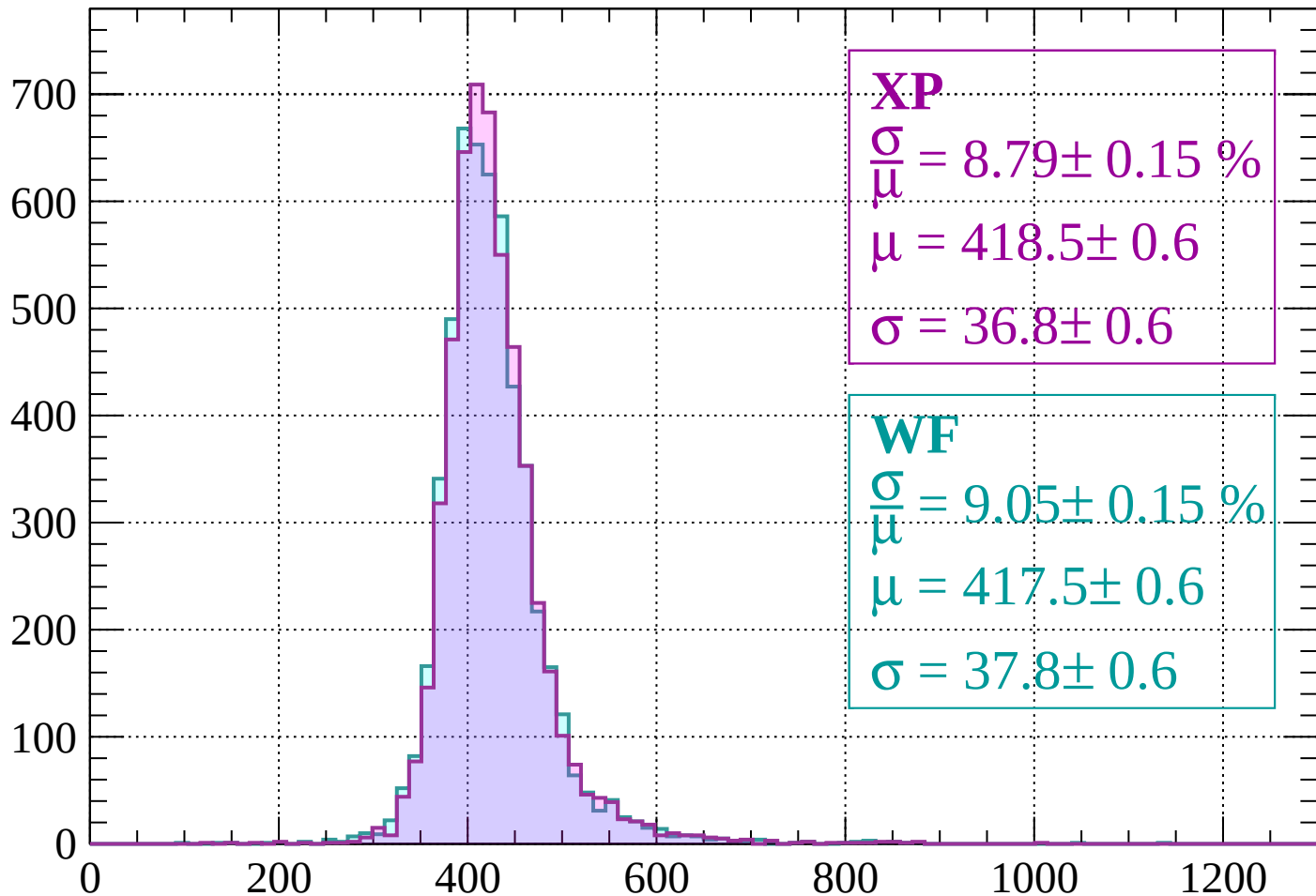
$$\mu = 413.5 \pm 0.7$$

$$\sigma = 37.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.15 \%$$

$$\mu = 418.5 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.15 \%$$

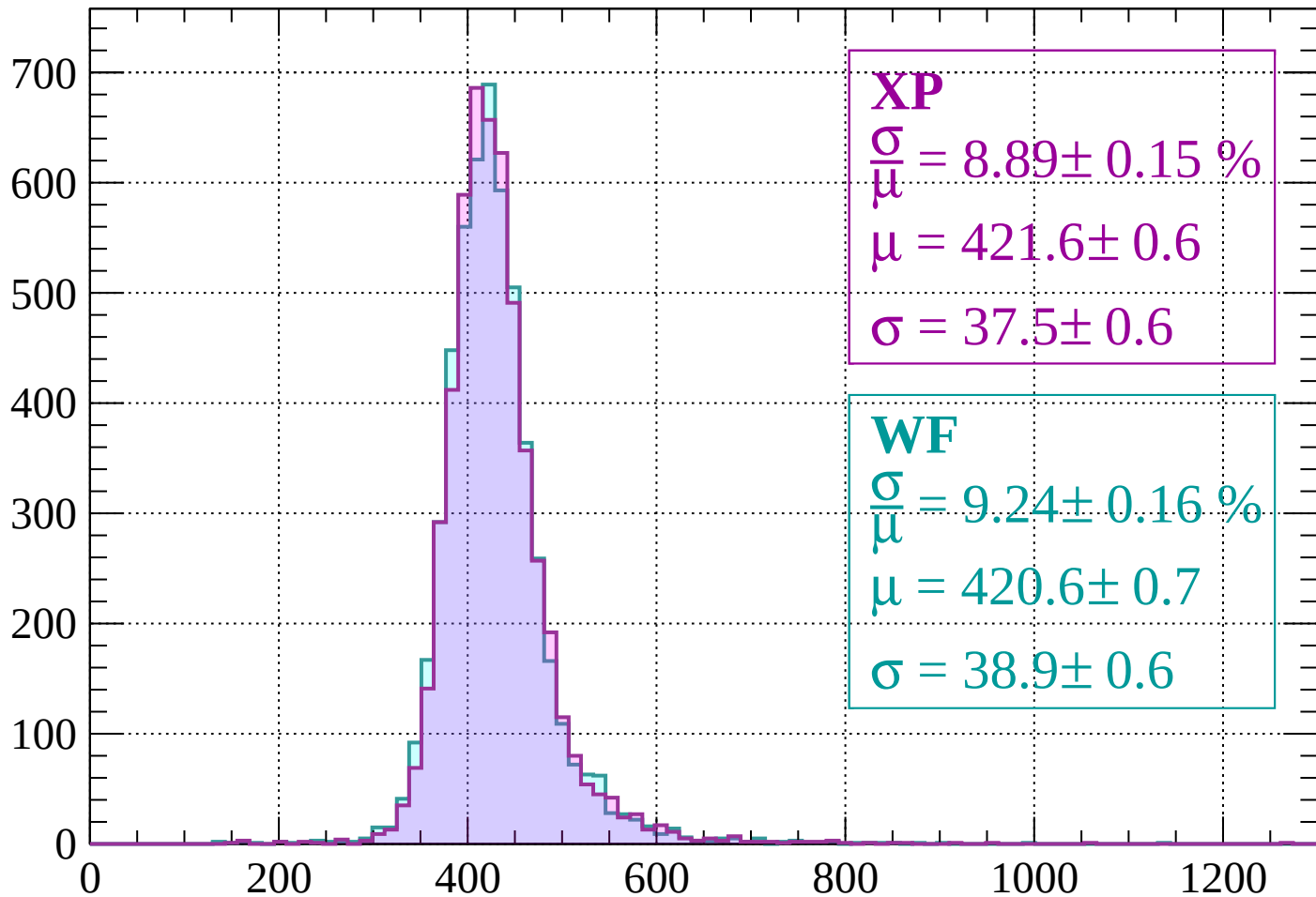
$$\mu = 417.5 \pm 0.6$$

$$\sigma = 37.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.15 \%$$

$$\mu = 421.6 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.16 \%$$

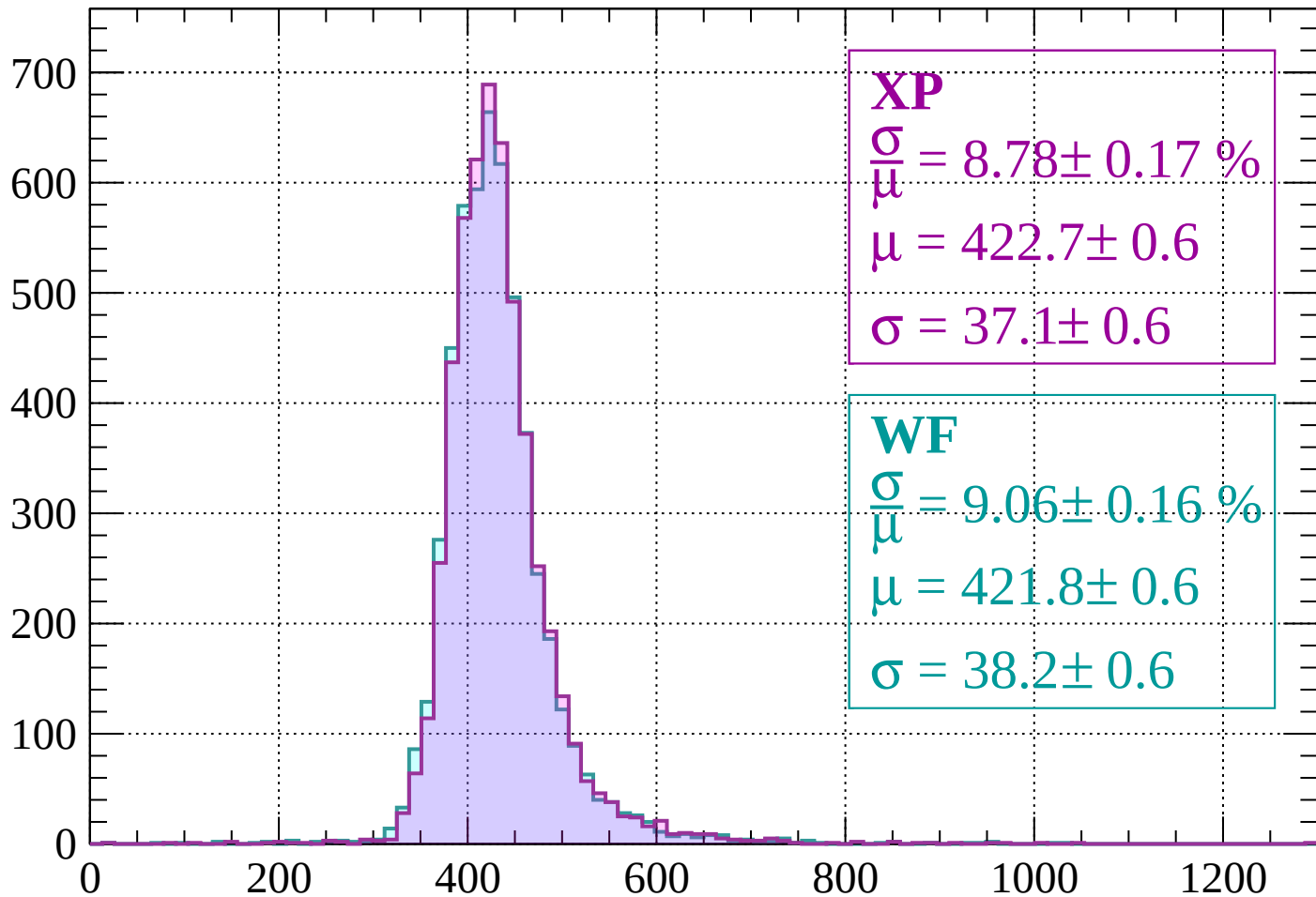
$$\mu = 420.6 \pm 0.7$$

$$\sigma = 38.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.17 \%$$

$$\mu = 422.7 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.06 \pm 0.16 \%$$

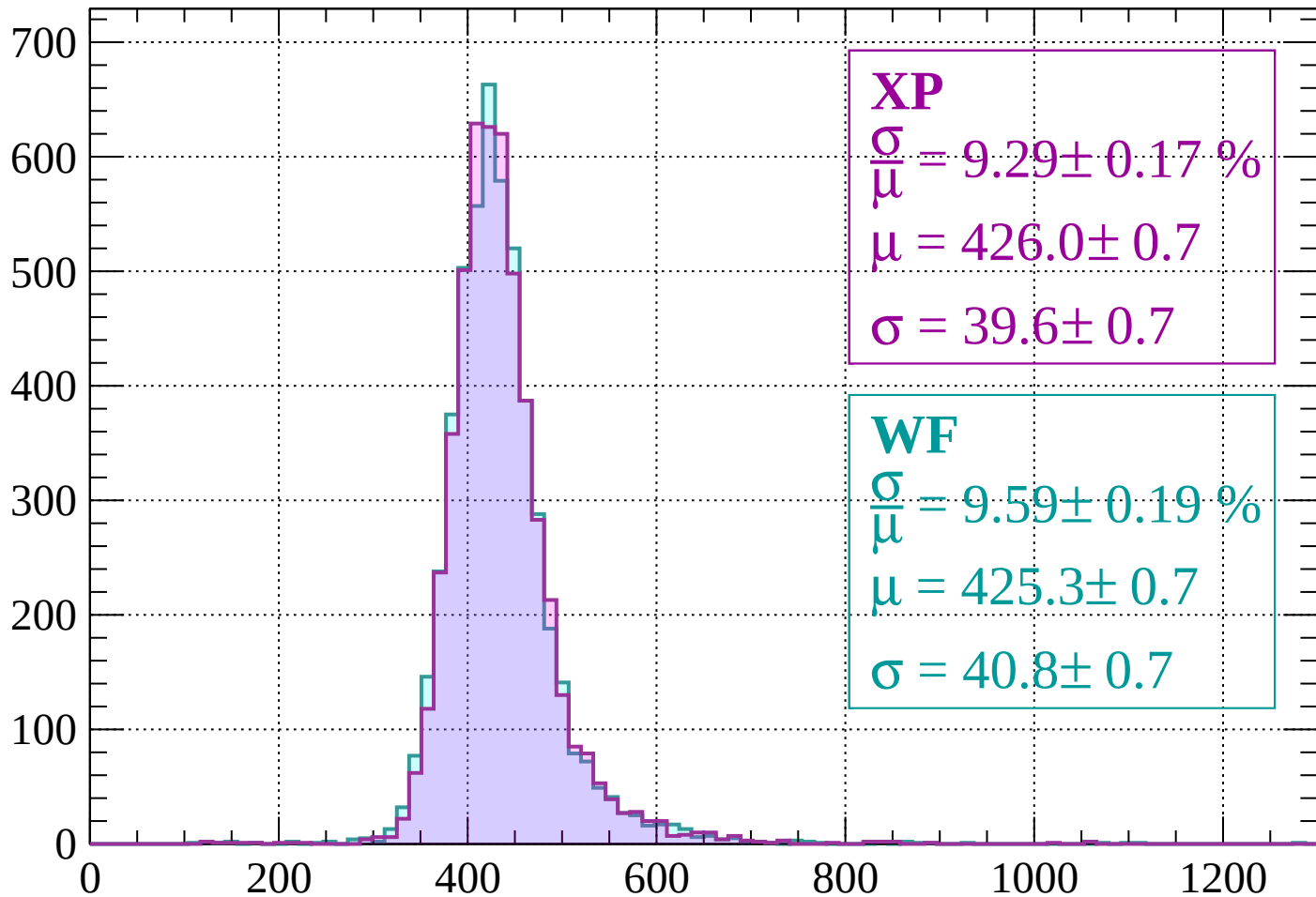
$$\mu = 421.8 \pm 0.6$$

$$\sigma = 38.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 9.29 \pm 0.17 \%$$

$$\mu = 426.0 \pm 0.7$$

$$\sigma = 39.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.19 \%$$

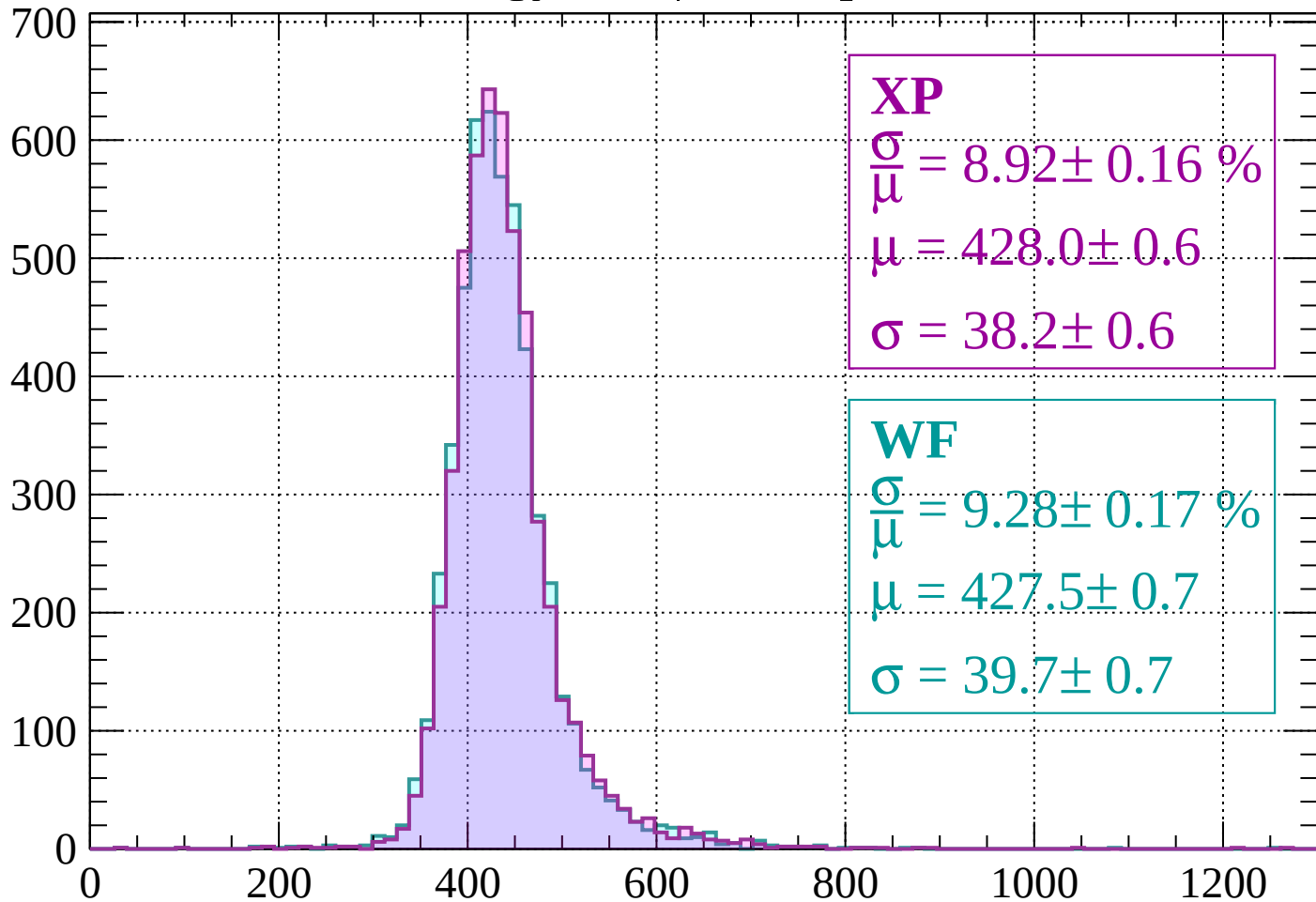
$$\mu = 425.3 \pm 0.7$$

$$\sigma = 40.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.16 \%$$

$$\mu = 428.0 \pm 0.6$$

$$\sigma = 38.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.17 \%$$

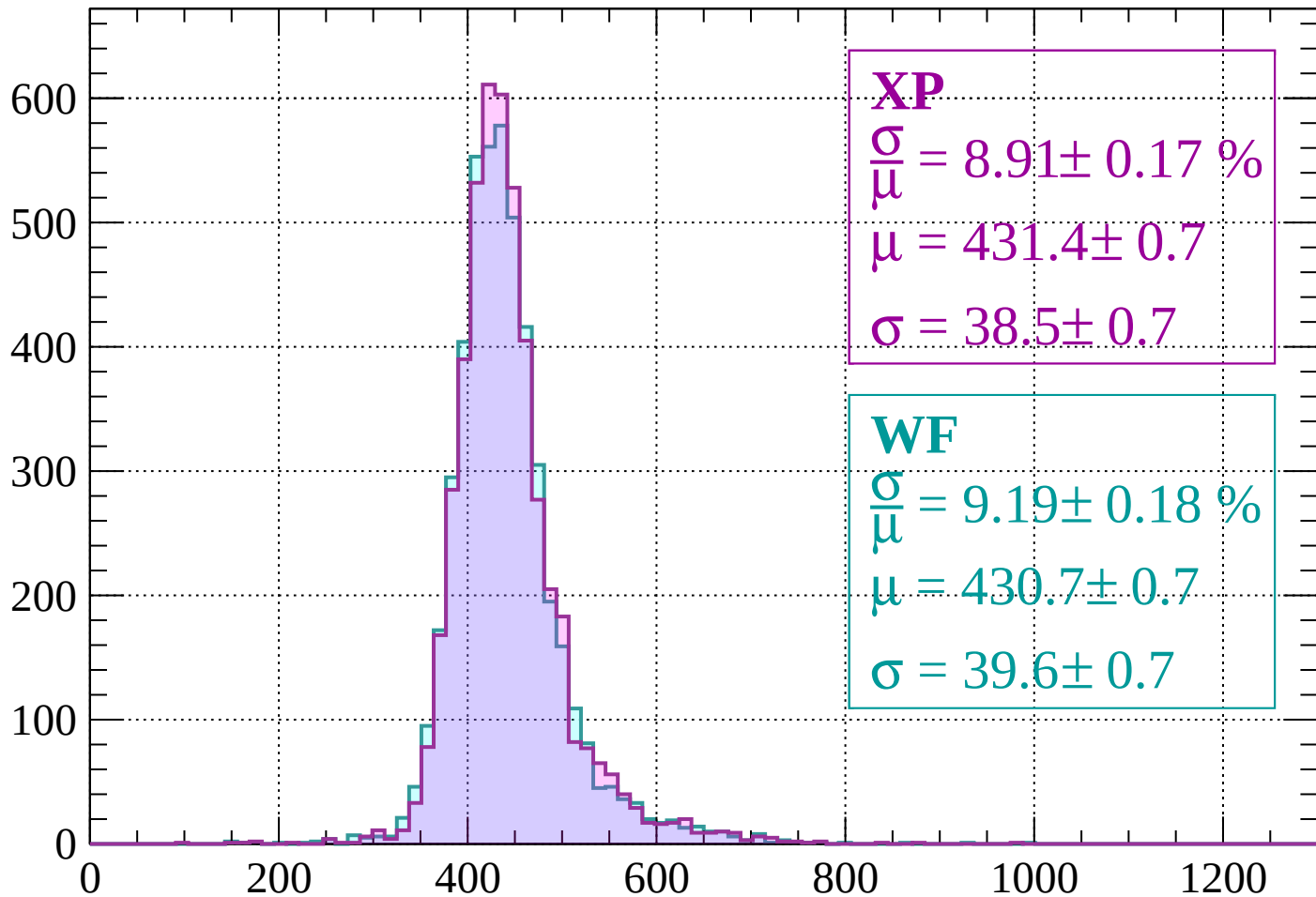
$$\mu = 427.5 \pm 0.7$$

$$\sigma = 39.7 \pm 0.7$$

dE/dx (ADC counts/cm)

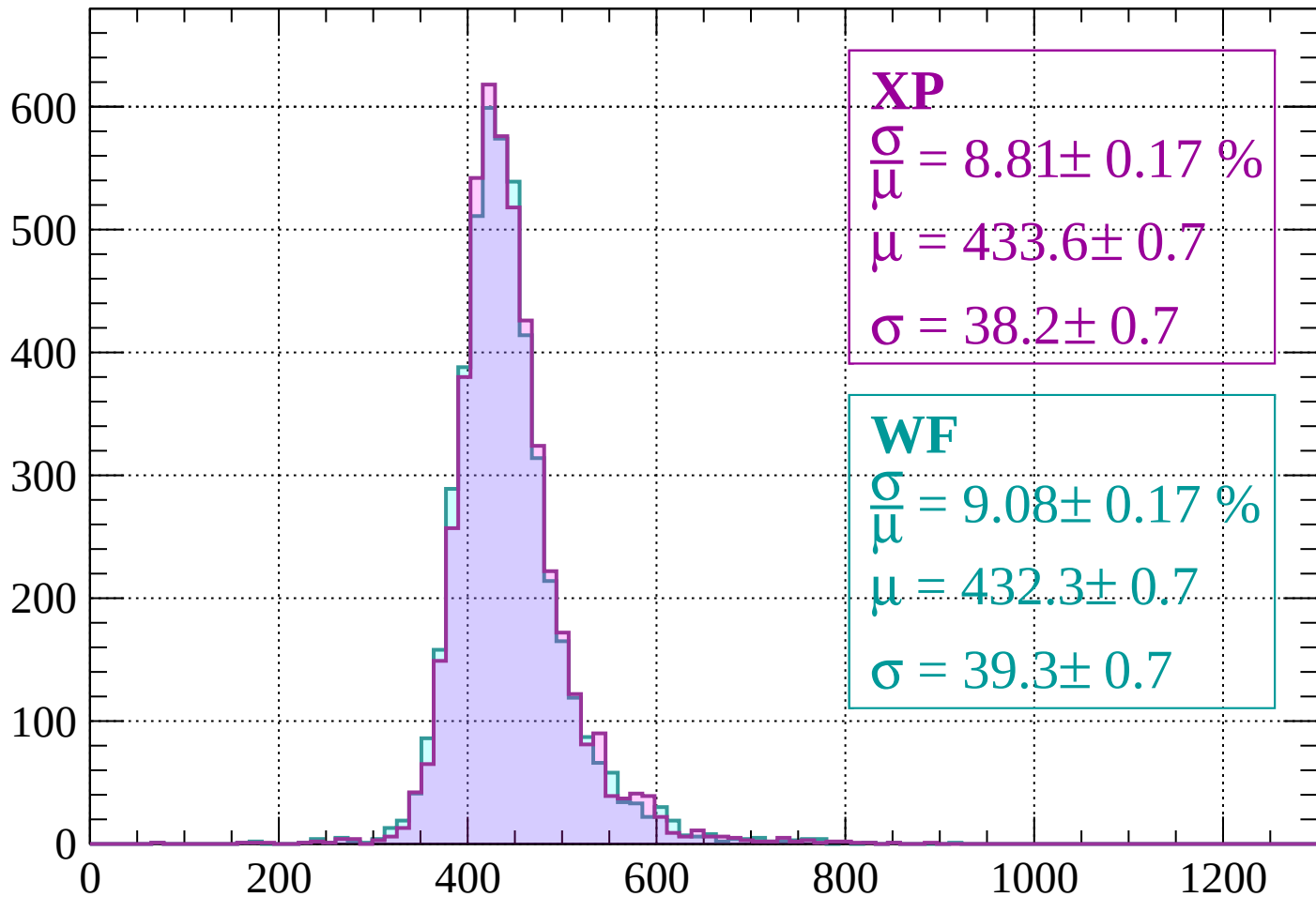
Energy loss | $960 < p < 1000$

Count



Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 8.81 \pm 0.17 \%$$

$$\mu = 433.6 \pm 0.7$$

$$\sigma = 38.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.08 \pm 0.17 \%$$

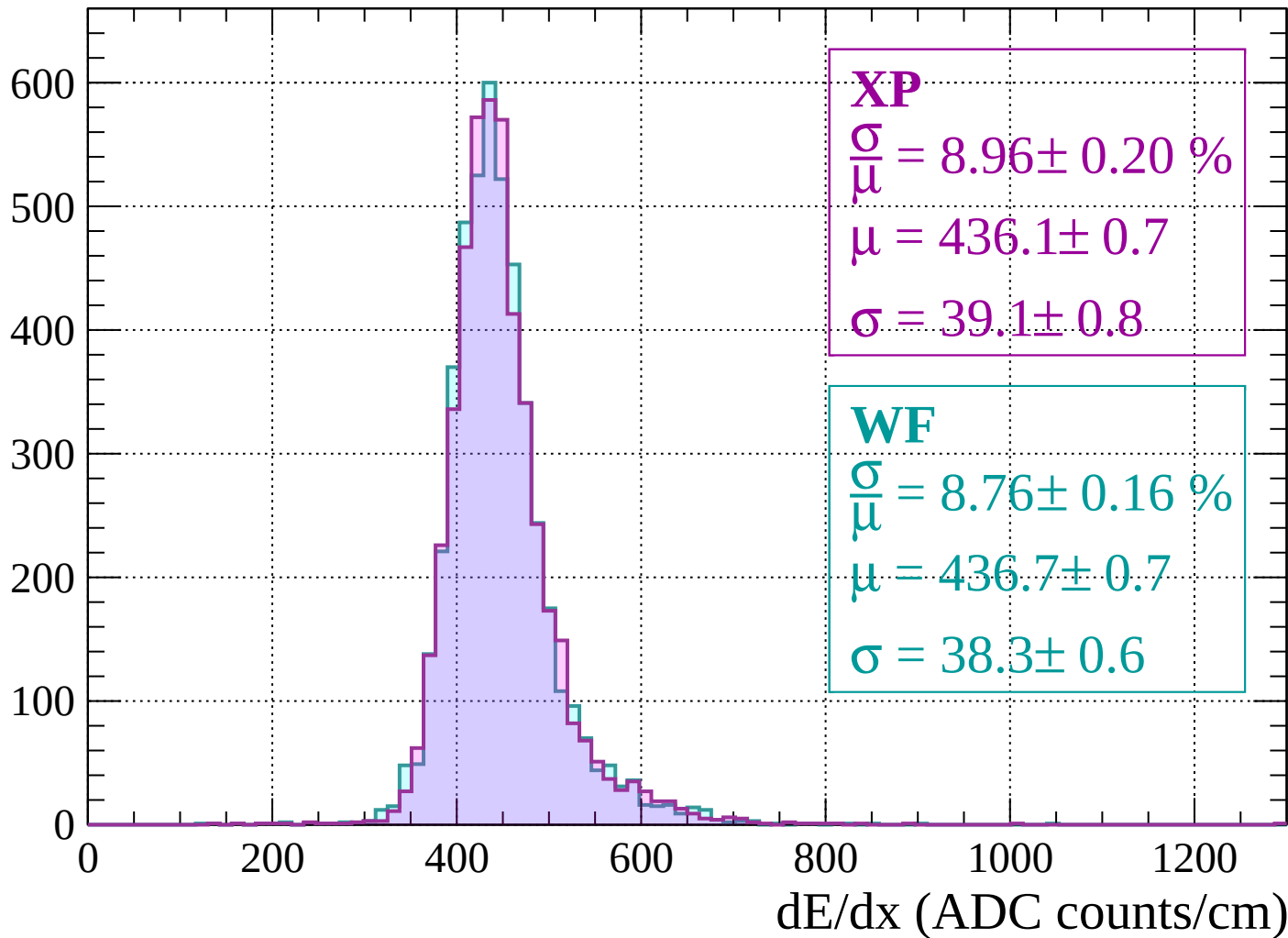
$$\mu = 432.3 \pm 0.7$$

$$\sigma = 39.3 \pm 0.7$$

dE/dx (ADC counts/cm)

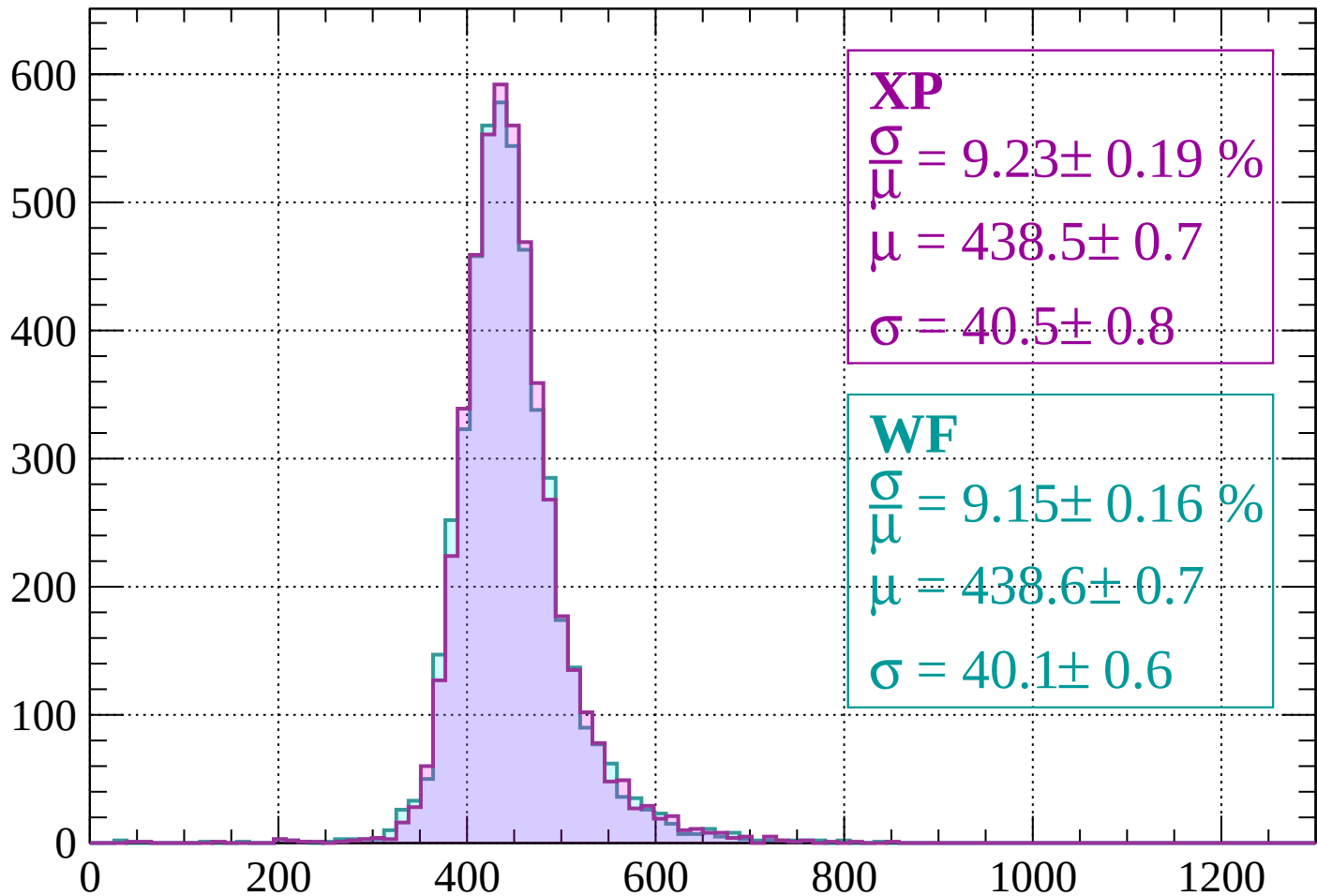
Energy loss | $1040 < p < 1080$

Count



Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 9.23 \pm 0.19 \%$$

$$\mu = 438.5 \pm 0.7$$

$$\sigma = 40.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.16 \%$$

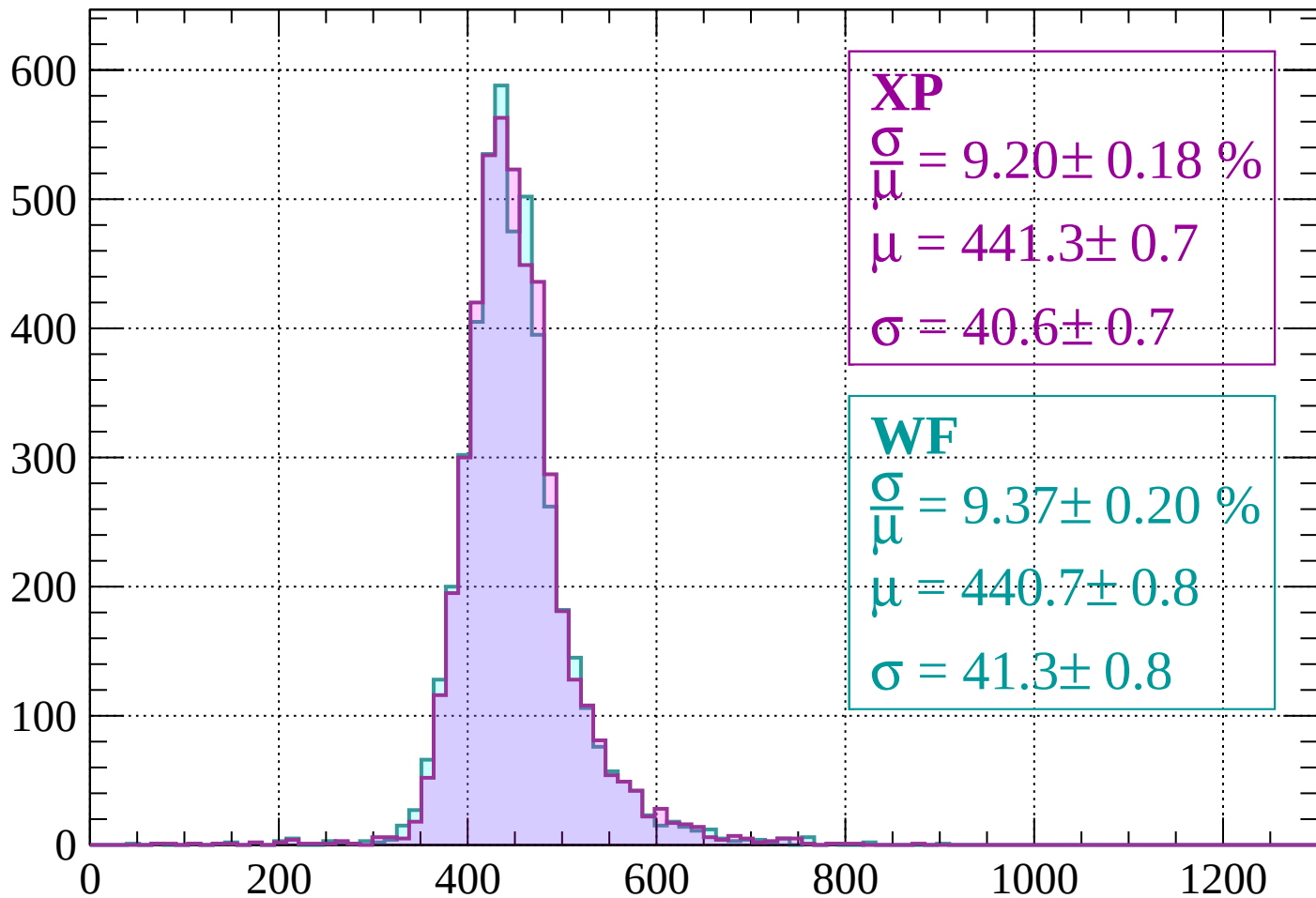
$$\mu = 438.6 \pm 0.7$$

$$\sigma = 40.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

Count



XP

$$\frac{\sigma}{\mu} = 9.20 \pm 0.18 \%$$

$$\mu = 441.3 \pm 0.7$$

$$\sigma = 40.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.20 \%$$

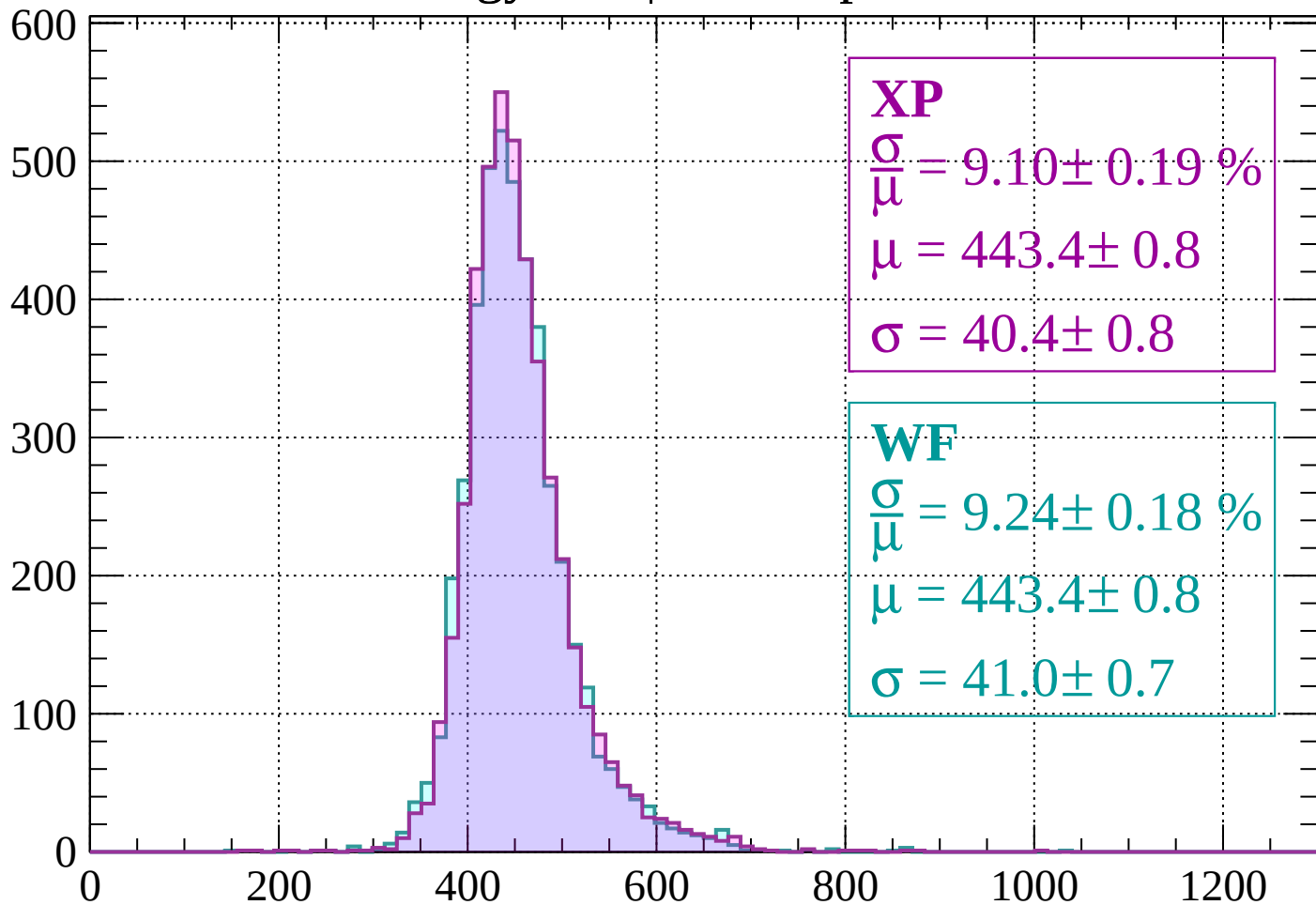
$$\mu = 440.7 \pm 0.8$$

$$\sigma = 41.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.10 \pm 0.19 \%$$

$$\mu = 443.4 \pm 0.8$$

$$\sigma = 40.4 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.24 \pm 0.18 \%$$

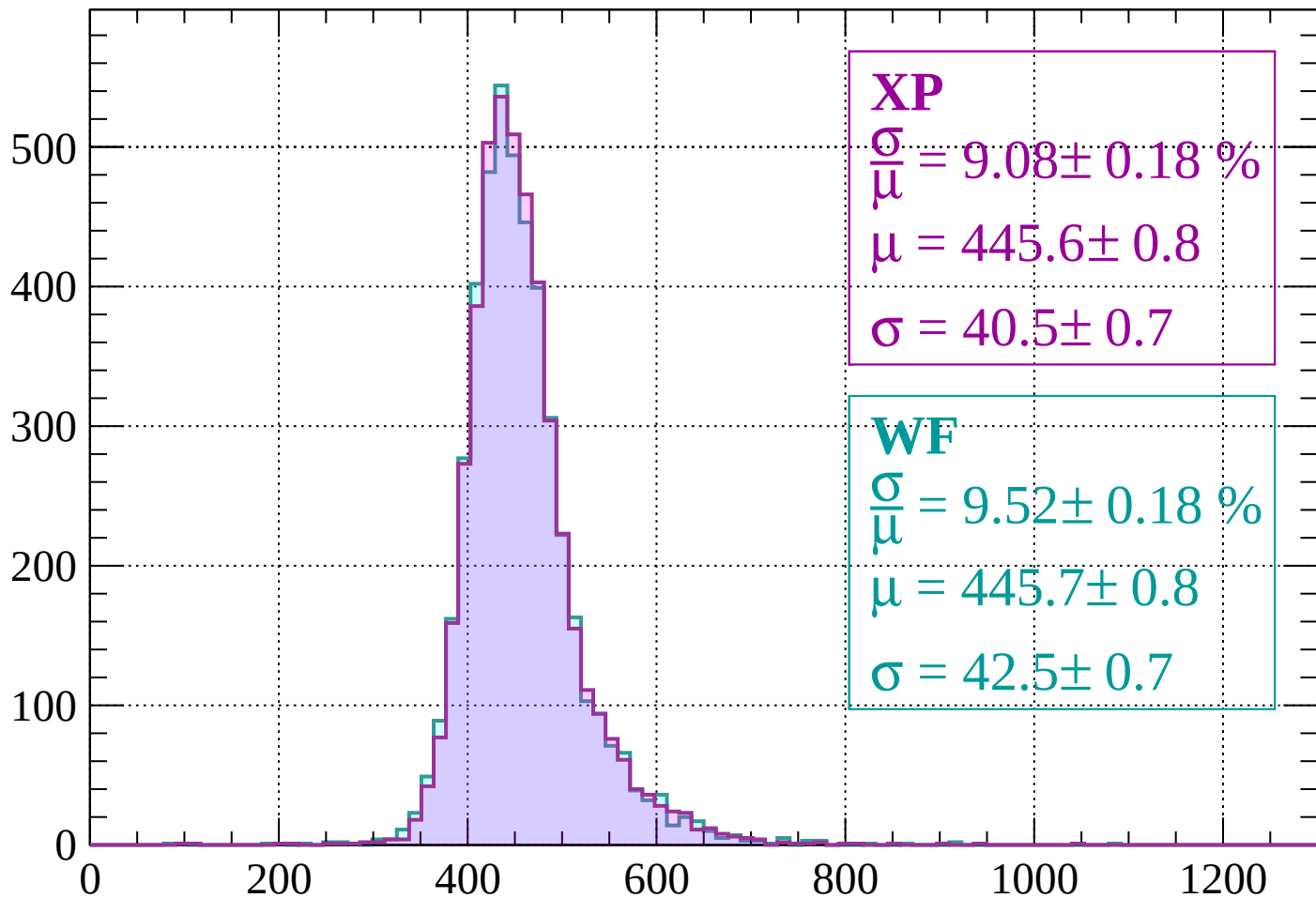
$$\mu = 443.4 \pm 0.8$$

$$\sigma = 41.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 9.08 \pm 0.18 \%$$

$$\mu = 445.6 \pm 0.8$$

$$\sigma = 40.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 0.18 \%$$

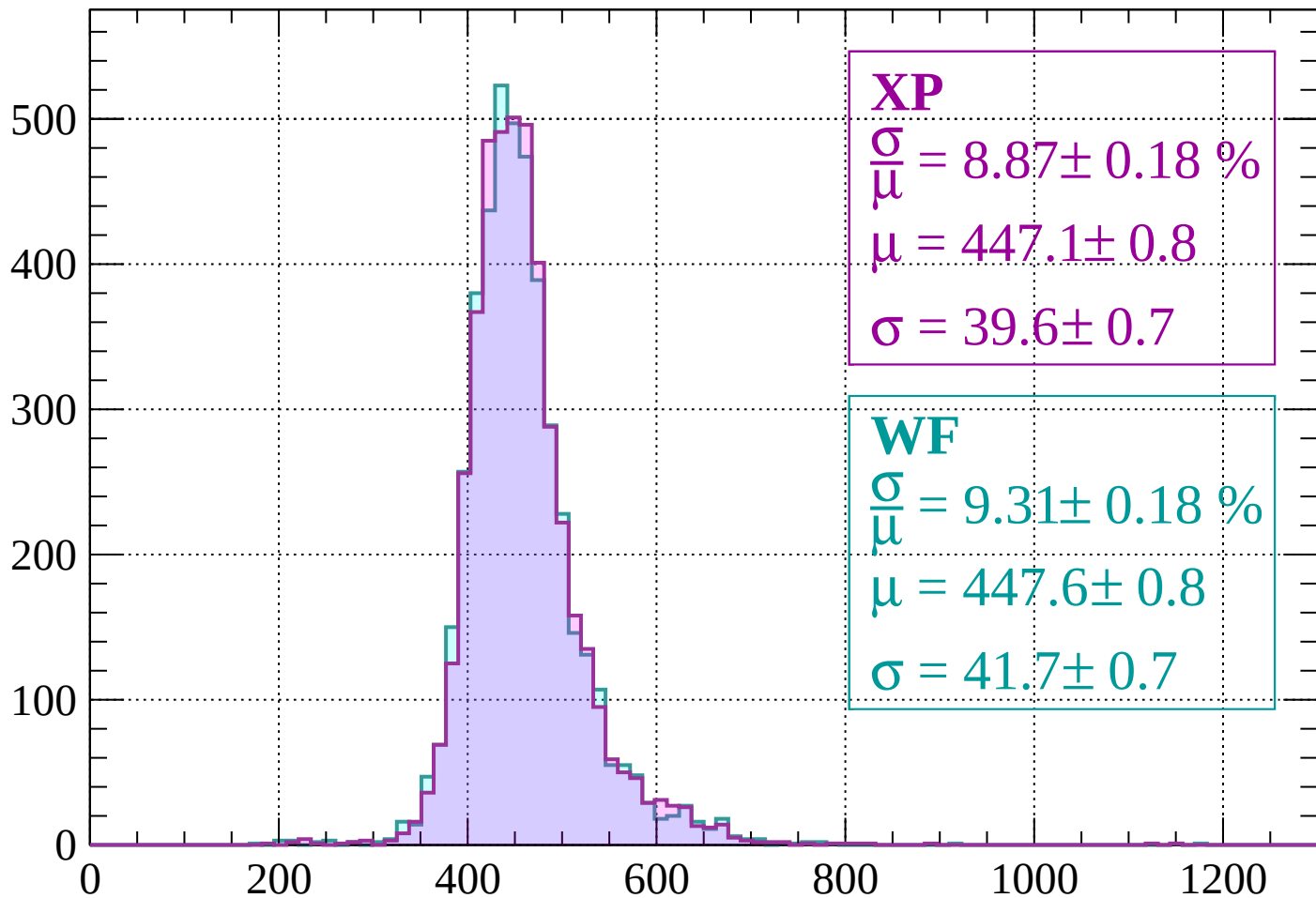
$$\mu = 445.7 \pm 0.8$$

$$\sigma = 42.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.87 \pm 0.18 \%$$

$$\mu = 447.1 \pm 0.8$$

$$\sigma = 39.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.31 \pm 0.18 \%$$

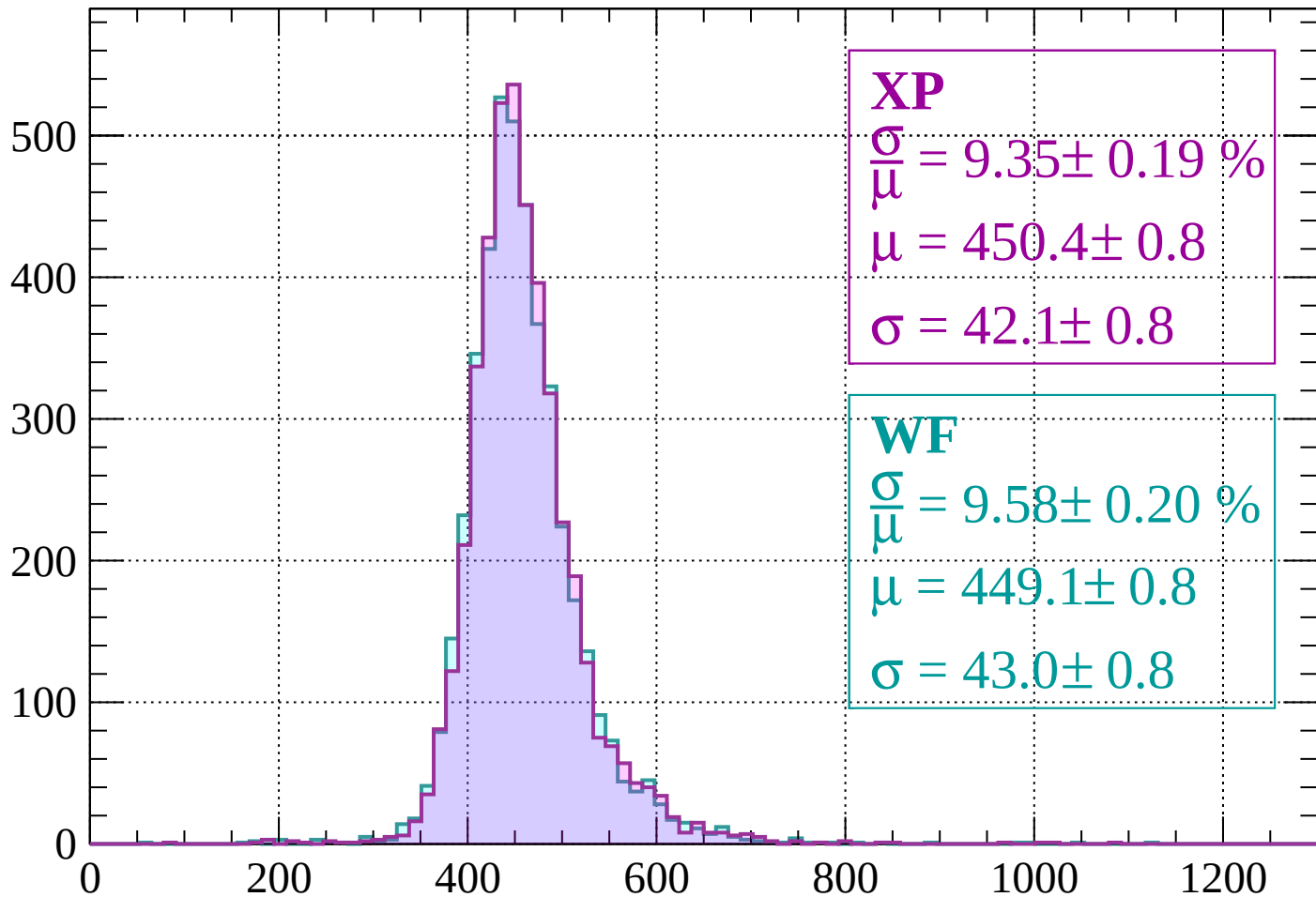
$$\mu = 447.6 \pm 0.8$$

$$\sigma = 41.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 9.35 \pm 0.19 \%$$

$$\mu = 450.4 \pm 0.8$$

$$\sigma = 42.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.58 \pm 0.20 \%$$

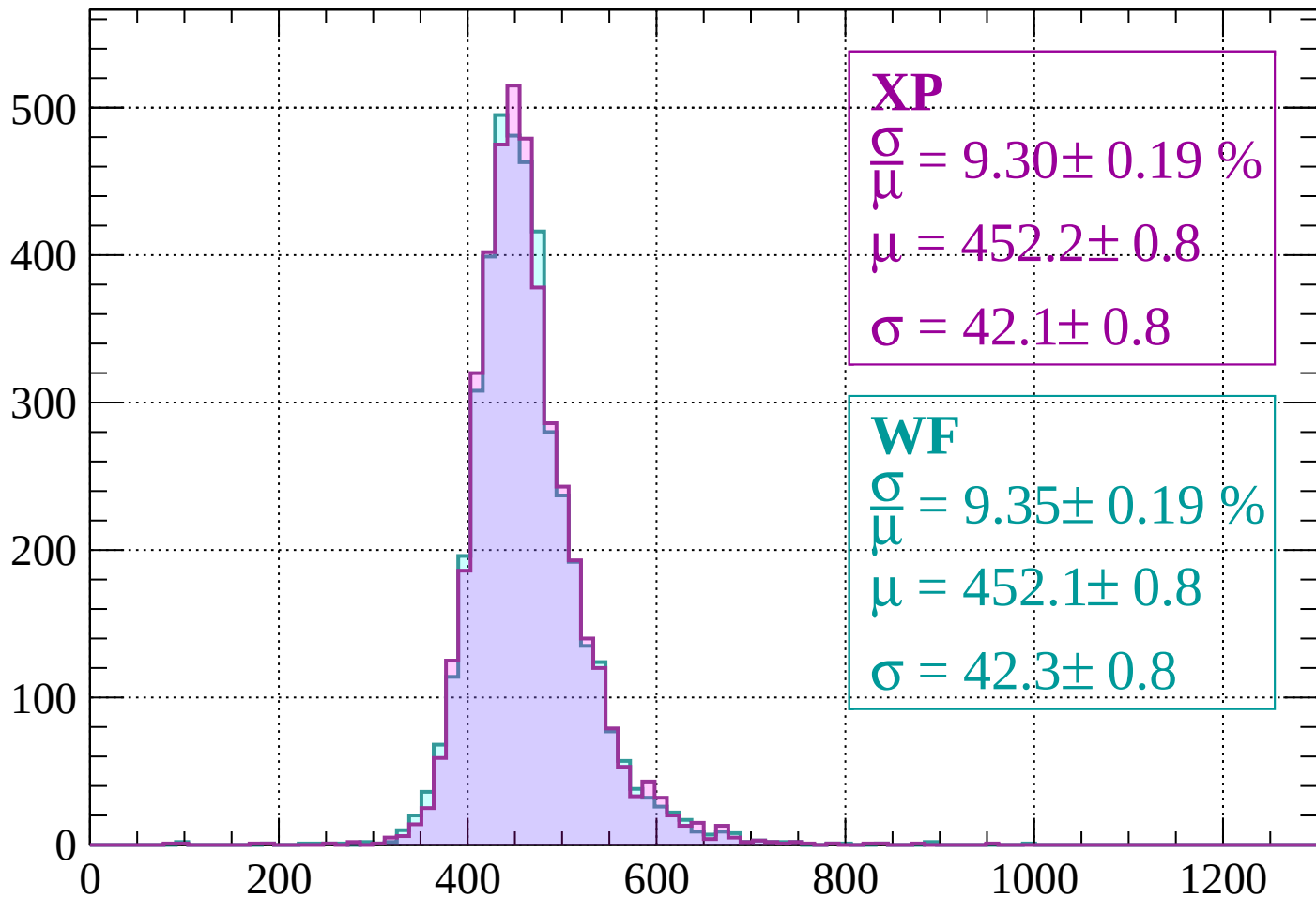
$$\mu = 449.1 \pm 0.8$$

$$\sigma = 43.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.19 \%$$

$$\mu = 452.2 \pm 0.8$$

$$\sigma = 42.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.35 \pm 0.19 \%$$

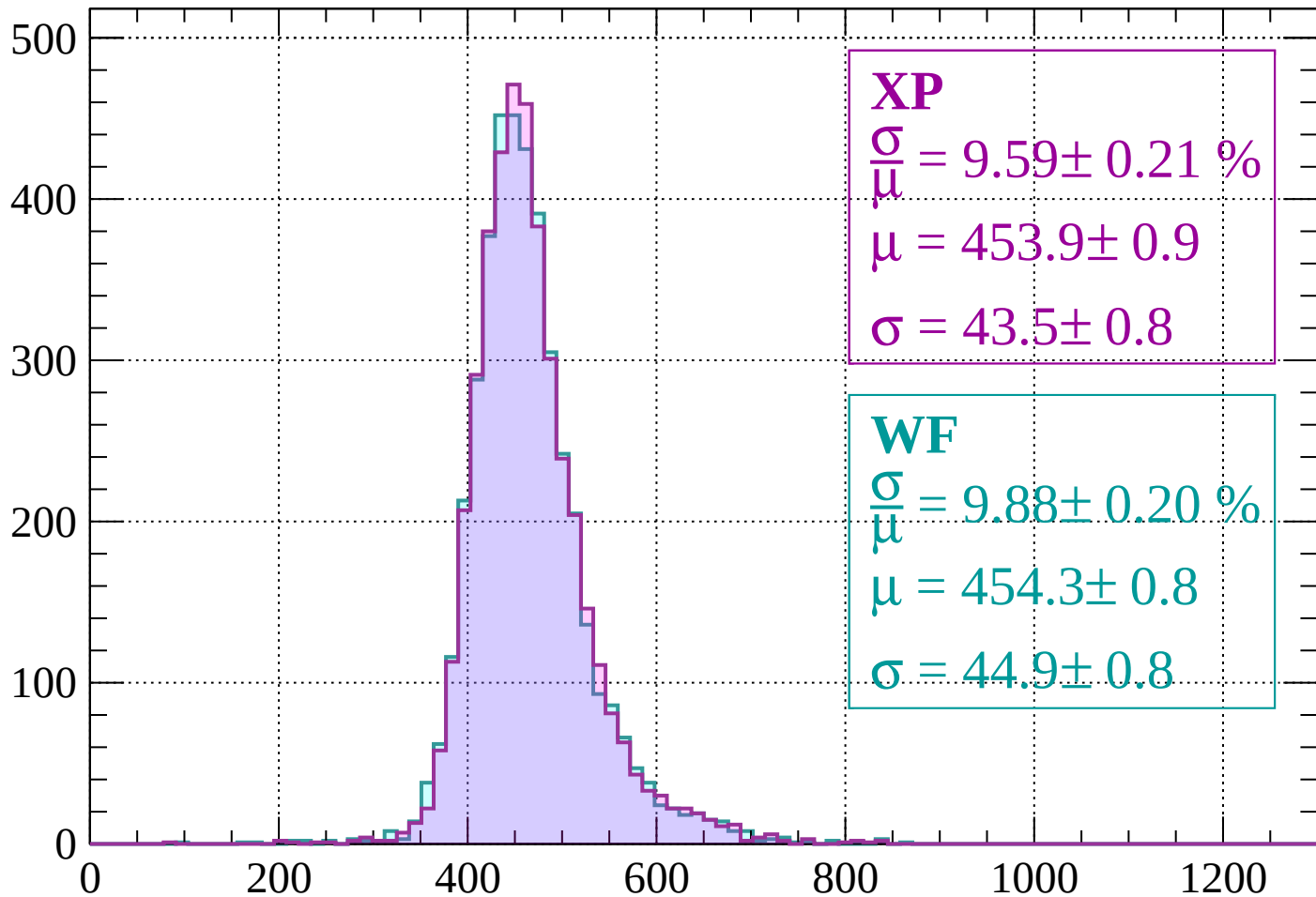
$$\mu = 452.1 \pm 0.8$$

$$\sigma = 42.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 9.59 \pm 0.21 \%$$

$$\mu = 453.9 \pm 0.9$$

$$\sigma = 43.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.88 \pm 0.20 \%$$

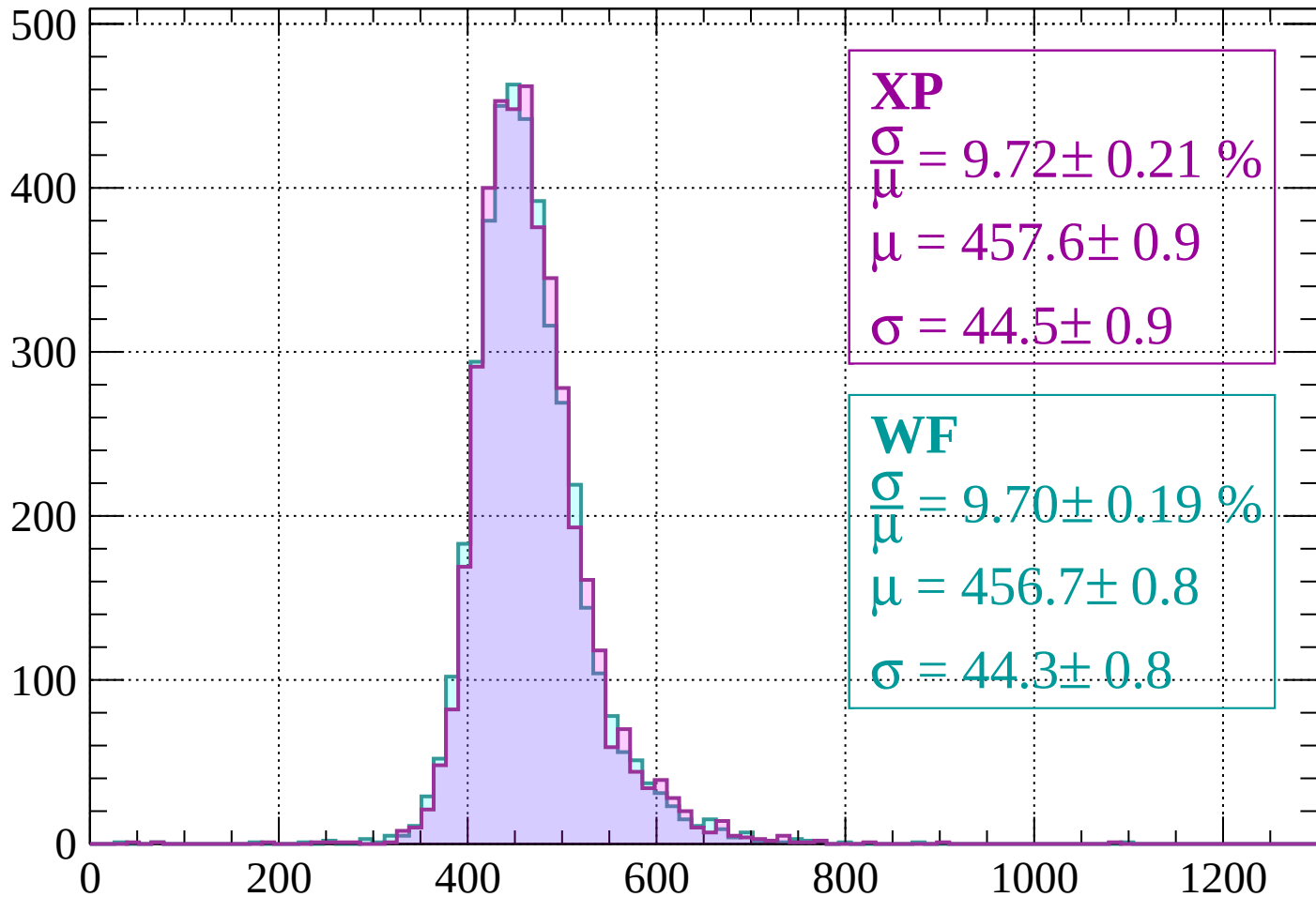
$$\mu = 454.3 \pm 0.8$$

$$\sigma = 44.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 9.72 \pm 0.21 \%$$

$$\mu = 457.6 \pm 0.9$$

$$\sigma = 44.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.70 \pm 0.19 \%$$

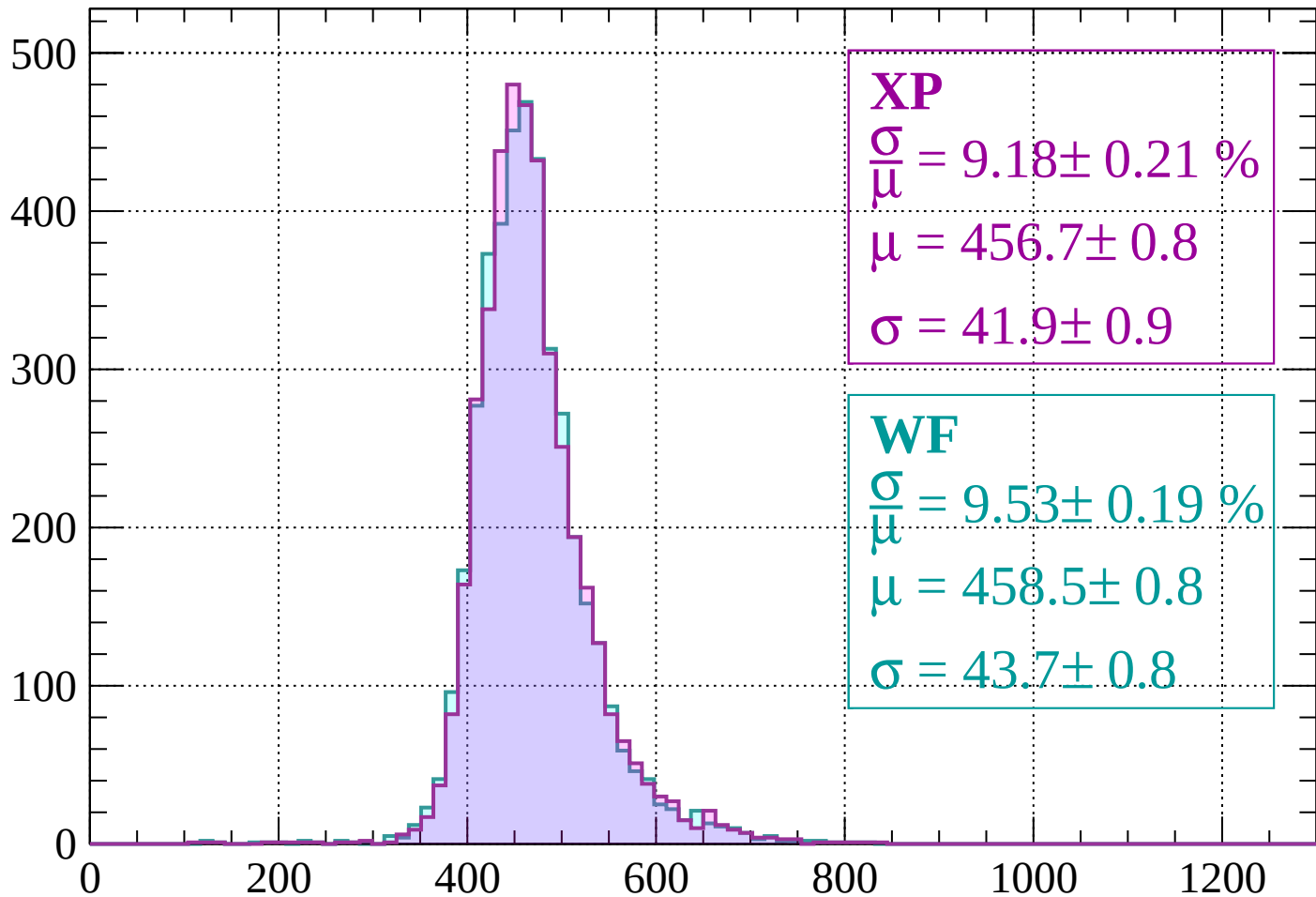
$$\mu = 456.7 \pm 0.8$$

$$\sigma = 44.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.21 \%$$

$$\mu = 456.7 \pm 0.8$$

$$\sigma = 41.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.53 \pm 0.19 \%$$

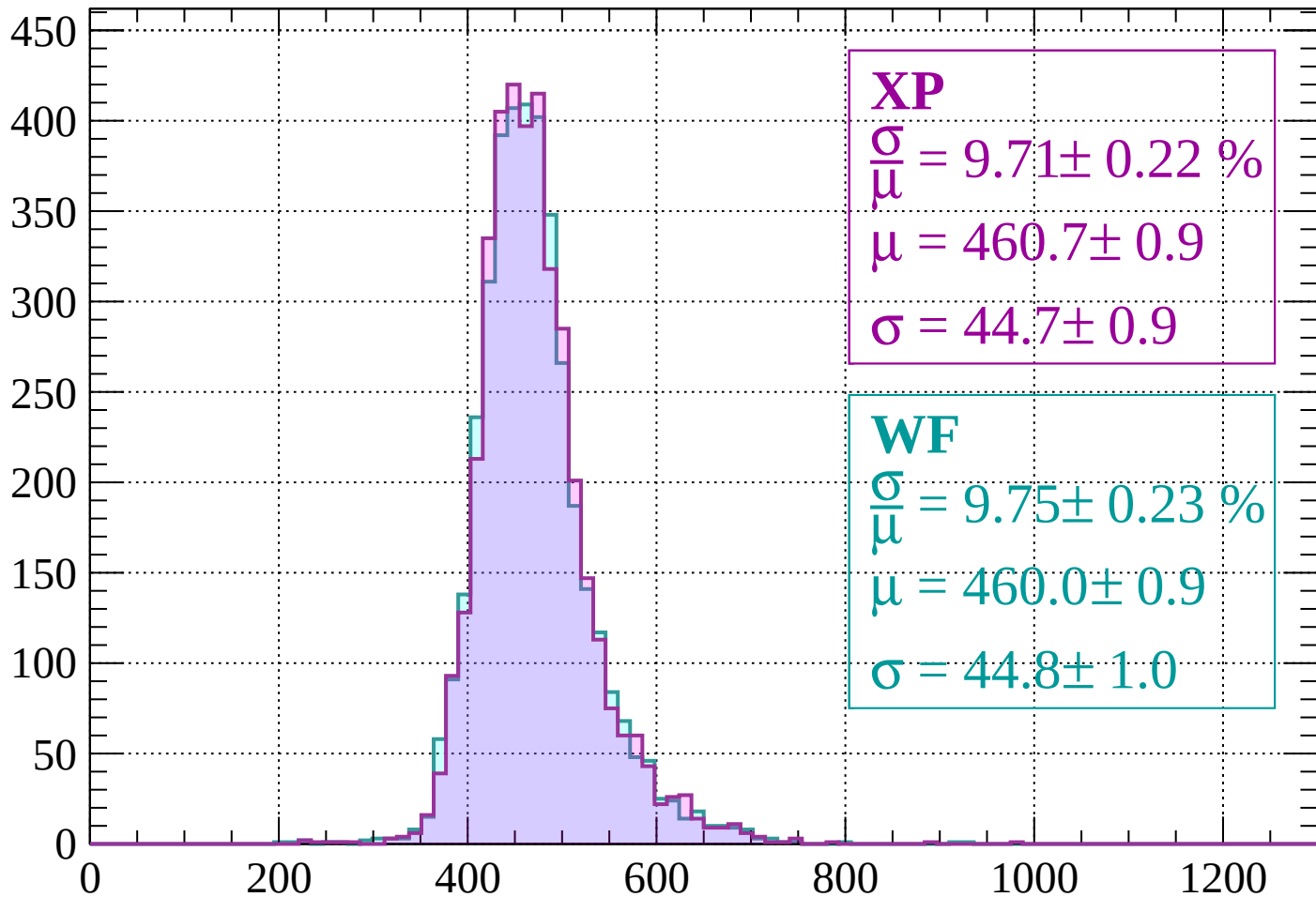
$$\mu = 458.5 \pm 0.8$$

$$\sigma = 43.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 9.71 \pm 0.22 \%$$

$$\mu = 460.7 \pm 0.9$$

$$\sigma = 44.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.23 \%$$

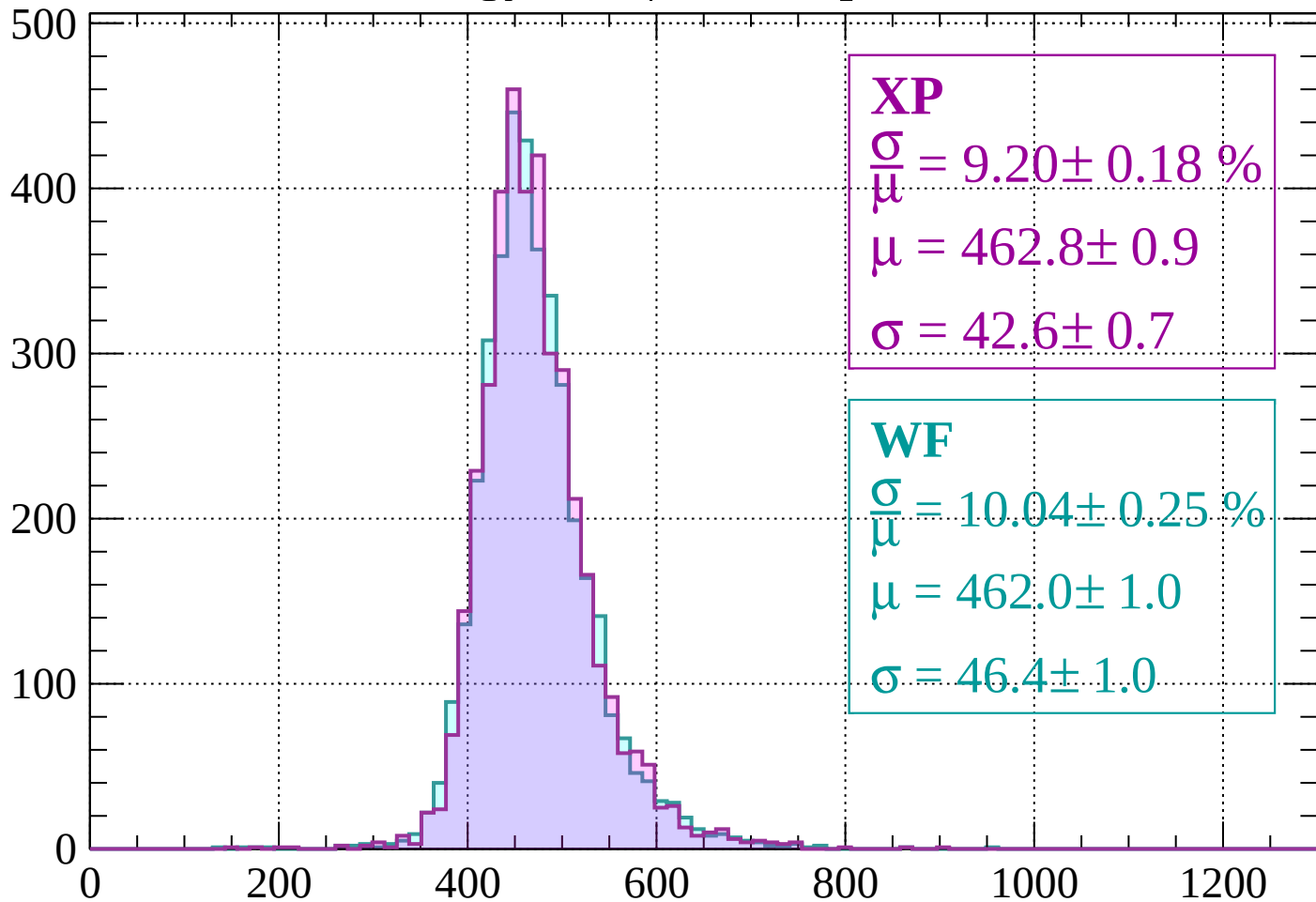
$$\mu = 460.0 \pm 0.9$$

$$\sigma = 44.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.20 \pm 0.18 \%$$

$$\mu = 462.8 \pm 0.9$$

$$\sigma = 42.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.04 \pm 0.25 \%$$

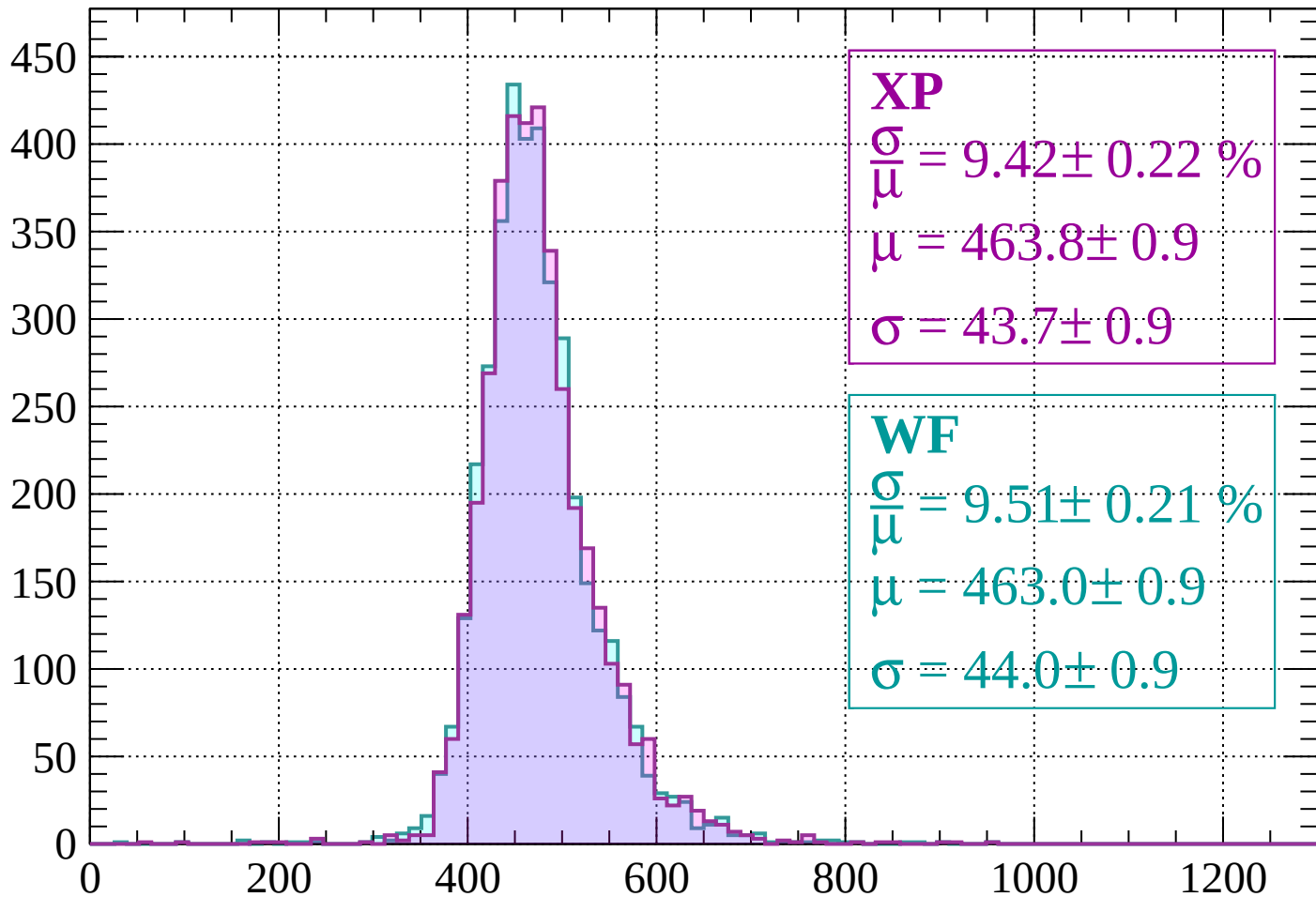
$$\mu = 462.0 \pm 1.0$$

$$\sigma = 46.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 9.42 \pm 0.22 \%$$

$$\mu = 463.8 \pm 0.9$$

$$\sigma = 43.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.51 \pm 0.21 \%$$

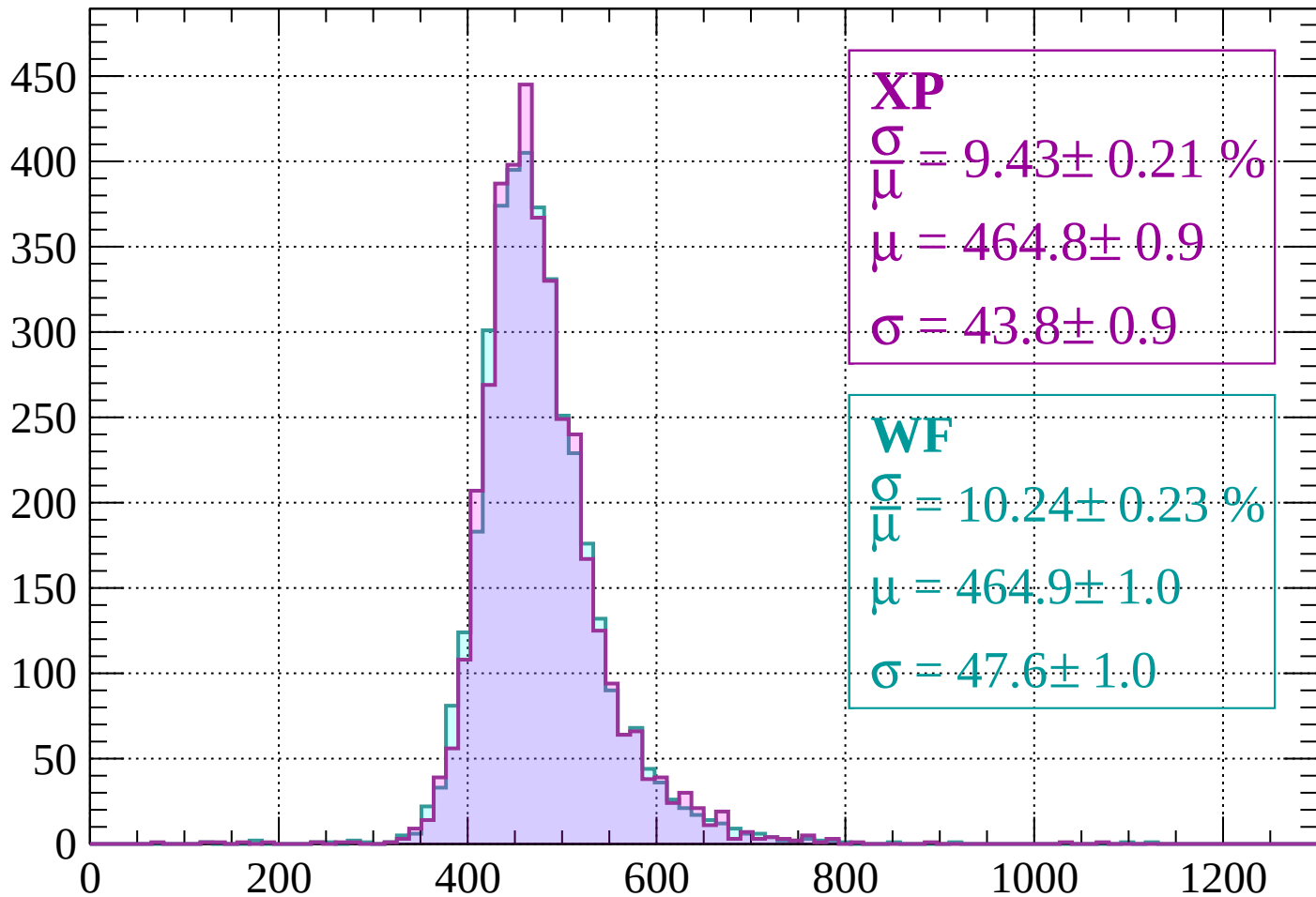
$$\mu = 463.0 \pm 0.9$$

$$\sigma = 44.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

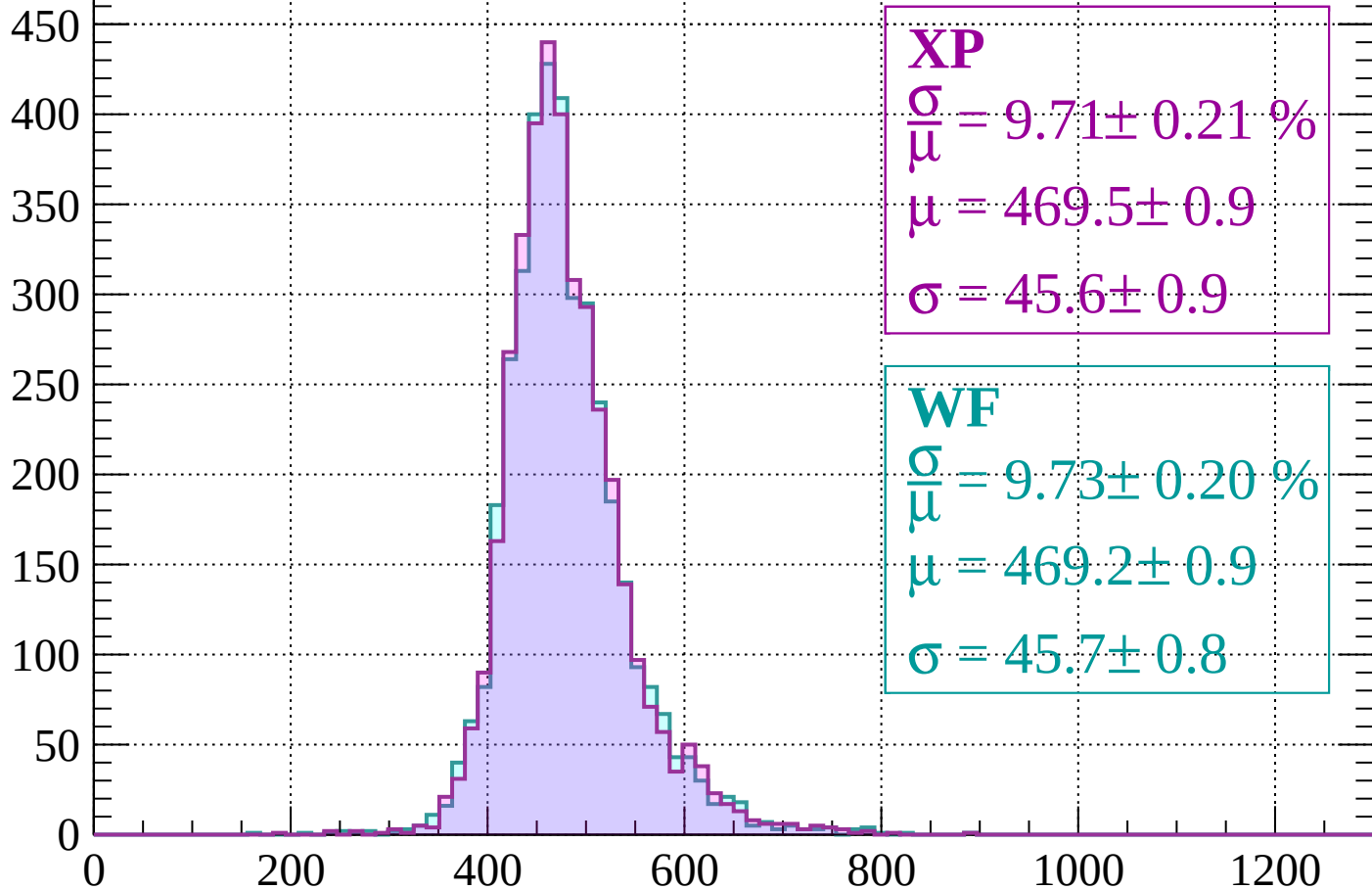
Count



dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.71 \pm 0.21 \%$$

$$\mu = 469.5 \pm 0.9$$

$$\sigma = 45.6 \pm 0.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.73 \pm 0.20 \%$$

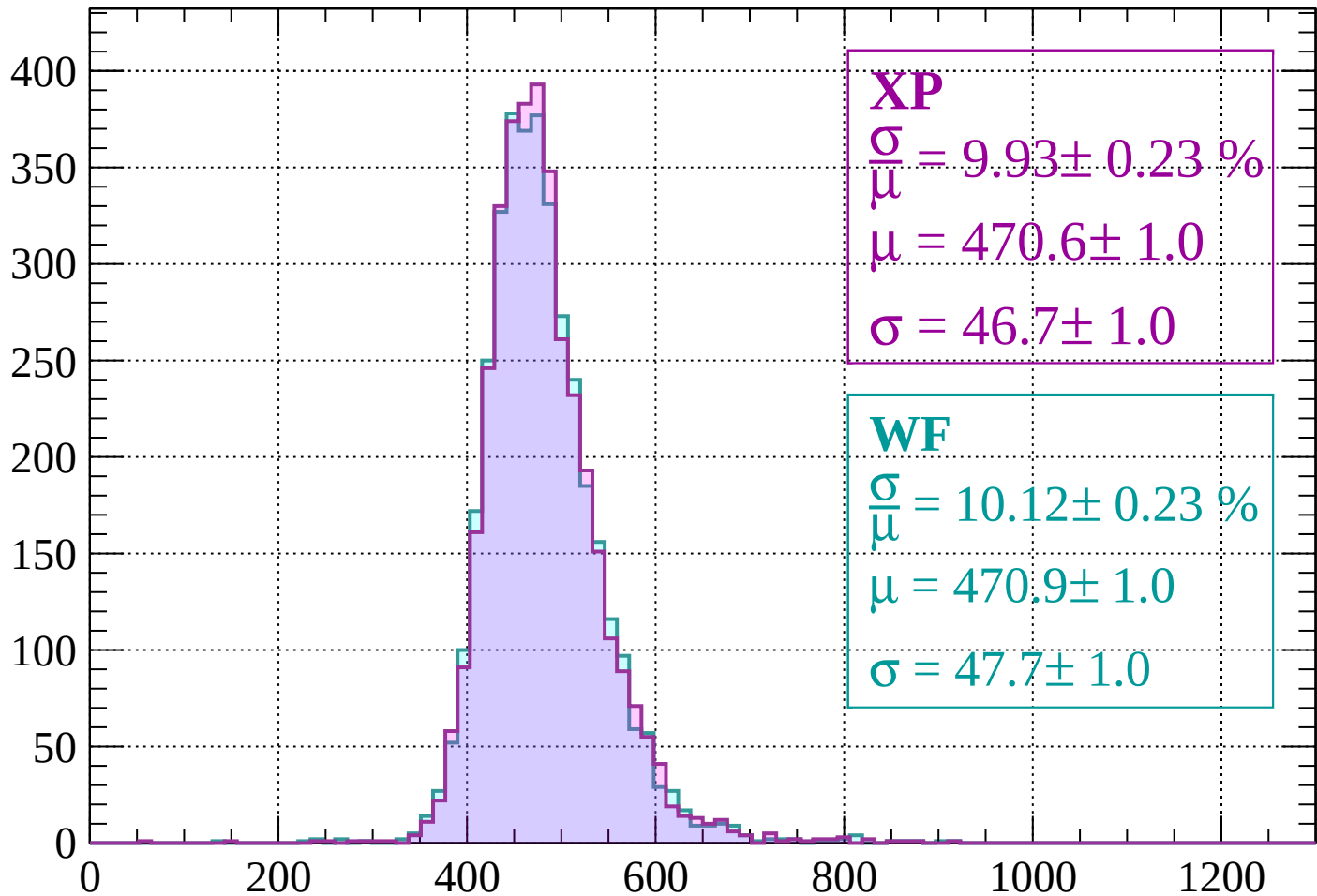
$$\mu = 469.2 \pm 0.9$$

$$\sigma = 45.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 9.93 \pm 0.23 \%$$

$$\mu = 470.6 \pm 1.0$$

$$\sigma = 46.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.12 \pm 0.23 \%$$

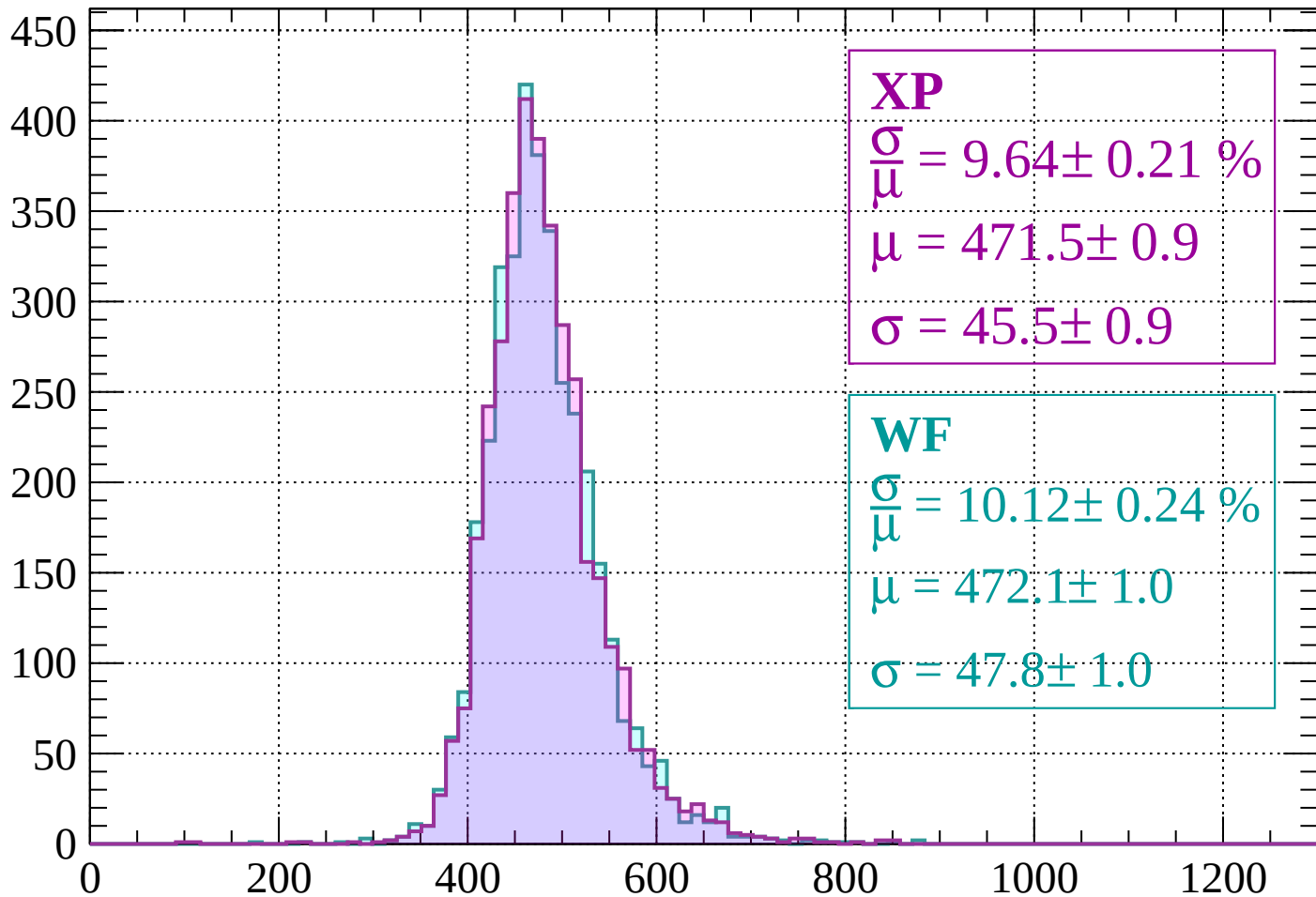
$$\mu = 470.9 \pm 1.0$$

$$\sigma = 47.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 9.64 \pm 0.21 \%$$

$$\mu = 471.5 \pm 0.9$$

$$\sigma = 45.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.12 \pm 0.24 \%$$

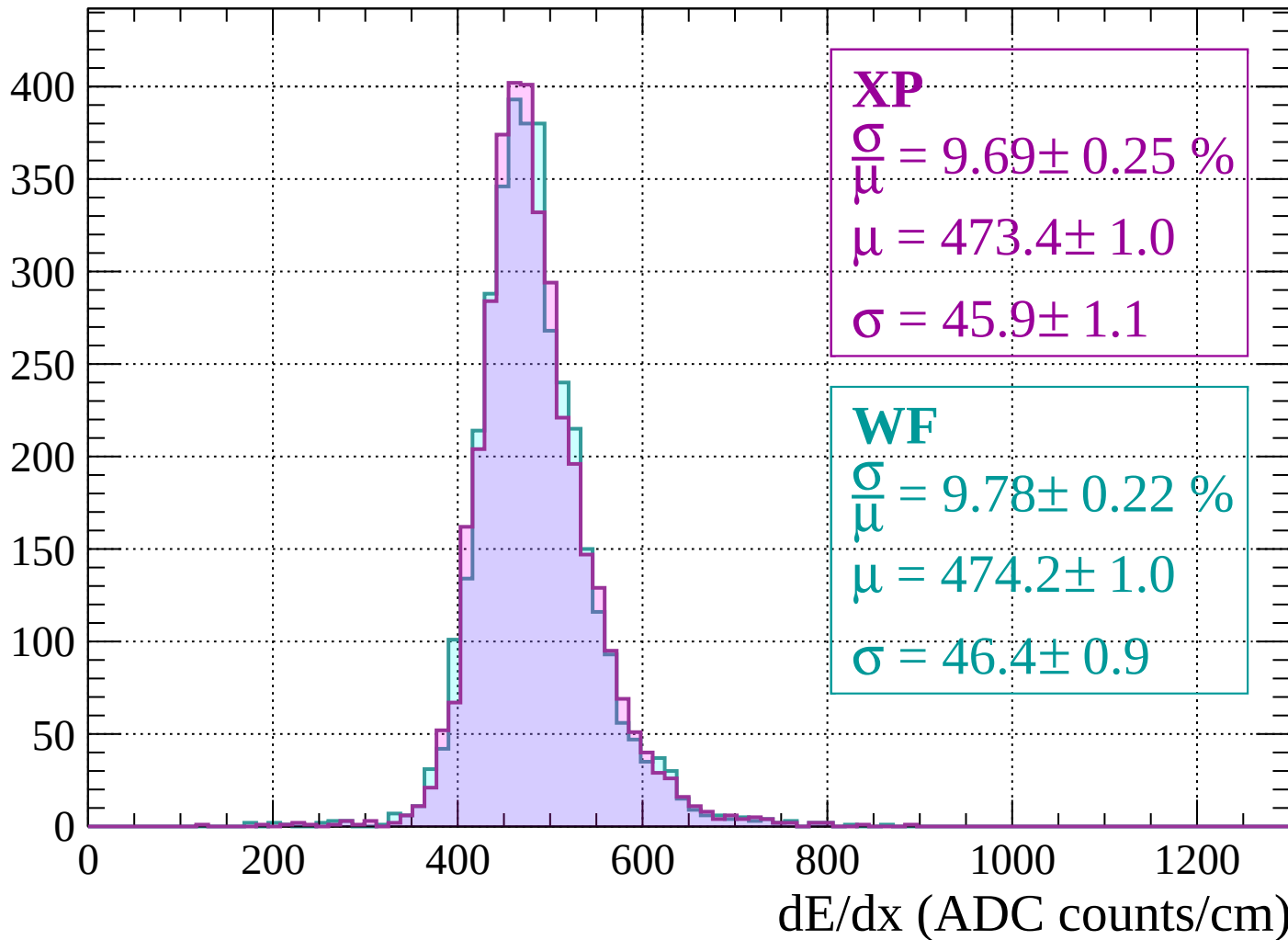
$$\mu = 472.1 \pm 1.0$$

$$\sigma = 47.8 \pm 1.0$$

dE/dx (ADC counts/cm)

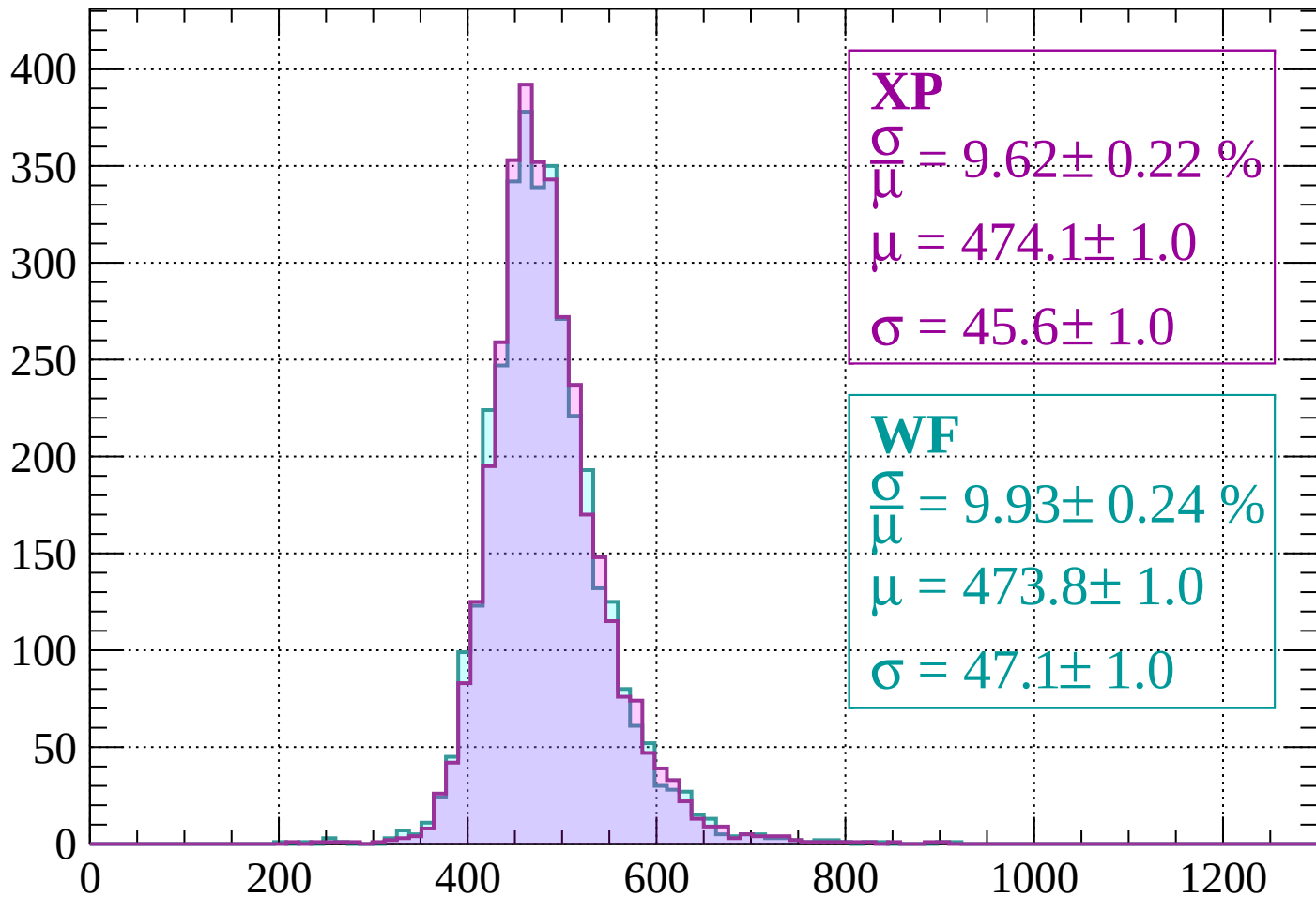
Energy loss | $1760 < p < 1800$

Count



Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 9.62 \pm 0.22 \%$$

$$\mu = 474.1 \pm 1.0$$

$$\sigma = 45.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.93 \pm 0.24 \%$$

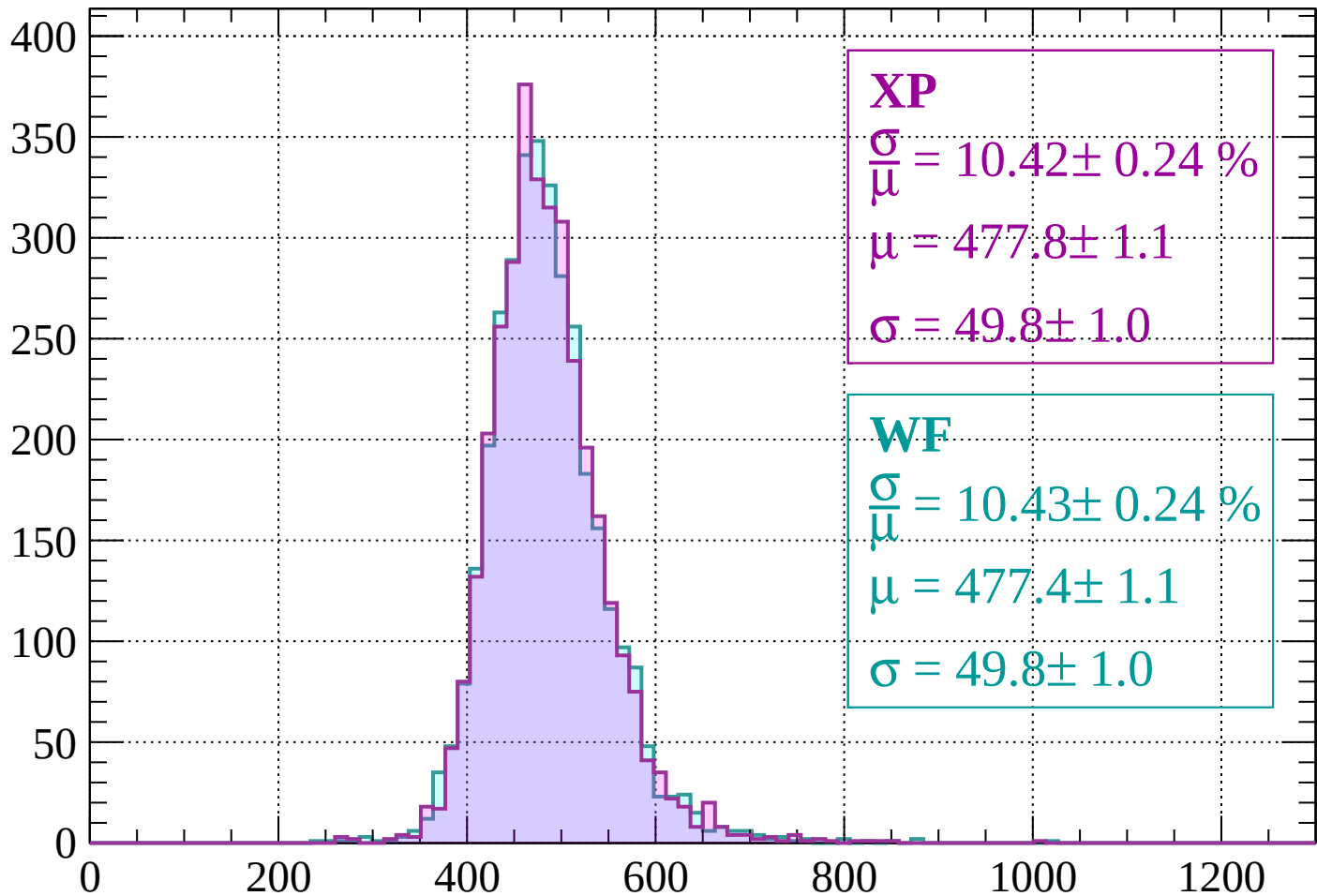
$$\mu = 473.8 \pm 1.0$$

$$\sigma = 47.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 10.42 \pm 0.24 \%$$

$$\mu = 477.8 \pm 1.1$$

$$\sigma = 49.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.43 \pm 0.24 \%$$

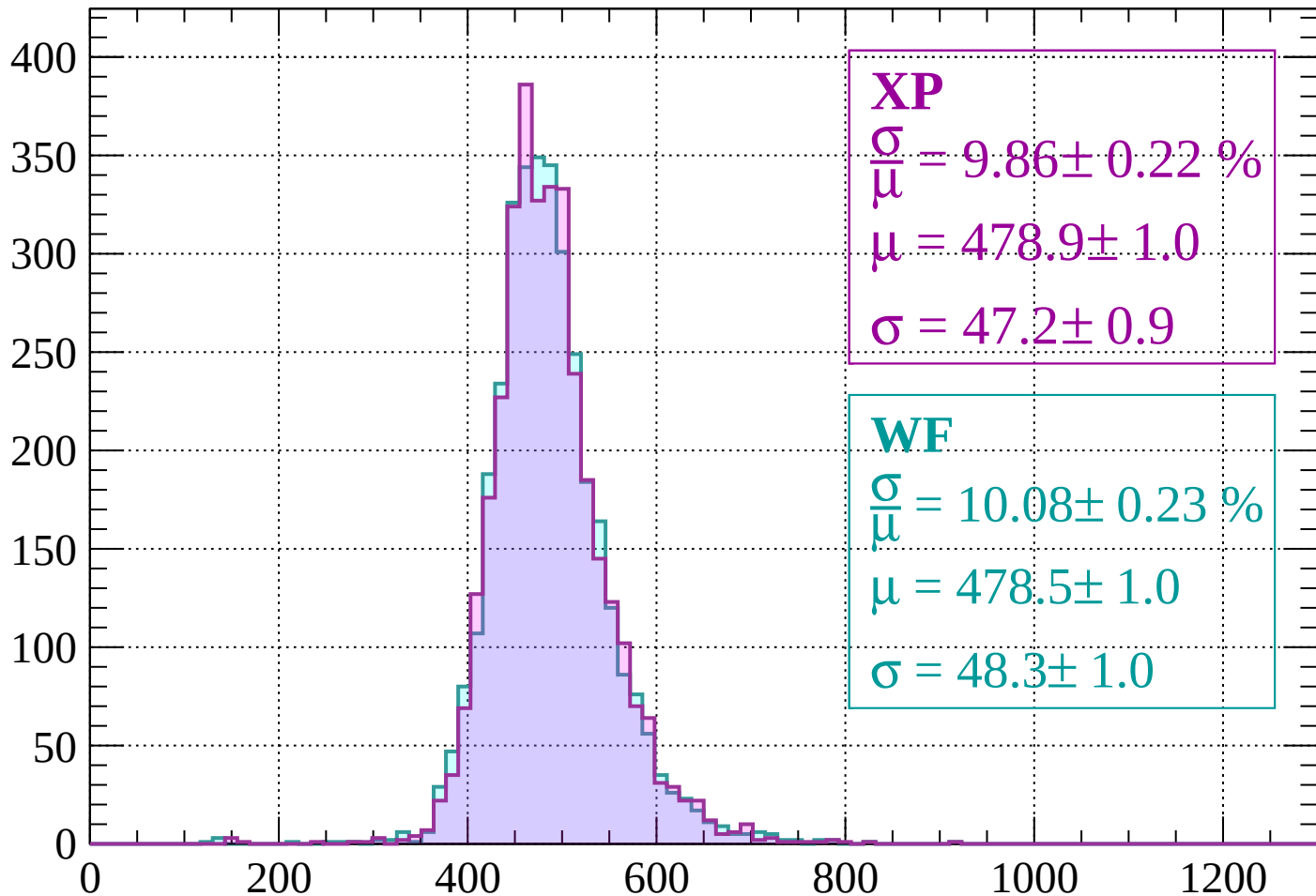
$$\mu = 477.4 \pm 1.1$$

$$\sigma = 49.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 9.86 \pm 0.22 \%$$

$$\mu = 478.9 \pm 1.0$$

$$\sigma = 47.2 \pm 0.9$$

WF

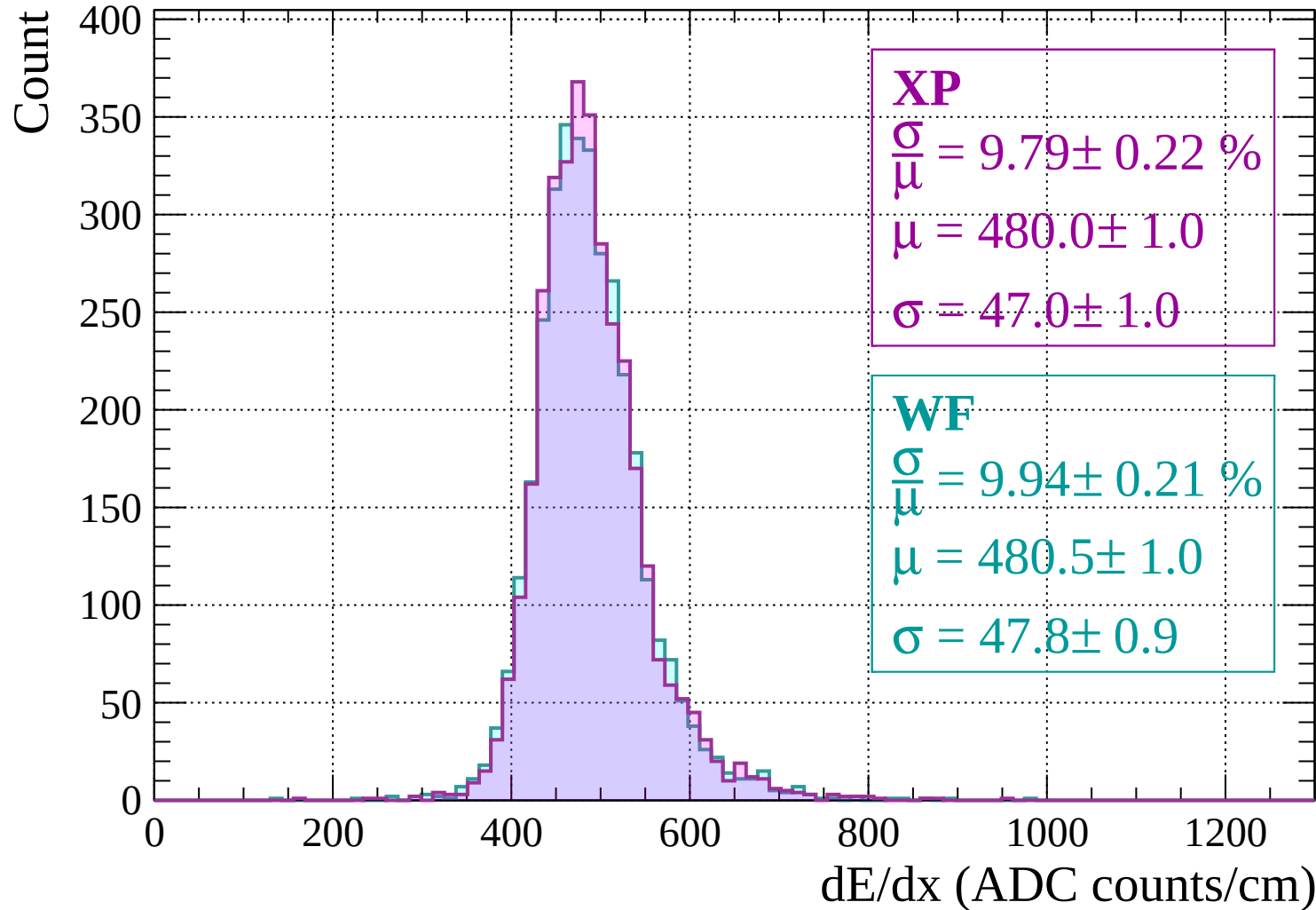
$$\frac{\sigma}{\mu} = 10.08 \pm 0.23 \%$$

$$\mu = 478.5 \pm 1.0$$

$$\sigma = 48.3 \pm 1.0$$

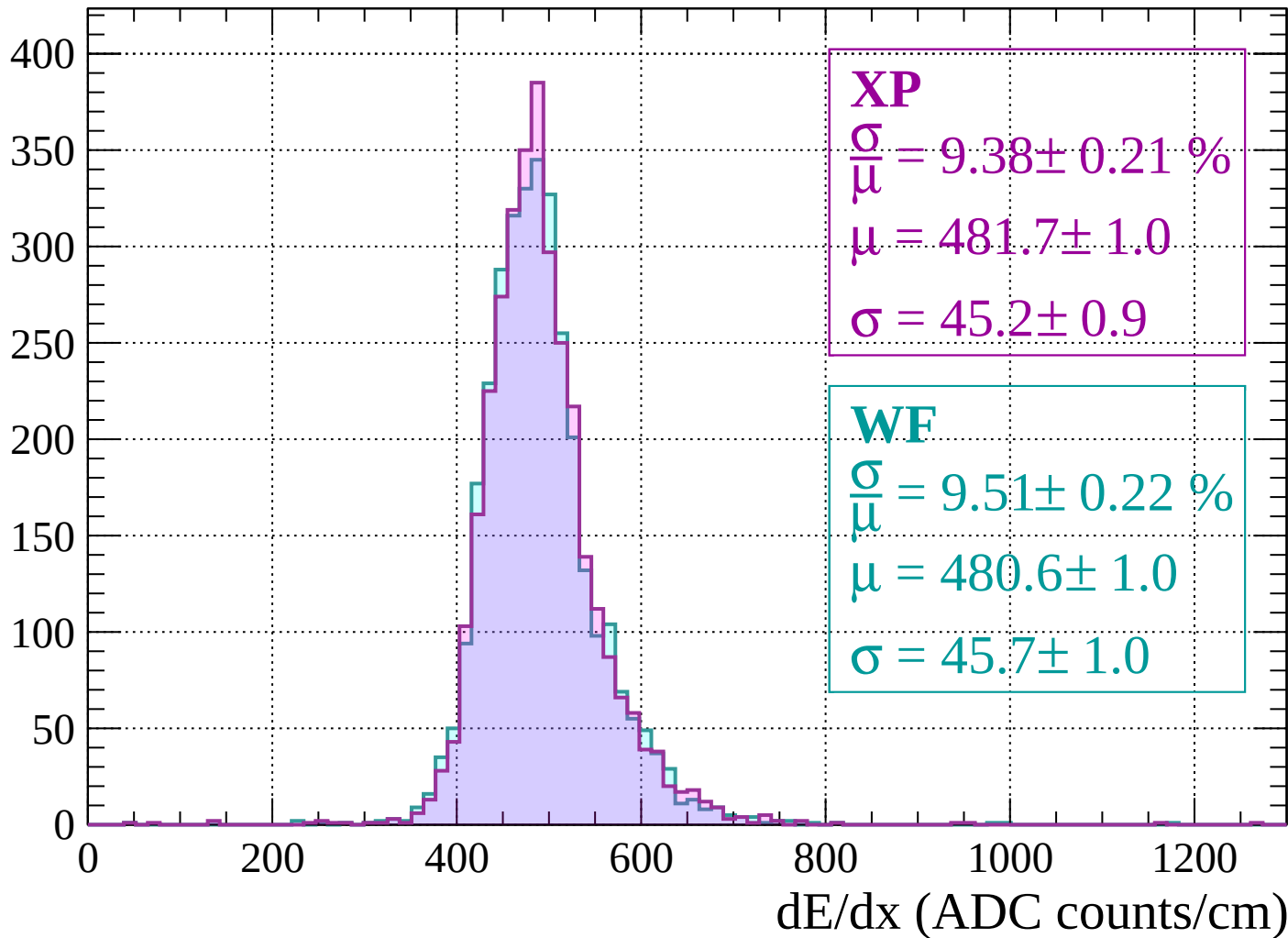
dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$



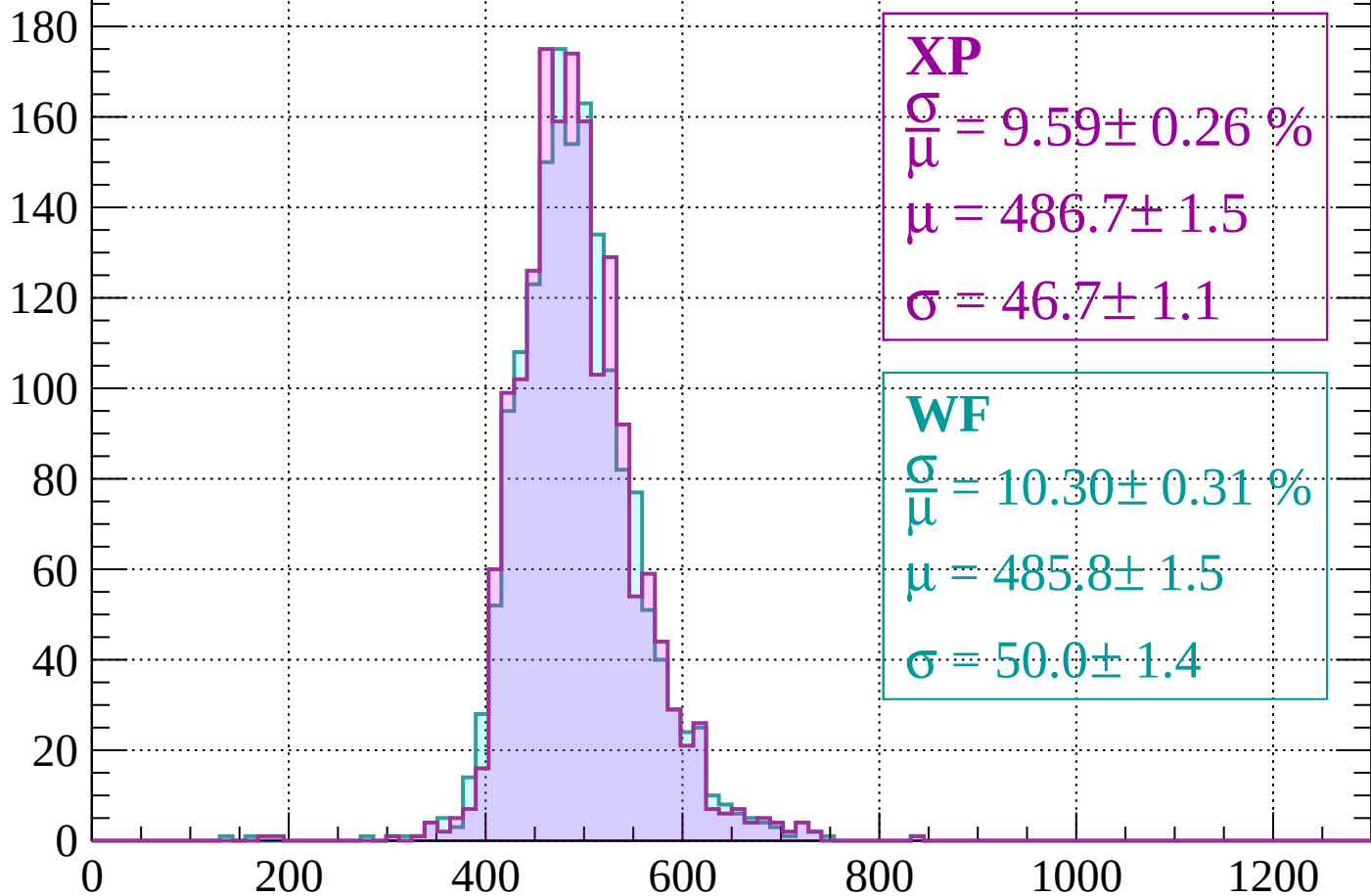
Energy loss | $1960 < p < 2000$

Count



Energy loss | $2000 < p < 2040$

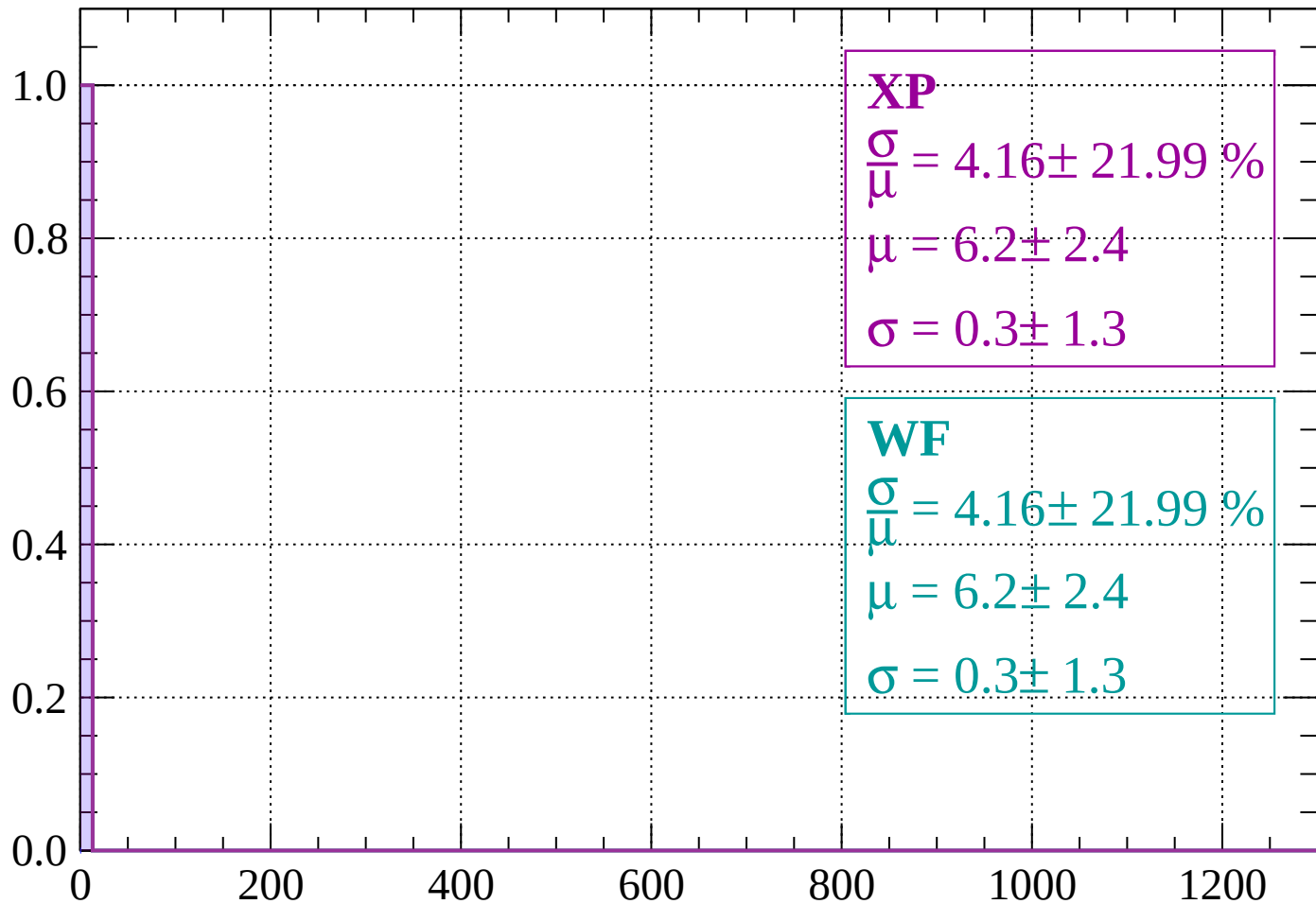
Count



dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

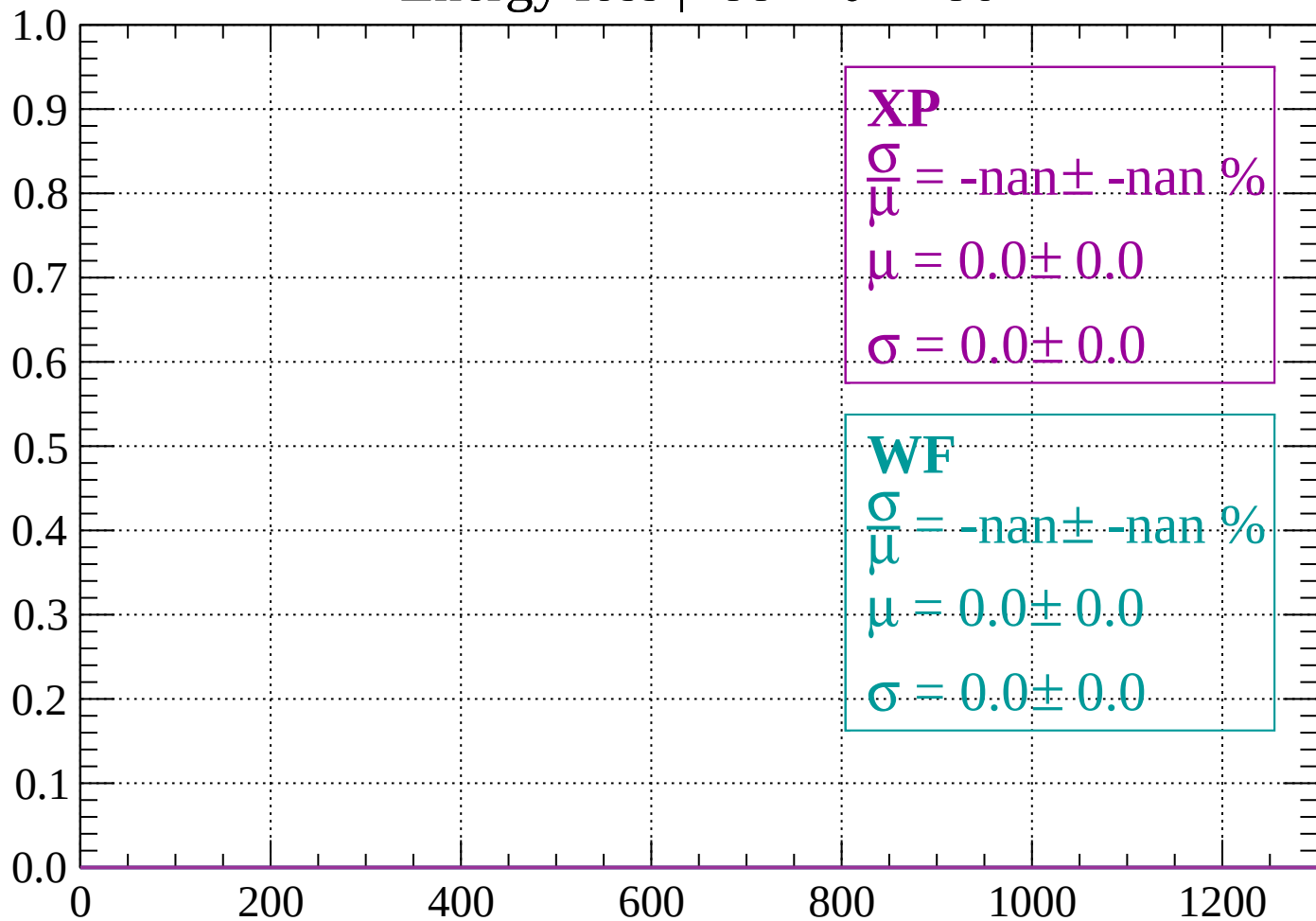
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

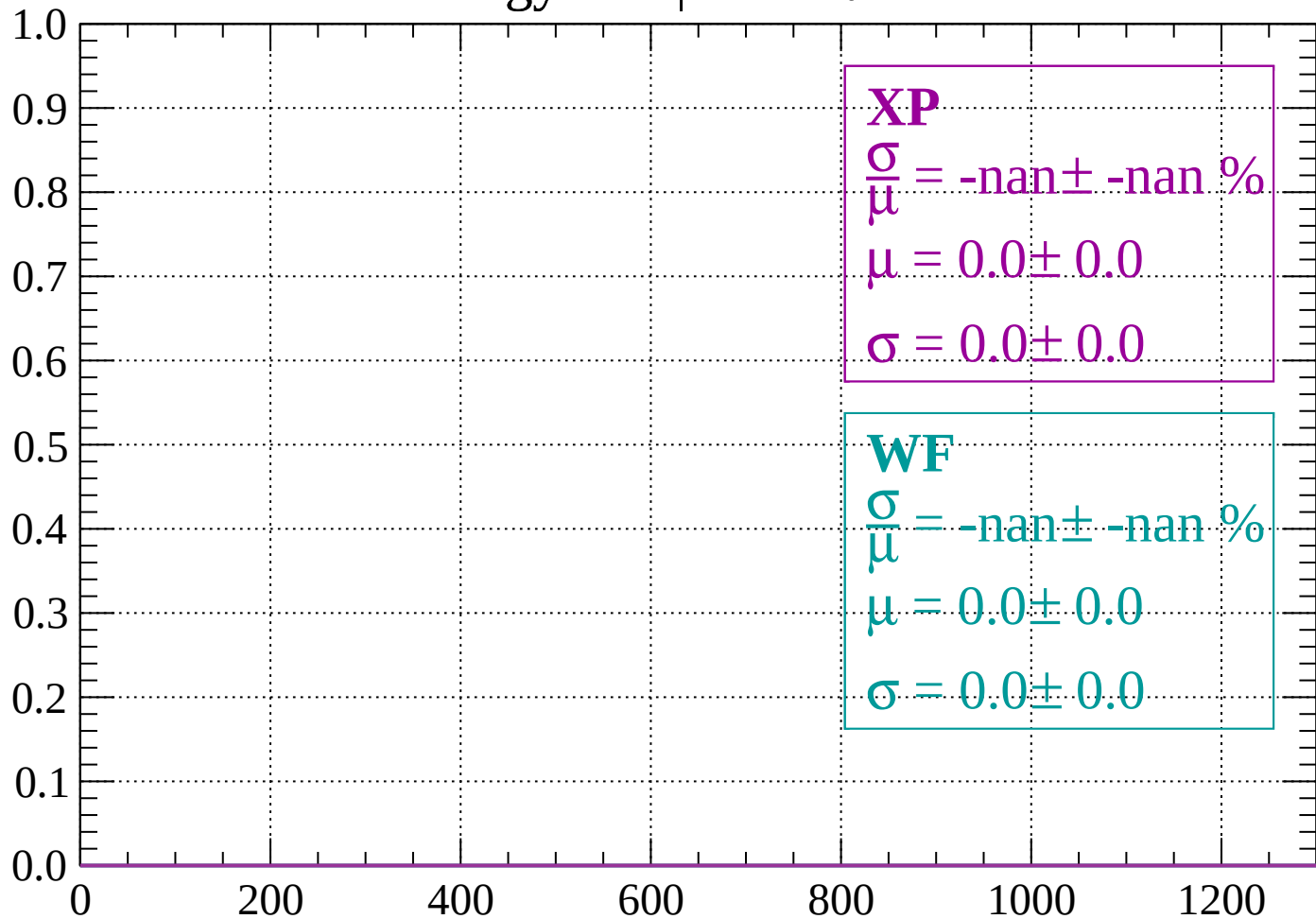
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

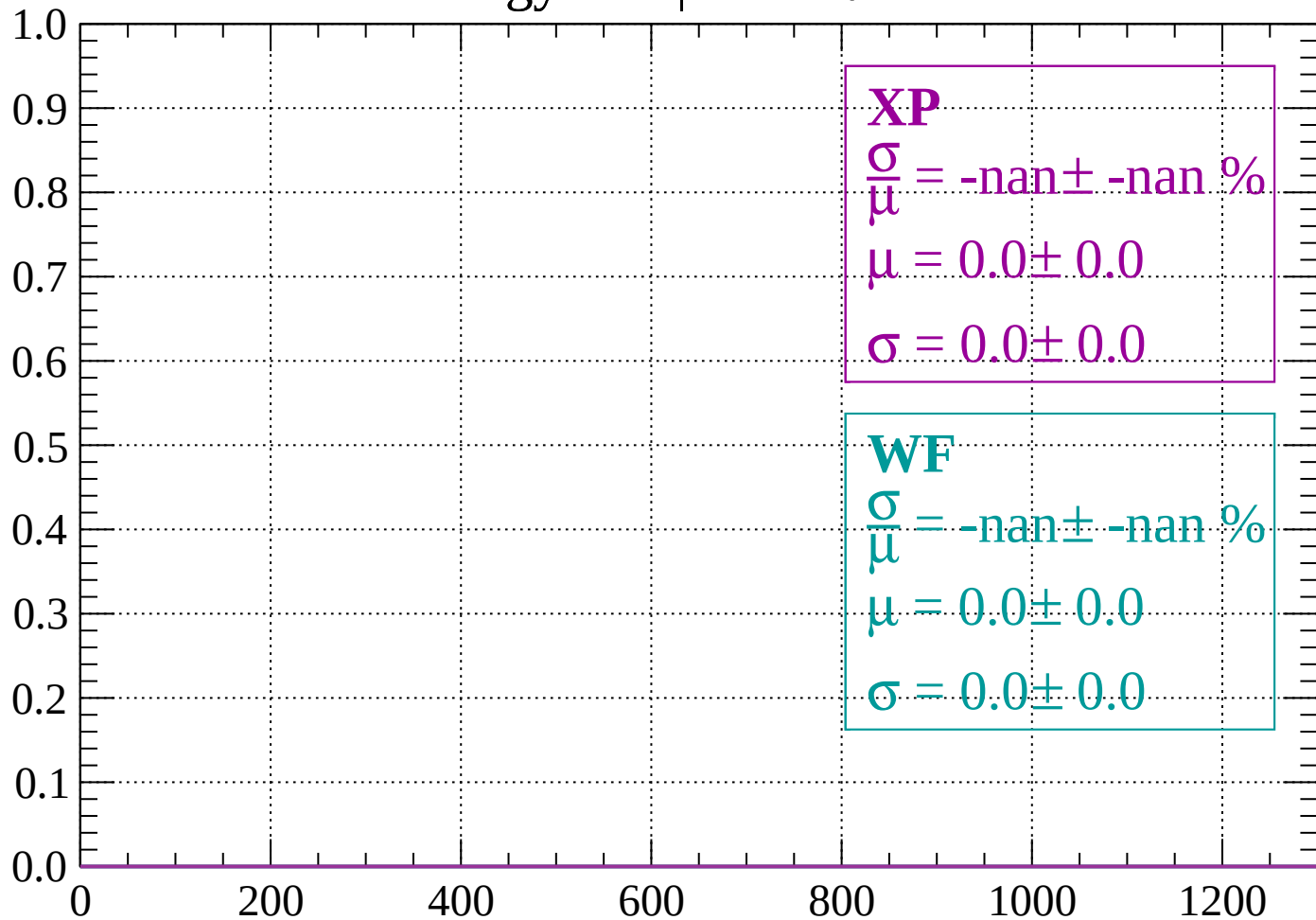
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

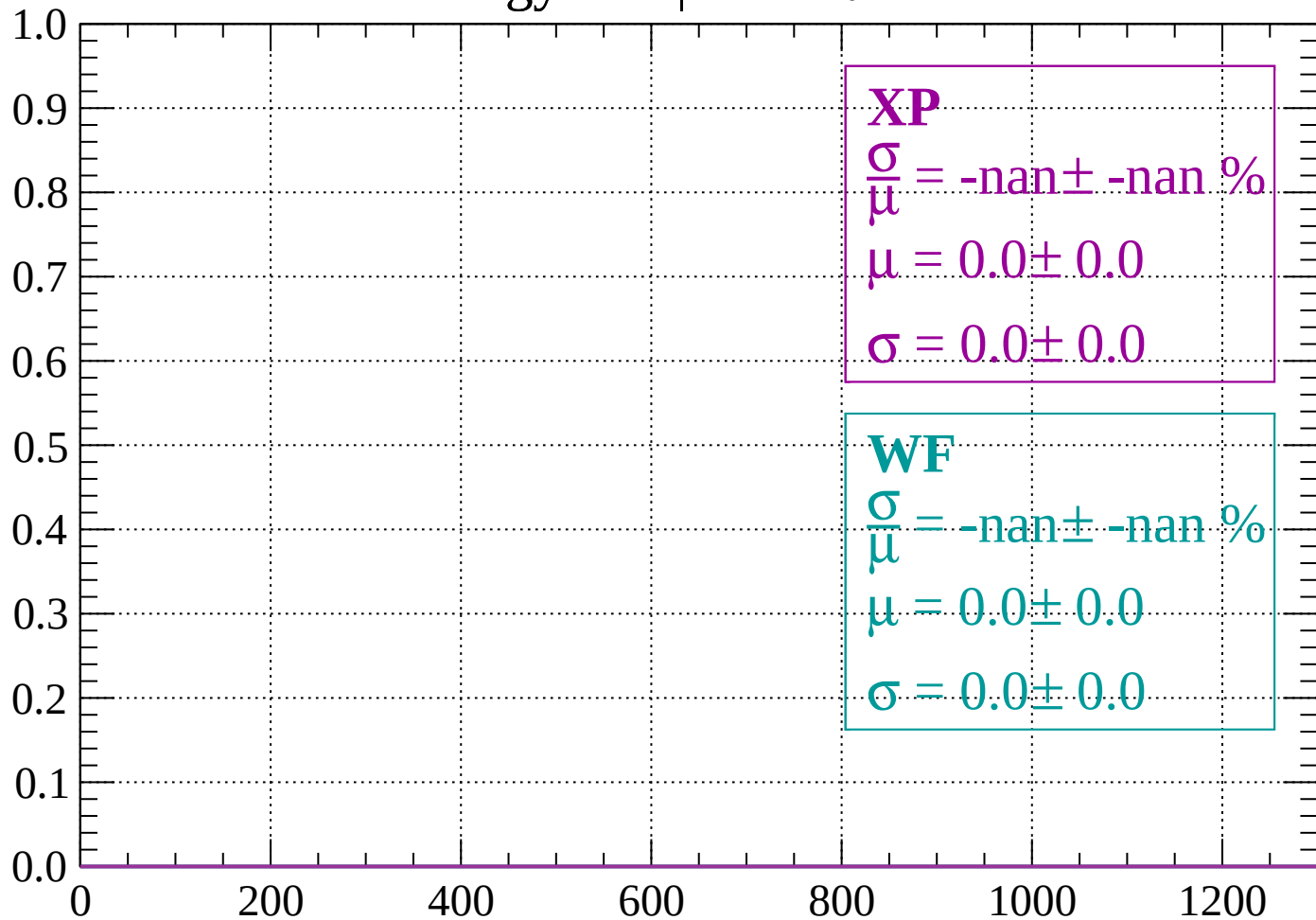
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

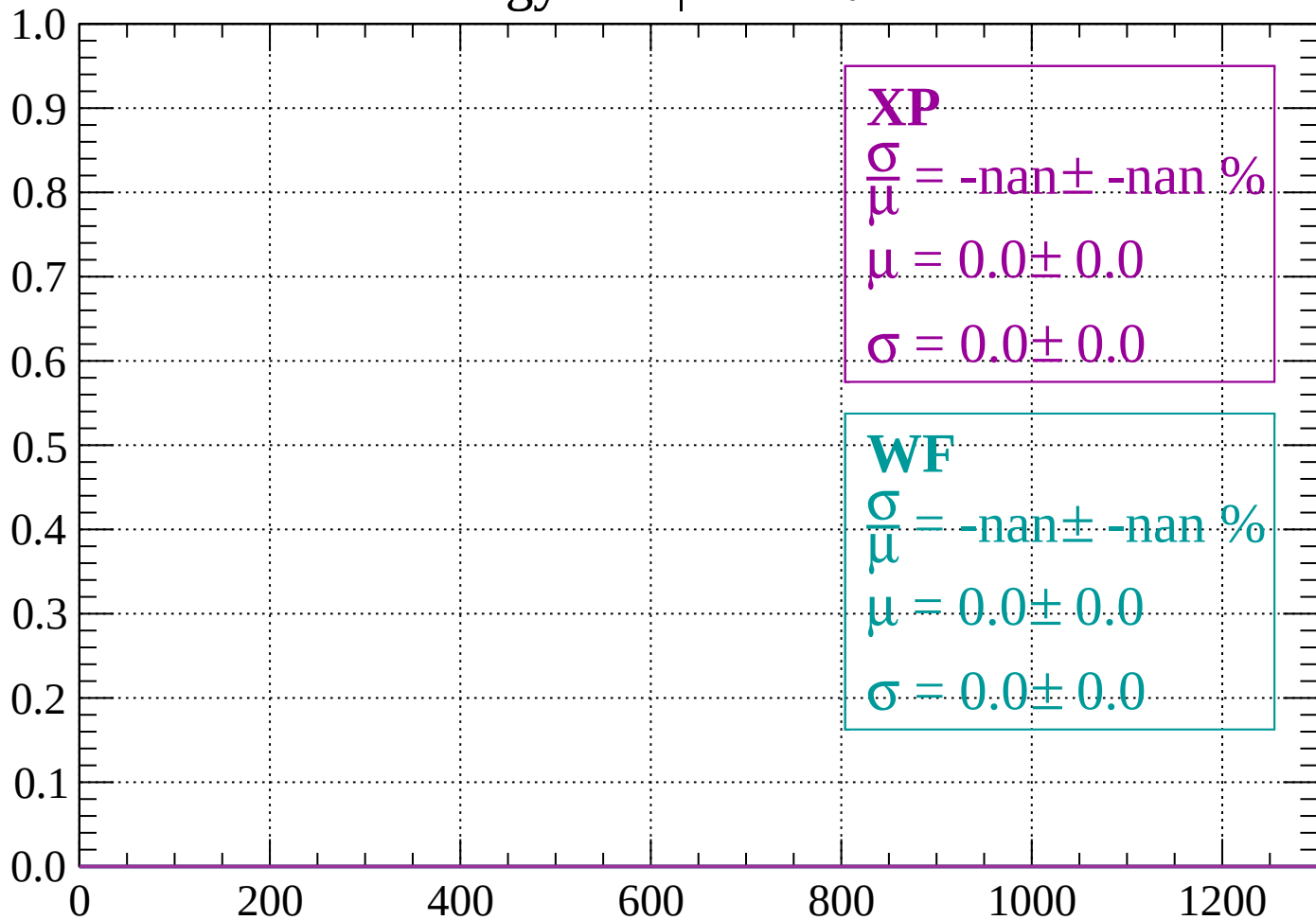
Count



dE/dx (ADC counts/cm)

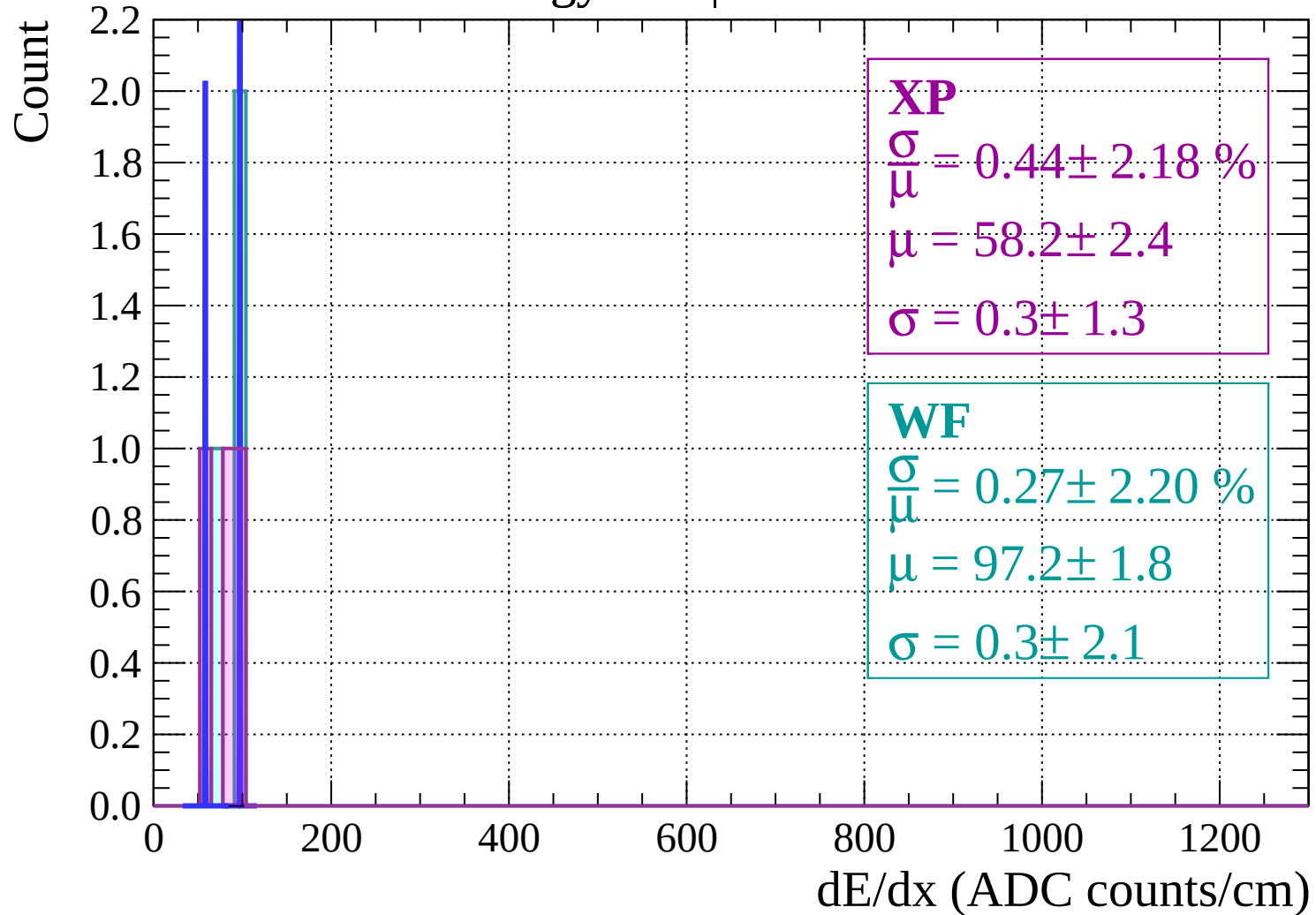
Energy loss | $-80 < \theta < -78$

Count



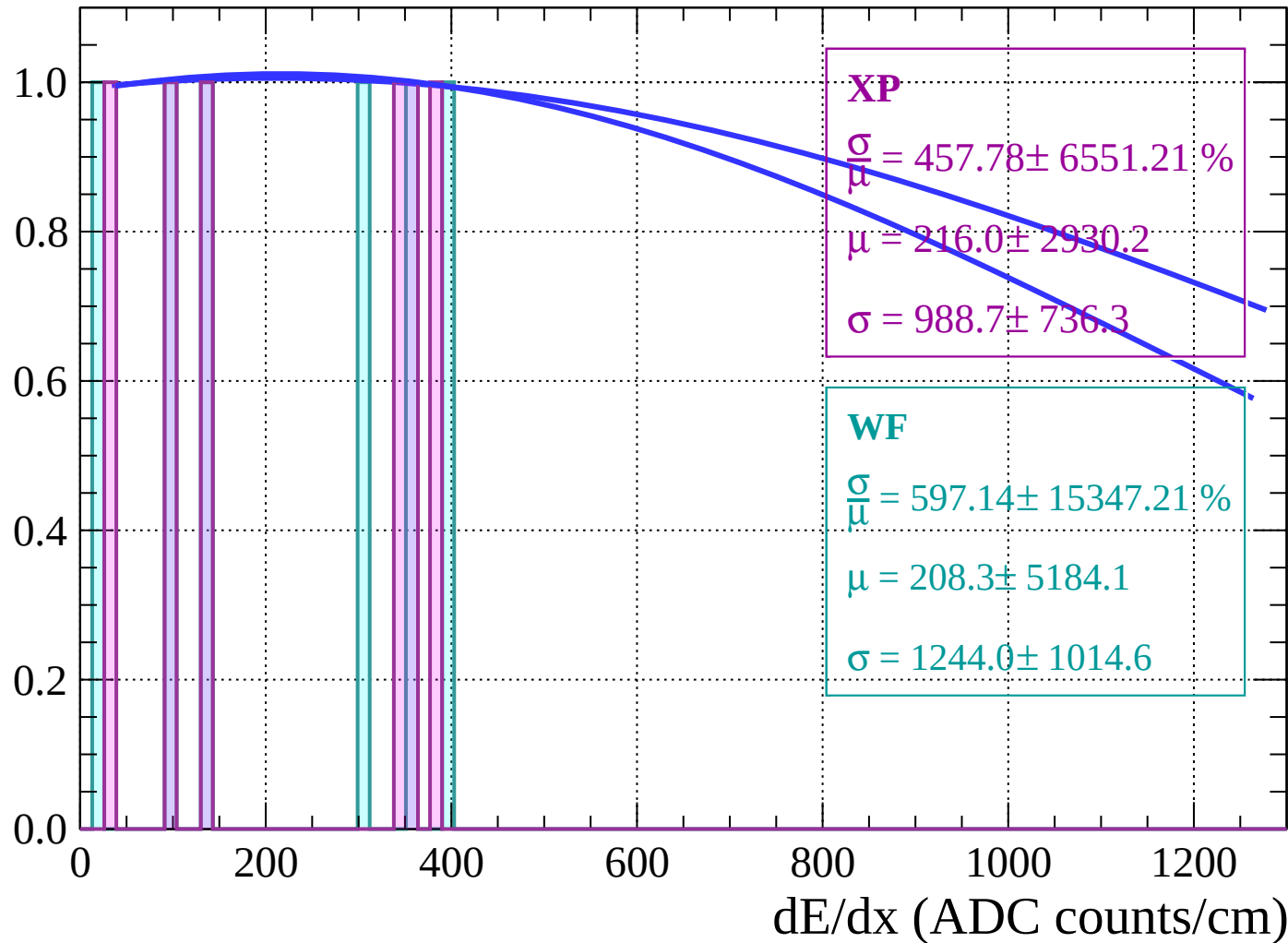
dE/dx (ADC counts/cm)

Energy loss $|-78 < \theta < -76$



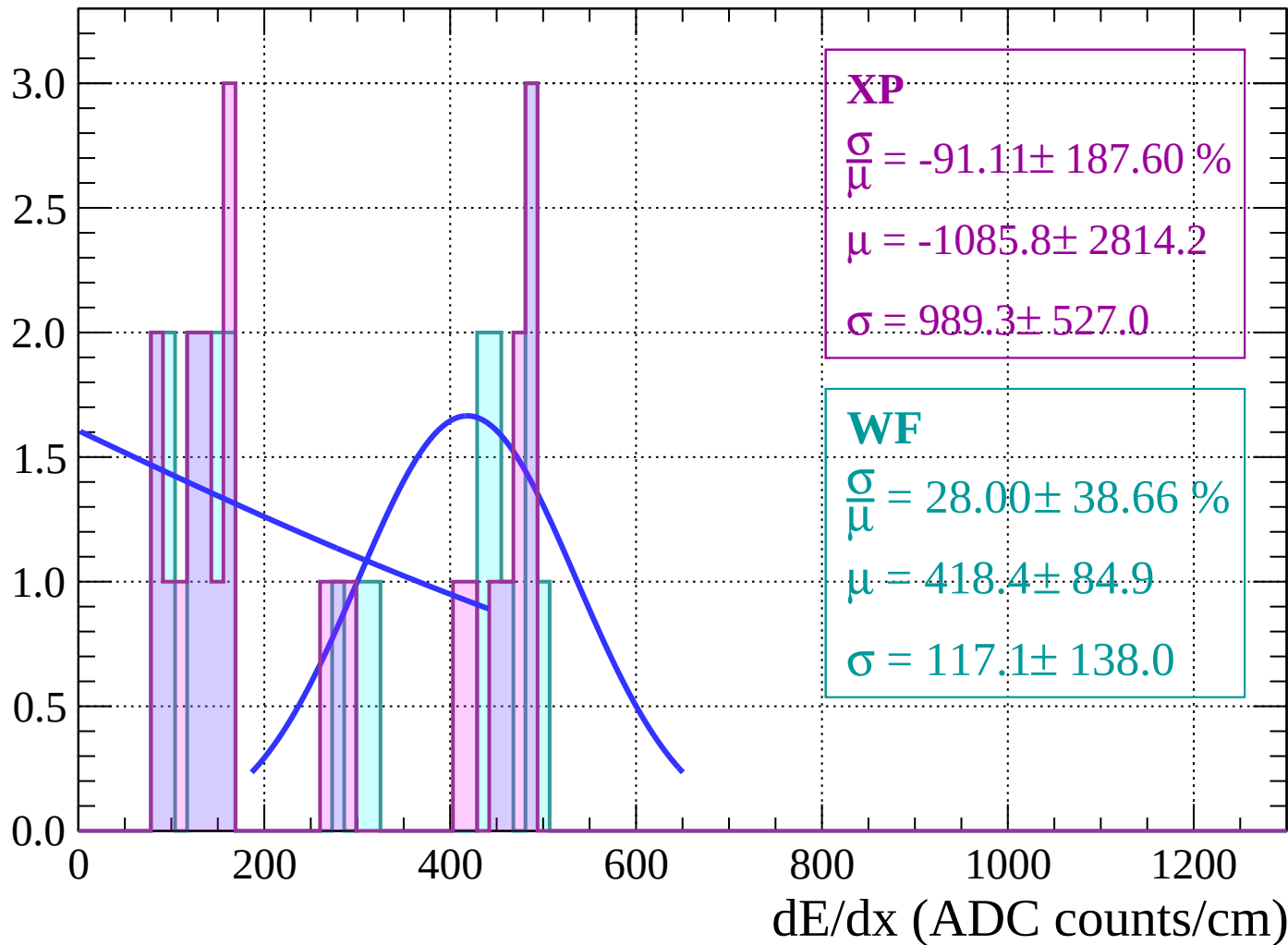
Energy loss | $-76 < \theta < -74$

Count



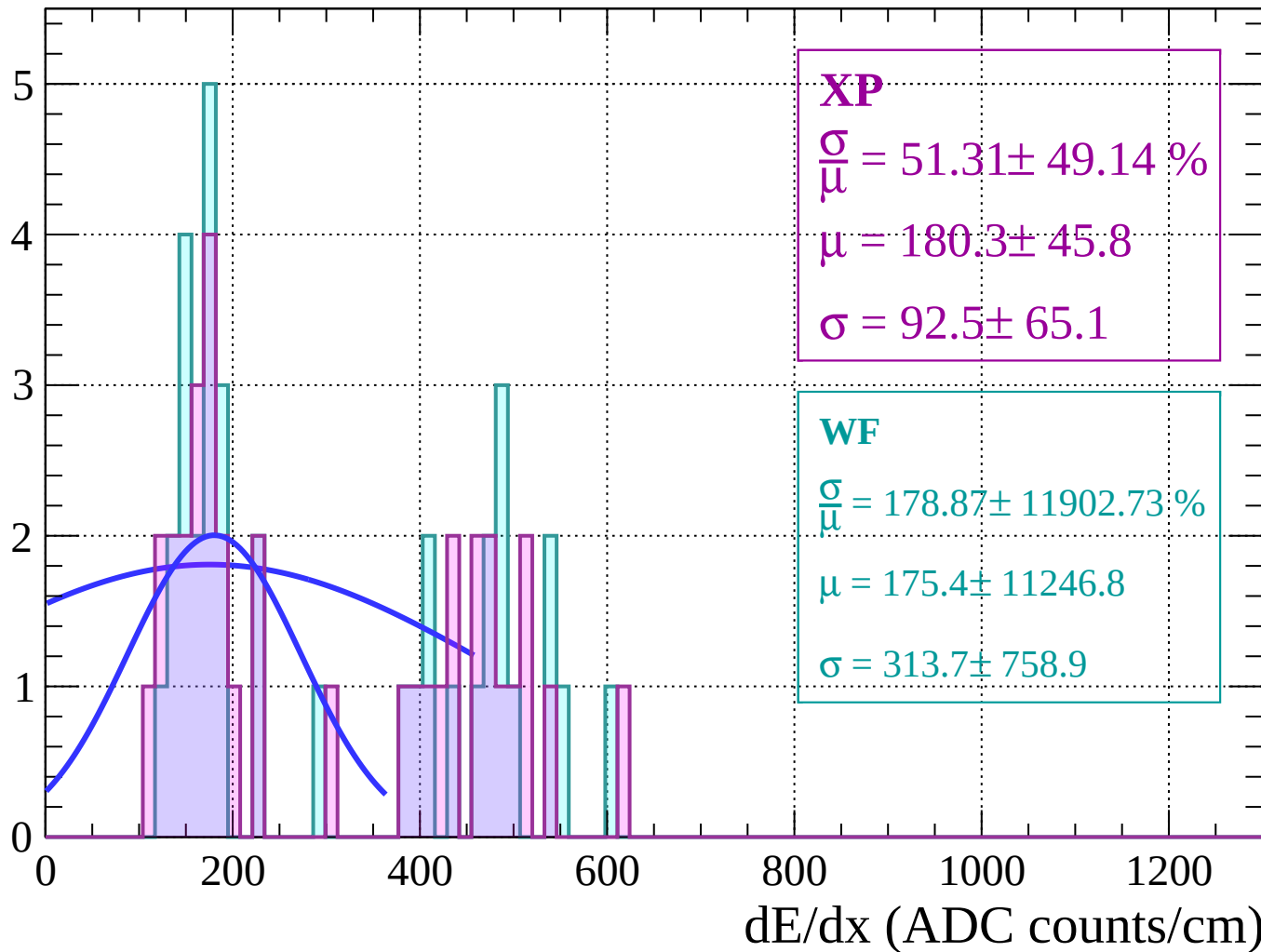
Energy loss | $-74 < \theta < -72$

Count



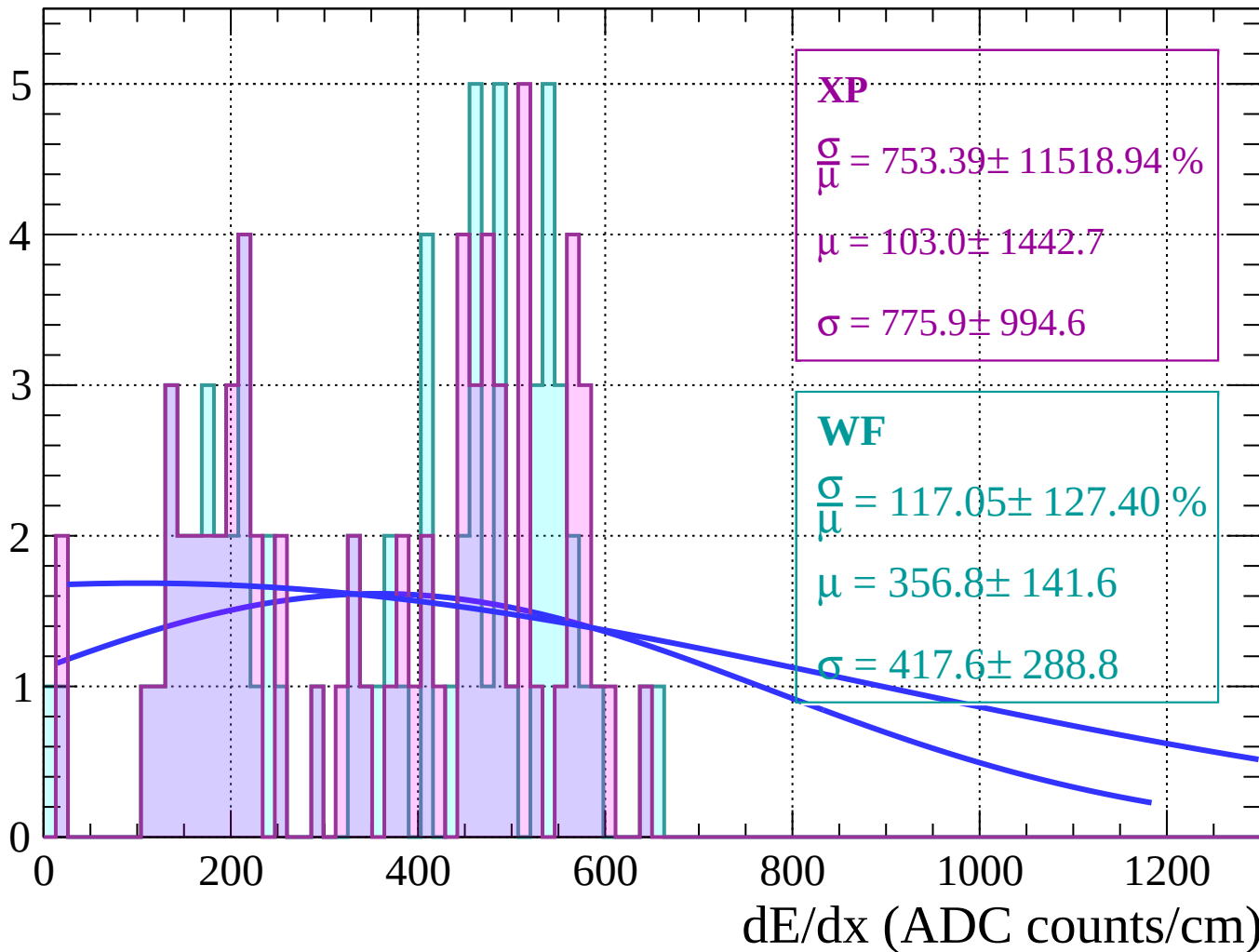
Energy loss | $-72 < \theta < -70$

Count



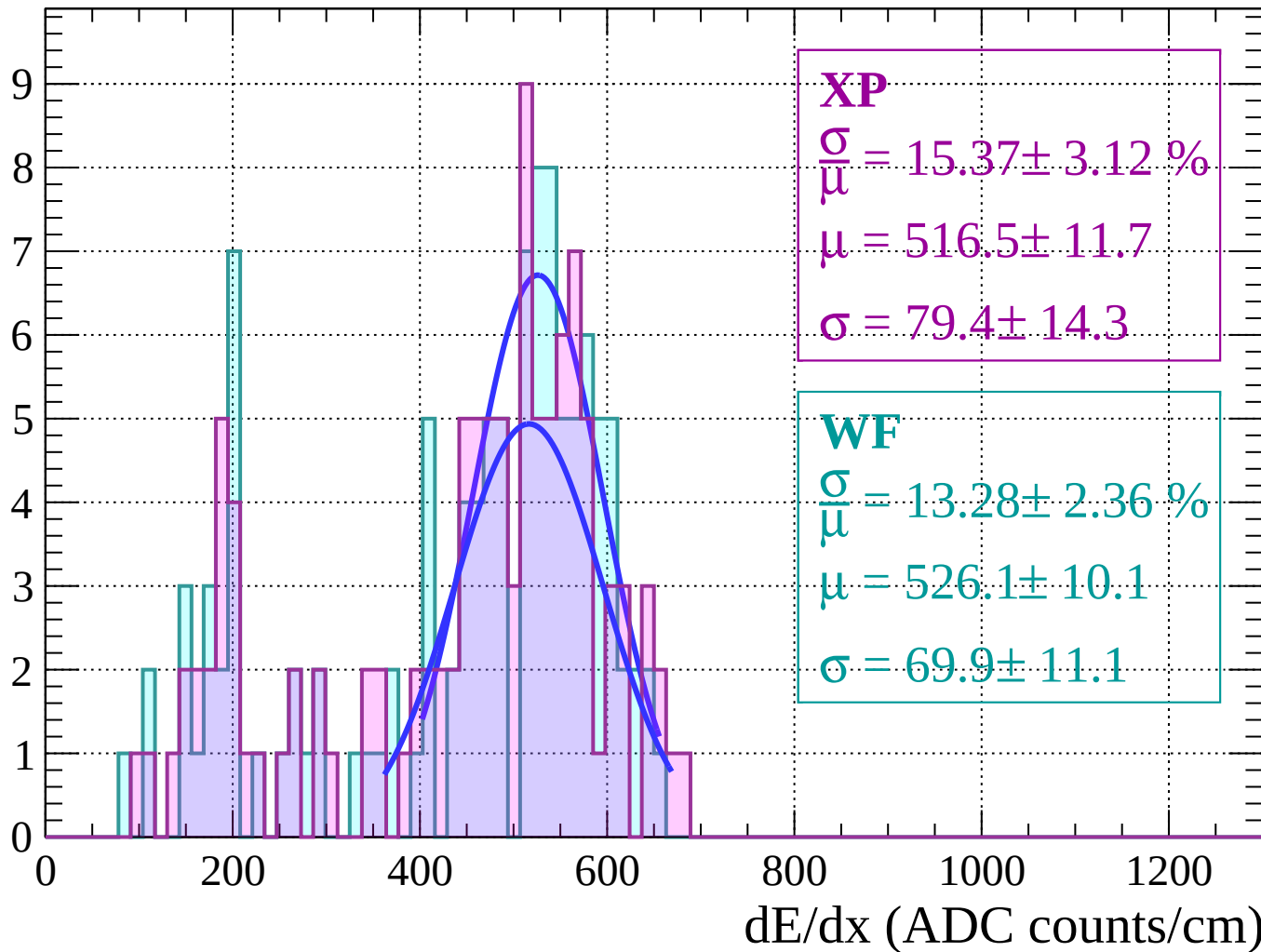
Energy loss | $-70 < \theta < -68$

Count



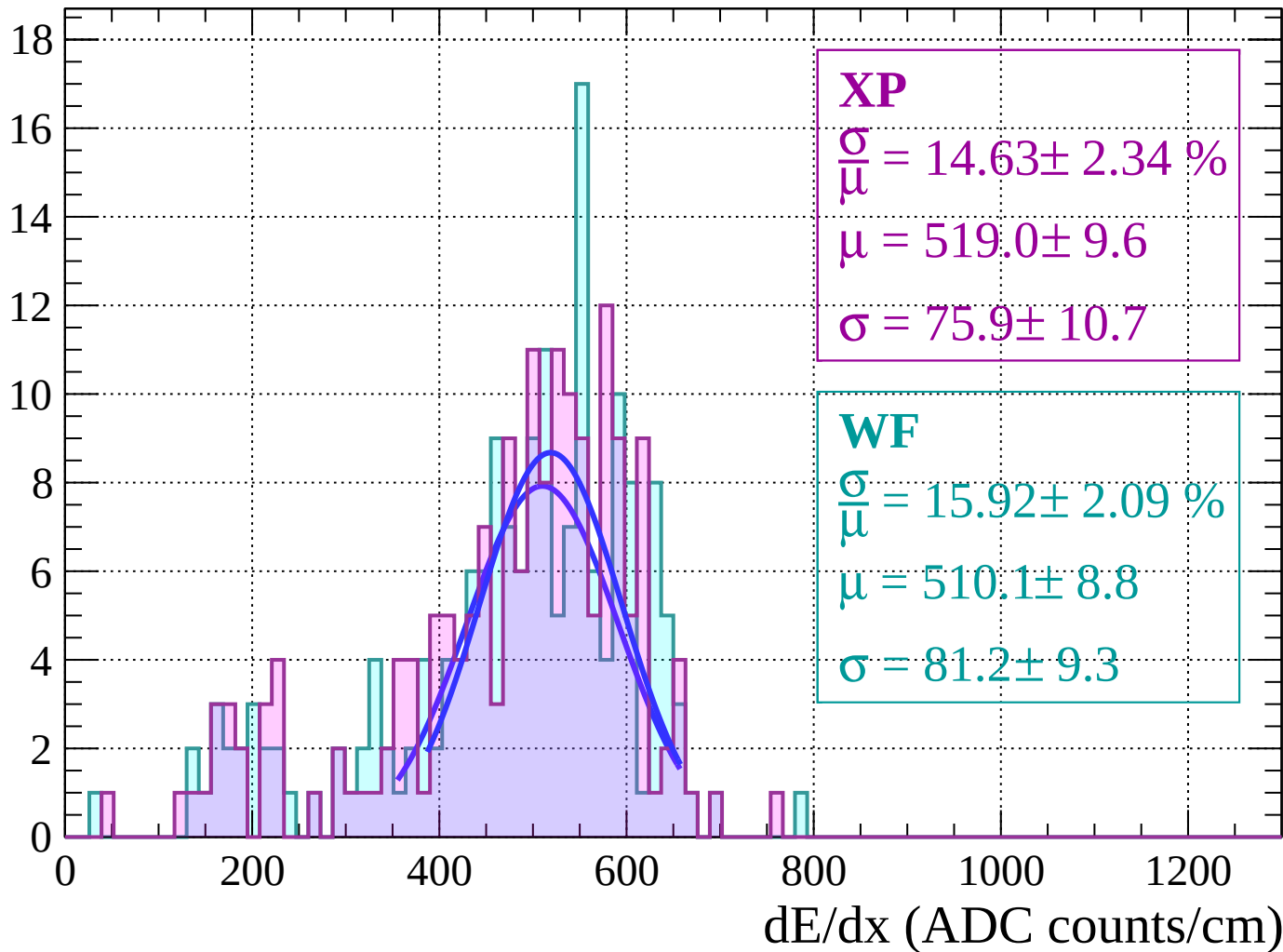
Energy loss | $-68 < \theta < -66$

Count



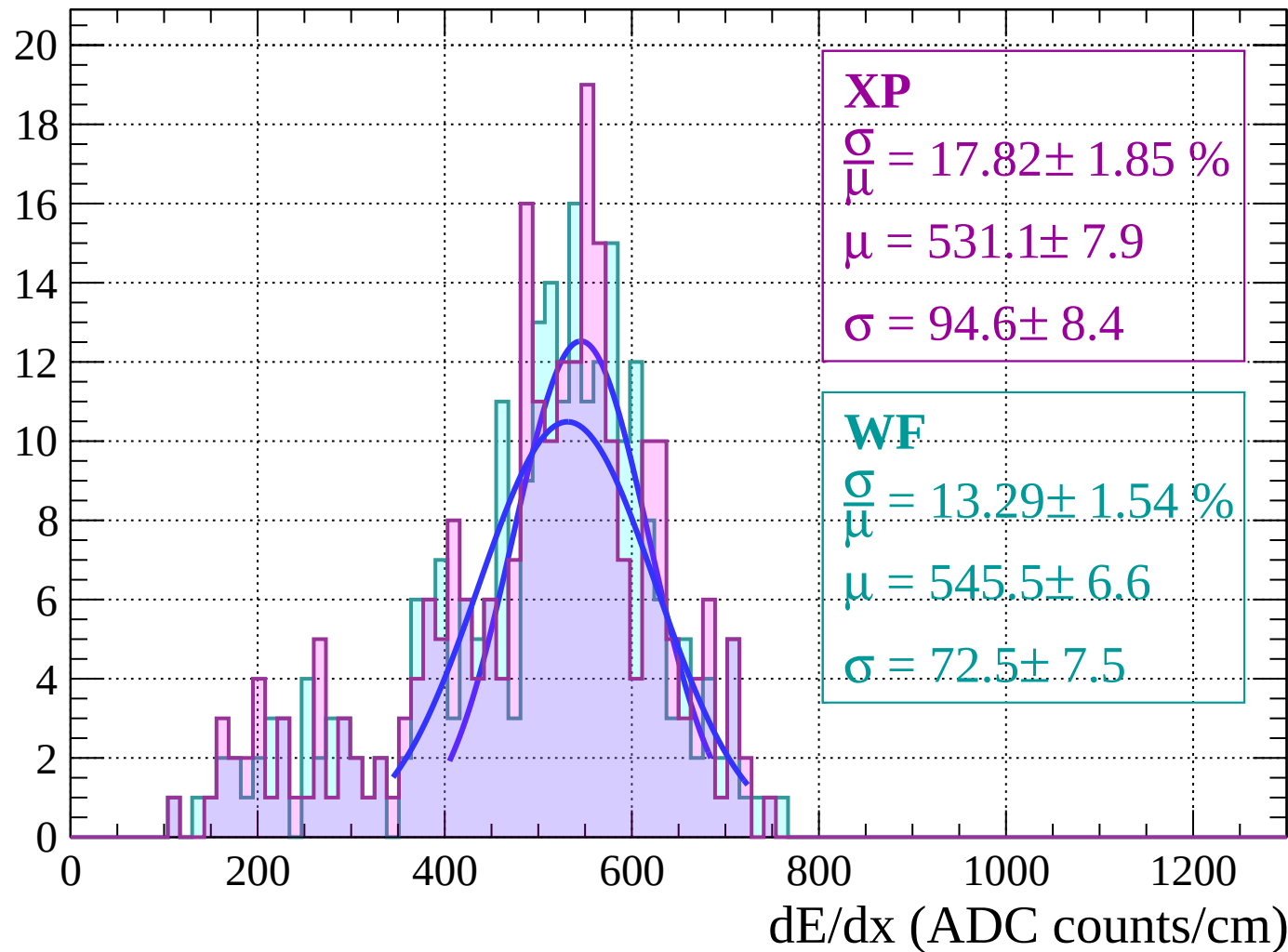
Energy loss | $-66 < \theta < -64$

Count



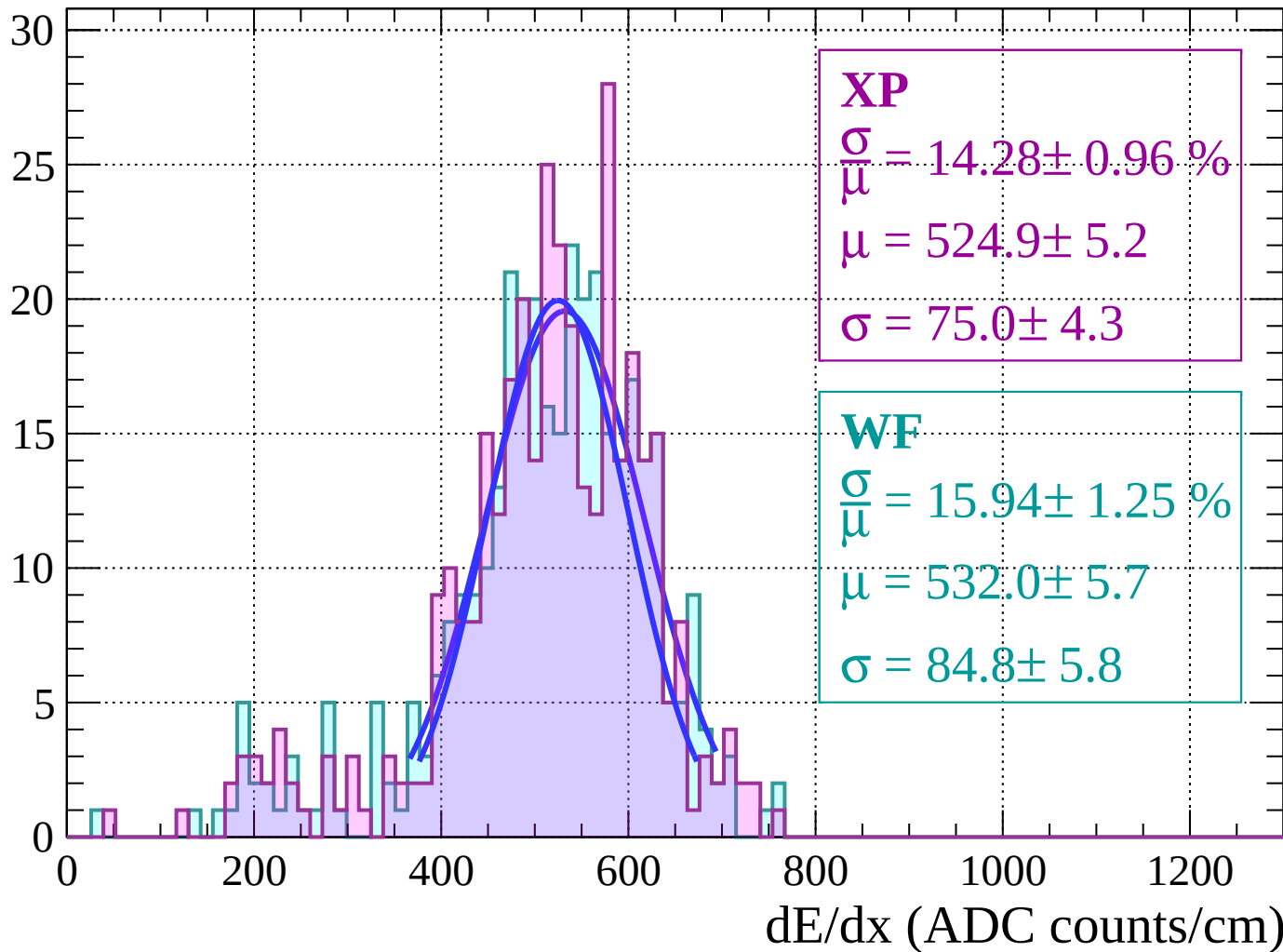
Energy loss | $-64 < \theta < -62$

Count



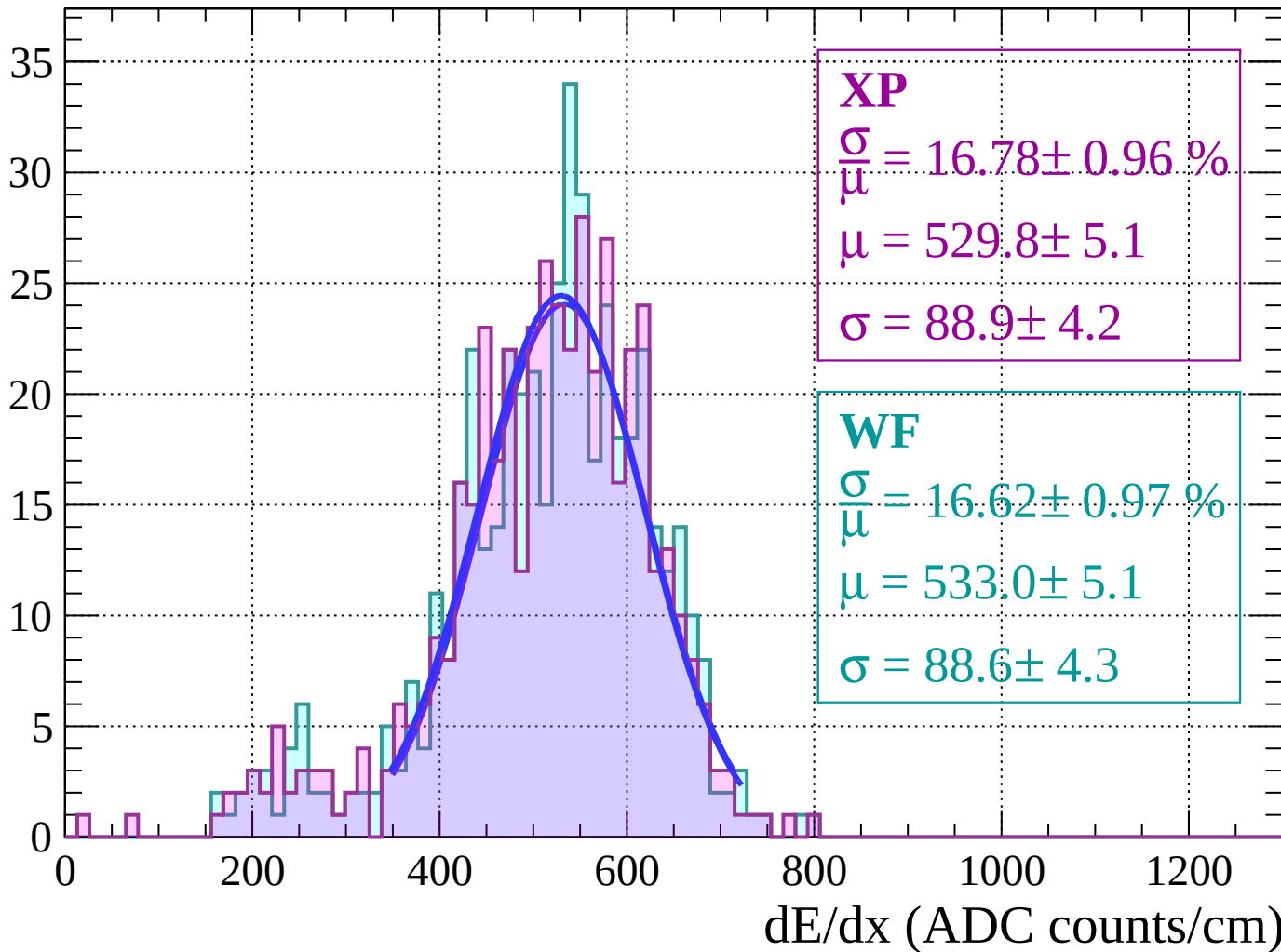
Energy loss | $-62 < \theta < -60$

Count



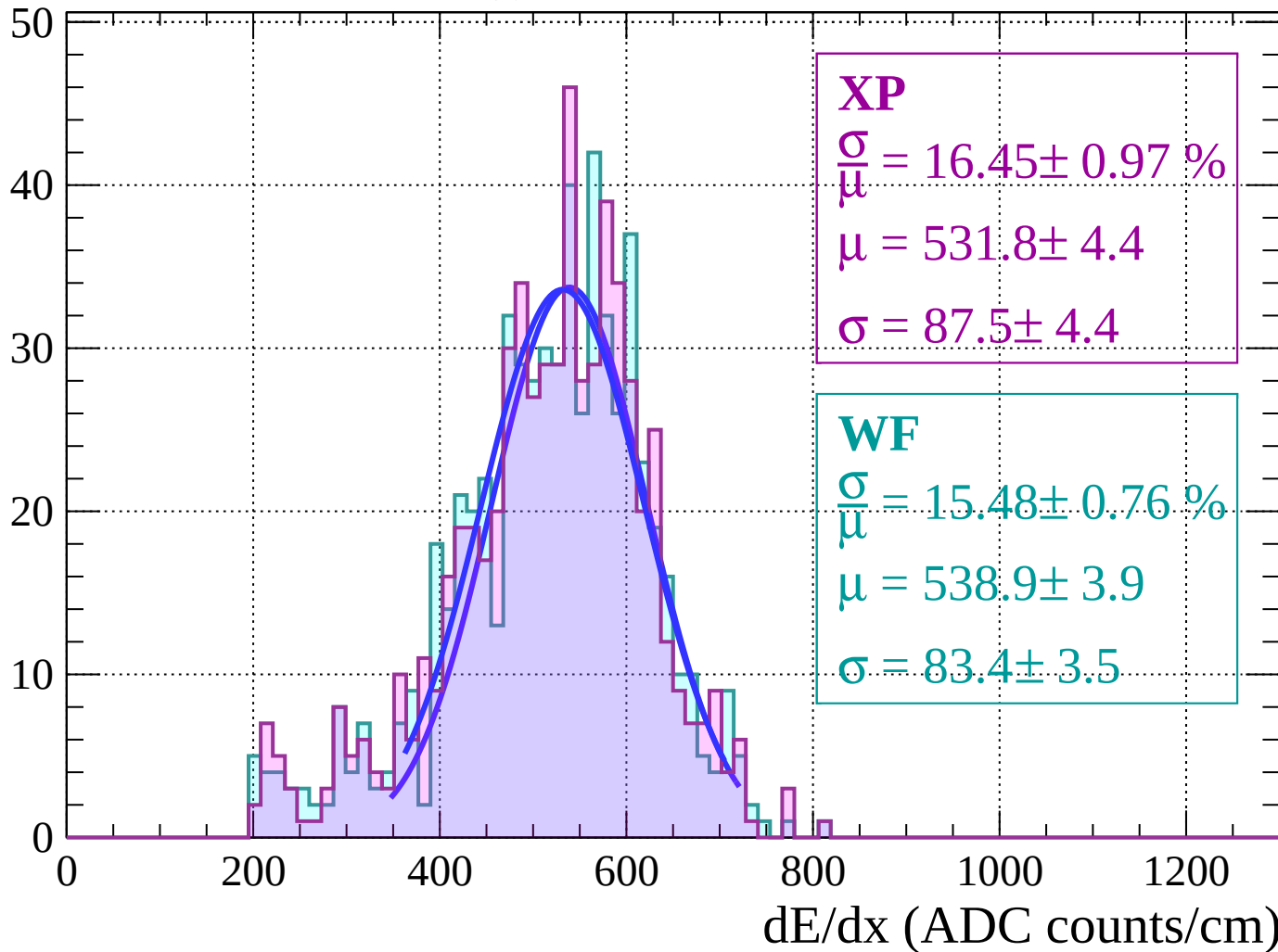
Energy loss | $-60 < \theta < -58$

Count



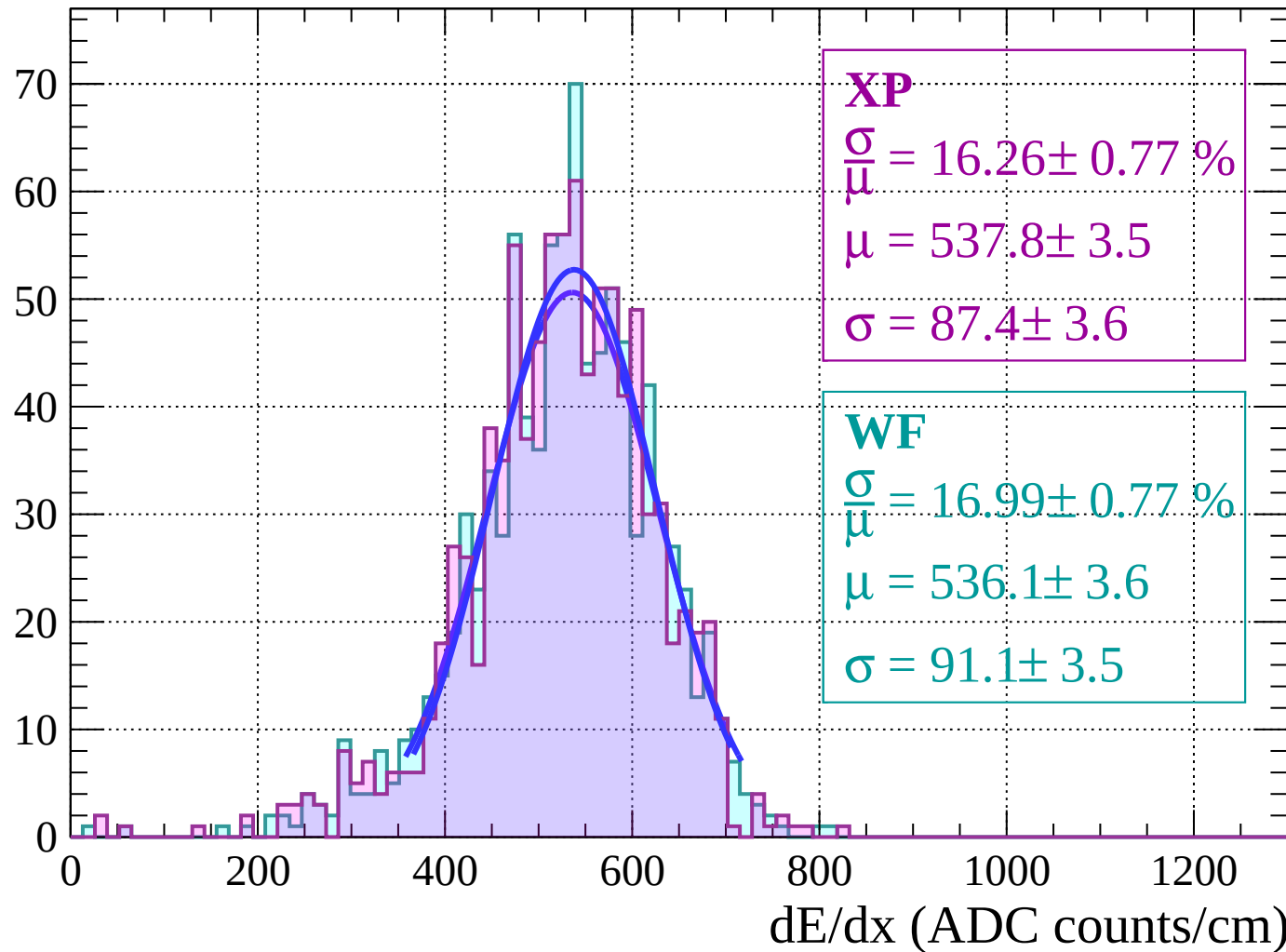
Energy loss | $-58 < \theta < -56$

Count



Energy loss $|-56 < \theta < -54$

Count



Energy loss | $-54 < \theta < -52$

Count

XP

$$\frac{\sigma}{\mu} = 16.90 \pm 0.77 \%$$

$$\mu = 536.6 \pm 3.4$$

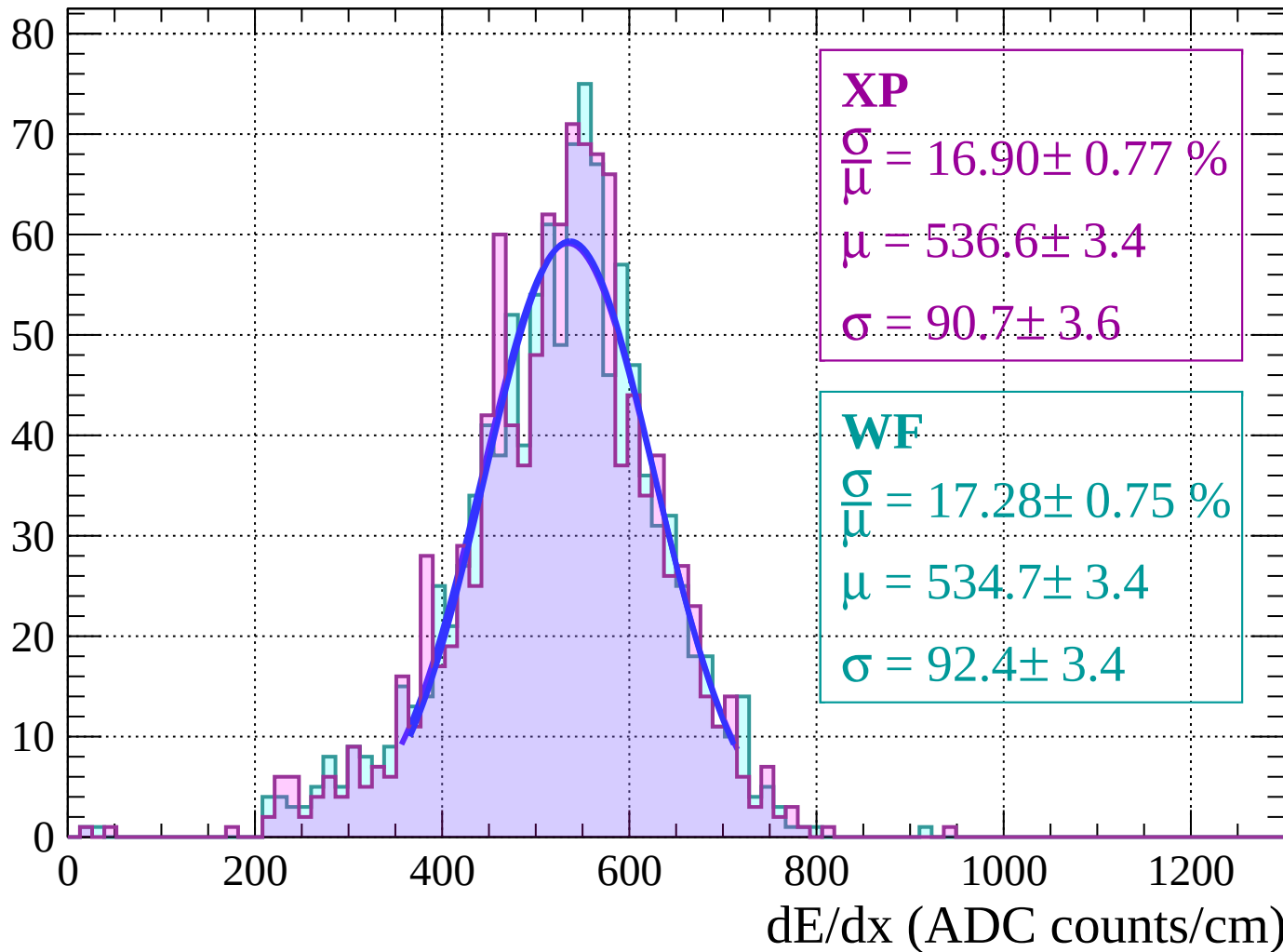
$$\sigma = 90.7 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 17.28 \pm 0.75 \%$$

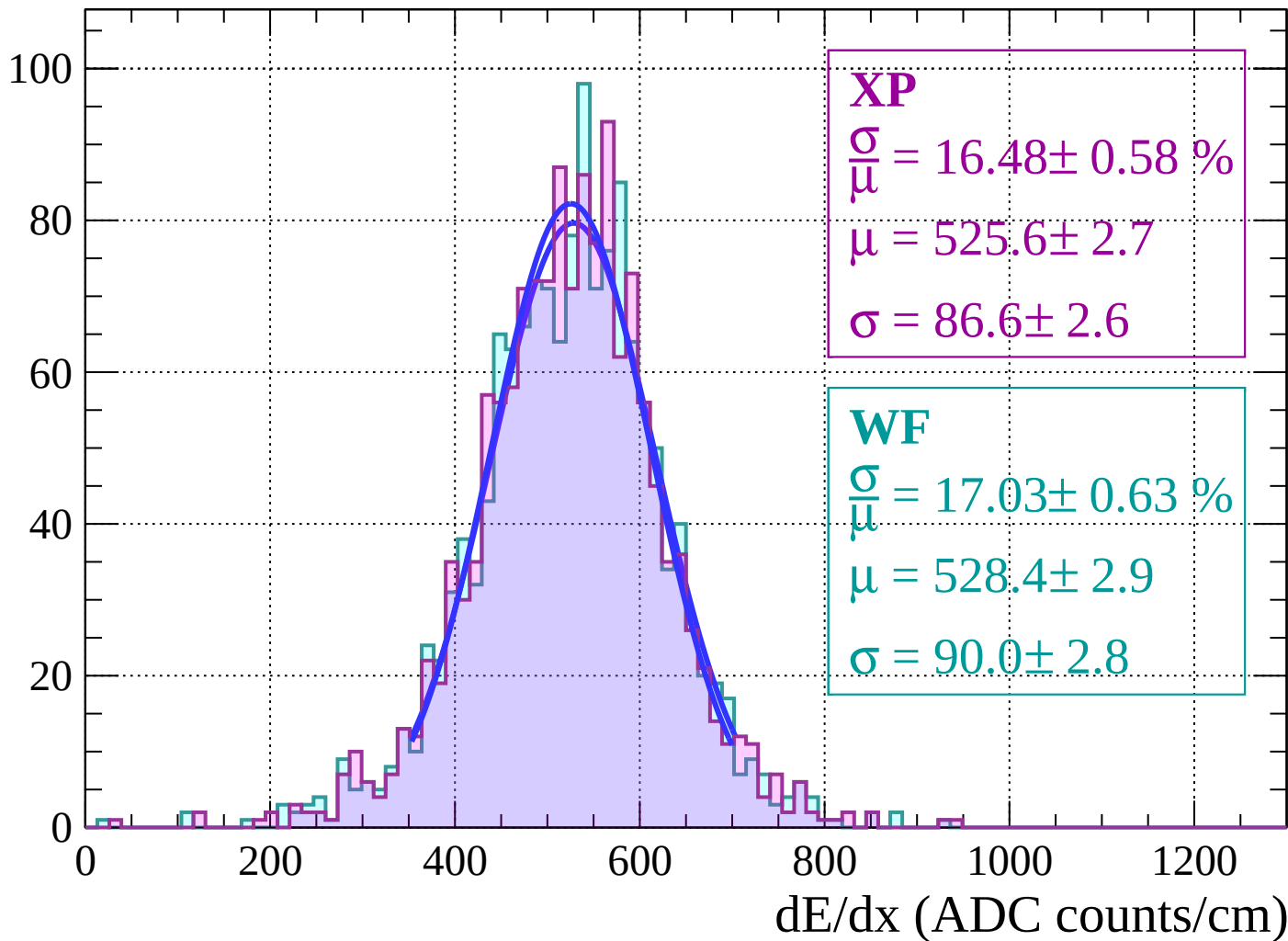
$$\mu = 534.7 \pm 3.4$$

$$\sigma = 92.4 \pm 3.4$$



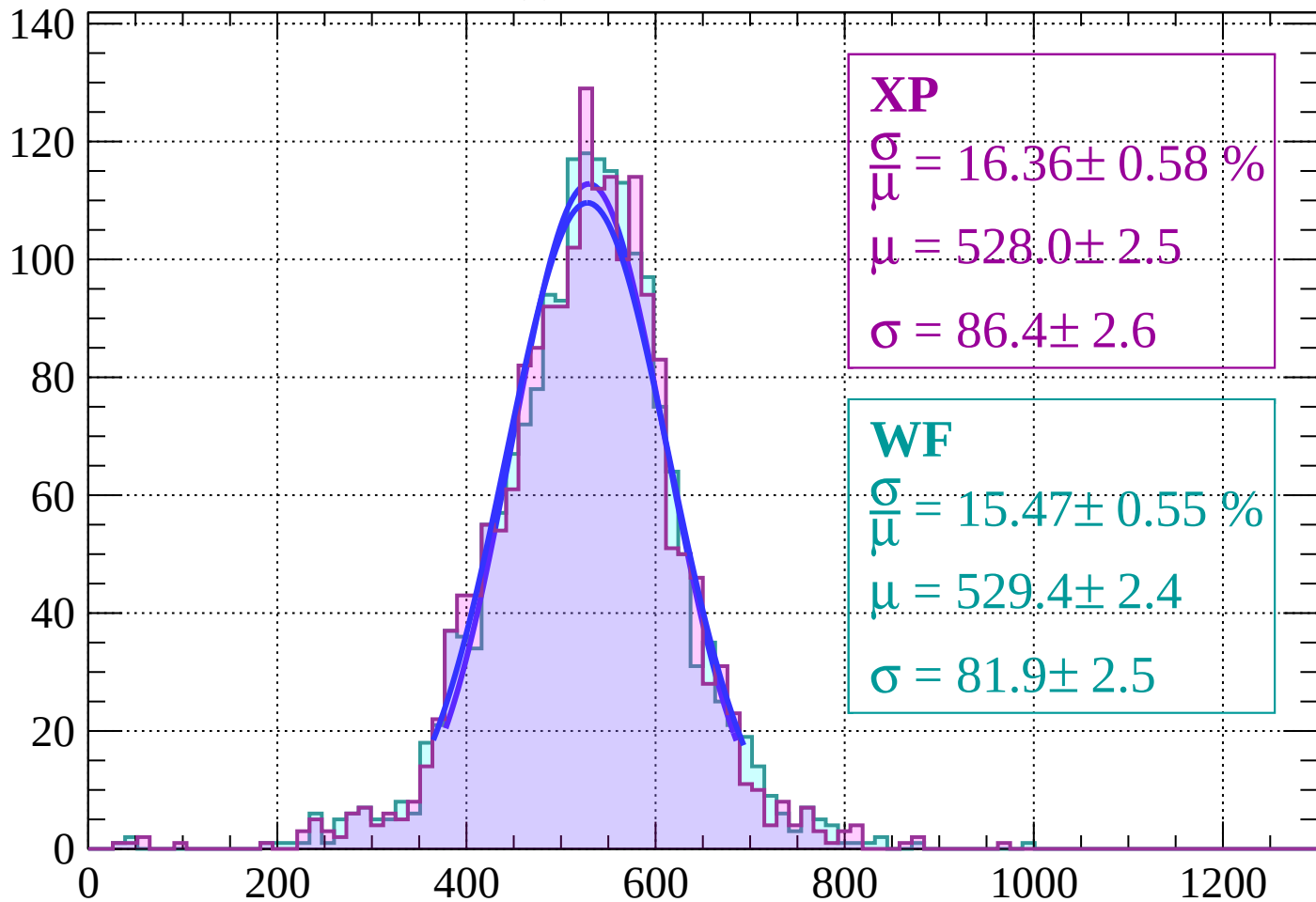
Energy loss | $-52 < \theta < -50$

Count



Energy loss | $-50 < \theta < -48$

Count



XP

$$\frac{\sigma}{\mu} = 16.36 \pm 0.58 \%$$

$$\mu = 528.0 \pm 2.5$$

$$\sigma = 86.4 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 15.47 \pm 0.55 \%$$

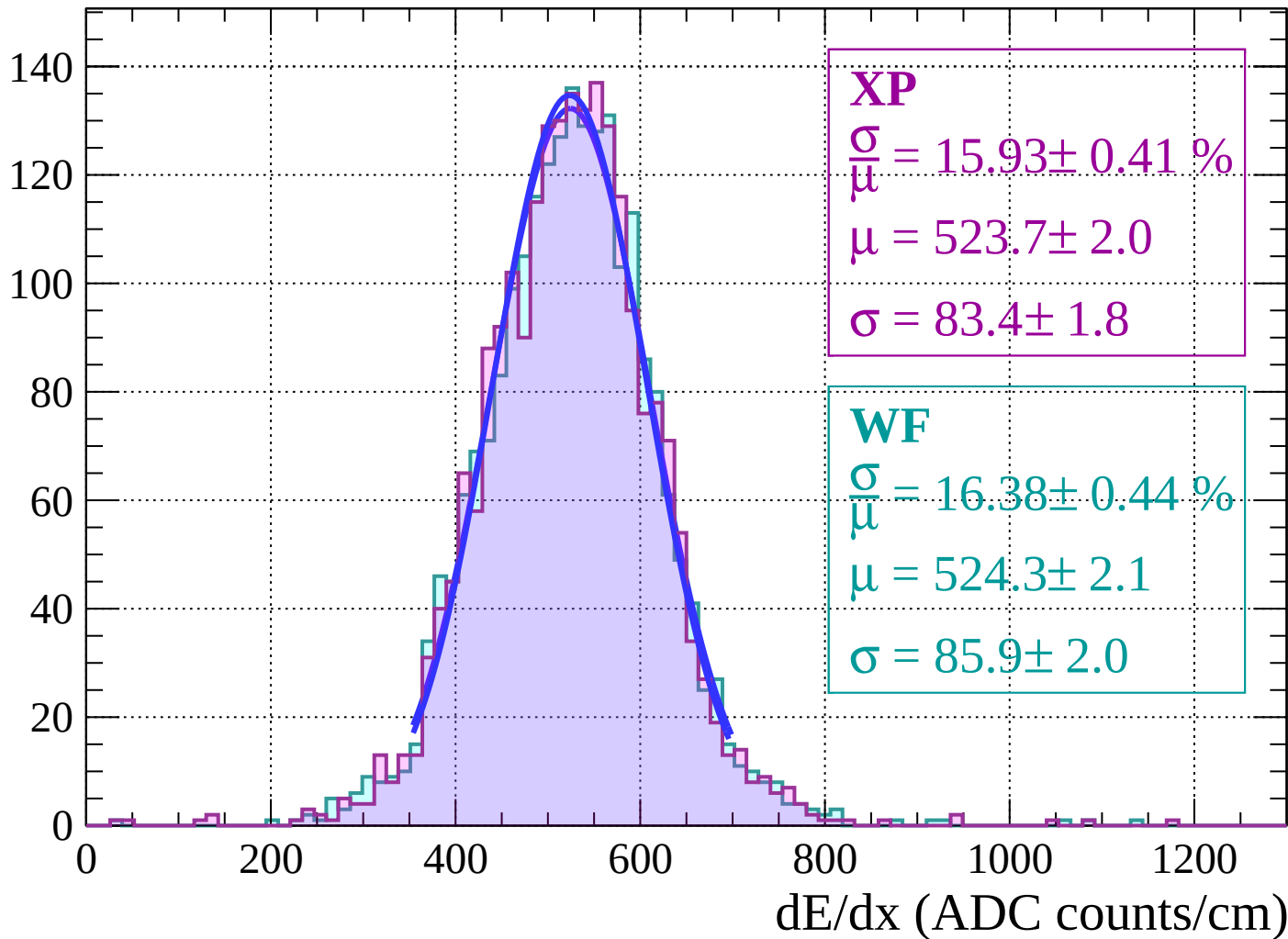
$$\mu = 529.4 \pm 2.4$$

$$\sigma = 81.9 \pm 2.5$$

dE/dx (ADC counts/cm)

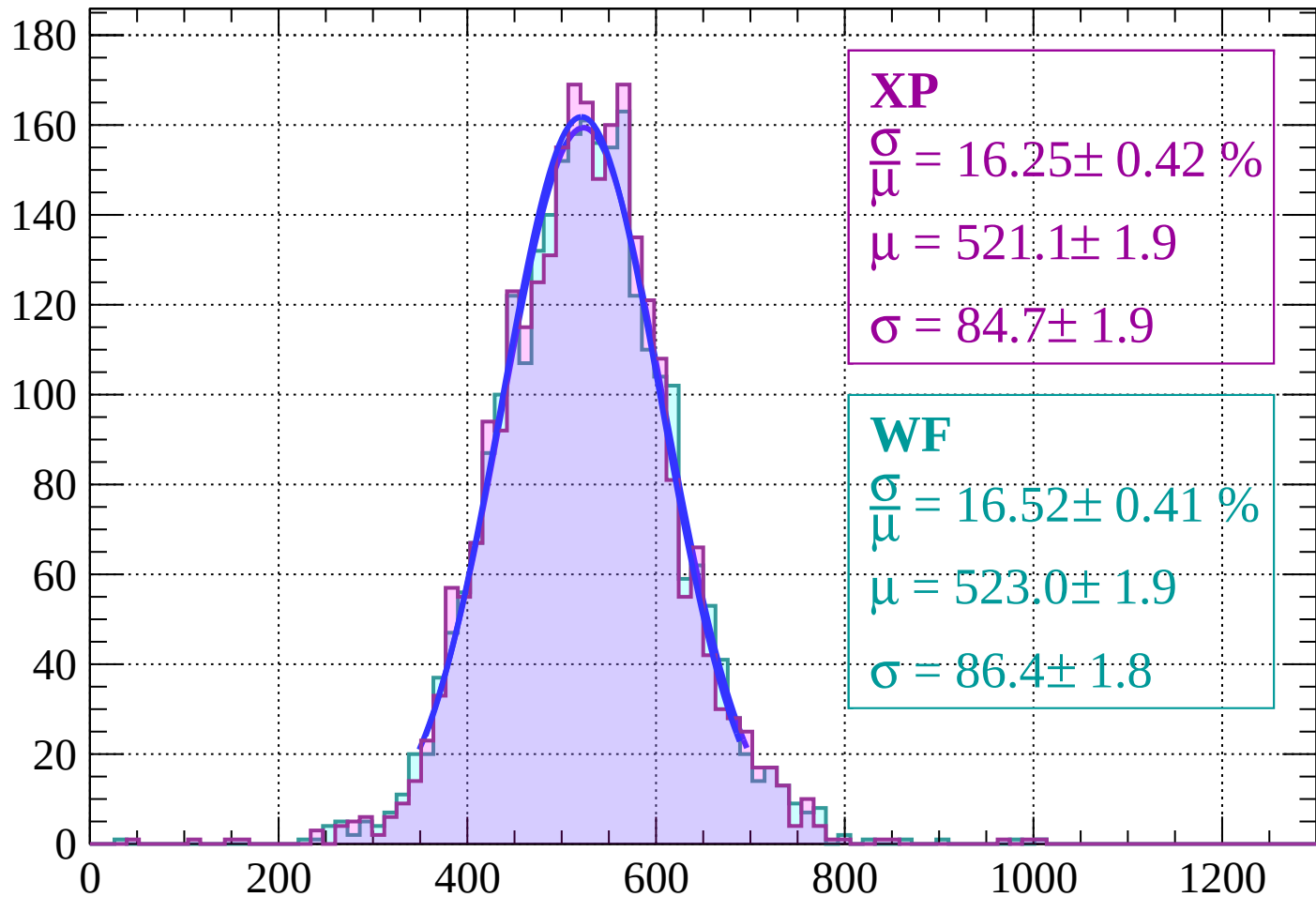
Energy loss | $-48 < \theta < -46$

Count



Energy loss $|-46 < \theta < -44$

Count



XP

$$\frac{\sigma}{\mu} = 16.25 \pm 0.42 \%$$

$$\mu = 521.1 \pm 1.9$$

$$\sigma = 84.7 \pm 1.9$$

WF

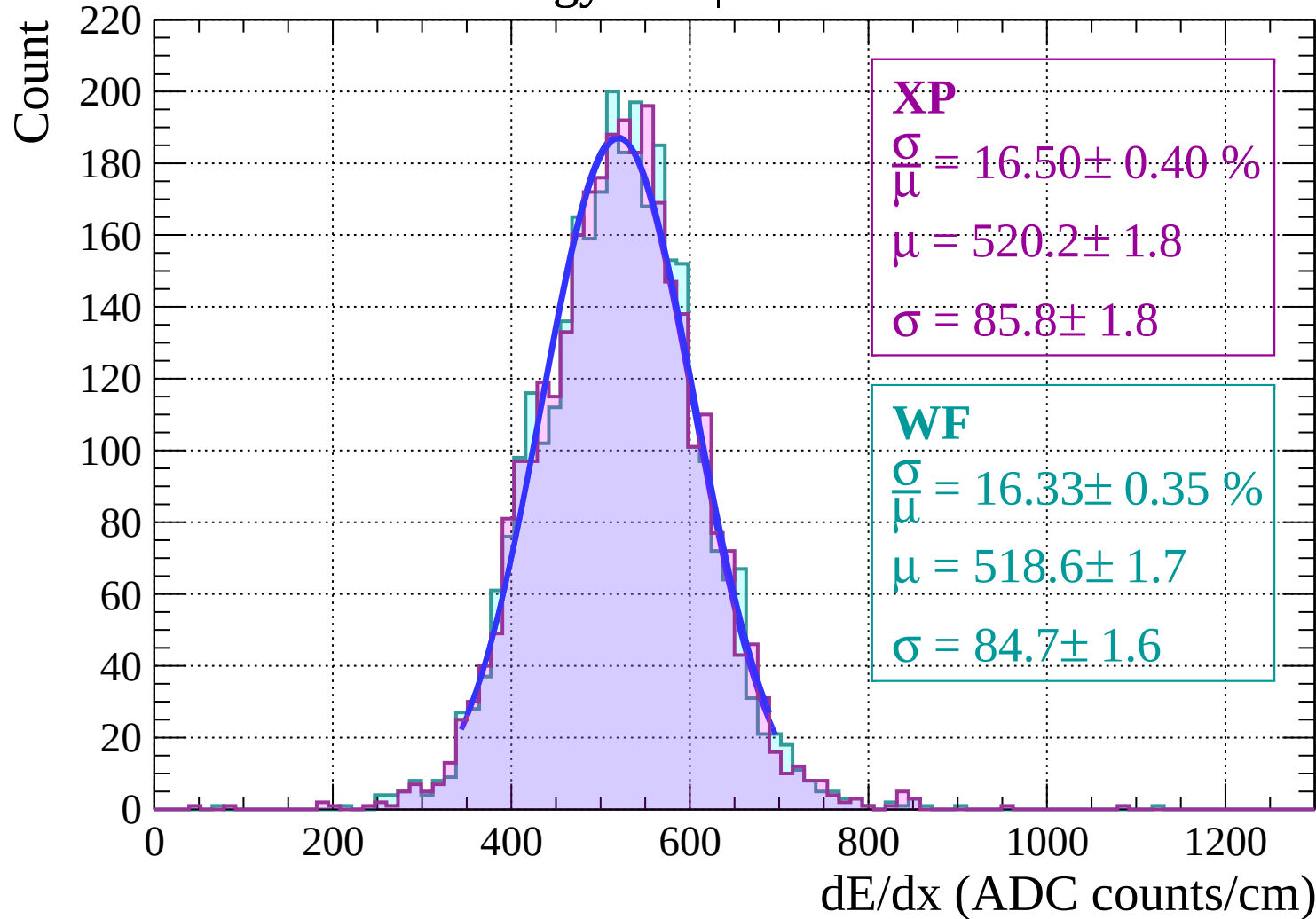
$$\frac{\sigma}{\mu} = 16.52 \pm 0.41 \%$$

$$\mu = 523.0 \pm 1.9$$

$$\sigma = 86.4 \pm 1.8$$

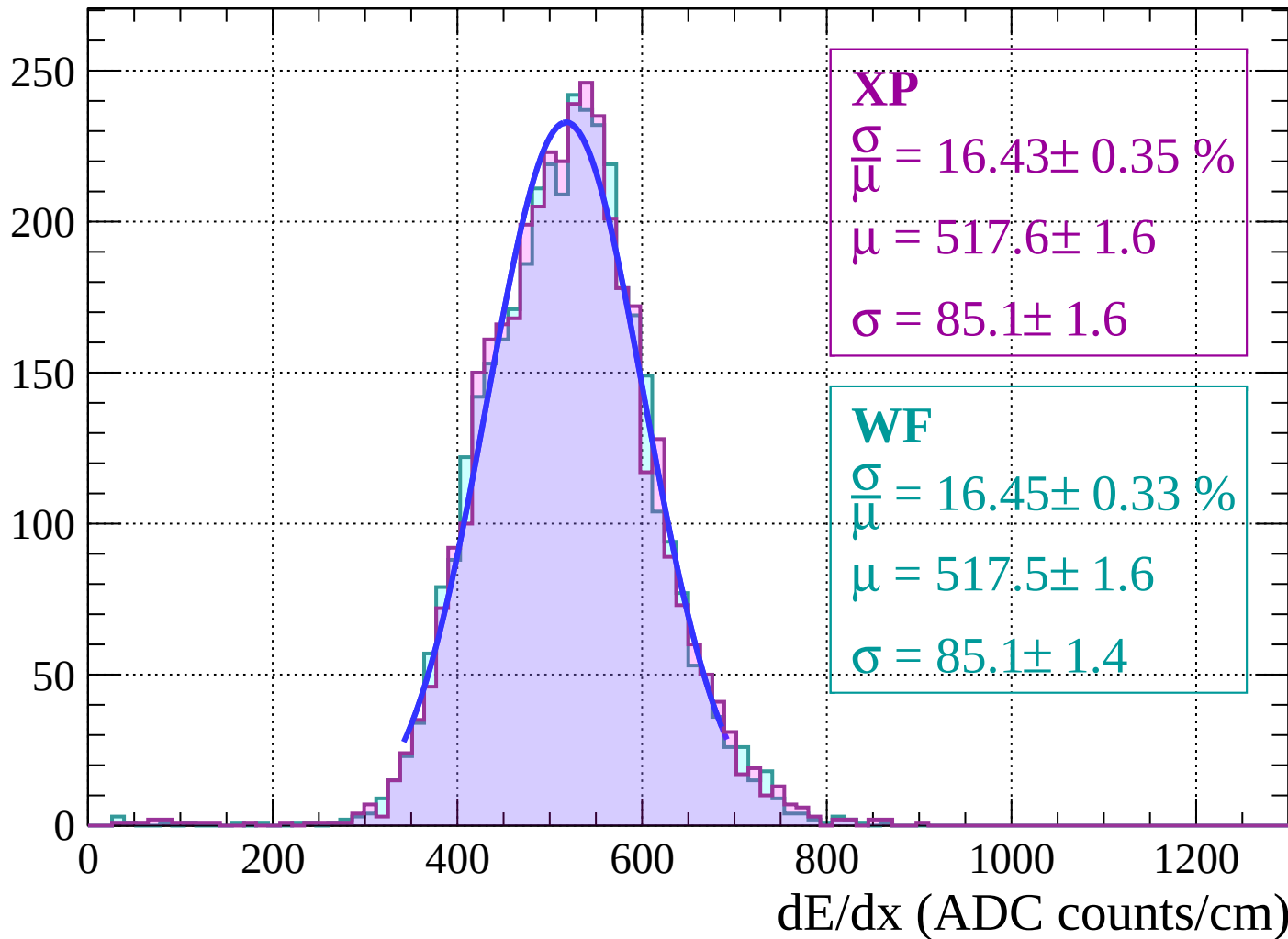
dE/dx (ADC counts/cm)

Energy loss | $-44 < \theta < -42$



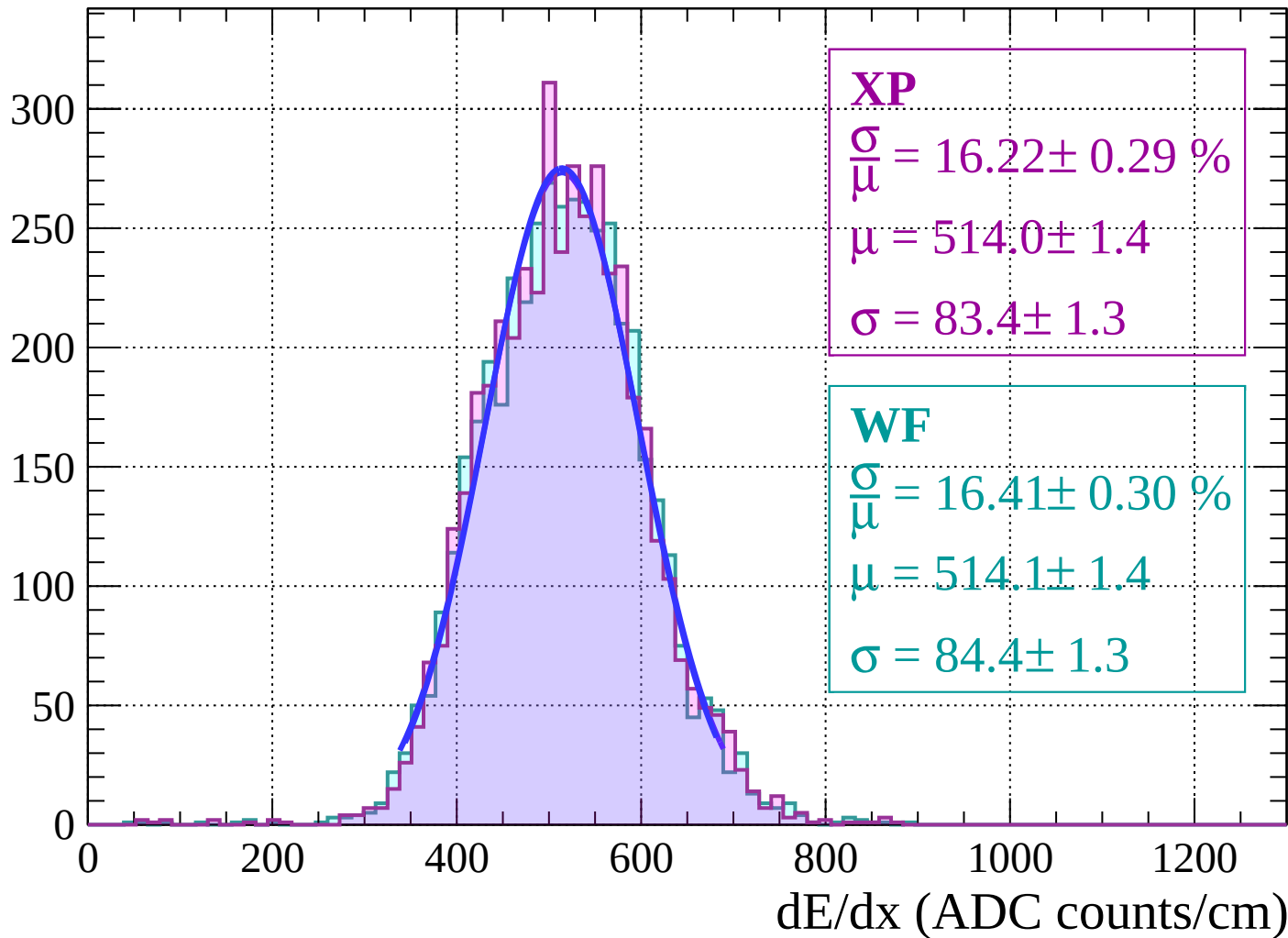
Energy loss $|-42 < \theta < -40$

Count



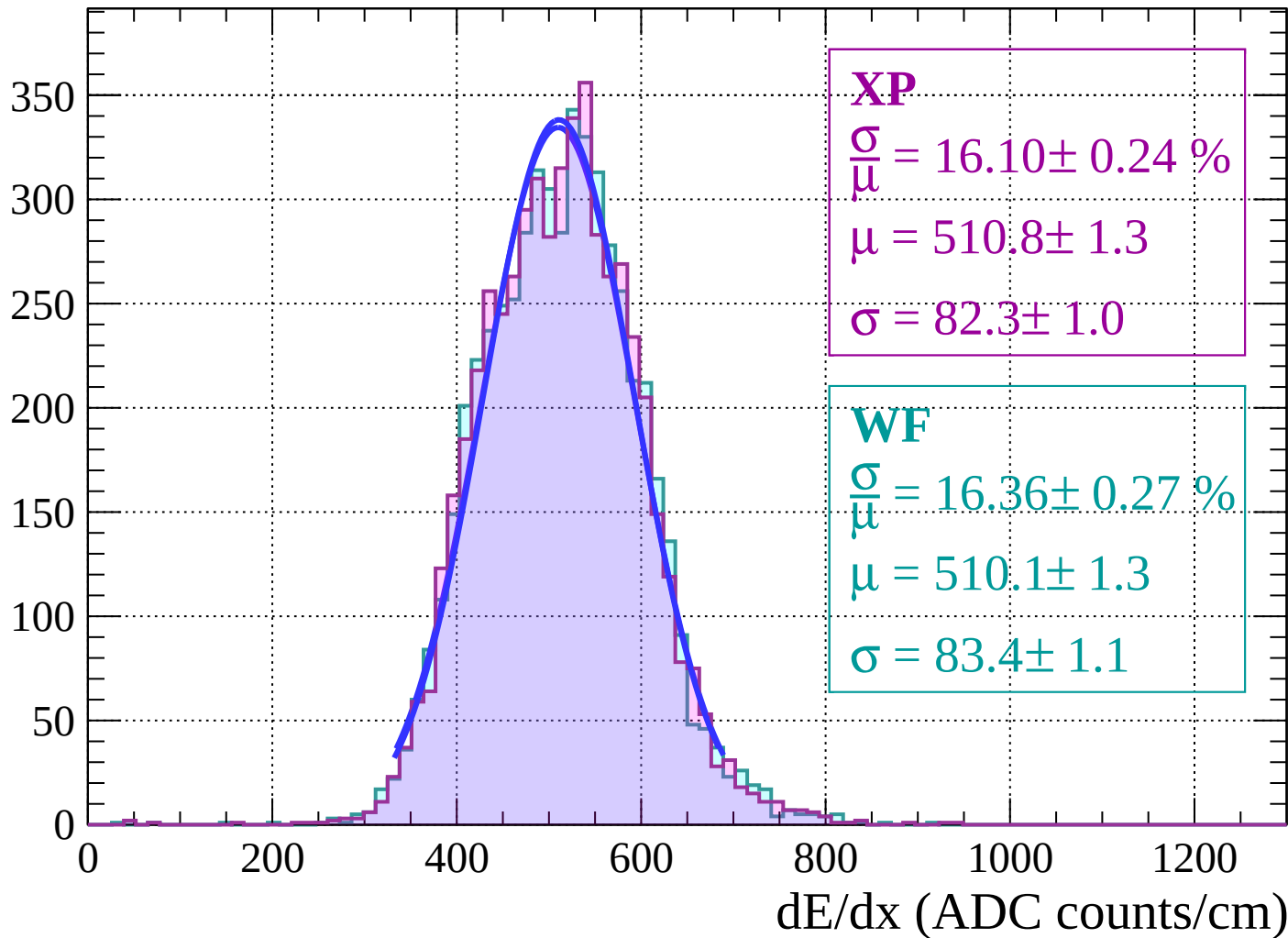
Energy loss | $-40 < \theta < -38$

Count



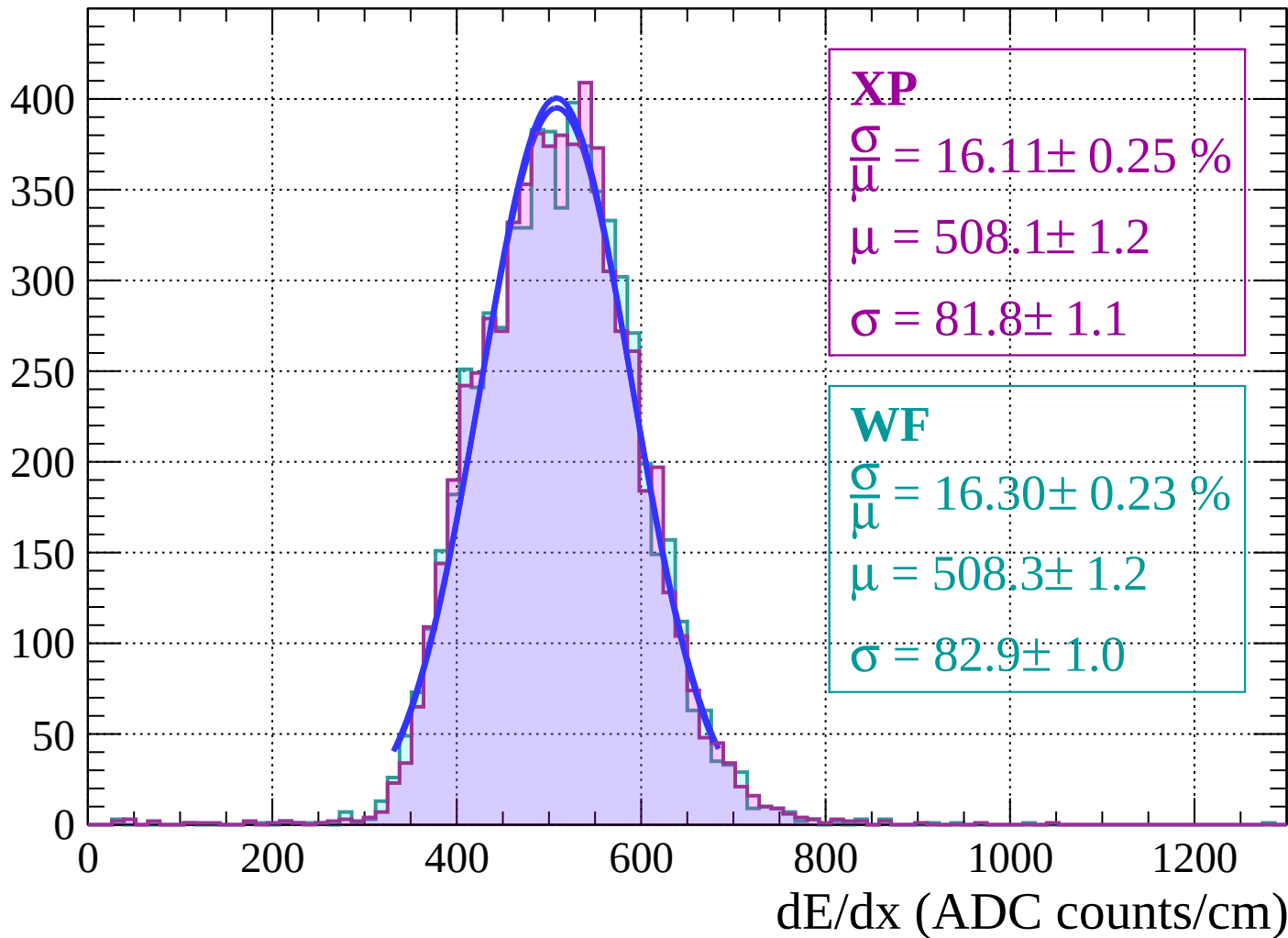
Energy loss $|-38 < \theta < -36$

Count

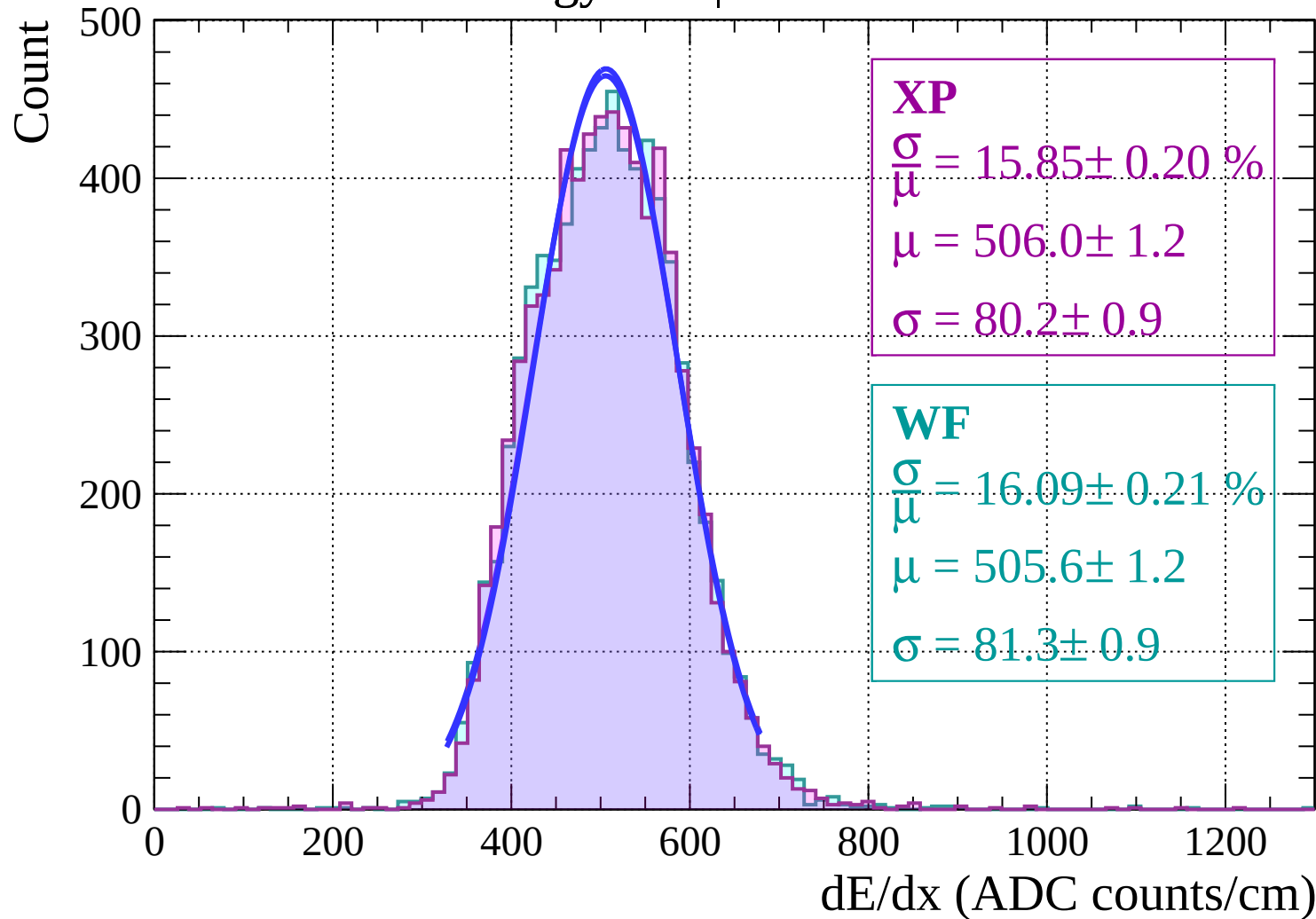


Energy loss $|-36 < \theta < -34$

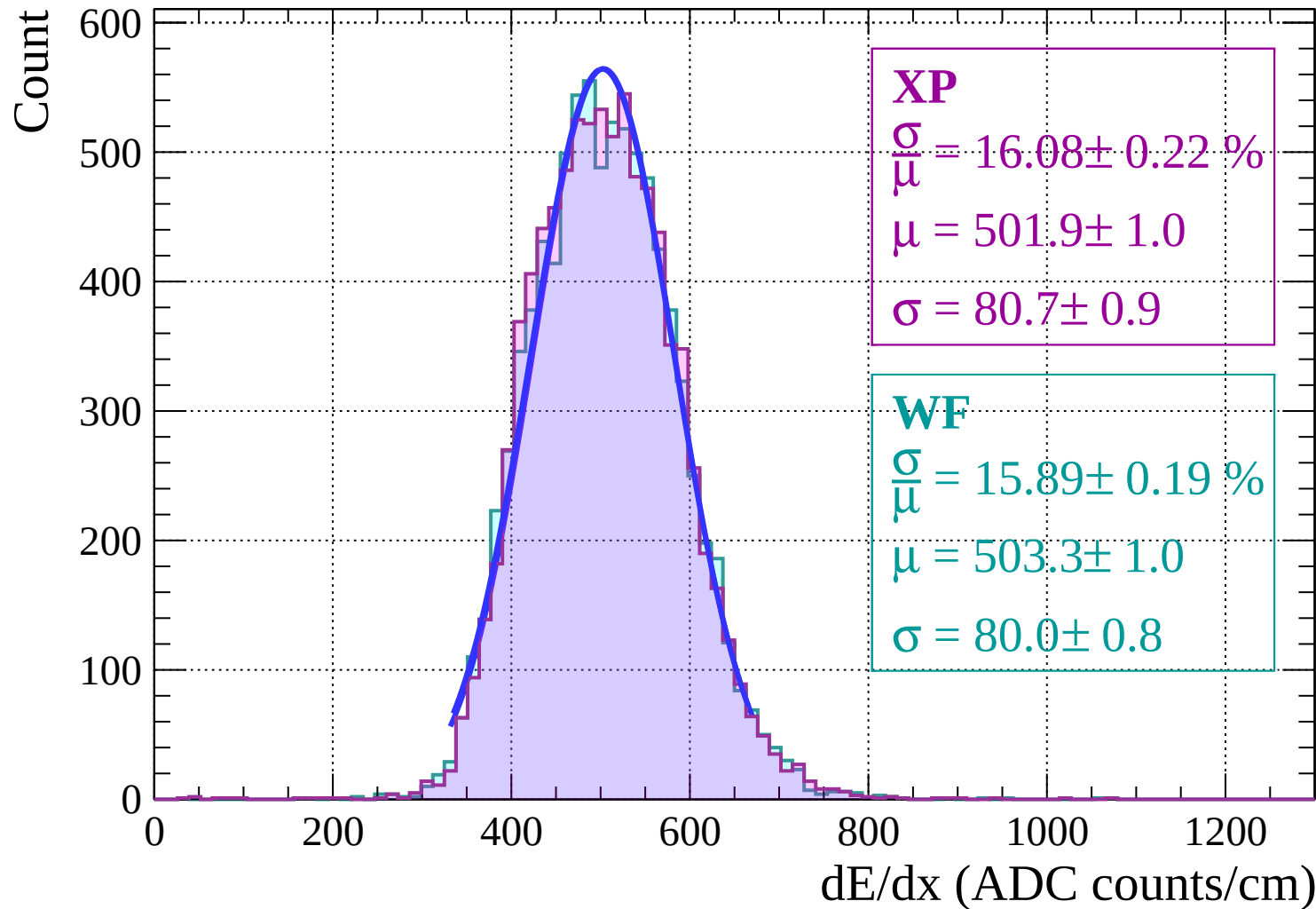
Count



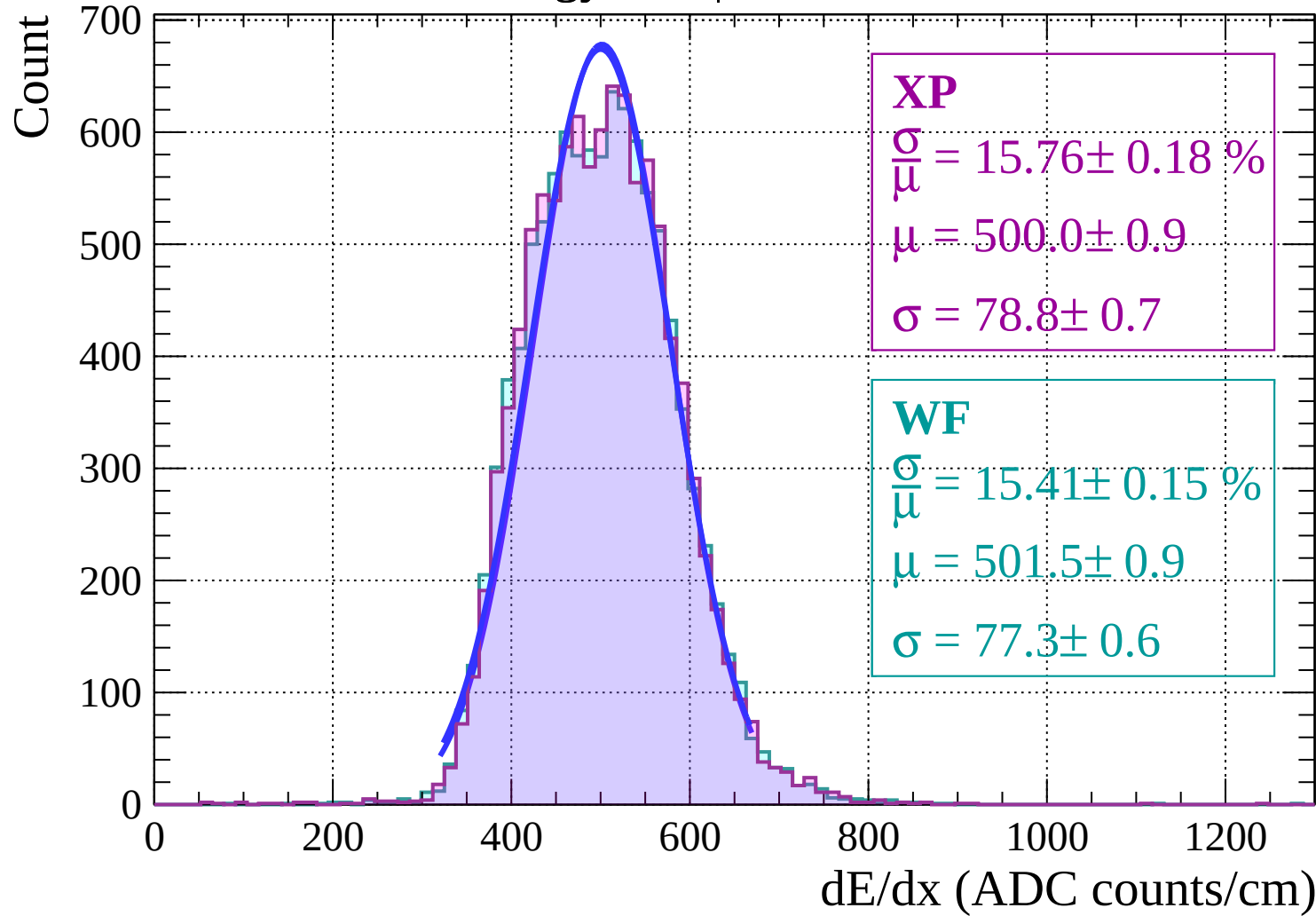
Energy loss $|-34 < \theta < -32$



Energy loss $|-32 < \theta < -30$

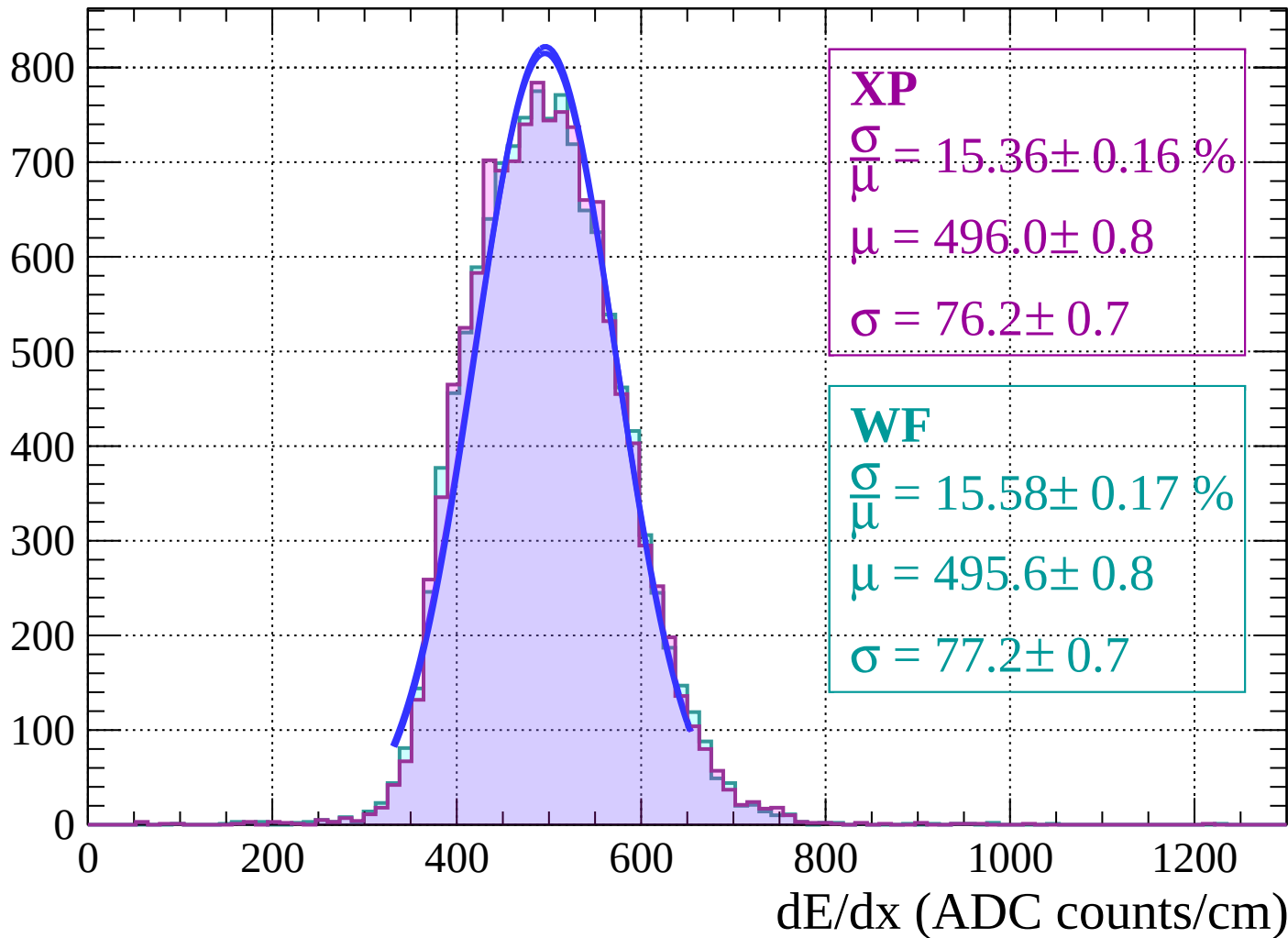


Energy loss | $-30 < \theta < -28$

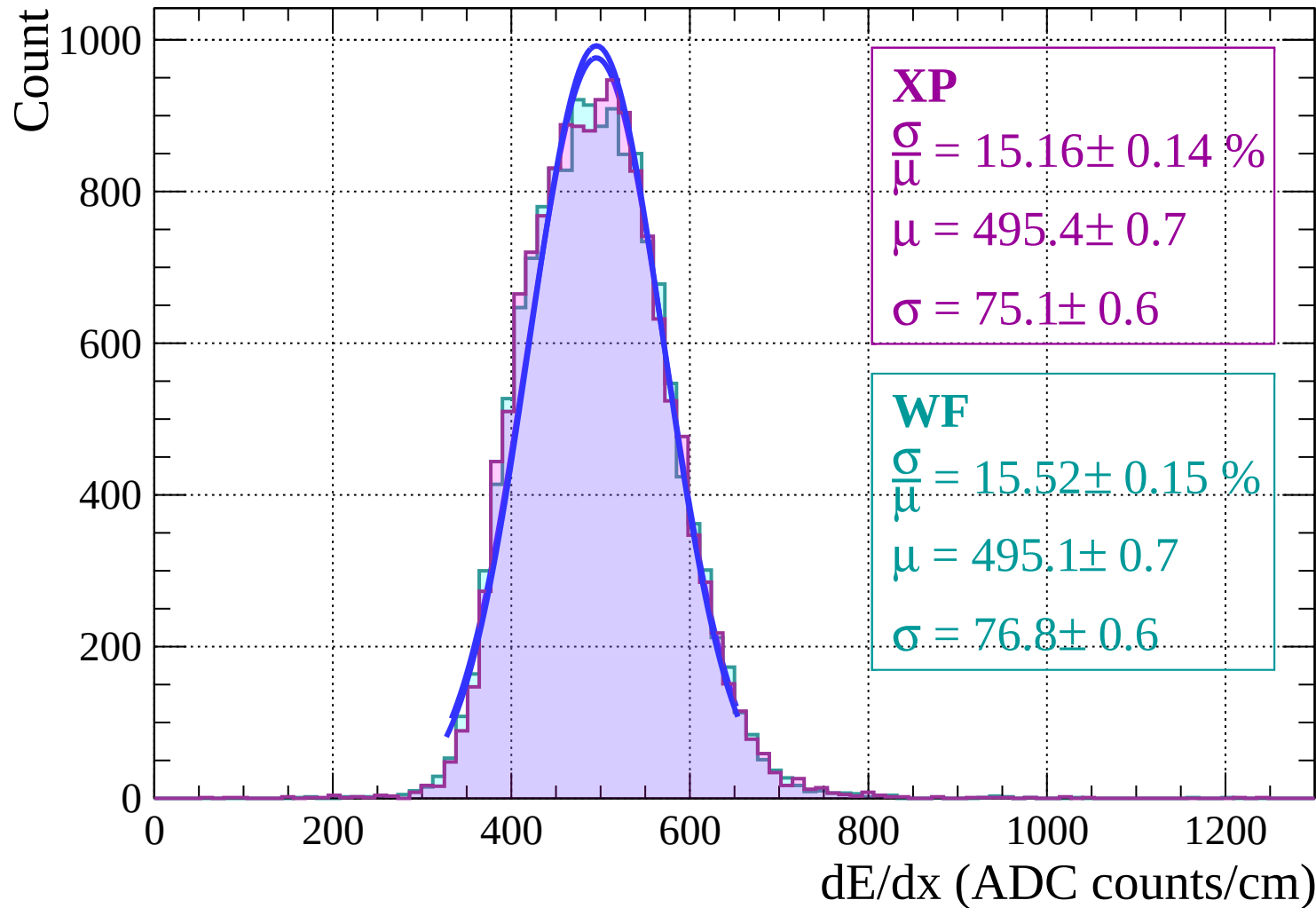


Energy loss | $-28 < \theta < -26$

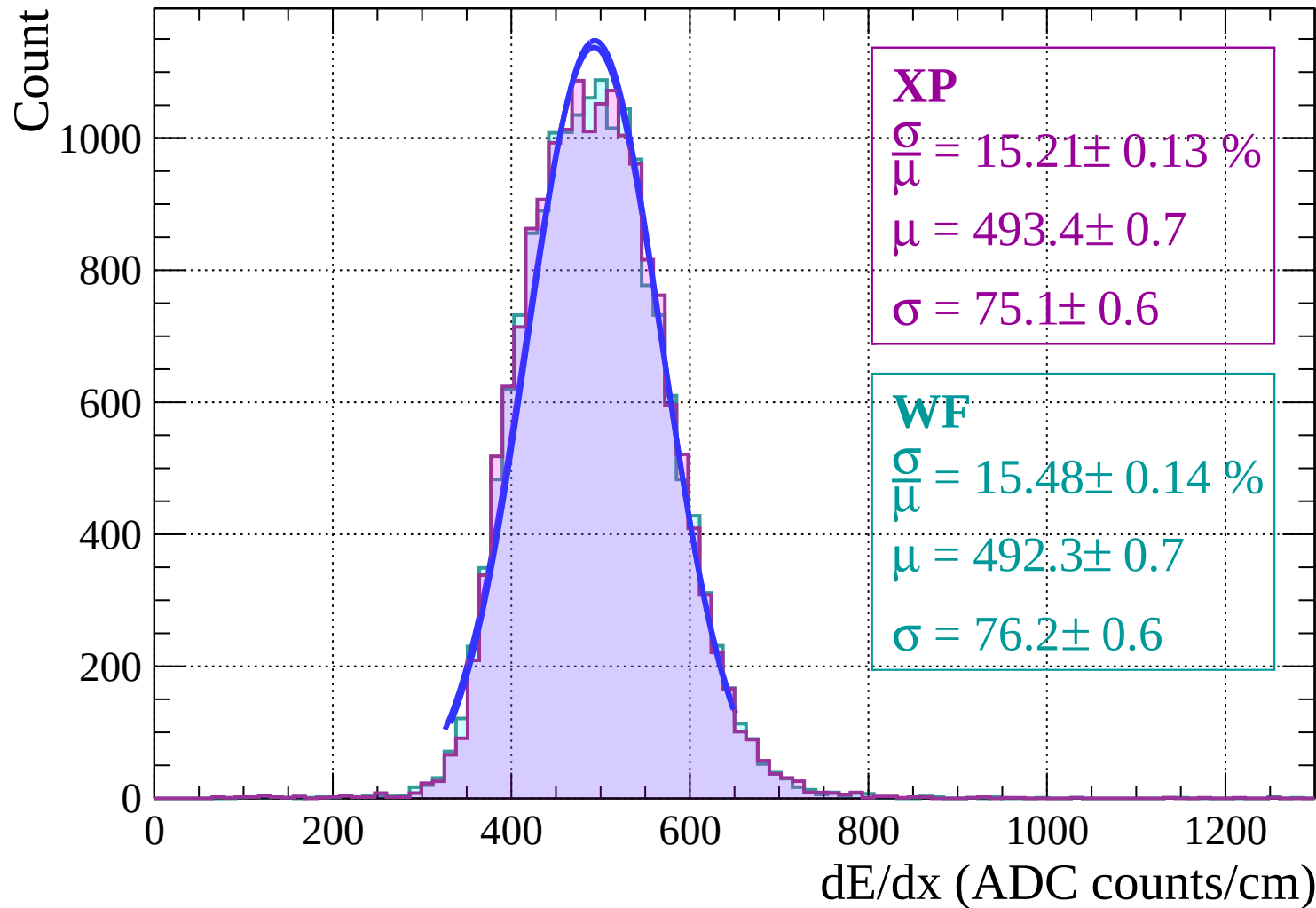
Count



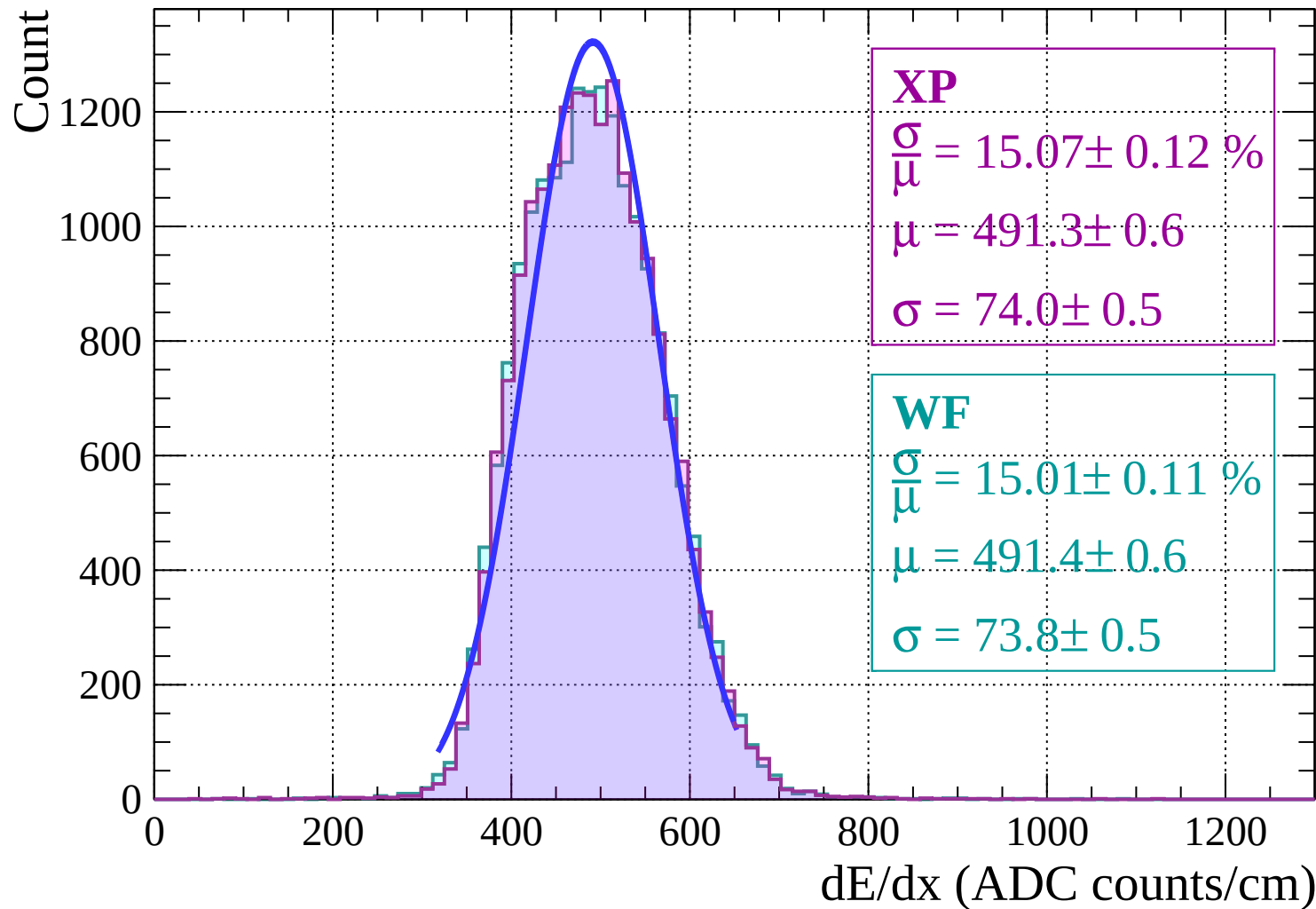
Energy loss | $-26 < \theta < -24$



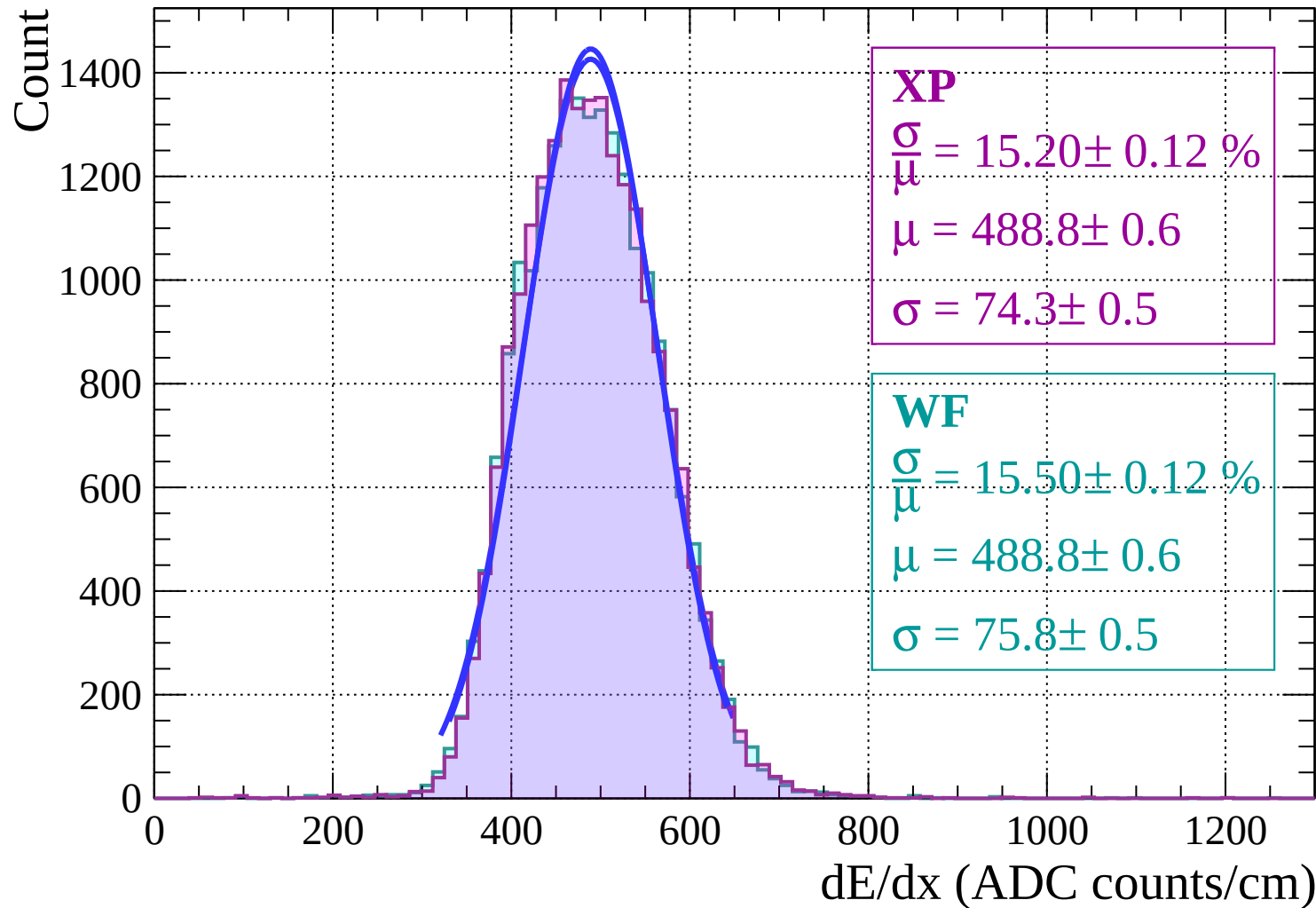
Energy loss $|-24 < \theta < -22$



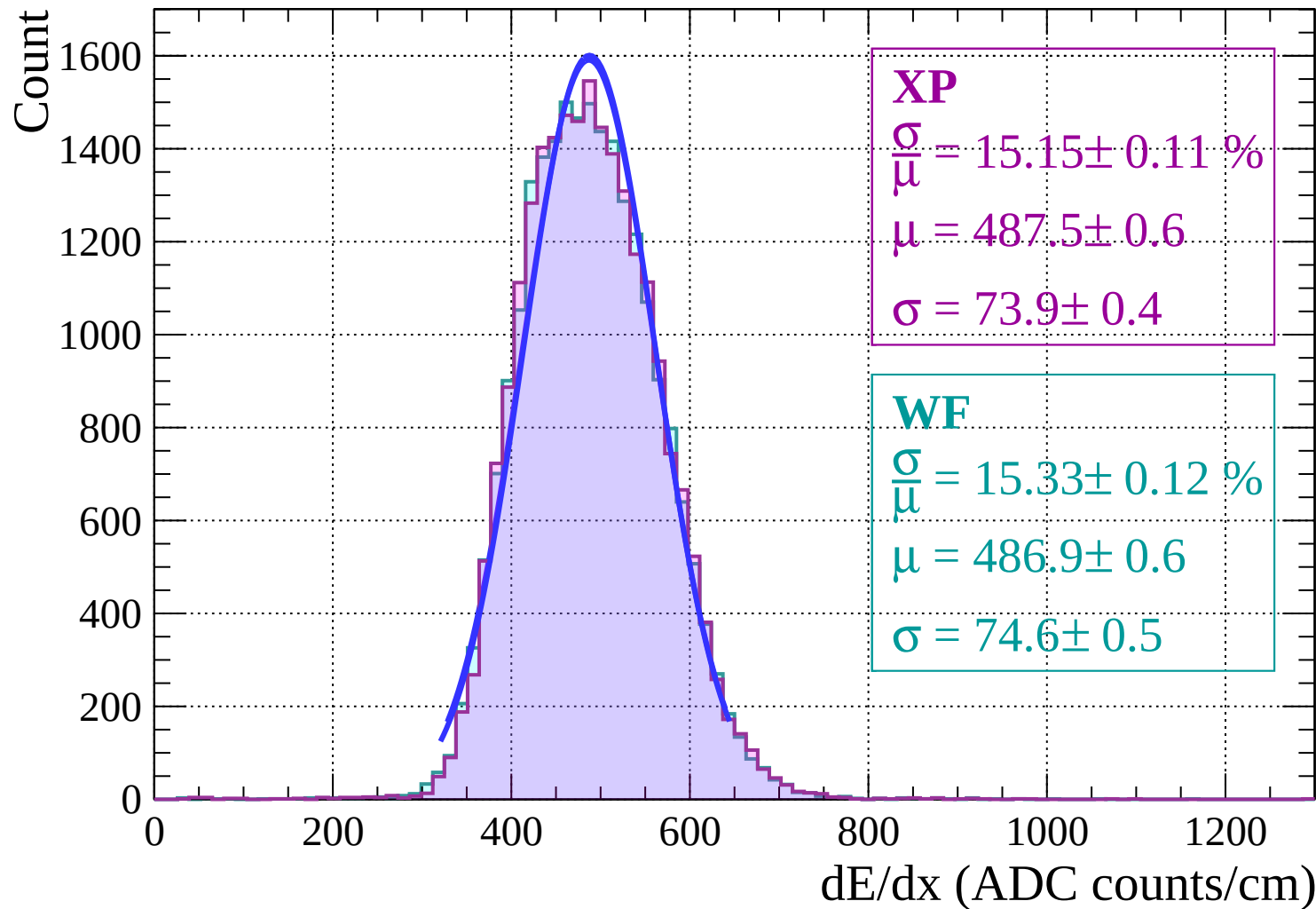
Energy loss | $-22 < \theta < -20$



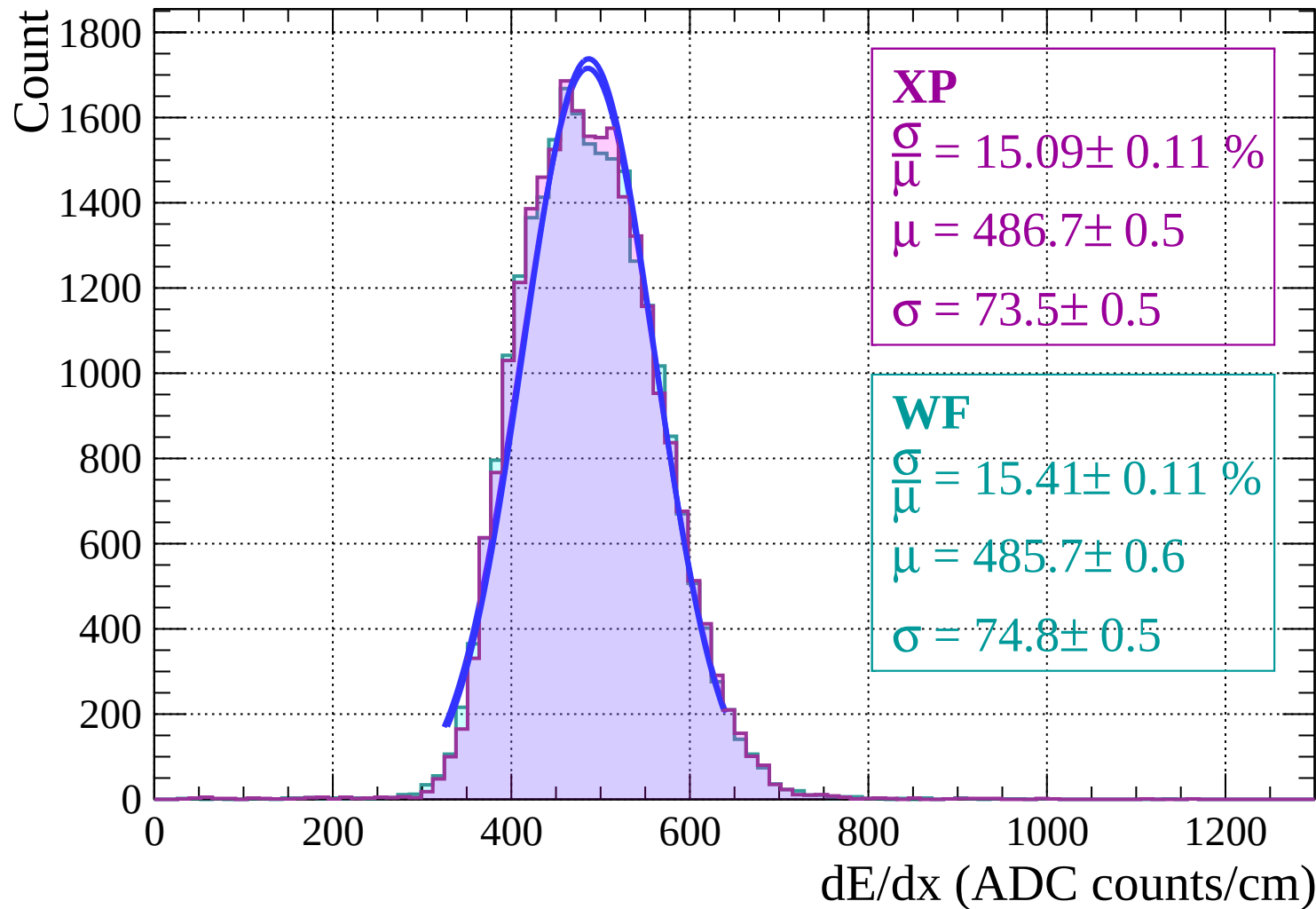
Energy loss | $-20 < \theta < -18$



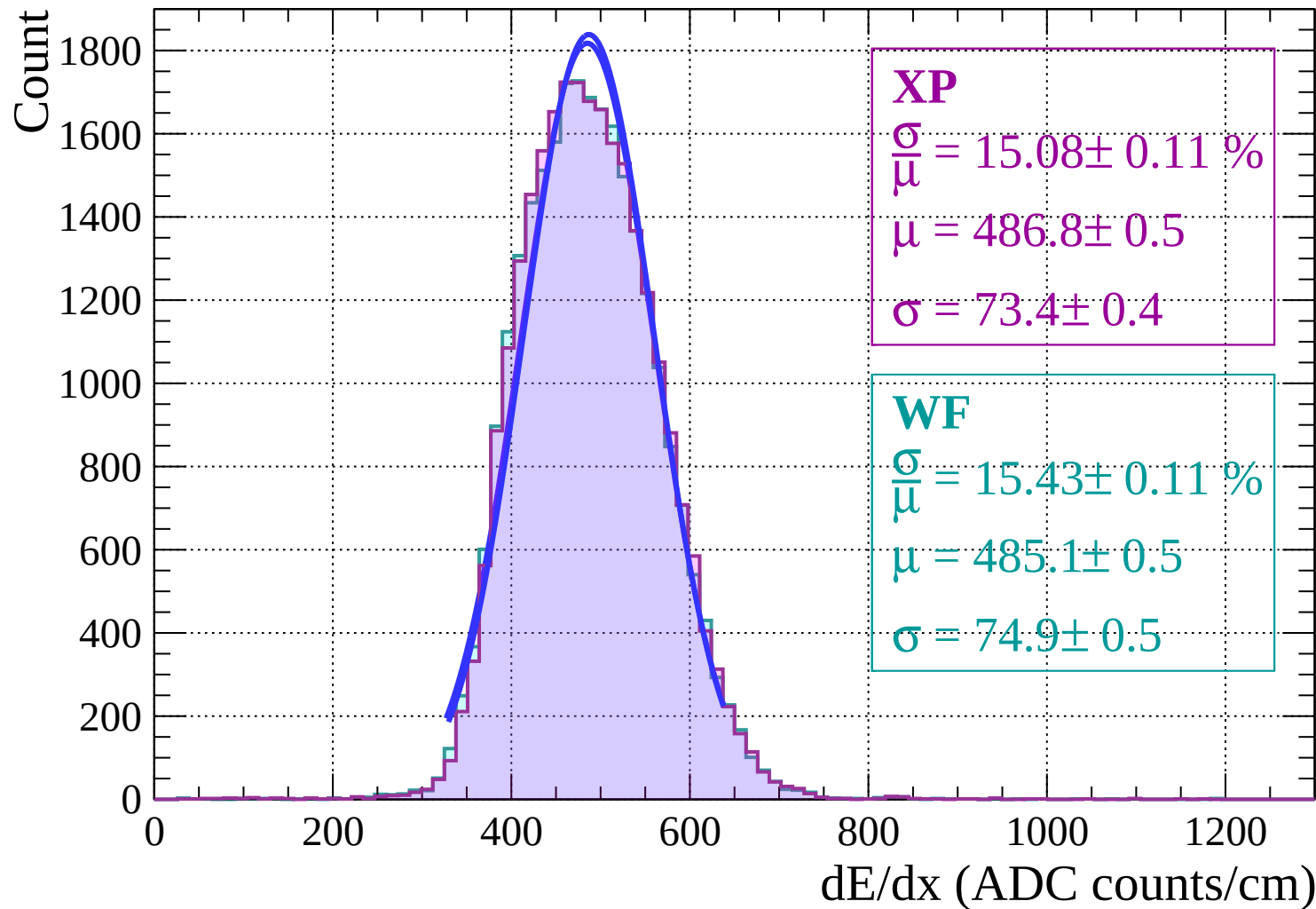
Energy loss | $-18 < \theta < -16$



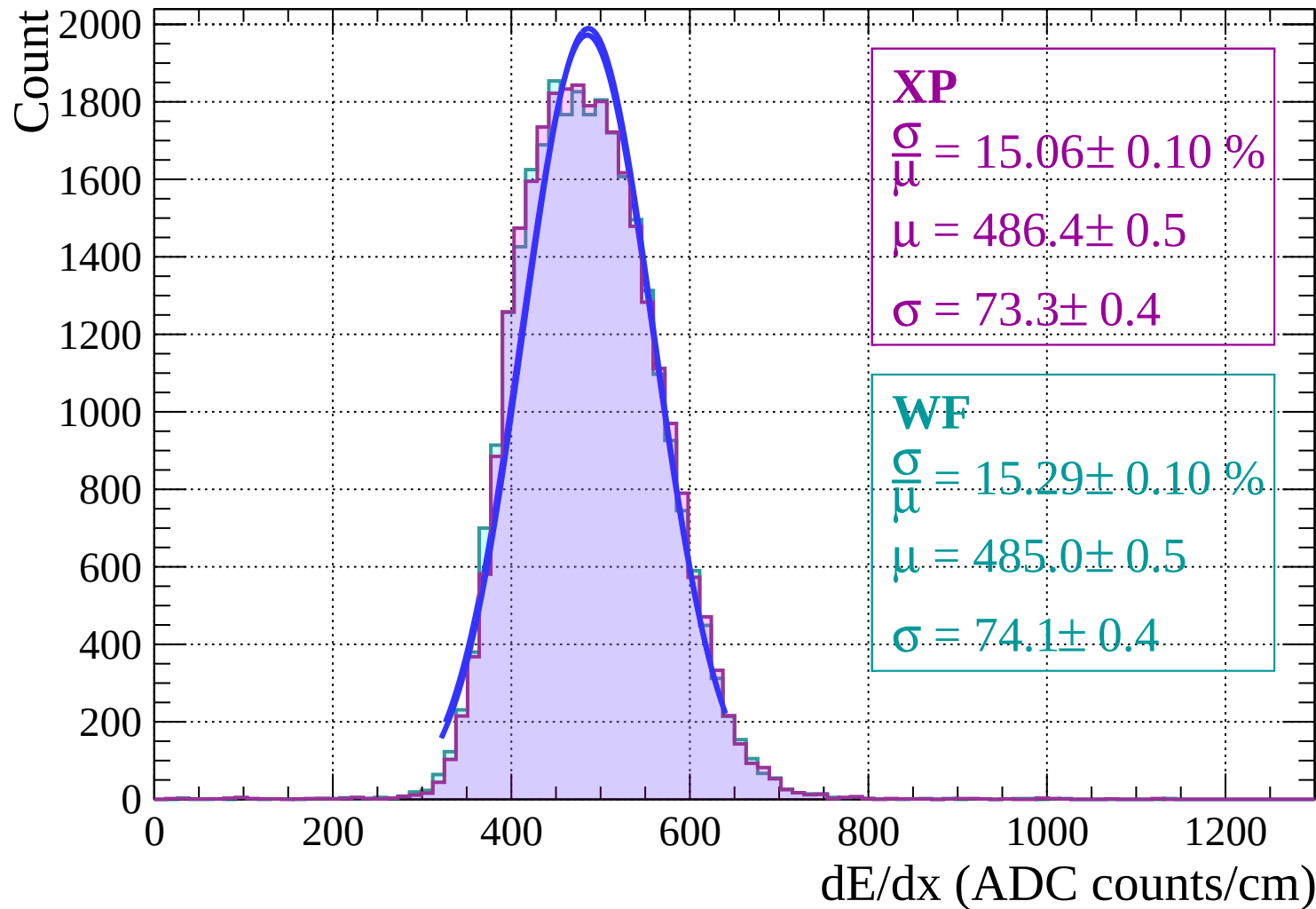
Energy loss | $-16 < \theta < -14$



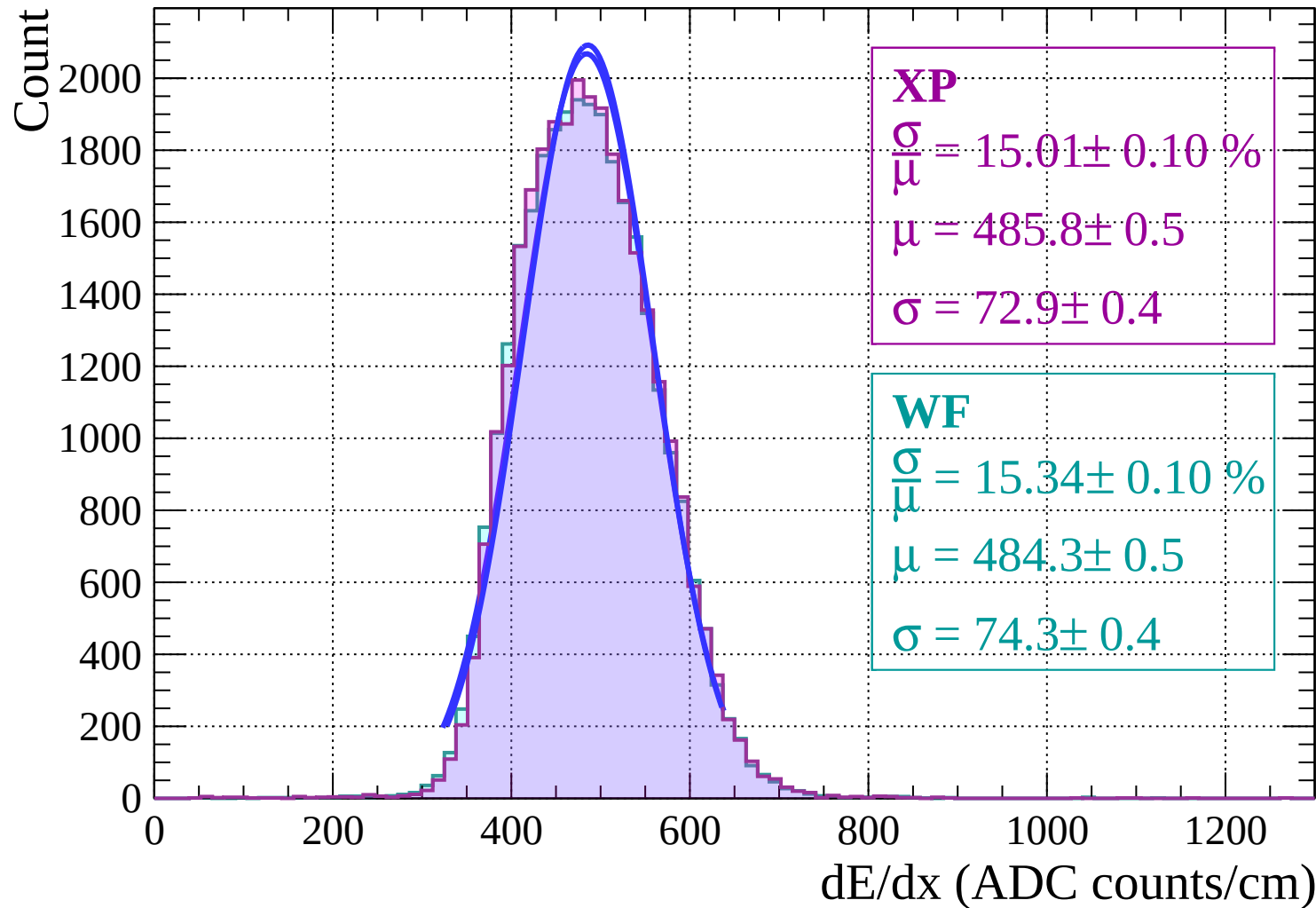
Energy loss | $-14 < \theta < -12$



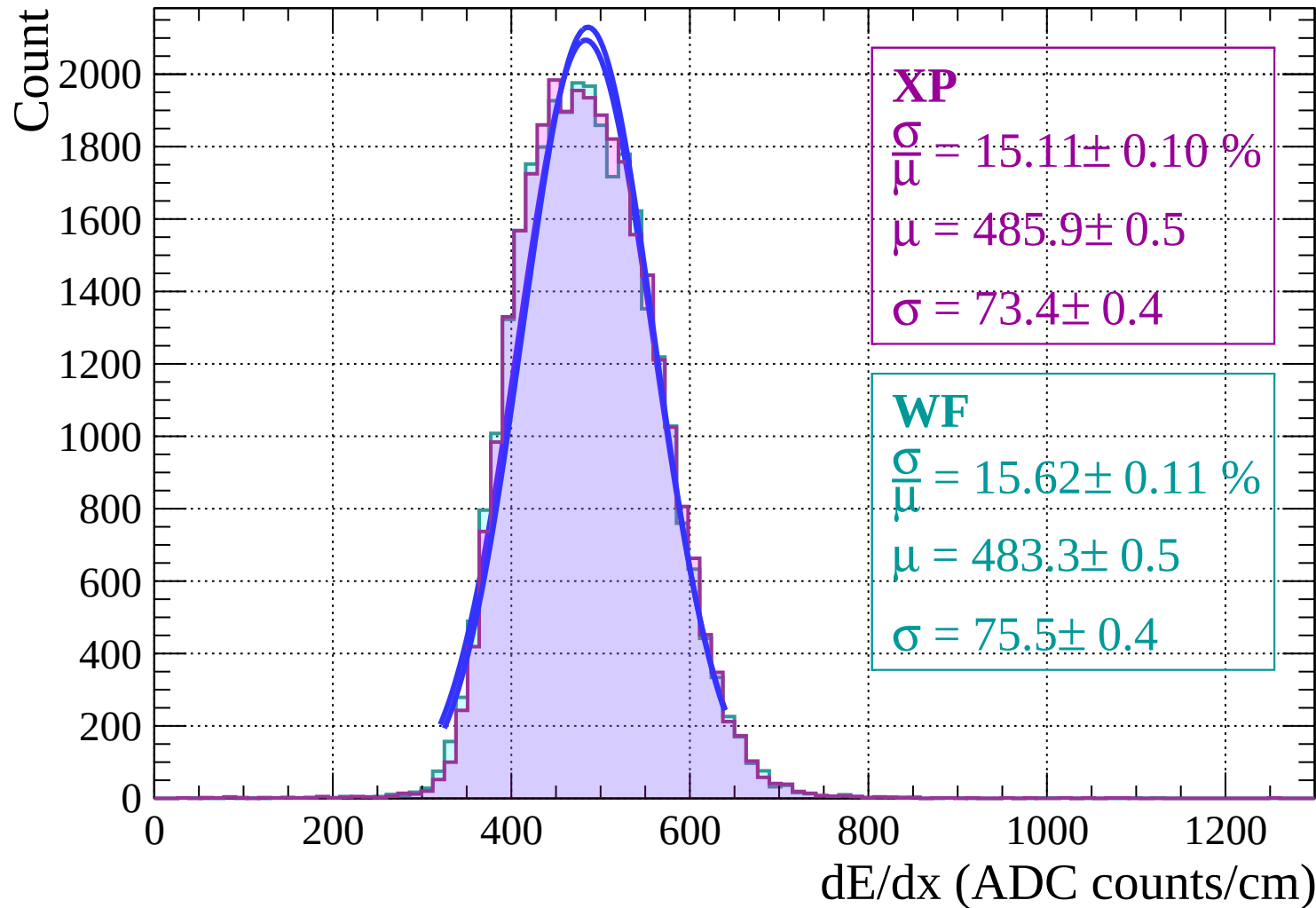
Energy loss | $-12 < \theta < -10$



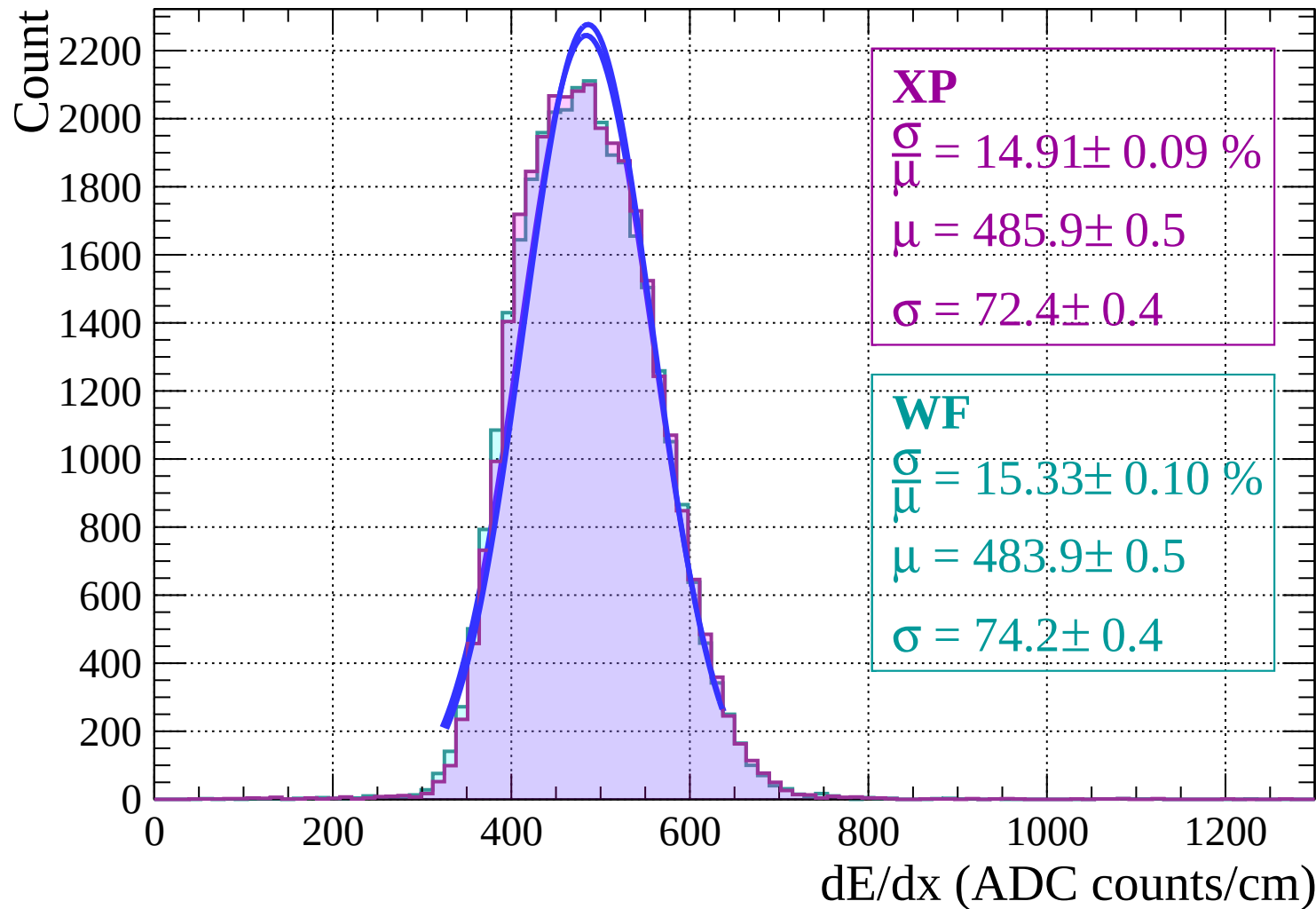
Energy loss | $-10 < \theta < -8$



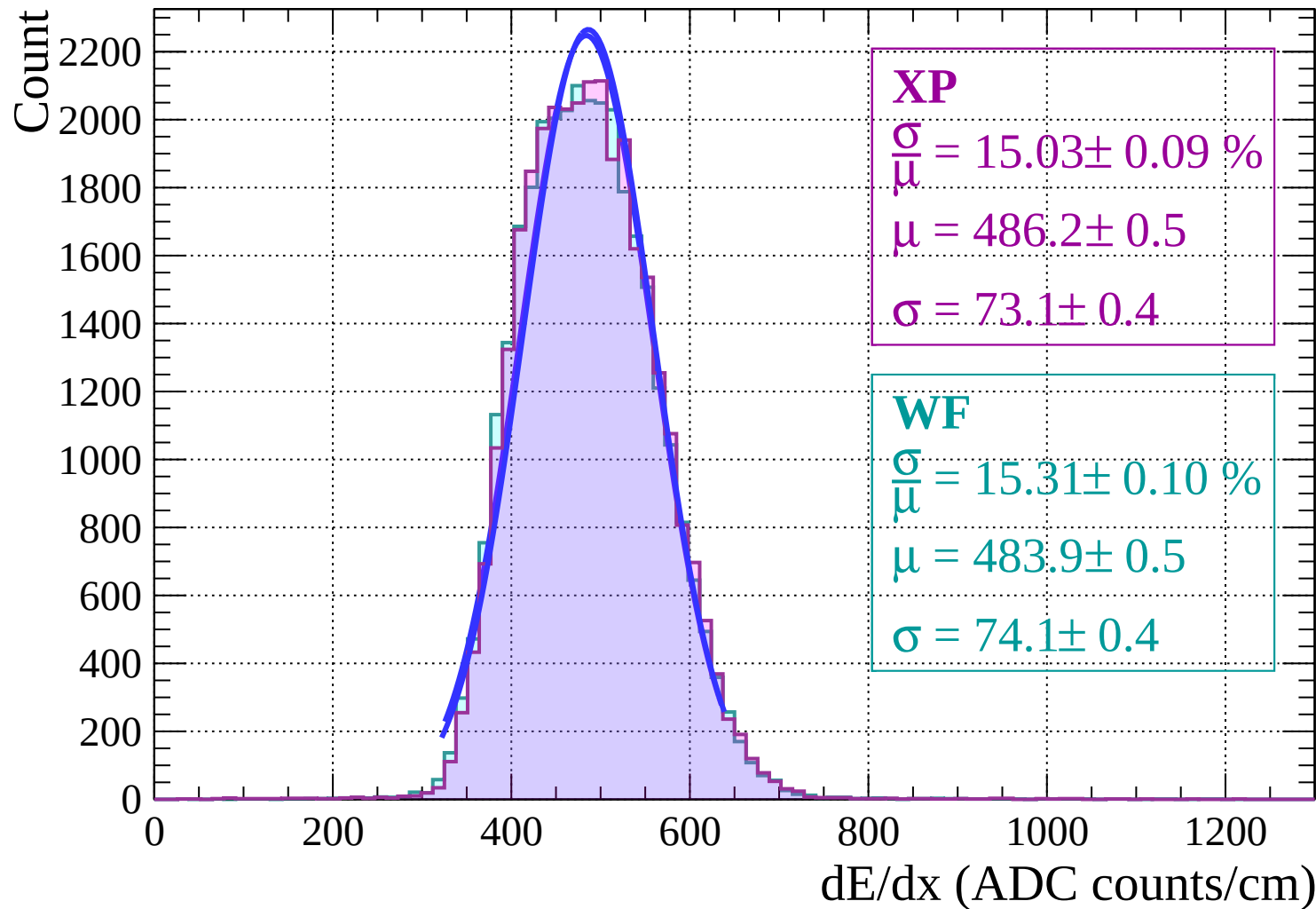
Energy loss | $-8 < \theta < -6$



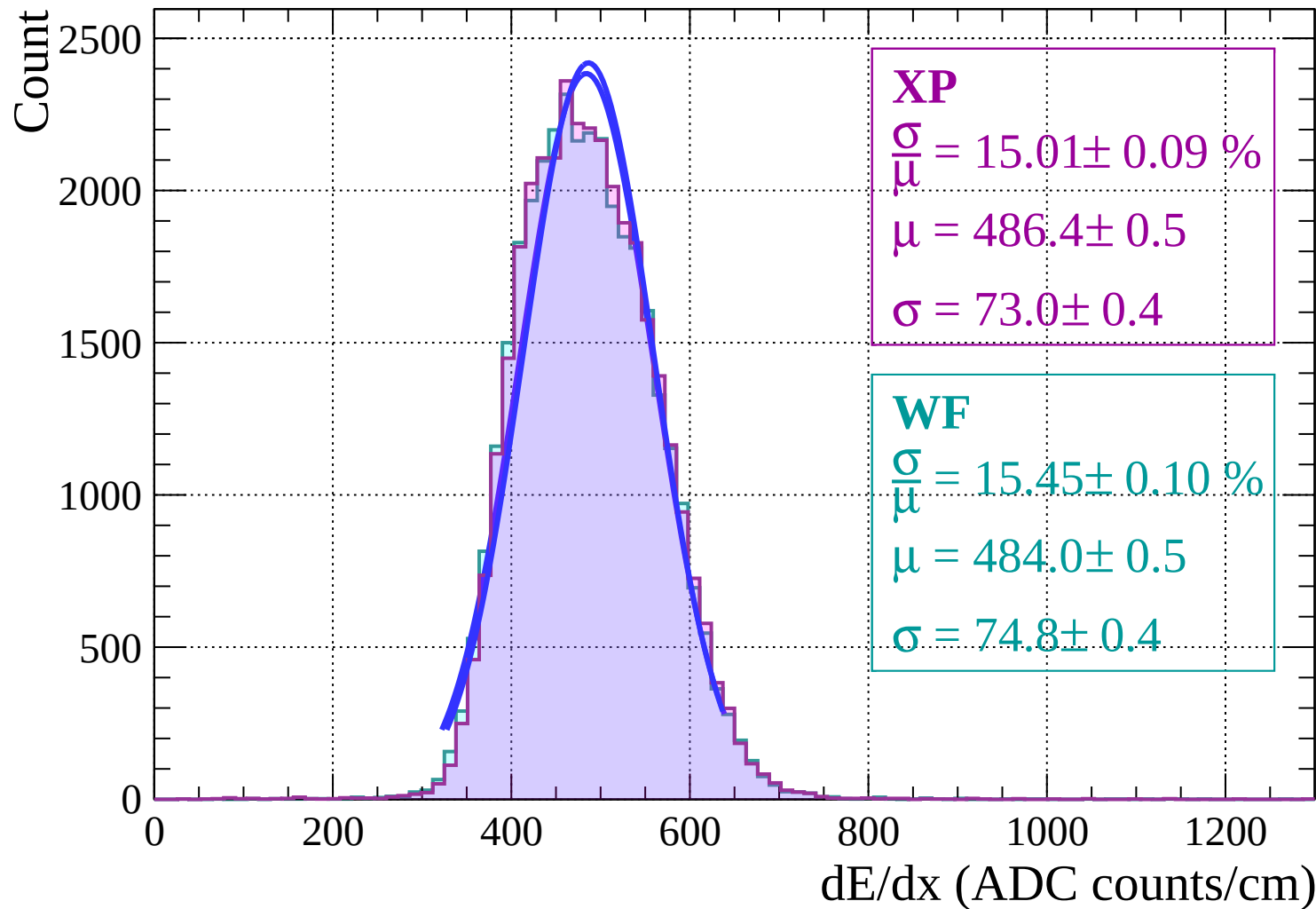
Energy loss | $-6 < \theta < -4$



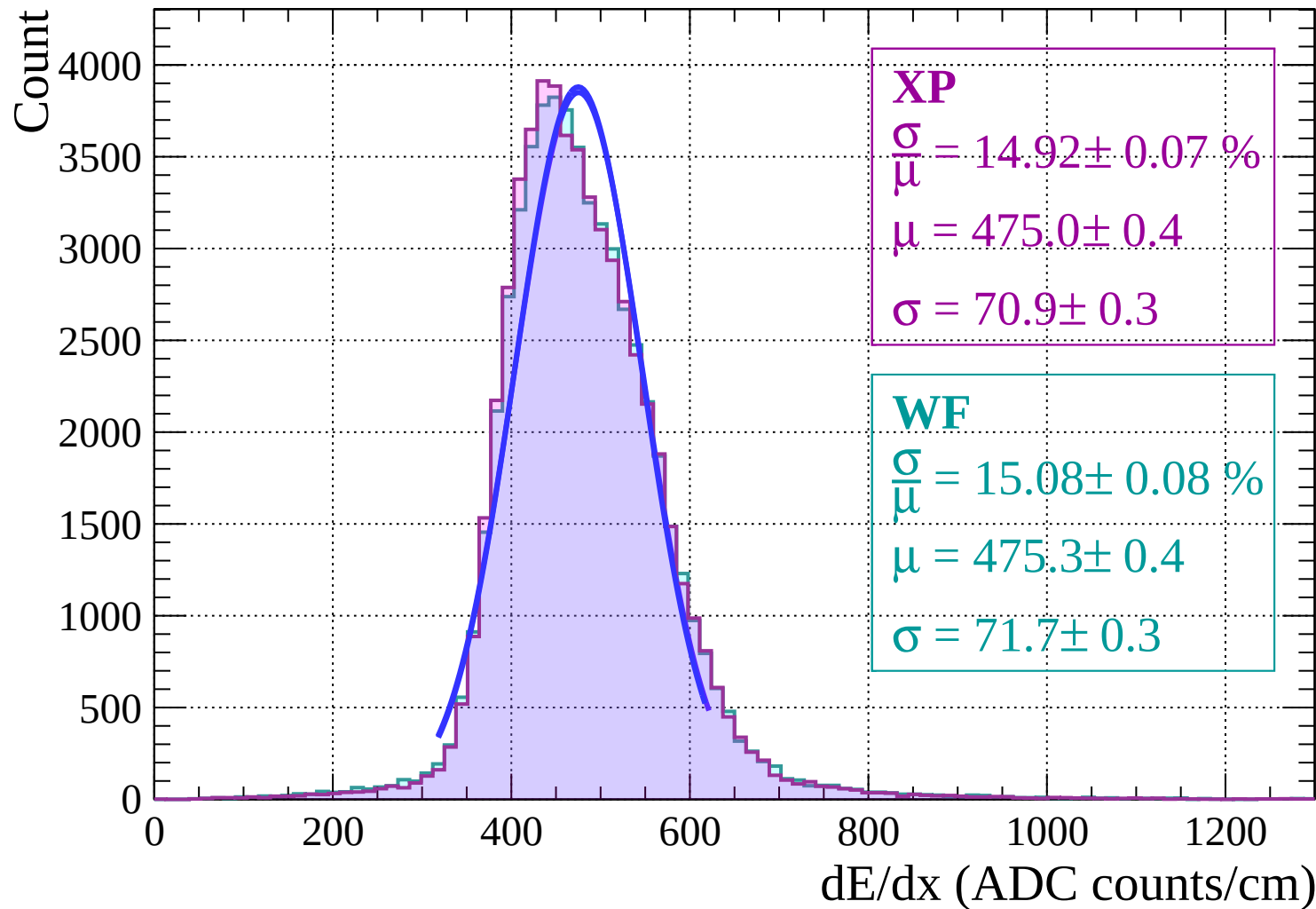
Energy loss | $-4 < \theta < -2$



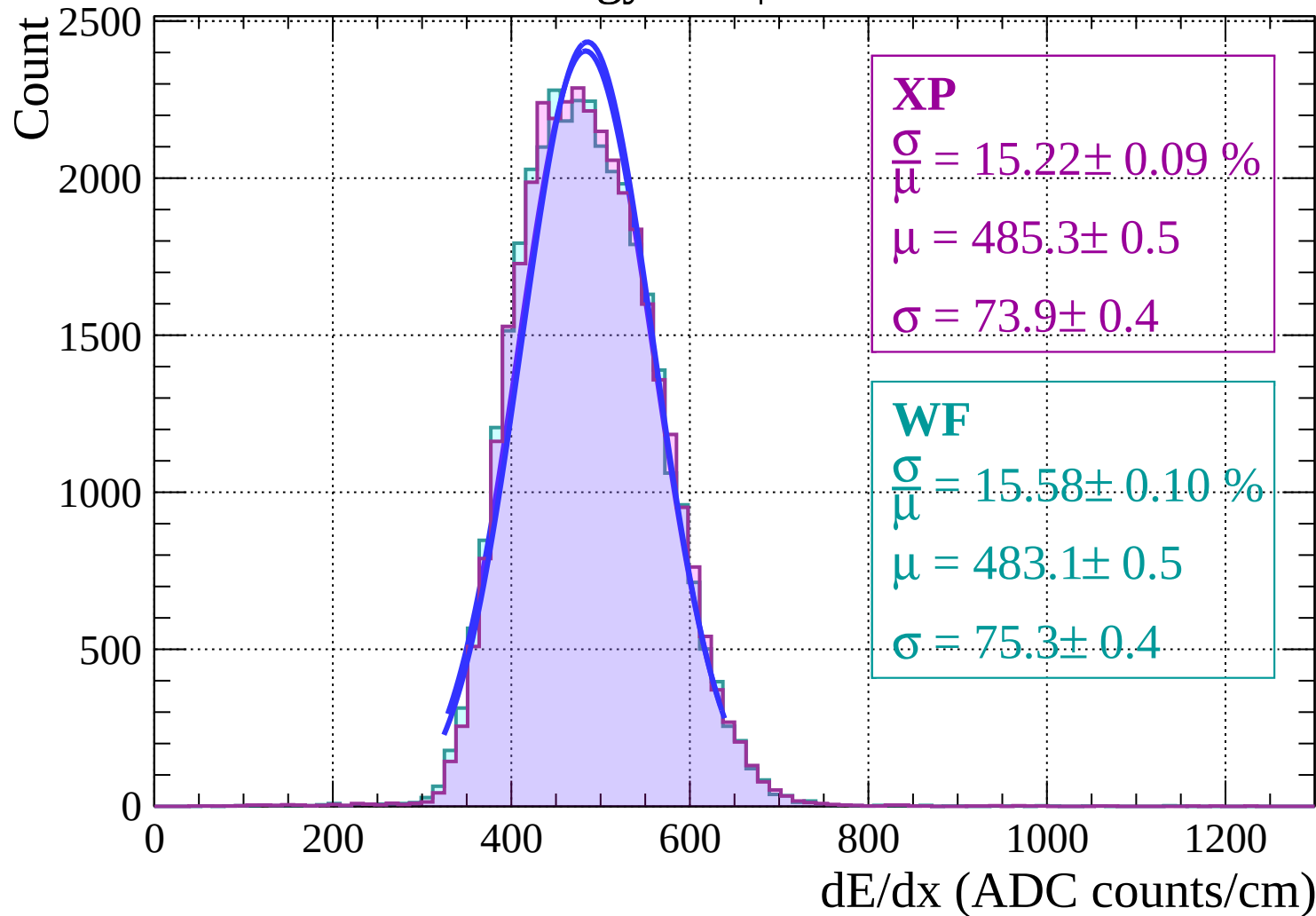
Energy loss | $-2 < \theta < 0$



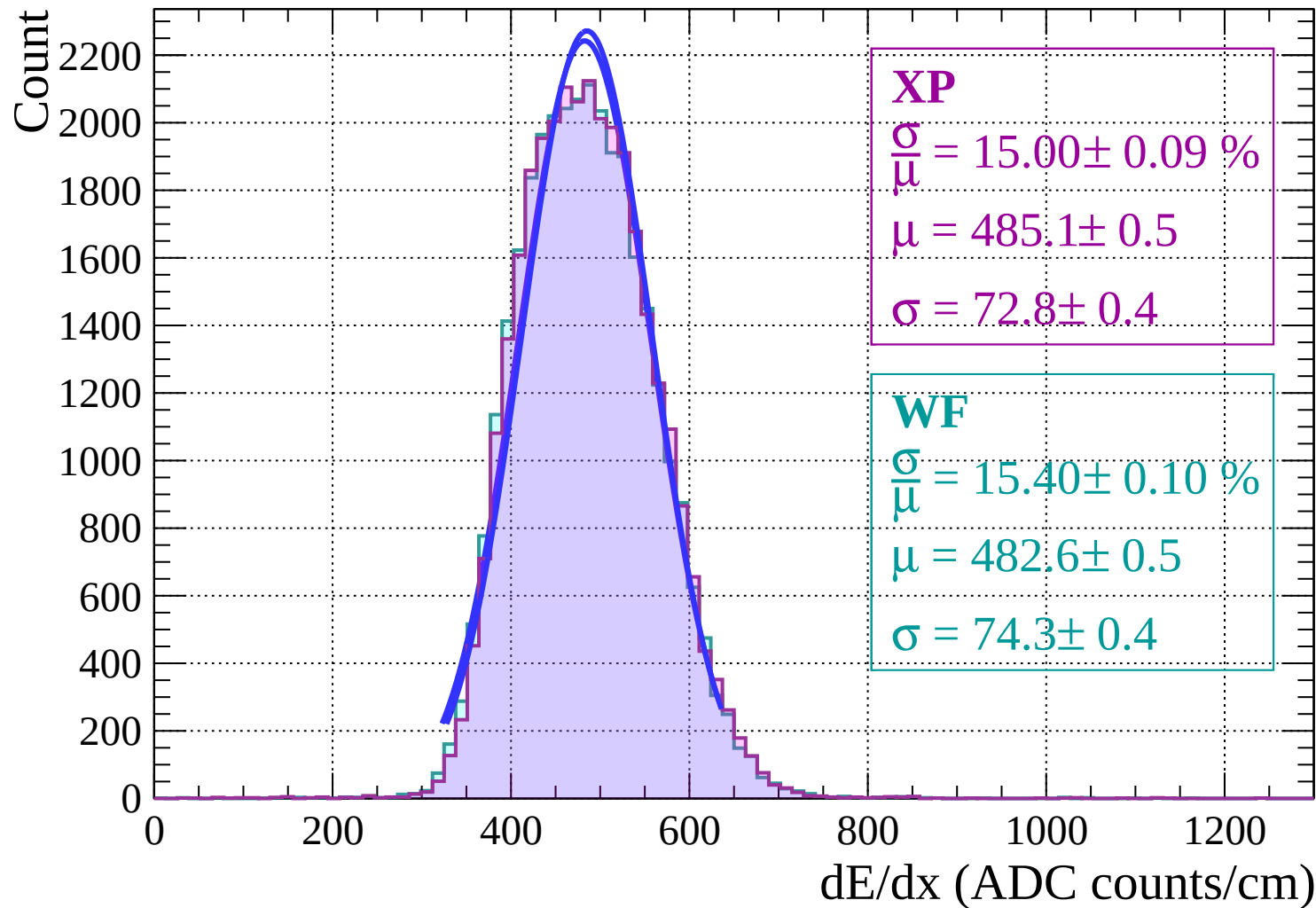
Energy loss | $0 < \theta < 2$



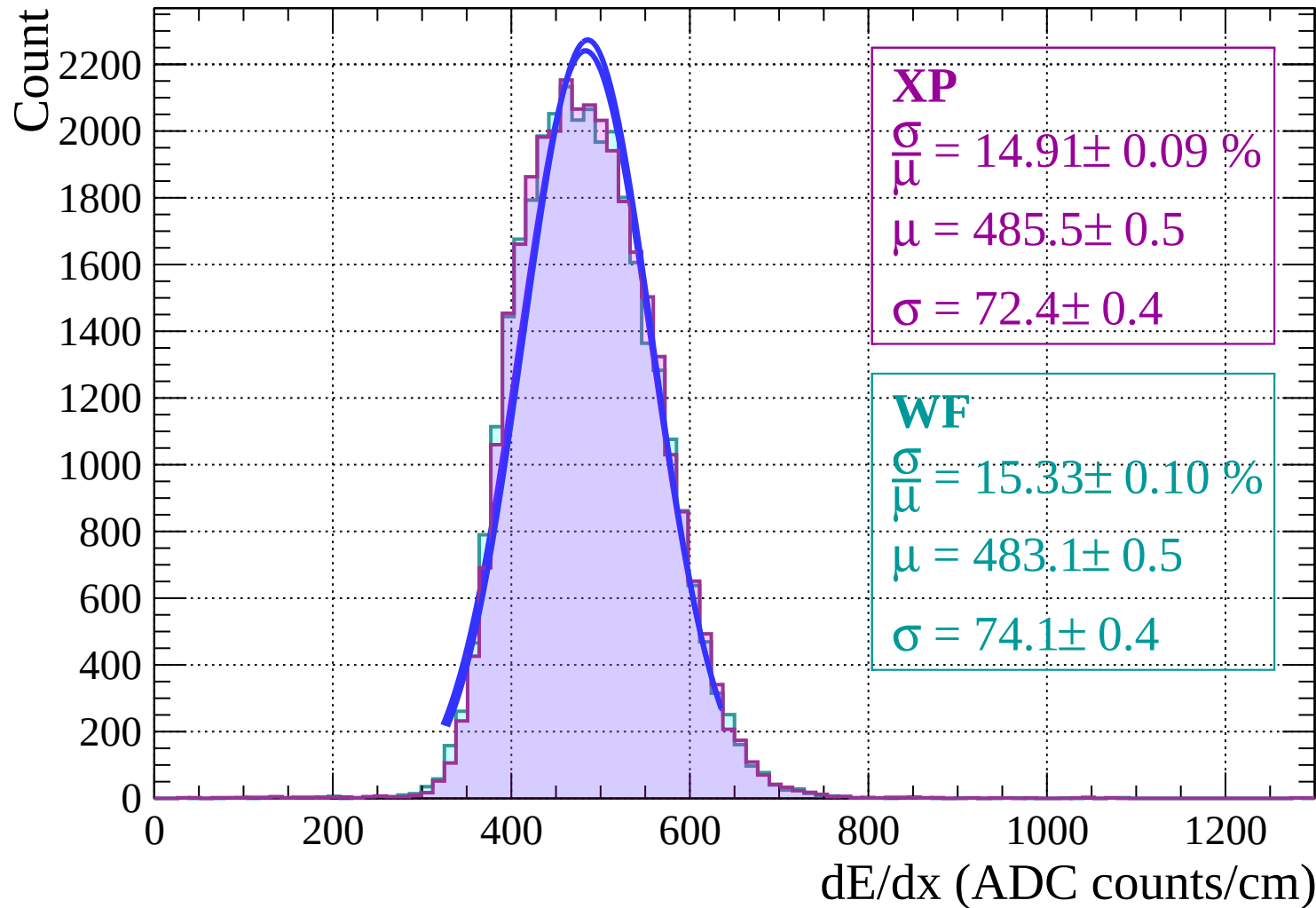
Energy loss | $2 < \theta < 4$



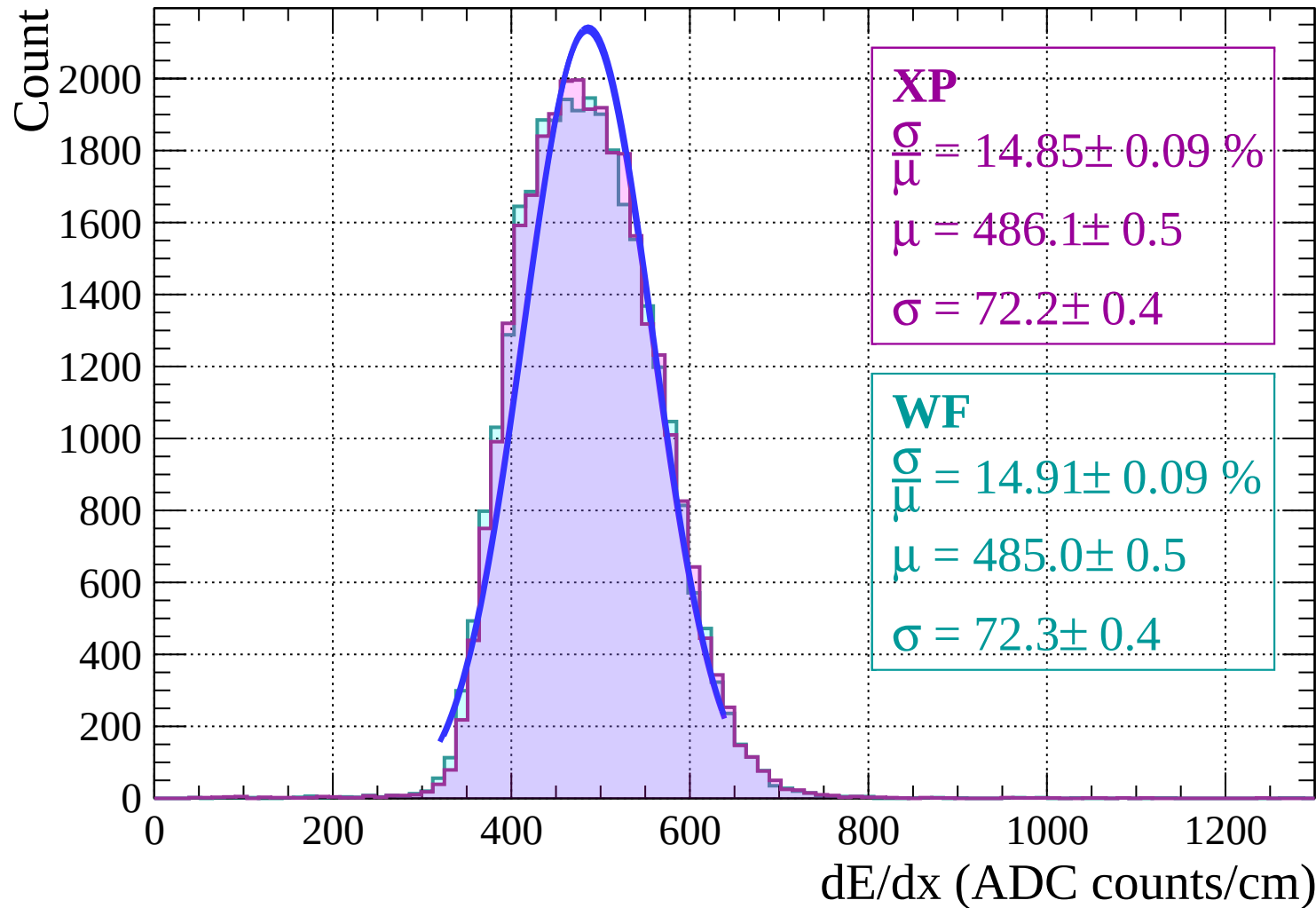
Energy loss | $4 < \theta < 6$



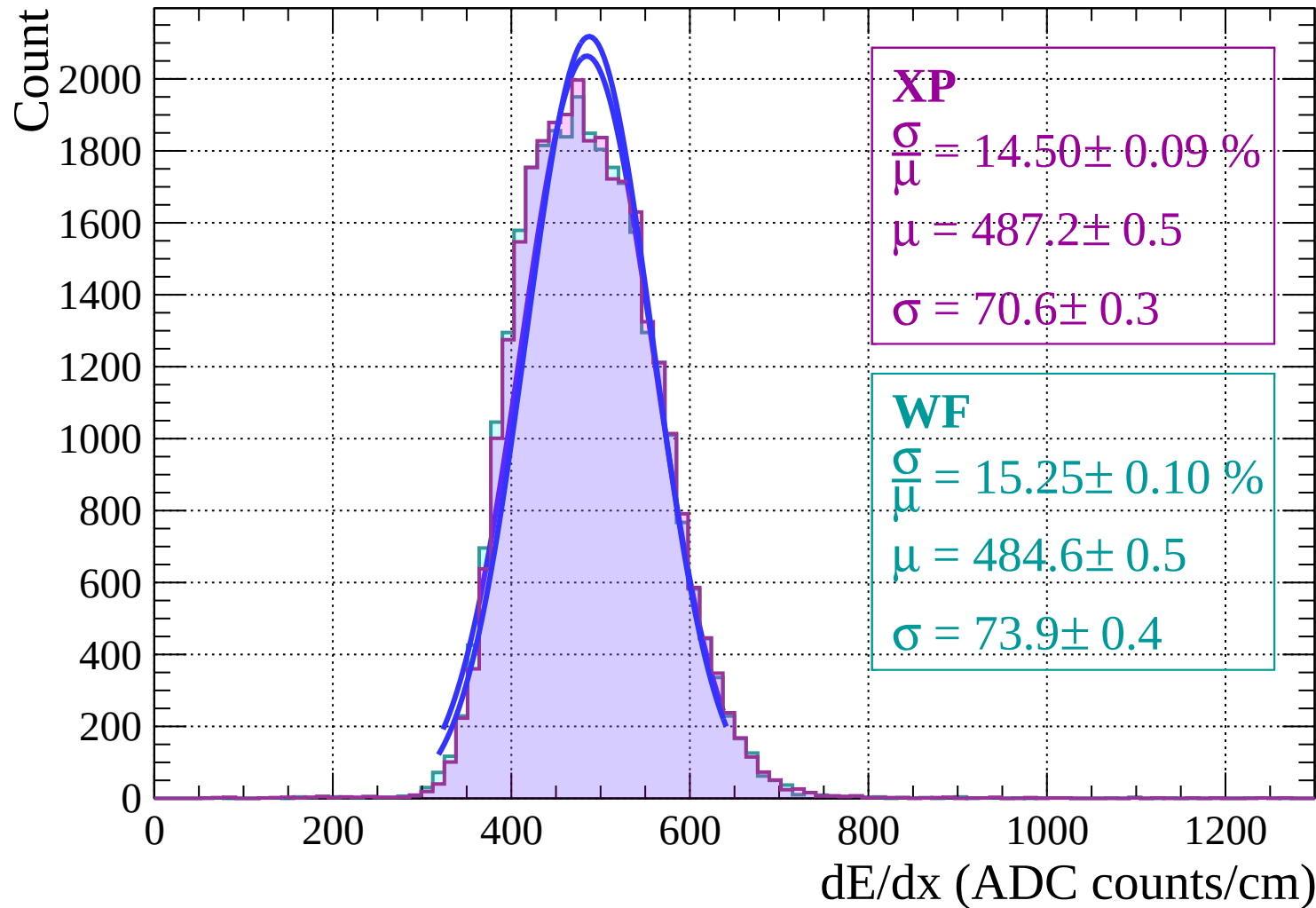
Energy loss | $6 < \theta < 8$



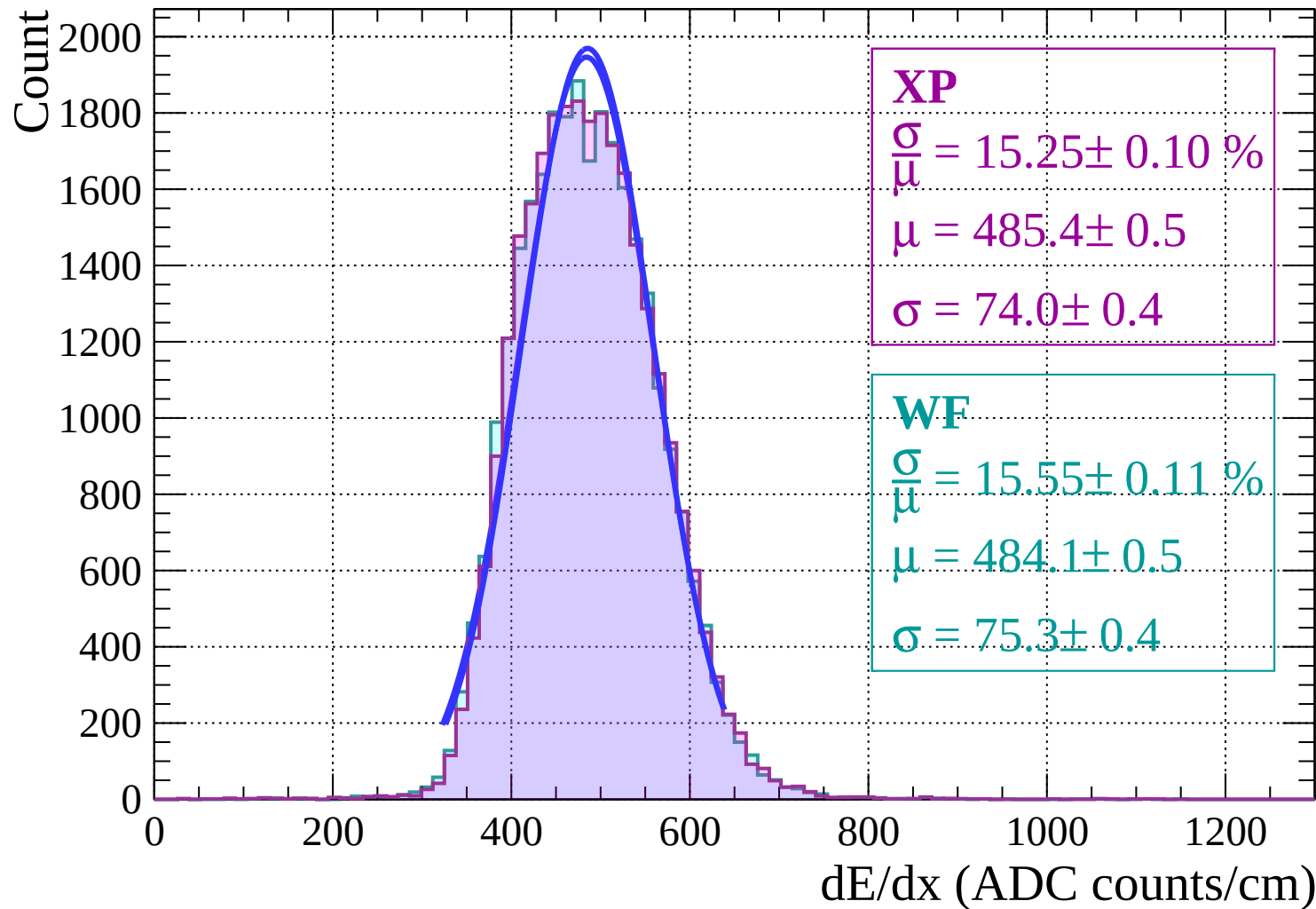
Energy loss | $8 < \theta < 10$



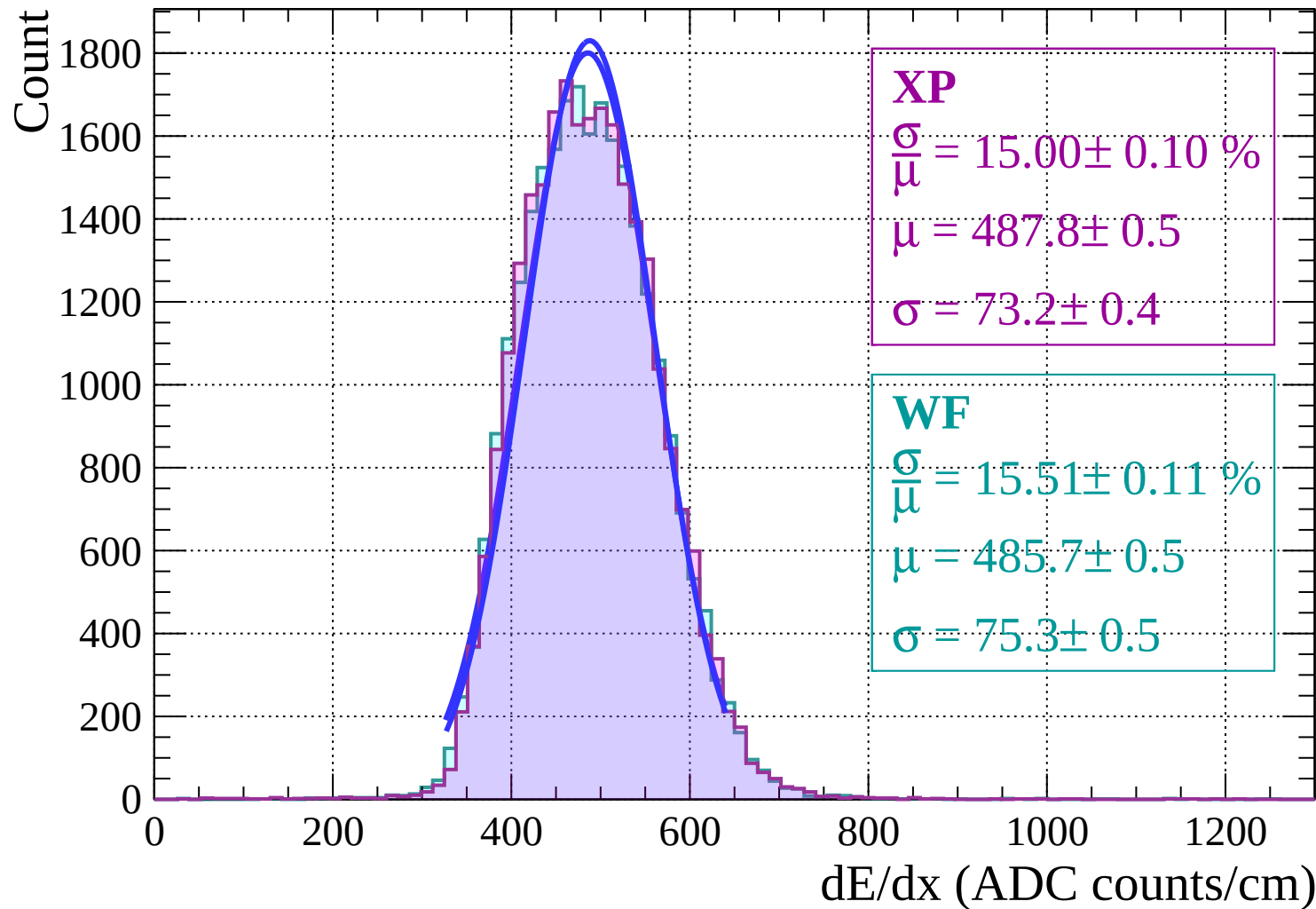
Energy loss | $10 < \theta < 12$



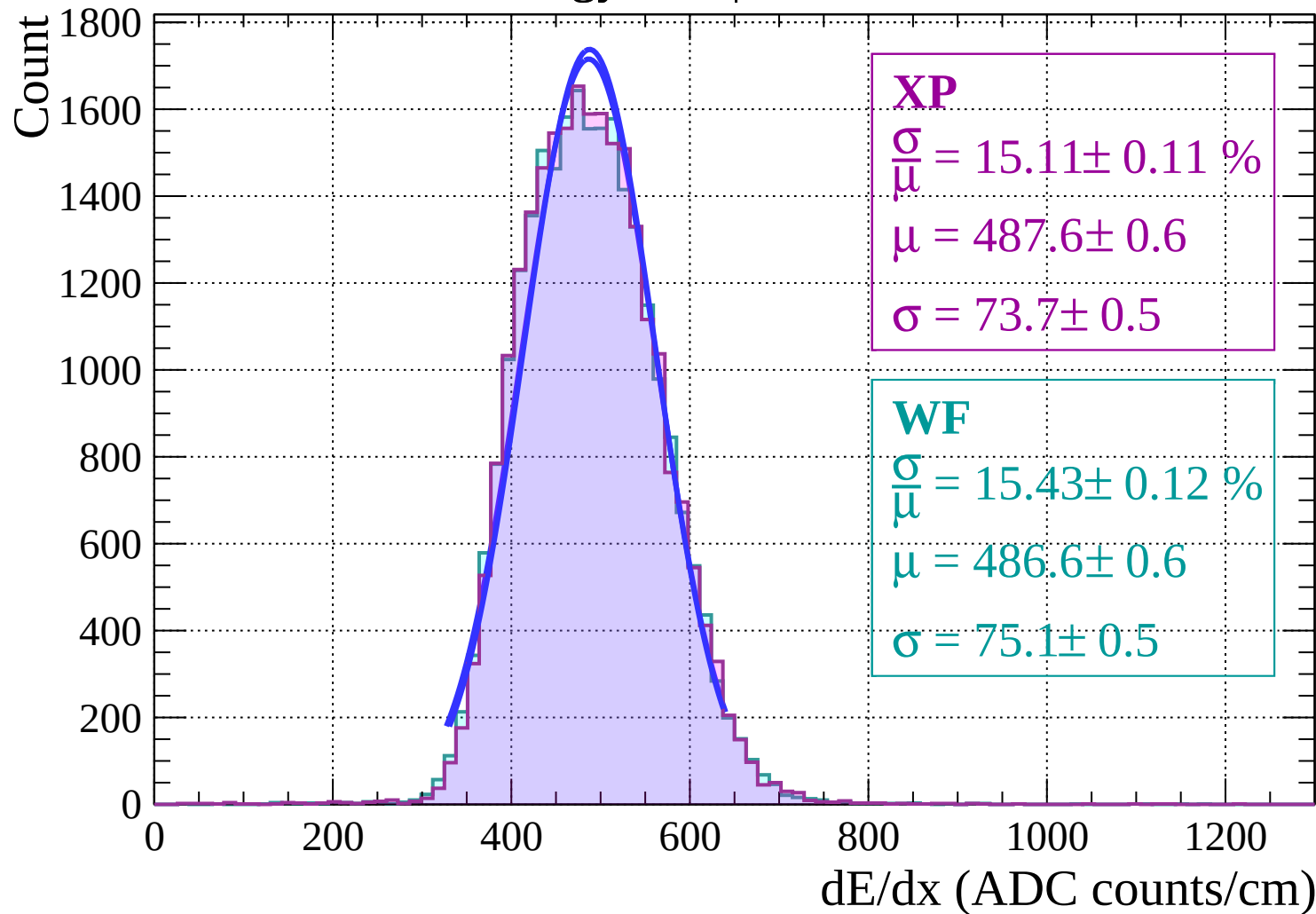
Energy loss | $12 < \theta < 14$



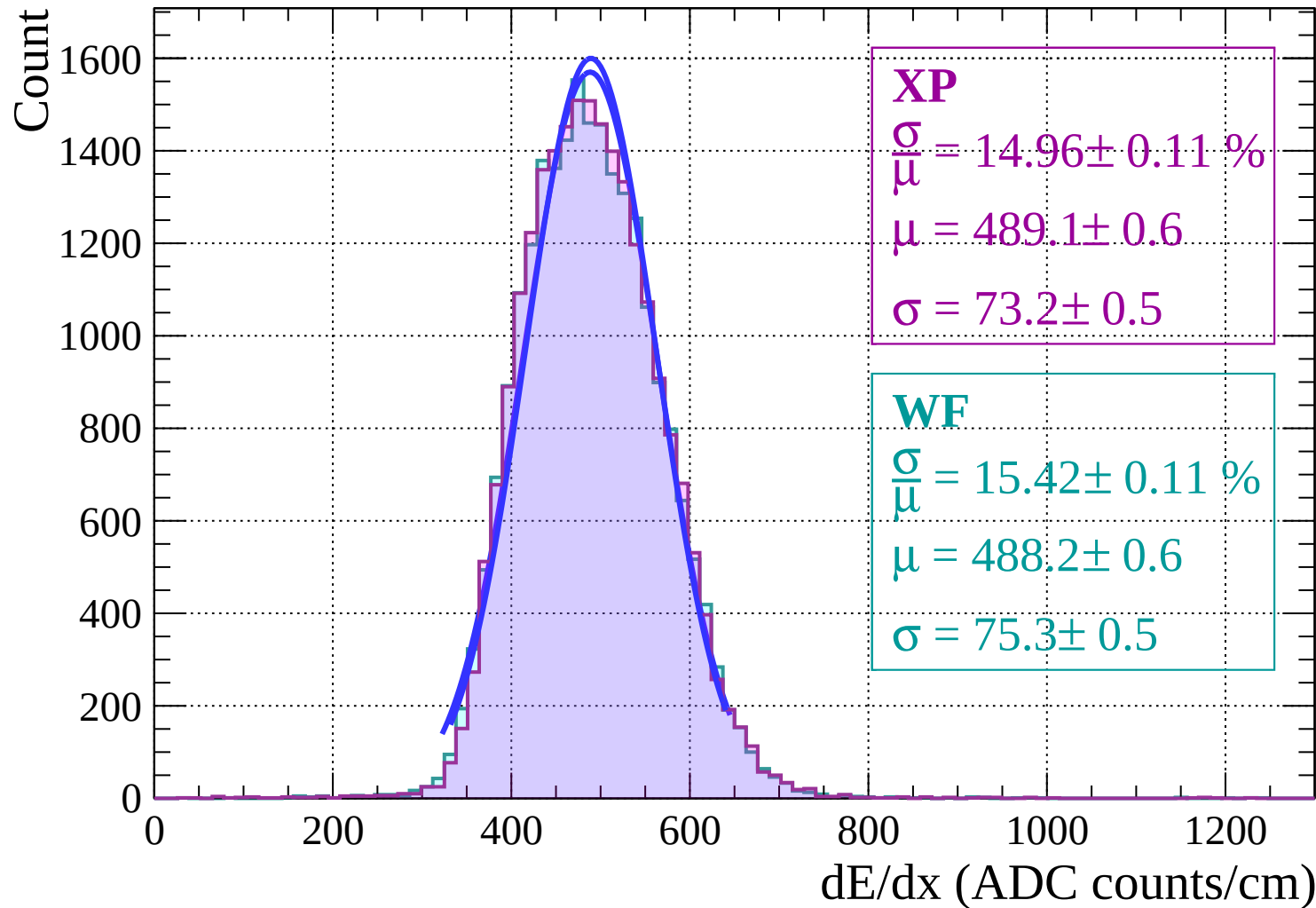
Energy loss | $14 < \theta < 16$



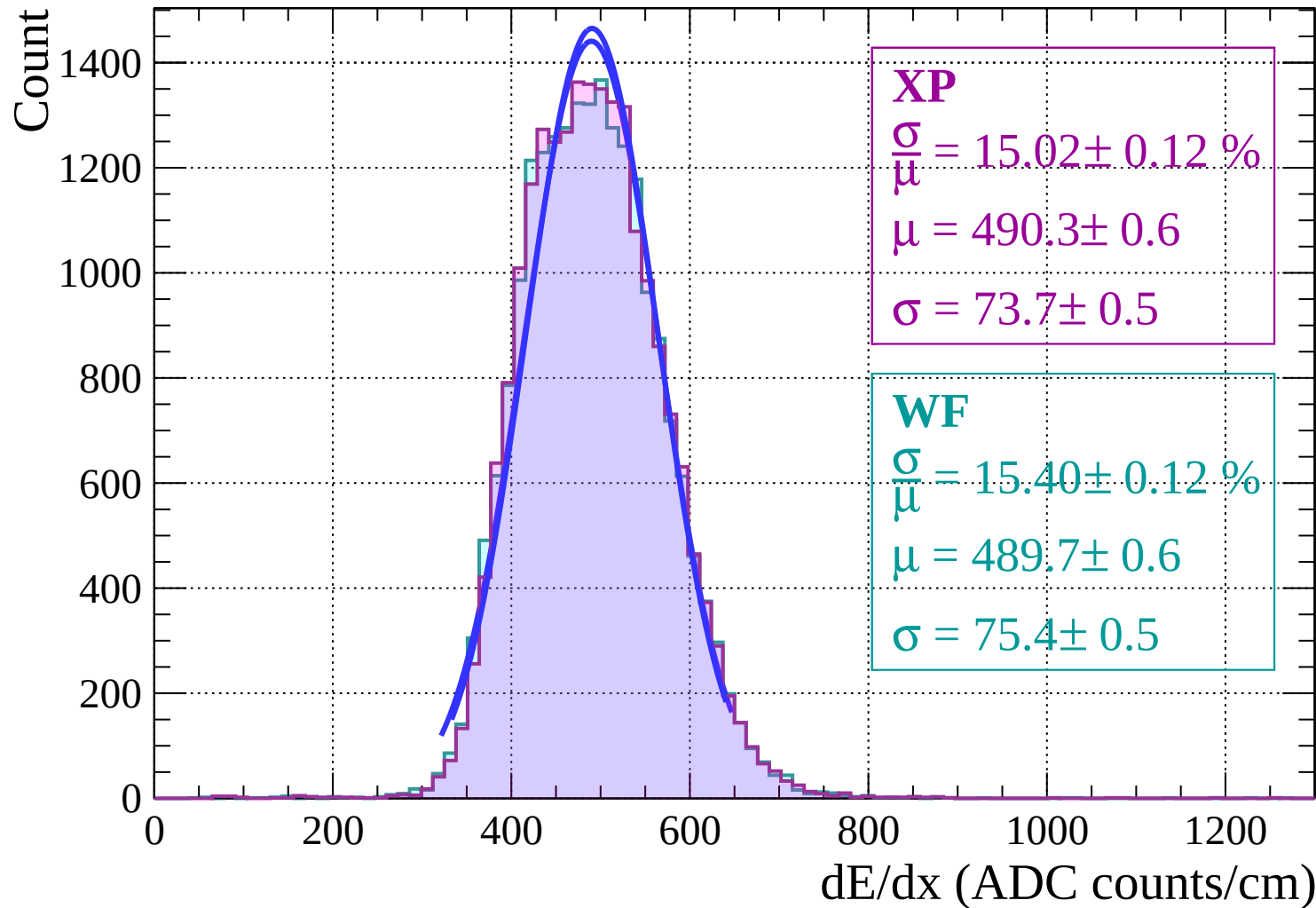
Energy loss | $16 < \theta < 18$



Energy loss | $18 < \theta < 20$

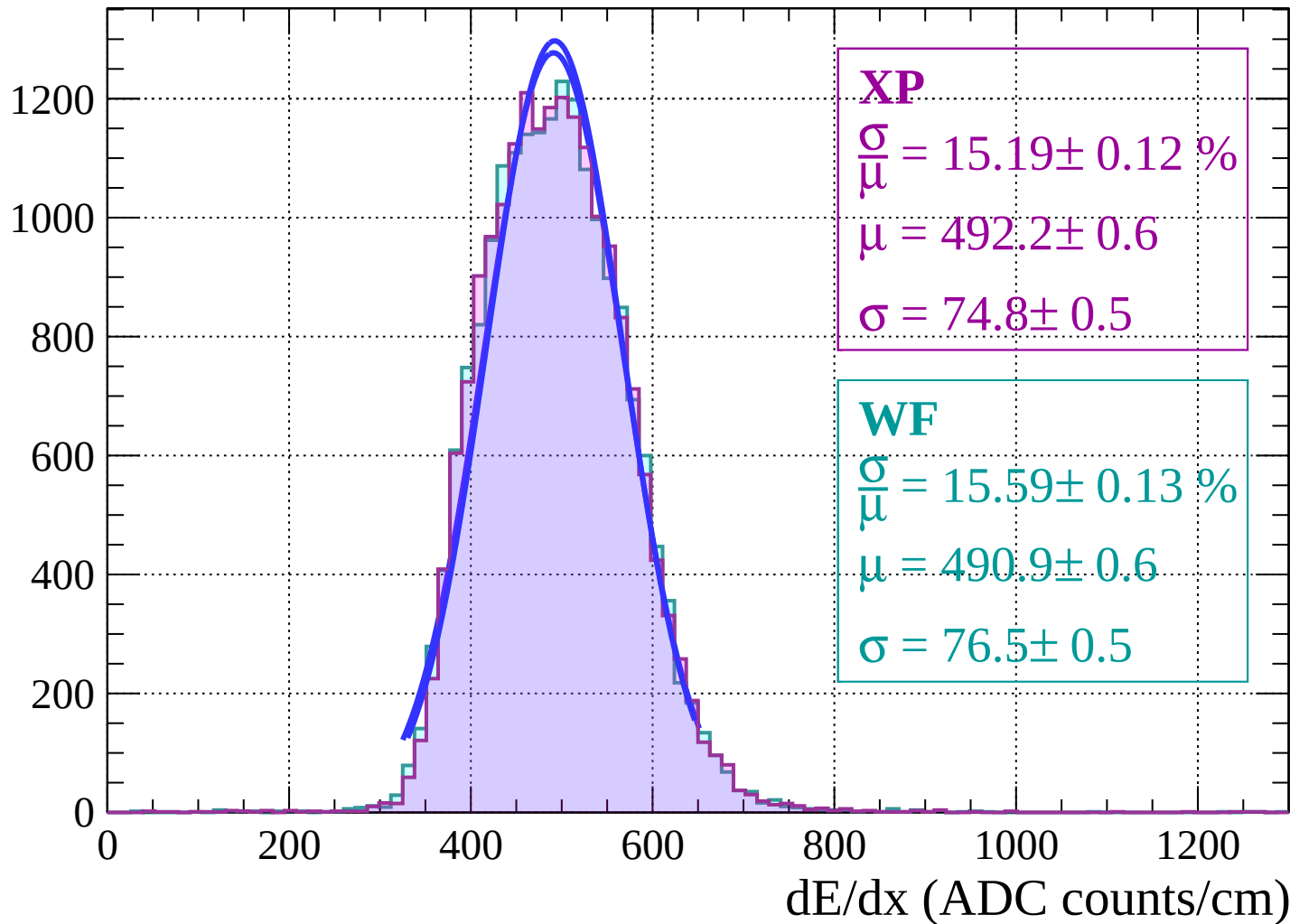


Energy loss | $20 < \theta < 22$

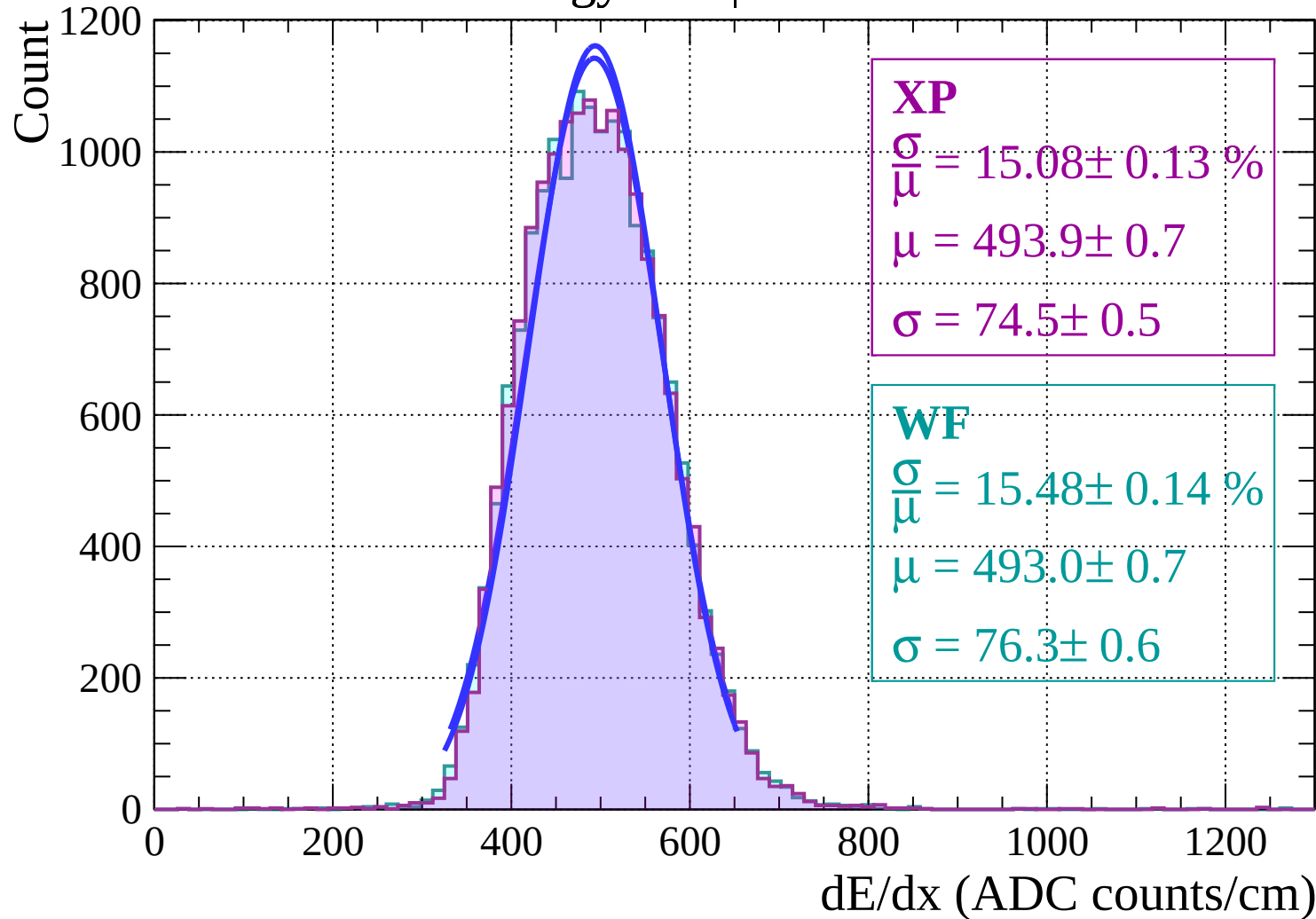


Energy loss | $22 < \theta < 24$

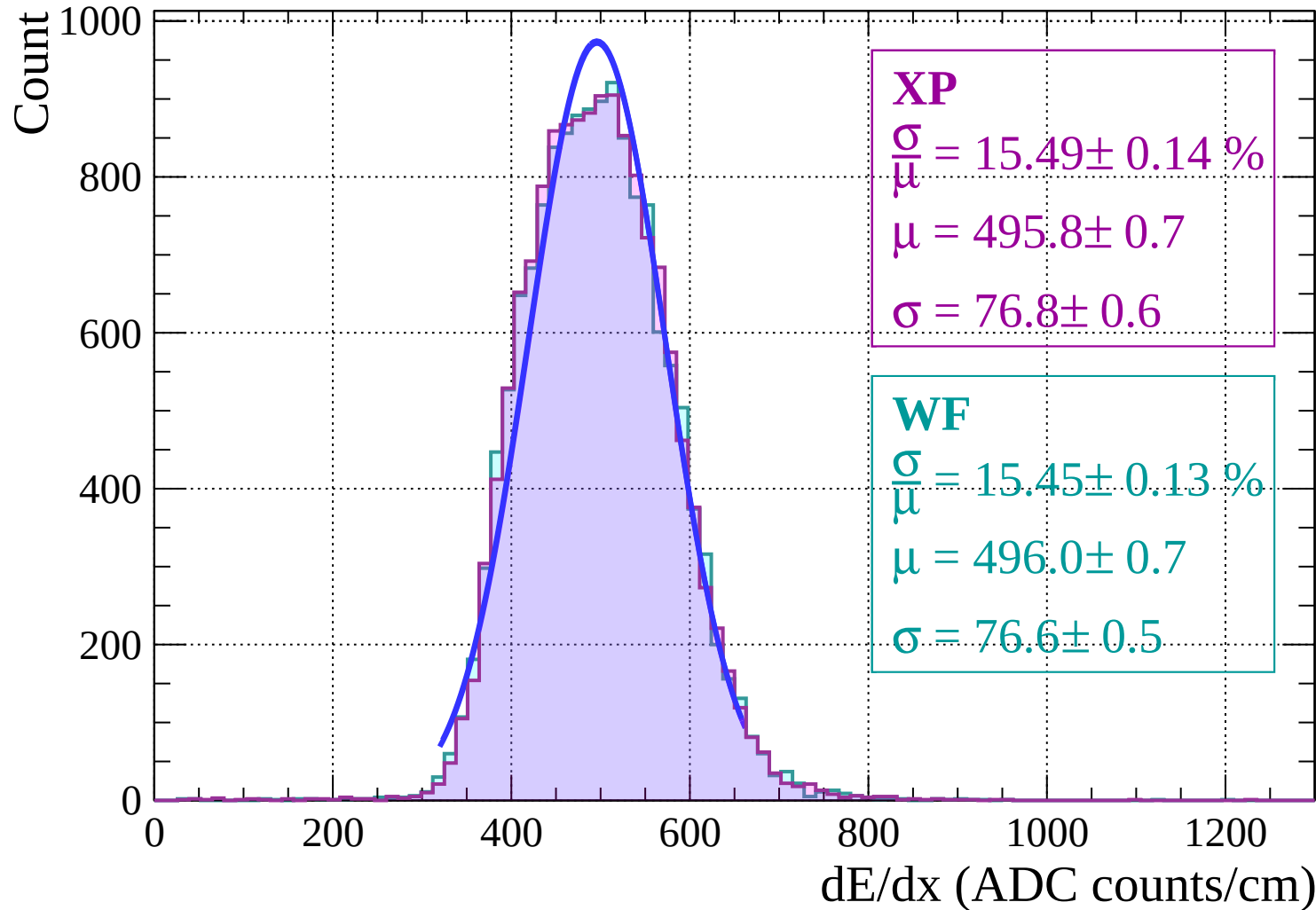
Count



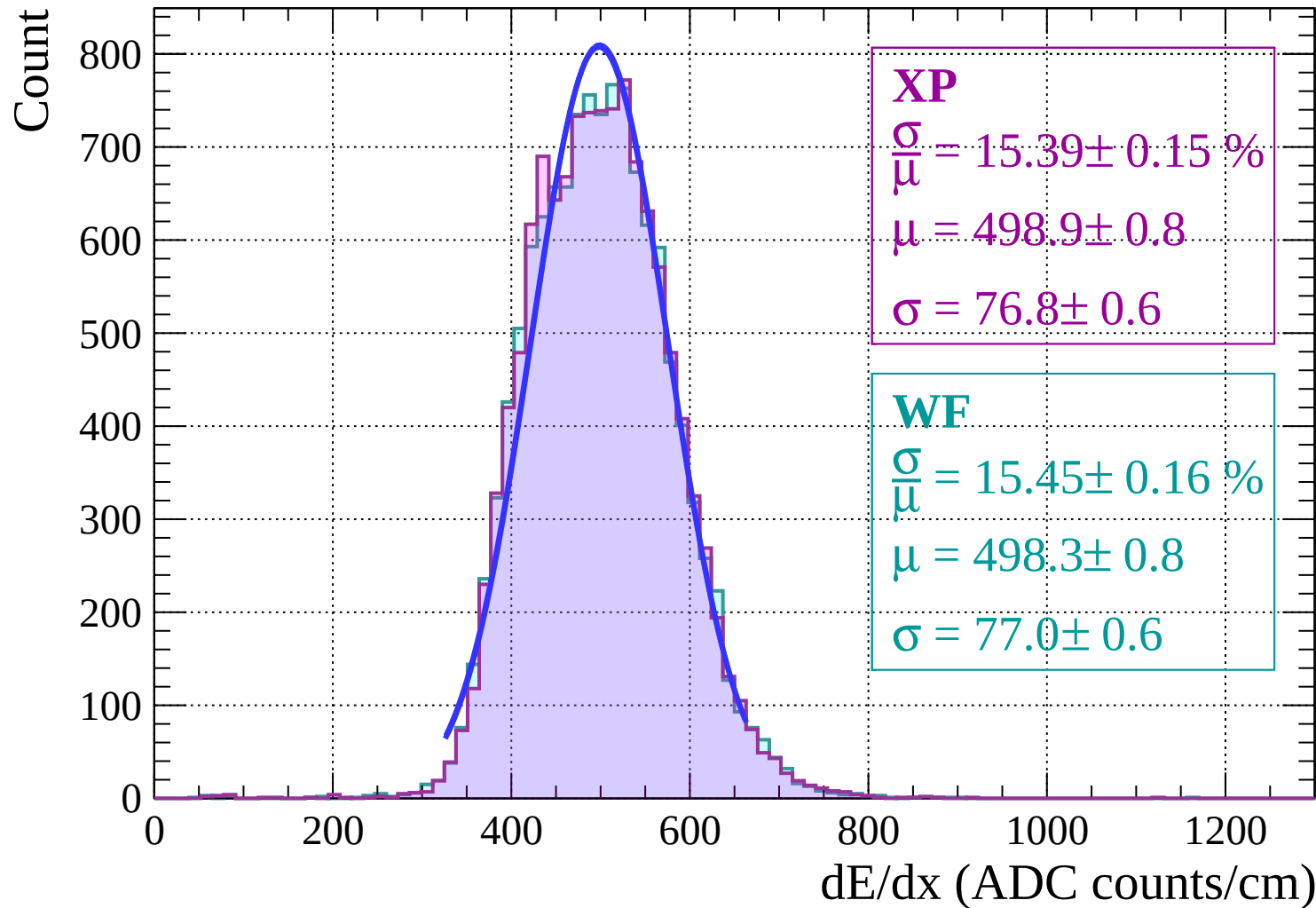
Energy loss | $24 < \theta < 26$



Energy loss | $26 < \theta < 28$

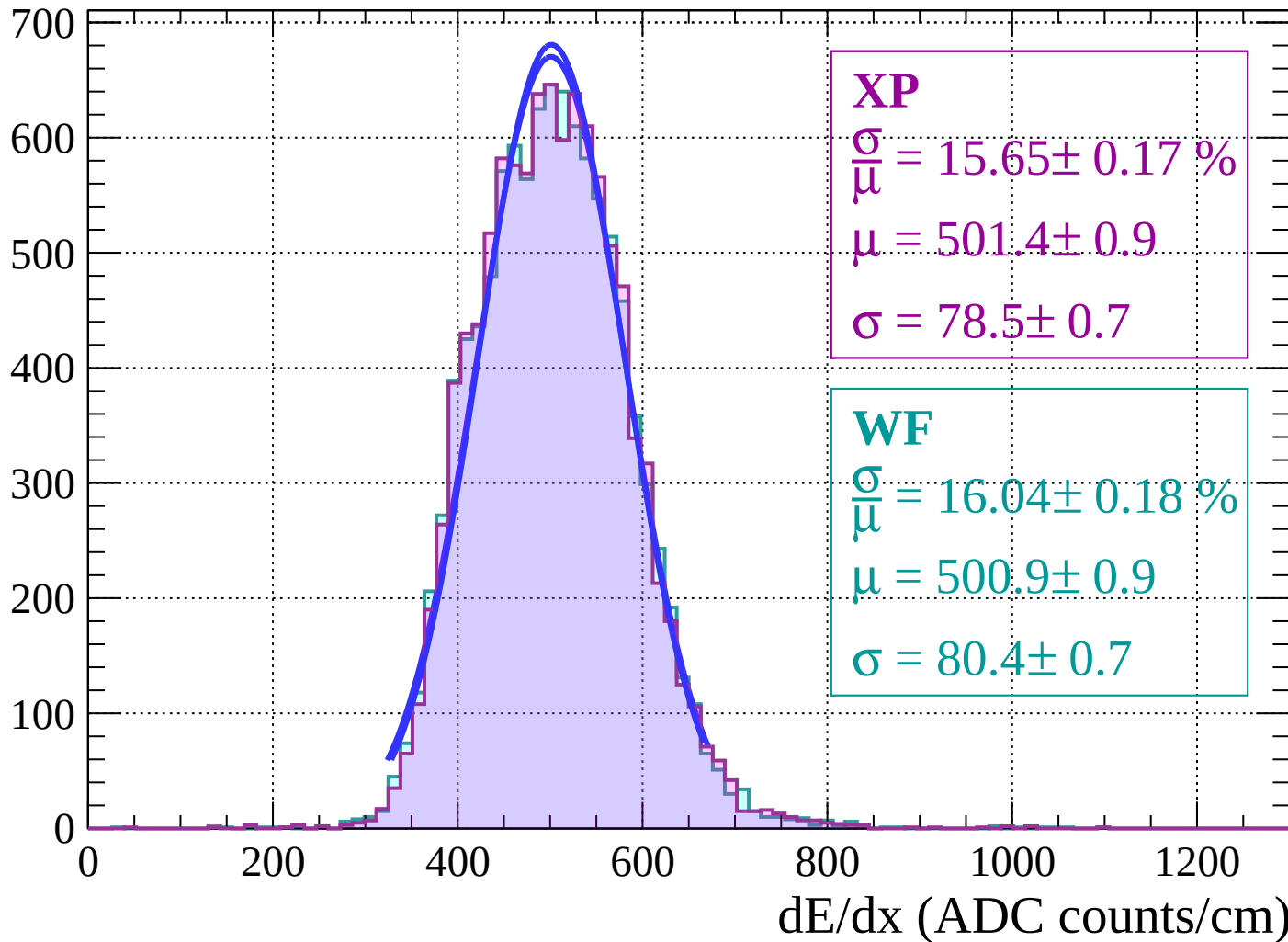


Energy loss | $28 < \theta < 30$



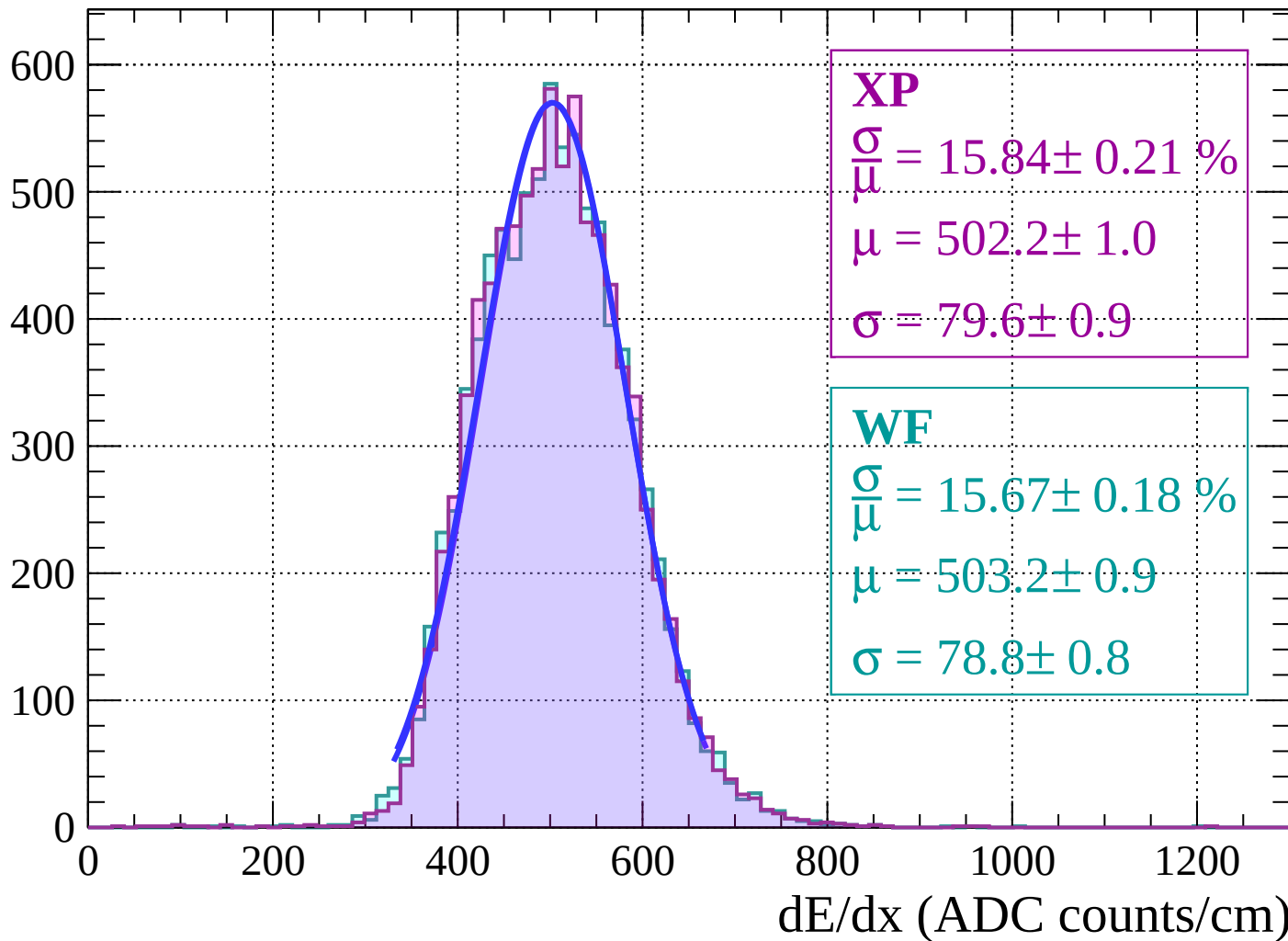
Energy loss | $30 < \theta < 32$

Count



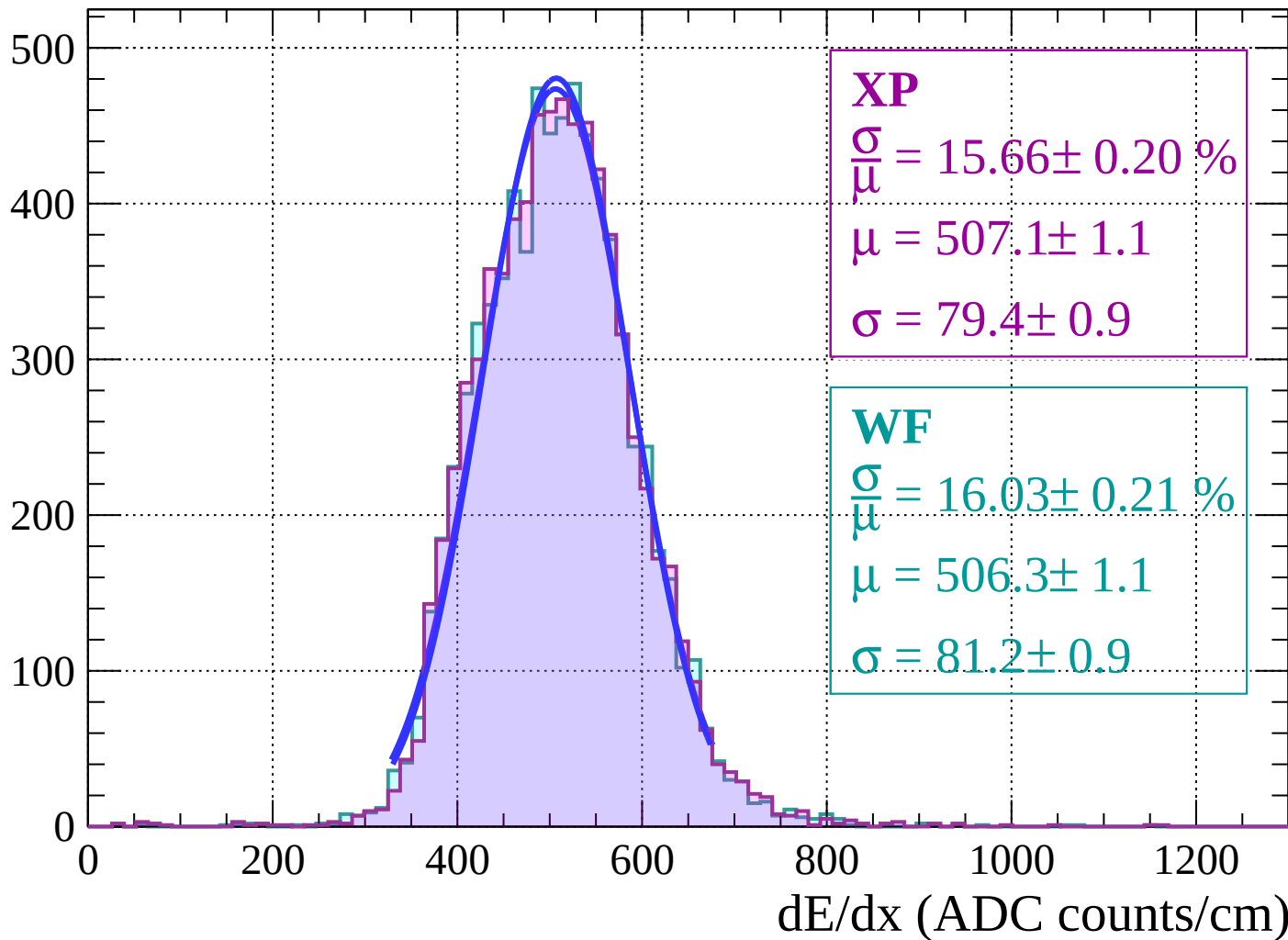
Energy loss | $32 < \theta < 34$

Count



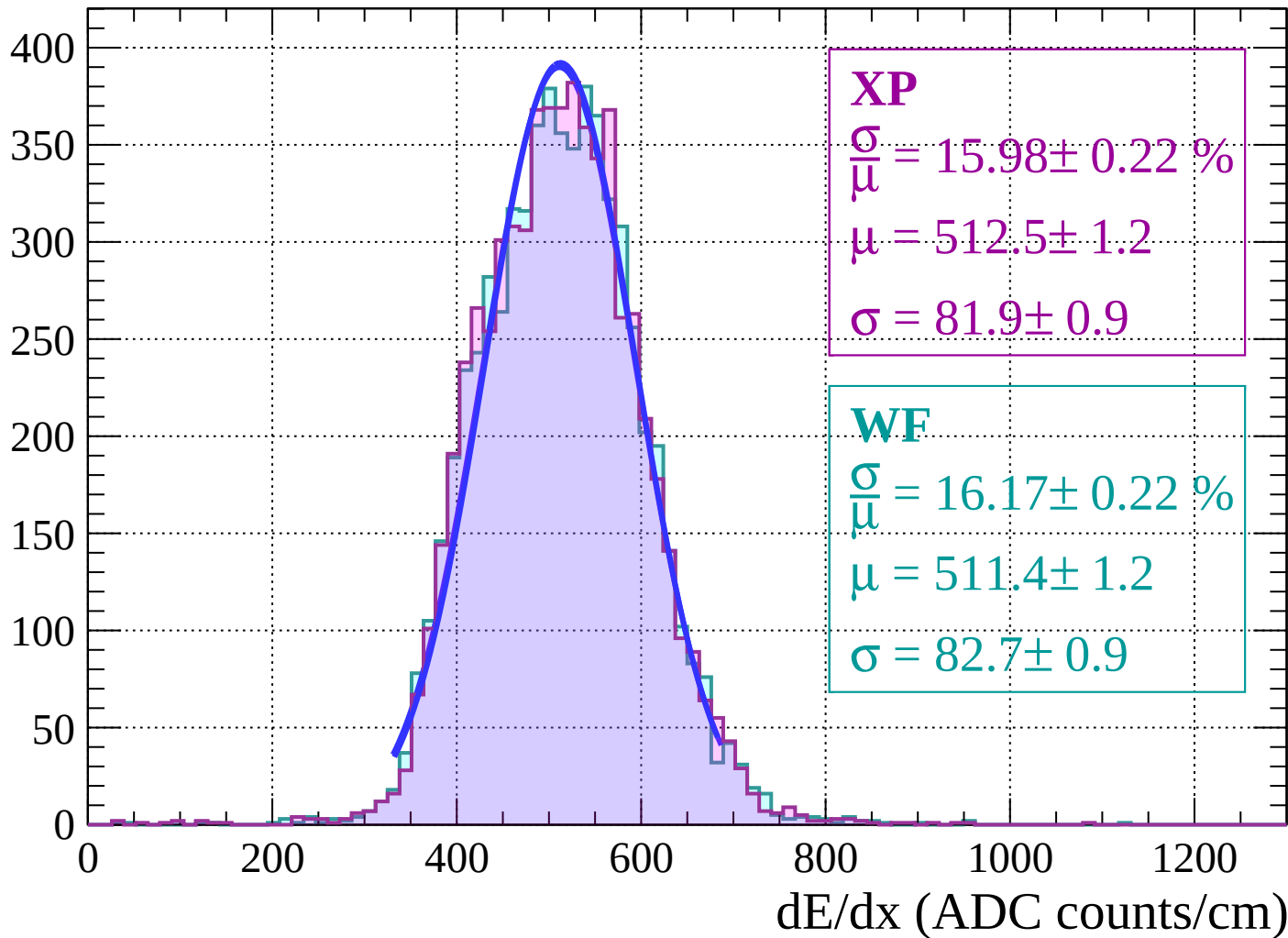
Energy loss | $34 < \theta < 36$

Count

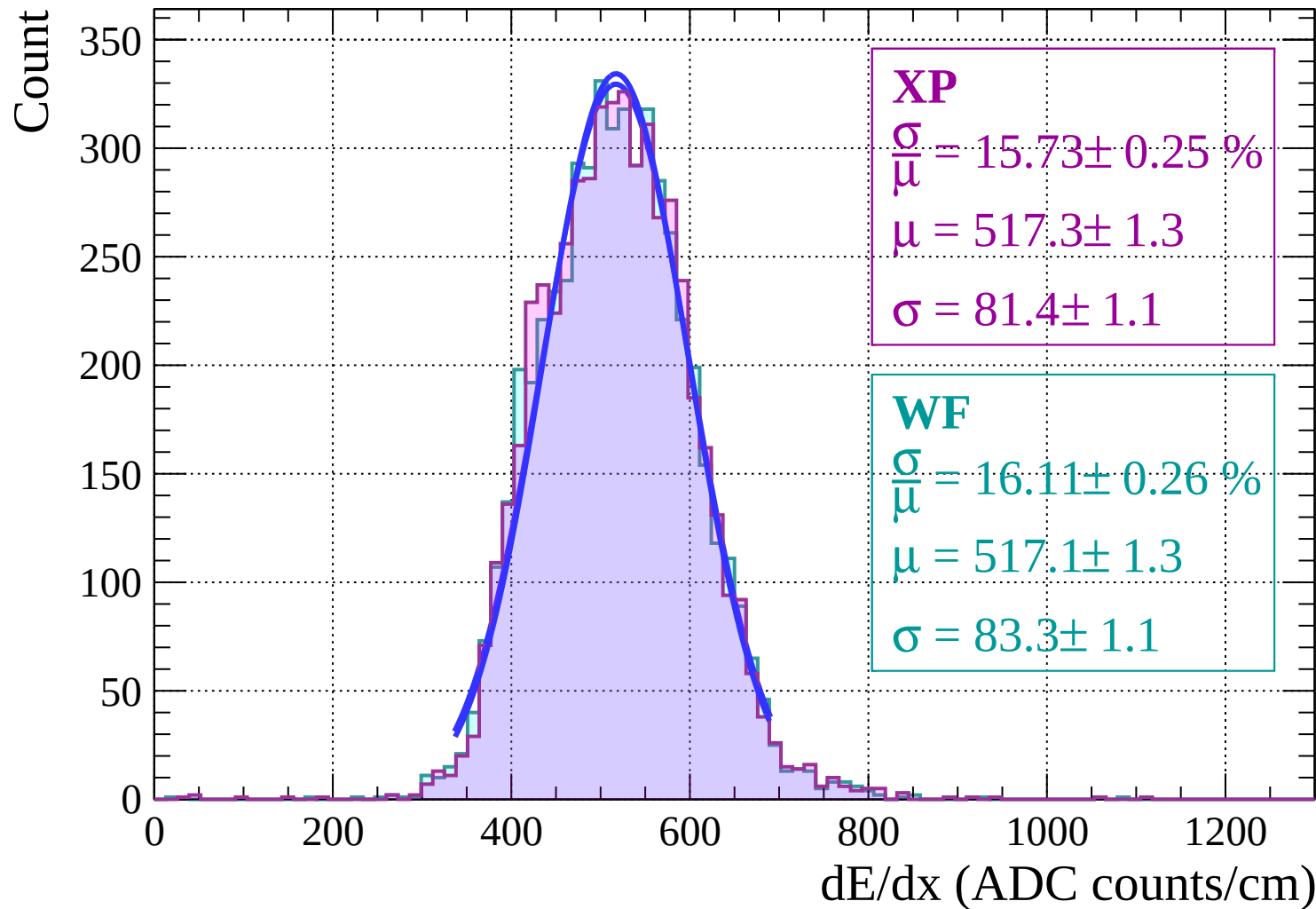


Energy loss | $36 < \theta < 38$

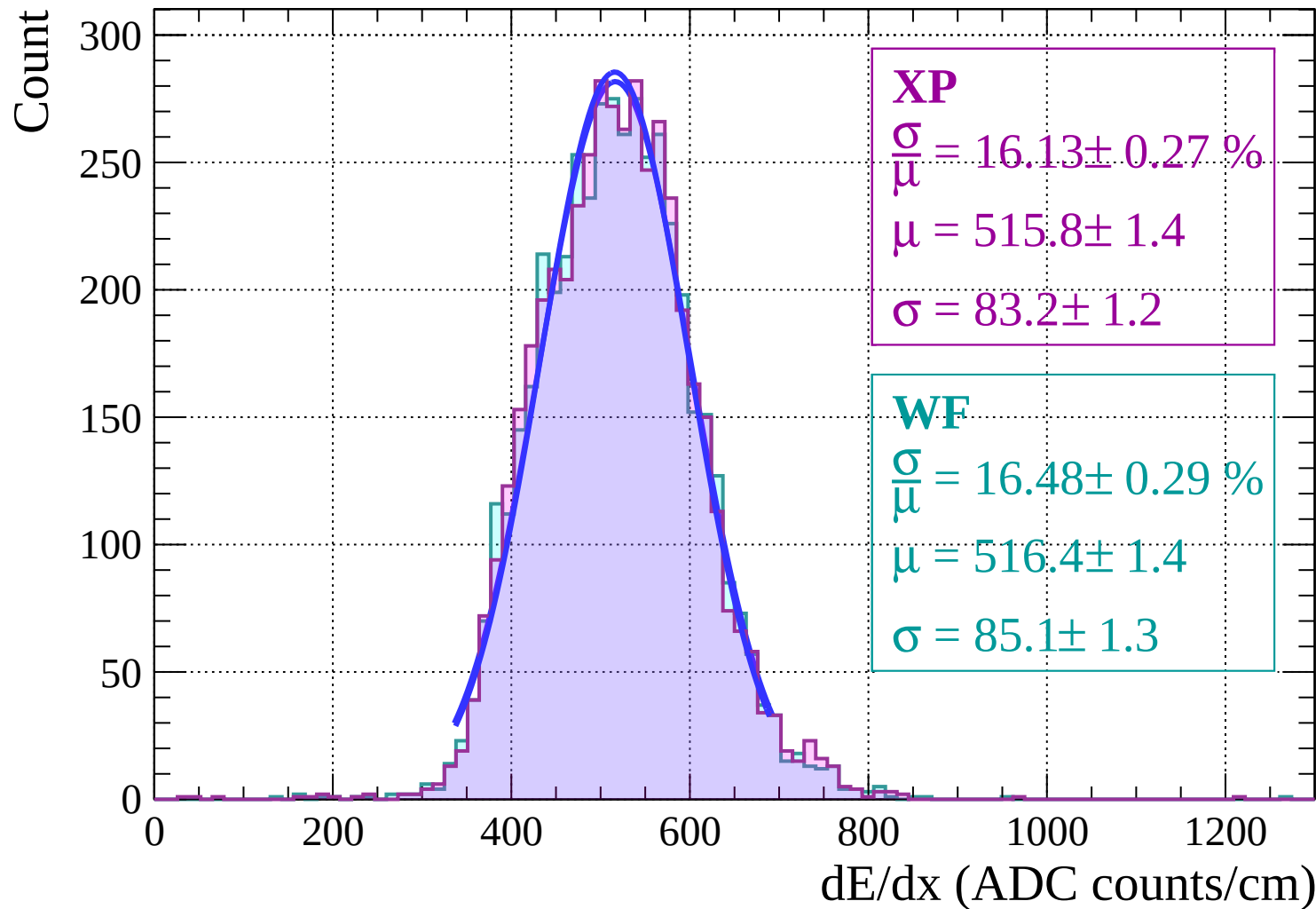
Count



Energy loss | $38 < \theta < 40$

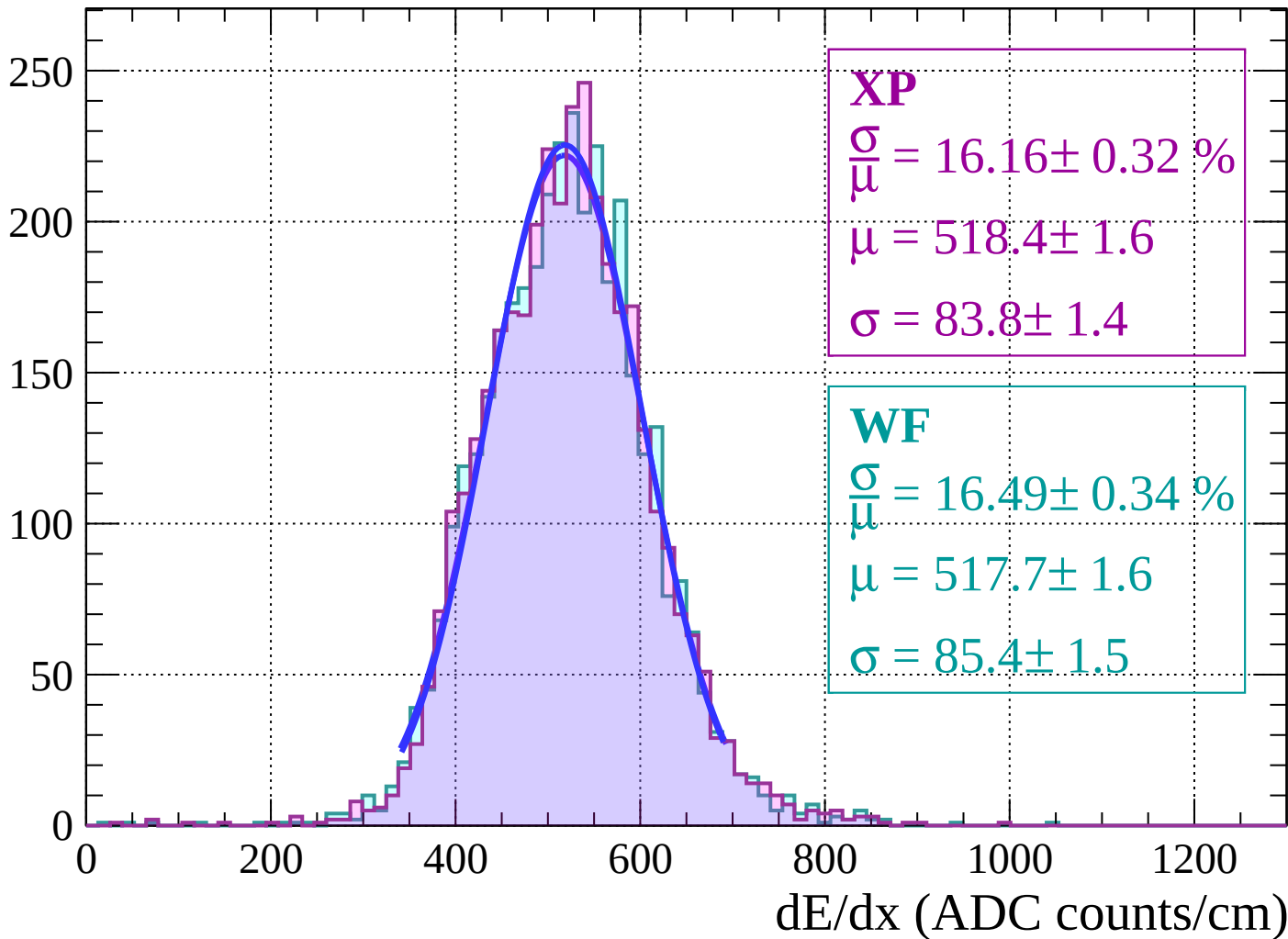


Energy loss | $40 < \theta < 42$

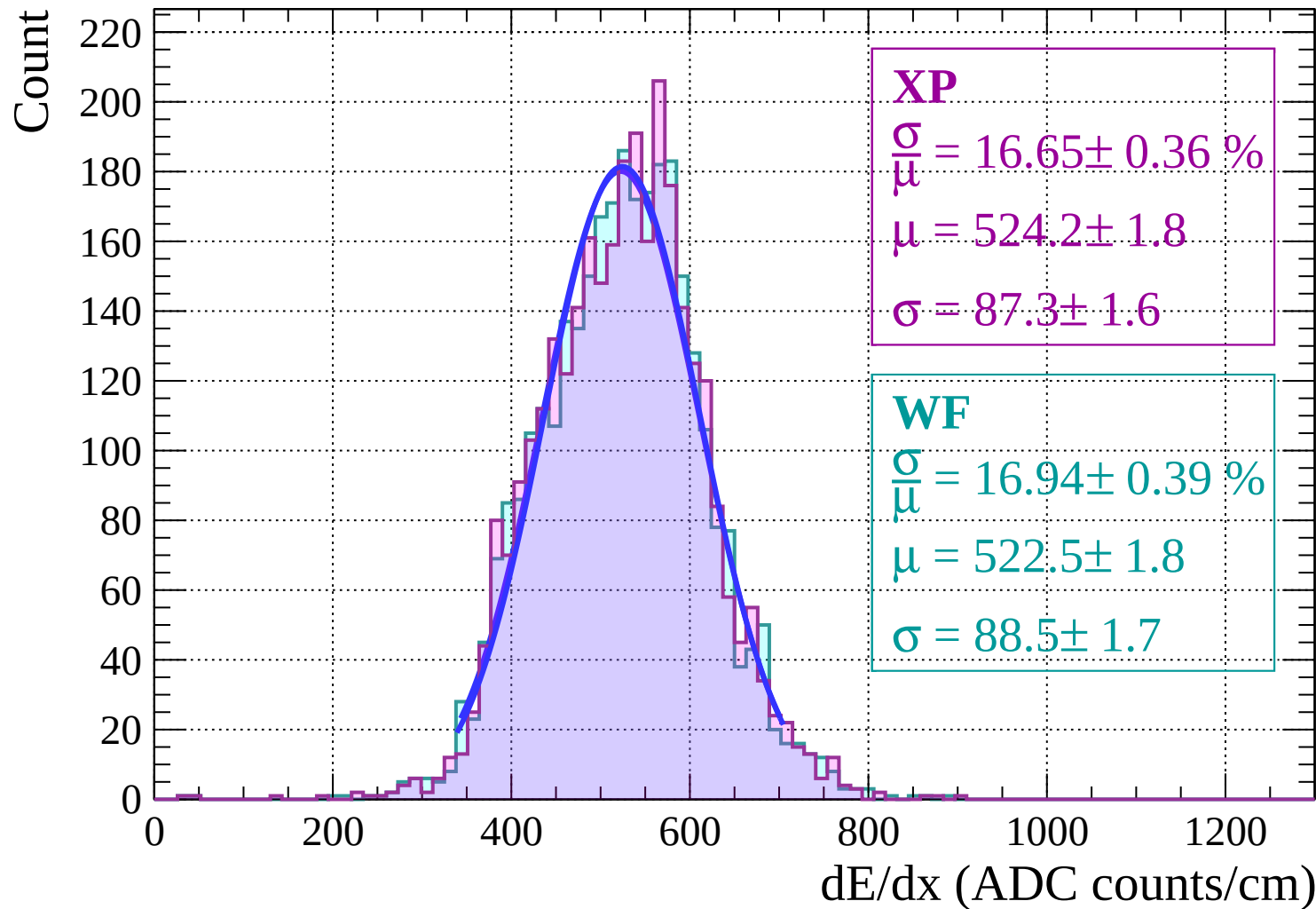


Energy loss | $42 < \theta < 44$

Count

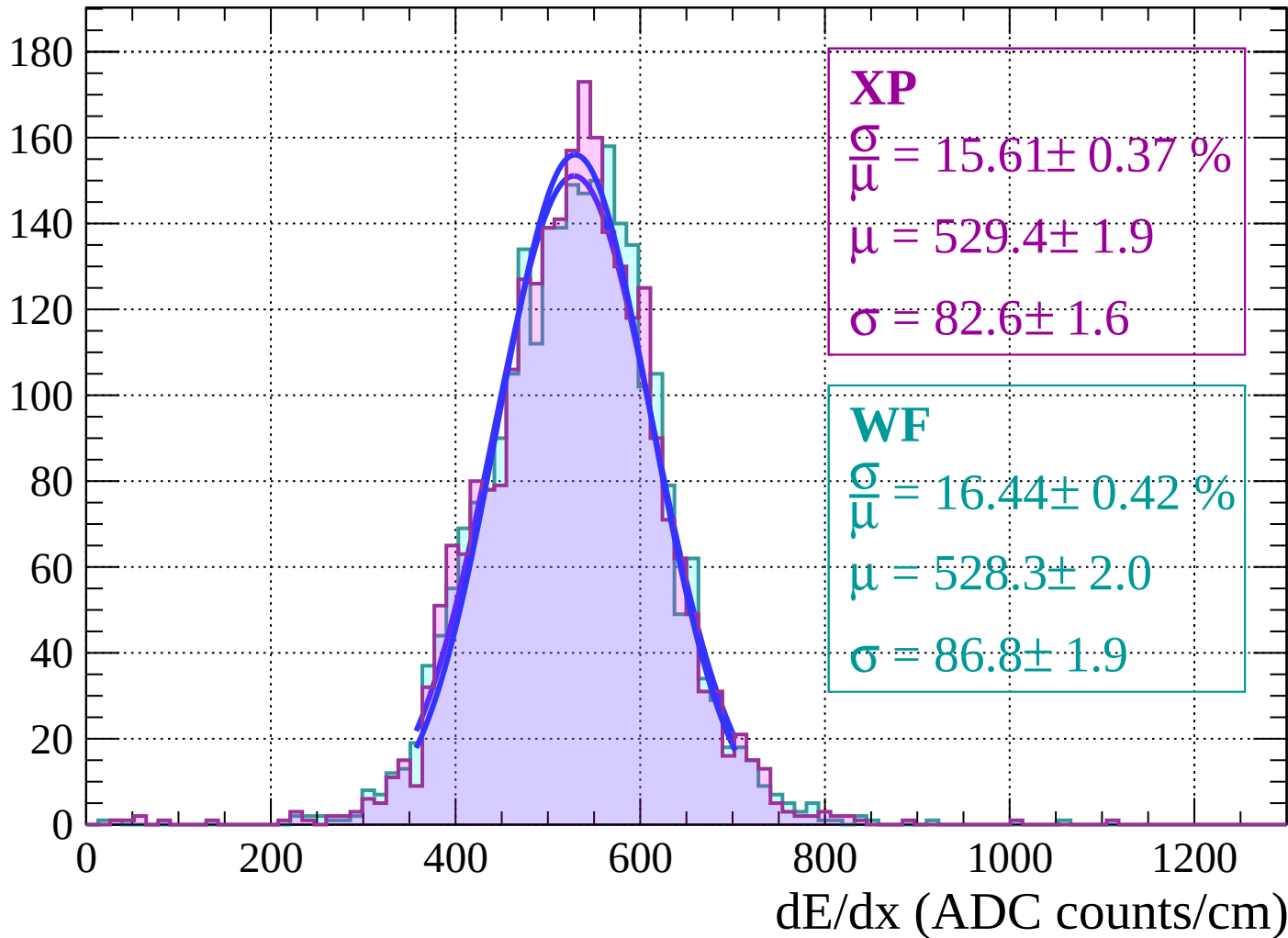


Energy loss | $44 < \theta < 46$



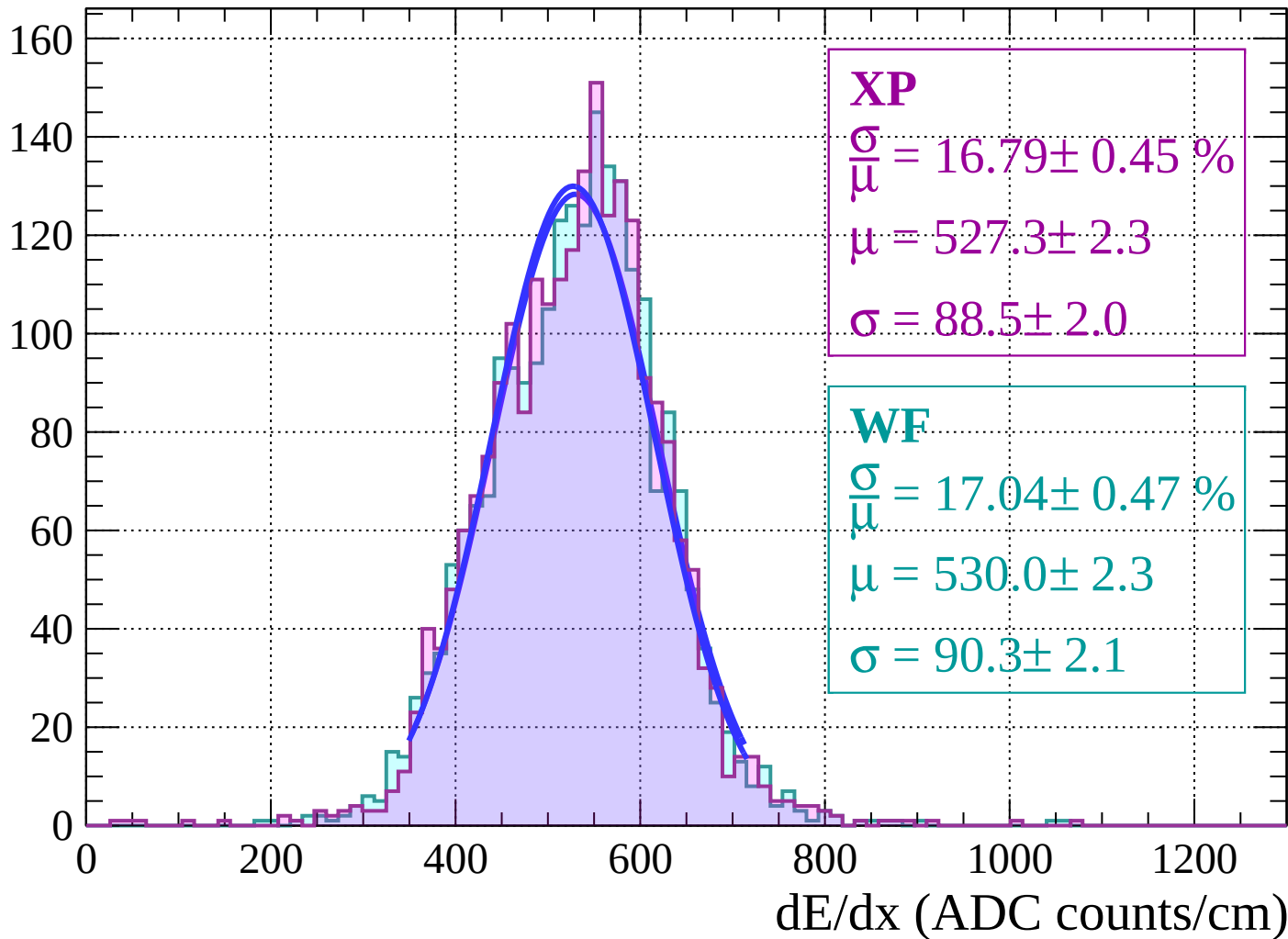
Energy loss | $46 < \theta < 48$

Count



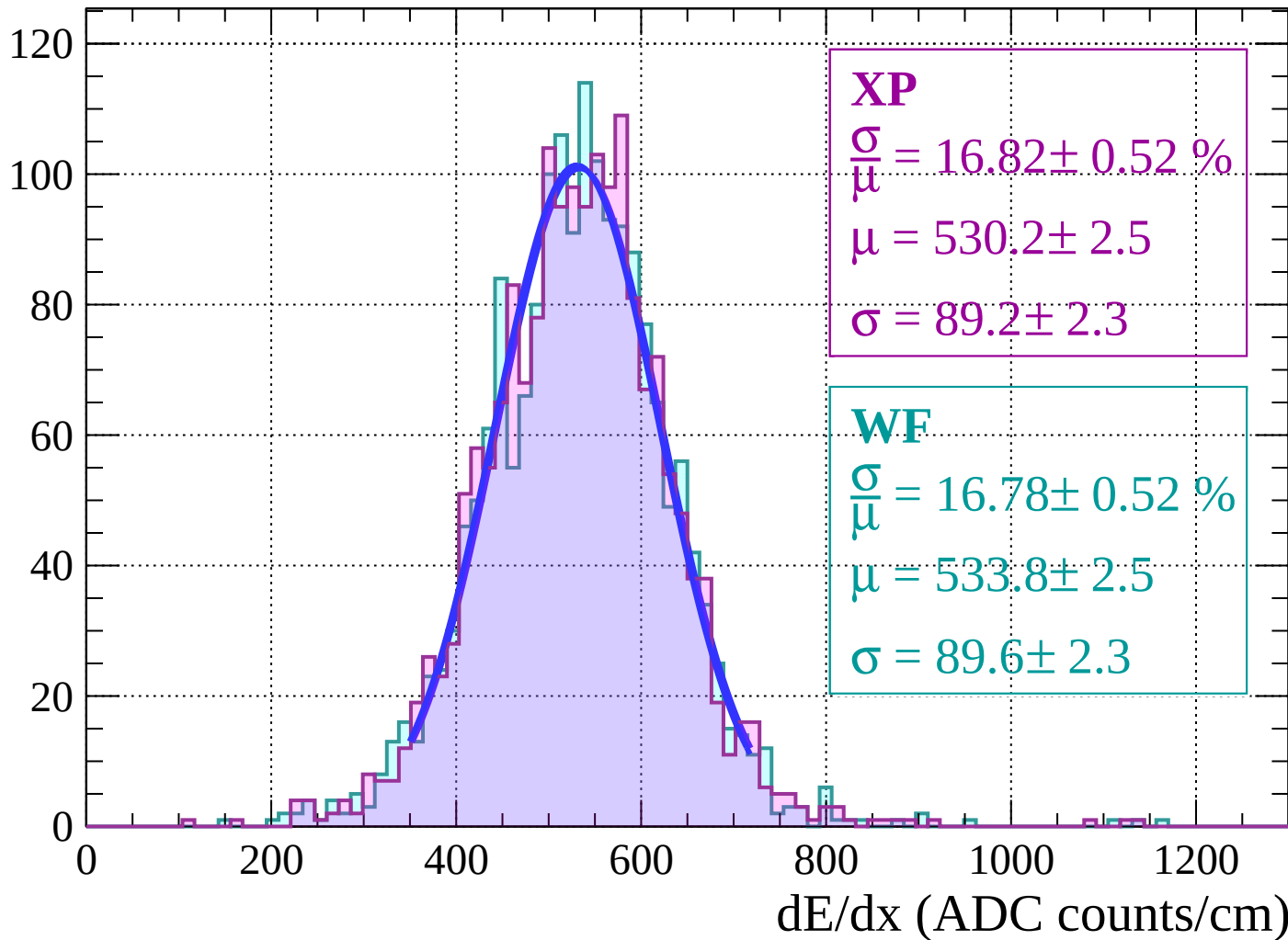
Energy loss | $48 < \theta < 50$

Count



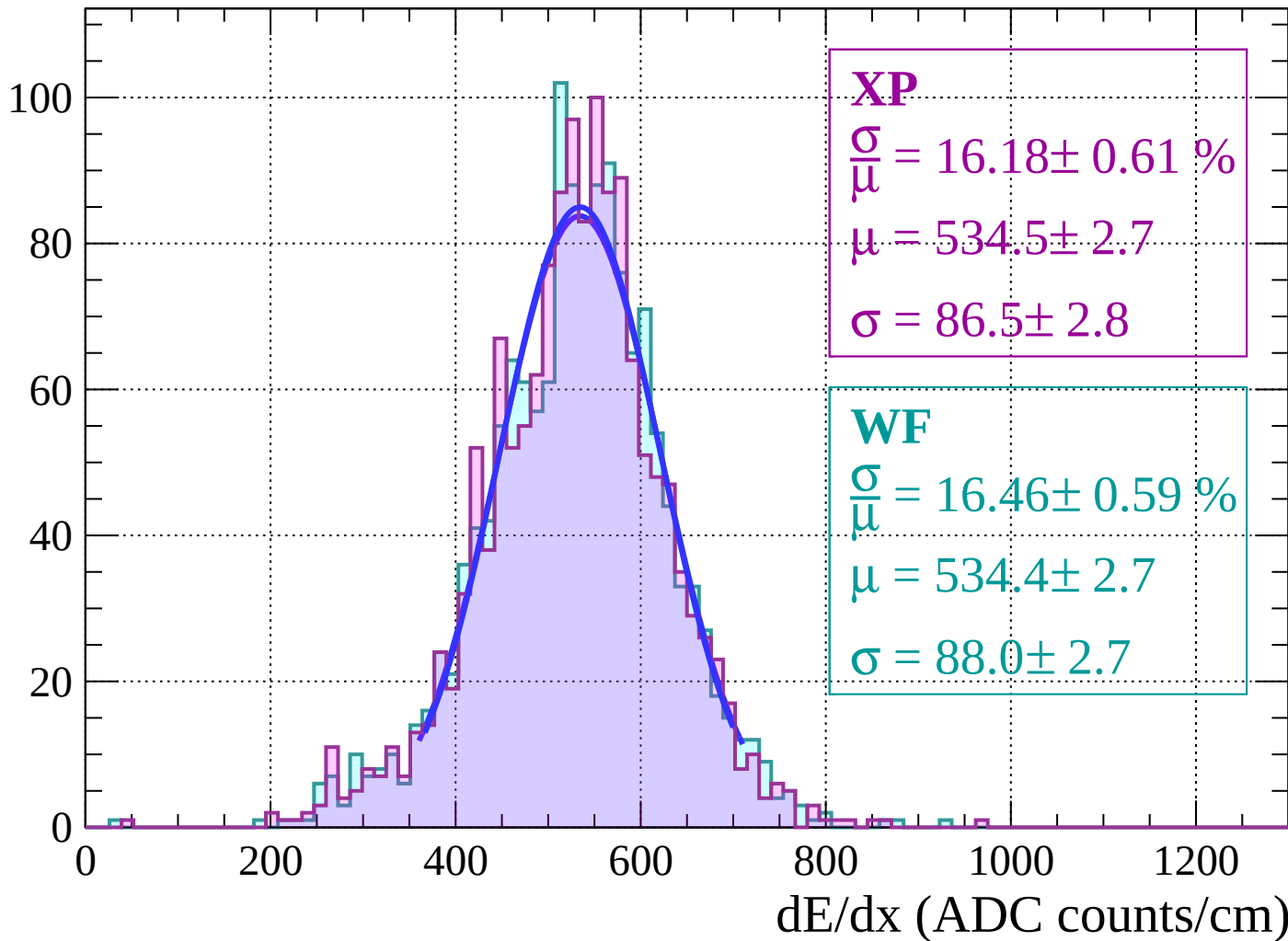
Energy loss | $50 < \theta < 52$

Count



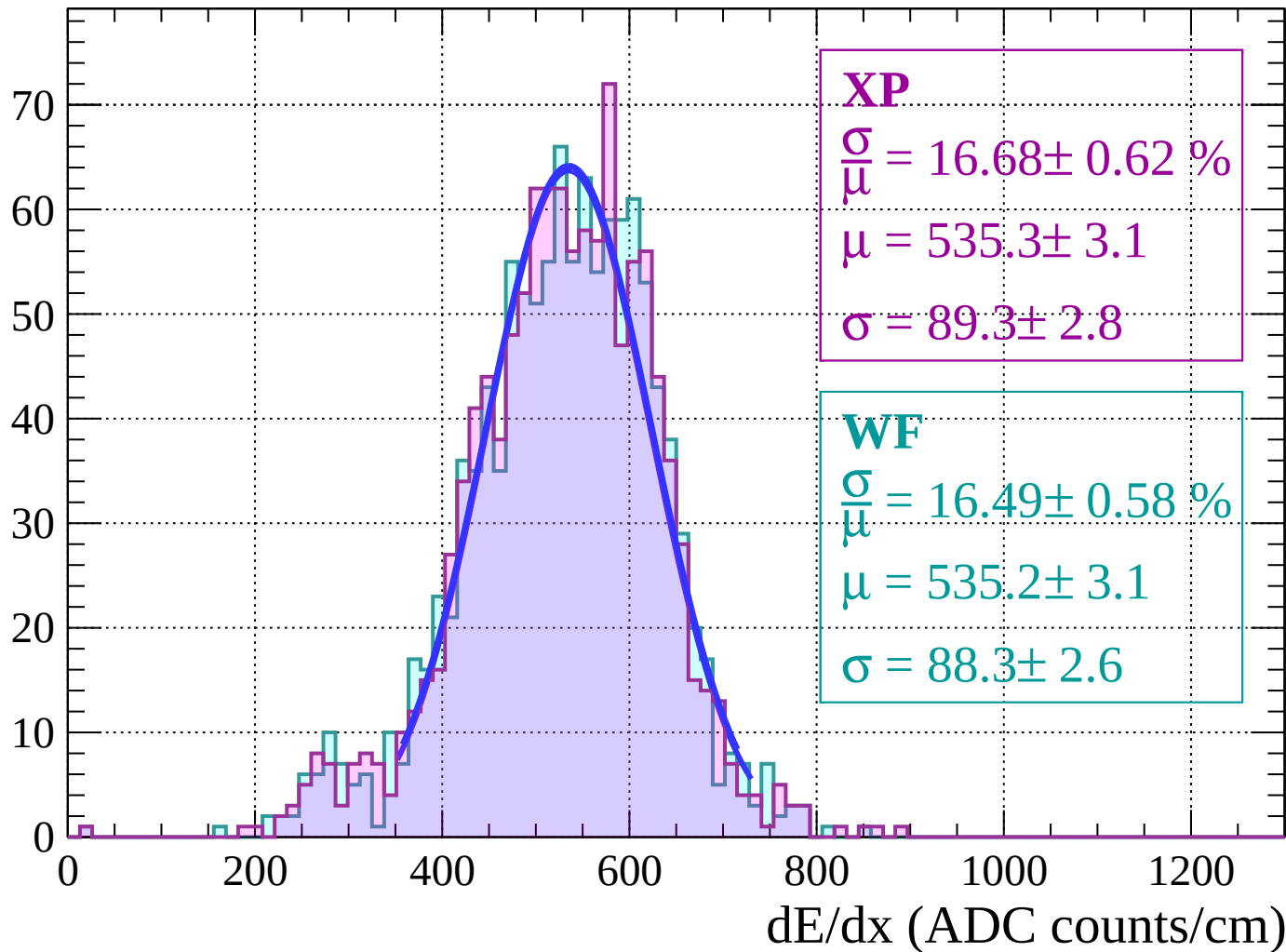
Energy loss | $52 < \theta < 54$

Count



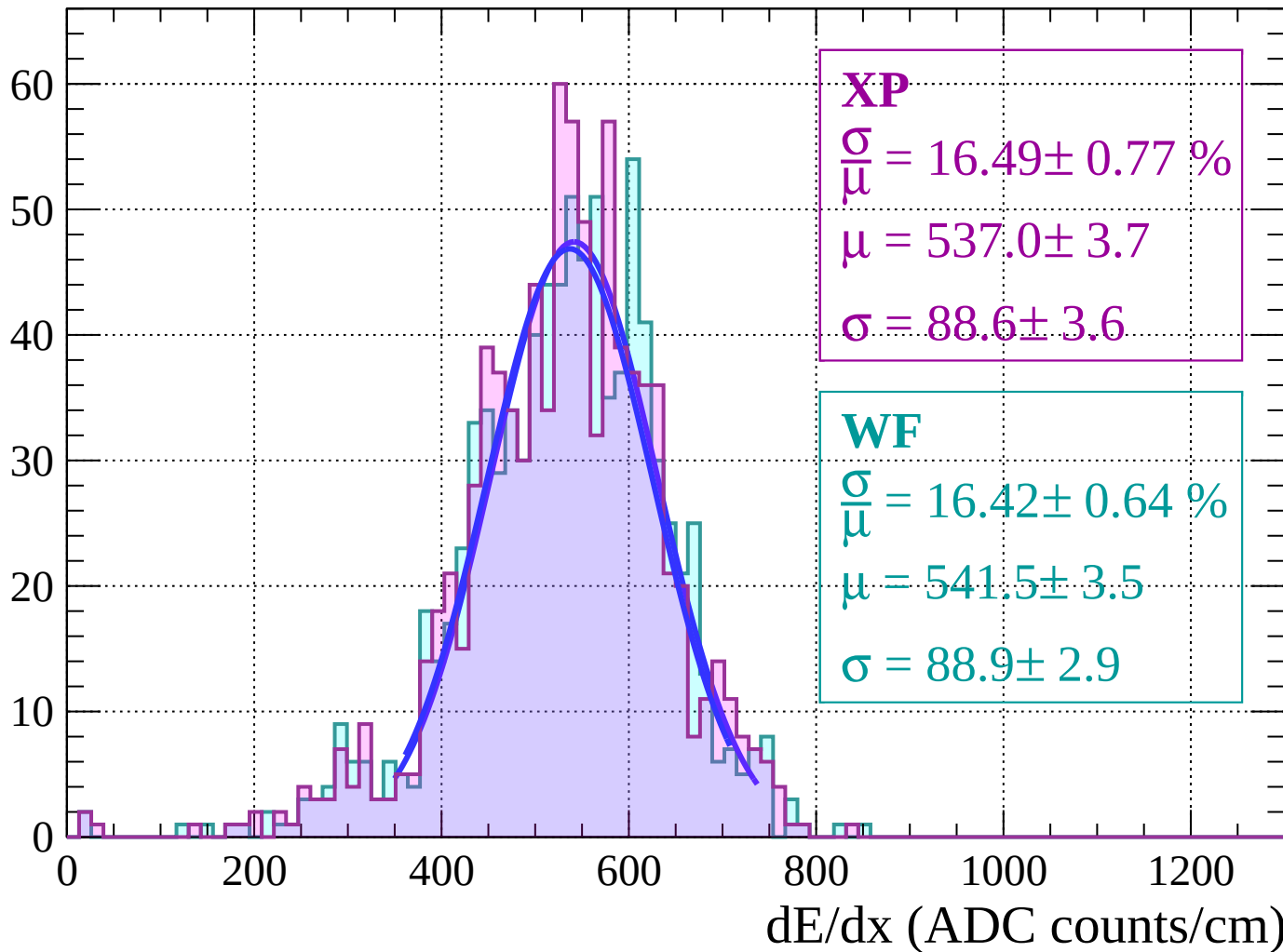
Energy loss | $54 < \theta < 56$

Count



Energy loss | $56 < \theta < 58$

Count



Energy loss | $58 < \theta < 60$

Count

XP

$$\frac{\sigma}{\mu} = 15.74 \pm 1.01 \%$$

$$\mu = 530.6 \pm 4.3$$

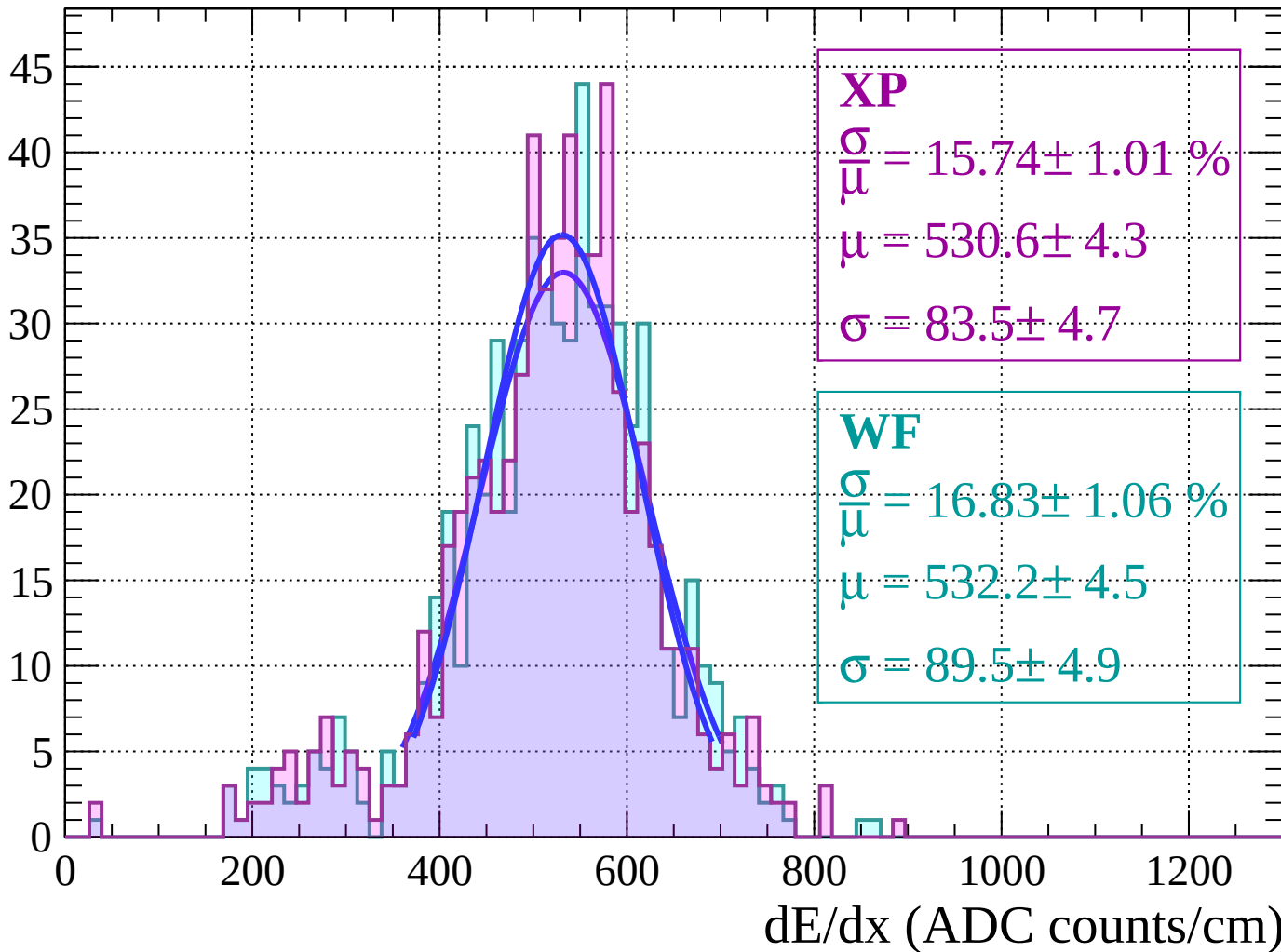
$$\sigma = 83.5 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 16.83 \pm 1.06 \%$$

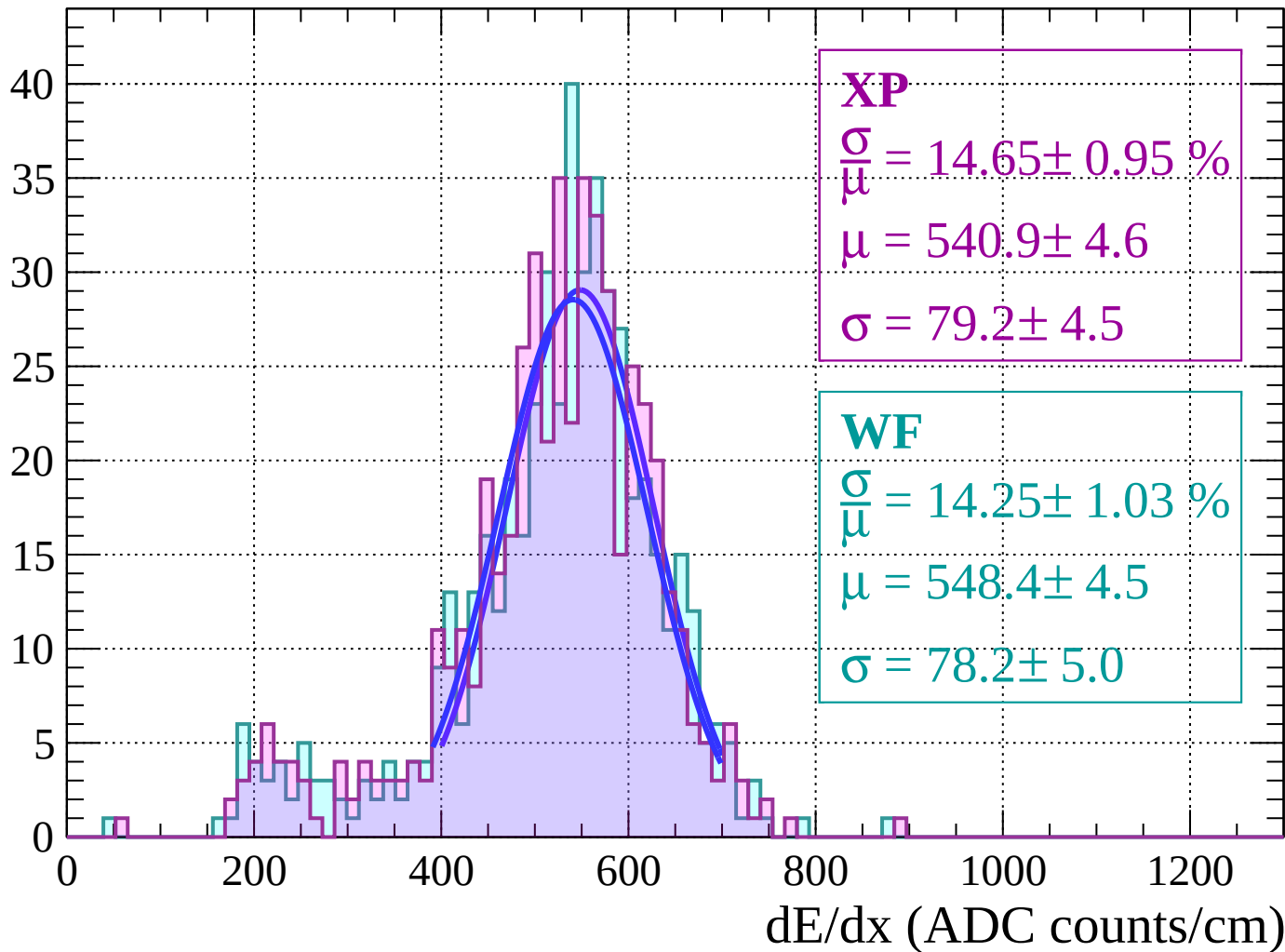
$$\mu = 532.2 \pm 4.5$$

$$\sigma = 89.5 \pm 4.9$$



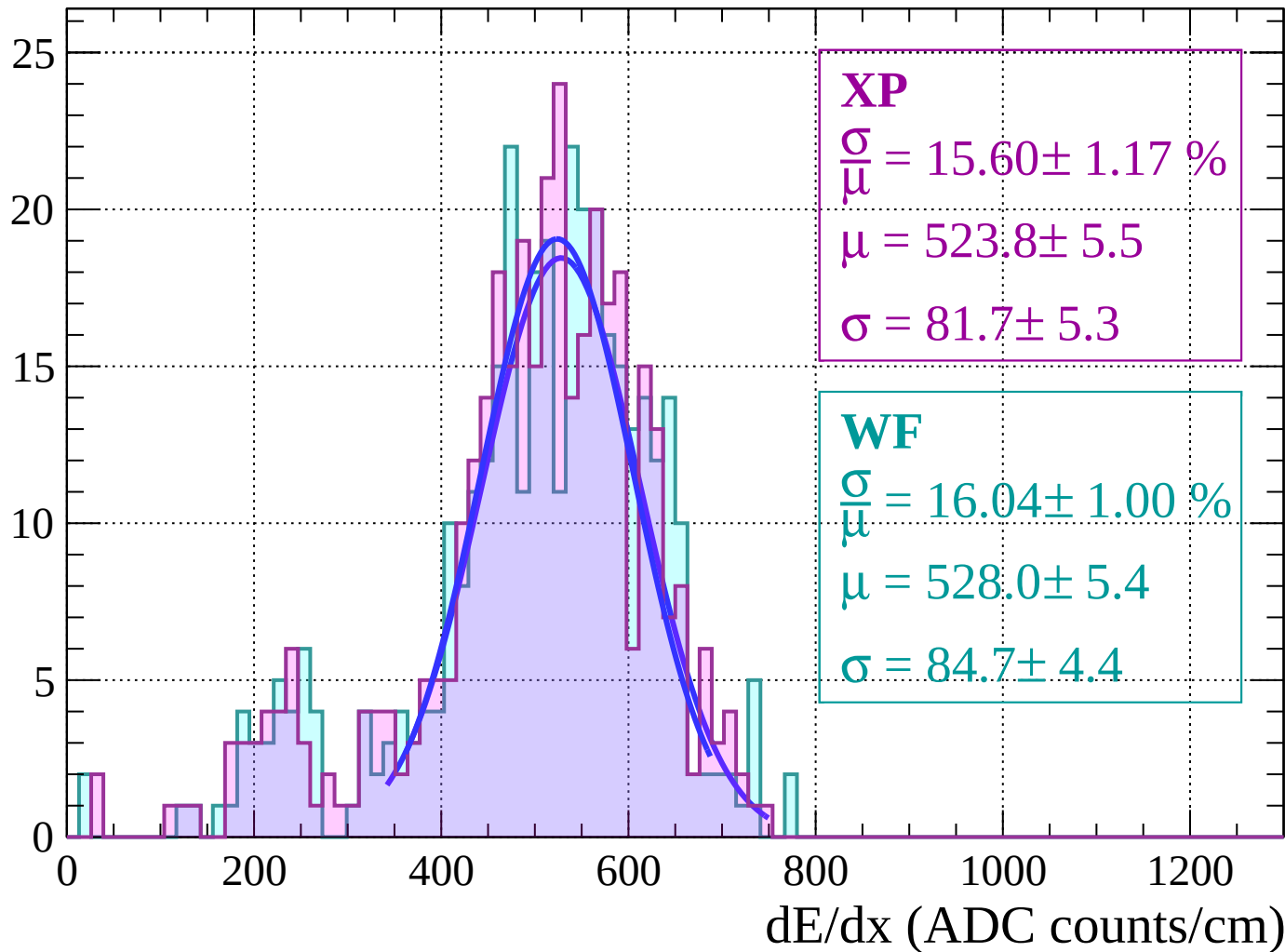
Energy loss | $60 < \theta < 62$

Count



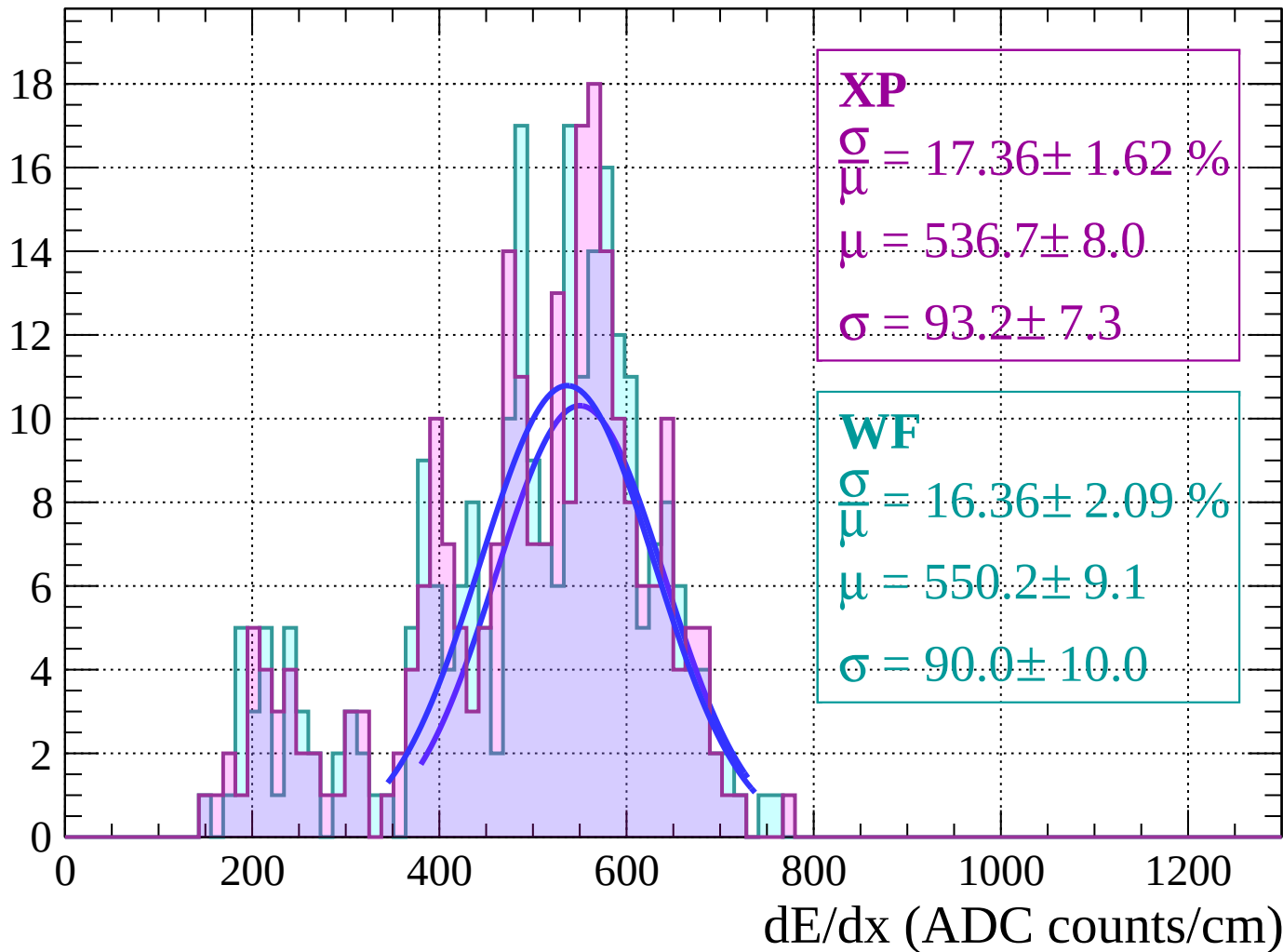
Energy loss | $62 < \theta < 64$

Count



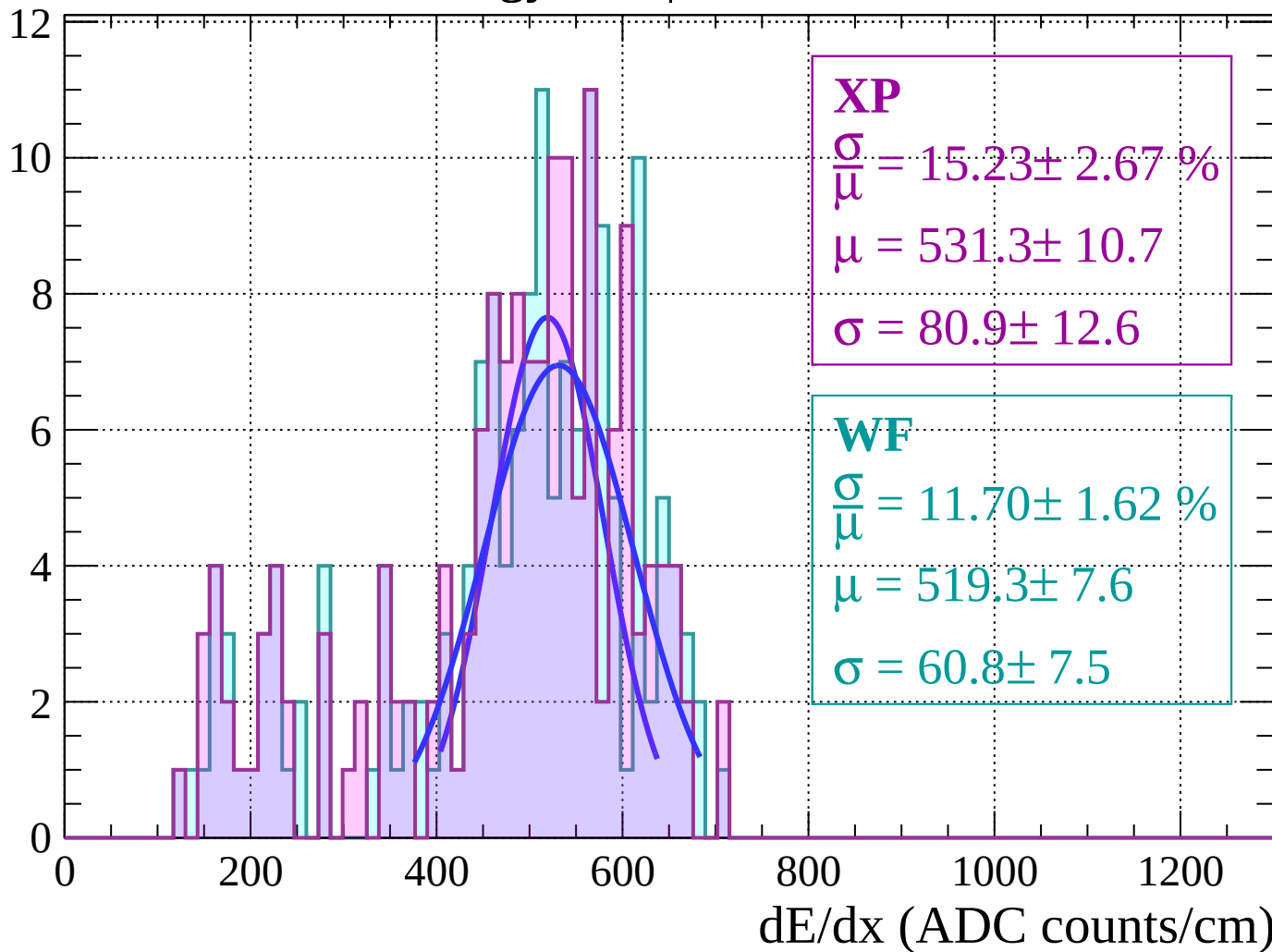
Energy loss | $64 < \theta < 66$

Count



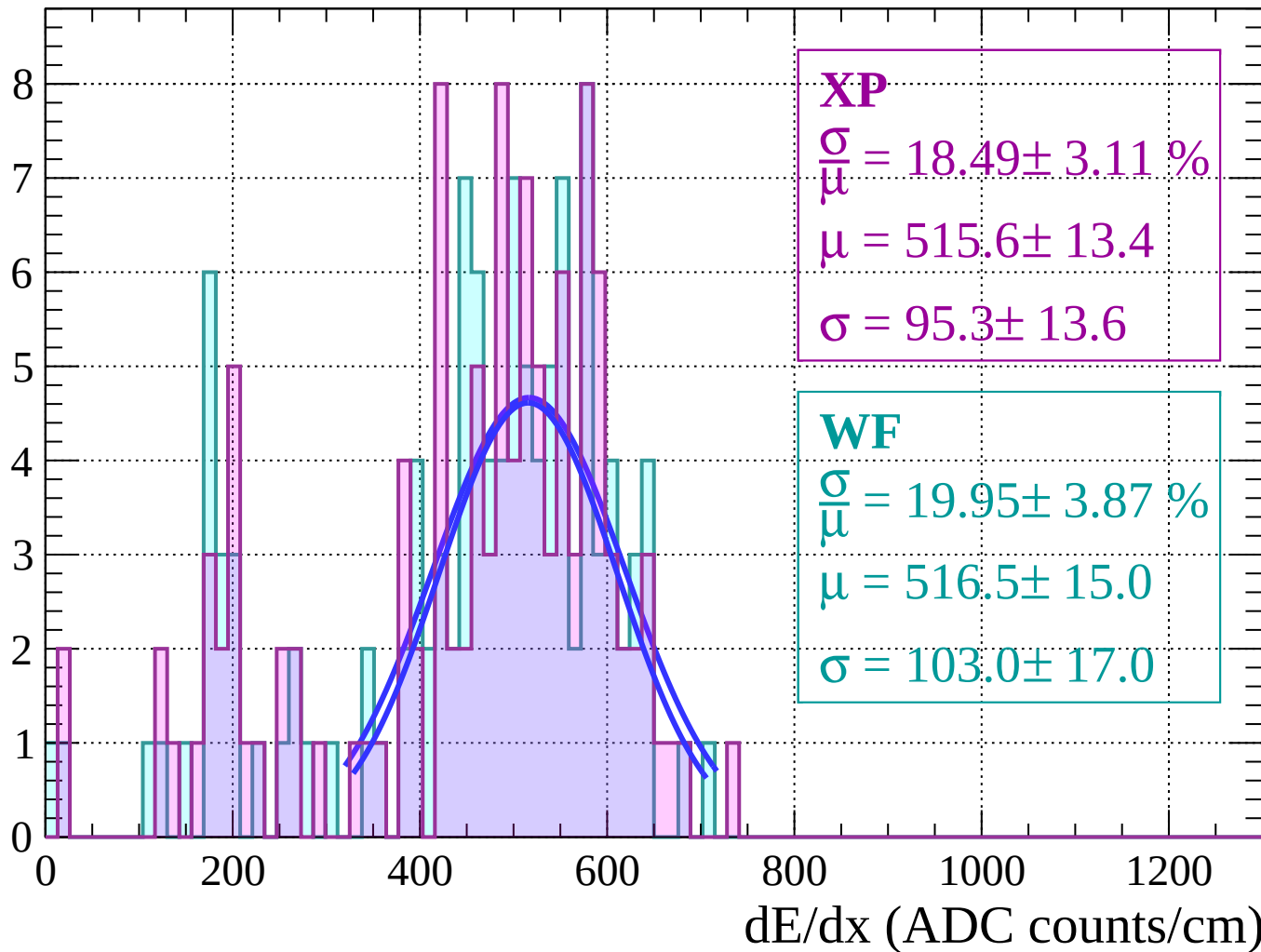
Energy loss | $66 < \theta < 68$

Count



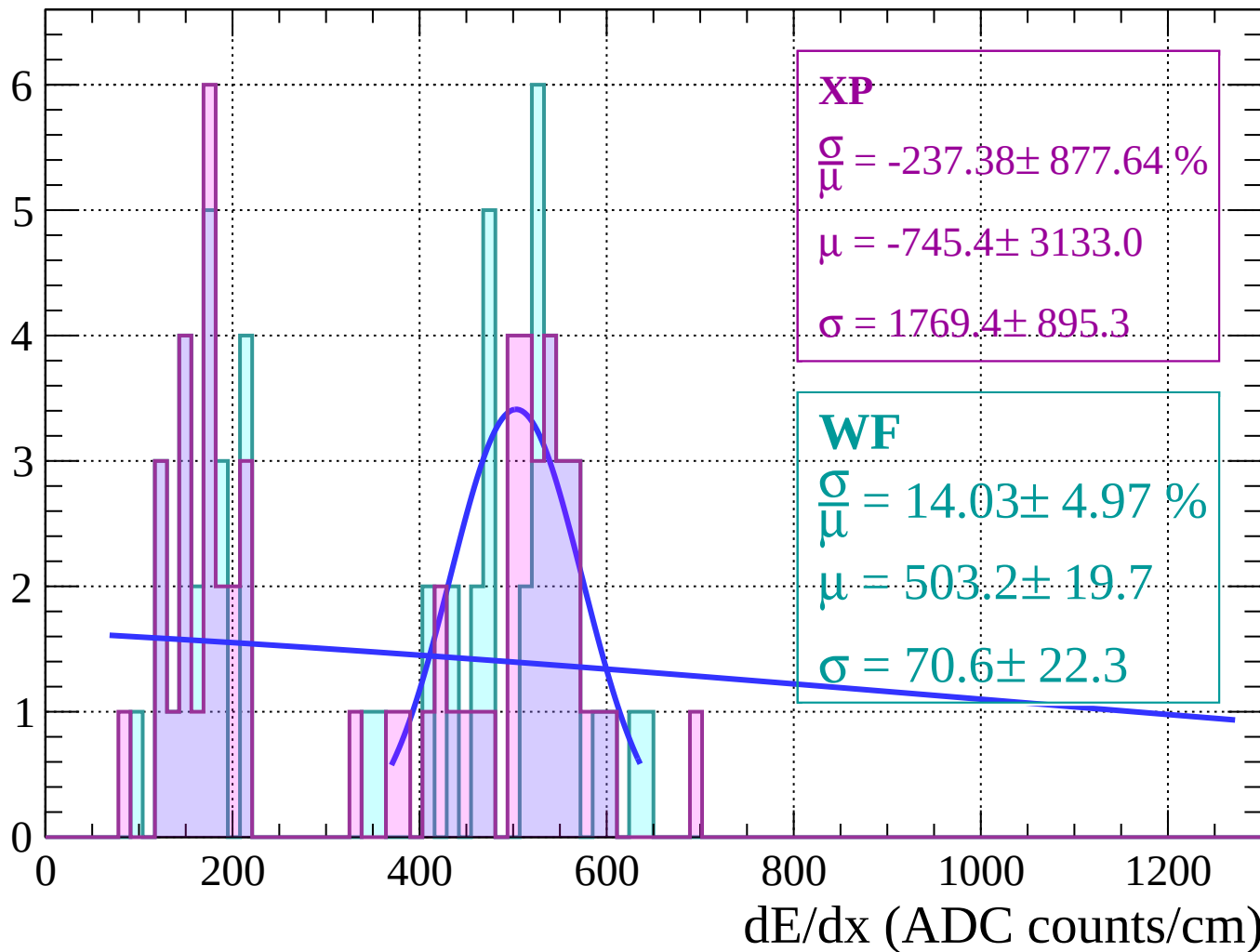
Energy loss | $68 < \theta < 70$

Count



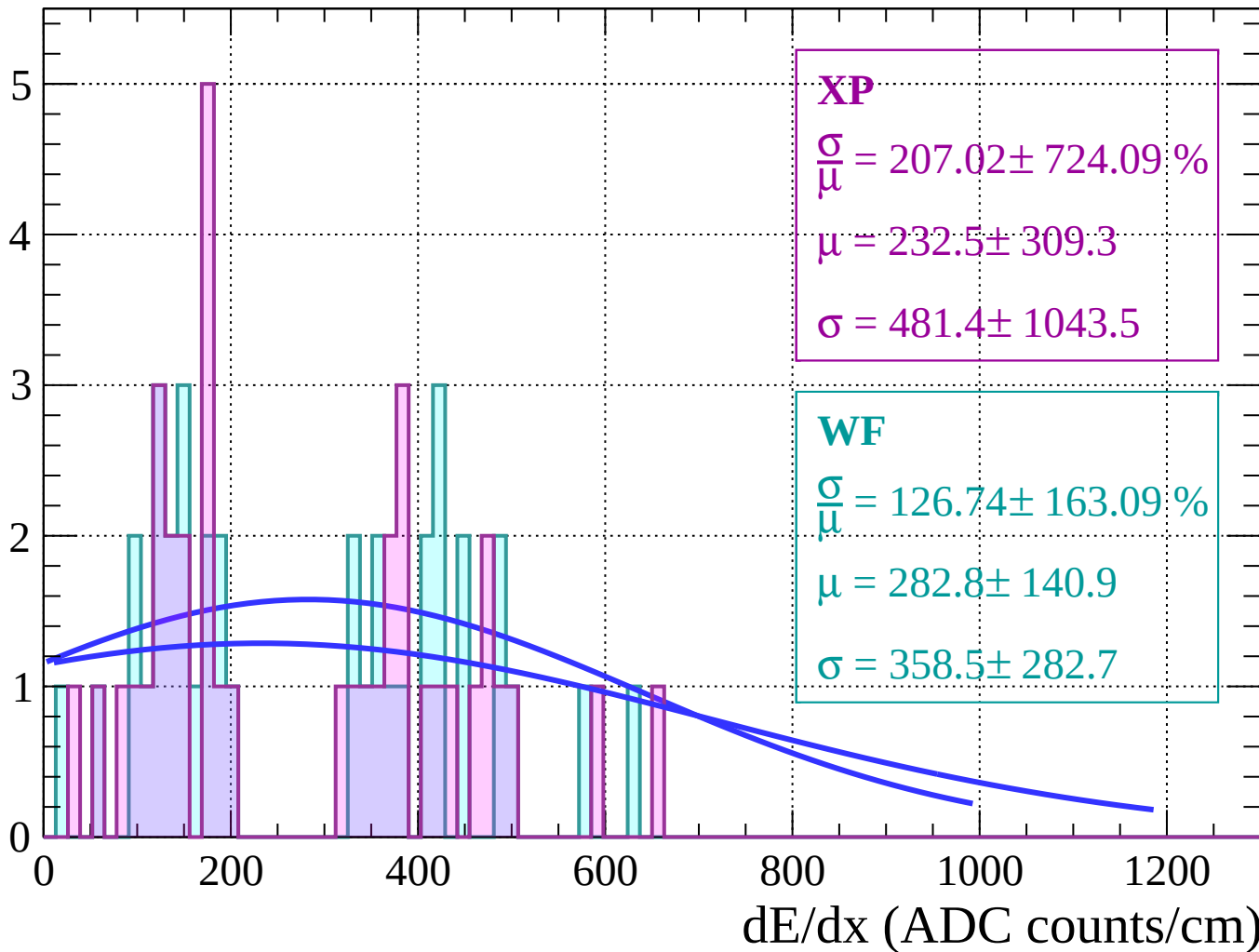
Energy loss | $70 < \theta < 72$

Count

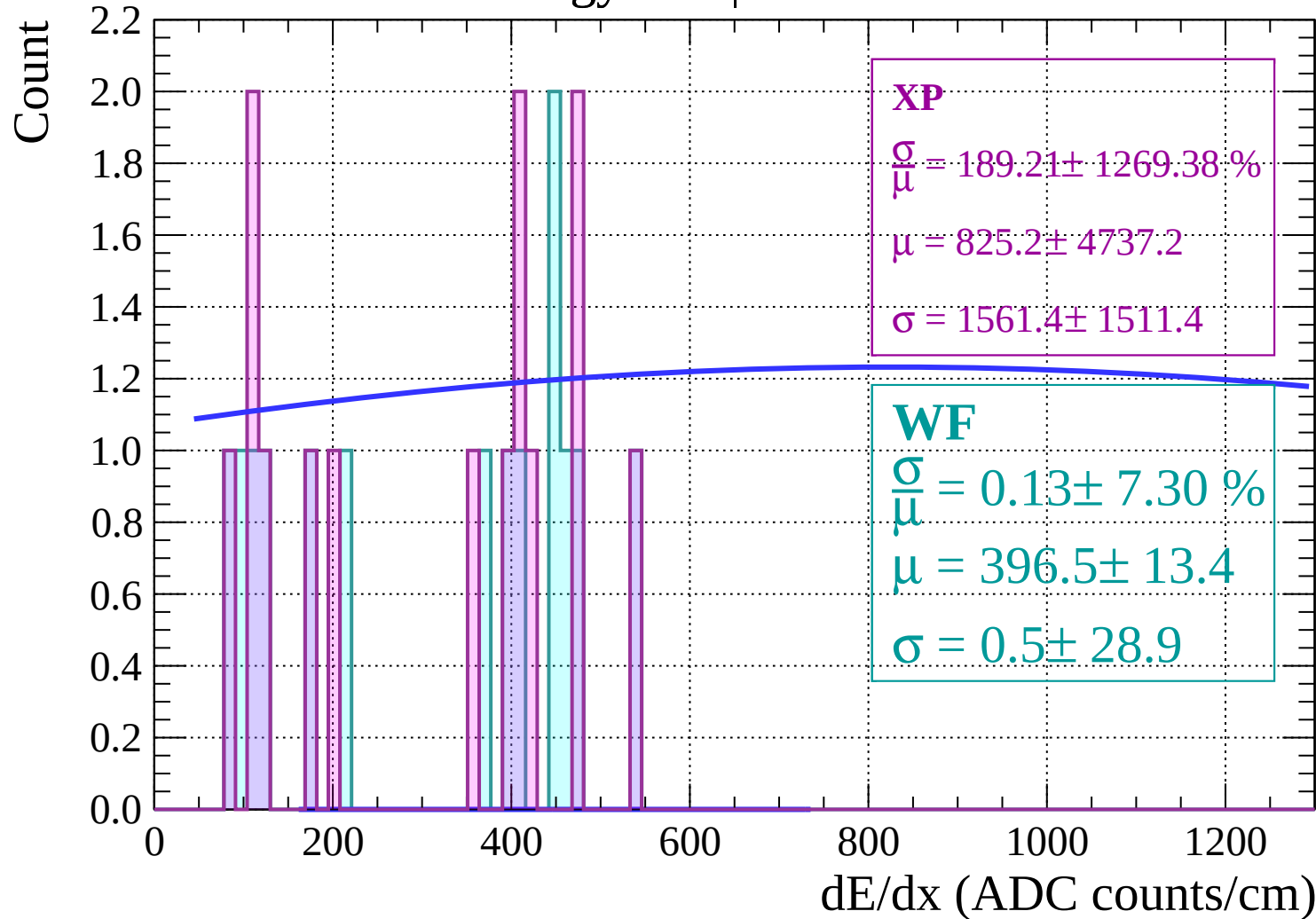


Energy loss | $72 < \theta < 74$

Count

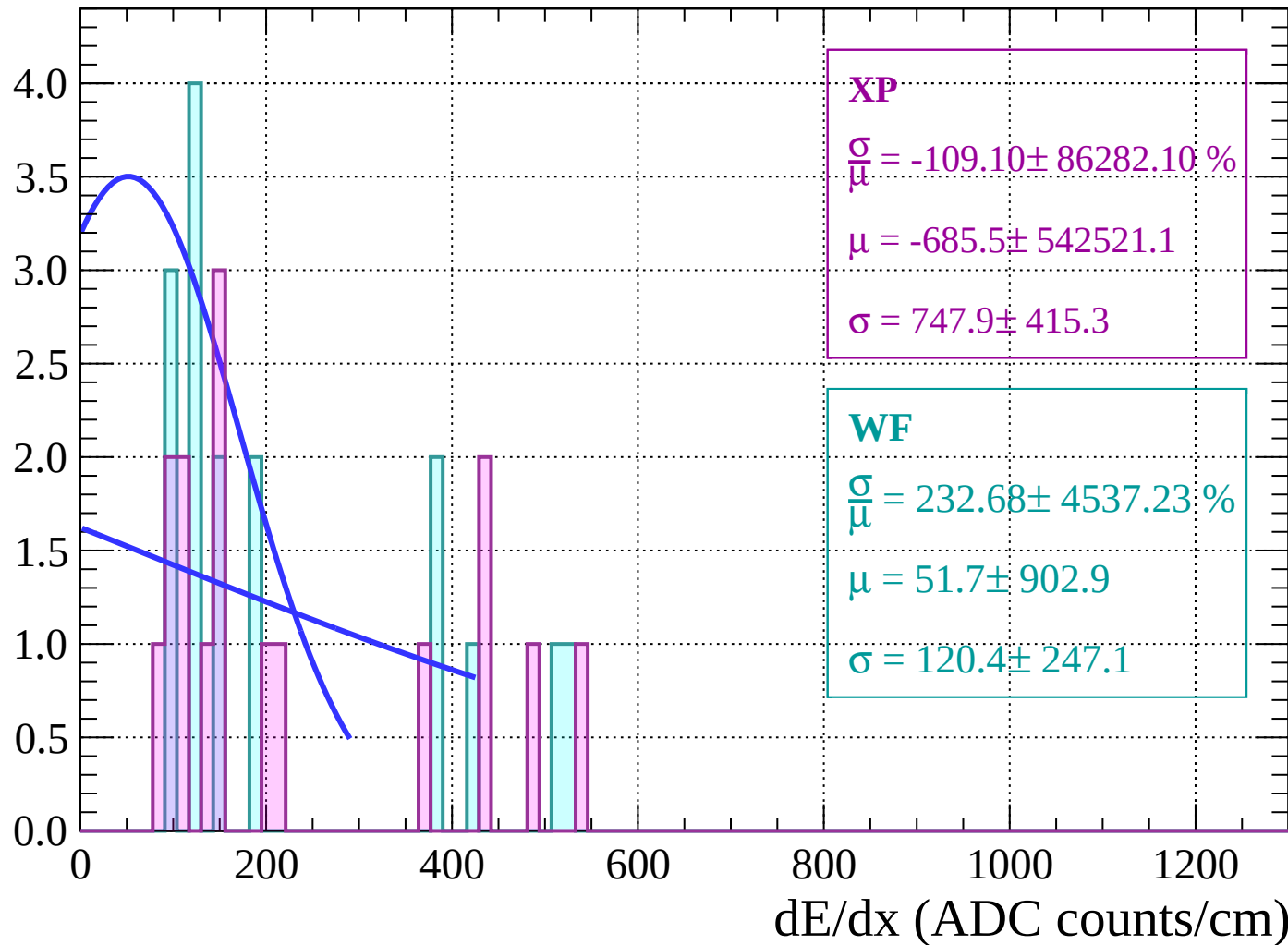


Energy loss | $74 < \theta < 76$

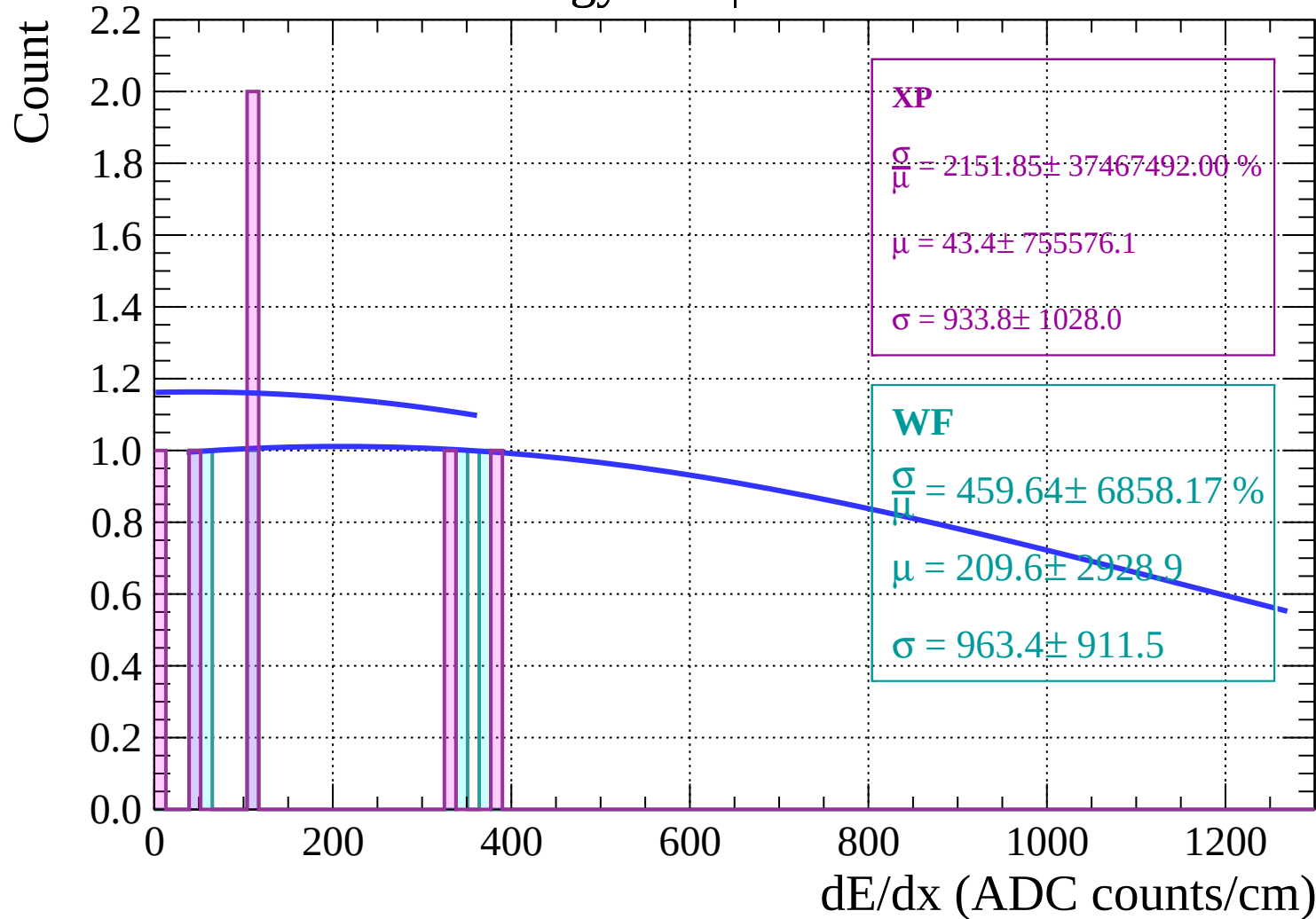


Energy loss | $76 < \theta < 78$

Count

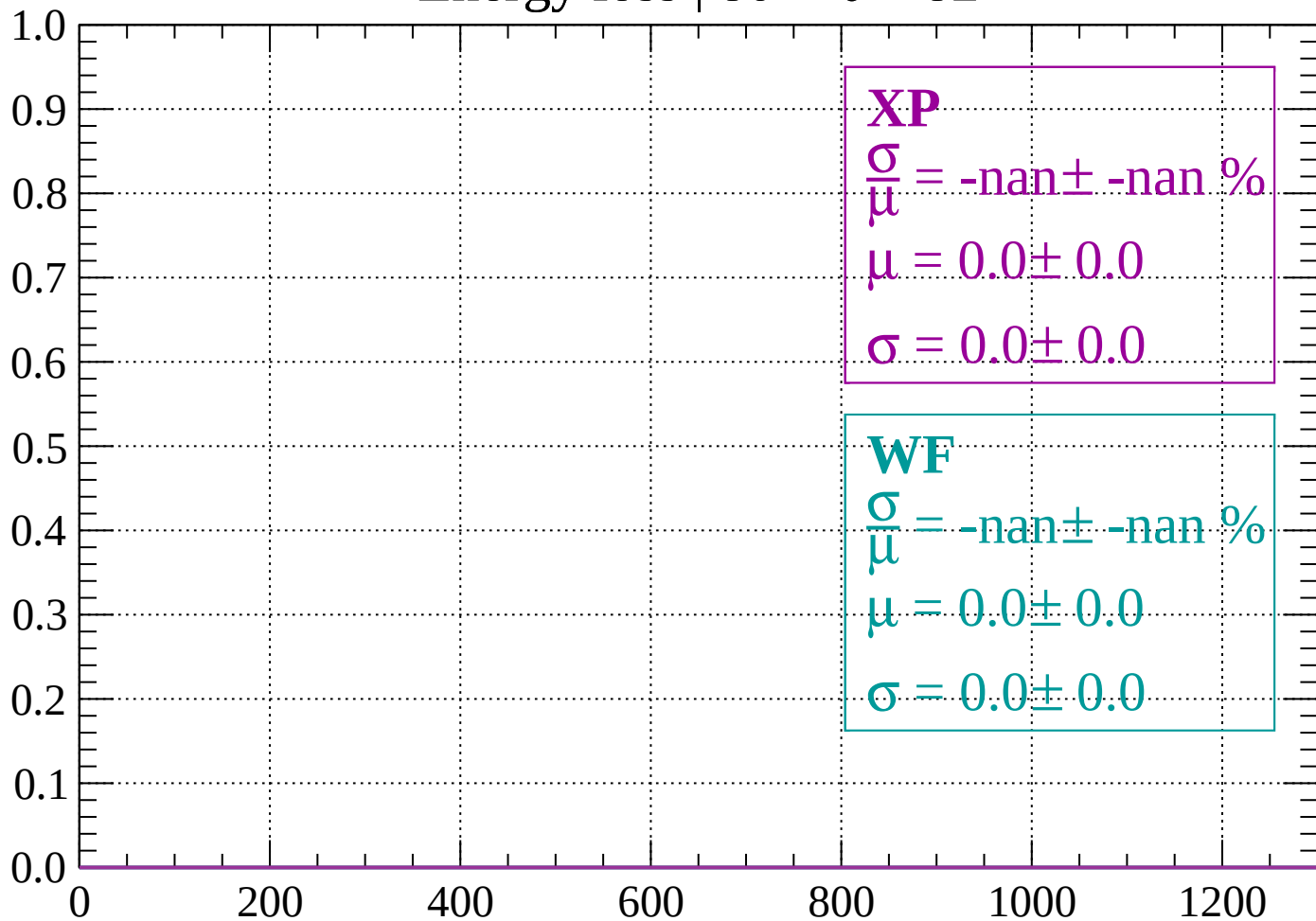


Energy loss | $78 < \theta < 80$



Energy loss | $80 < \theta < 82$

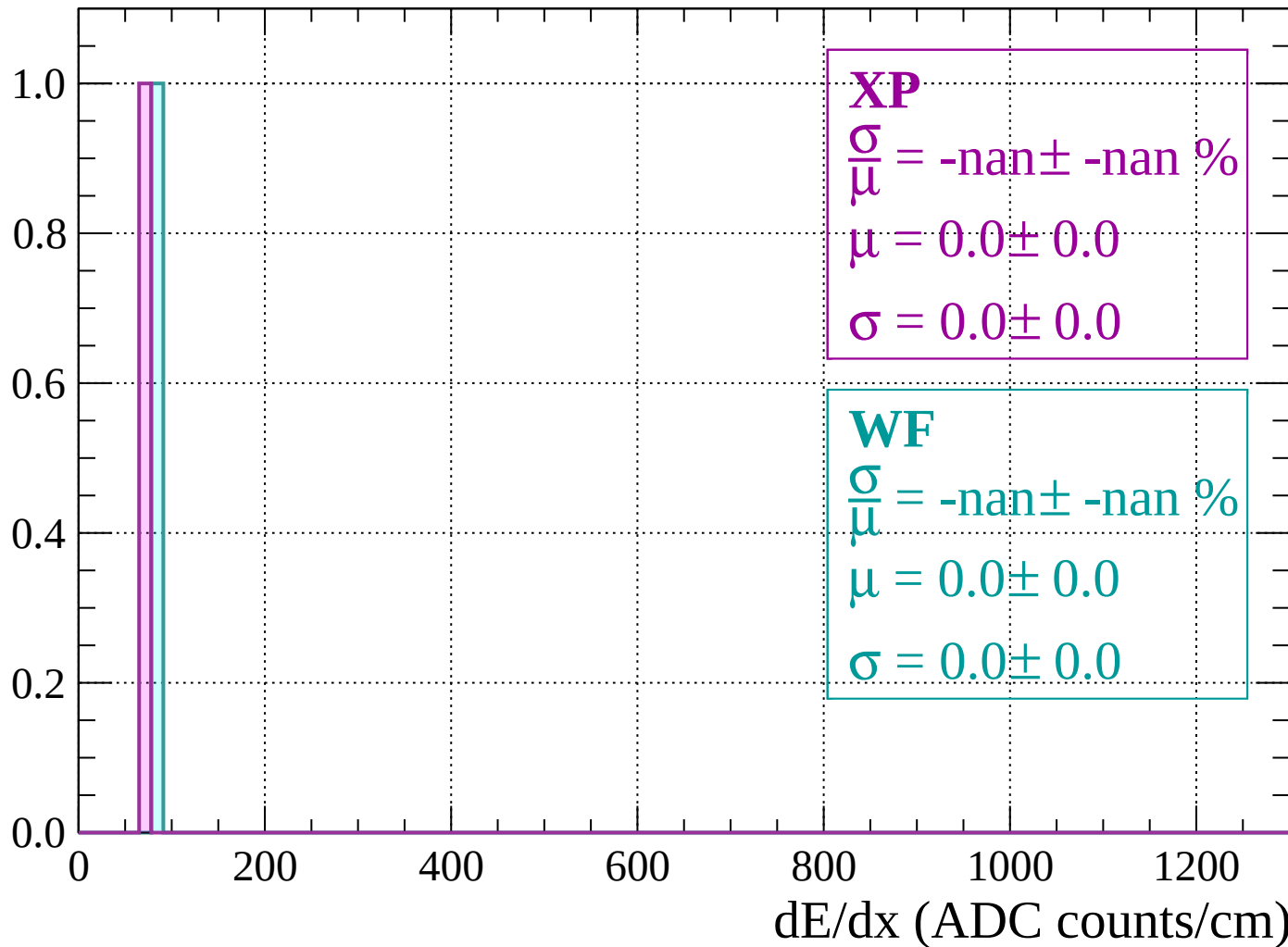
Count



dE/dx (ADC counts/cm)

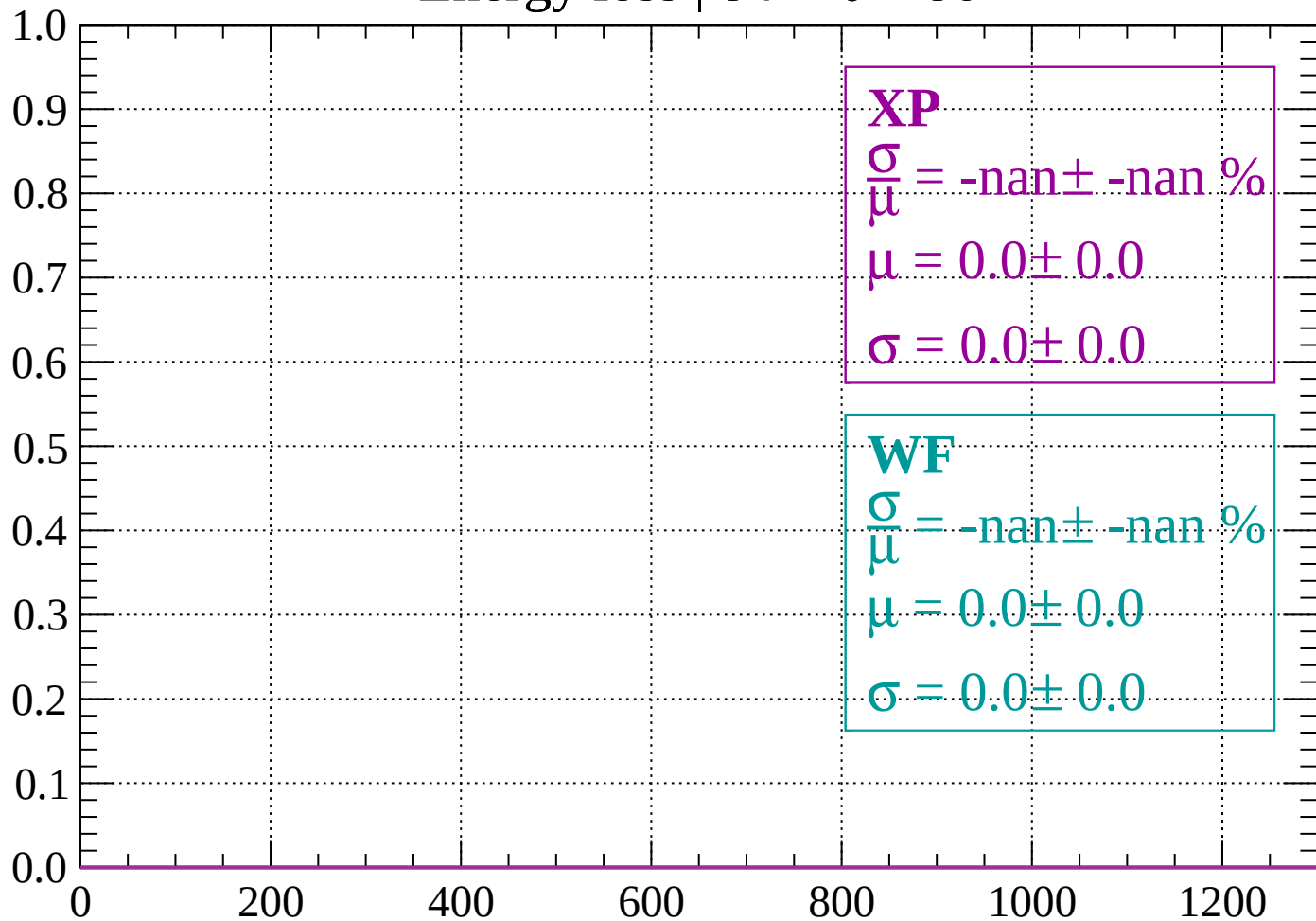
Energy loss | $82 < \theta < 84$

Count



Energy loss | $84 < \theta < 86$

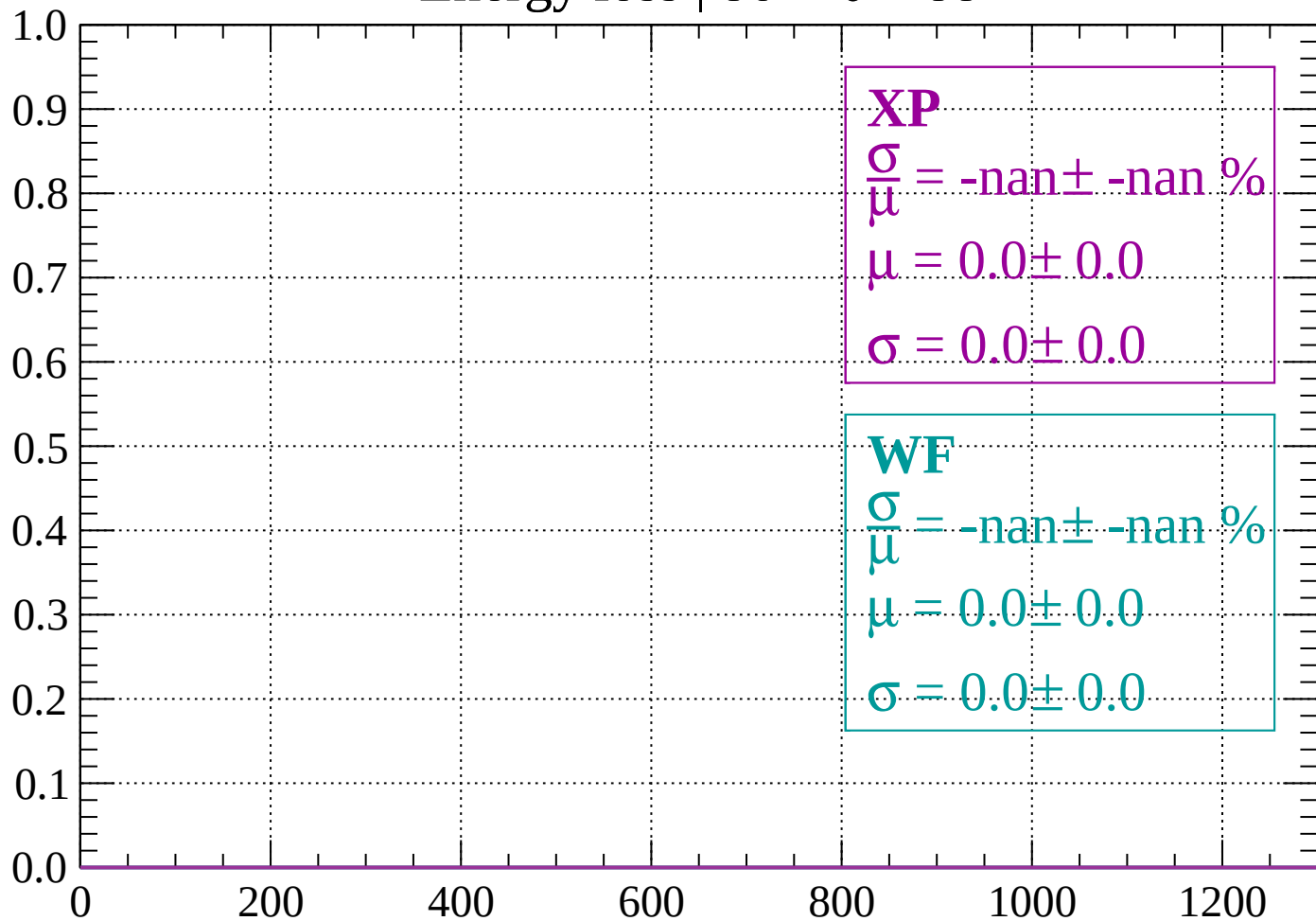
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

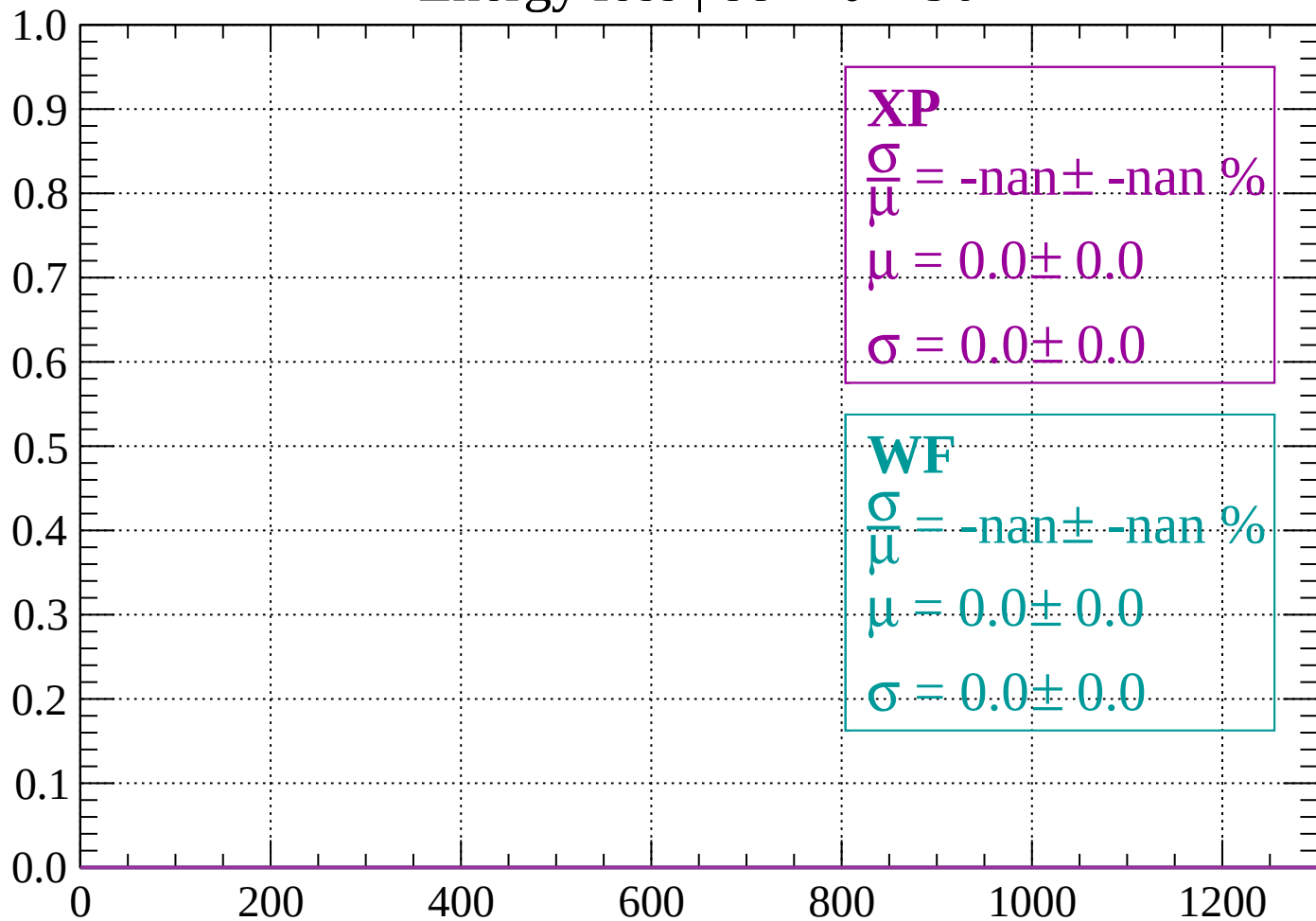
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

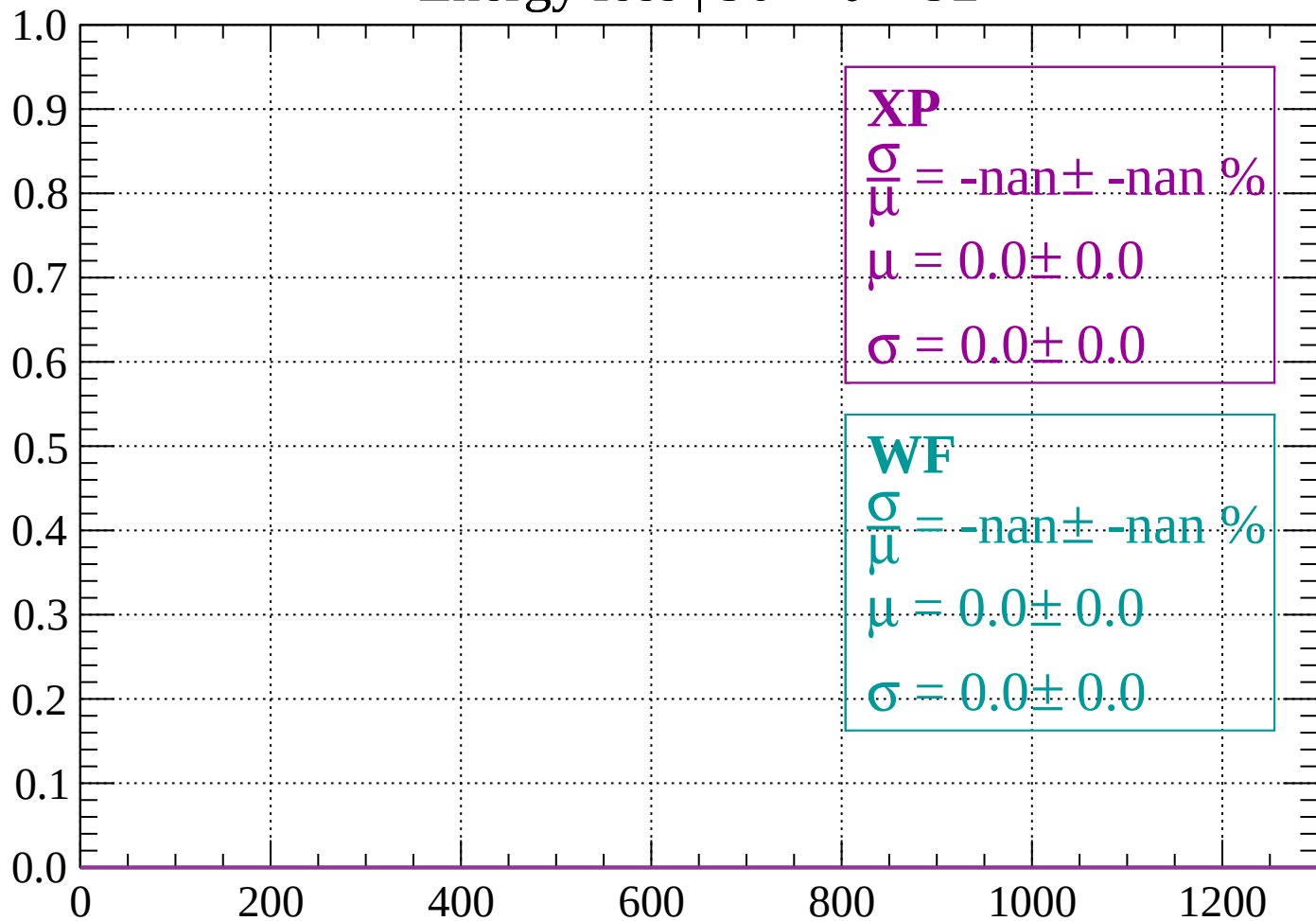
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

